





TENTH EDITION

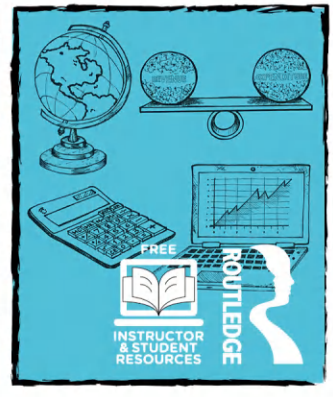
SUSAN CAPEL AND MARILYN LEASK'S

LEARNING TO TEACH IN THE SECONDARY SCHOOL

A Companion to School Experience

Edited by **SARAH YOUNIE, ELIZABETH HIDSON, JULIA LAWRENCE** and **NIKKI BOOTH**



LEARNING TO TEACH IN THE SECONDARY SCHOOL

This market leading text for beginning teachers on all undergraduate, postgraduate and school-based routes to qualified teacher status provides an essential introduction to the key skills and knowledge needed to become a successful teacher. Offering advice on all aspects of teaching and learning, this tenth edition has been thoroughly updated to reflect changes in the field and covers key new topics including artificial intelligence (AI), online pedagogies and working with your mentor. There are also expanded units on diversity and inclusion, teacher wellbeing and professional development.

The text includes a wealth of examples, resources and tasks to support you in successfully applying theory to practice, and in critically reflecting on and analysing your practice to maximise pupil learning. The wide range of pedagogical features supports both school- and university-based work up to Master's level. Written by experts in the field, the 42 concise units are underpinned by evidence-informed practice and focus on what you need to know to thrive in the classroom, including:

- lesson planning;
- curriculum;
- managing behaviour;
- digital technologies and AI;
- effective communication with pupils;
- how pupils learn, cognitive development, educational neuroscience;
- assessment, marking and feedback;
- neurodivergent pupils, diversity and inclusion;
- special educational needs and disabilities (SEND);
- building resilience, managing stress, workload and time;
- applying for jobs and early career development.

The book is extended and enhanced with digital resources available at routledgelearning.com/learningtoteach. These include downloadable chapter-specific resources, links to useful websites, interactive quizzes and flashcards. You'll also find animated explainer videos and author-created videos, as well as specific online-only units covering England, Northern Ireland, Scotland and Wales.

Supported by the subject-specific titles in the *Learning to Teach Subjects in the Secondary School* and *A Practical Guide to Teaching Subjects in the Secondary School* Series, this book is an essential purchase for every aspiring secondary school teacher.

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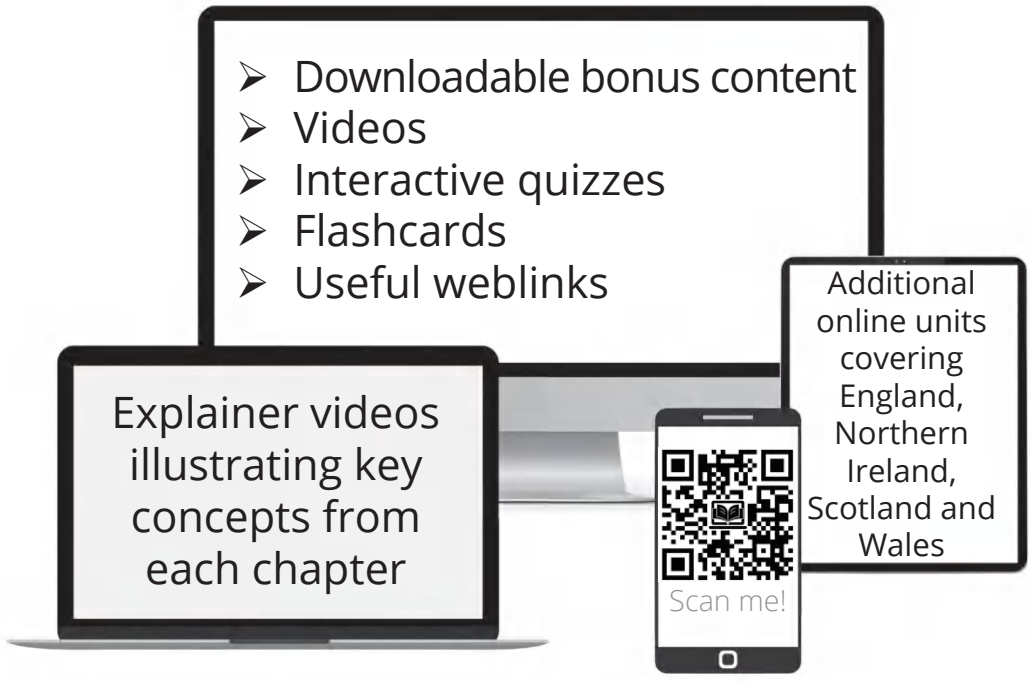
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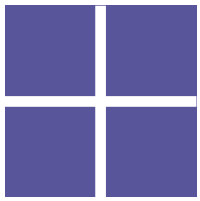
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INTRODUCTION TO THE TENTH EDITION

Sarah Younie, Elizabeth Hidson, Julia Lawrence and Nikki Booth

Teaching is a complex activity and is both an art and a science. In this book, we show that there are certain essential elements of teaching that you can master through practice and that help you become an effective teacher. An effective teacher is one who can integrate theory with practice, use evidence to underpin their professional judgement and use structured reflection to improve practice. They are comfortable in the presence of young people and interested in them as individuals as well as learners. Effective teachers motivate and encourage pupils by planning interesting lessons, linking their teaching to the life experiences of their pupils, whilst responding to diversity and respecting individual differences. In turn, teachers earn their pupils' respect, through the skills mentioned above and by maintaining firm but fair discipline as they seek to create an effective learning environment.

There is no one correct way of teaching, no one specific set of skills, techniques and procedures that you must master and apply mechanically. This is, in part, because your pupils are all different and each day brings a new context in which they operate. Equally, every teacher is an individual and brings something of their own unique personality to the job and their interactions with pupils.

We hope that this book helps you to develop skills, techniques and procedures and provides you with ways of understanding what you do and see. As you develop as a beginning teacher, you will integrate this knowledge in ways that reflect your unique personality and style, whilst also developing your personal philosophy of teaching and learning and gaining insight into how your values shape your professional practice. We also hope that the text provides the stimulus for you to want to continue to learn and develop throughout your career as a teacher.

This book introduces the professional knowledge, skills and understanding required by teachers, including general principles of effective teaching, covering enduring principles that underpin effective teaching. However, we acknowledge that teachers work in a political environment where governments may make reforms regularly to change requirements for curriculum content, assessment and the structure of the education system. It is important that you keep up-to-date with requirements in the context in which you work by taking responsibility for your continuing professional development (CPD) and making the most of the mentoring available to you as an early career teacher. Many teachers join their subject association to keep up to date (https://www.subjectassociations.org.uk/our-members/).

As a teacher it is important to recognise that reflecting on and adapting your teaching is part of your professional role and that good teaching provides the foundation for supporting pupils to make progress.

Developing your philosophy of teaching and learning

On your initial teacher education (ITE) programme much of your time is spent in school. You can expect your ITE not merely to provide *training* but also to introduce you to wider educational issues. What we mean by this is that ITE is a step on the journey of personal development in which your teaching skills develop alongside an emerging understanding of the teaching and learning process and the education system in which it operates. This is a journey of discovery that begins on the first day of your ITE and may stop only when you retire. Thus, we use the term initial teacher *education* rather than initial teacher *training* throughout this book.

The school-based element of your ITE provides the opportunity to appreciate at first hand the complex, exciting and contradictory events of classroom interactions without the immediacy of having to teach all the time. It should allow you time, both in the classroom and the wider school to make sense of experiences that demand explanations. Providing such explanations requires you to have a theory of teaching and learning.

By means of an organised ITE programme that provides for structured observation, practical experience and reflective activity suitably interwoven with theoretical inputs and evidence, you begin to develop your own theory of teaching and learning which is embedded in your practice. Theoretical inputs and evidence to underpin practice can come from a range of sources including tutors, mentors and teachers, lectures and print and web-based resources. Theory also arises from practice, to enable you to better understand what is going on in the classroom environment and to inform and develop your practice.

Everyone who teaches has a theory of how to teach effectively and of how pupils learn. The theory may be implicit in what the teacher does and teachers may not be able to explicitly articulate to you what their theory is; this is tacit knowledge garnered over time through multiple classroom experiences. For example, a teacher who is a disciplinarian is likely to have a different theory about the conditions for learning than a teacher who is liberal in their teaching style. Likewise, some teachers may feel that they do not have a philosophy of education.

There are several different theories and philosophies about teaching and learning. You need to be aware of what these are, reflect on them and consider how they help you to explain more fully what you are trying to do and why. Through the process of theorising about what you are doing, reflecting on a range of other theories as well as your own, reading relevant research studies and drawing on the evidence base, you understand your practice better and develop into a reflective practitioner. You become a teacher who makes conscious decisions about teaching strategies to employ and who modifies their practice in the light of experiences. It is recognised that ITE only enables you to start developing your own personal understanding of the teaching and learning process. Hence, this process must continue throughout your career.

An articulated, conscious philosophy of teaching emerges only if a particular set of habits is developed, in particular, the habit of reviewing your own teaching systematically. It is these habits that need to be developed from the start of your ITE. So, what is your philosophy? For example, do you consider that your job is to transfer the knowledge of your subject to pupils? Or are you there to lead them through its main features? Are you 'filling empty vessels' or are you the guide on a 'voyage of discovery'? On the other hand, perhaps you are the potter, shaping and moulding pupils?

How to use this book

Structure of the book

The book is laid out so that elements of appropriate background information and theory along with evidence from research and practice introduce each topic. These are interwoven with tasks designed to help you identify key features of the topic.

A number of different enquiry methods (tasks) are suggested for you to use to generate data, e.g. reflecting on reading and observation or on an activity you are asked to carry out, asking questions, gathering data, discussing with a mentor, tutor or another beginning teacher. Some of the tasks involve you in activities that impinge on other people; for example, observing a teacher in the classroom, or asking for information. If a task requires you to do this, *you must first seek permission from the person concerned*. Remember that you are a guest in school(s); you cannot walk into any teacher's classroom to observe. In addition, some information may be personal or sensitive and you need to consider issues of confidentiality and professional behaviour in your enquiries and reporting. We suggest you keep your responses to tasks in your professional portfolio to enable you to refer back to them.

An appendix on writing and reflection is included on the companion website to help you with the written assignments on your ITE programme. A glossary of terms is also included in the back of the book to help you interpret the jargon of education.

We call school children *pupils* to avoid confusion with *students*, by which we mean people in further and higher education. We refer to those learning to teach as *beginning teachers*. The important staff in your life are those in your school (we use the term *mentor* to refer to the person with whom you work most closely) and higher education institution or ITE provider (we use the term *tutor* to refer to the person with whom you work most closely). Your institution or provider will have its own way of referring to staff.

Meeting the requirements of your ITE

The range and type of requirements (standards) you are expected to meet during your ITE are derived from those, for all beginning teachers, in the country in which you are learning to teach. The units in this book are designed to help you work towards meeting these requirements. At appropriate points in the text you should relate the work directly to the specific requirements for your ITE.

Reflective practice, evidence-informed practice and your professional portfolio

As you read through the book, undertake other readings, complete the tasks and undertake other activities as part of your ITE, we suggest you keep a professional portfolio. You may want to keep a diary of reflective practice to record your reactions to, and reflections on, events, both good and bad, as a way of letting off steam! It enables you to analyse strengths and areas for development, hopes for the future, and elements of your emerging personal philosophy of teaching and learning.

Your portfolio holds a selective record of your development as a teacher, your strengths as well as areas for further development, and is something that you continue to develop throughout your teaching career. It is likely that your ITE provider has a set format for this. If not, you should develop your own. You can use any format and include any evidence you think appropriate. However, to be truly beneficial, it should contain evidence beyond the minimum required for your ITE. This further

evidence could include, for example, work of value to you, a response to significant events, extracts from your diary of reflective practice, good lesson plans, evaluations of lessons, teaching reports, observations on you made by teachers, outcomes of tasks undertaken, assessed and non-assessed coursework. It could include teaching reports written by teachers, mentors or tutors and yourself.

At the end of your ITE you can use your portfolio to evaluate your learning and achievements and as the basis for completing applications for your first post; and to take to interview. It can form the basis of a personal statement describing aspects of your development as a teacher during your ITE. It can also help provide the basis of your CPD as it enables you to identify aspects of your work in need of development and thus targets for induction and development in your first post.

The introduction of the initial teacher training and early career framework (ITTECF) in England means that early career teachers (ECTs) are entitled to a two-year support package, continuing to build on the core concepts from ITE and working to the same teachers' standards. This book underpins these core concepts and we hope that you continue to find it useful throughout your early career.

Ways you might like to use this book

There are likely to be times when you are faced with a problem in school which has not been addressed up to that point within your ITE and you need some help immediately, for example, before facing a class the next day or next week. This book is designed to help you address some of the issues or difficulties you are faced with during your ITE, by providing supporting knowledge interspersed with a range of tasks to enable you to link theory with practice.

The book may be useful in a number of ways. You may wish to use it alongside your ITE handbook, which outlines specific requirements of your ITE programme, agreed ways of working, roles and responsibilities. It is designed more for you to dip in and out of, to look up a specific problem or issue that you want to consider, rather than for you to read from cover to cover (although you may want to use it in both ways, of course). You can use it on your own as it provides background information and supporting theory along with evidence from research and practice about a range of issues you are likely to face during your ITE.

Using the tasks

Reflecting on an issue faced in school with greater understanding of evidence of what others have written and said about it, alongside undertaking some of the associated tasks, may help you to identify some potential solutions. The tasks can also be used for collaborative work with other beginning teachers or your mentor or tutor. Although you can complete many of the tasks individually, most tasks benefit from wider discussion, which we encourage you to do whenever possible. However, some tasks can be carried out only with other beginning teachers and/or with the support of a mentor or tutor. You should select those tasks that are appropriate to your circumstances.

Further reading and resources

This book will not suffice alone. We have attempted to provide you with guidance to further reading by two methods: first, by references to print and web-based material in the text, the details of which appear in the references; second, by further readings and relevant websites at the end of each unit.

There is much educational material on the Internet. Government, teaching council and subject association websites are useful. There are also many other websites but do look at the evidence underpinning the advice on offer: custom and practice in all professions is under scrutiny as expectations for professional practice to be evidence-based rise.

The website which accompanies this text is available at routledgelearning.com/learningtoteach and includes further information and links to more useful websites. It also has activities that are linked to the chapters and units in the book, as well as downloadable versions of many of the templates provided in each chapter. Video overviews and explanations by a range of our authors, are provided to help you to integrate the topics in the book into your practice.

We suggest you keep a record of useful websites in your portfolio or as a curated set of browser bookmarks that you can synchronise across the digital devices you use. Most browser software allows you to import and export bookmarked websites.

If you see each unit as potentially an open door leading to whole new worlds of thought about how teachers and societies can best educate their children, then you have achieved one of our goals: that is, to provide you with a guidebook on your journey of discovery about teaching and learning. Remember, teaching is about the contribution you make to your pupils, to their development and their learning and to the wellbeing of society through the education of its young people.

The book is further supported by subject-specific and practical texts (*Learning to Teach X Subject in the Secondary School* and *A Practical Guide to Teaching X Subject in the Secondary School*) in the *Learning to Teach in the Secondary School* series as well as the *Debates in (Subject)* and *Mentoring (Subject) Teachers* and by the online MESHGuides' research summaries on www.meshguides.org. Our companion text *Surviving and Thriving in the Secondary School: The NQT's Essential Companion* (S. Capel, J. Lawrence, M. Leask and S. Younie (eds.) (2019)) is designed to support you as an early career teacher in the next phase of your development. You may also find it useful now as well, as it covers aspects of teaching not included in this book which, nonetheless, you experience on your ITE programme.

Finally, we hope that you find the book useful, and of support in school. If you like it, tell others; if not, tell us, especially if there is something we can develop for the next edition or for inclusion on the companion website.

Sarah Younie, Elizabeth Hidson, Julia Lawrence and Nikki Booth
January 2026



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1 Becoming a teacher

The three units in this chapter explore the complexity and breadth of the teacher's role and the nature of teaching. Some of the expectations of you in becoming a teacher include (but are not limited to):

- Dress appropriately (different schools have different dress codes).
- Act in a professional manner; for example, be punctual and reliable; act with courtesy and tact; and respect confidentiality of information.
- Take active steps to ensure that your pupils learn.
- Discuss pupil progress with parents.
- Become familiar with and work within school procedures and policies. These include record-keeping, rewards and sanctions, uniform and relationships between teachers and pupils.
- Be open to new learning: seek and act on advice.
- Be flexible; for example, if there is a change in the timetable on a particular day.
- Accept a leadership role. You may find imposing your will on pupils uncomfortable, but unless you establish your right to direct the work of the class, you are not able to teach effectively.
- Recognise and understand the roles and relationships of staff responsible for your development.
- Keep up to date with your subject.

To become an effective teacher, you need to supplement your *subject content knowledge* with *pedagogical knowledge* (about teaching and learning) and to develop your professional knowledge, skills, and judgement. Ways of developing these provide themes running throughout the book. Evidence about effective practice is becoming increasingly easy to access to support your development. In the UK, you can find a wealth of material on websites, including government-supported and subject association websites, to support you as a teacher.

Each unit in this chapter examines different facets of the work of beginning (and experienced) teachers. You are posed questions about your values and attitudes because these influence the type of teacher you become, the ethos of your school, and the values and attitudes of pupils in your care.

Unit 1.1 is designed to give you an introduction to what teachers do. It first considers Teachers' Standards and Codes of professional practice in the UK, then your role as a teacher, including your

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role in raising attainment and improving life chances and your work in the classroom. It then considers professional knowledge for teaching generally, including your personal subject construct, followed by specific consideration of one aspect of general pedagogic knowledge – aspects of managing the learning environment. The unit then focuses on effective teaching, introducing classroom rights and responsibilities. Finally, it looks at your digital profile – what image do you want to project?

Unit 1.2 discusses your roles and responsibilities as beginning teachers. It considers first the key players in the school and your initial teacher education (ITE) programme. It then looks at how to navigate your school placement, taking an ecological approach to professional practice – looking at the policy, community, organisational, interpersonal, and individual levels. Finally, it focuses on your professional learning journey and the stages of development you go through. As your own confidence and competence in managing the classroom grow, you can expect the focus of your work to move from your self-image and the mechanics of managing a lesson, to whole class learning and, as you become more experienced, to the learning for the individual pupil.

Unit 1.3 focuses on helping you to understand the relevance of digital technologies and artificial intelligence (AI) for you and your pupils. Having focused on planning to teach using digital technology resources to enhance the pupil learning experience, the unit then looks at the potential of online learning. The unit moves on to consider artificial intelligence (AI) in schools and what you need to know as a teacher regarding digital technologies, introducing a framework for auditing your knowledge and understanding of digital technologies, which is important self-knowledge for enhancing your competence in using digital technologies in your lessons to enhance pupils' learning. Finally, the unit looks at your role and responsibility in promoting online safety for both yourself and the pupils you teach.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Watch the explainer video on the varied and professional role of a teacher by scanning the QR code or visit routledgelearning.com/learningtoteach.

1.1 What do teachers do?

Linda la Velle and Stephen Newman

Introduction

Education is probably the most powerful influence on the development of our society. The education that young people receive through schooling goes way beyond knowledge about a body of academic subjects: it is built upon and shapes the values, rights, and responsibilities of our society. That is why the form and content of education are so often the focus of scrutiny and are so keenly contested, and why you need to be able to articulate and develop your own personal philosophy of education.

Nobody entering the teaching profession does so as a blank canvas; everyone has experienced education, and this shapes their sense of what teachers are and do. The media, television, cinemas, and literature also provide people with many and varied representations of teachers. What examples spring to your mind?

Teachers, pupils, parents, carers, politicians, local authorities (LAs)/councils/academy trusts (in England), teachers' unions, professional subject associations, and educational researchers may have very different views of what teachers should do and how they should do it. Each of these perspectives needs to be considered. What a parent/guardian may expect from you as a teacher, what their child may expect, and what you as a teacher believe you should provide may differ significantly. What teachers do, therefore, is multifaceted: "teaching is a challenging, complex, intellectual and ethical endeavour" (Universities' Council for the Education of Teachers (UCET), 2020, p. 2). Some of this complexity is reflected in documents such as the *Initial Teacher Training and Early Career Framework* (DfE, 2024a). How to manage this complexity without compromising the needs of the individual pupil, the requirements of parents and guardians, or your own professional integrity is the focus of this unit.

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OBJECTIVES

At the end of this unit, you should be able to:

- describe your developing vision of yourself as a teacher;
- be aware of the values and ethical code/code of conduct that guide your work;
- describe various aspects of a teacher's academic, pastoral, and administrative roles;
- understand the multifaceted nature of teaching;
- explain how teachers can manage the learning environment;
- explain the rights and responsibilities of teachers and pupils within classrooms;
- demonstrate that your use of digital tools and your digital profile reflects the values and behaviours to which you aspire as a teacher.

Check which requirements of your initial teacher education (ITE) programme this unit relates to.

In the UK, different sets of Teachers' Standards/Codes of Practice or their equivalent provide sets of rules or criteria covering a wide range of teachers' work and responsibilities (see Table 1.1.1).

In addition, those who inspect schools, such as Ofsted in England, the Education Training Inspectorate in Northern Ireland, Education Scotland and Estyn in Wales, set out what they require of educational institutions that they inspect.

What you do as a teacher, therefore, has to meet a complex set of demands, and your ITE programme should provide you with opportunities both to explore individually the kind of teacher you wish to be, and also to understand the context (or, more likely, the contexts) in which you are working, and the demands placed upon you as a teacher to be both professional and collegial (see, for example, Wragg, 2004; Organisation for Economic Co-operation and Development (OECD), 2020; DfE, 2024a).

Given their importance, it is worthwhile making yourself familiar with any teacher standards that are applicable to you in your context. Task 1.1.1, which focuses on what you personally need to do to become a qualified teacher, will help you to do this.

Table 1.1.1 Teachers' Standards and Codes of professional practice in the UK

Department for Education (2021) *Teachers' Standards: Guidance for school leaders, school staff, and governing bodies*. Website: https://www.gov.uk/government/publications/teachers-standards
General Teaching Council for Northern Ireland (2018) *Code of Values and Professional Practice 2018*. Website: https://www.gtcni.org.uk/professional-space/professional-competence/code-of-values
General Teaching Council for Scotland (2021) *A guide to the professional standards*. Website: https://www.gtcs.org.uk/knowledge-base/articles/a-guide-to-the-professional-standards
Welsh Government/Llywodraeth Cymru (2019) *Professional Standards*. Website: https://hwb.gov.wales/professional-learning/supporting-professional-development/professional-standards



Task 1.1.1 Focusing on the Teachers' Standards/Code of Practice or equivalent

You need to be familiar with the Teachers' Standards (or equivalent) for the country in which you are learning to teach. Look at them now. What do they tell you about your role as a teacher? What do they mean to you at the beginning of your ITE programme? Do you feel they capture what you want to be as a teacher? Talk to your mentor or tutor about what achieving the standards might look like at the end of your programme.

Which aspects of these Teachers' Standards (or equivalent) do you feel most prepared to meet? Which do you believe you need more help to meet? How do you see yourself developing your capability over your ITE and career as a teacher? You may find it useful to discuss these areas with your mentor, thinking about how you can record evidence of your achievements and any areas where you may require additional support.

Your role as a teacher

The sorts of standards/codes mentioned above set out some of the 'rules' and expectations of teaching. However, each of those needs to be interpreted. This is an important aspect of your ITE, working with your school-based mentor, other experienced teachers, and university/SCITT tutors. The exact way in which this is done, and when it is done, can vary between countries, programmes, and contexts. For example, in recent times we have seen the development of teaching online, and other types of emergency remote learning, as schools responded to the Covid-19 pandemic. We have also seen the development of many online resources such as those provided by Oak National Academy. But, in one way or another, you will spend time observing experienced teachers where you will not only hear what they say and write but also see what they do. A teacher's primary job is to ensure that pupils learn. However, teachers approach this role in many different ways. Some of the teaching styles and strategies that you see you will like and relate to; others may not seem as appropriate or comfortable. Reflecting on these preferences and responses is important as you think about your own developing practice.

Acquiring understanding is not something that usually happens automatically or instantaneously. At first, you may tend to 'try out' what you have seen more experienced teachers do: imitation is an important aspect of learning. It may be that experienced teachers cannot always explain clearly what they are doing that makes the lesson successful. Teaching is an extension of the personality, and it will be important to understand those aspects of your own character that will help – or hinder – the development of your teaching persona.

One of the consequences of an understanding developing over time is that it is important not to dismiss anything too quickly. There is no single correct way to teach: a range of approaches from the didactic (formal, instructional, heavy on content) to the experiential (learning by discovery and doing) may all be appropriate, often in the same lesson (see Unit 5.3 on teaching styles and Unit 4.1 on ways to group pupils for learning). You might digitally record your practice so that you can analyse and improve your style as you develop as a teacher, but make sure you gain the necessary permission for digitally recording lessons in school. An alternative approach is 'micro-teaching'. This is a common practice in many ITE programmes and involves recording a

Table 1.1.2 Some of the activities that teachers undertake in their academic and pastoral roles

The academic role	*The pastoral role and spiritual and moral welfare*
This encompasses a variety of activities including: ■ subject teaching ■ lesson preparation ■ setting and marking of homework ■ monitoring pupil progress over time ■ assessing pupil progress in a variety of ways, including marking tests and exams ■ writing reports ■ recording achievement ■ working as part of a subject team ■ curriculum development and planning ■ undertaking visits, field courses ■ reporting to parents/carers ■ planning and implementing school policies ■ extra-curricular activities ■ keeping up to date with your subject (often through work with your subject association. See https://www.subjectassociations.org.uk/ ■ keeping up to date with technological advances as they relate to education ■ being an examiner for public examination boards	These roles vary from school to school. They include: ■ helping pupils with problems ■ being responsible for a form/tutor group ■ registering the class, following up absences ■ monitoring sanctions and rewards given to form members ■ reinforcing school rules and routines, e.g., on behaviour ■ writing reports, ensuring records of achievement and/or profiles are up to date ■ working collegially as part of a pastoral team ■ teaching personal, social, and health education (PSHE) and citizenship ■ house/year group activities (plays/sports) ■ liaising with parents and carers ■ ensuring school information is conveyed to parents via pupils ■ extra-curricular activities, e.g., educational trips ■ giving careers and subject guidance ■ taking part in a daily act of worship or religious observance if required by legislation ■ liaising with primary schools

short episode of teaching by a beginning teacher to the rest of the group. The recording is then analysed by the beginning teacher, often with the support and input of the tutor and/or other beginning teachers.

Over time, you will also come to develop an understanding of those aspects of the teacher's role concerned with pupils' pastoral and personal development. In approaching this, you have an important role to play in supporting the school ethos by reinforcing school values, rules, and routines, e.g., behaviour, dress, mutual respect, the right to learn, and in encouraging pupils to develop self-discipline. Table 1.1.2 lists the main activities in each of these areas that you are expected to undertake. All of these require you to get to know pupils as individuals.

Your role in raising attainment and improving life chances

The purpose of teaching is, of course, related to the role of the teacher. Here, it has to be said, opinions and emphases differ. Some, for example, take what others regard as too simplistic a perspective: namely that it is incumbent on you as a teacher to ensure that every pupil is given the very best opportunity to learn and fulfil their potential. The rationale for this perspective is based on the idea that the connection between educational attainment and life chances is significant; that pupils obtaining the best grades have more choices, are able to access the top-ranking universities and better employment and so on. It is thus clearly desirable that pupils attain to the very best of their ability through their schooling. Great pressure is often applied to teachers to demonstrate this.

Pupil attainment is compared internationally, and the results are of great interest and concern for governments (see Unit 7.4). A different perspective of the role of the teacher sees it as intimately bound up with the notion of social justice. On this view, teaching is not just about raising attainment, at least not in any simple sense, but in using education to redress some of the social and economic inequalities between different people and different groups. In seeking to raise the attainment and aspirations of pupils, wider issues of poverty, of inequality, and of discrimination cannot be ignored (Tett and Hamilton, 2019; Steinberg and Krumer-Nevo, 2022). In your role as a teacher, you may well have to consider your response to these perspectives and how you will deal with any tensions you find between them.

Your work in the classroom

On the surface, teaching may appear to be a relatively simple process. The reality is somewhat different. Classroom teaching is only the most visible part of a teacher's work. It may help to consider the cycle of pedagogic reasoning and action set out by Shulman (1987) in Figure 1.1.1, its elements and the underlying knowledge and skills that drive it.

The cycle begins and ends with an act of **comprehension**. The teacher knows about something to be taught, and then has to 'transform' that knowledge into a form that is learnable by the pupils. This **transformation** requires the deployment of several stages and skill sets. In Figure 1.1.1, the sub-cycle to the right shows the process of transformation with the teacher serially *preparing* (critical scrutiny and choice of materials of instruction); *representing* (consideration of the key ideas and how they might best be represented in the form of analogies, examples, etc.); *selecting* (choice of teaching strategies); and *adapting*, sometimes called differentiating: tailoring input to pupils' capabilities and characteristics. The teacher sequences a series of teaching/learning episodes to create a logical yet varied lesson. The teacher then teaches that lesson (**instruction**), during which there are 'in-flight' checks for pupil understanding as well as more formal assessments and feedback (which themselves require all the processes above). These provide opportunities for **evaluation**. Following the lesson, an effective teacher sets aside time for reconstruction, re-enactment, or recapturing of events and accomplishments; **reflection**, which is the critically important process

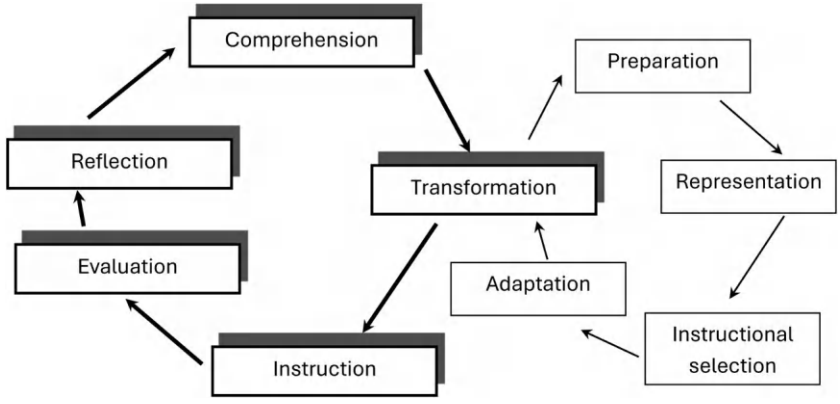


Figure 1.1.1 Shulman's pedagogic cycle

Source: After la Velle et al. (2000)

of analysis through which the individual teacher learns from experience. This brings the teacher to a new, more informed and nuanced level of comprehension about the topic of the lesson. Much has been written about reflection (see, for example, Schön, 1983; 1987; Newman, 1999/2018; 2020; Fraser et al., 2024; Pereira et al., 2024).

The pedagogic cycle should therefore not be thought of as a flat cyclic diagram, but rather as an upward, three-dimensional spiral in which professional knowledge and expertise are continuously built.

The work of the teacher in the classroom is underpinned by many considerations. One analogy that can be useful in this respect is that of thinking of a lesson as an iceberg. The work in the classroom represents the tip of an iceberg (20–30 per cent). Supporting this tip but hidden in the base (70–80 per cent), are the elements of teachers' professional expertise. These elements include:

- *planning* a sequence of lessons to ensure that learning progresses;
- *evaluation* of previous lessons;
- *planning and preparation* for individual lessons;
- *established routines and procedures* that ensure that the work of the class proceeds as planned;
- *personality*, including the teacher's ability to capture and hold the interest of the class, to establish their authority;
- professional knowledge, such as subject content knowledge (SCK); pedagogic knowledge about effective teaching and learning; knowledge of learners; knowledge about the educational context in which you work – local and national;
- *technological knowledge*: each element of the pedagogic cycle can be supported by the use of appropriate technologies, for example, pupils' use of tablets for research; simulation of practical science work; use of generative artificial intelligence (genAI) for lesson planning;
- *professional judgement* built up over time through reflection on experience.

Throughout your ITE programme, you should expect to develop confidence and new levels of competence in all these areas.

Professional knowledge for teaching

The knowledge teachers need for teaching has been described and categorised in different ways. Shulman (1986; 1987) identifies seven knowledge bases that form what he regards as the minimum knowledge for teaching. These are summarised in Table 1.1.3, then considered below.

Knowing a lot about your subject does not automatically make you an effective teacher. Teaching requires you to transform the knowledge you possess into suitable activities that lead to learning (sometimes called pedagogic content knowledge). Your professional knowledge comprises a number of different components.

Subject content knowledge (SCK)

This is a declared body of knowledge about your subject. The curriculum for your subject comprises the concepts and skills that pupils are expected to acquire. You amass knowledge about, and become encultured into, these concepts and skills from a variety of sources: your education at home, at school, and at university, as well as through personal study and reading. Together, these

Table 1.1.3 Forms of professional knowledge for teaching

1	*(Subject) Content knowledge*: the content that is to be taught. Schwab (1964) identifies two components of content knowledge: ■ substantive: core concepts and skills in the subject ■ syntactic: the way these concepts and skills are structured and organised within the subject
2	*General pedagogic knowledge*: broad principles and strategies of classroom management and organisation that apply irrespective of the subject
3	*Pedagogical content knowledge (PCK)*: knowledge of what makes for effective teaching and deep learning, providing the basis for teachers' selection, organisation, and presentation of lesson content, i.e., the integration of subject content and its related pedagogy. Grossman et al. (1989) break PCK into four components: ■ knowledge and beliefs about the *purposes* of teaching a subject at different levels; ■ knowledge of pupils' understanding, and the *concepts and misconceptions* of subject matter; ■ knowledge of *curriculum* materials available for teaching a subject and knowledge of horizontal and vertical curricula for the subject; ■ knowledge of *instructional strategies* and representations for teaching particular topics.
4	*Curriculum knowledge*: materials and programmes that serve as 'tools of the trade' for teachers.
5	*Knowledge of learners and their characteristics*: this comprises a variety of issues – child development; how children develop with age (empirical or social knowledge); cognitive development; knowledge of the needs of particular individuals or groups of pupils.
6	*Knowledge of educational contexts*: political, curricular, sociological, cultural, geographical, historical, and psychological factors may all be important here.
7	*Knowledge of educational ends (aims), purposes, values, and philosophical and historical influences*: both short- and long-term goals of education in general and of particular subjects.

Source: Adapted from Shulman (1986, 1987)

shape the quantity and quality of your SCK. Your explicit awareness of and engagement with these different sources of SCK vary, but as you begin your career as a teacher, this is likely to be the area in which you are most confident. You should actively and continuously seek to extend the range of your SCK; this is a career-long endeavour. This process supports your confidence for teaching and engages you with your subject on a personal level. A word of caution, however: you may see this body of knowledge as the key measure of your likely effectiveness as a teacher, but it is the way you transform that knowledge into effective teaching that is most important. Task 1.1.2 asks you to audit your SCK.



Task 1.1.2 Auditing your subject content knowledge

Analyse a copy of the curriculum for your subject, identifying areas where your SCK is good now, areas in which you require some additional knowledge, and areas in which totally new learning is required. Set yourself targets for developing your knowledge in the areas for development you identify. Discuss these areas for development with your tutor. Plan a course of action for this development.

General pedagogic knowledge (GPK)

This body of knowledge and understanding relates to what Shulman described as those "broad principles and strategies of classroom management that … transcend subject matter" (Shulman, 1987, p. 8) designed to guide class instruction, organisation, and management. These include, for example, settling a class, managing the classroom environment for effective learning, managing resources and other equipment, gaining and sustaining the attention and interest of the whole class, encouraging the disaffected, supporting the less able, extending the most able, and motivating each member of the class. By developing your general pedagogic knowledge, your classroom becomes a more varied, stimulating, and rewarding environment both for yourself and your pupils.

Pedagogical content knowledge (PCK)

PCK is that form of teachers' knowledge in which what they know about their subject is effectively transformed into a form that is learnable by the pupils. This is a combination of SCK and pedagogy that provides the specific knowledge you need for effective teaching and learning in your subject area(s). For example, the ways in which music teachers shape teaching and learning differ in some ways from the ways in which design and technology teachers shape teaching and learning.

Teaching requires you to adapt your PCK to the classroom demands of teaching. It also requires you to consider the processes of your subject and how you can teach these to your pupils. What, for example, are the language and writing demands of your subject, and how are you going to teach them? What research skills and processes should pupils adopt? What forms does assessment (both formative and summative) take in your subject area(s), and how does this relate to content and process? PCK is effectively what pedagogy looks like in your subject area(s). Shulman (1986, p. 9) sums it up as follows. PCK is:

> the most useful form of representation of … [the most regularly taught topics in one's subject area] … the most powerful analogies, illustrations, examples, explanations, and demonstrations – in a word, the ways of representing and formulating the subject that makes it comprehensible to others.

You should also think about the historical development of your subject and about how it came to be as it is. You will be familiar with the 'culture' of your subject, including its vocabulary, the environment in which it occurs, and why it is an important area of human knowledge. This dimension enhances your sense of what your subject is about and why it is studied.

Task 1.1.3 asks you to consider PCK.



Task 1.1.3 Pedagogical content knowledge

Look closely at the forms of PCK in Table 1.1.3. Consider carefully how you could apply your knowledge in each of the categories identified by Grossman et al. (1989) to your work with pupils to make them more reflective learners and to personalise their learning experience.

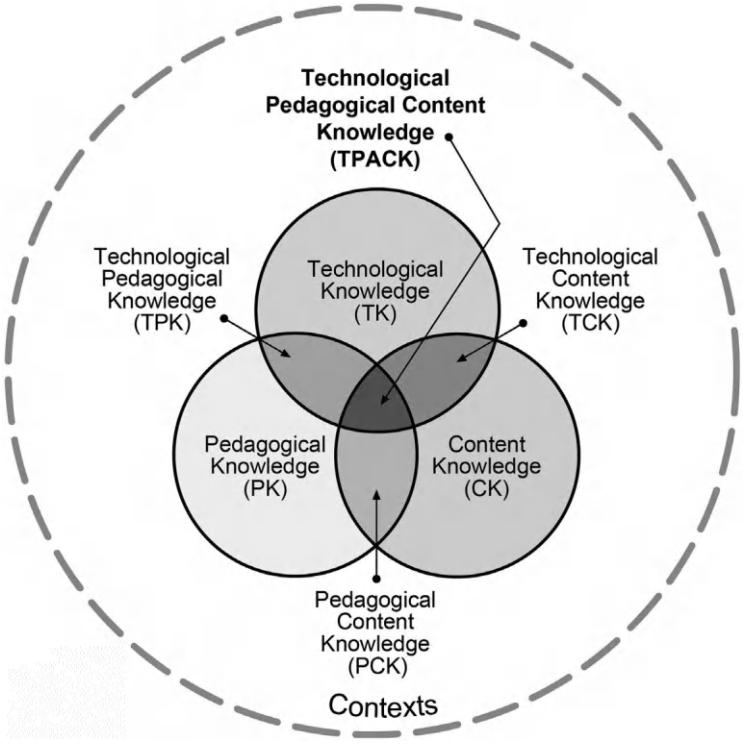


Figure 1.1.2 The Seven Components of Technological Pedagogical Content Knowledge (TPACK)
Source: http://tpack.org and reproduced by permission of the publisher, © 2012 by tpack.org

PCK has been transformed by the rise of information and communications technologies in education. Mishra and Koehler (2006) put forward their Technological Pedagogical Content Knowledge (TPACK) model (Figure 1.1.2), which shows the relationship between subject knowledge, pedagogy, and technology.

The rise of genAI has produced a range of new ethical dilemmas for education, particularly for the evaluation and assessment of what pupils are learning. Teachers now need to adapt their existing knowledge types and acquire new knowledge in order to use genAI tools effectively (Mishra et al., 2023).

Curriculum knowledge

This is your knowledge of the requirements and range of programmes for teaching your subject(s) across the age ranges that you are preparing to teach. Crucially, this knowledge base includes an understanding of the sequence in which the facts and concepts of a discipline should be taught and learned for greatest understanding. It also encompasses knowledge of the variety of instructional materials available to support the programmes of work. It includes knowledge of the curriculum required in your country and in your school(s), the public examinations they serve, and the requirements of those examinations.

Knowledge of learners and their characteristics

Pupils come with different kinds of knowledge. Shulman (1986; 1987) and Grossman et al. (1989) define this as including empirical and social knowledge, i.e., what children/adolescents of a particular age range are like, how they behave in classrooms and school, their interests and preoccupations, their social nature, how contextual factors such as the time of day, weather, or exciting events can affect their work and behaviour, and the nature of the pupil–teacher relationship, as well as cognitive knowledge.

You need to think about how children/adolescents develop and learn, and what they know. This consists of two elements: knowledge of child development that informs practice; and knowledge of a particular group of pupils, the kind of knowledge that grows from regular contact with these pupils, of what they can and cannot know, do, or understand.

Knowledge of educational contexts

Shulman (1986, 1987) and Grossman et al. (1989) define an educational context as any setting where learning takes place. This includes formal settings (nurseries, schools, universities, and colleges), informal settings (home, museums, concert halls, art galleries, etc.), and broader educational contexts (social, cultural, and communal). The context often has a significant effect on teaching and learning and the work of teachers. In the multicultural classrooms of today, you can expect to be teaching pupils from a wide range of different educational and cultural systems where the expectations of teachers and pupils may be quite different from those you have personally experienced. These issues mean that it is important to think about issues such as:

- the type and size of schools;
- their catchment areas;
- class size;
- the extent and quality of support for teachers;
- the amount of feedback that teachers receive on their performance;
- quality of relationships in schools;
- expectations, philosophies, and attitudes of the head teacher;
- schools' policies, curriculum, assessment processes, monitoring and reporting, safety, school rules and expectations;
- the 'co-', 'hidden' and 'informal' curricula, which include the values demonstrated to pupils through the way the school is run (see also Unit 7.2, 'The secondary school curriculum').

The units in Chapter 7 of this book and on the web provide an overview of educational contexts in the four countries of the UK (England, Northern Ireland, Scotland, Wales).

With the advent of genAI, contextual knowledge is expanded beyond its traditional meaning, described above. Typically, contextual knowledge in the TPACK framework refers to understanding the specific educational environment in which teaching and learning occur, including factors like the pupil population, school culture, and available resources. However, with the increasing use in education of genAI, Mishra et al. (2023) have argued that educators need a broader form of contextual knowledge, including an understanding of the wider societal and ethical

implications of AI technologies. These authors emphasise that teachers should focus not only on the immediate classroom context but also on how these tools might make an impact on society at large, including issues of privacy, bias, and the potential to shape future educational norms and practices.

Knowledge of educational ends (aims), purposes, values, and philosophical and historical influences

This includes the values and priorities that shape the education that pupils receive. Teaching is a purposeful activity, both in the short-term (goals for individual lessons or series of lessons) and the long-term (broader purposes, philosophies, and functions of education). Views as to the purposes of education vary widely. As you have seen above, some would argue that its long-term goal is to produce efficient workers to serve the needs of society. Others see education as being concerned with issues of social justice. Aims and purposes tend to be implicit rather than obvious and openly enacted. This is why it is so important to think about and develop your own personal understanding of what it is to teach.

Your personal subject construct

All of the above aspects of professional knowledge for teaching are brought together in your *personal subject construct* (Banks et al., 1999), which encapsulates your own values and assumptions about your subject. This construct provides the basis of your work as a teacher, your understanding of the nature of your subject and how to teach it. Think, for example, about how your political, philosophical, theoretical, ethical, and religious views shape the version of your subject you wish to teach. Within subject areas, specific questions may arise. What, for instance, is the role of sport in physical education? Should creationism be taught alongside evolution and the Big Bang in science lessons? And what about your wider role as a teacher, beyond your subject boundaries? What is your view of supporting language development or teaching mathematical skills as the need arises in your lessons? How do you think technological advances are making an impact on your subject? Such questions have a significant impact on the choices that you make as a teacher. You should ensure that the personal beliefs and subject constructs you use in the classroom do not exclude colleagues and pupils with different views.

Managing the learning environment: a key part of your general pedagogic knowledge

An important aspect of your job is managing the learning environment of your classrooms. *Learning to manage the classroom* is similar in many ways to learning to drive a car. At the outset, there seems to be so much to remember (using the clutch, brake, changing gear, watching other traffic, looking in the mirror, indicating, obeying the speed limit, and so on), but after a short time such skills become part of subconscious patterns of behaviour.

Much of what experienced teachers do to manage their classes has become part of their subconscious classroom behaviour, so much so that often teachers find it hard to articulate exactly what it

is they are doing or why it is successful. This situation, of course, does not help you as a beginning teacher. It also gives weight to the spurious notion that teachers are born rather than made and that nobody can tell you how to teach.

Some teachers may well begin teaching with certain advantages such as a 'good' voice, organisational skills, or 'tech-savviness'. Nevertheless, there are common skills and techniques to be learned that, when combined with an awareness of and sensitivity towards the contexts within which you are teaching, enable you to manage your classes effectively. Part of the joy of teaching is that *it is a continuously creative and problem-solving activity.* Pupils and groups of pupils each have their own characteristics and group dynamics, which experienced teachers take into account when planning for teaching and learning. Although lessons with different groups may have similar content, the same lesson is virtually never delivered in the same way twice because of the variety in relationships between individuals, the whole class and the teacher.

Rogers (2002, p. 5) comments:

> Day-to-day school teaching normally takes place in a rather unusual setting: a small room (for what is asked of it), often inadequate furniture and space to move, a 50-minute time slot (or less) to cover set curriculum objectives, and 25 to 30 distinct and unique personalities, some of whom may not even want to be there. Why should there not be some natural stresses and strains associated with a teacher's day-to-day role?

One of your most important roles is to bring together the various personalities of your classroom (including your own) to create the best possible context for learning. This requires careful thought, planning, and preparation. The majority of events and issues occurring in the classroom are foreseeable and can, therefore, be planned for. It is always better to be proactive than reactive.

Effective teaching

Here, as to be expected from what has come before, the notion of effective teaching is not one that can be considered devoid of context (Muijs and Reynolds, 2018; Maulana et al., 2023). Some useful ideas that may be helpful to consider include, for example, whether:

- the learning experience structured by the teacher matches the needs of each pupil and tasks effectively build on pupils' knowledge, skills, and attitudes;
- a balance has been struck between the pupils' chances of success against the level of difficulty required to challenge them.

The wider contexts of teaching include consideration of:

Classroom rights and responsibilities

It is important to think about rights and responsibilities in the classroom, including your own (NEU, 2023). Everyone should understand clearly that rights are counterbalanced by responsibility in terms of behaviour and participation and that in the best interests of everybody, clear and appropriate sanctions will be applied to those who do not comply.

The following are useful areas to consider in relation to the rights and responsibilities of your classroom:

- *Respect*: all pupils and teachers deserve personal respect; everyone should employ respectful language; it is important to respect the views and beliefs of others.
- *Attention*: every pupil has the right to receive a fair share of teachers' attention; when invited to address the class, pupils have the right to be heard; everyone must pay full attention to the requirements of the lesson; when the teacher speaks, all must pay attention.
- *Learning/teaching*: all pupils have the right to learn; teachers have the right to teach; everyone has the responsibility of cooperating so that effective teaching and learning can take place.
- *Safety*: everyone should expect to be safe; everyone must ensure that safety is not compromised. Remember that teachers are responsible for the wellbeing and safeguarding of their pupils. Think carefully about the activities with which pupils engage; consider their risks and take appropriate steps to ensure safe practice. Some subject areas, such as science, technology, or physical education carry more inherent risks, but all teachers need to take personal responsibility for ensuring safety and wellbeing in their lessons. Pupils should be made aware of the risks and take responsibility for acting safely.
- *Safeguarding* of children is a high-profile issue. There is guidance which is important for all those in education to follow – see, for example, DfE (2024b); Scottish Government (2023); Education Wales/Addysg Cymru (2022); and, in Northern Ireland, the Department for Education (2024c). Major areas that all teachers must consider include child protection issues; physical abuse and neglect (including female genital mutilation); mental abuse (including forced marriage); sexual abuse and exploitation; emotional wellbeing; e-safety; accident protection and prevention; drug and alcohol misuse; and mental health.

Teachers must be familiar with such issues, the common signs of problems, and procedures and channels for dealing with them. Each school is obliged to develop policies to support practice in these areas, and charities exist in some specialist areas. There may well be other rights and responsibilities that you wish to establish for your classroom. Task 1.1.4 asks you to think now about what these might be and how you are going to establish and maintain them. See Unit 8.5 on your legal responsibilities.



Task 1.1.4 Classroom rights and responsibilities

Working with fellow beginning teachers, if possible, consider the rights and responsibilities operating in classrooms that you have observed. Make a record of your notes and refer to and develop these during your ITE programme.

Your digital profile: what image do you want to project?

Finally, in developing your professional code of conduct and ethical stance, we suggest you consider what your digital footprint says about you. It is important that you review how you are presented on social media to check that you are portraying the values and behaviours that are consistent with 'being a teacher'. This will earn you the respect of your pupils, parents, and carers and your peers and employers.

SUMMARY AND KEY POINTS

So, let us return to the question that is the title of this unit: What do teachers do?

- There are many different views of what teachers do, should do and how they should do it. You will develop your own vision of your role as a teacher;
- Your own philosophy of teaching affects the way you approach your work and develops over time as you acquire further professional knowledge and judgement;
- As a teacher, your work will be informed and guided by various codes of conduct;
- The work of a teacher is multifaceted. As a teacher, you have different roles, such as academic, pastoral, and administrative, and you have responsibilities to your pupils, their parents and guardians, your head of department, your school, your head teacher and others;
- Whatever your own subject discipline, all teachers are teachers of literacy, numeracy, and digital skills;
- Being an effective teacher does not mean simply knowing your subject. It also means:
  - knowing how to teach lessons that are intellectually robust, challenging, and stimulating;
  - managing the classroom effectively and fairly; assessing and monitoring pupils' progress promptly and accurately;
  - modelling in your own behaviour and practice what you expect pupils to do; planning for inclusion and the needs of individual pupils;
  - managing the rights and responsibilities of the classroom;
  - upholding school policies and procedures;
  - responding to the pastoral and personal needs of your pupils;
  - completing administrative duties;
  - contributing to the wider life of the school;
  - knowing your responsibilities for safeguarding;
  - knowing your legal responsibilities.

As you progress through your ITE programme, you develop knowledge, understanding, and skills that enable you to fulfil your roles and responsibilities in all these areas. Through your experiences in school, you should move from knowing about skills to a position where you can use them flexibly and appropriately in a range of situations. In other words, you learn to do what teachers do – the school equivalent of plate-spinning – as you balance the many demands of the wonderful job that is teaching.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

Further resources

Association of American Educators (2015) *Code of Ethics for Educators*, viewed 29 July 2025, from https://www.aaeteachers.org/codeofethics

Chartered College of Teaching (2024) *Code of Professional Conduct*, viewed 29 July 2025, from: https://chartered.college/wp-content/uploads/2025/02/Code-of-Professional-Conduct-February-2025.pdf

These two sites provide examples of a code of ethics for educators as well as a range of other resources.

The Council for Subject Associations, viewed 29 July 2025, from: https://www.subjectassociations.org.uk/

This site points you towards a wide range of subject associations which can provide support and guidance as you develop as a teacher.

Education Support, viewed 29 July 2025, from https://www.educationsupport.org.uk/

Education Support is a charity providing mental health and wellbeing support to all education staff and organisations.

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1.2 Beginning teachers' roles and responsibilities

Rachel O'Sullivan and Moira Hulme

Introduction

Schools are busy places and teachers are often required to juggle many tasks at once. Unit 1.1 provides some insight into what it is to be a teacher. This unit looks at what it is to be a beginning teacher in a secondary school and considers the school placement. We consider the importance of understanding school location and culture as you prepare to join a school community for the first time as a beginning teacher. We look at the relational dynamics of learning to teach, the expectations of beginning teachers in school and offer some guidance about your roles and responsibilities. The unit then considers how your development as a professional is likely to pass through significant changes over your initial teacher education (ITE) programme.

OBJECTIVES

At the end of this unit, you should be able to:

- identify key individuals who will support your professional learning during your school placement;
- prepare for your school placement by researching the environment, both within and external to the school;
- collect key information on school organisation, policies and culture;
- identify expectations, roles and responsibilities of beginning teachers in school;
- work collegially with other staff to support your professional learning;
- chart your development over your ITE programme and into the early career phase.
- Check which requirements of your ITE programme this unit relates to.

The key individuals in the school and your ITE programme

Before you start any school placement, it is important to know the key individuals in the school and your ITE programme. Although nomenclature may differ between schools and regions, the terms in **bold** in the first column of Table 1.2.1 are those used in this unit/book.

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Table 1.2.1 Key individuals in your ITE programme

Key individual	*Roles*
Headteacher/Principal	Carries overall responsibility for the care of pupils, the quality of teaching and learning, strategic vision and school self-evaluation. Provides opportunities for professional learning for school staff. In England, has oversight of school-led ITE or chooses approved external partners in the collaborative delivery of ITE.
Professional mentor/ Lead mentor/ Coordinating mentor	Usually, a senior member of staff who holds a variety of additional roles. Overall responsibility for beginning teachers in the school. Liaises with subject mentors and university tutors. Supports mentor development. Organises regular school-wide training sessions. Key contact for school-level policies, organisational structure and lines of accountability. Supports arrangements for beginning teachers to observe classes and undertake in-house training; organises beginning teachers' teaching timetables (probably with the subject mentor) and access to professional activities outside timetable (e.g. staff meetings, pastoral work, extra-curricular activities and consultation evenings).
Subject mentor/General mentor/Subject tutor/ school-based mentor	Your first point of contact within the school. Organises your day-to-day learning in the department, timetabling and weekly meetings. Reviews your professional learning needs and provides subject-specific mentoring. Provides verbal and written formative and summative feedback. Identifies strengths, development priorities and sets targets. A key contact in terms of curriculum, lesson planning, schemes of work and teaching timetable. Provides support for your wellbeing.
Class teachers	Teachers whose classes you are given responsibility for during your time in school. Your subject mentor may also be one of the class teachers. Key contact for class- and pupil-level information to support planning. Discuss homework routines, access to digital technology and learning resources. Liaise on the use of teaching assistants and other support staff.
Head of Department/ Head of Faculty/ Subject lead	Responsible for running the subject department/curriculum area. Line manager for subject teachers. Key contact for department-level data, policies and resources.
University tutor/Link tutor/Liaison tutor (or equivalent in school-centred initial teacher training (SCITT)	Responsible for your ITE programme in the university. Reviews your targets and monitors your progress against the Standards. Maintains regular contact to support your in-school experience. Links university coursework with practical experience in school. Provides support for reflection and the development of your research and policy literacy.
Beginning teacher/ Trainee teacher/Student teacher	Yourself. Maintain professional attitudes and behaviour with school staff and pupils: punctual, respectful and fair. Be proactive in seeking sources of support. Adhere to school policies and uphold school values. Plan and teach lessons, assess learning. Prepare for weekly meetings, complete required documentation, including records of professional learning. Respond appropriately to feedback and challenge.
SENCO/SENDCO (Special educational needs and disabilities co-ordinator)	Responsible for overseeing the implementation of a school's special educational needs (SEN) policy and coordinating provision for pupils with special educational needs and disabilities (SEND). Key contact to discuss additional support needs and inclusion.

(*Continued*)

Table 1.2.1 (Continued)

Key individual	*Roles*
Pastoral and **safeguarding** leads.	Leads the pastoral team (form tutors) who provide pastoral and emotional support for all pupils, overseeing welfare and facilitating pupils' personal development. The safeguarding lead ensures teachers carry out their safeguarding duties. Liaises closely with children's social care and other services that support children and families. Key contact for child protection and safeguarding policy. This policy to be read in conjunction with other school policies for whom a range of staff have overall responsibility, including anti-bullying, e-safety, health and safety, relationships, sex and health education (RSHE), behaviour policy, attendance procedures and the staff code of conduct that addresses staff-pupil relationships and communications including the use of social media and online platforms.

Navigating your school placement

As you prepare for your first school placement, it is helpful to remember that you are joining a system of education that operates at many levels: individual, classroom, organisational, community and national policy levels. Home, school, community and culture all influence success in the classroom. Figure 1.2.1 uses a social ecological approach (Bronfenbrenner, 1979) to show how your development as a teacher is nested within the micro-, meso- and macro-systems of education. In the early stages of your school placement, you will initially be concerned with observing expert teachers, planning and preparation, developing your understanding of the classroom environment and evaluating learning. In addition to developing a repertoire of effective teaching and learning techniques, you will also be expected to demonstrate professional values and fulfil wider professional responsibilities. As you move from beginning to more experienced teacher, and possibly eventually to teacher leader, your involvement in whole school activity and community engagement will increase. That is, you move from micro- to macro-level.

Drawing on the ecological model above (Figure 1.2.1), the following four sections help you prepare for your school placement by taking an 'outside-in' view of your professional role. First, we consider the significance of the locality and community resources and ask you to undertake a pre-placement contextual analysis of the environment around your school (Task 1.2.1). Second, we focus on your first formal visit to your placement school as a beginning teacher and ask you to make an inventory of school policies and a note of the facilities you may need (Task 1.2.2). Third, in the first weeks of your placement we ask you to dig deeper in your analysis of school culture and values using school ethnography as a mode of enquiry (Task 1.2.3). Fourth, we reflect on the phases of professional development you are likely to experience during ITE and beyond. In particular, we emphasise the critical role of collaboration in supporting professional growth. We ask you to construct a map of all the connections and facilitators that can help to sustain your professional learning now and in the future (Task 1.2.4).

You can add to this resource as you progress through your ITE programme. As you gain experience, revisit the ecological model (Figure 1.2.1). As you work with and alongside communities of practice, including your department team, the pastoral team, support staff and external agencies, you will be more aware of the range of your professional responsibilities and develop a deeper understanding of your wider role and emerging teacher identity.

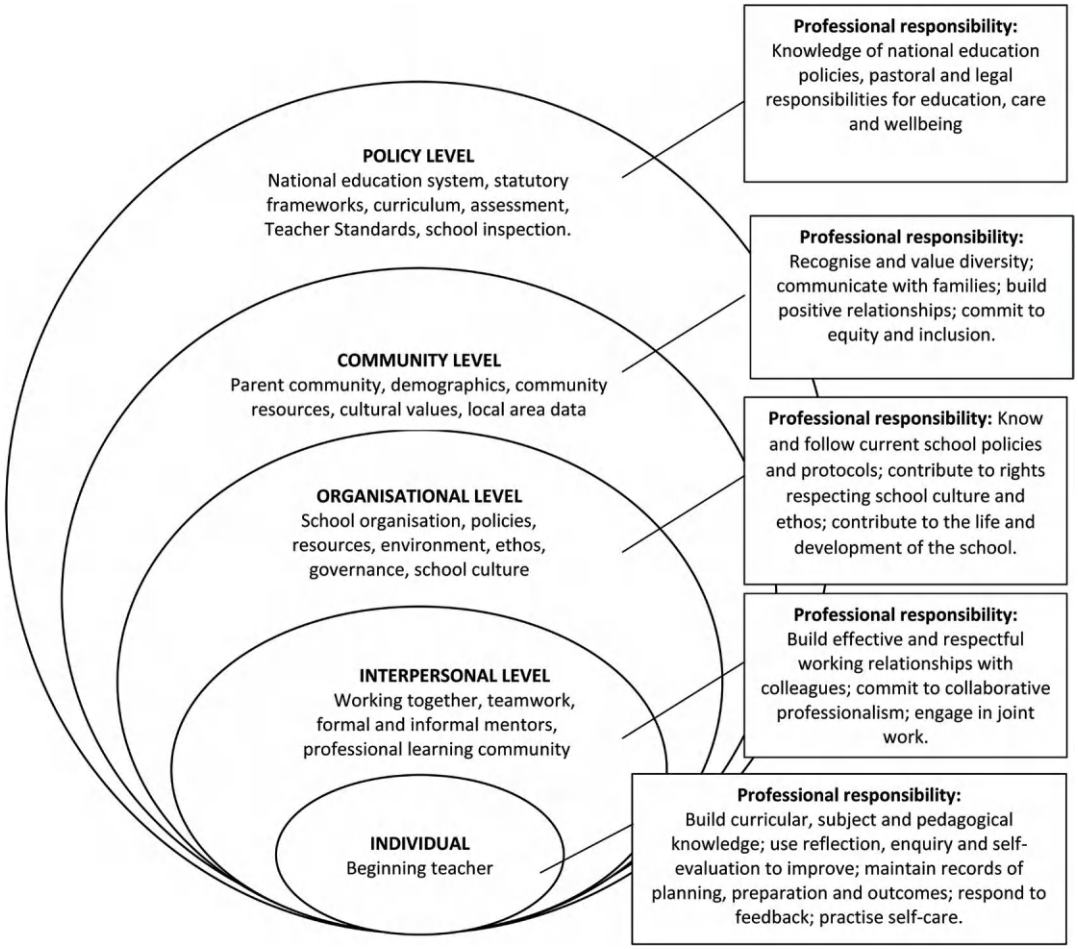


Figure 1.2.1 An ecological approach to professional practice

Community-level: school locality

We know that place and the communities built around them shape pupils' educational experiences and outcomes (Mon-Williams et al., 2023). Children's development and learning is influenced by the interaction of factors inside and outside school (Darling-Hammond et al., 2020). You should always approach notions of 'community' with care. Communities can be diverse and dynamic with overlapping membership and the intake of secondary schools will often draw on a wide and diverse catchment area. Understanding the community in which your school is located is important. Take time to learn about the community, it's values and how these translate into school culture. Find out what your school does to build community partnerships and how you may contribute to such partnerships. Avoid making assumptions which may consciously, or unconsciously, shape your expectations of and relationship with pupils. Context matters to teachers as pupils bring diverse cultural resources or 'funds of knowledge' into school (Esteban-Guitart, 2023). Skilled educators are attentive to such differences, value diversity and work with others to develop caring and culturally responsive learning environments (see Unit 4.4 Responding to diversity).

Table 1.2.2 Sources of information about schools and their localities

Indices of deprivation: England (https://www.gov.uk/guidance/english-indices-of-deprivation-2019-mapping-resources) Northern Ireland (https://www.nisra.gov.uk/statistics/deprivation) Scotland (https://www.gov.scot/collections/scottish-index-of-multiple-deprivation-2020/) Wales (https://gov.wales/welsh-index-multiple-deprivation)
Local area reports are available through Nomis, a service provided by the Office for National Statistics (https://www.nomisweb.co.uk/reports/localarea)
School data and inspection reports: England (https://www.gov.uk/school-performance-tables) Northern Ireland (https://www.etini.gov.uk/publications/type/inspectionreports and https://www.education-ni.gov.uk/services/schools-plus) Scotland, *School Information Dashboard* (https://education.gov.scot/parentzone/my-school/school-information-dashboard/) Wales (https://www.estyn.gov.wales/inspection)

As you prepare to join your placement school, the following questions posed by Thomson and Hall (2017) will help you to consider the interaction of place and school: 'How many neighbourhoods are inside this school? What is the relationship between the school and the surrounding area? What are the communities in this school? How are they connected?' (pp. 77–78). Thomson and Hall suggest three techniques you could consider when getting to know the locality of your placement school: (1) a *neighbourhood walk* at different times of the day; (2) a *windshield survey* or drive around to take note of community facilities, e.g. housing, retail, transport, community services; and, (3) *asset mapping* (pp. 90–92). Asset mapping is a mode of enquiry that looks for strengths in the physical environment, local organisations, community groups, services and networks. The focus on community assets is intended to displace deficit views of 'problem' neighbourhoods that are sometimes an unintended consequence of initiatives that link education and disadvantage. Schools across the UK and internationally are increasingly developing 'place-based' and 'place conscious' pedagogies that connect teaching and learning in schools with their social and environmental settings (Vander Ark et al., 2020).

In addition to direct observation, there is a wide range of open access online data sources that you can use to research the social and educational context of your placement school (see Table 1.2.2).



Task 1.2.1 Contextual analysis

Conducting a contextual analysis of the school and local community is an important component of pre-placement preparation. Contextual analysis enables us to make sense of our pupils and the factors that may affect their learning. Only when we fully understand our pupils and their context can we recognise and remove barriers to learning and plan engaging, relevant and culturally responsive lessons. Undertaking a contextual analysis involves reviewing factors that will help you better understand your school and your pupils, information which will inform your planning and practice. Use the list of contextual influences below to write notes relating to your placement school. Consider specifically where you may need to do further research or gather

additional information to help you develop a full picture of your school's context and who your pupils are. Add any additional categories that are relevant, e.g. information gathered from a neighbourhood walk or windshield survey. Use the information in Table 1.2.2 to help source data and online searches will provide a wealth of data relating to your school's context (e.g. Progress 8 and Attainment 8 accountability measures in England).

Contextual influences/factors to include (not an exhaustive list):

- Socio-economic catchment area: resident employment, housing and health profile;
- Resident population, household composition and languages, education outcomes;
- School national attainment measures over time;
- Number of pupils on roll and school capacity;
- Special features of the school, e.g. religious designation, school type;
- Number of pupils eligible for free school meals;
- Pupil premium funding;
- Number of pupils registered with SEND; number of pupils with an Education Health and Care Plan (EHCP);
- Looked after children;
- Number of pupils who are multilingual;
- Pupil destinations;
- Pupil absence data;
- Most recent inspection outcomes.

Consider the implications for your practice in the school.

Organisational level: school organisation and culture

Ideally, you will visit a school (in-person or virtually) before you start your placement. In addition to the information gathered before you set foot in school, you will have additional questions for the professionals who will be supporting you with practical teaching. It is important that you know how the school works, so you feel prepared when you start teaching. You will want to learn more about the department you will be working with and the activities in which you are going to be engaged. Every school operates differently and gathering school and department level information on policies and practices will enable you to learn about the particular dynamics of your placement school. For example, what is the structure of the school day? How is behaviour managed? What is the policy for homework? Who would you report to if a pupil disclosed an issue of concern to you? What is the leadership structure? Knowing information such as this will be significant in helping you settle in quickly.

The geography of the building is important, particularly if your placement is in a large school, perhaps with several different blocks or operating on more than one site. Secondary schools vary immensely in size and physical features, ranging from the small rural or special school with under 100 pupils to large schools with 2,000 pupils. Multi-school trusts may have Executive Headteachers who lead across more than one phase and across multiple sites. Some schools are newly built or refurbished while others have more restricted facilities. Each type of building has advantages and disadvantages. On your visits (and later when you start in school), be sensitive to the affordances of the school estate and the patterns and protocols that govern daily life. See task 1.2.2.



Task 1.2.2 School policies and facilities

When you first visit your placement school, gather and read the school policies, including on:

- behaviour management
- equal opportunities
- health and safety
- IT systems and records management
- marking and assessment
- SEND
- safeguarding and e-safety
- uniform

As you tour the school, consider:

- the layout of the building
- the location of the facilities and resources you may need, e.g. library/learning resources, subject base room, ICT technicians, site manager. organisational strategies and routines, e.g. daily briefing, meeting cycles, assessment calendar
- deployment of classroom assistants
- the existence of any extended services, e.g. breakfast club
- the range of extracurricular activities

Make notes as needed and ask your mentor about anything that is not clear to you.

When beginning teachers join a new school they are socialised into the ways of thinking and being that are appropriate in that setting. Its history, intake, locality, as well as the material resources, professional culture, learning culture and the external policy environment shape each school. In your role as a beginning teacher, it is your responsibility to 'read' or make sense of your school. Taking on the role of school ethnographer will enable you to look anew at what might have previously gone unnoticed. Ethnography is the study of culture; in this case, the norms and routines that shape school life. Taking an 'inquiry stance' will help you to make explicit any assumptions you may have about this new setting (Cochran-Smith and Lytle, 2009). A wide range of visual images and artefacts are available to the novice school ethnographer. Task 1.2.3 will help you to 'read the school' so that you can participate effectively as a new member of this learning community.



Task 1.2.3 'Reading' the school

Make time in the first few weeks of your placement to review the features of school life listed below. Consider/make notes on:

- School website and official social media;
- School reception foyer, front office and other public areas;
- School corridor displays, banners and posters;

- Classroom displays/learning walls;
- School assemblies;
- Whole school behaviour policy;
- Staff code of conduct;
- Dress codes for pupils and staff;
- Use of school outdoor space, grounds, parking, signage (permitted/prohibited spaces);
- Use of pastoral space and withdrawal/referral rooms;
- Staffroom and subject base rooms – layout, seating, interaction, resources;
- Staff meeting agenda and minutes.

Use the following prompts to help you reflect on school culture and norms.

- What do your observations suggest about what is valued in this school?
- How do visual cues reinforce what is acceptable and out of bounds?
- Where do pupils enter?
- Where do visitors and staff enter?
- What do visual and cultural artefacts suggest about the leadership style, priorities and learning culture of this school?
- Who is/is not represented in images and displays?
- What do the visual and cultural artefacts 'say' about this school's vision of a 'good pupil', 'good teacher', 'good school' and the purpose of schooling?
- Which behaviours and attitudes are celebrated, disparaged or ignored?
- What have you learned about community engagement, pupil wellbeing and home-school relationships?
- How do the school ethos, teachers' values and commitments align with national policy trends and your own beliefs and values? (See Unit 7.1 Aims of education and Unit 8.5 Accountability).

Try to suspend initial judgement and interrogate the content and meaning of what you see. Make notes and discuss your observations with other beginning teachers and your mentor.

Interpersonal-level: collaborative professional learning

Taking on the role of a beginning teacher means forging and managing professional relationships with adults, as well as pupils. Commit time to building a positive working relationship with all staff, particularly those who are supporting you, observing your teaching and assessing your progress.

During your initial days in school, introduce yourself to staff you did not meet on visits before you started your placement, including teaching and support staff in your subject department and key personnel outside the department such as the headteacher, deputies/assistant heads, pastoral leaders, safeguarding lead and the SENCO. You will encounter support staff with a range of administrative and non-teaching roles such as attendance officers, technicians, careers advisors, cover supervisors and counsellors. Senior teachers in school will liaise with a range of external agencies and specialist practitioners including children's social care, e.g. Looked After Children, community public health nurses, educational psychologists and school community police officers. Take time to learn about the important role undertaken by external agencies in providing holistic support for pupils.

You are joining the profession at a time when classroom practice is becoming de-privatised and collaboration is encouraged. In the past, teaching was caricatured as a private practice and early career teachers might experience isolation behind firmly closed classroom doors. In the classic study *Schoolteacher*, Lortie (2002) used the metaphor of an egg crate to describe how the social organisation of schools compartmentalised learning and individualised teachers' practice. In contrast, the value of collaborative learning in communities of enquiry is now widely recognised, if not always fully realised in practice (Hargreaves, 2019). As a result, classroom observation is no longer the preserve of the beginning teacher. Beginning and experienced teachers can expect to benefit from mutual observation, collegial reflection and constructive feedback (see Unit 5.4 Improving your teaching).

Secondary schools that are strong professional learning communities encourage within-subject and cross-curricular observation to support professional growth throughout the career course (Philpott and Oates, 2017). While subject identity is important to the secondary teacher, teachers in different departments often share common pedagogical concerns and can learn from one another. Multiple perspectives can be brought to bear on common teaching issues when pupil progress and wellbeing is regarded as a collective responsibility. During your school placement, you may encounter forms of structured observation and joint work including instructional or learning rounds (City et al., 2009; Philpott and Oates, 2017) (group observation), learning walks (short, focused observations), research lessons or lesson study (Kusanagi, 2022) (joint planning, observation and evaluation of lessons focused on a target area), guided viewing of digital video (Körkkö, 2021), as well as team, pair or co-teaching. As a beginning teacher you will be 'stirred in' slowly (Kemmis et al., 2017) perhaps through paired/trio/team teaching or taking part of a lesson or small groups, before moving on to take responsibility for whole-class teaching with the class teacher nearby (see the units in Chapter 2, Beginning to teach).

Professional Standards across the four nations of the UK acknowledge the significance of working with others inside (and outside) school to enhance pupil achievement and wellbeing (For a summary see Table 1.2.3). The Initial Teacher Training and Early Career Framework (ITTECF) in England (Department for Education, 2024) notes the role of teamwork in developing as a professional. The contribution of the wider community – teachers, support staff, parents and external agencies – is emphasised in Northern Ireland. The National Model of Professional Learning in Scotland promotes pupil and teacher learning as a collaborative practice. In Wales, collaboration is one of five Professional Standards – alongside pedagogy, leadership, innovation and professional learning (Welsh Government, 2019). In Scotland and Wales, experienced teachers are positioned as curriculum makers and are increasingly expected to work with others to generate department-, school-level and sometimes regional-level curricula resources (Priestley and Drew, 2019) (see Unit 7.2 The secondary school curriculum). From the start of your professional journey, you will use the relevant national Standards to assess your progress as an effective collaborator.

You are not alone on your ITE journey, support is available from a number of sources – the staffroom, your departmental colleagues, other beginning teachers, 'X' (formerly Twitter), online curriculum sharing sites, teacher blogs and Facebook groups and your ITE provider are just a few. The platformisation of schooling has greatly increased the number of online resources available to teachers. Do your research and remember to apply quality criteria when viewing material shared through social media and available through online educational marketplaces (Carpenter et al., 2022). National organisations provide support for beginning teachers as they progress to registration; these include the Chartered College of Teaching (England), General Teaching Councils (Northern Ireland and Scotland) and the Education Workforce Council (Wales). A wide range of specialist subject associations (e.g. the Council for Subject Associations, https://www.subjectassociations.org.uk/) support

Table 1.2.3 Collaborative practice identified in professional standards across the UK

England	*Northern Ireland*	*Scotland*	*Wales*
Initial Teacher Training and Early Career Framework (Department for Education [DfE], 2024) Teachers can make valuable contributions to the wider life of the school in a broad range of ways, including by supporting and developing effective professional relationships with colleagues (p. 26) Building effective relationships with parents, carers and families can improve pupils' motivation, behaviour and academic success (p. 26) Seeking challenge, feedback and critique from mentors and other colleagues in an open and trusting working environment (p. 26) Collaborating with colleagues to share the load of planning and preparation and making use of shared resources (e.g. textbooks) (p. 27) Teaching assistants (TAs) can support pupils more effectively when they are prepared for lessons by teachers, and when TAs supplement rather than replace support from teachers (p. 26)	*Teaching: The Reflective Profession* (General Teaching Council for Northern Ireland [GTCNI], 2011) Understand the significant roles of staff from school and from other agencies in the life of the child, and the need to interact with such individuals or agencies (p. 26) Consider the significance of parents as partners in the educative process and the need to communicate effectively with them (p. 26) Recognise the significance of the community in school life (p. 28) Teachers will contribute to the life and development of the school, collaborating with teaching and support staff, parents and external agencies (p. 39)	*Standard for Provisional Registration* (General Teaching Council for Scotland [GTCS], 2021) Enquiring and collaborative professionalism is a powerful force in developing teachers' agency and delivering our commitment to engaging children, young people, their families and communities in the education process (p. 4) Teachers in Scotland commit to lifelong learning through being reflective, enquiring, using evidence and working collegially with all members of the learning community. (p. 5)	*Professional Standards for Teaching and Leadership* (Welsh Government, 2019) The teacher takes opportunities to work productively with all partners in learning in order to extend professional effectiveness (p. 44) Organised and constructive work with a range of colleagues to enhance learners' experience is a consistent feature of the teacher's practice (p. 44) The teacher develops high quality relationships with colleagues in order to have a positive impact upon learners' experiences within the school (p. 45)

secondary teachers. Beginning teachers who network and work collaboratively talk of feeling supported emotionally and with their evolving pedagogy. The ease of connectivity through online platforms creates opportunities to 'e-meet', share and discuss practice. If opportunities already exist to join a community of practice, we encourage you to take advantage of these, e.g. TeachMeets. If not, consider how you might establish a community of practice with other beginning teachers at your placement school, ITE programme or from within your subject area. Effective collaboration will almost certainly be significant in supporting your progress through your ITE programme and beyond.

As demonstrated above, teaching is essentially a relational and ethical activity. As a beginning teacher, you will interact with pupils to assess their learning needs and wellbeing. You will work closely with a team of teachers and support staff to address identified needs and refer concerns to your mentor and senior professionals. You will participate in communities of enquiry with your peers and mentors to discuss your experiences and professional development. When you are appointed to your first post, you will take responsibility for classes and work closely with age/stage/department/

Table 1.2.4 Ten tenets of collaborative professionalism

1. Collective autonomy	Teachers are given authority to make decisions about professional practice.
2. Collective efficacy	Teachers believe that together they can make a real difference to pupils' learning and lives.
3. Collaborative inquiry	Teachers work together to problem solve and reflect critically on issues of common concern.
4. Collective responsibility	Teachers acknowledge mutual obligation and shared accountability.
5. Collective initiative	Improvement is a community effort to which all can contribute.
6. Mutual dialogue	Openness to constructive feedback and reciprocal learning.
7. Joint work	Working together to support learning and improvement.
8. Common purpose	Shared commitment to help pupils reach their potential and flourish.
9. Collaborating with pupils	Teachers involve pupils in decisions about change.
10. Big-picture thinking for all	Shared vision about the direction of change.

Source: Adapted from Hargreaves and O'Connor (2018, pp. 6–7)

faculty leads in regard to curriculum, assessment and pedagogy. You will strive to foster positive home-school links. Outcomes for children are enhanced by effective collaboration and partnership work between educators, administrators, children's services, parents/carers and families. Similarly, teacher learning and professional growth is enhanced when learning networks are strong and sustained across the career course. Hargreaves and O'Connor (2018) identify ten aspirational principles of collaborative professionalism in teaching (see Table 1.2.4; then complete Task 1.2.4).



Task 1.2.4 Network map

During your school placement, consider how far each of the tenets of 'collaborative professionalism' (Hargreaves and O'Connor, 2018) are evident. As you become familiar with your school placement environment, make notes and collect examples of the practices and places where collaboration takes place, e.g. department office, staffroom, corridors, meetings.

- Does collaboration have to occur formally for it to be most effective?
- Which of the ten principles do you recognise from your daily experiences in school?
- In what ways do teachers realise these principles in action?
- How do teachers use their time in collaborative ways?
- What are the main drivers of collaboration?
- What are the main barriers that inhibit greater collaboration and how might these be overcome?

Use your notes to draw a network map of all the connections you have with other adults in school and the links you have with other people and organisations that support your professional learning outside school.

- What does your professional learning community look like?
- Who or what is missing?

- How could your learning network be strengthened?
- What support do you need for this to happen?

Keep a copy of this map in your file. Discuss it with formal and informal mentors in school and add to it as you progress through your ITE programme.

Individual level: your professional learning journey

As you begin your ITE programme, remember it is 'initial' teacher education; the start of what we hope, will be a long and fulfilling career through which you will continue to learn, reflect, evolve and improve. Beginning teachers do not commence their ITE journey from the same starting point. For example, you may have spent several years working as a Teaching Assistant (TA) or you may have worked with young people outside a school context. You may have decided to change career after years in industry or you may be starting your ITE straight from your undergraduate degree with little experience of working with young people. Whatever your starting point, you have secured a place on your ITE programme because of the potential you have shown and you will take your own qualities, skills and talents into the classroom. Your ITE programme is designed to offer opportunities and experiences of, and insight into, the world of school and the busy, varied and challenging life of a teacher. No-one expects you to be an expert, be realistic in your expectations of yourself as you embark on your ITE programme.

As you start your placement, it is your responsibility to be organised. You must provide lesson plans in good time (usually at least 24 hours before the lesson), arrive for lessons with all the necessary equipment/resources organised and in place. You must communicate effectively and in a timely and appropriate manner with colleagues, mentors, pupils and parents. You will be required throughout your placement to evidence your progress against the standards relevant for your programme. Gathering evidence of your progress may start slowly, this is to be expected. It is important that you know and understand the expectations of your placement and work to demonstrate your progress against each aspect of the standards. This may include, for example, a write-up of having attended a consultation evening, your contribution to a whole-school event or a thank you card from a pupil or parent. Other evidence is drawn from your lesson plans, your evaluations and observations carried out by your subject or professional mentor. It may be possible to digitally record part of your lessons to provide additional evidence, but before doing so be sure to check this with your school's policy on recording. Many schools now use video platforms such as IRIS Connect (https://www.irisconnect.com/uk/) to record and share classroom video. Be proactive. It is better to have a wealth of multi-modal evidence gathered throughout your school placement(s) from which you can select your best practice than realise in the final weeks of a placement that you have insufficient evidence in some areas. Keeping your folders (e-portfolios and/or hard copies) in good order is essential. You are required, throughout your placements, to be reflective and reflexive as you plan, prepare, teach and evaluate your lessons. Cultivating habits of reflection, target setting and self-evaluation during ITE will help you to sustain your professional learning into induction and the early career phase (i.e., the first three years of teaching).

You are not expected to progress systematically, in an orderly fashion through the standards, moving from one to the next. Your progress will be unique to you; avoid drawing comparison with other beginning teachers and do not expect your progress to be linear. You may grasp some aspects of teaching quickly, for example, you may swiftly establish a positive climate for learning

Table 1.2.5 Phases of beginning teacher development

Self →	*Class* →	*Pupil*
Self-image and class management	Whole-class learning	Individual pupils' learning
How do I come across?	Are the pupils learning?	What are the different needs of my pupils?
Will they do what I want?	What are the learning outcomes?	How effective are my strategies for ensuring all pupils learn?
Can I plan enough material to last a lesson?	Am I achieving my objectives?	How can I find out?
	How do I know?	

and demonstrate your ability to incorporate effective assessment into your teaching. However, you may find that your subject and curriculum knowledge is limited and you may find it difficult to adapt your teaching to meet the needs of all pupils. Do not fear! Be honest with yourself about where your experience is limited and your knowledge is lacking. Discuss with your mentors and colleagues how you might address gaps and reflect on what you could do differently and do better, to support your progress. It is likely that you will be required to work on a subject knowledge audit as part of your course, if not we encourage you to create your own audit to help you identify where work needs to be done and to help you keep check of your progress.

As a beginning teacher you progress through phases of development throughout your ITE programme. In the first weeks, it is not unusual to experience insecurity with respect to your emerging teacher identity. As you gain experience and skill, your attention shifts from your exposition, timing and materials to noticing and appraising the impact of your decisions on pupil learning. Table 1.2.5 identifies three phases of development experienced by beginning teachers in the early stages of practical teaching. These reflect the transition from a concern with self and belonging, to a focus on the whole class, to a focus on individual pupils (Capel, 2019, p. 37). You progressively shift your attention from a narrow concern with self-identity, survival and control, to develop a repertoire of strategies that can be adapted to a range of circumstances to promote whole-class and individual pupil learning. With support and reflection on practice, you will move from a preoccupation with 'surviving' to assessing your efficacy and impact as an educator.

Make the most of your time on school placement to gain as much experience of school life as possible. Speak to many different colleagues, not just those you work with closely. Talk to TAs and non-teaching staff, the SENCO, and pastoral and safeguarding leads. Offer to support, or set up, extra-curricular clubs or activities and get involved in the wider life of your school. The experiences you gain outside the classroom are valuable and will help you establish positive relationships with your colleagues and pupils while at the same time helping you evolve into the teacher you want to be (see Chapter 8, Your professional development).

It is important that you make time and take time to get to know your pupils as individuals. Knowing your pupils as individuals will enable you to stretch and challenge all pupils across the age and ability range and in so doing meet their individual learning needs. Knowing your pupils will enable you, for example, to intervene in a timely manner to stop undesirable behaviour from escalating or you may use praise in a specific way to motivate a particular pupil. Do not be afraid to take risks, try new ideas and different approaches as you explore what works well for you, as you learn and progress. Take time to think about the teacher you aspire to be, the values you will uphold and the behaviours

you will endeavour to demonstrate every day, in every interaction. Knowing and revisiting your aspirations regularly, will help you develop and strengthen your teacher identity.

Whether you have significant school-based experience, or you have not been in a school since your own school days, your ITE is going to be a challenge. The workload is significant, your days long and the pressure, at times, is intense; you may even question whether you have made the right decision to embark on a career in teaching. It is for these reasons that it is of upmost importance that you make time to focus on your mental health and wellbeing and look after yourself. Establish boundaries and determine a time each evening when you will stop work. Your pupils will not benefit from a sleep-deprived teacher. Set aside time to do things that you know support your wellbeing, whether that be exercise, baking or catching up on a box set. Do not struggle on alone, talk to your colleagues, your peers and your mentors; they all understand something of your experience and will be able to offer advice and support. Teachers often find it difficult to switch off from work so prioritise a manageable work-life balance from the outset. Start as you mean to go on. Pupils need energetic, enthusiastic and positive teachers so plan time to rest, recharge and restore. Do not underestimate the importance of taking care of yourself, not doing so is likely to lead to burn-out, stress, illness and absence and no one wants that, least of all you (see Unit 8.4 Developing your resilience, managing stress, workload and time). There are a number of resources available to help you manage your social, physical and psychological wellbeing, e.g. the traffic light system identified by NHS Education for Scotland (2022), https://learn.nes.nhs.scot/30741.

Finally, hold firm to the reasons you decided to embark on your ITE programme. Teaching is a rewarding, fulfilling and wonderful career. This is an exciting time, do not forget to enjoy it.

Some advice from beginning secondary teachers approaching the end of their ITE programme is offered in Figure 1.2.2.

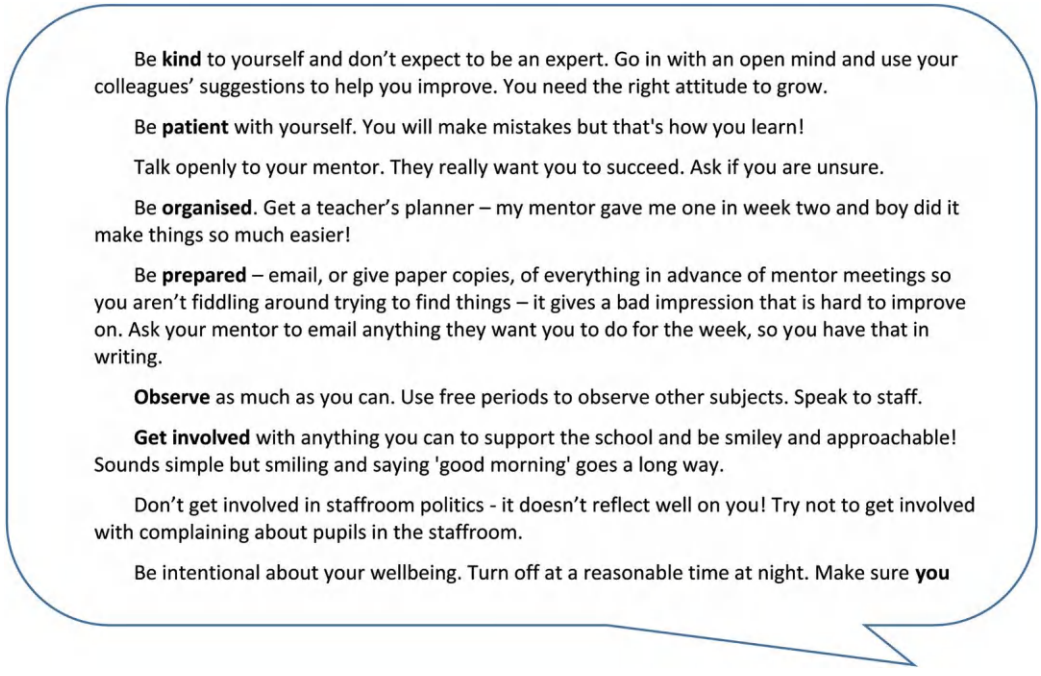


Figure 1.2.2 Advice from early career teachers on navigating your first placement

SUMMARY AND KEY POINTS

This unit has touched on your multiple, changing roles and responsibilities as a beginning teacher. We hope that it has given you a better understanding of:

- the preparation you can do prior to school placement;
- what you need to do and how you will be supported during school placement;
- the expectations of you, your roles and responsibilities in school and as a professional;
- the phases of beginning teacher development over your ITE programme.
- As a result, we hope this has given you a better understanding of your development as a teacher.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Capel, S., Lawrence, J. Leask, M. and Younie, S. (eds.) (2019) *Surviving and Thriving in the Secondary School: The NQT's Essential Companion*, Abingdon: Routledge.

Hulme, M., O'Sullivan, R. and Smith, R. (eds.) (2021) *Mastering Teaching. Thriving as an Early Career Teacher*, Buckingham: Open University Press.

These two books are designed to support early career teachers in the next phase of development as a teacher. However, you may find them useful as they cover aspects of teaching not included in this book that, nonetheless, you experience on your ITE programme.

Hargreaves, A. and O'Connor, M.T. (2018) *Collaborative Professionalism. When Teaching Together Means Learning For All*, Thousand Oaks, CA: Corwin/Sage.

This accessible book explains the importance of working with others in education to articulate and realise common goals. A wealth of classroom, school and district-level examples and strategies are used to illustrate the value of collaboration in teaching well.

Any additional resources and an editable version of any relevant tasks/tables in this unit are available on the companion website:

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1.3 Using digital technologies and artificial intelligence (AI) for professional practice

Andrew Connell

Introduction

The use of digital technologies will challenge you as a teacher to think about how you design opportunities for learning. Technology should do this. It should invite you to question when and when not to use it in your practice and what difference it will make to the progress of the pupils you teach. Fundamentally, digital technologies can shift the responsibility to the pupils themselves. They can demonstrate their understanding, opening up greater freedom and the development of skills for life-long learning and employment. In doing so, they "try new things, seek to learn from those experiences, and work to adopt and embed the practices that work best" (Sharples et al., 2019, p. 3). Digital technologies offer you opportunities to further enhance pupils' learning. Therefore, as a teacher you should look for opportunities to develop your ability to use digital technologies effectively.

Schools are adaptable and complex learning organisations which rely on an increasingly embedded digital infrastructure to support aspects of the curriculum (Selwyn, 2019). The technologies used within the classroom are often mirrored by the technologies popularised by a society, its access to funding and its skilled implementation by its workforce. "Ultimately, it doesn't matter how great an educational idea or intervention is in principle; what really matters is how it manifests itself in the day-to-day work of people in schools" (Sharples et al., 2019, p. 3).

OBJECTIVES

At the end of this unit you should be able to:

- understand the relevance of digital technologies and artificial intelligence (AI) for you and your pupils;
- plan to teach using digital technology resources to enhance the pupil learning experience;
- understand the potential of online learning;

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- use an appropriate framework for auditing your knowledge and understanding of digital technologies;
- understand your role and responsibility in promoting online safety and digital citizenship for both yourself and your pupils.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Understanding the relevance of digital technologies for you and your pupils

Digital technologies used innovatively in your subject teaching can facilitate and enhance learning for your pupils (this is demonstrated in Figure 1.3.1). Ultimately, the aim is to enable pupils to become both confident and competent in: learning to use the technology; trying and applying what they have learned in different contexts; then challenging that understanding. This cycle of application helps support a pupil's fluency in using digital technologies and supports their learning in the subject. Task 1.3.1 asks you to consider the use of digital technologies in your teaching.

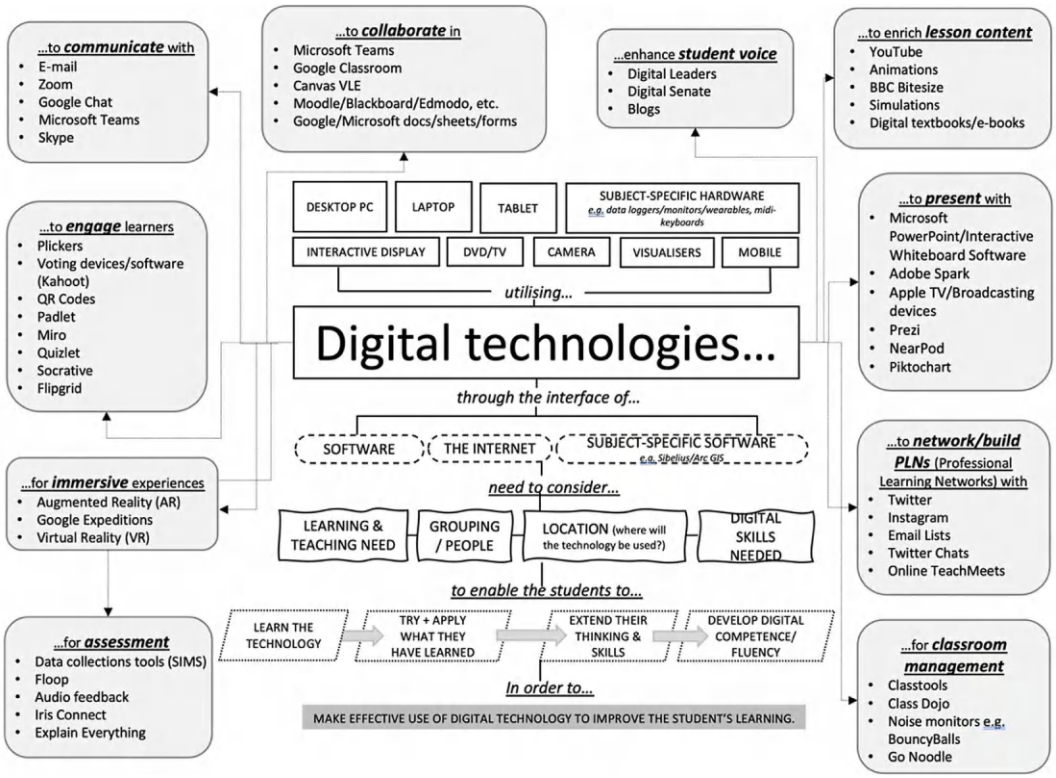


Figure 1.3.1 The scope of digital technologies in the classroom

Technology as a tool

Where the technology is ***a physical object*** used to support learning (e.g. a computer, a digital camera, mobile technology, Virtual Reality [VR]/Augmented Reality [AR]/Mixed Reality [MR], assistive technology, artificial intelligence [AI]). Learning to use the tool is the focus of the pupil activity.

Technology as a subject

Teaching the ***subject matter of technology*** (databases, graphic packages, spreadsheets, the educational programming language Scratch, desktop publishing, digital citizenship, digital health and wellbeing, e-safety). This type of lesson supports pupils in understanding specific content knowledge around the Computing curriculum.

Technology across the curriculum

The use of technology ***to support the teaching of a curriculum subject***. For example, how does technology assist you in teaching your subject?

Figure 1.3.2 The application of digital technologies into the classroom



Task 1.3.1 Consider the use of digital technologies in your teaching

Consider some lessons you are teaching. Identify what technological applications you are using and why you might be using them. Then consider how you might change your approaches, using Figure 1.3.1 to help you. The middle section encourages you to think about the technology and its classroom application, whilst the outer circles provide reasons why you use digital technologies as well as some suggestions of the tools you could implement within your teaching.

The application of digital technologies into your classroom can be considered in three ways (see Figure 1.3.2).

Many subject teachers are already making good use of digital technology and looking out for new ways of using digital technologies to stimulate pupils and extend their learning. These teachers understand that digital technologies are a tool to be applied selectively but they are not the complete solution to meeting pupils' needs. The role of the teacher when using technology is often multi-purpose: to teach; to facilitate; and to learn with the pupils.

Plan to teach using digital technology resources to enhance the pupil learning experience

There are different considerations to think about when integrating digital technologies into your lesson. As well as generic lesson planning (see Unit 2.2), the introduction of different hardware or software into your teaching embeds a tool which may change the way you teach that lesson. Using the bullet points below as prompts, complete Task 1.3.2 to support your thinking for a current lesson you are planning.

Task 1.3.2 Planning to teach using digital technology resources

Points to consider:

- Is a technology you are planning to use for teaching or for learning? Both are implemented in slightly different ways.
    - When incorporating a technology in your *teaching* you are considering the impact on the *delivery of the content*. Will this simulation strengthen my teaching point? Will using Google Maps assist the pupils in seeing the different parts of a river so they can understand how best to protect against flooding?
    - When incorporating technology for *learning* you are considering how the use of the technology enables pupils to enhance their understanding in ways they could not easily do before. For example, the use of the app 'Coach's Eye – Video Analysis' or the slow-motion feature of an iPad to assess a sports technique?
- Does the type of technology affect how you will *group* the pupils? If pupils are in pairs sharing the technology, are you monitoring that both pupils are engaging with the technology and not just the more confident pupil?
- Do the pupils already know how to use the technology, or will extra time be needed for them to learn specific procedures or techniques?
- Does your school adopt a certain model of technology provision that affects the way you plan? A Bring Your Own Device (BYOD) approach implies that the use of digital technologies is a constant presence in the classroom whilst the use of bank laptops relies on the individual teacher to organise that opportunity (https://edfutures.net/).

Plan an activity using digital technologies in a lesson. Discuss and critique your planned activity with another beginning teacher. What amendments, if any, should be made? Why?

The potential of online learning

The debate around the use of digital technology and its potential for online learning stretches back to the early 2000s (O'Leary and Ramsden, 2002; British Educational Communications and Technology Agency (Becta), 2004; Laurillard et al., 2011; Salmon, 2011). Beginning teachers entering the profession need to be familiar with the range of approaches as outlined within the Online Learning Spectrum (Figure 1.3.3) as each school's approach is likely to adopt one or a combination of these within their setting.

Artificial intelligence (AI) in schools

There is much in the media about AI and its potential impacts, good and bad. This applies just as much in education as elsewhere. AI is, however, just another technological tool, which when used well can support learning and teaching, but also can be misused. There are a range of AI tools and applications that could be used. For example, see https://www.carnegielearning.com/blog/21-ai-tools-for-teachers/. It is worth identifying other teachers who are ahead of the game, such as Matthew Wemyss on LinkedIn.

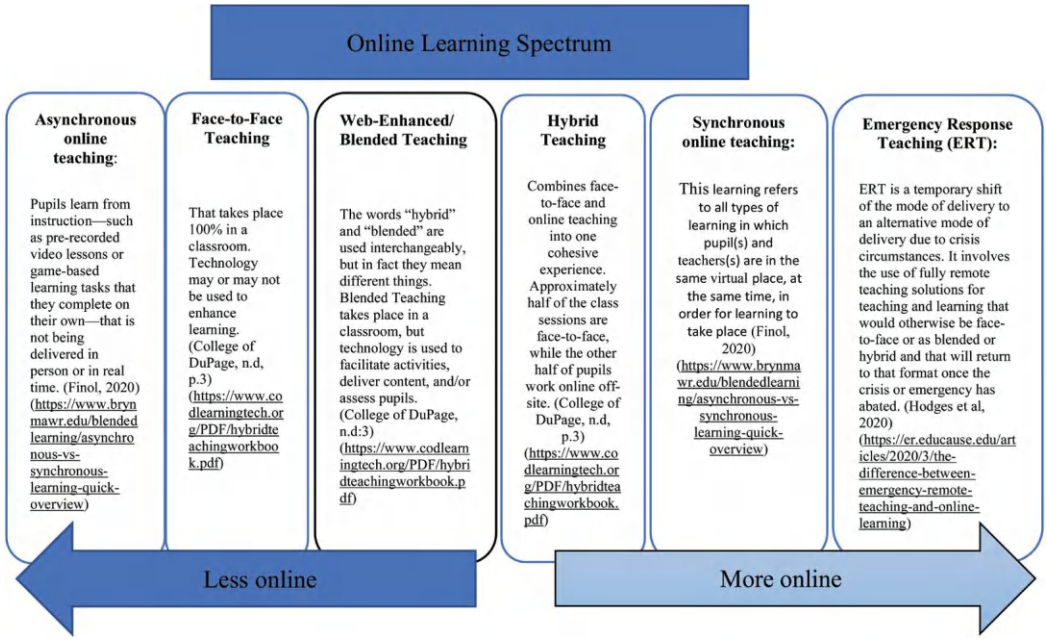


Figure 1.3.3 The Online Learning Spectrum

Source: Adapted from College of DuPage (n.d., p. 3)

Your pupils will meet AI throughout their lives, for example, Chat Bots, so they need to know about it and how to use it appropriately and how to avoid its pitfalls. Many jobs will use AI to some extent. Everyone needs to know how to recognise when AI is used, especially when it is misused.

There is some evidence that pupils are using AI more than teachers already (Wemyss, 2024; Department for Education [DfE], 2025a). Schools are beginning to create AI policies (Fullan et al., 2024) and the DfE has published a paper looking at how Generative AI (GenAI) is used at the moment in schools, which includes consideration of opportunities, impact and barriers to using GenAI (DfE, 2025a).

If you are going to use AI tools in your classrooms consider these points (which equally apply to other tools):

1. Why are you using it? Will it help learning? Is it the best AI tool for the purpose?
2. Will the pupils see the point of the particular application/tool and engage with it? Discuss this with them.
3. Show your pupils how to use it. Point out common errors and limitations. Teach them about using prompts effectively.
4. Be clear about which activities AI can/could be used with (and which it cannot) and when other methods would be better. Help them understand how to make this choice.
5. Have open conversation with your pupils about the ethics of AI, and how it can support their learning.
6. Stress that AI is not a substitute for human thinking and problem solving. It can help, it can provide alternative solutions, but it cannot replace the need to critically evaluate solutions.

Jacob (2024) presented a model of four principles that is useful to share with pupils and staff, for them to apply if thinking about using AI. This is an adaptation of that model.

1. Understand what AI can and cannot do – get the pupils to research this. What are the legal and ethical implications of using AI?
2. After the research, choose the right AI tool(s) and method(s) and prompts to create some output.
3. Critique the output. Is it true, is it complete (critically evaluate)? Does another tool produce a better/different output (compare and contrast)? Refine the better output.
4. When you are happy with the output, use it effectively, ethically and appropriately. Acknowledge the use of AI.

AI might also be used to support teaching and reduce workload (Goldman et al., 2024; Hashem et al., 2024; DfE, 2025a). There are ways it can support planning, research, resource creation and assessment. However, as for the pupils, be aware of the ethics, limitations and possible pitfalls of using it. Always evaluate outputs to check they are accurate, complete and meet your pupils' needs and then adapt the outputs to make sure they do.

As a teacher, what do I need to know regarding digital technologies?

We live in a world that is influenced by our engagement with digital technologies for the way in which we live, play and work. The following section outlines what secondary school teachers need to know regarding the implications of, and guidance for, using digital technologies for teaching and learning in the classroom, monitoring pupils' progress and their own professional development:

Data Protection: Be aware of the General Data Protection Regulation (GDPR), which was implemented in May 2018 (superseding the Data Protection Act), and its implications for secure storage and usage of both electronic and manual data related to pupils. See https://www.gov.uk/data-protection for the latest on data protection in the UK. You need to understand how your placement schools meet the requirements of GDPR and the impact of GDPR upon your teaching, including the use of any online digital tools that you might use.

Online Learning: Be aware of different approaches to working online with your pupils. Know the difference between online *learning* and online *teaching*?

Digital Tools: Be aware of and able to use digital tools to support teaching and learning inside and outside the classroom. Examples include digital tools for formative assessment, such as Kahoot or Yacapaca, and digital resources such as TED-ED (https://ed.ted.com/) to enhance and enrich teaching and learning.

Digital Champions: Know who the digital champions are in your subject area; subscribe to their social media. This can keep you up-to-date with developments regarding how digital technologies are being used in your subject.

Digital Resources: Check out what your subject association identifies as digital resources to support teaching and learning.

Digital Literacy: Ensure you are able to use and promote critical digital literacy in your subject, just as you would with literacy and numeracy. You are expected to model good practice. This

includes the use of applications for teaching, learning and administrative tasks, e.g. creating resources, planning or tracking pupils' progress.

Digital Footprint: Ensure that you are aware of your own digital footprint, especially across the social media that you use.

Some of these topics are explored in the following sections.

The relevance of digital technologies for you and your pupils

You need to be clear about why you might use digital technologies in teaching your subject lessons. Is it to entertain, to engage their curiosity, enthuse them, enlighten them about a difficult topic or concept or to empower their learning in your subject? It should be the last four.

Subject associations, museums, art galleries, Ofsted and technology companies (such as Apple, Google and Microsoft), provide excellent examples of how existing and emerging technologies can be used to support teaching and learning within and outside the classroom. For example, some museums and art galleries have digitalised their collections and placed them online and offer virtual tours, so that they can be used as resources for teaching and learning.

STEM (science, technology, engineering and mathematics) Ambassadors delivered remote sessions using video conferencing, and subject matter experts create and post videos on YouTube. Badger et al. (2019), reported the use of technology to promote pupil metacognition through 'think alouds', whereby pupils (and their teacher) used digital technologies to help them produce their own screencasts of equations, calculations, complex abstract concepts as well as demonstrating their examination question technique. This promoted an understanding of learning strategies as well as engaging the pupils in metacognitive talk. Their use of digital technologies helped deepen their analysis and understanding more. Pupils were encouraged, when correcting their work in practice examinations, to explain the error in their original thinking and then describe how they arrived at the correct answer.

Google Expeditions (https://artsandculture.google.com/project/expeditions) enables a class to be taken on an immersive virtual reality field trip without the need to leave the classroom! For example, in geography, your pupils can explore coral reefs; in RE, pupils can experience festivals from around the world; and, in science, pupils can explore human anatomy.

The key points of the above examples of good practice are that:

- the best use of digital technologies as a resource is context-driven;
- if a pupil sees the benefit, then they are more likely to use the same technologies again elsewhere and more importantly within a different subject.

What is common about the 'good practice' cited is that the digital technologies were applied in both a contextual and in a practical manner. If you are going to develop your digital confidence, digital capability and digital competence to mirror the good practice presented, understanding the theoretical work that underpins these approaches should help you to do so. In these cases, the approaches are underpinned by the work of Vygotsky (1978) and his theory of the Zone of Proximal Development (ZPD) and Bruner's (1961) theory of discovery learning (see Unit 5.1). Now complete Tasks 1.3.3 and 1.3.4.



Task 1.3.3 Using digital technologies to support teaching and learning in your subject

In Task 1.3.1, you were asked to consider digital technologies you use in your teaching to support pupils' learning. You are now asked to identify how digital technologies, including AI, are being used to support teaching and learning in your subject more generally.

Ask other teachers in your subject area, and check your subject association for what they identify as digital resources to support teaching and learning.

Discuss with another beginning teacher how you can use digital technologies to engage, enthuse and empower pupils in the subject you teach.



Task 1.3.4 How are digital technologies used in an examination specification in your subject?

Look at an examination specifications for your subject and identify where digital technologies can be used to enhance the teaching and learning experiences. Remember, not all topics lend themselves equally to the use of digital technologies.

Using an appropriate framework for auditing your knowledge and understanding of digital technologies

In order to use digital technologies effectively to enhance pupils' learning, it is important that you are clear about your own level of knowledge, skills and understanding; this section is designed to help you achieve this by presenting digital competence frameworks that you can use to self-assess your own digital competence as a teacher.

One starting place is to self-review your knowledge, skills and understanding of how digital technologies can support your teaching, and your pupils' learning in your lessons. There have been a number of initiatives to support teachers to develop their digital literacy competences, including:

- The European Framework – Digital Competence for Educators (DigCompEDU) framework (Redecker, 2017; Caena and Redecker, 2019)
- Digital Teaching Professional Framework (Education and Training Foundation, 2018)
- Learning Wales Digital Competence Framework (Welsh Government, 2018)
- The DigiLit Leicester project (Fraser, Atkins and Hall, 2013)
- JISC Digital Capabilities Framework (JISC, 2018).

For example, The Digital Teaching Professional Framework (https://www.et-foundation.co.uk/supporting/edtech-support/digital-skills-competency-framework/) created by the Education and

Training Foundation, is a competency framework for teaching and training teachers comprising seven areas:

a. Planning your teaching;
b. Approaches to teaching;
c. Supporting pupils to develop employability skills;
d. Subject-specific and industry-specific teaching;
e. Assessment;
f. Accessibility and inclusion;
g. Self-development.

Although written for the further education sector in England, it is applicable to teachers in secondary schools throughout the UK. The DigCompEdu framework, derived from the DigComp 2.1 Framework (Carretero, Vuonkari and Punie, 2018), aims to capture these teacher-specific digital competences. It consists of six different competence areas with a total of 22 competences which focus on:

1. professional environment;
2. sourcing, creating and sharing digital resources;
3. managing and orchestrating the use of digital tools in teaching and learning;
4. digital tools and strategies to enhance assessment;
5. the use of digital tools to empower pupils;
6. facilitating pupils' digital competence.

Areas 2 to 5 of this framework are the pedagogic core of the framework, which detail the competences that teachers require to foster effective inclusive and innovative learning strategies using digital tools.

Arising out of the DigCompEdu framework is the SELFIE for Teachers tool (https://education.ec.europa.eu/selfie-for-teachers) which allows teachers to self-reflect on their own digital competencies. This tool is not designed to assess an individual teacher's performance, but to empower them to reflect on their own use of digital technologies in the classroom. Teachers can then personalise their own learning pathways to further develop their digital competence, confidence and capability (Task 1.3.5).



Task 1.3.5 Your digital competence SELFIE

Complete your own self-reflective digital competence selfie by using the SELFIE for Teachers tool (https://education.ec.europa.eu/selfie-for-teachers). Once you have completed the SELFIE statements you will automatically receive a report on your proficiency level in each of the digital competence areas and suggested next steps. You will also receive a certificate and digital badge.

Revisit and review your SELFIE certificate and update your knowledge at regular intervals during your ITE programme.

Understanding your role and responsibility in promoting online safety for both yourself and the pupils you teach

In 2018, the UK Council for Child Safety (UKCCIS) Education Working Group produced *Education in a Connected World*, a framework for online safety. This framework details the knowledge and skills pupils need to navigate the internet safely. In 2019, the DfE published *Teaching Online Safety in Schools* (DfE, 2025b), which provides guidance for teachers teaching pupils how to stay safe online. Both documents provide a useful reference point for the expectation and content to be covered in secondary schools.

Online safety or e-safety (DfE, 2025b) falls under the remit of safeguarding within a school and your placement school will have a designated safeguarding team. Some online safety issues are addressed in dedicated personal, social and health education (PSHE) class sessions or within a year assembly. Issues that place pupils at risk that may be discussed include online bullying, online grooming, sexual exploitation, changes to sexual behaviour due to sexting, being misled by fake news and AI, radicalisation, online gambling, addiction to online gambling.

As a teacher, it is important to make your pupils aware of not only the benefits of digital technologies, including social media, but also their potential risks. You should speak openly and honestly about the negative impact of digital technology. Even the most diligent of teachers cannot prevent the misuse of social media and peer pressure amongst secondary-school-aged pupils. For example, as a positive, TikTok as a platform houses vast amounts of learning material that pupils can access asynchronously, encouraging them to fill in gaps in their knowledge and provides the ability to watch the introduction of concepts repeatedly at their own pace if the input was missed during class; however, there is always a real concern about inappropriate or misleading content. Although, many issues are addressed centrally, as a teacher you are responsible for online safety in your lessons and also reporting any issues that you become aware of. Seek support from your school's designated safeguarding team for their advice and guidance if you are unsure of how to proceed with complex issues. The school may also have its own resources, and we can highly recommend classroom resources produced by Child Exploitation and Online Protection Centre (CEOP), BBC, Vodaphone's Digital Parenting and Safer Internet Day to help you with delivering digital safeguarding topics. These resources range from animations to short films. Usually, pupils take these seriously. Pupils can be asked to reflect on what their thoughts are about a video they have seen. Informing pupils that there are no right or wrong answers provides an environment in which a mature debate and discussion can take place. You can then ask pupils to devise some general rules or advice based on the videos that other pupils should follow regarding online safety.

Online safety cannot simply be addressed in one lesson or one form period each year. All technology-related issues have intertwined behavioural elements to them. In the same way you plan for and manage behaviour and expectations in the classroom, the same considerations should be in the forefront of your mind when addressing pupil's digital safety. Continually asking yourself how you can develop pupils' awareness of the issues as well as responding to incidents of digital safety as they arise in a proportional way is something that all curriculum subjects can address as the opportunity arises.

If an online safety incident does occur and you are aware of it, then it should be dealt with seriously and you should follow your school's safeguarding procedures as online safety is a whole school issue. As soon as it is practically possible, inform the teacher whose class you are taking what has been disclosed to you. Also, ensure that your mentor, the designated school safeguarding lead(s) and, if you are on an ITE programme in partnership with a university, your university tutor

are made aware of your discovery so that they can advise and support you. Task 1.3.6 asks you to explore safeguarding procedures in your placement school.



Task 1.3.6 Safeguarding procedures in your placement school

Ensure you are familiar with the safeguarding procedures specific to each placement school. Summarise this procedure as a flowchart and record who are the designated school safeguarding lead(s).

As part of your safeguarding training during your ITE programme, you may undertake online safeguarding training, for example, CEOP. You may also wish to undertake your own self-assessment regarding your digital literacy skills in relation to e-safety and online identity. The self-assessment of E-Safety and Online Identity from the Digital Literacy Competency Framework (Fraser, Atkins and Hall, 2013), is reproduced in full on the companion website.

Being part of world-wide digital community

As soon as either you or your pupils begin to use the Internet, you and they are connected to a global digital community in which everyone has a digital voice and leaves a digital footprint of what they have accessed, listened to and everything they have posted on online platforms as a conversation, image or video. Not only do we have physical citizenship but by using digital technologies then we have a digital citizenship too. Digital citizenship refers to the individual's responsible use of digital technologies to engage with society on any level. One aspect of teaching with digital technologies is to promote good digital citizenship for your pupils. As teachers you promote the following seven key concepts of digital citizenship:

1. empathy;
2. how the Internet works;
3. understanding user data;
4. practicing critical digital literacy;
5. acknowledge the digital divide that exists in society;
6. practicing digital wellbeing;
7. securing digital devices.

As schools take preventative measures to ensure pupils' online safety, for example, asking pupils to read, sign and abide by an Acceptable User Policy (AUP), adhering to guidance for behaving responsibly online and asking them to select a secure password for their login to the school's network, so should you. You will be required to sign the staff version of your school's AUP at the start of a school placement, prior to being allowed to access the school's network. In addition, you might want to consider your own digital footprint by completing Task 1.3.7. One reason for this is that pupils may well search the web to discover what they can about you. You might want to consider amending your profile name, so that it is harder for your pupils to find you. You never know who might read (or screenshot) your postings! The DfE encourages schools to check the profile of job applicants.

Task 1.3.7 Reviewing your digital footprint

It is important, before you start your ITE programme (and school placements), to ensure photographs and references to you on the internet represent you as the professional you want to be. In order to discover and review your online presence, take the following steps:

1 Google your name and see how many search results relate to you.
2 Which search results would you want the pupils you teach, or a potential employer, to see? (If there are any you would not want them to see, take them down.)
3 Review your privacy settings on any social media platform that you belong to and any social media apps. You should amend your settings on these so that only trusted friends can see what has been posted.
4 Review your social interactions. Do they set the image you would want to portray as a professional in school?

SUMMARY AND KEY POINTS

The dialogue around the use of different hardware and software can really be reduced to two questions which have always underpinned the use of digital technologies: what difference does its use make to the progress of my pupils and how might it support my teaching?

This unit has touched on different aspects of digital technology, however its use is a vast area of subject knowledge, so you are encouraged to follow and develop your interests. Remember to experiment with different ways of incorporating technology into your lessons. Figure 1.3.1 identifies different avenues for you to explore.

Technology, including AI, is just a tool you can use in your toolkit. As a carpenter knows the right tool to use at the right time, the same is true with technology; ultimately you are in control. We hope that this chapter has given you a better understanding of:

- the relevance of digital technologies for you and your pupils;
- planning to teach using digital technology resources to enhance the pupil learning experience;
- the potential of online learning;
- using an appropriate framework for auditing your knowledge and understanding of digital technologies;
- your role and responsibility in promoting online safety and digital citizenship for both yourself and the pupils you teach.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

For bonus digital resources designed to enhance the book materials, head to routledgelearning.com/learningtoteach.

Further resources

The suggested resources have been selected to encourage you to think about opportunities for using all types of digital technologies in both an innovative and effective manner to enhance teaching and learning in your subject. They also seek to demonstrate that inspiration can come from an unexpected range of sources, and so pose the question: 'That looks interesting: could I adopt that to enhance my subject teaching?'

Atherton, P. (2018) *50 Ways to Use Technology Enhanced Learning in the Classroom: Practical Strategies for Teaching*, London: SAGE/Learning Matters.

This practical guide explores different ways to the use of technology enhanced learning (TEL) in the classroom. Areas such as gamified learning, formative assessment, social media, the flipped classroom, instant feedback tools and many more are introduced as tools to support learning.

Bradshaw, P. and Younie, S. (2018) 'Understanding online ethics and digital identity', in S. Younie and P. Bradshaw (eds.) *Debates in Computing and ICT Education*, Abingdon: Routledge, pp. 114–136.

This chapter provides an understanding for both pupils and teachers of ethical issues concerning digital technologies and an awareness of your online identities.

Fraser, J., Atkins, L. and Hall, R. (2013) *DigiLit Leicester: Initial Project Report*, Leicester: Leicester City Council (CC BY-NC 3.0), viewed 21 July 2025, https://cabinet.leicester.gov.uk/documents/s64164/Building%20Schools%20for%20the%20Future-Update%20-%20Appendix%201a.pdf

This early report from the 'DigiLit Leicester' research project contains a digital literacy competency framework with all of its descriptors and statements, which allow you to assess your own digital literacy skills. This also enables you to identify which competences you may need to develop further.

Kukulska-Hulme, A. and Traxler, J. (2020) 'Design principles for learning with mobile devices', in H. Beetham and R. Sharpe (eds.) *Rethinking Pedagogy for a Digital Age – Principles and Practices of Design*, Abingdon: Routledge, pp. 181–196.

This chapter explores ways mobile technology is making learning more personal, informal and pupil centred and the challenges teachers have designing for individuals.

Luckin, R. (2018) *Enhancing Learning and Teaching with Technology: What the research says*, London: UCL IOE Press.

This book brings together research findings to provide an evidence-based approach to using digital technologies in the classroom and highlights effective practice.

You may find the following resources and websites sources of inspiration as you seek to develop your own digital competence.

Child Exploitation and Online Protection (CEOP) (https://www.ceop.police.uk/safety-centre)

CEOP is a command of the United Kingdom's National Crime Agency (NCA) tasked to bring online sex offenders to the UK courts. One of CEOP's roles is that of public awareness campaigns and educational programmes, including the ThinkUKnow education programme (https://www.thinkuknow.co.uk/), which is currently used in UK schools.

Digital Enhanced Education Webinars.

A collection of useful YouTube videos. Mainly HE focussed, but some of it is transferrable to other sectors. https://www.youtube.com/channel/UCDs-c_TX9GUIA3ODjlbayYQ

Digital Parenting (Vodaphone) (https://www.vodafone.co.uk/mobile/digital-parenting)

The Digital Parenting website produced by the mobile phone company, Vodaphone, provides guidance for parents on the use of technologies and how to have difficult conversations regarding improper use of technology. As well as an electronic magazine, they also produce blogs and guidance on digital wellbeing.

EdFutures (https://edfutures.net/Main_Page)

The EdFutures website provides both case studies and analysis on models of technology provision which you might adopt in your classroom.

European Schoolnet (EUN) (www.eun.org)

EUN is a network of 31 Ministries of Education in Europe, which runs subject-specific projects and online CPD courses for teachers, conducts and publishes research into innovative teaching and supports teachers and pupils working on collaborative projects across Europe (i.e. e-twinning).

Fullan, M., Azorin, C., Harris, A. and Jones, M. (2024) 'Artificial Intelligence and school leadership: Challenges, opportunities and implications', *School leadership and Management*, 44(4), 339–346.

Jacob, M. (2024). Critical thinking at the heart of Gen AI literacy. Presentation 20/11/2024 as part of the webinar 'Helping Students Develop Critical Thinking When Using Generative AI'. *Digitally Enhanced Education Webinars YouTube Channel*. https://www.youtube.com/channel/UCDs-c_TX9GUIA3ODjlbayYQ

OAK National Academy (https://www.thenational.academy)

This is a DfE funded initiative to provide an online portal for access to online resources for teaching and learning which were developed by teachers during Covid-19.

Safer Internet Day (https://www.saferinternetday.org)

This website curates teaching and classroom resources that can be used on a Safer Internet Day, which is used to promote more responsible use of online technology and mobile phones.

TED-Ed Lessons Worth Sharing (http://ed.ted.com/)

This website provides digital resources to enhance and enrich teaching and learning.

UK Council for Child Safety (UKCCIS) (https://www.gov.uk/government/groups/uk-council-for-child-internet-safety-ukccis)

UKCCIS (2018) wrote Education in a Connected World, a framework for online safety, which details the knowledge and skills that a pupil will need in order to navigate the internet safely.

Any additional resources and an editable version of any relevant tasks/tables in this unit are available on the companion website:?

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2 Beginning to teach

The previous chapter explored the role and responsibilities of the teacher and how you might manage those requirements. In this chapter, we examine how you can learn from observing experienced teachers and then consider key aspects of planning and preparing lessons, taking responsibility for whole classes and developing effective working relationships with your mentor.

For most beginning teachers, there is a period during which you observe other teachers working, participate in team teaching and take responsibility for part of a lesson before teaching whole lessons independently. During this period, you use observation and critical reflection to build your professional knowledge about teaching and learning and develop your professional judgement about managing learning environments. Unit 2.1 is therefore designed to help you actively 'read' classrooms rather than simply watch teachers at work, focusing your attention on the detailed dynamics of effective teaching and learning. Experienced teachers make it look easy, but if you have ever learned to drive, you will appreciate that experienced drivers also make it look easy even though it takes many hours to become proficient.

It can be challenging for a beginning teacher to become fully aware of the extensive planning that underpins each lesson. Schemes of learning (long-term programmes of work) are usually developed collaboratively by teams of staff over time and remain in place for extended periods. The detailed planning for individual lessons may also appear less visible. Experienced teachers often internalise their planning processes, making their lesson notes appear deceptively brief compared to what beginning teachers require. Unit 2.2 explains the planning processes across three key levels: schemes of learning, sequences of learning and individual lesson plans.

Taking responsibility for whole classes represents a significant step in your journey as a beginning teacher. Unit 2.3 addresses this transition by introducing the PROPS framework (Personal attributes, Routines, Outcomes, Planning and Subject knowledge), which provides practical guidance for developing your teaching persona, establishing effective classroom routines and building confidence in lesson delivery. This unit recognises that developing these skills requires careful preparation and reflection.

The quality of planning remains crucial to your success in enabling pupils to learn effectively. Defining clear and specific learning objectives and intended learning outcomes for pupils' learning is an area that many beginning teachers initially find challenging. Without clear direction and purpose, lessons can become a series of activities that keep pupils busy but don't necessarily advance

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their learning. Effective planning ensures that every element of your lesson from the opening activity to the final plenary contributes meaningfully to pupils' progress towards specific learning goals.

Unit 2.4 focuses on developing working relationships with the professionals whose responsibility it is to mentor your development. The importance of mentoring in teacher preparation has been increasingly recognised, with effective mentoring now established as a formal entitlement in initial teacher education. This unit explores not only what you can expect from your mentor, but crucially, what it means to be 'mentorable' and how you can actively contribute to building a co-operative and productive working relationship that supports your growth as a beginning teacher.

By the end of this chapter, you should be able to plan lessons in which both you and your pupils know exactly what they are meant to be learning, understand how to work effectively with other adults in schools, develop your teaching persona and classroom management skills and engage productively in the mentoring relationship. Explicitly sharing your learning objectives with pupils provides them with clear goals and, potentially, a sense of satisfaction as they achieve the outcomes you have set for their learning. You will be able to unpick your own progress by engaging in professional dialogue reflecting on how well your teaching leads to pupils making progress in your lessons.



ONLINE RESOURCES

Watch the explainer video on observing, planning, routines and working with your mentor by scanning the QR code or visit routledgelearning.com/learningtoteach.



2.1 Reading classrooms

How to maximise learning from classroom observation

Ana Redondo

Introduction

How do you actively *read* a classroom rather than simply *watch* a teacher at work?

During your school experiences you will observe experienced teachers and fellow beginning teachers, both within and outside of your subject specialism.

When observing a lesson, you may be one of several adults in a lesson, each with a different role, e.g. Learning Support Assistants (LSAs). Pupils are aware of everyone's presence and can expect support from them all. You will gain additional learning from these situations.

The more observations you complete, the more comparative information you can gather between pupils, classes, teachers and pedagogical approaches. Asking questions of the local context and how that may impact on learning is an important part of the exercise. Avoid being judgmental about what you observe whilst remaining a critical observer.

Identifying key points in the lesson where learning happens will help you get the most out of the experience of lesson observation. Identifying what learning takes place, when and how, by individuals and by the class as a whole is crucial to your development as a teacher.

Teachers and pupils have a working relationship in which both parties work within codes of behaviour and boundaries. Classroom ethos and good orderly conduct are very much due to the lead teacher who ensures everyone adheres to the school's behaviour code and expectations. Beneath that order there is a history of carefully nurtured practice by the experienced teacher, establishing an appropriate atmosphere conducive to learning and supporting pupils.

Sometimes pupils challenge these boundaries for a whole host of good reasons, and you will see how teachers restore the working atmosphere. Each teacher has their own way of dealing with this challenge and this will widen your own repertoire of skills in managing learning and learners.

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OBJECTIVES

At the end of this unit you should be able to:

- define the focus of your observations to achieve specific learning goals;
- use different strategies for recording your observations in forms that lend themselves to subsequent analysis;
- use observation to analyse teaching strategies and pupil learning behaviours to enhance your ability to plan and teach your own lessons;
- begin to understand the teaching and learning process and gain insights into how you wish to teach.

Check which requirements of your initial teacher education (ITE) programme relate to this unit.

Preparing to observe: some general points

Qualified teachers expect to be observed in the context of continuing professional development (CPD), appraisal, inspections and preparation for inspections. Good practice includes peer observation between colleagues in order to support pupils more effectively. See Dudley (2016) on Lesson Study, which draws on cognitive theory referring to how people best learn by participating in social situations. You should look for opportunities to observe fellow beginning teachers and experienced teachers.

Teachers may be less used to being observed by people who are analysing what is happening in the lesson rather than measuring it against predetermined criteria such as teachers' standards. As an observer, it is advisable to engage in dialogue with the teachers you observe, both before and after the lesson. Before the lesson you could agree on areas for observation and what you plan to do with the data. After the lesson you may wish to clarify aspects you have observed, discussing the rationale for actions taken to develop a deeper understanding of learning. Equally, ascertaining their perception of the lesson enables you to contextualise events and actions you observed, which may not always be obvious.

Remember that observation is fundamentally about enquiry: seeking to know what is happening, why it is happening and what its impact is likely to be. It is not passive. It should focus on perception—making sense of what you see—rather than judgement. Take an open, positive approach to observation but also be well prepared by reading and discussing key elements of lessons with other beginning teachers and your mentor. Developing your professional 'noticing' skills creates a kind of literacy for 'reading' and recognising what takes place in a lesson. This skill develops gradually, becoming more perceptive over time.

The form in which you record your observations varies according to your selected focus. Many schools use their own observation proformas or you may be given one by your tutor. You should adapt these to your own needs or create a new one, particularly for collecting specific data that you might otherwise forget during the lesson. Lesson observation in pairs has great advantages as sharing an observation activity with another beginning teacher allows you to compare notes and discuss the significance of what was observed. Having a specific structure to collect information

enables more focused dialogue and exchange, as well as suggestions to feed into your own teaching. Unions such as the National Education Union (NEU) and NASUWT provide guidance on lesson observation.

Who should you be observing and why?

Initially, you are naturally concerned with familiarising yourself with class dynamics: how pupils interact with each other and the teacher, and how they respond to learning. These observations are particularly valuable for preparing you to teach these specific classes and eventually others in different contexts. However, you should not observe with a view to imitating the teacher's style, but to analyse what they do effectively and decide what to adopt in your own practice (see Unit 2.3 on taking responsibility for whole lessons). Building rapport with pupils develops over time as you gain experience and spend more time with the same classes.

Through observation you will develop professional practices that feel comfortable and enable you to be effective in your teaching methods, approaches and strategies to positively impact pupils' experiences. It will also help you determine the best ways of fostering positive attitudes and creating learning environments conducive to academic progress (see Unit 4.2 on adolescent wellbeing and Unit 5.1 on how pupils learn).

Observing experienced teachers exposes you to a range of styles and strategies for teaching, learning and assessment from which you can develop your practice (see Unit 5.5 on personalising learning).

Many aspects of a lesson can form the focus for observation. Teachers modelling respectful attitudes towards pupils is an important part of professional practice observation, as this is a prerequisite for earning pupils' respect. The learning environment and ethos the teacher creates are central. Teachers establish rules and routines often reinforced by single gestures or signals—a raised eyebrow when a pupil speaks out of turn. As a beginning teacher, note and annotate these for your future practice.

When you start teaching, adhere to existing classroom routines as part of the overall school expectations, particularly initially, so pupils perceive you as a regular staff member. For specific routines you wish to introduce, consult with the regular teacher and/or mentor, then wait for an appropriate period before implementing your own approach.

You should also observe what and how pupils learn. Use guidance in Units 4.1 on adaptive teaching, 5.2 on active learning and 5.5 on personalising learning to construct observation schedules. Observe small groups working with specialist support teachers for pupils with learning difficulties, and LSAs supporting individual pupils in mainstream classrooms (see Unit 4.6 on creating an inclusive classroom and Blamires, 2014). This provides opportunities to understand individual pupils' learning needs in greater depth. Following a class or pupil across the curriculum to observe teachers in other subject disciplines provides insight into pupils' school day experience. All opportunities to observe teachers working with pupils offer learning experiences: you may note similar strategies across disciplines and teaching approaches. Teachers in assembly, on duty at break or working with pupils in extra-curricular activities are professionally engaged, and their work in these contexts directly relates to classroom practice. Be prepared to make notes about your observations and questions for later follow-up with the teacher, mentor or tutor. Effective observation is an intellectual exercise requiring critical engagement, relevant questioning and support from your mentor or tutor through appropriate reading and discussion. Table 2.1.1 gives examples of questions to address through observation, perhaps prior to teaching a class. Add your own questions.

Table 2.1.1 Examples of questions that can assist you in focusing your lesson observations (a printable version of this table is available on the book's companion website)

Briefing and preparing for observation

- Note the date, time and place of the lesson.
- Are you briefed on the topic being taught in the lesson and the composition of the class?
- With respect to professional ethics, have you agreed upon your role: a participant or non-participant?
- Have you agreed upon how and what you will observe?
- Have you agreed upon how you are going to provide feedback to the teacher as well as any future use you may make of your notes?

Teaching and learning questions

- What was the plan/structure/shape of the lesson?
- How was the lesson introduced? How did the pupils know the intended learning outcomes planned for the lesson?
- What were the different learning activities that the pupils undertook?
- Was there group or pair work?
- Did pupils receive a range of tasks, degrees of help and variety of resources?
- What were the different ways that pupils recorded or presented their learning?
- How did the teacher direct the pace of the lesson? 'Pace' refers to the appropriate allocation of time to complete tasks set.
- What form of question and answer episodes did the teacher initiate?
- How were pupils encouraged to ask and answer questions?
- What resources were used to assist in learning and how were they used?
- How did the teacher provide a range of teaching strategies to ensure all pupils can access what has been taught?
- Were digital technologies used during the lesson? How?
- Was the teacher handling the technology used?
- Did the pupils have a hands-on experience to complete tasks?
- What learning was supported through the use of technologies (see Unit 5.3 on teaching strategies)

Management of learning and pupils' dimensions

- What were the teacher's expectations about pupils' responses to what has been taught?
- How did the teacher promote behaviours for learning?
- Were there established routines and codes of conduct? What were they?
- Were issues of health and safety considered during the lesson? What were they? How? When?
- Did the teacher use strategies that promote behaviours conducive to learning? What form did that take and what were the outcomes?
- Is there a rationale for the teacher to use seating plans? Why? Do these favour particular pupils?
- How did the teacher use voice and gesture in the lesson to encourage pupil participation?
- How did the teacher make use of assessment during the lesson and how was this fed back to the pupils?
- Were pupils involved in the assessment process? How? What were the outcomes?

Other professional issues

- Was a teaching assistant or special educational needs teacher also in the room? What was their role and how did they work with the pupils? How did they work with the teacher before and after the lesson?
- Have you identified any gaps in your subject knowledge through observing this lesson? How are you going to address them?
- How has this observation made you consider your future professional practice as a teacher? What professional issues has it raised for you? Discuss these with other beginning teachers and your mentor or tutor in relation to the importance of becoming a reflective teacher.

If you are undertaking a piece of classroom research for your ITE programme, data from classroom observations will be useful. Unit 5.4 provides an introduction to practitioner research. You will also find details about observation as a data collection method in any research methods textbook. Many of the questions and prompts in this unit can be tailored towards gathering data for your research.

What do classrooms look like?

As part of lesson observation, it is useful to 'read' the impact of the room's appearance and layout on pupil and teacher learning. The classroom is more than a room with chairs, desks, boards and screens—it is a learning environment that should promote and support effective teaching and learning (see Unit 5.7 on critical thinking and 5.8 on the language-rich classroom).

Classrooms often express the school's values and ethos and can reflect and promote specific subjects. Task 2.1.1 provides a planned approach to analysing learning environments.



Task 2.1.1 The learning environment

Consider a classroom or teaching space you are going to be teaching in at your placement school and how you can manage this effectively. Is this a specialist room or a general classroom? Is the space used mainly by one teacher and discipline?

- Sketch the layout and seating arrangements. Identify the light source and how this affects visibility of the board, screen and display boards you will use to support your teaching. Note other features, resources and the teacher's desk. Check how much space there is for you to move amongst pupils to evaluate their work and support them. Consider space required for pupil movement and potential health and safety obstacles (e.g. tripping over chairs, bags). Consider the acoustics (see Underwood et al., 2016).
- Note your perceptions of the room layout's advantages and limitations for pupil and teacher learning. Does the layout support collaborative group working?
- Describe displays, noting proportions of pupil work versus teacher/published material. Are displays colourful and well maintained? Do they encourage pupils to value their own and others' work? What values do displays portray? Are they interactive, up to date and relevant to pupils' learning experiences? How might displays prompt pupils to think?
- What technological tools are available and what is their intended use?
- Comment on whether you would like to be taught in this room and whether the environment promotes the subject and pupil learning. Give reasons for your response.

Record your observations and responses.

How lessons begin and end

Lessons are structured into different parts after careful consideration of time allocation to maximise learning opportunities. How a lesson begins is significant to its success. It is usual to begin with a starter task to engage pupils from the moment they enter the classroom, progressing through

activities that end with a plenary to summarise key points and provide pupils with a chance to reflect on what and how they have learned (see Unit 2.2). This offers an opportunity for skilful questioning in a short timeframe—between five and ten minutes—highlighting key points and enabling pupils to understand links between prior and future learning. Questioning (see Unit 5.2) also provides teachers with useful information about pupils' understanding of tasks and whether they have made sense of their work. Effective questioning includes appropriate support for pupils who may struggle to articulate answers coherently.

Skilful teachers demonstrate good pupil management and learning skills, have strong subject knowledge and often make it appear effortless. When first observing lessons, focusing on each part is a useful approach to analyse how timings, teaching materials and learning organisation are managed. You can also focus on task types and how these encourage participation, higher-order thinking and acquisition of new concepts with clear understanding and application. You might be asked to contribute actively to one lesson episode as a way of beginning to teach your allocated group.

Routines

Routines are important for ensuring smooth transitions between lesson tasks and providing a backdrop of familiarity to pupils (see Unit 3.3). They are particularly important when taking on new classes. Observe closely how this is done. Whilst you should follow routines in the same way initially, you can later develop your own approaches to organising lessons, following school and department structures and policies whilst introducing your own ideas as you become more confident. Your mentor will encourage you to become more assertive and independent and take some risks, albeit agreed beforehand. Experienced teachers are very familiar with their pupils and vice versa, creating a specific dynamic you cannot immediately reproduce. Be observant, discuss and ascertain your mentor's views to develop confidence in adopting, adapting and aligning approaches with your own style after a period of practice.

At the end of lessons, after summing up (often called a plenary), allow enough time for pupils to be dismissed in an orderly and safe manner, ensuring a well-organised exit. Therefore, observe techniques for dismissing pupils.

Timing

When planning lessons, careful attention needs to be paid to how long individual episodes should last. When observing lessons, you are advised to pay attention to the amount of time allocated to individual activities (Task 2.1.2).

The structure of a lesson and transitions

Lesson structure is very important to its effectiveness (see Units 2.2 and 2.3). When experienced teachers teach, they work to a plan but deviate when new learning needs become apparent. Good teaching is flexible and responsive. Experienced teachers use transitions to summarise learning at key points and check for understanding before moving to the next activity.

Task 2.1.2 Analysing the beginning, middle and end of a lesson

A comprehensive observation framework for analysing the beginning, middle and end of lessons is available on the companion website, along with detailed recording sheets. A partial example is shown here:

Observer's name:						
Date and time:						
Class name and subject:						
Notes on class composition:						
Real time	*Location*	*Pupil actions*	*Teacher actions*	*Pupil talk*	*Teacher talk*	*Notes*
9:00	Corridor	Lining up outside	Greeting pupils at door	Quiet chatter	'Good morning, please line up quietly'	Orderly entry
9:02	Classroom	Taking seats	Taking register	Responding to names	Calling register	Assigned seating
9:05		Starting books	Introducing starter	Discussing task	Giving instructions	Quick mental math activity

Key parts are:

- learning objectives and intended learning outcomes shared with pupils, linking these to longer-term objectives;
- presentation of key concepts and tasks, explaining how to complete these and providing examples, adding visual support as necessary to ensure all pupils understand expectations;
- pupils complete tasks either individually or collaboratively;
- pupils can be invited to extend task outcomes if time and expectations have been considered in your original planning.

Teacher talk and oral feedback

It is useful to focus on teacher talk and feedback to pupils during a lesson to estimate its impact on learning and their responses (see Units 3.1 and 5.8; also Alexander, 2020; Education Endowment Foundation, 2015; University of Cambridge, 2015). There are various ways of doing this, including analysing audio or video recordings of a lesson. However, on most occasions, written notes are sufficient. Task 2.1.3 is designed to help you analyse the way teachers talk to pupils. Unit 5.8 provides detail on the use of language in classrooms.



Task 2.1.3 Analysis of teacher talk

Record examples of different types of teacher talk during a section of the lesson and note their impact on pupil learning. For each category below, record specific examples you observe and consider how pupils respond. A detailed recording table is available on the companion website.

- Giving information
- Correcting errors or misapprehensions
- Praising
- Questioning to check understanding
- Questioning to deepen understanding
- Asking pupils to focus on specific aspects
- Summarising learning
- Encouraging pupil reflection
- Coaching in skills
- Answering pupil questions
- Correcting inappropriate behaviour
- Guiding pupils back on task
- Outlining next learning tasks

A detailed recording table is available on the companion website.

Discuss your observations with the teacher after the lesson.

Questions are often classified as 'closed' and 'open'. Closed questions are about facts with a single correct answer. Open questions may not have a right answer or may have several ways of responding. They require pupils to speculate about events or anticipate new ideas or explanations (see also Unit 3.1).

What types of question do teachers ask? Are they simple questions with one-word answers or more complex ones involving explanation? Investigate the frequency of different question types. Task 2.1.4 asks you to focus specifically on questioning (see also Units 5.1, 5.2 and 5.8).



Task 2.1.4 Teachers' questions and the development of questioning skills

The following questions may help to focus your observation. Does the teacher:

- Ask mainly closed questions?
- Ask both open and closed questions according to purpose and circumstance?
- Accept only right answers or dismiss wrong answers?
- Give enough time for pupils to answer?
- Encourage pupils to frame replies?
- Provide scaffolding to encourage responses?

How does the teacher respond to right and wrong answers given by pupils?

Discuss your responses and your interpretation with your mentor.

Pupil talk and interaction

It is equally interesting and highly relevant to analyse pupil talk and interaction, which, if you follow a class or group to lessons in different subjects or with different teachers, can change in significant ways.

You could simply use a class list and place a red tick next to each pupil's name when they ask for information and a black tick when they answer a question or offer information to gauge pupil involvement.

More detailed observation with a small group, one pupil or the entire class can be undertaken using Task 2.1.5 over a 20-minute period, recording minute by minute. A detailed recording template with examples is available on the companion website.



Task 2.1.5 Analysis of pupil talk

Make notes in individual, group or whole-class pupil talk over a 20-minute period, recording pupil contributions and behaviour patterns.

Discuss your observations with the teacher after the lesson.

Observing management of pupils and encouraging learning behaviours that maximise learning

Before beginning lesson observations, familiarise yourself with the school's and department's policies about managing learning and pupils. These policies achieve a common approach to fostering appropriate learning behaviours. It is essential that you work within these policies (see also Unit 3.3). Make time to discuss your initial ideas about managing pupils drawn from your reading so that you can implement and develop your own ways of interacting with pupils and manage learning effectively.

Focus upon pupil learning

Beginning teachers often focus on the teacher teaching rather than how and what pupils are learning. This is not surprising. How individual pupils respond to the work set needs to be closely observed and then related to their different attainment level, interests and experiences in that subject. This information provides clues about their learning journey and how the teacher makes decisions about adapting teaching for the diverse needs of all pupils. Hence, you need to shift your focus and observe what and how pupils are learning. Task 2.1.6 provides some examples of how to do this.



Task 2.1.6 Analysis of pupil learning

Before you start, write down the intended learning outcomes for the lesson (see Unit 2.3) and identify the task/activities on which you are going to focus.

As a participant observer, you can take notes as the lesson unfolds, look over pupils' work and actively engage with coaching and guiding their learning activities. This enables you to appreciate what strategies and resources are working effectively and why.

As a non-participant observer, your task becomes subtler. If you can move around once pupils are involved in an activity, you can oversee their task completion. If it is appropriate, you can ask them brief questions, but you must not disengage them from their task. Do not be tempted to do the work for them! Always lead them to think through the task with your help.

In highly active lessons there may be considerable chatter and activity, but there is equally considerable learning taking place. Quiet lessons where pupils seem attentive are not necessarily lessons where most learning occurs. Different teachers have different expectations about pupil talk depending on the subject and specific tasks. Observe closely whether pupil talk relates to the tasks, and how the teacher responds.

The most flexible way to record pupil learning is to make bullet points as you notice things; a proforma may not be appropriate because pupil learning is complex. Once the lesson is over, ask the teacher if you may review the work the pupils have completed.

Discuss your observations with the teacher after the lesson and record both your analysis and discussion.

Teacher positioning is critical in supporting learning. The teacher's position influences interactions, so you should observe where teachers stand and how they move during lessons. Task 2.1.7 asks you to consider this aspect of classroom management.



Task 2.1.7 Teacher positioning and movement

Teacher movement reveals much about their relationship with pupils and classroom management. Draw a simple classroom map marking key features. Track teacher movement at regular intervals, noting time and lesson activity. Consider:

- Where is the teacher positioned most often and why?
- How does the teacher monitor all pupils and keep them on task?
- Were some pupils given more attention than others?
- How does movement relate to pupil behaviour and subject content?
- Does the teacher use furniture as barriers or move freely amongst pupils?

Summarise your findings and reflect on 'teacher territory' and 'pupil territory'. Share observations with other beginning teachers.

Observing assessment for learning

A common weakness identified in school inspections in England by Ofsted (Harford, 2018) is a lack of focus on assessment for learning (see also Unit 6.1). Ensure that you know your school's assessment policies in relation to teaching and learning as well as national assessments and those of the subject department. Task 2.1.8 asks you to focus on assessment for learning (see Wiliam, 2009 for an introduction).



Task 2.1.8 Assessment for learning

Undertake some research about Assessment for Learning choosing reputable sources. Observe a class and write detailed notes about the activities the teacher uses to assess pupils' understanding. Focus on:

1 How does the teacher find out whether pupils:
   (a) know what they are supposed to be doing?
   (b) understand what they are doing?
   (c) know the possible learning outcomes?
2 Does the teacher:
   (a) talk to individual pupils as they work?
   (b) talk to small groups?
   (c) intervene with questions or suggestions when progress is slow?
   (d) ask challenging questions to move learning on?
   (e) address the whole class when common problems emerge?
   (f) accept responses in various forms to suit all pupils' preferences?
3 How does the teacher survey the whole class when working with small groups or individuals? Common strategies include direct questions, discussion, pupils presenting work, reviewing work on devices or in books, role play, setting tasks to test understanding and posing problems to evaluate learning.

Also note down what the teacher does to correct misconceptions and misunderstandings or to deal with a lack of understanding and to advance learning. How does the teacher reassure and motivate? (See Units 3.2 and 6.1). How does the teacher consolidate learning?

Discuss your observations with the teacher or your mentor.

How does the teacher use learning resources and aids during the lesson?

Teachers use resources and aids to help pupils learn effectively by employing varied teaching strategies and differentiating for different attainment levels, interests and experiences. Pupils learn best when exposed to a variety of activities at different levels and through many formats and approaches, be they practical, open-ended tasks that require exploring possible solutions or accessing and using audio-visual tools (see Keuchel et al., 2015, on Visual Literacy). The important point here is that all pupils are learning and making measurable progress. Providing a range of tasks and formats allows the teacher to be creative and embed variety (see Units 4.1 and 5.5) in their approach to supporting pupils to meet the learning objectives.

Subject content-focused observation

All teachers during the early stages of their career have to work hard to fill gaps in subject content knowledge and, as schemes of work and the curriculum change, experienced teachers also need to learn new content. You may want to focus an observation on lesson content concerning a topic in which there is a gap in your own subject content knowledge (see also Unit 1.1).

Using video to support lesson observation

Lesson observation and the subsequent analysis is a challenging activity. In five minutes of a lesson so much can happen that is of significance and worthy of discussion. Recording part of a lesson for detailed analysis with the teacher is an ideal way of learning. By the same token, taking still images can be a very useful stimulus for post-lesson discussions about aspects of teaching and learning. However, this needs to be discussed and approved by the teacher you are observing, and ethical issues need to be addressed.

Video recording your own lessons enables you to analyse your teaching and to identify ways to improve your practice. You will notice aspects of the lesson that otherwise you may have missed. This process is an effective way to reflect on your practice and set goals for improvement. You can work with peers to inform the discussion about how to improve your professional practice and the experience of self-evaluation (Blomberg et al., 2011). Initial identification of what to look at and how to learn to 'notice' (Sherin and van Es, 2005) whilst 'reading' the learning space (Nind et al., 2015) are significant aspects of the process.

Established ethical procedures and guidelines about taking photographs or videos of children need to be adhered to: familiarise yourself with school policy and ensure you have formal consent from the school to video record. The use of photographs or video recordings of pupils normally requires parental consent. Videos of pupils at school need to be kept confidential and within school systems. Many schools use video recording of lessons as part of their programme of CPD for all teachers and, in these circumstances, pupils learn to ignore the cameras. Software can be used to annotate videos in real time so that key aspects of the lesson, e.g. questions asked or explanations given, can be easily grouped for playback and analysis.

SUMMARY AND KEY POINTS

- Lesson observation is something you can expect to take part in throughout your career. You may observe others in order to learn yourself, you may be observed as part of performance management in your school or the inspection process or when you are a supporting beginning, newly qualified and early career teachers.
- Observation is an important CPD activity for all teachers. It is also a key tool for reflective practice and practitioner research (see Unit 5.4).
- Your observation skills will develop with practice and focused effort using some of the methods suggested here over your whole career.
- The tasks in this unit were designed to help you develop your understanding and get the most out of observation; more importantly, they should enable you to enquire as you seek solutions to challenges arising in your own teaching.

Check which requirements for your ITE programme you have addressed through this unit.

FREE
INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.

Further resources

Dudley, P. (2016) *Lesson Study: Professional Learning for Our Time*, Abingdon: Routledge (see also http://lessonstudy.co.uk/) (Viewed 31 July 2025).

Lesson study is widely practiced in schools as an improvement strategy.

Ofsted (2024) *School inspection handbook*, Department for Education, effective from 16 September 2024. Available at: https://www.gov.uk/government/publications/school-inspection-handbook-eif/school-inspection-handbook-for-september-2023 (Viewed 31 July 2025).

This can provide you with information about what OFSTED is focusing on in relation to Assessment for Learning.

O'Leary, M. (2020) *Classroom Observation: A Guide to the Effective Observation of Teaching and Learning* (2nd edn), Abingdon: Routledge.

This book should help you to develop your ability to observe in a classroom.

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2.2 Schemes of learning, sequences of learning and lesson planning

Sophy Bassett-Dubsky, Mark Bowler and Bethan Coles

Introduction

> Planning is critical and underpins effective teaching, playing an important role in shaping pupils' understanding and progression. It is the area of work where teachers can bring their passion for a subject and their desire to make a difference together.
>
> (Department for Education (DfE), 2016, p. 5)

In order to plan effectively, teachers must consider a number of factors including pupils' prior learning, the ways that pupils learn (see Unit 5.1), the requirements of the curriculum (see units in Chapter 7) and appropriate methods of teaching to suit the needs of all pupils and resources available (see Units 4.1 and 5.9). In addition to these points, lesson evaluations inform the planning process (see Figure 2.2.1). Notwithstanding these factors, we are aware that planning is one of the main contributors to increased workload and stress for teachers (Churches and Fitzpatrick, 2023; DfE, 2024a). This unit will balance the critical role that planning takes, whilst supporting teachers to manage their time effectively. This may include collaborative planning with other teachers, and the

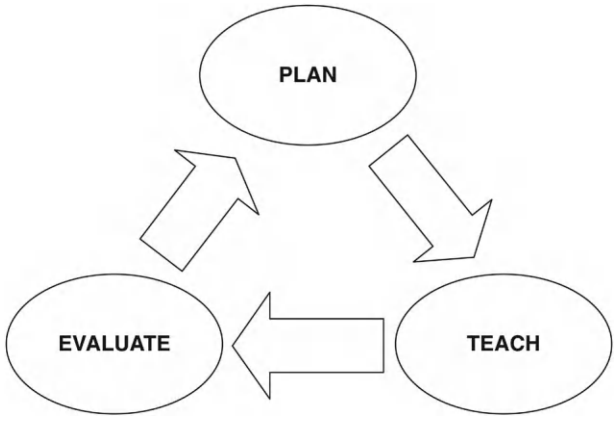


Figure 2.2.1 Plan–teach–evaluate cycle

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purchasing of pre-prepared materials and adapting existing plans (Churches and Fitzpatrick, 2023). These factors and approaches to planning will be considered in this unit.

Three levels of planning are particularly relevant to your work in the classroom – a *scheme of learning* which outlines lessons for a term or a year or so, a sequence *of learning*, which outlines a group of lessons around a particular topic and a *lesson plan* for individual lessons. You quickly gain experience in planning as you prepare lessons and sequences of learning at your placement schools. Evaluation is integrally linked to planning and teaching, and lesson evaluations form the basis of any changes or developments to your plans, both within and between lessons. For this reason, a flexible approach must be adopted.

OBJECTIVES

At the end of this unit you should be able to:

- understand what is meant by a scheme of learning, sequence of learning and lesson plan;
- explain what is meant by the terms: 'learning outcomes', 'learning intentions' and 'adaptive teaching';
- construct sequences of learning and lesson plans to progress learning over time.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

Planning what to teach and how to teach it

The factors influencing *what* should be taught (content of lessons) are discussed in Unit 1.1 and units in Chapter 7, but how much you teach in each lesson and *how* you teach it (see Units 5.3 and 5.9) are the teacher's own decisions based on an understanding of the pupils they are teaching. Throughout your teaching career it is extremely beneficial to your own practice and pupil learning to plan collaboratively with other teachers (Cordingley et al., 2015; Dudley, 2019; Churches and Fitzpatrick, 2023).

Content of lessons

The knowledge, skills and attitudes appropriate for a young person entering the world of work in the 21st century are vastly different to those that were considered appropriate even 15 years ago. Ideas about what teachers should teach change regularly and the curriculum is under constant scrutiny by those responsible for education (see, for example, the overview of educational research by The Office for Standards in Education, Children's Services and Skills (Ofsted, 2019).

Before you start planning, you will require an overall picture of what learning is planned for the pupils over a period of time. As a beginning teacher, you are usually given clear guidelines about what to teach and the goals for pupils' learning within your subject. These goals are often set out in nationally produced documents; for example, the English National Curriculum, the Northern Ireland Curriculum, the Scottish Curriculum for Excellence, the Welsh School Curriculum, school documents and specifications for accredited courses as issued by examination boards. See Units 7.3 (England), 7.4 (Northern Ireland), 7.5 (Scotland) and 7.6 (Wales) for nation-specific policy and

curriculum overviews. You should become familiar with the curriculum requirements and the terminology relevant to your subject. Notwithstanding the statutory requirements for your subject, it is also prudent to design curricula which consider local needs and contexts (Ofsted, 2019), such as socio-economic factors, geographical location, school facilities and access to expertise.

Teaching strategies

To emphasise the earlier point, although the learning content may be prescribed, the decision about which teaching strategies to use is usually yours (see Unit 5.9). As you become more experienced as a teacher, you acquire your own personal teaching style. People learn in different ways and different teaching strategies are suitable for different learning intentions and different types of material. You should become familiar with a range of ways of structuring learning experiences in the classroom. For example, you might choose to use discussion, discovery learning or role play to achieve particular learning intentions. Units 5.3 and 5.9 give detailed advice on teaching styles and strategies that help you achieve different learning intentions. Task 2.2.1 asks you to reflect on your preferred approaches to learning.



Task 2.2.1 Reflecting on effective teaching and learning

Spend a few minutes identifying the teaching strategies and learning activities that you have found most effective in your own learning experiences (see Unit 5.1). Consider approaches such as active learning, effective feedback, scaffolding techniques and retrieval practice. Add notes identifying situations where learning was less effective. Compare these notes with those of other beginning teachers. Consider how different learning intentions require different teaching strategies rather than different approaches for different learners. Plan to use a diverse range of evidence-based teaching strategies to support all pupils in achieving learning intentions and to match the demands of different learning tasks. This helps you adapt your teaching methods to the learning context (see Units 2.1 and 4.1).

Schemes of learning, sequences of learning and lesson plans

There are three main stages to planning for pupil learning:

1. *the scheme of learning;*
2. *the sequences of learning;*
3. *the lesson plan*.

A number of formats for schemes of learning, sequences of learning and lesson plans are used. We suggest that you read the advice given for the teaching of your subject in the subject-specific texts in this *Learning to Teach* series and gather examples from the teachers and beginning teachers you are in contact with. However, whilst the level of detail may vary between different approaches, the purpose is the same – to identify learning intentions in the long term (scheme of learning), in the medium term (sequence of learning) or in the short term (lesson plan) and the learning content and

strategies to achieve these intentions. Your initial teacher education (ITE) provider may require you to use a specific proforma. However, it would be beneficial to try different approaches to planning in order to find those most appropriate to your situation. The best plans are ones that support you in your teaching so that your pupils learn what you intend them to learn over a period of time, rather than simply completing a series of tasks with no clear progression. The examples in this unit can be used for you to trial and modify.

Scheme of learning

This might also be called the 'programme of study' or 'scheme of work' and is usually designed at the department level. Different terms may be used in your school or in your subject, but the purpose is the same – to devise a long-term plan for the pupils' learning, usually across a term, a year or period of years. The purpose of a scheme of learning is to ensure the continuity of pupil learning and build on the learning that has gone before. In some departments, schemes of learning are very detailed and include a framework for teaching, learning and assessment as well as safety issues, rather than just the topic to be taught. Therefore, a beginning teacher should use the scheme of learning as a starting point for their planning.

Sequence of learning

As with the scheme of learning, different terms may be used for medium-term planning. A sequence will normally provide a plan for pupils' learning from a few weeks to an entire term. Sequences of learning are often informed by a department's scheme but will generally be written by an individual teacher. Initially, beginning teachers should co-construct sequences of learning with expert others.

Both schemes and sequences of learning will plan for the progression of pupils' knowledge, skills and attitudes over time.

Using a scheme of learning to plan sequences of learning

Often at your placement school you are given the school's sequence of learning to inform your planning (see, for example, Table 2.2.1).

The sequence itself may be quite brief, but whatever the format, the following questions should be considered when designing sequences of learning:

1. What has been taught before?
2. What do you want the pupils to know, understand and be able to do?
3. How much time is available to do this work?
4. What resources are available?
5. How will you assess what learning has taken place?
6. How does this work align with the work pupils are doing in other subjects?
7. What is to be taught later?

Each question is now discussed in turn. As you will see from reviewing these questions, you should consider the relationship between the curriculum (what to teach), planning (what and how to teach) and the assessment (how to measure progress). Whilst this unit considers these elements

Table 2.2.1 Sequence of learning proforma

Sequence of learning		
Topic:		
Related prior learning:		
Class:	No in class: No of lessons:	Duration:
Learning Outcomes: (in relation to what pupils will know/understand/be able to do from the NC or accredited course specification) Common misconceptions/challenges when delivering this topic: Cross curricular links:		
Framework of lessons:		Ref. (NC/Spec)
Resources:		
Assessment strategies:		
Out of class learning: Wider reading and research to inform planning:		

separately, they are inextricably linked and your ability to expertly combine them in your practice will ensure the success of your pupil's learning.

1. *What has been taught before?* This information should be available from school documentation (for example, the schemes of learning) and from staff. In the case of pupils in their first year of secondary education, there is usually a member of staff responsible for liaising with feeder primary schools who may have this information. See Unit 3.4 on primary-secondary transitions.
2. *What do you want pupils to know, understand and be able to do?* A sequence of learning will include statements about what pupils should know, understand and be able to do over a period of time. These are normally referred to as either learning intentions or learning outcomes, although there are important differences between the two. *Learning* intentions are specific statements that set out what pupils are expected to learn across the lesson. Learning outcomes are assessable learning objectives which describe the action or behaviour of pupils that will provide evidence of their progress over a sequence of lessons. In devising each sequence of learning, a small aspect of the whole curriculum is selected and a route planned through it to provide the best opportunities for pupils to learn. Progression in pupil learning should be considered and built into sequences of learning.
3. *How much time is available to do this work?* The department or school in which you are working decides the number and length of lessons devoted to a topic. Don't forget that out of class learning has a valuable role to play in enhancing learning, and some time may be taken up by such things as assessments, revision, special events, bank holidays and school training days.
4. *What resources are available?* Investigate what pre-existing plans, digital technologies, teaching and human resources are available to both enhance your planning and teaching whilst helping to reduce your workload. Familiarise yourself with procedures for accessing these

resources and critically review their suitability for your intended classes, making adaptations where required. You may find there are resources outside the school to draw upon, e.g., parents, governors, charities and subject associations. Many organisations provide schools with speakers on current topics. There may be field study centres or sports facilities nearby. Before planning to use any resources, you should check whether there are any safety issues to consider. See Unit 8.3 on your Health and Safety responsibilities.

5. *How will you assess what learning has taken place?* Teaching, learning and assessment are interlinked. Most of the work you are doing with pupils is either teacher or pupil assessed; however, some is externally assessed. A good deal of assessment is formative, often referred to as Assessment *for* Learning – to check pupils' knowledge, skills and attitudes in relation to learning intentions and to guide the next steps of learning. This takes the form of teacher observation, teacher feedback, questioning or peer and self-assessment.

Assessment may also be summative – undertaken at the end of a piece of work to assess the progress achieved. You should keep good records of the pupils' progress (out of class learning, class work, assessment results) as well as providing these in the form required by the school or department. Unit 6.1 focuses on assessment issues.

Task 2.2.2. asks you to consider how you might record your assessment of pupils.



Task 2.2.2 Record keeping and assessment

Ask staff in your department how they expect pupil assessment records to be kept and what forms of assessment you should use for the work you are doing. Make notes and compare practice with other beginning teachers.

6. *How does this work align with the work pupils are doing in other subjects?* There are many areas of overlap where it is useful to discuss pupils' work with other departments. For instance, if pupils are having difficulty with measurement in technology, it is worth checking whether and when the mathematics department teaches these skills and how they teach them. Cross-curricular dimensions to the curriculum (see Unit 7.2) are usually considered by the school and responsibilities for different aspects shared among departments. Ask staff in your department what responsibilities the department has in this area.
7. *What is to be taught later?* Progression in pupil learning has to be planned and schemes and sequences of learning are drawn up for this purpose. From these documents, you know what learning is to come and the contribution that each lesson is to make.

Task 2.2.3 asks you to draw up a sequence of learning.



Task 2.2.3 Planning a sequence of learning

In collaboration with your mentor, review existing resources to co-create a sequence of learning to last about six to eight lessons. Focus on one particular class that you are teaching. Use the format provided by your ITE programme (or the one in Table 2.2.1). Ensure you consider questions one to seven above in designing the sequence of learning.

Lesson plan

The lesson plan provides an outline of one lesson within a sequence of learning which covers a number of lessons. In planning a lesson, you are working out the detail required to teach one aspect of the sequence of learning. Co-planning at this stage is fundamental in your development as a beginning teacher (DfE, 2024b). To plan the lesson, it is useful to use a model or framework, an example of which is provided in Table 2.2.2. It is important to note that a range of planning models are available and can be reviewed to find a structure that suits your needs (Capel et al., 2019).

The following questions should be considered when planning lessons:

1. What is the range of attainment of the pupils?
2. What do the pupils know about the topic?
3. What are the learning intentions for this lesson?
4. What time is available for the lesson?
5. What resources are available?
6. What approaches to classroom management should I use?
7. What teaching strategies should I adopt?
8. How do I assess pupil learning?
9. What risks are associated with the work?

Table 2.2.2 Planning a lesson: one possible approach

Date:	*Class:*		*Topic:*	
Lesson: (e.g. 2/8)	No. pupils:		Focus:	
Duration:	Resources:			
Learning intentions:				
Class context – Specific learning support requirements and strategies:				
Assessment *for* learning strategies:				
Time	*Teacher Activity/Strategy/Adaptation*		*Organisation/ Resources*	*Key Learning Points/Questions*
	Starter			
	Lesson episodes			
	Review/Preview			
Lesson evaluation:				
Were intentions achieved by pupils? What evidence do you have to answer this question with certainty? What went well? What didn't go well? What needs to be addressed next time? What are your priorities for pupil learning in the next lesson? What are your priorities for improving planning, organisation and transitions? What are your teacher performance targets?				

Each question is now discussed in turn.

1. *What is the range of attainment of the pupils?* As you develop as a teacher, you are expected to adapt to different levels of pupil attainment in your planning. Adaptive teaching, sometimes referred to as differentiation, is the need to consider pupils' attainment in your planning and teaching so that all pupils are challenged and extended by the work (Parsons et al., 2018; Ofsted, 2019). This can be achieved in different ways depending on the material to be taught. Adaptive teaching may, for example, be achieved by *outcome* (that is, different types or qualities of work may be produced) or by *task* (*where tasks are adapted for different pupils*) or by *teacher input* (that is, the level of support or teaching strategy applied) (Unit 4.1 provides further information). You provide continuity of learning for the pupils by taking account of, and building on, their existing knowledge, skills and attitudes.
2. *What do the pupils know about the topic?* As your experience of the curriculum and of pupils' learning develops, it becomes easier to answer this question. You should consider what has been taught before as well as the experience outside school that pupils might have had. It may be appropriate to do some form of testing or analysis of knowledge, skills and attitudes or to have a discussion with pupils to discover their prior experience regarding the work in question. As a beginning teacher, you should seek advice from the staff who normally teach your classes, as well as consulting national guidance materials.
3. *What are the learning intentions for this lesson?* Learning intentions are statements identifying what you hope pupils will know, understand and be able to do by the end of a lesson. Drawing up effective learning intentions requires considerable thought. When writing learning intentions, you should include a verb (what your pupils will do to evidence their learning e.g. demonstrate or explain), a context (the situation in which the learning will take place e.g. individually, in a group, written or verbally) and in some subjects a quality (the characteristics or features of the behaviour e.g. with accuracy, creativity or confidence) that ensures your intention is measurable. Verbs that help you be precise are those such as state, describe, identify, prioritise, analyse, explain, create and justify. These verbs force you to write intentions that can be observed or measured. If you think that your learning intentions are vague, ask yourself whether they make it clear what the pupils have to do to demonstrate their learning. Task 2.2.4 supports your writing of measurable intentions or outcomes.



Task 2.2.4 Writing measurable learning intentions or outcomes

Using Table 2.2.3, based on research into learning domains, collate a list of useful subject-specific verbs that will help to ensure your intentions/outcomes are measurable and progressive. Some generic verbs are provided to help you.

4. *What time is available for the lesson?* On the example lesson plan provided, a timeline is drawn down the left-hand side. You should plan for short learning episodes in order to maintain the engagement of the pupils. At the planning stage, think practically about how long it is likely to take to set up, complete and review each task. During the lesson, the timeline enables you to see easily if it is necessary to adapt the original plan to fit the time available.

Table 2.2.3 Selecting verbs based on learning domains to ensure measurable learning intentions or outcomes

Cognitive domain (based on Anderson and Krathwohl, 2001)		
Progressive levels	*Example verbs*	*Subject-specific verbs*
Creating	Design, plan, invent	
Evaluating	Judge, assess, identify strengths/ weaknesses	
Analysing	Analyse, compare, contrast	
Applying	Apply, use, select	
Understanding	Explain, discuss, identify	
Remembering	Define, list, label	
Affective domain (based on Krathwohl, Bloom and Masia, 1973)		
Progressive levels	*Example verbs*	*Subject-specific verbs*
Characterising by value	Influence, defend, support	
Organising	Modify, relate, integrate	
Valuing	Accept, initiate, endorse	
Responding	Agree to, reply, react	
Receiving	Listen, acknowledge, watch	
Psychomotor domain (based on Dave, 1975)		
Progressive levels	*Example verbs*	*Subject-specific verbs*
Naturalisation	Manage, create, design	
Articulation	Adapt, reorganise, combine	
Precision	Master, refine, control	
Manipulation	Complete, improve, make	
Imitation	Copy, repeat, follow	

5. *What resources are available for the lesson?* It is important to select and make available the most appropriate resources to achieve the learning intentions. Check how resources are reserved in your department and book them early because other staff may need them at the same time. You should also be prepared to adapt resources to suit different classes and to evaluate their suitability.
6. *What approaches to classroom management should I use?* You should plan for how you group pupils, integrate resources and manage transitions between activities and stages of the lesson (see Units 2.3 and 4.1).
7. *What teaching strategies should I adopt?* Teaching strategies should be selected as the best method to achieve your learning intentions (see Units 5.3 and 5.9). Where possible, a range of learning strategies should be planned to engage and support all pupils' learning (see Unit 5.2). Relevant questions should be planned for every lesson to assess pupils' knowledge and understanding during the lesson and to develop their higher order thinking skills. Phrasing appropriate questions is a key skill for a teacher (see Unit 3.1).

8. *How do I assess pupil learning?* The assessment methods selected should enable the learning intentions to be accurately assessed. It is therefore important to choose reliable assessment methods and you should seek advice from the teacher of the class. Ensure that you allocate sufficient time to carry out your chosen assessment methods effectively (see Unit 6.1).
9. *What risks are associated with the work?* Safety is an important issue in schools. In some subjects, the assessment of risk to the pupils and incorporation of strategies to minimise this risk are a necessary part of the teacher's planning. Departmental and national guidelines are provided and should be followed to ensure the safety of pupils. As a beginning teacher, you should consult your head of department, class teacher or mentor for guidance on safety issues. Never undertake an activity until risk assessment has been considered. For examples of risk assessments see your subject association guidance (e.g. The Association for Science Education; The Association for Physical Education; The Design and Technology Association). Subject associations are listed in Appendix 2.

Lessons have a structure and a rhythm to them. As you read this next section, think about the overall pattern to a lesson and the skills you use at each stage.

Planning parts of a lesson

Planning is an important part of the teaching process. Figure 2.2.1 illustrates the planning, teaching and evaluation cycle that should occur for every lesson that you deliver.

Lessons can be divided into a number of sections or episodes. Each of these must be planned carefully in order that your pupils can achieve the learning intentions. Most lessons will typically include a starter and review of learning in addition to the main lesson content.

Starter activity

In order to actively engage pupils from the outset of the lesson, an interesting and relevant starter activity must be planned. Think of this as the 'hook' to draw pupils into the lesson. This activity may draw upon and/or assess previous understanding (for instance, a retrieval type activity such as a low-stakes quiz) or might present a challenge to pupils. Current research suggests that the regular recalling of previous learning improves long-term memory, retention of information and a deeper understanding (Jones, 2019; Agarwal, Nunes and Blunt, 2021). Having such an activity displayed at the start of the lesson will give you time to organise your main activities as the class arrives.

Main episodes

Before you begin the main sections/episodes of the lesson, you need to clarify your learning intentions for the pupils. Sometimes you may elect to do this before the starter activity, whereas at other times you may complete the starter before explaining the learning intentions. It is considered good practice to display these visually to your pupils. Pupils must be aware of the purposes of the learning activities/episodes and how they will contribute to their learning journey within the lesson. Activities/episodes should be 'chunked' as you must bear in mind that pupils' concentration span is 10 to 15 minutes. You must also plan for the transitions between each activity/episode, and it may be necessary to conduct mini-plenaries/reviews at the end of each episode to consolidate learning to ensure a smooth and well-paced lesson. Each episode should build upon the previous one.

Review/Preview

This is a very important part of the lesson during which learning should be summarised. This involves referring back to the main learning intentions for the lesson. It is an important time for the teacher to check pupil learning and this requires skilful questioning (see Unit 3.1). At this point, it may be appropriate to give a brief preview of the next lesson. Enough time should be set aside to record out of class learning tasks and clear away any equipment.

General points to consider

Previous sections of this unit outline the key issues in effective planning. The following section highlights some general points that enhance the likelihood of overall success.

- *Resources*: Make sure your plan identifies the exact number of any resources that are required for the lesson. Ensure that you familiarise yourself with the equipment and know how to use it. If possible, set it up prior to the lesson.
- *Activities*: It is better to plan too many activities than too few. If one of your planned activities does not work or pupils complete the tasks more quickly than you anticipated, additional activities can be used.
- *Out of class learning and forward planning*: Ensure that you plan enough time to set out of class learning and/or remind pupils of any materials that they might require for the next lesson. This can then be recorded in their planner.
- *Group management*: Learn pupils' names as quickly as possible (Beech, 2018). Consider whether there is a need to design a seating or work space plan for your class. This can have both logistical and learning benefits.
- *Routines*: Classroom routines are important to ensure effective organisation of pupils and resources. These might include distribution and collection of resources; arrival and exit of pupils and movement around the classroom. Ensure that you are at your teaching area before the lesson to greet the pupils on arrival. At the end of the lesson, pupils' departure should be supervised to ensure that they leave your teaching area in a safe and orderly manner. In subjects where there is a level of risk (for example, science or physical education), planning a safe routine is of paramount importance. Where effective routines are already in place, adhere to these: pupils will expect you to use the routines that their teacher will have worked hard to establish. When required, you may need to implement new routines after consultation with your mentor.

Note: the lesson plan should not be regarded as a rigid blueprint for action.

Making adaptations during the lesson when necessary should be seen as a positive action rather than a failure (Capel et al., 2019).

Evaluation and planning future lessons

As soon as you can after the lesson, evaluate its success. In the early stages of your practical teaching, this could be in collaboration with your mentor. Were the learning intentions achieved? What went well? What didn't go well? What evidence do you have that allows you to answer these

questions with some degree of certainty? Sometimes it is useful to wait until you have reviewed their work before evaluating the lesson (see Unit 5.4). What are your priorities for pupil learning in the next lesson and how do they fit in with the sequence of learning? What are your priorities for improving planning for organisation and transitions? What are your priorities for your performance as a teacher? If you develop the practice of reflecting on your work as a matter of course, then modifying future practice on the basis of this reflection becomes second nature.

Task 2.2.5 challenges you to plan, teach and evaluate a series of lessons to support pupil progression.

Task 2.2.5 Planning for pupil progression

Plan a sequence of learning to last between six and eight lessons. Plan, teach and evaluate lesson one in conjunction with your mentor. Review the quality of your planning and teaching using Brookfield's (1995) four critical reflection lenses of self-review, pupil review, peer review and theoretical literature. Identify your areas of strength and development in relation to promoting pupil progression through effective planning. Identify the key points that should be considered when planning lesson two. Repeat this plan–teach–evaluate cycle (see Figure 2.2.1) for the sequence of learning. During the sequence, use the lenses approach to identify areas where pupils have made significant progress and where they still require support. Consider any adaptations you will need to make to your planned sequence and when planning further sequences of learning for the class.

Discuss this with your mentor and store in your portfolio for future teaching.

SUMMARY AND KEY POINTS

You should now be able to:

- understand what is meant by a scheme of learning, sequence of learning and lesson plan;
- explain what is meant by the terms: 'learning outcomes', 'learning intentions' and 'adaptive teaching';
- construct sequences of learning and lesson plans to progress learning over time.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

Further resources

DfE (Department for Education) (2024) Improve workload and wellbeing for school staff, London: DfE, viewed 10 September 2024, https://improve-workload-and-wellbeing-for-school-staff.education.gov.uk/

Jerome, L. and Bhargava, M. (2015) *Effective Medium-Term Planning for Teachers*, London: Sage.

This text provides advice and suggestions for effective medium-term planning. It includes examples of planning, justifications for the importance of medium term plans and explores links between pupil attainment and effective medium-term planning.

Kyriacou, C. (2014) *Essential Teaching Skills* (4th edn), Cheltenham: Nelson Thornes.

This provides an excellent and readable overview of the key skills that underpin effective teaching.

Lesson plans are widely available on the web but the quality varies. See examples on the Times Educational Supplement (www.tes.co.uk) or the *Guardian Teacher Network* (www.theguarduan.com/teacher-network). Always check the strength of evidence of any advice you find.

Mutton, T., Hagger, H. and Burn, K. (2011) 'Learning to plan, planning to learn: The developing expertise of beginning teachers', *Teachers and Teaching*, 17 (4), 339–416.

This article discusses planning at a variety of levels and recognises the importance of subject knowledge. It suggests that collaborative planning and teaching can allow beginning teachers to access the knowledge and understanding of more experienced professionals.

Thompson, C., Spenceley, L., Tinney, M., Battams, E. and Solomon, A. (2024) *The Ultimate Guide to Lesson Planning; Practical Planning for Everyday Teaching*, London: Routledge.

This book covers all aspects of planning such as aims, starting a lesson, resources and activities and assessment.

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DfE (Department for Education) (2024a) *Improve Workload and Wellbeing for School Staff*, London: DfE, viewed 31 July 2025, https://improve-workload-and-wellbeing-for-school-staff.education.gov.uk/

DfE (Department for Education) (2024b) *Initial Teacher Training and Early Career Framework*, London: DfE, viewed 31 July 2025, https://assets.publishing.service.gov.uk/media/661d24ac08c3be25cfbd3e61/Initial_Teacher_Training_and_Early_Career_Framework.pdf

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Jones, K. (2019) *Retrieval Practice: Research and Resources for Every Classroom*, Woodbridge: John Catt Educational Ltd.

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2.3 Taking responsibility for the whole class

Alexandra Brown and Kate Duffy

Introduction

This unit addresses possible issues that you may experience before you are asked to take on the responsibility of the whole class. Teachers are expected to lead by example, and this may feel overwhelming as you take sole responsibility for the first time. You will have observed many experienced teachers at the start of your training, and they may have made this look easy. However, they would all tell you that their confidence grew over time, and the trick to gaining that confidence was to try and do as much planning as possible prior to the lesson.

The process of teaching and learning is a complex one as you navigate the expectations of subject knowledge, professional knowledge and judgement, along with the practical elements of classroom routines and previous lesson evaluations. These expectations are more tangible than the 'hidden' aspects of a teacher's work as you may recall in the iceberg image from Unit 1.1. To get the planning right before you take on the whole class, sometimes it can be helpful to remember the key elements of complex ideas using an acronym: PROPS. It gives you some prompts to consider when you are taking charge of whole lessons by addressing the following:

P – Personal attributes – being the teacher
R – Routines in the classroom
O – Outcomes
P – Planning
S – Subject

We will address each element separately throughout the unit, although some may overlap. For example, the subject and the outcomes may influence the routines that are adopted and followed in the classroom. The unit will address areas that you can reflect upon and prepare for before entering the classroom. First, we discuss *personal attributes* and how you personify the role of 'the teacher' and how you can use verbal and non-verbal communication strategies to develop your confidence to deliver lessons. Then we address *routines* for pupils, other adults and for you as a teacher. A key element is to be clear about what you are trying to achieve in a lesson and the learning *outcomes* for pupils. We then explore effective *planning* to ensure that learning takes place and introduce you

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to the concept of the 'LOOP' for planning purposes. You also need to audit your *subject knowledge* – see the Learning to Teach textbook for your subject for this.

OBJECTIVES

- have a greater awareness of your own persona as 'a teacher' and understand the impact that this can have on others;
- have an understanding of the power of routines to support the delivery of effective lessons;
- recognise and apply the core principles of lesson planning that ensure pupils achieve the planned outcomes and make progress;
- begin to build your confidence to prepare for classroom activity.

Reflecting on your personal attributes

Many beginning teachers find it difficult initially to identify with the role of 'teacher'. McLerney, Korpershoek, Wang and Morin (2018) explain that teacher attributes are both professional and personal. Professional qualities such as organisational and interpersonal skills are expected requirements. However, personal qualities like enthusiasm and passion, while equally important for fulfilling your role, can be more challenging to demonstrate. In Stephen Spielberg's film Catch Me If You Can (2002), the character Frank Abagnale arrives at a new school wearing a smart blazer. When another student mistakes him for a substitute teacher, he decides to assume this role. Abagnale confidently positions himself at the front of the classroom, adopts an authoritative tone, writes his name on the board and instructs the pupils to sit down and open their books. While teaching involves far more than appearance and voice, you may still wonder: How should I dress? Will pupils listen to me? Where should I stand? How much should I talk? What do I do with my hands? These concerns are common among beginning teachers. Start by following your mentor's lead and checking school policies, particularly the dress code. The activities below help you explore your perceptions of being 'a teacher' and assess yourself against these expectations. You will identify your strengths and development areas while considering how to build positive relationships with pupils and colleagues.

Understanding your personal and professional attributes is essential for becoming a teacher. Your attributes and behaviours directly influence classroom dynamics. First, complete Task 2.3.1 to establish what you think makes a great teacher.



Task 2.3.1 What makes a great teacher?

Reflect on the teachers you have observed. What key attributes make them effective? Which qualities most positively impact pupils' learning?

Discuss these with others and record the outcomes.

When we talk about personal attributes, we mean the mannerisms, values and behaviours that all human beings have. Some people are generous, humorous, animated, smiley, kind, etc. In your description of the 'teacher', you probably had lots of positive words such as these. It may be that

some people who enter teaching have a set of personal attributes that pre-dispose them to behaving in certain ways which make for being a great teacher. In an online 'new teacher' event, Hywel Roberts asked 'What do you want it to be like to be taught by you?' Some of the comments were:

- *Fun, exciting, varied and engaging*
- *Inspiring a desire to explore and to ask questions*
- *Creative and fair*
- *I want to be approachable, engaging and know my subject inside out*
- *I want my teaching to be meaningful*
- *Challenging and enjoyable*

(Roberts, 2023, p. iv)

These new teachers recognised the positive dispositions and attributes needed for teaching. These, along with others such as the ability to give clear explanations and instructions, and remaining calm in difficult situations will likely need developing over time. Unit 3.1 provides detailed guidance. Don't expect to have everything perfected for your first lesson. You do, however, need a plan to enhance your existing personal attributes and to develop others. Begin with realistic self-assessment of your starting point. A tool used in management is the 360° self-evaluation. Complete Task 2.3.2 to apply this tool to your development.



Task 2.3.2 Personal strengths and areas for development: 360° self-evaluation

To complete your own self-evaluation, ask four trusted people to provide you with a list of your top five personal attributes as a teacher. Ask them to identify three areas for development. They might be, for example, colleagues in school, friends, your mentor or tutors. This gives you a balanced view of how others perceive you. Think back to the attributes of a great 'teacher' you developed for Task 2.3.1. Use the feedback from the 360° to assess how close you are to this. What do you need to do to close the gap and how will you do this?

Record this information and update it as you develop.

Confidence

Awareness and self-reflection are useful tools in bringing the attributes of a teacher to the forefront of your thinking and planning. When you take responsibility for a class for the first time, this can feel daunting, as all eyes and expectations are upon you. You are no longer assisting in the learning process but are now responsible for leading learning. To help you feel confident before you take sole charge of a class, you need to know:

- the routines and expectations of the classroom and the wider school – particularly the format and requirements of lessons that pupils are used to;
- about the pupils – ability levels and any additional needs they have;
- about previous lessons and the focus for future lessons;

- what subject content you will be teaching;
- how you and the pupils will know they have made progress;
- the resources needed and available;
- which other adults are available to support your lesson and how you will utilise them.

(See Forms of Professional Knowledge for Teaching in Table 1.1.2 for the knowledge needed for teaching).

Planning your first solo lessons takes time and you should seek advice and guidance from your mentor, who will have a lot of experience from which you can learn. Allow plenty of quality time for discussion and modifications. Being able to take professional advice means that you are developing personal attributes valuable for a teacher such as patience, perseverance, resilience, ability to listen and take advice and understanding the value of reciprocity in professional life. A thorough lesson plan is a key component to confident delivery (see Unit 2.2 on lesson planning) as effective planning and evaluation of lessons makes you think through and visualise in advance, what will happen during each part of the lesson and why. Kidd and Czerniawski (2010, p. 175) explain that 'The best lessons are performances that are meticulously planned, scripted and resourced. They are enthusiastic and carried out with conviction. Underpinning all of these performances … is time spent planning and preparing'. Planning is both your guide and a resource which you can control.

For Frank Abagnale in the movie cited earlier, his confidence enabled him to take responsibility of the class. His non-verbal and verbal communication helped him grab the attention of the students as he entered the class. Below are some of the areas where he was able to do this.

How do I dress?

You will be expected as a professional to conform to the school's dress code so check this in advance of starting. Schools will have different expectations for their staff: some may ask for formal, office-style clothing, with more casual clothing only allowed on certain occasions. You will spend most of your lessons standing up, so clothing, footwear and jewellery should always be comfortable, practical and in line with school norms.

Where should I stand?

Your positioning in the classroom is crucial. Many beginning teachers position themselves almost exclusively at the front of the class. This is not helped in classrooms where technology is controlled from a central location, but you can control the computer from anywhere in the classroom using a remote pointer or wireless mouse. You may also be constrained by the layout of the classroom. Although it may not be easy to change immediately, reflect on how the classroom layout impacts learning.

The most important thing is to position yourself where you and your pupils can see each other. Consider whether you need to move pupils to facilitate this, e.g. bringing pupils closer to you at the start of a lesson and when requiring their full attention on you. Observe and talk to experienced teachers about where they stand and why. Notice how they move to scan pupil activity when they are on-task (see Unit 2.1 on observing classrooms). Also, discuss with them their rationale for where they seat their pupils. Research shows seating plans based on learning needs rather than friendships are most effective for progress (Haghighi and Jusan, 2012).

Consider your classroom as a stage where you can work anywhere, and pupils work where you direct them. Like a theatre director, control elements that enhance the learning experience. Ensure good acoustics so you and pupils can hear each other clearly (see advice in Underwood, Turner, Whyte and Rosenberg, 2015).

How loudly do I have to talk? Will the pupils listen to me? Will they understand my accent?

Verbal communication is vital for effective teaching. Your voice is a primary tool for pupil communication, so develop awareness of how pace, volume and tone impact learning. Teachers need to speak clearly and assertively, as Evans and Savage (2017, p. 7) note: 'being more skilled, enthusiastic and confident with your voice will allow you to develop your pedagogy in new and innovative ways'. Both what you say and how you say it matter.

Record yourself to identify how nerves affect your voice and notice any verbal habits like ending sentences with 'ok', 'erm' or 'umm'. While you need not change your natural speech patterns, 'talk done well can be inspiring and engaging' (Evans and Savage, 2017, p. 12). Consider voice coaching or drama workshops if offered, and protect your voice during cold and flu season.

How much should I talk?

The aim of every lesson is for pupils to make progress. You have to assess whether they have done this, and if you dominate the talk in the classroom, it can be difficult to work out what a single pupil or groups of pupils have learned. A good lesson has a balance of teacher and pupil talk and engaging activities that will keep pupils 'on task' (see Unit 5.2 on active learning and Unit 5.9 on pedagogy). Teacher and pupil communication plays a significant part in hindering or encouraging pupil progress.

What do I do with my hands?

Gesture is a powerful form of communication. Think about observing a play. Actions are as powerful as words. For example, shielding your torso by crossing your hands or arms in front of your body can be perceived as a sign of stress, defensiveness or vulnerability (Navarro and Karlins, 2008). Similarly, pointing gestures can be perceived as aggressive and confrontational. Think about the eye contact you make with pupils. When you are nervous you might not even look at the pupils, which may be perceived as you not being interested in them. Maintain eye contact with pupils, as looking away when nervous can suggest disinterest. Combine eye contact with appropriate facial expressions – simple changes in mouth or eyebrow position convey pleasure, disappointment, surprise or concern. A nod and smile are effective classroom management tools that reinforce your teacher persona. A seminal text piece of research undertaken in the late 1950s by Erving Goffman (2022) showed that how we *intend* our persona to be received may not always be perceived by others. It's important to become aware of how your pupils may perceive your gestures and actions and adjust accordingly.

Building relationships

Beginning teachers often focus on developing their relationships with pupils and this is very important to enable you to plan and deliver an effective lesson. You need to know their abilities, personalities and additional needs. You will be developing your communication strategies with

pupils, and other adults who may be in your lessons such as teaching assistants (TAs). As a teacher you are the leader of learning, which means you need to lead everyone. Now complete Task 2.3.3, which asks you to consider the role of the TA.



Task 2.3.3 The role of the teaching assistant (TA)

Observe an experienced TA working with a teacher. Consider:

- What do they do?
- How is their work effective in supporting learning?
- How does the teacher include them in the lesson?

Record your observations.

First, establish which adults are available to help you and what their role is. You may have a general TA or a TA to support pupils with additional needs. You may have a technician if you teach a science or technology-related subject. These skilled professionals will help you plan and deliver lessons, but you must specify their tasks before, during and after each lesson. Support staff follow the same school routines and policies, so they will reinforce your behaviour management and learning decisions. In early lessons, ask your mentor to act as a TA to provide feedback on your collaboration skills and offer discreet interventions when needed. For example, they might spot the pupil who finishes the last page in their book and can intervene – you might not even know yet where to get new books! This strategy can also give your mentor a valuable and defined role in what is *their class*.

In this section, you have reflected upon your personal attributes and ways in which these can be enhanced. You have also begun to think about how to develop your confidence to deliver lessons as the lead teacher. From this you should have a greater awareness of your own persona as 'a teacher' and the impact that this can have on others.

Routines

In this next section, we will explore the power of routines to support the delivery of effective lessons. Your job is to ensure that other adults are well-utilised in supporting pupil learning. You need a routine for communicating the lesson plan in a timely and appropriate fashion and for gaining feedback from them. Do not expect a TA to be able to do their own preparation if you only give them the lesson plan as the lesson starts! Adopt a *firm*, *fair*, *friendly* approach to build positive relationships while maintaining clear expectations. Ensure your routines align with school-wide practices so pupils experience consistency across subjects and teachers.

Effective classroom management

Effective classroom management is essential for quality teaching and learning. While beginning teachers often focus on reward systems and praise, pupils behave well when they find learning relevant and engaging.

Classroom routines set the behavioural tone by providing clear structure where rules are understood by all. These routines become instinctive with practice. Pupils expect teachers to establish norms for classroom work, talk and movement. Since you're joining an existing department and school culture, adapt to established practices rather than introducing completely new approaches.

Key routine areas include:

- Classroom management: operational management of physical space;
- Classroom relationships: how people interact and relate;
- Learning expectations: standards for learning behaviour.

Consider how classrooms operate. Completing Task 2.3.4 will help you to identify current practices in your school, and to create a reference checklist for leading lessons.



Task 2.3.4 Classroom routines

Observe and document the following routines:

- entering and leaving the room;
- gaining attention (pupils to teacher and teacher to pupils);
- pupil movement around the classroom;
- equipment distribution and tidying;
- leaving the classroom (toilet/illness);
- routine administration – taking registers/keeping records;
- lesson time management.

Keep this list as a reference throughout your placement.

For your early lessons, one of your main goals is to get the pupils down to work promptly. To do this, you need to be confident in the operational routines of the classroom so it's important you work with your mentor to understand these as your early lessons will go more smoothly if you fit in with established routines.

Routines for behaviour management

In terms of behaviour, pupils expect a teacher to be fair and consistent in applying the school rules. They expect those who comply to be acknowledged (perhaps rewarded) and those who do not to be reprimanded (or sanctioned). How will you do this? Schools already have established routines to reward and sanction pupils. You should use these and ensure that you match the reward or sanction to the pupil's action. One of your routines should be to reward rather than sanction to try to create a positive environment for learning. Often, proximal praise highlighting desired behaviour can work more effectively than sanctions. Ask your mentor to observe your ratio of rewards to sanctions in your lessons. Behaviour management is cited as an area of concern by beginning teachers more often than any other area and Units 3.2 and 3.3 provide more detailed guidance.

One of the more challenging areas is how to manage low-level disruptive behaviour such as calling out. This is where a thorough understanding of the school policy on behaviour is vital. Prepare a note of how you will respond before the lesson – do not leap straight into giving detentions and avoid confrontations in front of the whole class. You could use the following ladder of interventions:

- first, try to use non-verbal gestures to remind a pupil to keep working. Use your voice sparingly;
- next, a quiet, individual conversation with the pupil;
- try proximal praise where you reward a pupil nearby who is on-task;
- try positioning yourself or another adult near the pupil who needs more encouragement to stay on-task;
- try a more detailed conversation with a pupil after the lesson where you highlight the issue and give them clear options about how to correct the behaviour and what the consequences of continuing their behaviour will be.

Now complete Task 2.3.5, which asks you to consider what you would do in this scenario.



Task 2.3.5 Dealing with swearing

Discuss the following two scenarios with your mentor or tutor and identify an appropriate response:

- You overhear a pupil use swear words in conversation with another pupil.
- A pupil swears at you.

Record the outcomes of your discussions.

You should always try to remain positive and calm; try to hide any frustrations or anger that you feel in the situation. This can be difficult but remember mentors understand that off-task behaviour occurs and they want to see that you have a routine for managing behaviour and that you are consistent. Mentors will have experienced off-task behaviour from students in their career. Prior to beginning your teaching ensure you are confident with your school's behaviour policy including rewards and sanctions – completing Task 2.3.6. will help with this.



Task 2.3.6 School policies: rewards and sanctions

Find out about the policies on rewards and sanctions at your placement school. Make notes of the key issues that affect your work. Check your understanding of the application of these policies with your mentor. Write down the approach you are going to take with respect to rewards and sanctions and discuss this with fellow beginning teachers and/or your mentor.

Routines for gaining attention

Gaining pupils' attention is a crucial skill that experienced teachers make look effortless. Decide on your approach and use it confidently:

- Verbal signals: clear phrases like 'Stop and look this way';
- Non-verbal signals, such as a raised hand or other established gesture;
- Auditory signals, such as a specific sound or chime.

Follow up with individual focus if needed ('That includes you, Robin') to reinforce expectations for all pupils.

Use attention signals strategically: when pupils encounter difficulties, for addressing misconceptions, or highlighting safety points. Avoid interrupting productive work.

Establishing silence is key. A fundamental classroom rule is that pupils don't speak when you're speaking. Wait for complete silence before continuing saves time and establishes clear expectations.

When observing experienced teachers, notice:

- Verbal cues – 'Eyes forward', 'Pens down' (avoid repeating 'quiet' or 'shush');
- Non-verbal cues – strategic use of eye contact, gestures or simply waiting.

Establish clear routines for how pupils gain your attention: this is typically hands up and no calling out. Learn your school's practices and apply them consistently.

Expectations of and for learning

A good teacher has high expectations of and for learning. They plan for learning to take place. You need to establish a routine in your classroom whereby pupils are aware of the intended learning objectives for the lesson (see Unit 2.2 and Tables 2.3.1 and 2.3.2), the activities that allow the pupils

Table 2.3.1 Planning lessons: general guidance

- Plan the lesson discuss your plan with the classroom teacher to understand pupils' next learning steps. Keep your plan accessible throughout the lesson.
- Plan your resources (and alternatives) – are you using worksheets, pictures, cue cards, video, textbooks, PowerPoint, the interactive whiteboard? Always have backup plans for equipment failures or unavailable specialists.
- Prepare extension work for early finishers (see Table 2.3.2).
- Link tasks to earlier work (this is known as sequencing) and set tasks that are relevant to the pupils' lives and locality.
- Make the content challenging. Ensure the challenge is realistic and pupils are productive; do not make the work so difficult that pupils give up.
- Know the class as much as possible. Observe beforehand, learn names quickly using registers or seating plans. Keep consistent seating until names are learned.
- Decide how you are going to monitor and give feedback on pupils' work during the lesson.
- Check which other adults will be in the room, and any materials are needed for particular pupils. Share your lesson plan so that they are aware of what you are trying to do.
- Keep track (in your post-lesson evaluation notes) of which activities 'worked' (or did not work) with the class and why.

Table 2.3.2 Backup plans in case pupils finish work quickly

- Let the pupils peer-assess each other's work and feedback against the learning objectives.
- Walk around and look at pupils' work, give constructive comments and share this with the class, if appropriate.
- Use your lesson objectives and intended learning outcomes to devise questions about the work or get pupils to devise them. For example, devise a quiz related to the lesson; develop a spider diagram for summarising the key points in a topic, produce a mnemonic to aid the recall of key issues.
- Homework (either past or just set) can be discussed in more detail. You may allow the pupils to discuss this together.
- Pupils' existing knowledge on the next topics could be discussed through question and answer. (Learning is more certain when you, as the teacher, build on pupils' existing knowledge and experience.)

to engage with and achieve the learning objectives and the method of assessment. Checking and marking pupils' work routinely during the lesson helps improve their work. It also helps you to set specific, well-matched and challenging targets for all.

Offering equitable attention to all pupils to ensure pupil progress

Pupil progress is the fundamental driver of all effective lessons and requires skilful planning and delivery. Pupil progress can be evidenced through the outcomes of a lesson. Put simply, outcomes are what pupils achieve in the lesson. Some outcomes are easily measured, for example, the amount of correctly completed mathematics problems. A variety of outcomes can be planned for in advance and then adapted during teaching (see Unit 4.1).

As a beginning teacher you must demonstrate how you have met the learning outcomes for your ITE programme and particular professional standards. Your outcomes are likely to be closely scrutinised to ensure that the best teachers enter the profession. Focusing on pupil outcomes and the extent to which pupils achieve the learning objectives set for them are the main focus as you develop into an effective teacher.

Planning

Planning to ensure learning takes place means understanding this LOOP acronym:

L – Learning starting point – what pupils already know
O – Objectives – the learning for the lesson
O – Outcomes – how you will know the learning has happened
P – Progress – how you will know what learning has taken place

Your lesson plan serves two purposes: enabling your mentor to review your intentions and helping you map lesson delivery for effective learning. The plan should reflect your research, thinking and evaluation of previous lessons (see also Unit 2.2 on lesson planning). If a lesson is to be effective, you need to be clear about what you are trying to do with your class and what you will be expecting of them. It is easy to fall into the trap of planning lessons that are relatively superficial where pupils look busy with activities that are not really advancing learning so you must be explicit about what pupil progress looks like and how you will know that it is happening. Use the checklist in Table 2.3.1 to support your planning.

Now complete Task 2.3.7 on comparing lesson plans.



Task 2.3.7 Comparing lesson plans

Exchange a lesson plan with a colleague. Discuss the questions below.

- Are the learning objectives and intended learning outcomes clearly specified?
- Are different learning needs considered?
- Are the pupils likely to be actively and appropriately engaged throughout?
- Are the tasks relatable to the pupil's own experiences?
- Have you planned appropriate time sequences into your lesson?
- Is there scope for pupils to demonstrate to you what they have learned this lesson?

Record the outcomes of your reflections and discussions.

Never start your planning just from an activity – start with objectives and outcomes and plan activities which achieve these. You should always ask yourself if the activity that the pupils are doing is directly related to the learning objectives and intended learning outcomes. You will spend a good deal of time thinking about the learning you want to take place and carefully crafting your learning objectives and intended learning outcomes. Unavoidable incidents do occur to interrupt lessons. We discuss below some of the more common incidents and possible solutions.

Challenges to anticipate

Judging the timing during a lesson is difficult initially and following a timeline on your lesson plan can help you gauge how the lesson is progressing and adjust accordingly. Monitoring the timing is useful for you and the pupils: consider using a displayed timer for activities. It may be that you have too much or too little material. Pupils must reach their next lesson on time, so end punctually. Begin wrapping up five minutes early (more if equipment needs tidying). Summarise achievements against learning objectives, set homework expectations and preview the next lesson. See Table 2.3.2 for strategies when pupils finish early.

With experience, you acquire the skill of fitting work to the time available.

Managing behaviour generally has been covered earlier (see also Unit 3.3). Inevitably, some pupils try to deflect you from your lesson goals. This needs to be addressed, but ignore deliberately provocative remarks like 'This is boring' to avoid confrontation (Rogers, 2015). Motivate disengaged pupils by linking work to their interests and ensuring relevance – if you cannot explain a lesson's relevance to pupils' lives, maintaining motivation becomes difficult. Seek advice from experienced colleagues about persistently challenging pupils, as they likely affect other staff too.

Answering questions: You may be worried that you will be asked a question and you do not know the answer. This is bound to happen. You can admit you do not know – 'What an interesting point, I've not thought of it that way, Chris', or celebrate the moment: 'Yasmin, that is a really good question. Where might we find an answer?'. Whichever strategy you use, make arrangements for the answer to be found.

Lack of equipment: You may have pupils who are not able to engage with the lesson because they do not have the appropriate equipment. Get most of the class working so that you can then address individual needs. Many schools have systems in place for dealing with pupils' lack of kit and equipment. Find these out so that you follow the school system. You can loan items to maintain lesson flow, but retrieve them and emphasise that pupils must bring their own equipment.

Checking equipment: You must check equipment in your teaching room beforehand and have a backup planned if your lesson is dependent on specific equipment or software. The more sophisticated your use of technology, the more likely you will encounter technical difficulties. Ensure that backup plans still allow pupils to achieve the lesson's intended learning outcomes.

Schools have different approaches to dealing with loss and breakage of equipment by pupils. Managing your lesson so that there is sufficient time at the end to check that equipment and books are returned saves you time in the long run.

Subject knowledge and pedagogy

To use LOOP, you need to know the subject knowledge and specific pedagogic practices linked to your subject. You can audit your subject knowledge against the curriculum for your classes and then fill in any gaps. Unit 5.9 on pedagogy would be a good next step. We also suggest you join your subject association to access the latest knowledge and to network with colleagues (www.subjectassociations.org.uk).

SUMMARY AND KEY POINTS

In this unit, we wanted you to think about your teaching 'persona', the power of routines and the importance of good planning so that you could build your confidence as a beginning teacher. Here are our key points as a summary:

- Teaching success builds on your personal qualities and ability to form strong relationships with diverse pupils (remember PROPS).
- Lesson observation helps you identify strengths and areas for growth, and with experience comes greater confidence and self-belief.
- First impressions set a positive tone – create an organised environment through careful planning (remember LOOP) and purposefully develop your professional teacher persona.
- Most beginners successfully develop their confidence and classroom management skills over time.
- One challenging lesson is simply a learning opportunity that helps you analyse and improve your practice. Embrace a reflective, problem-solving approach to grow from each experience.
- The key transition is recognising yourself as the teacher you're becoming – pupils, parents and colleagues already see your potential and support your development in this important role.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Coe, R., Aloisi, C., Higgins, S. and Major, E.L. (2014) *What Makes Great Teaching? Review of the Underpinning Research*. Durham: The Centre for Monitoring and Evaluation, Durham University and the Sutton Trust, viewed 22 July 2025, https://www.suttontrust.com/wp-content/uploads/2014/10/What-Makes-Great-Teaching-REPORT.pdf

Reviews research to identify the greatest impact on pupil progress. Looks at feedback and managing behaviour as well as teacher knowledge.

DfE (Department for Education) (2024) *Further Guidance and Resources for Supporting Behaviour in Schools*, viewed 22 July 2025, https://www.gov.uk/government/publications/behaviour-in-schools--2/further-guidance-and-resources-for-supporting-behaviour-in-schools

Education Endowment Foundation (2025) *Deployment of Teaching Assistants*, viewed 22 July 2025, https://educationendowmentfoundation.org.uk/education-evidence/guidance-reports/teaching-assistants

This guidance is designed to help you make decisions about how to deploy your teaching assistants. Its five recommendations are drawn from the evidence base on teaching assistant (TA) deployment practices and how these can lead to improved outcomes.

Westwood, P. (2021) *Commonsense Methods for Children with Special Educational Needs*, 8th edn. Abingdon: Routledge. Many of the approaches used with pupils with SEN work well with all pupils. This text outlines approaches to effective teaching and provides research to support different strategies.

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2.4 Working effectively with your mentor

Victoria Crooks

Introduction

Beginning teachers embark on their teacher training with several common concerns, for example, you might be worried about having sufficient subject knowledge or being able to manage pupil behaviour. You might also be thinking about your school-practice placement and the future relationship you will form with your mentor. This unit is intended to enhance your knowledge and understanding of how to navigate this relationship to ensure you are well supported in your development as a teacher.

The importance of the mentor in the preparation of beginning teachers has been recognised since the 1980s (Little, 1990; McIntyre & Hagger, 1996). More recently, the mentoring role has become increasingly formalised in England, with the right to receive "clear, consistent and effective mentoring" from a trained mentor now established as an entitlement in the *Initial Teacher Training and Early Career Framework* (IT TECF) (DfE, 2024, p. 7). The mentoring relationship really does matter; mentors often have the greatest influence upon the development of beginning teacher identity (Izadinia, 2015; 2016). It is therefore understandable that beginning teachers seek to establish a positive relationship with their mentor.

As a beginning teacher, you may very well have an idea of the kind of mentor you would like. This may have been influenced by your own experiences of teachers who have inspired and supported you, or shaped by your experience of mentoring in an employment or extra-curricular context. Your conversations with serving teachers about their mentoring experiences may also have influenced your expectations for your initial teacher education (ITE) programme. The reality is likely to be somewhat different; it is important to remember that often what we think we want in a mentor is not actually what we need!

An effective mentor will provide their mentee with a steady introduction into the realities of teaching and working with pupils. They will show you how to put into practice the theory you have been reading and discussing on your ITE programme. Mentors will tailor their support to your specific needs, considering your prior knowledge and experience, helping you develop the skills you need for independent practice. Additionally, they will support you to grow as a critically reflective and adaptable practitioner, able to work successfully with and alongside colleagues.

However, what your mentor is like and what they do is only part of the equation. So much of your growth during your ITE will be down to your own engagement with the mentoring process. This unit

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will therefore also explore what it means to be *mentorable* and will help you to consider your own role in cultivating a co-operative working relationship with your mentor.

OBJECTIVES

At the end of this unit you should understand:

- the role of your mentor.
- what you can expect of your mentor, and what your mentor will expect of you.
- how you can contribute fully to the development of an effective mentor–mentee relationship.
- how to engage in effective professional dialogue with your mentor.
- that sometimes issues arise between a mentor and mentee, and how these can be addressed in a professional manner.

The role of your mentor

There are many different models of mentoring. Depending on the context of your ITE programme, your mentor's prior experience of mentoring, and maybe their own experience of being mentored, they may view their mentoring responsibilities to be primarily one, or a combination, of the roles in Table 2.4.1.

Mentoring can also take different forms, for example, peer mentoring (Cornu, 2005), group mentoring (Mitchell, 1999), instructional coaching (Knight & van Nieuwerburgh, 2012; Knight, 2023) or as one element of a tripartite relationship between mentee, mentor and a tutor (Ball-Smith, 2020). This helps to explain why mentors may take a different approach to their role and is especially important for helping you understand why your experience of mentoring can differ when moving between school placements.

For our purposes here, mentoring is defined as the one-to-one support provided to a beginning teacher by a more experienced school-based colleague (mentor). In this context, a mentor's support will "[bridge] both the disciplinary and the affective domains" of their mentee's development (Crooks, 2024, p. 26), being as concerned with the development of the mentee's professional dispositions and values, as with developing their subject knowledge for teaching (La Velle & Flores, 2018). Consequently, a mentor will usually assist you in successfully navigating your way into the profession (Pitton, 2006) by balancing the need to develop your subject knowledge and understanding of

Table 2.4.1 Mentoring roles and responsibilities

■ supervisor ■ assessor ■ instructor ■ teacher ■ trainer	■ guide ■ collaborator ■ colleague ■ counsellor ■ supporter	■ advocate ■ role model ■ colleague ■ friend

Source: Clutterbuck (1998), Cordingley et al. (2005), Brondyk and Searby (2013) and Ambrosetti, Knight and Dekkers (2014)

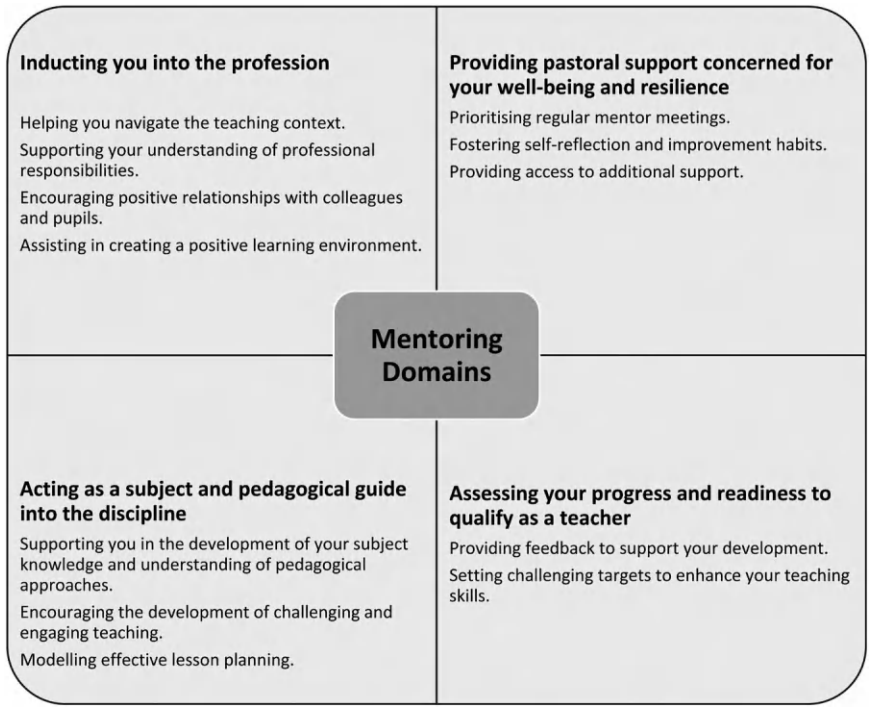


Figure 2.4.1 Mentoring domains

Source: Crooks (2024)

core classroom competencies with the need to develop your professionalism, confidence, independence and teacher identity.

Mentors are therefore likely to adopt some of the approaches outlined in Figure 2.4.1 to support you as a beginning teacher in the different domains of the mentoring role.

Task 2.4.1 will help you consider how your previous experience of mentoring has shaped the expectations you have of the mentoring you will receive as a beginning teacher.

Task 2.4.1 Considering your starting points and expectations of mentoring

Reflect on a time when you were mentored (for example, in another professional setting or sports team).

What was the purpose of the mentoring in this context? How was the purpose achieved and how did it shape your experience?

What did you find helpful about that experience? What did you find difficult about the mentoring relationship?

What expectations do you have of the mentoring you will receive as a beginning teacher? What three qualities would you value most in a mentor?

Discuss your reflections with other beginning or early career teachers.

What you can expect from your mentor

Mentors are usually experienced teachers who are interested in supporting the development of beginning teachers. Mentors may be just a few years further into their career than their mentee, or they could be a Head of Department or hold other additional responsibilities in the school. Some will have actively sought the role because they view having a long-lasting impact on another teacher's career, and being able to contribute to school community, as a professional privilege (Glover, Jones, Thomas & Worrall, 2024). Others see mentoring as offering opportunities for their own continuing professional development. Schools may also assign the mentoring role to a teacher at short notice.

Mentors are generally volunteers and nearly always busy people; mentoring duties are usually additional to their contracted role and without remuneration. It is important, therefore, that your expectations are realistic. As an emerging professional, you will increasingly need to take initiative and responsibility for your own learning and development as you progress through the course, demonstrating that you are ready to take on the full teaching role. While it is natural at times to want immediate support, your mentor may not always be able to respond instantly. It is essential to be patient, professional and proactive in addressing your concerns, seeking alternative solutions or returning to the issue later when your mentor has more capacity.

You can expect your mentor to engage with you, structure your teaching, observe and provide feedback on your teaching. Mentors are typically responsible for:

- allocating an appropriate teaching timetable, including teaching, observation, planning, preparation and assessment.
- conducting weekly meetings.
- providing regular informal and formal feedback on observed lessons.
- setting specific targets for improvement of professional practice.

What your mentor will expect of you

Navigating the complexities of a new school environment can be daunting, especially when transitioning between placements. Understanding the unwritten rules and meeting your mentor's expectations can feel particularly overwhelming.

Professionalism is central to the Teacher Standards (DfE, 2011). Part 1 emphasises teaching and learning, while Part 2 focuses on personal and professional conduct. Your mentor's first expectation will be that you quickly learn to adopt the professional behaviours required of a teacher in the classroom, the wider school and the community. Your mentor will be looking for you to demonstrate the following:

- Professionalism: Adhering to the school's dress code and mobile phone policies, demonstrating personal integrity and having due regard for the law.
- Reliability: Meeting deadlines, maintaining punctuality, good attendance and following absence reporting procedures.
- Interpersonal Skills: Developing positive relationships with pupils, colleagues and parents, and actively participating in school life.

However, these are just a few of the many professional qualities, dispositions and responsibilities expected of beginning teachers. Your mentor will likely also want to see you display:

- Commitment to teaching and professional growth – preparedness to enhance your specific subject knowledge and pedagogic skills (see Unit 5.9).
- An inclination to develop teacher dispositions such as humility, open-mindedness and self-awareness (Osguthorpe, 2013) – engaging in honest self-reflection and possessing a readiness to learn from the professional experience and expertise of your mentor.
- Resilience in the face of professional critique.
- Accountability for your work and interactions.
- Cultivation of a confident teacher identity and classroom persona – communicating effectively, taking initiative and being active in school life.

Task 2.4.2 asks you to consider the expectations mentors have of their mentees.



Task 2.4.2 Mentor expectations of their mentee

What do you understand to be the key expectations mentors have of their mentees?

Reflecting honestly on your character, dispositions and attributes:

- Where do your strengths lie in relation to these expectations?
- What aspects of these expectations do you think you may need particular support to meet or develop an understanding of in context?

Consider what actions you would take if the following situations occurred:

- You were running late or unable to attend school.
- You were unable to meet a deadline for lesson planning.
- You have an urgent question about your practice, but your mentor is unable to meet with you that day.
- You have a personal commitment on the same night as open evening.

What procedures or approaches do you need to clarify with your mentor that would help you negotiate these scenarios?

Discuss your ideas with other beginning or early career teachers.

Contributing to the development of an effective mentor–mentee relationship

Orland-Barak and Wang (2021) outlined four distinct approaches to teacher mentoring: personal growth, situated learning, core practice and critical transformation. These approaches influence the nature of the mentoring relationship and the learning experiences of mentees during school placements. Some mentors emphasise the affective and emotional aspects of mentoring, as exemplified by Turnbull's CARE model (2009). Others adopt a more coaching-oriented approach, focusing on target setting, skill rehearsal and observation of expert practice (Farndon, 2019).

The mentor–mentee relationship typically begins as more hierarchical in nature and, in the early stages, it resembles an apprenticeship model where the mentor assumes a more directive role, acting as a knowledge broker, shaping the mentee's learning experience and setting development objectives (Thompson, 2016). Gradually, as the mentee's knowledge and skills grow and their teacher identity develops, the relationship shifts towards a more collaborative partnership (Cordingley et al., 2005; Carnell, 2006). As you take increasing ownership of your own learning journey, it is possible for a more shared and reciprocal professional relationship to emerge in which both parties gain from their work with one another. This is another reason why your mentoring experiences can vary between placements.

As Adey (1997) suggests, establishing a trusting personal relationship in the early stages of the placement is essential for beginning teacher development. Mentees should strive to build positive relationships with their mentors (Marsh, 2002), while recognising the relationship is founded upon the professional expectations previously articulated in this unit.

Effective mentor–mentee relationships require open communication, mutual respect and a willingness to understand each other's limitations and find solutions for moving on when things don't go according to plan. This is not always easy for beginning teachers who occupy a liminal space (Cook-Sather, 2006; Wood, 2012) where they simultaneously feel like a student but are expected to act like a 'real' teacher (Pillen, Beijaard & den Brok, 2012). Regular and transparent communication is the key to beginning and maintaining a successful professional relationship with your mentor (Graves, 2010). As one mentee I worked with in my role as PGCE subject-lead commented: "Misunderstandings are usually what makes the relationship with your mentor challenging, whether they are about your mentor's expectations, instructions you've been given, or personal and professional circumstances". Fostering a respectful and understanding relationship is key to navigating times when one, or both, of you are experiencing additional pressure which makes misunderstandings more likely to occur.

Agreeing how and when you will communicate with your mentor is especially important. Your weekly meeting provides a dedicated space for you to review and reflect upon your progress and plan for the next stage of your development. Additionally, some mentors may opt for brief post-lesson discussions or provide brief written feedback notes for discussion in the main mentor meeting. Establishing clear expectations regarding availability for meetings and how you both prefer to communicate is essential. Both parties should consider their respective commitments and respect boundaries around contact in the evening and at weekends. It is also advisable to agree on a timeline for submitting lesson plans in advance to facilitate effective support. If this is your second placement, you may encounter differing expectations from your previous mentor. For example, while a previous mentor may have been flexible with lesson plan submission deadlines, your current mentor may need you to submit plans earlier in the week because they have a different work pattern. Schools also vary in their organisational cultures and communication norms. Some institutions prioritise face-to-face communication, while others rely heavily on email. It is essential to familiarise yourself with these unwritten rules and seek clarification from your mentor if you are unsure.

From the beginning of your placement you want to demonstrate that you are keen to develop a strong relationship with your mentor. Table 2.4.2 describes actions you can take to prepare for your first meetings with your mentor to lay the foundation for a positive relationship, while Task 2.4.3 gives you the opportunity to reflect on effective mentor and mentee relationships.

Table 2.4.2 Building an effective relationship with your mentor

Suggested preparation for building an effective relationship with your mentor:
■ Research the School: Familiarise yourself with the school's ethos, culture, recent inspection reports and local news. ■ Share Your Background: Discuss your previous experiences, strengths, weaknesses, goals and any relevant personal circumstances with your mentor. ■ Review School Policies: Carefully review school policies, especially those related to safeguarding – it is imperative you know how to share safeguarding concerns. You should also ensure you understand school procedures around, for example, behaviour management, homework, registration of pupils and pupil management system, use of the school network, how to access resources (including reprographics) and absence procedures. ■ Communicate with Your Mentor: Discuss key information with your mentor to ensure you understand expectations and can access support.



Task 2.4.3 Reflecting on effective mentor and mentee relationships

What are the key indicators of an effective relationship with your mentor?

Identify three things you can do to develop a good relationship with your mentor from your first meeting.

What information might you need to share with your mentor to help them better understand your context and starting points and to help you both to avoid misunderstandings?

To help you have a productive conversation with your mentor about your context and starting points, note down any relevant personal circumstances it would be helpful for them to know to avoid potential misunderstandings.

Engaging in effective professional dialogue

As a beginning teacher, you will participate in a wide range of professional dialogues concerning planning, teaching, assessment and pastoral issues. Nevertheless, conversations centred on lesson observations will constitute one of the most significant and transformative professional exchanges for your ongoing development (Lofthouse & Wright, 2021).

Your observations of teaching and learning

Observing lessons is a privilege and valuable experience for beginning teachers, providing a 'frame of reference' for their own practice (Hagger, Burn, Mutton & Brindley, 2008, p. 169) (see Unit 2.1). Observation provides the opportunity to evaluate different teacher personas and pedagogical approaches you may wish to adopt for yourself. Later on in your development it offers a valuable lens through which to reflect upon your own classroom practice. However, uncontextualised observation can feel like a distraction from the 'real business' of practising your own teaching (Crooks, 2022; London, 2024).

When meeting with your mentor, engage in professional conversations about the observations you have undertaken, prioritising inquiry into teacher decision-making – your role is not to make

judgements about the teaching of this more experienced colleague. Your mentor may ask you questions about your observations to spark professional dialogue and identify additional learning activities to help your development, for example, joint planning, team-teaching or conducting a filmed self-observation. Taking a proactive approach to observation also demonstrates that you are taking responsibility for your progress, further develops the dialogue between you and your mentor, and strengthens your relationship.

Observations of your teaching by your mentor

One of your mentor's key roles will be to observe your teaching practice and to engage with you in a post-lesson reflective conversation, focusing on what pupils have learnt and how this was achieved. They will also help you to set targets for your ongoing development.

The asymmetrical nature of the mentor–mentee relationship is never more exposed than during the observation process (Puttick, 2022), in part because it is also connected to the responsibility the mentor has as an assessor of competency against professional frameworks such as the Teachers' Standards (London, 2024). While it is an expected part of the process of learning to teach, the cycle of mentor observation, post-lesson discussion and target setting can cause concern and anxiety for beginning teachers (Burn, Hagger & Mutton, 2015). It can be challenging, but it is important to understand that your mentor's feedback is a constructive critique of your teaching practice, not a personal criticism. It is this process of professional dialogue that will support your reflective practice, helps you identify areas for development and challenges limiting beliefs (Korthagen, 2014).

You will get the most out of these conversations by being an active participant in them. Typically, your mentor will invite you to conduct a self-evaluation of the lesson prior to sharing their own observations. Before you meet with your mentor, it is beneficial to reflect on the lesson, identifying both strengths (so you can repeat them) and areas for development in your teaching practice. Task 2.4.4 is a good starting point for these reflections. Adopting a balanced perspective in this reflection process may be difficult, and employing techniques such as identifying equal strengths and weaknesses or annotating your plan with questions to foster critical reflection can help you achieve this. As you gain more experience, you will increasingly focus on pupil learning over your own teaching 'performance'. Reviewing pupil work helps to provide important insights into a lesson's impact on learning.



Task 2.4.4 Reflecting on a lesson you have been observed teaching

Review a lesson you have recently been observed teaching.

- What did you intend for the pupils to know or learn to do by the end of the lesson? Did they secure this knowledge and learning? How do you know?
- Identify three aspects of the lesson that supported pupil learning or demonstrated an improvement on previous areas for development.
- Identify three aspects of the lesson that needed further development to ensure the pupils could better access the lesson and make progress in their learning.
- How might you improve this aspect of your practice? What questions will you need to ask your mentor to help you identify your next steps?

Discuss your reflections with your mentor.

Learning to positively receive feedback during these professional dialogues is challenging, especially if you perceive it as criticism rather than constructive advice. You will invest significant effort in preparing for observed teaching episodes and may therefore also experience the disappointment of falling short of your own expectations for the lesson. It is natural that you may feel defensive about your work, however it is important to remember the purpose of the feedback is to improve your teaching. Being unreceptive to the guidance of your mentor may also damage your relationship with your mentor who will likely feel less able or inclined to provide the feedback necessary for your development. Consider whether your approach to these conversations demonstrates your openness to being mentored.

Drawing on Osguthorpe's (2013) concept of teacher dispositions, you'll need to cultivate professional humility, resilience, and being non-judgemental of self to engage in constructive, sometimes robust, professional dialogues. This will mean responding openly and thoughtfully to your mentor's questions, which are designed to encourage self-reflection rather than seeking the external evaluation of your mentor's assessment (Hobson & Malderez, 2013; Crooks & London, 2022). It will also involve actively listening to feedback and asking clarifying questions to deepen your understanding, rather than viewing it as an adversarial process where you need to defend your work. For instance, you might ask, 'What specific steps can I take to improve this aspect of my practice?' Receiving support from your mentor is not a weakness, in fact it is vital to your progress.

After the post-lesson conversation, you and your mentor will develop targets and identify specific actions you need to take to achieve these targets. Initially, your mentor will guide target setting, but as your placement progresses, this will become a collaborative process, with the aim that you will eventually be able to independently identify and plan for your own development. Your targets must be specific and measurable for effective improvement. The post-lesson conversation provides a great opportunity to clarify expectations and ensure mutual understanding of how the targets can be implemented. It is your responsibility to ask clarifying questions to develop specific targets and action steps; refer to Table 2.4.3 for an example of how to conduct this kind of professional dialogue. If you feel overwhelmed by your targets, discuss this with your mentor. You may need help to prioritise certain targets or temporarily shift your focus to achieve quick wins before returning to more challenging development areas. Following feedback, you should demonstrate your commitment to improvement by modifying your practice to meet your targets. It is also important to hold follow-up discussions with your mentor to review your progress, celebrate achievements and seek additional support as needed.

Table 2.4.3 Example of professional dialogue

Mentee: My target says to 'improve classroom management'. Could you give me an example of how I could do that please?
Mentor: Well I think I really mean you need to address low-level behaviour interruptions during the lesson, especially when you are speaking. You could do this by using countdown routines to get everyone's attention and by using the warnings and behaviour system to be clear about your expectations.
Mentee: Ok, so my target could be to address low level behaviour interruptions, by using routines and the behaviour system. Could I also add positive noticing to that? I think I could do that more.
Mentor: Great idea!
Mentee: Thank you, I understand what I need to do next lesson.

Concerns about progress

It can be difficult to discuss challenges, including personal issues affecting your professionalism, with your mentor, especially if you fear negative judgment or impact on your assessment. Your mentor may also find it challenging to address concerns due to a desire to avoid upsetting you or damaging your confidence. However, it's crucial to address these concerns with your mentor promptly and professionally to prevent minor issues from escalating. Before a meeting with your mentor, you may find it helpful to attempt Task 2.4.5 to help you focus on the key issues you wish to raise. This demonstrates your commitment to self-reflection, your resilience and determination to resolve the issue with their support. It is also common to receive differing advice from different teachers. If this leaves you feeling confused or unsure, remember to talk to your mentor for guidance on how to best focus your targets to ensure progress. Raising concerns openly and professionally to find mutually acceptable resolutions is essential for effective teamwork.



Task 2.4.5 A framework for raising issues with your mentor

Reflect on your concerns about your progress. Work through the questions below to help you gather your thoughts and prepare for a professional conversation.

- What aspect of my progress am I concerned about?
- Are there other factors, beyond school, that are making this situation more difficult for me to manage?
- How is this situation making me feel and what impact is that having on my progress and professional capacity?
- What solutions have already been suggested?
- What have I already done to carry out these suggestions?
- What have I not been able to achieve?
- What new solutions can I suggest?
- What support do I need to succeed?
- When can I review whether this issue has been resolved?

Discuss your responses with your mentor.

When issues occur between the mentor and mentee

While most mentor–mentee relationships are positive, issues can arise. A common concern for mentors is their mentee's professionalism and integrity. Taking responsibility for your actions and demonstrating commitment to your development is crucial to maintaining a professional relationship.

As discussed earlier in the unit, mentors operate in diverse contexts and hold varying philosophies but, even so, most demonstrate high levels of emotional intelligence and mentoring competence, despite their numerous professional commitments. Potential mentor–mentee conflicts often stem from differing perspectives and perceptions. Most situations can be resolved by assuming positive intent from the other party. For example, you might feel your mentor is nitpicking, but by assuming positive intent, you'll understand that they're likely trying to ensure you meet the necessary requirements for qualification. Table 2.4.4 presents scenarios that highlight potential

Table 2.4.4 Seeing potential conflict from both sides

Scenario 1: Aasma and Adam

Mentee perspective

Aasma is frustrated with Adam, her mentor. At the end of the day, when she wants to just have a quick word about the lessons she is teaching tomorrow, he is always curt and impatient and cuts her off. Aasma is anxious about getting things wrong because she's had a run of dreadful lessons where she'd used outdated departmental resources and hadn't understood the lesson rationale. She's disappointed that Adam, who offered to mentor her, seems impatient and unwilling to provide the support she needs.

Mentor perspective

Adam finds mentoring Aasma challenging. Despite his busy schedule (and no timetabled mentoring period), he commits to a full hour of mentoring each week. While he's willing to provide feedback on lesson plans given 48 hours in advance, Aasma often seeks impromptu end-of-day chats. Adam feels he was clear about his time constraints due to needing to get home promptly to take over childcare so his partner can get to her nursing shift, yet Aasma seems to expect immediate availability.

Scenario 2: Sam and Jon

Mentee perspective

Sam enjoys teaching and engaging with students but believes extensive lesson planning can be unnecessary and time-consuming. Before beginning teacher training, Sam worked as a cover supervisor in a school with a centrally planned curriculum, including standardised resources. Sam observed that students benefited from the consistency provided by these pre-planned lessons. During this time, Sam witnessed a beginning teacher being reprimanded for deviating from the prescribed lesson plan.

Mentor perspective

Jon is puzzled by Sam's reluctance to plan lessons from scratch. Despite collaborating on joint planning and team-teaching, Sam consistently relies on pre-existing departmental resources, often without considering the broader context or purpose of the lessons which is impacting on pupil learning. Sam's defensive response to feedback and insistence on using standardised lessons leads Jon to question Sam's commitment to the full scope of teaching responsibilities.

Scenario 3: Annie and India

Mentee perspective

Annie had a positive experience with her TP1 mentor, Hardeep, who was supportive and understanding. When Annie said she didn't feel confident to teach a group because of their behaviour, Hardeep wouldn't push too hard, and instead took over the teaching. In contrast, Annie's current mentor, India, is more direct. Annie feels overwhelmed and unsupported by India's approach, which often involves encouraging her to 'just get on with it' rather than offering emotional support or practical advice.

Mentor perspective

India is an experienced mentor with a proven track record of supporting her mentees through initial teacher education. She's encountered a variety of personalities and work ethics, and she's committed to helping all her mentees succeed. India recognises that Annie will need to take on more responsibility as an early career teacher. While she's supportive, she won't allow Annie to avoid challenging situations, as she believes it's essential for her professional development. India's ultimate goal is to prepare Annie for a successful career as a qualified teacher.

misunderstandings between mentors and mentees, demonstrating how easily conflicts can arise and can therefore be avoided.

Building a strong relationship with your mentor is crucial. Communicate openly and honestly about any concerns and prioritise speaking with your mentor first before involving others. Face-to-face communication is always better than an email; as one beginning teacher I tutored memorably commented, "The problem with an email is it doesn't smile". If the problem cannot be resolved between you, then you should consult your school ITE coordinator to gain guidance about how to navigate the situation. You can also seek support from your ITE provider who will have experience in resolving these difficult situations.

Now use Task 2.4.6 Thinking through your response to disruptions to the mentor and mentee relationship to read and reflect on the scenarios in Table 2.4.4.



Task 2.4.6 Thinking through your response to disruptions to the mentor and mentee relationship

Read and reflect on the scenarios for potential conflict.

- How can assuming positive intent help both the mentor and mentee resolve their conflict?
- What questions can the mentee ask to better understand the mentor's concerns about their progress?
- What can the mentee share with their mentor to foster mutual understanding and empathy?
- What strategies could the mentee employ to repair the relationship? What lessons can you learn from these scenarios to improve your relationship with your own mentor?

Discuss your thoughts with other beginning or early career teachers.

SUMMARY AND KEY POINTS

This unit has focused on how you can work effectively with your mentor and has helped you to develop your understanding of:

- the role of a mentor and the characteristics of effective mentoring relationships.
- how to build strong mentor–mentee relationships through effective communication and trust.
- the need to take ownership of your professional development by actively listening to and implementing your mentor's advice.
- How to engage in professional dialogue to foster trust and honesty to help you make effective progress.
- How to address potential issues between yourself and your mentor in a professional and constructive manner.

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Further resources

Wright, T. (Ed.). (2017). *How to Be a Brilliant Trainee Teacher* (2nd ed.). Abingdon: Routledge, is a clear and practical book that helps you understand how pupils learn and how you can enhance your own progress as a beginning teacher.

The subject-specific books in the Learning to Teach (Subject) series, the Practical (Subject) Guides, Debates in (Subject) and Mentoring (Subject) Teachers extend the information in this textbook, as do the MESHGuides' research summaries on www.meshguides.org.

Although the series of texts published by Routledge entitled Mentoring (subject) Teachers in the Secondary School: A Practical Guide are written for mentors, the guidance provided for mentors would enable you to both see subject- and phase-specific mentoring practices from a mentor's perspective and help you understand how to make best use of the mentoring relationship.

The blog Becoming a History Teacher provides support to mentors and beginning teachers navigating the initial and early career teaching years. Despite the name of the blog, much of the advice and support is applicable to all subjects and phases. https://www.routledge.com/Mentoring-Trainee-and-Early-Career-Teachers/book-series/MTECT and https://uonhistoryteachertraining.school.blog/

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3 Classroom interactions and managing pupils

Effective classroom interactions and management are essential to effective learning. Classroom interactions are those with (and between) pupils (and others) in the classroom to support learning. Classroom management refers to arrangements made by the teacher to establish and maintain an environment in which learning can occur, e.g. effective organisation and presentation of lessons so that pupils are actively engaged in learning. Classroom management skills and techniques are addressed throughout this book in a number of different chapters and units. This chapter includes units about different aspects of classroom management related to interacting with pupils. Together they give an insight into the complex relationships which are developed between teachers and pupils, and emphasise the need for well-developed skills and techniques that you can adapt appropriately to the demands of the situation. They reinforce the fact that, although you must plan your lessons thoroughly, not everything you do in the classroom can be planned in advance, as you cannot predict how pupils will react in any situation on any given day.

One commonality of teachers from whom we have learned a lot is their ability to communicate effectively with pupils to enhance their knowledge, skills and understanding. Most of us tend to think we communicate well. However, communication is a complex process. Unit 3.1 is designed to help you communicate effectively in the classroom. It looks first at verbal communication, including gaining pupils' attention, using your voice and the language you use. It then considers types of communication, including explaining, questioning, discussion and active listening. Aspects of non-verbal communication are then considered, including showing enthusiasm, confidence and caring and your appearance, gesture, posture, facial expression and mannerisms, particularly in relation to how you present yourself as a teacher.

Some pupils are motivated to learn and maintain that motivation; others are inherently motivated to learn but various factors result in them losing motivation, and still others may not be inherently motivated to learn but their motivation can be increased. A study of motivation, therefore, is crucial to give you some knowledge and insight into how you can create a motivational climate that helps to stimulate pupils to learn. Unit 3.2 looks at what motivation is, presents a number of theories of motivation and considers how these can inform your teaching and pupils' learning. It looks at how the motivational climate in your lessons can impact on pupil motivation. It also looks at some specific factors that influence pupils' motivation to learn; specifically, personal achievement

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(success), rewards, including the use of praise, punishment and feedback are discussed. Finally, it considers motivating individuals and the class as a whole.

We recognise student teachers' concerns about managing behaviour and misbehaviour. Unit 3.3 looks at adopting a positive approach to behaviour, which focuses on positive relationships with pupils and a positive classroom climate in which all pupils can learn effectively. This approach is more consistent with an inclusive schooling approach. Thus, the focus of the unit is on preventing misbehaviour as far as possible rather than on a reactive approach that focuses on 'discipline' for misbehaviour. The unit considers a whole school behaviour policy before interrogating the term 'unacceptable behaviour' and factors that may cause unacceptable behaviour. The unit then looks at a key principles of a behaviour for learning approach. This highlights the importance of getting the simple things right, building positive relationships in the classroom, structuring your lessons to promote positive behaviour, and rights, responsibilities, routines and rules in the classroom.

The final unit in this chapter, Unit 3.4, focuses on the transition between primary and secondary school, recognising that this is only one of the many transitions pupils experience as they go through compulsory schooling. It considers conceptualisations of primary-secondary transitions, highlighting successful transitions. The unit then focuses on discourse around transitions which have an influence both on the messages that might consciously or subconsciously give to pupils and families and how transitions support is provided. The unit moves on to the theorisation of transitions, specifically the Multiple and Multi-dimensional Transitions (MMT) Theory, before looking at planning and preparation for primary-secondary transitions. It then considers factors having an impact on pupils' transition experiences including pupils' relationships with family, peers and teachers. The unit concludes by highlighting what seems to work for pupils (including voice and opportunities to discuss concerns; familiarisation and knowledge of the new context; rehearsing in a safe environment; active learning, learning agency, autonomy and choice).

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3.1 Communicating with pupils

Paula Zwozdiak-Myers and Susan Capel

Introduction

Communication is a complex two-way process involving the mutual exchange of information and ideas – written, verbal and non-verbal. Clear and effective communication includes delivering and receiving information, which involves listening, observation and sensitivity. Although most of us tend to think we communicate well, when we study our communication skills systematically, most of us can find room for improvement. You need well-developed communication skills combined with awareness and sensitivity to the diverse needs, experience and level of understanding of your pupils. This unit focuses on the quality of communication between a teacher and pupils, which is critical to effective learning.

We first consider aspects of verbal communication, including using your voice (volume, projection, pitch, speed, tone, clarity and expressiveness), the language you use and the importance of active listening. We next consider aspects of non-verbal communication (appearance, gesture, posture, facial expression and mannerisms), particularly in relation to how you present yourself as a teacher.

OBJECTIVES

At the end of this unit, you should be able to:

- appreciate the importance of effective verbal and non-verbal communication skills and understand their inter-relationship;
- vary your voice consciously to enhance your teaching and pupils' learning;
- appraise your use of language and use questioning more effectively as a teaching and learning tool;
- gain awareness and control over how to effectively present yourself as a teacher.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

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Verbal communication

Gaining attention

You need to establish procedures for gaining pupils' attention at the beginning of a lesson and when you want the class to listen again after they have started an activity. This latter skill is especially important if there is a safety risk in the activity. Before you start talking, make sure all pupils can see and hear you, they are silent and paying attention. To initiate and sustain group attention: 'a centre-front position, facing the class group, standing relaxed and scanning the faces of the pupils while cueing for attention will normally (and positively) signal the teacher's readiness and expectation' (Rogers, 2015, p. 62). Establish a means of getting silence (say, 'quiet please', raise your hand, blow a whistle in physical education or bang on a drum in music), and use this technique each time to ensure consistency of approach. Wait for quiet and do not speak until there is silence. Once you are talking, do not move around. This distracts pupils, who may pay more attention to the movement than to what you are saying.

Using your voice

A teacher's voice is a crucial element in classroom communication. It is like a musical instrument, and if played well, pupils will be an appreciative and responsive audience. Certain qualities are fixed and give your voice its unique character. You can however alter the volume, projection, pitch, speed, tone, clarity and expressiveness to use it more effectively and lend impact to what you say. These verbal dynamics are important elements of paralanguage, and, if sensitively tuned, your voice can become a powerful agent of expression and communication.

The most obvious way to vary your voice is by altering the *volume*. It is useful to have the whole volume range available, from quiet to very loud, but rarely a good thing to be loud when not necessary. Of course, you have to be heard, but this can be accomplished through projection rather than volume. You *project* your voice by making sure it leaves your mouth confidently and precisely. This needs careful enunciation and breath control. If projected well, you are able to make a whisper audible at some distance. Equally, good projection brings considerable volume to your ordinary voice without resort to shouting.

Each group of spoken words has its own 'tune' that contributes to the meaning. A person may have a naturally high or low voice, but everyone can vary the 'natural' *pitch* to good effect. Generally speaking, deep voices sound more serious and significant; high voices sound lively and more exciting. To add weight to what is being said, the pitch should be dropped; to lighten the tone, the pitch should be raised. A voice with a lower pitch can create a sense of importance as it comes across as more authoritative and confident than a high-pitched voice. It can also be raised more easily to command attention, whereas raising a naturally high-pitched voice may result in something similar to a squeak, which does not carry the same weight.

Speed variations give contrast to delivery. You can use pause to good effect. It shows confidence if you can hold a silence before making a point or answering a question. Equally, have patience to wait for a pupil to respond. Research (for example, Muijs and Reynolds, 2018, p. 60) suggests 'three seconds or slightly longer is the optimal wait-time' for a closed, lower-level factual recall question, whereas a longer wait-time (up to 15 seconds) is required for 'open-ended, higher-level questions'. On occasion speaking quickly can be a valuable skill; however, this needs concentration and careful enunciation.

To use your voice effectively, these elements need to work together. You do not communicate effectively if the pitch is right, but you are not enunciating clearly or the volume is wrong; for example, shouting at a group, or pupils at the back cannot hear what you are saying. It is also important to put feeling into what you say to engage pupils. Often, pupils respond to *how* you say something rather than *what* you say. If praising a pupil, sound pleased, and if disciplining a pupil, sound firm. If you deliver all talk in the same way, do not be surprised if pupil response is undifferentiated since the intended meaning behind the message may not be readily understood. Now complete Task 3.1.1.



Task 3.1.1 The quality of your voice

Record your voice either reading from a book, newspaper, natural monologue or conversation. Listen to the recording with another beginning teacher. Your voice may sound different from the way you hear it; normally, you hear your voice coming back from the soft tissue and bone in your head. Most of your audience hear it coming forward. As you become used to listening to yourself, try to pick out the good points of your voice. Is it clear? Is it expressive? Is the basic pitch pleasant? Also, consider areas for improvement. Do you normally speak too fast? Is the tone monotonous?

Repeat the task, but this time vary the verbal dynamics. For example, try reading at your normal speed, then faster, then as quickly as you can. Remember to start each word precisely and concentrate on what you are saying. Then try varying the pitch of your voice. Listen to the tape with another beginning teacher and consider strategies and helpful advice for improving. Try these out in your teaching and store your thoughts for future reference/experimentation.

Language of the teacher

Teaching involves communicating with pupils from a variety of backgrounds and with different needs. To effectively support pupils with English as an additional language (EAL), for example, requires specialist knowledge and skill for teaching within the mainstream context in a way that promotes language learning alongside content learning. The learning of English for pupils with EAL takes place as much in the arts, humanities, mathematics and science as it does in the 'subject' of English, as well as the 'hidden curriculum', thus all teachers are teachers of language (see Flynn, Pim and Coles, 2023).

To develop pupils' language skills, a teacher's language must be accessible. Subject-specific vocabulary can be introduced gradually as you introduce your class to the language of the subject. To do this, you must not assume everyone knows the words or constructions that you do, including simple connecting phrases such as 'in order to', 'so that' or 'keep in proportion'. Start with simple, direct language that makes no assumptions.

It is easier for pupils to understand a new concept if you make comparisons or use examples, metaphors or references to which they can relate. Where appropriate, use a variety of words or explanations that ensure the meaning of what you intend to convey is understood by all pupils. As a teacher, your language must be concise. When you are speaking, stress or repeat important words or phrases. Placing an accent on certain syllables of the words you use gives rise to rhythmic patterns that affect the meaning of your message. These are important techniques in teaching. If they

help learning, repetition, accentuation and elaboration are valuable, but filling silence with teacher talk is generally unproductive. It is generally accepted that pupils understand something and learn it better if they hear it a number of times and it is explained in different ways. Task 3.1.2 focuses on the language of your subject.



Task 3.1.2 The language of your subject

Create a list of specialist words and phrases used in your subject or topic you may be teaching. How many of these might be understood or used by your pupils? In your lesson planning, how might you introduce and explain these words and phrases? Consider opportunities which enable pupils to practise their use of these words or phrases during the lesson. Record a lesson you are teaching, replay the recording and review your use of language. Also, listen with a beginning teacher learning to teach another subject who does not have your subject-specific vocabulary and knowledge and may be nearer to your pupils' experience of the subject. Discuss how you might improve your use of language in future lessons. Make a note of your thinking.

As well as conveying content, a teacher's language is used to create and develop interpersonal relationships with individual pupils which makes them more interested and motivated in learning. Using pupils' names, acknowledging their work and progress from previous lessons, showing interest in their lives outside the classroom and valuing their experiences are important in building mutual respect and creating a positive atmosphere for classroom learning.

Teachers use language to impose discipline. Often, negative terms are used for this. This is not inevitable, and a positive approach may have more success. For example, can you suggest a constructive activity rather than condemning a destructive one? Could earlier praise or suggestion have made later criticism avoidable? Rogers (2015) considers the complex, dynamic relationship between language and behaviour management, particularly in relation to communication with challenging pupils and those with emotional and behavioural difficulties.

Types of verbal communication

Verbal communication is used in many different ways in teaching. Explaining, modelling, demonstrating, questioning and discussion are core pedagogical tools for teachers and are briefly considered in the following subsections.

Explaining, modelling and demonstrating

Teachers spend a lot of time explaining to pupils. In some teaching situations it can be the main form of activity in the lesson, thus being able to explain something effectively is an important skill to acquire. Pupils learn better if they are actively engaged in the learning process and a good explanation actively engages pupils, and, therefore, is able to gain and maintain their attention. To accomplish this you can mix an explanation with tasks, activities or questions rather than relying on long explanations, dictating notes or working something out on the board.

Explaining provides information about *what, why* and *how*. It describes new terms or concepts or clarifies their meaning. Pupils expect teachers to explain things clearly and become frustrated when they cannot understand. A good explanation is clear and well structured. It takes account of pupils' prior knowledge and understanding, uses language pupils can understand and relates new work to concepts, interests or work already familiar to pupils. An explanation should be rephrased if it is not understood by pupils; the use of analogy or metaphor can also help an explanation.

Table 3.1.1 on the companion website identifies characteristic features of effective explanations. This checklist and sample questions may be useful when analysing your own and another beginning teacher's explanations.

Teachers often reinforce verbal explanations with a visual prompt, demonstration or model. Modelling is an effective learning strategy that allows pupils to ask questions and hear explanations related to each stage of the process as it happens, since the teacher can:

> 'think aloud', making apparent and explicit those skills, decisions, processes and procedures that would otherwise be hidden or unclear; expose pupils to the possible pitfalls of the task in hand, showing how to avoid them; demonstrate to pupils that they can make alterations and corrections as part of the process; warn pupils about possible hazards involved in practical activities, how to avoid them or minimise the effects if they occur.
>
> (Department for Education and Skills - DfES, 2004, p. 3)

Showing pupils what to do while talking them through the activity and linking new learning to old through questions, resources, activities and language is often referred to as *scaffolding*. Here, the teacher assists pupils to achieve tasks 'they cannot yet master on their own' (Muijs and Reynolds, 2018, p. 86) then gradually withdraws this support. Over time this develops independent learners. Combining verbal and visual explanations can be more effective than using verbal explanations exclusively.

Questioning

One technique in the scaffolding process for actively engaging pupils in their learning is questioning, which is a valuable part of interactive teaching. Teachers use a lot of questions: 'every day teachers ask dozens, even hundreds of questions, thousands in a single year, over a million during a professional lifetime' (Wragg and Brown, 2001, p. 1). Hence, questioning needs to be conducted skilfully (see, for example, Doherty, 2018).

Asking questions effectively

To use questioning effectively requires planning as you need to decide:

- what type of question(s) you are going to ask;
- how you are going to ask the question(s);
- when you are going to ask the question(s);
- why you are asking the question(s);
- who you are going to ask the question(s).

You may also consider how you expect the question to be answered, how to respond if the pupil does not understand or gives an inappropriate answer, and how long you are prepared to wait for an answer. Allowing thinking time is an important aspect of questioning.

You cannot plan your questioning strategy too rigidly and follow a pre-set agenda, as this may result in pupils asking fewer questions themselves, producing undeveloped responses, rarely discussing ideas with peers, presenting few thoughts of their own or demonstrating confusion. You must therefore exercise flexibility and adapt your plan to take account of how the lesson develops.

Asking questions is not a simple process. Questions are asked for many reasons; for example, to check understanding of an instruction or explanation, to reinforce or revise a topic, to deepen understanding, encourage thinking and problem-solving or to develop a discussion. Wragg and Brown (2001, pp. 16–17) classified the content of questions related to learning a particular subject as one of three types: *empirical questions* require answers based on facts or experimental findings; *conceptual questions* elicit ideas, definitions and reasoning in the subject being studied; and *value questions* investigate the relative worth and merit, moral and environmental issues. These categories often overlap, and some questions may involve elements of all three.

Another classification for planning questions with specific purposes in mind is Bloom's (1956) taxonomy of educational objectives within the cognitive domain, which captures six levels of complexity and abstraction. This was adapted by Anderson and Krathwohl (2001) to represent the nouns as verbs and make it more action oriented. Lower-level questions usually demand factual, descriptive answers, whereas higher-level questions are complex and require more sophisticated critical thinking. Pupils' cognitive abilities and levels of achievement can be increased when they are challenged and have regular access to higher-order thinking (Fisher, 2013; Muijs and Reynolds, 2018). Table 3.1.2 on the companion website links Bloom et al.'s (1956) taxonomy and Anderson and Krathwohl's (2001) adapted version with what pupils might be expected to do and provides possible question stems you can draw upon to help them realise those tasks.

Closed and open questions

The questions: Where is Mount Everest?, What does the Latin expression, 'Veni, vidi, vici' mean, who first used it and when? and How do you spell 'magnificent'? are *closed* questions. There is only one correct answer, which limits the scope of response. A short, closed question-answer session might be used to reinforce learning, refresh pupils' memories, provide a link to new work or help pupils remember what they have been taught at the end of a lesson.

Conversely, the questions: How can we save our planet?, What are the underlying principles of equity and social justice? and What do we know about the 'Big Bang' theory? are *open* questions, which broadens the scope of response since they have several possible answers. Open questions are designed to extend pupils' understanding of a topic, stimulate thinking and learning and challenge pupils' capacity to frame ideas in words. Pupils need to manipulate information by reasoning or apply information and use knowledge, logic, creativity and imagination. Open questions cannot be answered quickly as pupils need time to gather information, sift evidence, advance hypotheses, discuss ideas and plan their answers.

An example from a science lesson exemplifies the difference in purpose between closed and open questions. 'What is the chemical formula of carbon dioxide?' is a closed question requiring factual knowledge, whereas 'How does carbon combine with oxygen during the respiration process?'

requires understanding. The question 'In what way/s can we reduce our carbon footprint to slow down global warming?' requires a deeper level of reflection and research.

You can ask closed or open questions or a combination of the two as *a series of questions*, which takes time to build up if they are to be an integral part of the learning process. They must therefore be planned as an integral part of the lesson, not as a time-filler at the end where their effect is lost.

Other aspects of questioning to consider

Questions can be asked to the whole class, groups or specific individuals. Questions can be spoken, given out on printed sheets, written on (an electronic) whiteboard or by using digital technology (e.g. Kahoot, a free game-based learning platform which allows you to ask pupils questions in any subject, any language and on any device). Answers can be given at once or produced after deliberation, either spoken or written. For example, you may set a series of questions for homework and either collect the answers in to mark or go through them verbally with the class during the next lesson.

Effective questioning requires you to be able to ask clear, appropriate questions, use pause to allow pupils to think before responding and prompts to help those experiencing problems in answering. Key tactics identified by Wragg and Brown (2001, p. 28) for asking questions include: structuring; pitching and putting clearly; directing and distributing; pausing and pacing; prompting and pacing; listening and responding to replies; and sequencing. Muijs and Reynolds (2018, p. 60) identify three types of prompts to help pupils answer questions: *verbal prompts* (cues, reminders, instructions, tips, references to previous lessons or sentence starters); *gestural prompts* (pointing to an object or modelling a particular behaviour); and *physical prompts* (guiding pupils explicitly through motor skills).

Follow-up questions can be used to: probe further; encourage pupils to develop their answers; extend their thinking; change the direction of the questioning; and distribute questions to involve the whole class. Ensure the vocabulary is appropriate for the pupils' level of understanding and check your questions are not too long, complex or ambiguous. You can also encourage pupils to actively participate in questioning by listening and responding to answers, praising good answers, being supportive and respecting answers and not making pupils feel they will be ridiculed if they answer incorrectly.

Non-verbal aspects of communication such as eye contact, gesture, body language, tone of voice, humour and smiles are important in effective questioning because they should align with the words being used.

Discussion

Questioning may lead naturally into discussion to explore a topic further:

> For a fruitful discussion that allows pupils some significant say over what is discussed, while at the same time covering the ground that teachers judge to be important, it is best to think of questions that may be perplexing, intriguing or even puzzling to pupils. Skilfully chosen encouraging, broad questions are often effective in sparking off animated conversations. The process may begin with recall questions to extend and activate knowledge and then thought questions to lift the discussion.
>
> (Wragg and Brown, 2001, p. 44)

To maximise pupils' learning through discussion you need to be able to chair a discussion effectively. Plan how you are going to stimulate the discussion and respond should it drift from the main theme, for example, by interjecting suggestions or key questions. Seating arrangements and pupil grouping help develop a less structured atmosphere, which may encourage all pupils to contribute. Before using discussion in your classes, it might be wise to observe another teacher use this technique in their teaching.

Active listening

For effective communication, *being able to actively listen and take account of pupil response* is as important as sending the message effectively. Learn to recognise whether a message has been received as intended; for example, you get a bewildered look or an inappropriate answer from a pupil. Be able to react appropriately; for example, repeat or rephrase the same question. However, also reflect on why the communication was not effective: was the pupil not listening to you? If so, why not? Perhaps you asked a question and 'switched off' while an answer was being given, to think about the next question or part of your lesson. This lack of interest may convey boredom or indifference to the pupil, which negatively impacts the tone of the lesson.

Effective listening is an active process, with a range of non-verbal and verbal responses that convey the message to the pupil speaking that you are listening to what is being said. This is associated with conveying enthusiasm and generating interest, by providing reinforcement and constructive feedback to pupils. For example, looking alert, making eye contact with the pupil talking to you, smiling, nodding and using verbal signals to show you have received and understood the message or encourage the pupil to continue by saying 'yes', 'I see what you mean' or 'go on'.

Online communication

There are numerous channels in the virtual world for interacting with pupils through writing and speech, for example, emails, instant messaging, text messaging on mobile phones, discussion boards, video conferencing, chat rooms, electronic portfolios and online surveys. Whatever the medium, effective online communication still needs to be thought about to ensure that what you want to say is neither 'lost in translation' nor misinterpreted.

For example, your written message requires some planning and careful thought due to the absence of body language and its mis/interpretation will be based on your use of *lexicon* (the words of a language), *semantics* (the meaning of these words) and *syntax* (how the words and symbols are put together). The use of expression and way in which your message is constructed can, however, also reveal non-verbal cues such as tone and volume. Tubbs and Moss (2008, p. 136) write 'when individuals want to create an impression or express feelings, or convey variations in tone or volume, they use capital and lowercase letters differently, typing errors, exclamation points, and other punctuation marks and emoticons (sometimes called emoji's) along with their verbal message.'

Consider, for example, how the following sentence, although the same, may be interpreted differently based upon the use of verbal/text and non-verbal/text:

> **YOU MUST SUBMIT YOUR ASSIGNMENT BY 3.00PM ON FRIDAY AT THE LATEST!!!**
>
> ***You must submit*** your assignment by 3.00pm on Friday at the latest 😳!
>
> You must submit your assignment by 3.00pm on Friday at the latest 😳

Online communication etiquette (netiquette) suggests we can avoid 'shouting' or appearing to be 'angry' if we do not use exclamation marks, underlining and capital letters exclusively. It is also good practice to give your message a title; personalise your communications; be concise and ensure your choice of words accurately convey your intended message; be clear and only provide information that is timely and relevant to avoid confusion; think about your spacing, presentation and style of font (especially important for pupils with dyslexia); consider ways to monitor whether pupils have received your message; and, always provide opportunities for pupils to ask questions if they are confused or do not understand your message. Planning how, when and why to use online communication with your pupils should ensure consistency of approach and lead to productive mutual engagement since expectations become well established.

Non-verbal communication

Non-verbal cues can have as great an impact on the listener as the spoken word. The communication model devised by Albert Mehrabian (1972) is often referenced to highlight the crucial role non-verbal communication plays in teaching when you seek to convey meaningful messages, establish positive relationships and engaging learning experiences for your pupils. During face-to-face interactions, he found 93 per cent of the meaning behind verbal messages was received through non-verbal channels: notably, 55 per cent through such non-verbal cues as body language, facial expressions, gestures, mannerisms and posture; 38 per cent through pace, pitch, tone of voice and way the words are spoken; and 7 per cent from the actual words used.

Anderson (1998) considers the essence of eloquent, passionate and spirited communication is comprised of body position, facial expressions, gestures and movement to differentiate, extend, modify or transmit verbal messages and delivery of the lesson content. Friedman, Riggio and Casella (1988) also align non-verbal 'expressiveness' with 'charisma' and 'spirit' which can be demonstrated by emotional states and interpersonal attitudes that are captivating, exciting, inspiring and moving. The capacity to send non-verbal cues expressively and encode them accurately is therefore an important component of interpersonal communication and social interaction in your teaching.

Non-verbal communication supports or detracts from verbal communication, dependent upon whether verbal and non-verbal signals match each other; for example, if you are praising someone and smiling and looking pleased or reprimanding them and looking stern and sounding firm, you are sending a consistent message and are perceived as sincere. On the other hand, if you are smiling when reprimanding someone or looking bored when praising someone, you send conflicting messages that cause confusion and misunderstanding. When non-verbal cues do not reinforce meaning 'it communicates instead the speaker's lack of involvement. Rather than being the message about the message, it becomes the message about the messenger' (Robertson, 1996, p. 94).

Non-verbal communication can have a considerable impact without verbal communication; for example, looking at a pupil slightly longer than you would normally, communicates your awareness they are talking or misbehaving. This may be sufficient to gain the pupil's attention. You can probably think of a teacher who stands at the front of the class leaning against the board with arms crossed waiting for silence or the teacher who sits and listens attentively to the problems of a particular pupil. The meaning of the communication is clear and there is no need to say anything, which demonstrates that actions can speak as loudly, if not more loudly, than words. Non-verbal signals are important for good communication, classroom management and control.

Presenting yourself effectively

There is no single correct way to present yourself as a teacher since you are an individual and bring your own unique set of characteristics and personality to the job. This said, initial impressions are important, and how you present yourself to a class on first meeting can influence their learning over a period of time. Having prepared the lesson appropriately, the pupils' impressions of the lesson, and of you, are important. Part of the impression created relates to your appearance and pupils do value adults who are well dressed and take pride in their appearance. Pupils expect teachers to wear clothes that are clean, neat and tidy, and certain teachers to wear certain types of clothes; for example, it is acceptable for a physical education teacher to wear a tracksuit but not a history teacher.

How teachers follow up the first impression is equally important, for example, whether you treat pupils as individuals, how you communicate with pupils, whether you have any mannerisms such as constantly clicking a pen or saying 'OK' – which may reduce or prevent effective communication (pupils may focus on mannerisms rather than what is being said and even count the number of times you do it!). Effective teaching depends on and is enhanced by self-presentation that is *enthusiastic*, *confident* and *caring*.

Enthusiasm

Before many pupils make an effort to get to grips with something new, the teacher needs to 'sell' it to them as something interesting and worthwhile. Your enthusiasm should be sustained throughout a lesson, and in relation to each activity – not only when presenting material but also when a pupil has persevered to achieve a goal.

Your enthusiasm for your subject is infectious. However, there could be a danger of 'going over the top' when showing enthusiasm. If you are overexcited, it can portray a sense of triviality, so enthusiasm has to be measured. There are three principal ways to communicate enthusiasm verbally and non-verbally. The first is via *facial expression*:

> An enthusiastic speaker will be producing a stream of facial expressions which convey his excitement, disbelief, surprise or amusement about his message. Some expressions are extremely brief, lasting about one fifth of a second and may highlight a particular word, whereas others last much longer, perhaps accompanying the verbal expression of an idea. The overall effect is to provide a running commentary for the listener on how the speaker feels about the ideas expressed. In contrast, a speaker who is not involved in his subject shows little variation in facial expression. The impression conveyed is that the ideas are brought out automatically and are failing even to capture the attention of the speaker.
>
> (Robertson, 1996, p. 86)

The second is via *use of your voice*. The manner in which you speak gives a clear indication of how you feel about the topic being taught and is readily picked up by pupils. Your voice needs to be varied and indicate your feelings about what you are teaching. As you are engaged in something akin to a 'selling job', your voice has to show this in its production and delivery – it has to be persuasive and occasionally show a measure of excitement: 'Enthusiastic teachers are alive in the room; they show surprise, suspense, joy, and other feelings in their voices and make material interesting by relating it to their experiences and showing that they themselves are interested in it' (Good and Brophy, 2007, p. 385).

The third is via *poise and movement*. An enthusiastic speaker has an alert, open posture and accompanies speech with appropriately expressive hand and arm gestures – sometimes to emphasise a point, at other times to reinforce something being described through indicating relevant shape or direction, for example, an arrangement of apparatus or a tactical move in hockey. If enthusiastic, you are committed and involved, all aspects of your posture and movement should display this.

Think back to teachers you have worked with and identify some whose enthusiasm for their subject really influenced your learning. How did these teachers convey enthusiasm? How do you convey enthusiasm? To what extent does poise and movement reflect people's attitudes, emotions and intentions?

Confidence

As a teacher, it is very important to present yourself with confidence. This is easier said than done because confidence relates to your sense of knowledgeable mastery of the subject matter and assurance of being in control, which takes time to develop and establish within the context of building interpersonal relationships.

There is an irony in pupils' response to teacher confidence. Expression of authority is part of the role pupils expect of a teacher, and where exercised with confidence, pupils feel at ease and reassured. However, if they sense a teacher is unsure or apprehensive, it is in young people's nature to endeavour to undermine authority (see Rogers, 2015).

In many cases, experience brings confidence, yet pupils seldom allow this to influence their behaviour. Although the key to confident self-presentation is to be well planned, in terms of material and organisation, without the benefit of experience, your excellent plans may not work, and you may have no alternative 'up your sleeve'. Whatever happens, you need to cultivate a confident exterior, even if you are feeling far from assured inside.

Confidence can be conveyed verbally in clear, purposeful instructions and explanations that are not disrupted by hesitation. Instructions given in a direct and business-like manner, such as 'Sian, please collect the notebooks and put them in the blue box', convey a sense of confidence. Conversely, the same instruction phrased as a question, 'Sian, will you collect the notebooks and put them in the blue box?' can convey a sense of being less assured, not confident that, in fact, Sian *will* cooperate. There is also the possibility of the pupil saying, 'No'!

Your voice needs to be used in a firm, measured manner. A slower, lower, well-articulated delivery is more authoritative and displays more confidence than a fast, high-pitched mode of speaking. Use of voice is particularly important in giving key instructions, especially where safety factors are involved and in taking action to curtail inappropriate behaviour. This is perhaps the time to be less enthusiastic and animated, and more serious and resolute in your manner.

Non-verbally, confidence may be expressed through posture, movement and eye contact, in their own right and as accompaniments to verbal language. There is nothing agitated about the movement of confident people. They tend to stand still and use minimal arm gestures to reinforce the message being conveyed.

Eye contact is a crucial aspect of conveying confidence to pupils. A nervous person avoids eye contact, somehow being afraid to know what others are thinking, not wanting to develop a relationship that might ultimately reveal their inability or weakness. Clearly, it is your role as a teacher to be alert at all times to pupil reaction and strive to develop a relationship with all pupils 'encouraging them to seek your help and advice. Steady, committed eye contact is usually helpful for these

objectives. You must, however, also recognise that eye contact is perceived differently by people of different cultures: in certain parts of Africa, Asia and Latin America, avoiding eye contact is the way to show respect, whereas extended eye contact may be perceived as disrespectful or a challenge to authority. You should therefore take into account cultural norms and sensitivities. This applies to other aspects of non-verbal communication, such as dress, gesture, physical contact and spatial proximity to another person. (for guidance on cultural difference and diversity, see Knapp, Hall and Horgan, 2021, as well as the NALDIC website, 2025).

Caring

Your commitment to pupils' wellbeing and learning should be evident in all aspects of your manner and self-presentation. While this goes without saying, it is not as straightforward as it sounds since it requires sensitivity and flexibility. You, as the teacher, may need to modify your behaviour in response to pupils, rather than always expecting pupils to fall in line with everything you ask for. There is a potential conflict, and balance to be struck, between firm confidence and flexible empathy. One of the challenges is to find this balance and be able to respond suitably at the appropriate time.

A caring approach is evident in many aspects of teaching, from efficient preparation through to sensitive interpersonal skills such as active listening. Teachers who put pupils' interests above everything have taken the time to prepare work thoroughly in a form appropriate to the class. Similarly, the classroom environment shows thoughtful design and organisation. In the teaching situation, caring teachers are fully engaged with the task at hand, observing, supporting, praising, alert to the class climate and able to respond with an appropriate modification to the programme, as necessary. Above all, caring teachers show real sensitivity to pupils' individual needs and clearly communicate that each pupil's learning and success are valued.

Now complete Tasks 3.1.3 and 3.1.4.



Task 3.1.3 Communicating effectively

Select in turn each aspect of verbal and non-verbal communication identified above; your use of voice, language, explaining/questioning/discussion, listening, presenting yourself effectively, enthusiasm, confidence and caring. Prepare an observation sheet for your mentor or another beginning teacher to use when observing you teach a class. Use this as a basis for evaluation and discussion about how you can further develop each of these aspects of your teaching.



Task 3.1.4 Studying one aspect of communication in depth

Select one aspect of communication to study in greater depth. Review the relevant literature, design and conduct a piece of action research on that aspect with one of your classes. Report critically on your research, identifying issues related to the assumptions underlying the methods of data collection and analysis. Reflect on the outcomes and impact of your study as well as your own learning about effective communication.

SUMMARY AND KEY POINTS

- Good communication is essential for developing sound interpersonal relationships with pupils, a positive classroom climate and effective teaching and learning.
- This unit has aimed to help you identify strengths and areas for development in your verbal and non-verbal communication skills and presentation of self as a teacher.
- Developing your professional knowledge, understanding and judgement in this area should enable you to communicate more effectively, sensitively and to best advantage.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.

Further resources

Hall, G. (2018) *The Experiences of Secondary School Students with English as an Additional Language: Perceptions, Priorities and Pedagogy*, London: British Council.

This paper brings together perspectives of secondary EAL pupils and their teachers from two schools in Tyneside, England. Findings reveal that the relationship between language, access to the curriculum and identity are central issues and for many, perceived needs and priorities change over time. Conclusions also suggest that EAL pupils are multilingual and multicultural resources from whom others can learn and through which schools might celebrate diversity.

Knapp, M., Hall, J. and Horgan, T. (2021) *Non-Verbal Communication in Human Interaction* (9th edn), Dubuque, IA: Kendall Hunt Publishing Company.

Several chapters within this text explore non-verbal elements involved in any interaction, including the environment within which the interaction occurs; physical features of the participants; and their behaviour (for example, gestures, touching, territory and personal space, facial expressions, eye gazing and vocal sounds).

National Association for Language Development in the Curriculum (NALDIC) (www.naldic.org.uk).

The national subject association for (EAL) is a professional forum for learning more about (EAL) and how to support bilingual pupils in schools across the UK, to raise achievement and promote the development of research, policy and practice.

Patzer, R. (2024) *Questioning in the Classroom: How to Improve Your Skills*, viewed 01 August 2025, blog.irisconnect.com/uk/how-to-improve-questioning-skills#:~:text='Key%20Questions%20as%20Learning%20Objectives'%20Initiate%20thinking,thinking%20about%20what%20they're%20going%20to%20learn.

This blog considers why questioning in the classroom is important, effective questioning and discussion techniques, how to improve your questioning skills and includes a YouTube video clip (10.39 minutes) to demonstrate effective questioning in practice during a 12th grade English lesson in America.

Rosas, J. (2020) *12 Examples of Nonverbal* Communication (And How to Use Them), Knesix Institute, viewed 1 August 2025, <www.youtube.com/watch?v=uqV2vtI-oO0>

This 8-minute film considers the use and impact of such non-verbal cues as facial expressions, manipulator gestures, personal space and physical reactions to help boost your people analysis skills.

Rosenshine, B. (2012) 'Principles of instruction: Research-based strategies that all teachers should know', *American Educator*, 36 (1), 12–19 and 39.

This article provides an overview of key instructional strategies that can be used by teachers to support pupil's thinking and associated cognitive development. Reading this will enhance your understanding and application of questioning, in particular.

Shabatura, J. (2022) *Using Bloom's Taxonomy to Write Effective Learning Outcomes*, viewed 1 August 2025, https://tips.uark.edu/using-blooms-taxonomy/

This website provides further information about Bloom's taxonomy and explains ways in which it can be used to: aid in course design; work with learning outcomes; work with Quality Matters standards; course level and lesson level outcomes.

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3.2 Motivating pupils

Ros McLellan

Introduction

Pupils' overall motivation towards learning in school is a result of a number of factors, including past experiences, future expectations, school ethos, class climate, peer group, teachers, gender, family background, culture, economic status and class. As a result, the link between motivation and educational performance and achievement is complex.

Some pupils tend to be more motivated to learn; for example, if it is valued at home or they see a link between education and a job. Such pupils are more likely to work hard, demonstrate positive learning behaviours in the classroom and be more successful. Many pupils want to learn but depend on teachers to get them interested in a subject. Even though some pupils may not appear to be inherently motivated to learn, the school ethos, teachers' attitudes, behaviour, personal enthusiasm, teaching style and strategies in the classroom can increase their motivation to learn (see also, for example, Unit 5.3). On the other hand, pupils who do not feel valued at school are, in turn, unlikely to value school. Therefore, although some pupils may be inherently motivated to learn, they may become demotivated or have low motivation because of a learning environment that does not stimulate them, because a task is too difficult or because there is a negative impact of factors such as those identified above. Pupils for whom the motivational climate is not right are more likely to become disinterested and misbehave. If the teacher does not manage the class and their behaviour effectively, the learning of all pupils in the class can be negatively affected. Thus, a central aim as a teacher is to create a motivational climate that helps stimulate pupils to learn.

OBJECTIVES

At the end of this unit, you should be able to:

- understand the role and importance of motivation for effective teaching and learning and classroom management;
- appreciate some of the key elements of motivation for effective teaching;

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- understand how to motivate pupils effectively;
- understand how to create a classroom environment that challenges pupils.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

What is motivation?

Motivation is typically defined as a 'set of interrelated desires, goals, needs, values and emotions that explain the initiation, direction, intensity, persistence, and quality of behaviour' (Wentzel and Miele, 2016, p. 1).

Behaviour is readily observable as a teacher so focusing on this can immediately provide some insight into the motivation of your pupils. Activities that pupils start (initiation and direction), the amount of effort they put into those activities at any particular time (intensity) and continue with (persistence) and how appropriate their chosen activity is for what the situation requires (quality) reflect their motivation. Persistence has been found to be particularly critical for achievement. Duckworth (2016) describes this as 'grit', concluding that a willingness to persevere despite encountering initial failure or difficulties is what predicts success. Indeed, this has been found to predict success more reliably than measures of IQ (Duckworth, Peterson, Matthews and Kelly, 2007), and is something that can be learnt.

As teachers, we want to create a learning culture in our classrooms whereby pupils are encouraged to see mistakes as an opportunity for development is an integral part of this process and is discussed further below. Fundamentally, it is about developing a 'Growth Mindset' (Dweck, 2006). If pupils are able to learn with a growth mindset, then they embrace challenges, recognise that effort is essential to succeed, understand that feedback gives opportunities to learn and improve and that setbacks simply test perseverance, translating to better academic performance (Dweck and Molden, 2017).

The definition above identifies that motivation encompasses a range of elements, including desires, goals, needs, values and emotions which drive motivated behaviour. Different theories of motivation highlight specific elements. However, it is important to distinguish between intrinsic (motivation from within the person; that is, engaging in an activity for its own sake for pleasure and/or satisfaction inherent in the activity; for example, a sense of achievement at having completed a difficult piece of work) and extrinsic (motivation from outside; that is, engaging in an activity for external reasons; for example, to receive a reward, such as praise, or to avoid punishment). Research has found that those intrinsically motivated are more likely to persist and continue with an activity than a person extrinsically motivated. For instance, some pupils succeed at school/in a subject despite the quality of the teaching because they intrinsically enjoy learning and/or are interested in the subject, However, pupils are asked to do activities that are new to them, difficult, which they may not be immediately successful or that they may perceive to be of little or no relevance to them. Further, they may have low self-confidence and feel they have limited capacity to learn. In such circumstances, they are less likely to be intrinsically motivated and will need encouragement to become engaged and to build confidence. This may take the form of written or verbal praise for effort, making progress or success, or feedback on how they are doing or an explanation of the

relevance of the work. Teachers can deliberately plan extrinsic motivators into their lessons with a view to enhancing both self-confidence and intrinsic motivation. To get you started in thinking about motivation, Task 3.2.1 asks you to reflect on what motivates you as a learner.



Task 3.2.1 What motivates you as a learner?

Reflect on your own experiences at school. What was it that motivated you in the subject you are now learning to teach? Identify another subject in which you were less motivated and reflect on why this was the case. Use the terms identified above and discuss your reflections with another beginning teacher. Identify anything you can learn from these experiences to build into your own teaching. Keep your notes for future reference.

Theories of motivation

There are many theories of motivation. In addition, we adopt our own, often unconscious, theories. Selected key theories of motivation that are particularly pertinent to the learning context, along with some of their implications for you as a teacher in determining learning activities, are given in Table 3.2.1 and discussed further in the text below.

Applying theories of motivation to your teaching

It is often difficult to identify the exact reason for a particular pupil's behaviour at a particular time, and, therefore, what is motivating them. Likewise, it is often difficult for a pupil to identify exactly what is motivating them. At times you can infer whether or not a pupil is motivated by observing their behaviour. Although there may be other reasons for a pupil not listening to what you are saying, talking, looking bored or staring out of the window, one reason may be that the motivational climate is not right and, therefore, the pupil is not motivated to learn. Not all factors contributing to motivation can be directly influenced (such as prior experience), however teachers can do much to create a motivational climate that is conducive for learning. This section starts to consider how this can be done by applying the motivational theories summarised in the table to the classroom learning context.

Achievement motivation theory; central to this is the need to achieve, which can be encouraged by creating a learning environment in which 'the need for achievement in academic studies is raised' (Child, 2007, p. 254). Each individual sets for themselves a standard of achievement, according to their level of aspiration. Research has shown that when people experience an activity as 'challenging', they have higher levels of self-confidence, feel more in control, experience more positive emotions and have reduced levels of the stress hormone cortisol. In such circumstances, individuals might enter a state of 'flow', when they are so engrossed in what they are doing that nothing else seems to matter, they lose track of time and will continue what they are doing for the sheer sake of it, clearly exhibiting intrinsic motivation (Csikszentmihalyi, 2008). Intrinsic motivation leads to deep learning (Entwistle, 1990; see also deep and surface learning in Units 5.2 and 5.3). Deep learning means that learners try to understand what they are doing, resulting in greater understanding of the subject matter. This is a prerequisite for high-quality learning outcomes (that is, achievement).

Table 3.2.1 Important theoretical perspectives and their implications for you as a teacher

Theory	*Source*	*Main points*	*Implications for teachers*
Achievement motivation	Atkinson (1964) and McClelland (1961)	Motivation to perform an achievement-oriented task is related to the: (i) need to achieve on a particular task; (ii) expectation of success on the task; and (iii) strength of the incentive after the task has been completed successfully. This results in individuals setting themselves standards of achievement.	Create a learning environment that raises the need for achievement by raising levels of aspiration. Plan tasks that are challenging but attainable with effort. Work should be differentiated according to individual needs.
Achievement goal theory	Ames (1992a, 1992b), Dweck (1986), Dweck and Leggett (1988) and Nicholls (1984, 1989)	A social-cognitive perspective that identifies determinants of achievement behaviour, variations of which result from different achievement goals pursued by individuals. In a task- (or mastery-) oriented setting, the focus is on skill learning and exerting effort to succeed, and success is judged by self-improvement and mastery of a task. In an outcome- (ego- or performance-) oriented setting, individuals compare their performance and ability with others and judge success by beating others with little effort to enhance social status.	Pupils' goal orientation may be influenced by the motivational climate created by what teachers do and say. Therefore, plan a task-oriented learning environment that encourages pupils to improve their performance by trying hard, selecting demanding tasks and persisting when faced with difficulty, rather than an outcome-orientated learning environment that encourages pupils to select easy tasks that they can achieve with minimum effort and on which they are likely to give up when facing difficulties.
Attribution theory	Weiner (1972)	Success or failure is attributed to ability, effort, difficulty of task or luck, depending on: (i) previous experience of success or failure on the task; (ii) the amount of work put in; or (iii) a perceived relationship between what is done and success or failure on the task.	Reward effort as well as success, as pupils are more likely to try if they perceive success is owing to effort; for example, you can give two marks for work, one for the standard of the work, the other for effort. Use teaching and assessment that is individualised rather than competitive.
Expectancy theory	Rosenthal and Jacobson (1968)	A person uses a range of cues to form expectations (high or low) of another. That person then behaves in a way that is consistent with their expectations. This influences the other person's motivation, performance and how they attribute success or failure. The other person performs according to the expectations, thus creating a self-fulfilling prophecy.	In order to avoid the self-fulfilling prophecy of pupils performing according to the way teachers expect them to perform (high or low), do not prejudge pupils on their past performance. Rather, encourage pupils to work to the best of their ability all the time.
Self-determination theory	Deci and Ryan (1985) and Ryan and Deci (2000a)	A broad framework that considers both social and environmental factors that contribute to either enhancing or diminishing intrinsic and extrinsic motivation. Key to intrinsic motivation is fulfilment of the needs for autonomy, competence and relatedness. Extrinsic motivation operates by engaging pupils in tasks so that they can willingly accept the value of the task even if the task itself may not hold any specific interest to them. This can lead to greater persistence and better engagement in the task. Key to understanding extrinsic motivation are its different forms, e.g. internalisation and integration.	In a teaching context, it is acknowledged that interpersonal interactions (for example, communication and feedback) between teachers and pupils are critical in facilitating or impeding feelings of competency in pupils. As a consequence, intrinsic motivation can be elevated or diminished. Elevated intrinsic motivation does not occur unless also accompanied by a sense of autonomy. Research shows that autonomy-supportive teachers catalyse pupils' intrinsic motivation (Deci et al., 1981).

In contrast to this, when people experience an activity as 'threatening', their self-confidence is reduced, they perceive themselves to have limited control, experience more negative emotions and cortisol levels increase, which ultimately reduces their performance (Jones, Meijen, McCarthy and Sheffield, 2009). Thus, setting tasks that are challenging but achievable for each individual pupil (that is, individualised tasks), can be used to raise aspirations, as long as pupils understand that these are achievable and hence are not perceived as threatening.

Expectancy-value theory (Eccles, 1983) develops these ideas further, suggesting that individuals will be motivated to engage if they have a reasonable expectancy of success and that success is valued. Pupils' expectancy of success clearly links to how confident they feel about the task in front of them, also known as their self-efficacy (Bandura, 1997) so this will vary from pupil to pupil. Tasks on which pupils expect to achieve/succeed approximately 50 per cent of the time are the most motivating. However, this means that pupils are likely to fail on the task approximately 50 per cent of the time, so it is important to plan to reduce loss of motivation when pupils fail (for example, by praising effort, giving feedback on performance and so on). Furthermore, expectancy-value theory flags up the importance of understanding what pupils value in designing learning tasks. Tasks may be seen to have attainment (important for one's identity) or intrinsic (enjoyment/interest) value, both of which foster intrinsic motivation. Or they may be seen to have utility value (usefulness/relevance) which acts as an extrinsic motivator but may also lead to intrinsic motivation (Hulleman, Durik, Schweigert and Harackiewicz, 2008). Teachers therefore should get to know all of their pupils as individuals to know what is important to them and take this into consideration with an individualised approach.

Achievement goal theory; research has shown that people's learning behaviours (deep vs. surface, Entwistle, 1990) and achievement are directly related to the types of achievement goal they adopt in achievement situations (Covington, 2000; Elliot and McGregor, 2001). Essentially, there are two key considerations when understanding the type of goal being deployed. First, is it a mastery or performance-referenced goal? And second, is it an approach or avoid type of goal? These different components combine to create four possible types of goal: mastery-approach; mastery-avoid; performance-approach; and performance-avoid (see Figure 3.2.1). There seems to be a relationship

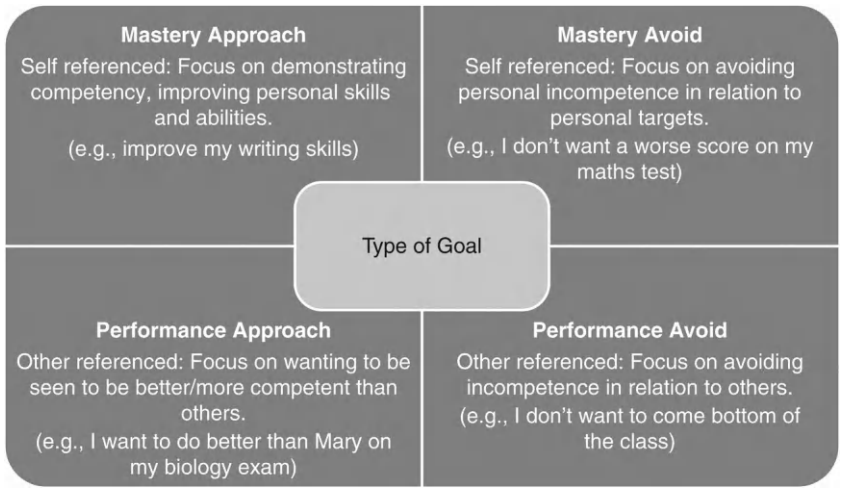


Figure 3.2.1 A representation of the different types of goals

Source: Adapted from Elliot and McGregor (2001)

between the types of goal adopted and whether pupils find an activity challenging or threatening. Elliot and McGregor (2001) identified that pupils who set mastery-approach goals perceived an examination situation to be challenging (positive), whereas those who adopted performance-avoid goals perceived it as threatening (negative).

Individuals who set mastery-approach goals have the most control over their achievement, and consequently understand that their efforts determine their success. Furthermore, the literature on expertise indicates that practice (i.e. the expending of effort associated with the setting of mastery-approach goals), in a purposeful way is the most critical factor in whether someone becomes an expert or not (Ericsson and Pool, 2016). Purposeful practice encourages a person to leave their comfort zone, to risk making mistakes and to challenge themselves. Gervis and Williams (2017) have shown how purposeful practice through the 3Fs ('Focus' – on mastery-approach goals, 'Feedback' – use all available and 'Fix-it' – by solving problems and finding a solution) develops expertise in coaching. Thus, this approach can help pupils learn and develop their skills. It can also be applied to you to develop your expertise as a teacher.

Given the importance of pupils setting mastery-approach goals, the question arises as to whether teachers can help pupils in this respect. Although individuals have a propensity towards setting particular goals or combinations of goals, which is termed their achievement goal orientation profile (Niemivirta, Pulkka, Tapola and Tuominen, 2019), the motivational climate in the classroom (sometimes termed goal orientation/structure of the classroom) can influence the achievement goal adopted (Midgley, 2014). Lessons perceived to be task-oriented are linked to adaptive motivational responses, including the adoption of mastery-approach goals. Such an environment supports pupils' academic and social growth by increasing intrinsic motivation to succeed at school, self-confidence, perceived competence and self-esteem. On the other hand, lessons perceived to be outcome-oriented are linked to maladaptive motivational responses. Such lessons focus on individual achievement and competition between individuals. This may foster extrinsic motivation and discourage hard work and effort to achieve success by pupils who fail to achieve the outcome. Further, controlling environments (teachers attempt to guide pupils' thinking by providing specific guidelines for their academic and personal behaviours in class) have been found to have a negative effect on perceived competence and participation, which results in decreased intrinsic motivation.

Teachers, therefore, can most definitely influence the goals pupils set. The acronym TARGETS: the academic ***T***ask, ***A***uthority pupils have over their learning, what pupils are ***R***ecognised for, criteria by which pupils are ***G***rouped, standards and procedures of pupil ***E***valuation, how ***T***ime is used and the nature of ***S***ocial relationships and interactions (Ames, 1992b), sums up the aspects of the learning environment that the teacher needs to focus on to encourage pupils to set mastery-approach goals. Thus, teachers who set moderately challenging tasks, give pupils opportunities to take decisions about their learning, recognise effort and improvement, do not group on ability grounds, use criterion-based rather than normative standards in evaluation (see below), use time flexibility and foster supportive and warm relationships can encourage pupils to set mastery-approach goals.

Attribution theory is concerned with how pupils attribute their successes and failures in achievement situations. There is a link between achievement motivation theory and attribution theory in that pupils with a high need to achieve tend to attribute their success to internal causes (for example, aptitude and effort), while they attribute failure to lack of effort. On the other hand, pupils with a low need to achieve attribute their failure to external factors (for example, bad luck). Although their motivations change with age, it is generally accepted that pupils are more likely to try harder if they can see a link between the amount of effort they make and success in the activity, therefore,

as a teacher, you should design activities that encourage pupils to attribute success or failure to effort. However, this is not always easy. Postlethwaite (1993) identified the difficulty of determining how much effort a pupil has made on a piece of work (especially that done at home) and hence the problems of marking the work. In 'norm-referenced' marking, a certain percentage of the class is given a designated category of mark, no matter how good each individual piece of work. Thus, each pupil's mark for a piece of work is given solely for their performance compared to that of the rest of the group. This encourages success or failure to be attributed to ability or luck and can encourage the adoption of performance and avoid rather than mastery-approach goals. In 'criterion-referenced' marking, all pupils who meet stated criteria for a particular category of mark are marked in that category. Thus, pupils are given a mark that reflects how closely the criteria for the assessment have been met, irrespective of the performance of other pupils. Although this overcomes some of the disadvantages of norm-referenced marking and can foster mastery-approach goals, it does not reflect how much effort the pupil has put into the work. Further, Postlethwaite (1993) suggests that effort can best be judged by comparing different pieces of the same pupil's work, as the standard of work is likely to reflect the amount of effort put in (that is, ipsative assessment). Giving two marks for the work, one for content and standard of the work and one for effort and presentation, can encourage effort. Thus, even if the content and standard are poor, it may be possible to praise the effort. This praise can motivate the pupil to try harder, especially if the pupil values the mark for effort. He suggested that another way of encouraging pupils to attribute success to effort is to ask them to write about the way they tackled the task (see also Units 6.1 and 6.2 on assessment).

According to expectancy theory, a teacher forms an impression of a pupil on which they base their expectations of that pupil and the teacher's verbal and non-verbal behaviour is based, consciously or unconsciously, on those expectations. Consequently, the pupil recognises, consciously or unconsciously, the teacher's expectations of them from their behaviour and responds in a way that matches these expectations. Thus, there is a self-fulfilling prophecy. When teachers expect high achievement and good behaviour, pupils perform to the best of their ability and behave well. If, on the other hand, teachers have low expectations of pupils' achievement and behaviour, pupils achieve little and behave badly. In the same way, teachers can develop stereotypes of how different groups of pupils perform or behave; stereotypes can direct expectations (see also Unit 4.4). However, it is possible to break this cycle if teachers explicitly recognise this process and ensure they have high expectations of all pupils, which research suggests will result in improved pupil attainment overall (Rubie-Davies, Peterson, Sibley and Rosenthal, 2015).

Self-determination theory comprises a number of sub-theories (Ryan and Deci, 2002) contributing significantly to the current understanding of motivation to learn in a classroom setting. The theory assumes that we all have a natural tendency to learn through, firstly, being intrinsically motivated to engage in things we find interesting and challenging, and secondly, through the human tendency to internalise and assimilate (i.e. integrate) things that we may not find interesting (i.e. fuelled by extrinsic motivation) (Ryan and Deci, 2016). Much of the research examines the conditions that support or thwart these two complementary processes. There is now quite a robust evidence base that teachers can be instrumental in creating the 'right' conditions for intrinsic motivation to flourish, if they engage in what is termed 'autonomy-supportive' behaviour (Núñez and León, 2015). An autonomy-supportive environment is one in which the teacher gives increasing responsibility to pupils (for example, for choices/options about what they want to do); encourages pupils' decision-making by spending less time talking, more time listening, making less directive comments, asking

more questions and not giving pupils solutions; allows pupils to work in their own way; and offers more praise and verbal approval in class. In such contexts, intrinsic motivation is increased, pupils demonstrate high-quality learning and high levels of creativity, both of which are desirable and should be cultivated by teachers in the classroom.

In contrast to achievement goal theory, which is premised on cognition (i.e. the thought process in this case determining achievement goal selection), Ryan and Deci (2000b) indicate that three innate needs underpin motivation. The idea of underlying needs driving behaviour is not new, as was seen in the discussion of the need for achievement in achievement motivation theory. Before we delve in more detail into the needs highlighted in self-determination theory, it is worth mentioning Maslow's (1970) well-known hierarchy of needs, as this forms a useful backdrop to consider motivation in relation to learning. Maslow classified all human needs, which extend well beyond learning situations to general survival and flourishing as humans, into one of five groups and contended that these could be placed in a hierarchy in terms of importance for human development, Higher order needs, which are more important for development cannot be satisfied if lower order needs are not met. Thus, physiological needs (hunger etc., level 1) must be satisfied before safety needs (level 2) can be considered. The needs for belonging (level 3) form the next level of the pyramid, with esteem needs (level 4) above these and, finally, self-actualisation (level 5) needs sitting at the apex. Clearly, if pupils are to be motivated to learn to realise their potential (level 5 need), they cannot have lower levels needs unmet. This is the rationale behind schools providing, for example, breakfast clubs (level 1 needs) and routines to support safety (level 2). This theory, while helpful is somewhat general, whereas theories like self-determination theory are more targeted towards explaining learning and motivation.

In self-determination theory, when the three core needs of competence, autonomy and relatedness, identified by Ryan and Deci (2000b) as underpinning motivation, are met, pupils will be intrinsically motivated. While each of these may appear to be separate and distinct, they interact with each other and mediate behavioural outcomes. Therefore, the motivation of an individual is a function of these factors. In order to enhance motivation in a pupil, the teacher needs to consider how they are working in relation to each of these and ensure their practice supports these needs, i.e. to be autonomy supportive. So, for example, effective positive feedback can enhance an individual's feeling of competence and, consequently, intrinsic motivation is increased. However, this does not occur unless accompanied by an increased sense of autonomy. The role of a teacher can be instrumental in enhancing feelings of competence through their interpersonal interactions with pupils also supporting the need for relatedness.

Extrinsic motivation is also considered to be an important factor in regulating behaviour in self-determination theory and can, in certain circumstances, complement intrinsic motivation when tasks are not perceived as interesting. The key factor is the degree to which an individual has 'internalised' the extrinsic factor in terms of consciously valuing the activity and seeing it as congruent to their goals (going back to what pupils value) as this directly impacts autonomy. The greater the internalisation, the more autonomous the individual feels, stimulating engagement and learning. Specifically, it is the relationship between 'internalisation' and 'integration' that is seen to be essential in contributing to how individuals perceive their own competence and self-determination. Figure 3.2.2 relates self-determination theory to flying a kite; it is important for a teacher to focus on developing both competence and autonomy at the same time (as it is important to pull both strings on a kite to keep it in the air), while not forgetting about the need for relatedness.

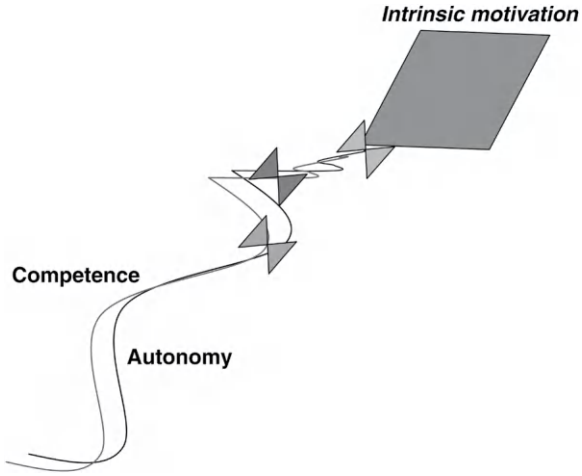


Figure 3.2.2 Developing competence and autonomy to enhance intrinsic motivation

Task 3.2.2 asks you to think about the application of these theories to your teaching.

Task 3.2.2 Using theories of motivation in your teaching

Review a range of literature on theories of motivation. Write a reflective commentary on these.

Work in a group with other beginning teachers. Each selects one theory of motivation. Use the information and the implications for teachers in Table 3.2.1 and in the text above as a basis for identifying practical implications of the theory for your lessons. Use this as the basis for planning for motivating pupils in one of your lessons based on this theory of motivation. Ask your mentor to observe the lesson and give you feedback on the effectiveness of your motivational approach. Record your own reflections. Meet with other beginning teachers and discuss the theory, your lesson and reflection, then identify ways that you can improve. Repeat the cycle of planning, teaching the lesson, reflecting and evaluating your effectiveness. Record the outcomes of your own and other beginning teachers' work. From this identify some key strategies that appeared to support pupils' motivation.

You might then like to try this using other theories of motivation to develop your toolkit of useful strategies.

Influencing the motivational climate in the classroom

Having considered some of the key theories of motivation and their implications, it is time to focus in more detail on some of teaching practices that are important for facilitating a positive motivational climate.

Strategies to facilitate personal achievement (success)

Personal achievement (generally called 'success' in an outcome-oriented learning environment) is generally motivating in itself, as mastery is a key source of self-efficacy (Bandura, 1997), as

is outlined in the achievement motivation and expectancy-value theories of motivation. Some pupils struggle to succeed, whereas others succeed much more quickly. There are many ways to help pupils succeed, such as using whole-part-whole teaching. In this, pupils are shown the whole activity first so that they know what they are trying to achieve. The activity is then broken down into small, self-contained, achievable parts, which allows pupils to receive reinforcement for each small, successful step. The separate parts of the activity are gradually put together until the whole activity has been built up. Pupils are given appropriate feedback at each stage (see giving feedback below); therefore, when they attempt the whole, they are more likely to succeed. You may relate to this by thinking about when you learned (or tried to learn) front crawl in swimming. You probably practised your arms, legs and breathing separately before you tried to put it all together (see Task 3.2.3). What other techniques can you use to help pupils succeed? (See also Unit 2.2 on lesson planning.)



Task 3.2.3 Whole-part-whole teaching

As part of your normal lesson planning with a class, select one activity you can break down into small, self-contained, achievable parts, which can be put together to build up gradually to the whole. Ask your mentor or another beginning teacher to observe you teaching this activity using whole-part-whole teaching. At the end of the lesson, discuss with some of the pupils how this went. Discuss with the observer how the pupils responded and how well they learned the task. Evaluate the lesson yourself and build on this experience in future lessons.

Self-assessment and goal setting can help pupils identify their personal successes, and form an important part of the self-regulation process that is linked to motivation (Zimmerman, Bandura and Martinez-Pons, 1992). Performance profiling techniques can be a valuable tool to aid this process. Performance profiling is based on Kelly's (1955) personal construct theory, which maintains that each individual holds unique cognitive constructs about the world and the self. It is through these that we all construct our own reality, and, consequently, we are the best experts on ourselves. As we develop, these constructs are changed and modified through our experiences and knowledge and understanding of the world. This theory suggests that, in order to understand another, first we have to be able to see the world from their perspective. Thus, for the teacher to enhance motivation, they need an understanding of their pupils' perspectives.

The process entails first asking pupils to identify key factors that they think are essential to be successful learners; these could be subject-specific or more generic. Through class discussion, you can agree on the final list. This can be as long or short as you want.

Once identified, these key factors can be added to the profile chart (see Figure 3.2.3). Each pupil can then complete their own profile by identifying their perceived strengths and weaknesses in relation to each of the key factors. The teacher can also complete their own chart for each pupil.

This creates a means to compare the pupil perspective with that of the teacher and can be a starting point for discussion. Through discussion, pupil and teacher can gain better insight and understanding into how the pupil sees themself. So, for example, if a pupil has rated themself as a '4' on one factor and the teacher rated them as a '7', this provides a useful framework for mutual

	1	2	3	4	5	6	7	8	9	10
Concentration										
Reading										
Spelling										
Writing skills										
Listening										

Figure 3.2.3 Example performance profile (1 = very poor, 10 = excellent)

understanding. It also provides a clear visual representation of the pupil's strengths and weaknesses, which makes it tangible (see the companion website, for example, Table 3.2.2).

Once a profile has been created, it can become an effective monitoring system and a focal point for goal setting. By agreeing the actions that need to be taken to address the profile and set goals, the pupil is able to take ownership of the process. In accordance with self-determination theory, we know that, by increasing autonomy, intrinsic motivation is increased. This process creates a platform from which this can develop. Furthermore, we know that changes in behaviour can only be initiated with the complete agreement of the individual, and this process ensures that the pupil is able to 'buy into' their own action plan to achieve success. Progress can be monitored through repeated profiles, which can provide a direct comparison to previous profiles, and serve to visibly map change and achievement towards the agreed goals.

Rewards and praise

Although personal achievement is motivating, pupils may not be immediately successful and might therefore need some form of extrinsic motivator to engage and persist. Behavioural learning theory (Skinner, 1953) suggests that activities or behaviour that result in a pleasurable outcome are more likely to be repeated, thus a positive reinforcement through reward can be used to shape behaviour and is likely to increase pupils' motivation to learn. Rewards can take different forms including social rewards, such as praise and comments of encouragement and approval, and more tangible rewards such as tokens (house points, certificates) and material rewards (objects such as vouchers). As praise is a social reward, it is apparent why pupils respond more positively to praise and positive comments about their work or behaviour, than to criticism and negative comments.

Regular use of praise may produce a motivational learning environment in which pupils work harder and behave better (Cameron and Pierce, 1994). If pupils misbehave in a classroom in which there is a positive motivational learning environment, Olweus (1993, p. 85) suggested that the use of praise makes pupils feel appreciated, which may make it easier for them to accept criticism of inappropriate behaviour and to attempt to change. However, research has regularly shown that teachers make insufficient use of praise (Gable, Hester, Rock and Hughes, 2009). Praise is given more often for academic than social behaviour, and social behaviour is more likely to be criticised than praised. One reason for this may be that teachers expect pupils to behave appropriately in the classroom.

Some teachers use very few different words to praise pupils; for example, 'good', 'well done', 'OK'. What other words can you use to praise someone or give feedback? Try to develop a list of such words because if you use the same word to praise pupils all the time, the word loses its effect. The range of words must be accompanied by appropriate non-verbal communication signals (see non-verbal communication in Unit 3.1 and Task 3.2.4).



Task 3.2.4 The language of praise

Using the observation schedule found on the companion website (insert link to Table 3.2.3)

Observe a class taught by an experienced teacher. Sit in a place where you can hear everything that is said. Record the number of times the teacher gives praise and makes negative comments to individuals, groups and the whole class in relation to academic work, effort, perseverance and behaviour. Observe the same experienced teacher in another lesson. This time write down the different words, phrases and actions the teacher uses to give praise and negative comments in each of these categories and the number of times each is used.

Ask someone to conduct the same observations on your lessons. Discuss the differences with your mentor and, if appropriate, develop strategies to help you improve the amount of praise you give and the range of words, phrases and actions you use to give praise (you might be surprised to find that you use a phrase such as 'good' or 'OK' very frequently in your teaching). Gradually try to incorporate these strategies into your teaching.

Although it is generally accepted that praise aids learning, there are times when it may not be appropriate. For example, pupils who become lazy about their work as a result of complacency may respond by working harder if their work is gently criticised on occasion. If praise is given automatically, regardless of the work, effort or behaviour, pupils quickly see through it and it loses its effect. Praise should be used discriminatively to reward relevant and appropriate achievement, effort or behaviour.

It is important to bear in mind that some pupils do not respond positively to praise; for example, they may be embarrassed, especially if they are praised in front of their peers. Others perceive praise to be a form of punishment; for example, if they are teased or rejected by their peers for being 'teacher's pet' for behaving themselves in class. Thus, conforming to the behaviours and values promoted in school results in negative social consequences. Therefore, although pupils know that they will be rewarded for achievement, effort or behaviour, they may also be aware of the norms of the peer group, which discourage them from achieving academically, making an effort or behaving well. The implication of this is that teachers need to bear in mind that pupils are motivated not only academically but socially with the latter underlying some of what is termed self-handicapping behaviour that might be seen in the classroom (Yu and McLellan, 2019).

Other pupils do not know how to respond to praise because they have not received much praise in the past; for example, because they have continually received low marks for their work. They may have developed a helpless reaction to failure, an attributional style where failure is attributed to lack of ability which is seen as out of their control therefore there is no point in trying (Dweck, 2000). Some of these pupils may want to attribute failure to not caring or not trying to succeed to

protect their sense of self-worth. This may be reflected by them not making an effort with work, or by misbehaving in the classroom.

Thus, pupils respond differently to praise. In the same class, some pupils will work hard to receive praise from the teacher or a good mark on their homework, while others do not respond to praise or work hard to avoid praise. Use your judgement when giving praise; for example, if you praise a pupil who is misbehaving to try to encourage better behaviour, you may be seen to be rewarding bad behaviour, thereby motivating the pupil to continue to misbehave in order to get attention. If you are not immediately successful in your use of praise, do not give up using it, but consider whether you are giving it in the right way; for example, would it be better to have a quiet word, rather than praise pupils out loud in front of their peers? As your professional knowledge and judgement develop, you become able to determine how best to use praise appropriately to motivate pupils in your classes.

Sanctions

As well as using praise, teachers also use sanctions to try to change behaviour. Where rewards are generally considered to be more effective as the increase appropriate behaviour, sanctions aim to deter inappropriate behaviour. If pupils are sanctioned, they know what behaviour results in sanctions and, therefore, what not to do, but may not know what behaviour avoids sanctions.

However, there are times when sanctions are needed. At such times, make sure that you use sanctions to best effect; for example, avoid sanctioning a whole class for the behaviour of one or a few pupil(s), always make it clear which pupil(s) are being sanctioned for what behaviour and always give sanctions fairly, consistently and in proportion to the offence. Make sure the sanction does not include the behaviour you want exhibited (for example, do not sanction a pupil by requiring them to run around the football pitch if that is what you had wanted them to do). This sends mixed messages and is likely to put the pupil off that activity, and always ensure the sanctions is carried out.

In order to increase appropriate behaviour, identify to the offender any positive aspects of the behaviour being punished and explain the appropriate behaviour (see also Units 3.3 and 4.5).

Feedback

Feedback is a formative process that gives pupils information about how they are doing and whether they are on the right track when learning something. This motivates them to make an effort and to continue. A pupil is more likely to learn effectively or behave appropriately if feedback is used in conjunction with praise. A sequence in which feedback is sandwiched between praise (that is, praise-constructive feedback-praise) is designed to provide encouragement and motivation, along with information to help the pupil improve the activity or behaviour. Giving praise first is designed to make pupils more receptive to the information and, afterwards, to have a positive approach to try again.

Try combining feedback with praise in your teaching. Be careful how you use the word 'but' when giving feedback, especially when using the sandwich approach. The power of this small word has the effect of devaluing everything that comes before it. So, for example, if you said, 'Well done for your concentration in class today *but* your writing needs attention', the pupil will not acknowledge the praise; it is as if you have not said it. If, however, you replace 'but' with 'and', it changes the message and the emotional response to it. If you said, 'Well done for your concentration in class today *and* your writing needs attention', the praise element will still be acknowledged.

By observing pupils very carefully, you are able to spot small changes or improvements, which allows you to provide appropriate feedback (Unit 2.1 looks at observation techniques and Unit 6.1 focuses on assessment for learning). Feedback can be used effectively with the whole-part-whole teaching method (see above). If you give feedback about how a pupil has done on each part, this part can be improved before going on to the next part. If you give feedback immediately (that is, as an attempt is being finished or immediately after it has finished, but before another attempt is started), pupils can relate the feedback directly to the outcome of the activity. Thus, pupils are more likely to succeed if they take small steps and receive immediate feedback on each step. This success can, in turn, lead to increased motivation to continue the activity.

One problem with giving immediate feedback is how you can provide feedback to individual pupils in a class who are all doing the same activity at the same time. There are several ways to achieve this; for example, getting pupils to work through examples in a book that has the answers in the back, or setting criteria and letting pupils evaluate themselves against these, or having pupils assess one another against set criteria (Unit 5.3 covers teaching strategies and styles, including the reciprocal teaching style of Mosston and Ashworth, 2002). If they have been properly prepared for it, pupils are generally sensible and constructive when given responsibility for giving feedback. In Task 3.2.5, the focus is on pupils giving feedback to each other.



Task 3.2.5 Pupils giving feedback

As an integral part of your lesson planning, select one activity in which pupils can observe each other and provide feedback. Devise a handout with the main points/criteria to be observed. Plan how you are going to introduce this activity into the lesson. Discuss the lesson plan with your mentor. Ask your mentor to observe the lesson. Discuss the effectiveness of the strategy afterwards, determining how you can improve its use. Also, try to observe teachers who plan for pupils to observe and give feedback to each other. Try the strategy at a later date in your school experience. Identify other ways in which you can get more feedback to more pupils when they are doing an activity. Include these in your lesson plans, as appropriate (see also teaching styles in Unit 5.3).

Feedback does not just come from a person (for example, the teacher or another pupil); feedback also comes from the activity itself. Activity-based feedback may be easier to identify for some activities than others; for example, a pupil gets feedback about their success if an answer to a mathematics problem matches that given in the book or the wicket is knocked down when bowling in cricket. In other activities, right or wrong, success or failure, is not as clear-cut; for example, there is often no right or wrong answer to an English essay. In the early stages of learning an activity, pupils find it hard to use feedback from the activity; for example, they may notice that they were successful at the activity, but not be able to identify why. Normally, therefore, they need feedback from another person. This immediate, external feedback can be used to help pupils become more aware of what they are doing, how they are improving, why they were successful or not at the activity and therefore to make use of feedback from the activity. However, it is important to bear in mind that teacher feedback will also influence pupils' motivational beliefs feeding back into their general motivation

for learning (Katsantonis, McLellan and Torres, 2023). Later in the learning (for example, when refining an activity), pupils should be able to benefit from feedback from the activity itself, and, therefore, it is better to encourage this internal feedback by, for example, asking appropriate questions, such as, 'How did that feel?' In this situation, the teacher should not give immediate feedback.

Finally, to be effective, feedback should be given about pupils' work or behaviour, not about the pupils themselves. It must convey to pupils that their work or behaviour is satisfactory or not, not that they are good (or bad), per se. Feedback and motivational loop it creates is central to self-regulated learning (Zimmerman and Moylan, 2009) that in turn enables pupils to achieve.

Motivating individuals and the class as a whole

As the discussion above has highlighted, there is no one correct way to motivate pupils to learn. Different motivation techniques are appropriate and effective in different situations; for example, pupils of different ages respond differently to different types of motivation, reward, punishment or feedback. Likewise, individual pupils respond differently. Further, any one pupil may respond to the same motivator differently at different times and in different situations.

Pupils need to feel that they are individuals, with their needs and interests taken into account, rather than just being a member of a group. Pupils need to be given opportunities to take ownership of the tasks in which they are engaged. If pupils are not motivated or bored, do not let them avoid doing an activity, but try to find ways of motivating them, such as by relating it to something in which they are interested. You can motivate pupils most effectively by using motivation techniques appropriate for a particular pupil in a particular situation.

Therefore, you need to get to know pupils as individuals. Learning pupils' names quickly gives you a start in being able to motivate pupils effectively (Unit 2.2 provides advice on learning pupils' names). Observation of pupils, talking to them and discussing a pupil with the form tutor or other teachers all help. As you get to know your pupils well, you can identify what motivates them by finding out what activities they enjoy, what they choose to do and what they try to avoid and what types of reward they work for and to what they do not respond, as well as their needs and interests.

The sooner you can relate to pupils individually, the sooner you can manage a class of individuals effectively. As a beginning teacher, you are at a disadvantage here because you do not usually spend enough time in one school to get to know the pupils well, and, therefore, you can only try to motivate individual pupils by using your knowledge and understanding of pupils of that age.

However, there is one further element to motivation to consider briefly here. The same principles can also be applied to enhance the collective motivation of a class. By engaging pupils in discussions about 'what kind of a class' they want to be, you can encourage pupils to elicit qualities that they think are necessary to enable all members of the class to succeed; for example, everyone to be on time and with the right materials, to respect others (and hence to listen quietly when someone else is talking), to work hard, to do their homework and so on. Thus, in essence, they create a class profile (similar to that in Figure 3.2.3 created for an individual pupil) that enables the collective perspective to be represented promoting a sense of relatedness (we are a team). Giving pupils ownership of the profile they create contributes significantly towards collective motivation and behaviour modification in line with the desired outcome. Consequently, if there is collective buy-in, pupils monitor each other's behaviour and work to reinforce the key aspects identified on the profile. The benefit for you is that the pupils can help you with class management.

SUMMARY AND KEY POINTS

This unit has identified some theoretical underpinnings, general principles and techniques for achieving an appropriate motivational climate in your lessons to increase pupils' motivation to learn. However, you need to be able to use these appropriately. For example, if you praise a group for working quietly while they are working, you may negatively affect their work. It is better in this situation to let the group finish their work and then praise them.

- In addition, pupils are individuals and therefore respond differently to different forms of motivation, reward, punishment and feedback. Further, the same pupil responds differently at different times and in different situations.
- To motivate each pupil effectively, therefore, requires that you know your pupils so that you can anticipate how they will respond.
- Motivation is supported by good formative assessment techniques (see Unit 6.1). Your developing professional knowledge and judgement enable you to combine theory with practice to motivate pupils effectively in your classes, which raises the standard of their work.

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Further resources

Kyriacou, C. (2018) *Essential Teaching Skills* (5th edn), Oxford: Oxford University Press.

This book contains chapters on lesson management and classroom climate, both of which consider aspects of motivation, such as whether lesson management helps to maintain pupils' motivation and whether the opportunities for learning are challenging and offer realistic opportunities for success.

Reeve, J., Ryan, R., Cheon, S. H., Matos, L., and Kaplan, H. (2022) *Supporting Students' Motivation*, London: Routledge.

Drawing heavily on self-determination theory, this highly practical book focuses on how teachers can shape the motivational climate in their classrooms through adopting an autonomy-supportive style. There are a number of chapters devoted to specific strategies teachers can implement and in each case a step-by-step guide is provided, which expand on the ideas in this chapter.

Schunk, D H., Meece, J.L. and Pintrich, P.R. (2014) *Motivation in Education: Theory, Research, and Applications* (4th edn), Harlow: Pearson Education Limited.

This is a comprehensive guide to theories of motivation in educational contexts. Following an introductory chapter providing background and historical context, the remaining chapters each focus on a specific theory or approach to motivation. These chapters provide a detailed account of the theories we have introduced and provide examples of how they can be applied to the classroom context.

Wentzel, K.R. and Miele, D.B. (eds.) (2016) *Handbook of Motivation at School* (2nd edn), New York, NY: Routledge.

This edited collection covers the major theoretical perspectives so is a good alternative to the Schunk et al. (2014) book. In addition, there are a number of chapters on contextual and social influences such as teacher-pupil relationships, culture and identity and teacher motivation, which will extend your understanding and suggest a range of approaches for facilitating a supportive motivational climate in the classroom. There is also a section on new directions including the role of mindfulness and contributions from neuroscience.

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3.3 Managing classroom behaviour

Adopting a positive approach

Philip Garner

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Introduction

This unit is designed to enable you to enhance your knowledge and skills in classroom management, and especially to support the development of positive approaches for dealing with pupil behaviour. It takes account of some of the most recent shifts in thinking and policy in this important dimension of your initial teacher education (ITE) programme. This unit should be read in conjunction with emerging new guidance resulting from a national consultation on revised guidance on behaviour in schools (DfE, 2024). This emphasises that 'All staff should be aware of the measures outlined in the school's behaviour policy and how they should implement these measures' (p. 7).

There has been a noticeable move away from reactive approaches to dealing with unwanted behaviour over the last 20 or more years. Such approaches were characterised by a preoccupation with 'discipline' being something that the teacher imposes on pupils (Robertson, 1996). Instead, a greater awareness of 'behaviour for learning' has become apparent, which is consistent with an accompanying quest to develop inclusive schooling for all pupils (Lauchlan and Greig, 2015). Even so, it should be clearly understood that such an approach to pupil behaviour does not diminish the importance of clear and explicit classroom rules to govern pupil behaviour – and their consistent application. You need to ensure that pupils are under no illusions that, as class teacher, you are in control; it is, after all, what they expect of you!

A 'behaviour for learning' approach emphasises the teacher's role in creating an appropriate climate in which all pupils can learn effectively. It encourages you to link pupil behaviour with their learning via three interlinked relationships: how pupils think about themselves (their feeling of self-regard as a learner); how they view their relationships with others (both teachers and fellow pupils); and how they view the curriculum (their relationship with the learning they are undertaking). Recognition of the interplay between these is seen as the basis of a positive approach to behaviour

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and emphasises the role of the pupils themselves in learning to manage their own behaviour (Ellis and Tod, 2018).

Understanding this way of working is an important aspect of a teacher's role in managing pupil behaviour. The most recent advice continues to balance this with a requirement that all teachers – and especially beginning teachers – understand some basic guidelines regarding the way in which unacceptable behaviour by pupils in a classroom can be minimised. Such behaviour has been a regular cause of concern on the part of teachers, educational professionals, parents and politicians for many decades; some aspects of these are dealt with in the section dealing with 'whole-school behaviour policy'.

Approaches to 'behaviour' in ITE programmes reflect the shift in emphasis in the way that the issue is being tackled. The focus is now on a middle ground between the earlier preoccupation with 'control' and 'discipline' and a more recent holistic approach that confirms the link between behaviour and pupil achievement. This is neatly summarised by the EEF (2021).

This unit concentrates on the principles underpinning the links between pupil behaviour and the taught curriculum, as well as some of the important practical dimensions involved in effectively promoting a positive ethos in the classroom in which all pupils can have the best chance to succeed. Wherever you are learning to teach, you need to ask your mentor about the current statutory regulations that are being applied. You will be made aware that, with an increasing emphasis on the inclusion of a wider diversity of pupils in mainstream schools, there will be those whose behaviour requires more specific intervention. Information about these can be found elsewhere (for example, Allen, 2021).

OBJECTIVES

At the end of this unit you should be able to:

- recognise the importance of the 'whole-school behaviour policy';
- interrogate a definition of the term 'unacceptable behaviour' and understand the significance of its underlying causes;
- recognise the importance of the links between pupil behaviour and their curriculum learning;
- understand the classroom implications of the current guidance to teachers regarding pupil behaviour;
- develop positive approaches to unacceptable behaviour that are based on relationships with pupils.

Check the requirements for your ITE programme to see which relate to this unit.

Whole-school behaviour policy

An important feature of a school's promotion of a 'positive climate', as well as its response to unacceptable behaviour, is the 'whole-school behaviour policy' (DfE, 2024). Moreover, everyone has a role to play in creating an appropriate behaviour culture in the school: 'The consistent and fair

Table 3.3.1 Content of a whole-school behaviour policy

1. a consistent approach to behaviour management;
2. strong school leadership;
3. classroom management;
4. rewards and sanctions;
5. behaviour strategies and the teaching of good behaviour;
6. staff development and support;
7. pupil support systems;
8. liaison with parents and other agencies;
9. managing pupil transition;
10. organisation and facilities.

implementation of the measures outlined in the behaviour policy is central to an effective whole-school approach to behaviour' (DfE, 2024, p. 10). As a matter of course, you should therefore make sure that you familiarise yourself with the content of the whole-school behaviour policy in your placement school. Clear guidelines regarding its content are now widely understood (see Table 3.3.1); however, you should also recognise that individual schools vary according to the behaviour routines and rules they put in place.

Now complete Task 3.3.1, which asks you to consider behaviour policies in your school.



Task 3.3.1 School policy on pupil behaviour

Familiarise yourself with your placement school's whole-school policy on behaviour and attendance. Discuss it with another beginning teacher who is placed in a different school. Consider both the similarities and differences in the two policies. What are the implications of the document for you as a beginning teacher, particularly in respect of classroom management?

A whole-school behaviour policy provides some of the basic building blocks that you use to help establish a positive classroom 'climate'; this term refers to the character or 'feel' of your classroom, as experienced by all those who come in contact with it – teachers, pupils and any classroom visitors. A major influence on this is the class teacher's own repertoire of knowledge, skills and understanding about pupil behaviour and classroom management. To assist you in acquiring these attributes, you will undoubtedly be supported by your mentor and experienced teachers and other professionals who provide practical support in positively managing behaviour.

Behaviour policies aim, among other things, to help teachers to promote positive behaviour and to support them in tackling issues of low-level unacceptable behaviour. They also point to the importance of providing creative and positive learning environments for all pupils whilst providing a framework that you can use to develop a set of rules and routines based on the development of a positive relationship between yourself and the pupils. This is the first step in helping to insulate pupils from those factors (discussed later in this unit), which might cause them to behave inappropriately and, in consequence, fail to thrive as learners.

What is unacceptable behaviour?

National advice and guidance on behaviour management in England has emphasised the development of appropriate, positive behaviour that brings significant benefits for all pupils. It is noted that 'Staff, particularly in secondary schools, emphasised the value of teaching desired behaviours and making them routine' (HMCI, 2019). Current guidance invites teachers to be clear about what behaviour they want pupils to engage in, and to model this as part of their teaching.

However, the term 'behaviour' has traditionally been taken to mean unacceptable behaviour. The Elton Report (Department of Education and Science (DES), 1989) refers to *misbehaviour* as 'behaviour which causes concern to teachers'. The term is one that can variously be replaced by a range of other expressions that teachers use to describe unwanted, unacceptable behaviour by pupils. Thus, disruptive, challenging, antisocial, off-task, unwanted 'acting out' and withdrawn behaviours are all terms that are widely used, according to the personal orientation of the teacher concerned and to the type of problem behaviour being described.

The term 'unacceptable behaviour' is well-established, and is used as a catch-all expression for pupil behaviours that span a continuum (DfE, 1994). The continuum ranges from low-level unacceptable behaviour at one end (such as talking out of turn, distracting others, occasionally arriving late in class) to more serious, sometimes acting out behaviour, at the other (such as non-attendance, verbal or physical aggression, wilful disobedience and bullying, including cyberbullying).

One of the major difficulties in defining what inappropriate behaviour constitutes is that it varies according to the perception, tolerance threshold, experience and management approach of individual teachers. It can also vary from one school to the next. As a result, what might be an unacceptable behaviour in your own classroom may be viewed in another context, or by another beginning teacher, as quite normal. Alternatively, what you accept as normal may be seen as unacceptable in another context or by another beginning teacher. This leads to confusion in the minds of pupils, and to potential tension between individual teachers in a school or between a beginning teacher and mentor or other experienced teacher. So, it is important to recognise that: (a) pupil behaviour should be described explicitly in terms of observable actions; and (b) responses to it need to take full account of a school's policy concerning behaviour and apply it with consistency. When you describe a pupil behaviour, you should always ensure that your definition is of the behaviour itself and not a description of the pupil as a whole. This avoids any likelihood of the pupil being labelled as a disruptive pupil or a problem pupil.

The Qualifications and Curriculum Authority (QCA, 2001) usefully identified 15 behaviours by which a pupil's emotional and behavioural development might be defined and assessed. These were divided into learning behaviours, conduct behaviours and emotional behaviours. Each group is subdivided into sets of criteria, depicting desirable and undesirable behaviours (see companion website – www. – for further details).

Task 3.3.2 helps you to develop a definition of unacceptable behaviour.



Task 3.3.2 What is unacceptable behaviour?

It is important that you arrive at a personal definition of what comprises unacceptable behaviour. Divide a blank sheet of paper into three. Head the left-hand section 'Totally unacceptable'; the right-hand section 'Acceptable'; and the middle one 'Acceptable in certain circumstances'.

Now examine your own classroom teaching, and complete each section. Remember, behaviour is as much about positive learning behaviour as it is those pupil actions that you regard as unacceptable or challenging. Reflect on your responses and discuss with your mentor. Should the opportunity arise, you might wish to undertake this exercise with your pupils, to gather their thoughts. Comparing your list to theirs is likely to prove very revealing! You may also wish to discuss your responses with another beginning teacher.

Scoping the causal factors

As Ayers and Prytys (2002, p. 38) noted, 'The way in which behaviour is conceptualised will determine the treatment of emotional and behavioural problems'. There are a number of causal factors that assist in explaining unwanted behaviour, disaffection and disengagement among some pupils; these are often multivariate and overlapping. The attribution of a cause can frequently result in the acquisition of a negative label by the pupil. But, on the other hand, understanding and recognising the causes can give you clues as to what might be successful interventions. A brief outline of causal factors is given in the following section. There is more exhaustive coverage in a variety of other sources (for example, Garner et al., 2014).

Factors that may cause unacceptable behaviour

Unacceptable behaviour may be caused by several of the factors identified in Table 3.3.2.

Table 3.3.2 Some causes of unacceptable behaviour

Individual factors
■ A pupil believes that the work is not within their grasp and, as a result, feels embarrassed and alienated and lacks self-esteem as a learner. ■ A pupil may well experience learning difficulties. ■ A pupil may have mental health, stress and possible drug misuse issues, all of which are important factors explaining underachievement and inappropriate behaviour in adolescence.
Cultural factors
■ Adolescence can be a period of rebellion or resistance for many young people. ■ Possible tension between societal expectation and the beliefs and opinions of the pupil. ■ Group/peer pressure can result in various forms of alienation to school. ■ Negative experience of schooling by parents, siblings or other family members.
Curriculum relevance factors (linked to both individual and cultural)
■ The curriculum may be seen by a pupil to be inaccessible and irrelevant. ■ The school may give academic excellence more value than vocational qualifications or curriculum options.

(*Continued*)

Table 3.3.2 (Continued)

School ethos and relationships factors

- Some schools can be 'deviance provocative'; their organisational structures and procedures are viewed by pupils as oppressive and negative.
- Some schools are less inclusive, both academically and socially, to pupils who behave 'differently'.

External barriers to participation and learning factors

- Family breakdown or illness usually impacts negatively on a pupil's mental health, and often on their sense of priority.
- Poverty and hardship can mean that a pupil's physiological needs are not met – such pupils may be tired, hungry and consequently easily distracted (see also Maslow, 1970 in Unit 3.2).
- Sibling and caring responsibilities may mean that some pupils arrive late in your lesson – or not at all.

One aspect of causality that needs further consideration is that the influences on pupil behaviour have been interpreted as forming three interlinked relationships, first identified in what Bronfenbrenner (1979) called the *ecosystemic theory of relationships*. In the case of a pupil who is consistently behaving inappropriately, it is suggested that there has been a breakdown in one (or more) of these three relationships:

- pupils' relationship with themselves (how pupils feel about themselves, their self-confidence as learners and their self-esteem);
- pupils' relationship with others (how they interact socially and academically with all others in their class and school);
- pupils' relationship with the learning they are undertaking (the curriculum) (how accessible they feel a lesson is and how best they think they learn).

This inter-relationship was subsequently adapted in strategies to address pupil behaviour, in what was termed 'behaviour for learning' (Ellis and Tod, 2009; EEF, 2021). This approach argues that all three 'relationships' need to be taken into account when planning your strategy to tackle unacceptable behaviour. The emphasis upon positive relationships is an integral component of the approach, with individual pupils as well as whole classes. This effort needs to begin from your first encounter with a group of pupils; over time, it will enable you to establish a classroom climate in which pupil learning can flourish.

Task 3.3.3 links causes to possible teaching strategies.



Task 3.3.3 Linking causes to possible teaching strategies

Consider one pupil that you are teaching who sometimes exhibits behaviour(s) unacceptable to you. Write a brief description of each of the behaviours, making sure that your language is clear and describes clearly observable pupil actions. Taking each behaviour in turn and referring to the causal factors identified earlier, assess which factors you feel might underlie

that particular behaviour. Consider how amenable to change each of the causal factors you have identified is. Also, identify any other teachers (for example, the special educational needs co-ordinator, SENCO) who might be able to provide you with advice and support. Finally, reflect on how your interpretation of cause might inform the way in which you choose to address the behaviour(s) shown.

Make notes and discuss your responses with your mentor.

Key principles of a behaviour for learning approach

There has been an increasing interest among educators in the 'social and emotional aspect of learning' (SEAL) over the last 15 or more years. This approach incorporates a long-standing awareness of aspects of 'emotional intelligence' (Goleman, 1995), which can assist teachers in helping to create a positive climate or culture for learning in which good relationships between them and their pupils are paramount (see also Unit 3.2 on motivation). It is based, in part, on the premise that 'generally a punitive approach tends to worsen or sometimes even create the very problems it is intended to eradicate … punishment alienates children from their teachers and does nothing to build up trust that is the bedrock of relationships' (Weare, 2004, p. 63). Crucially, and linking SEAL to the notion of 'behaviour for learning', an understanding of 'self' is a cornerstone of pupil motivation. As a result, pupils can:

- be effective and successful learners;
- make and sustain friendships;
- deal with and resolve conflict effectively and fairly;
- solve problems with others or by themselves;
- manage strong feelings such as frustration, anger and anxiety;
- recover from setbacks and persist in the face of difficulties;
- work and play co-operatively;
- compete fairly and win and lose with dignity and respect for competitors.

Although each of these represents a complex undertaking, your use of them as guidelines in developing your thinking about pupil behaviour will pay rich dividends as you progress in your teaching career.

A behaviour for learning approach accepts that most SEAL are learned and, therefore, can be taught or modelled by the teacher. Evidence strongly suggests that the most successful strategies for developing a positive learning environment are those that incorporate the promotion of positive relationships. As has been suggested, each of the three (interlinked) relationships is important in developing a positive learning environment in the classroom, and, as a teacher, you are at the very heart of orchestrating them. Although some pupils have relatively advanced 'social and emotional' skills when they arrive at school, others (often pupils who can sometimes behave unacceptably) might need support and the direct teaching of the specific skills they have not yet learned. So, your task is to focus on helping to develop appropriate skills that enable each pupil to learn within a variety of learning contexts. This can be in whole-class or small-group situations in the classroom

and elsewhere in the school. Some of the basic principles which inform the way in which this can be done are:

- behaviour for learning is a positive description. It tells pupils what you want them to do and why this helps them to learn, rather than focusing on behaviours that you do not want in your classroom;
- it requires that you place value on (and praise appropriately) pupil behaviour that enables and maximises learning;
- effective behaviour for learning strategies can range from high-level listening and collaborative learning skills to remaining seated for two minutes. The emphasis is upon setting targets that are reachable by pupils.

Finally, the importance of recognising the 'social and emotional' dimensions of pupil learning was highlighted during the health crisis caused by Covid-19, resulting in schools becoming more attuned to the impact on pupil mental health as a result and the possible connection with some unacceptable pupil behaviour and have built this into their overall management approach.

Getting the simple things right!

So far in this unit, the focus has been placed upon the underlying principles of creating a positive learning environment. But your understanding of these must coincide with a parallel focus on the strategies that you can implement practically to establish yourself as a teacher who can manage pupil behaviour effectively.

It is important that, whilst you recognise that managing a class is complex, you should concentrate on getting the simple, but essential, things right (DfE, 2011) (see also Bennett, 2016; DfE, 2019). Bennett (2017) emphasises the importance of planning for each individual pupil especially when setting behaviour goals, rather than just reacting to misbehaviour when it occurs. As a beginning teacher, you need to fit in with the practice in the classrooms you work in. However, to assist you in avoiding some of the more obvious pitfalls and to aid planning, complete Task 3.3.4.



Task 3.3.4 Relationships and pupil behaviour

As noted above (EEF, 2021), promoting positive behaviour requires the teacher to understand three sets of relationships, with *self*, *others* and with the *curriculum*. Further, that you need to take some practical steps in order to make these relationships happen. Examine the DfE (2011) behaviour checklist (https://dera.ioe.ac.uk/id/eprint/25115/1/charlie_taylor_checklist.pdf) and allocate each of the items in it to one or more of these relationship areas. Reflect on their distribution, compare and discuss your findings with another beginning teacher.

Incorporating both a set of clear principles and some straightforward actions enables you to prevent the occurrence of unacceptable behaviour by pupils. It also assists you in giving a firm and consistent response to any instances of problem behaviour that might arise. As a beginning teacher, keep things simple and avoid overuse of sanctions by giving pupils ways of getting out of a situation

leading to sanctions. This also enables you to present yourself in a classroom leadership role. Most pupils will come to your classroom wanting to learn, although there will be times when some will either be unable or unwilling to learn on account of some of the factors described earlier in this unit. In dealing with this situation, it is important to develop certain classroom leadership skills that contribute to your being able to establish a well-organised environment for learning, forge positive relationships with all pupils and establish a classroom ethos that allows pupils to demonstrate positive behaviour and optimum attainment.

A classroom leader needs to address three broad elements that help to define the ethos of the classroom. Whilst these issues are important for all pupils, they are essential elements for promoting the engagement and positive behaviour of pupils who are at risk of misbehaviour. They are:

- motivation: you need to provide time at the start of each lesson to tell pupils what they are learning and why. Pupils need to be involved at every stage in assessing whether these learning intentions have been met (Unit 3.2 looks at motivation in more depth);
- emotional wellbeing: to help reduce pupil anxiety, you should share the lesson structure with pupils at the start, so they know what is going to happen during the lesson;
- expectations: you need to give time at the start of the lesson and before each new activity to make clear what behaviours are needed for this piece of learning to be successful.

These three underpinning elements are embedded in more specific teacher actions that allow you to demonstrate your role as the classroom leader to your pupils. These include:

- good communication between yourself and your pupils (see Unit 3.1);
- secure subject knowledge;
- providing lively, well-paced lessons;
- understanding and meeting the learning needs of all pupils in your class;
- acting on your reflections and evaluations of previous lessons (feedback loop) (see also Unit 2.1);
- demonstrating confidence and direction in managing pupils;
- modelling desired behaviours yourself.

It is unlikely that, as a beginning teacher, everything clicks into place straight away; some actions develop with experience.

Building positive relationships in classrooms

As has been discussed earlier in this unit, promoting positive behaviour places emphasis on the relationships you form with your pupils. Ineffective interventions are usually the product of unsatisfactory relationships with individual pupils. These interventions, even though they are ultimately unsuccessful, take up valuable teaching time and impact negatively on the learning of an individual pupil, the rest of the class and also on your own confidence. Most interventions should take the form of positive actions that fit somewhere on a continuum from *positive reinforcement* through to *positive correction*. The actions you select should be those that enable learning to continue. They usually include eye contact, use of pupil name, description of the appropriate behaviour

you would like the pupil to demonstrate, praise and affirmation (see also Units 3.1 and 3.2). For example:

- modelling appropriate behaviour;
- positive reinforcement and the appropriate use of targeted praise;
- consistent and firm application of rules;
- use of verbal and non-verbal communication;
- listening to pupils and respecting their opinions;
- remaining vigilant (pre-empting unacceptable behaviour);
- dealing decisively with lateness and non-attendance.

By consistently using these approaches in your teaching, you are more likely to forge meaningful and positive relationships with your pupils. In sum, effective relationships mean that there is common ground between pupil and teacher. This is as vital in securing appropriate conditions for learning as it is for managing those behavioural issues that may be potentially problematic.

Structuring your lessons to promote positive behaviour

The design of effective lessons is fundamental to high-quality teaching and learning. Effective lesson design takes into account behavioural differences between pupils as much as it does their levels of achievement or the subject or skill they are learning. Your teaching should be characterised by:

- focus and structure so that pupils are clear about what is to be learned and how it fits with what they know already;
- actively engaging pupils in their learning so that they make their own meaning from it;
- developing pupils' learning skills systematically so that their learning becomes increasingly independent (see also Unit 5.2);
- using assessment for learning to help pupils reflect on what they already know, reinforce the learning being developed and set targets for the future (see Units 6.1 and 6.2);
- having high expectations of the effort that pupils should make and what they can achieve (see also Unit 3.2);
- motivating pupils by well-paced lessons, using stimulating activities that encourage attendance;
- creating an environment that promotes learning in a settled and purposeful atmosphere.

You can further promote a positive approach to behaviour by building individual teaching sequences within an overall lesson. The lesson (or a sequence of lessons) needs first of all to be firmly located, in the mind of the pupil, in the context of: (a) a scheme of work; (b) pupils' prior knowledge; and (c) their preferred ways of learning. It is also important to identify clear learning outcomes. Structuring your lesson as a series of 'episodes' by separating pupil learning into distinct stages or steps, and then planning how each step should be taught, enables those pupils who are at risk of distraction or lack of concentration to regard the lesson as a series of 'bite-sized chunks'. Finally, you can secure overall coherence by providing: (a) a stimulating start to the lesson;

(b) transition 'signposts' between each lesson episode, which reviews pupil learning until now and launches the next episode; and (c) a final plenary session that reviews learning (lesson planning is covered in Unit 2.2).

Rights, responsibilities, routines and rules

A framework for promoting positive classroom behaviour has been commonly constructed around the so-called 4Rs: rights, responsibilities, rules and routines (Rogers, 2007). You should recognise that such a focus operates best within the context of a fifth 'R', already encountered in this unit – 'relationships'. In applying aspects of the 4Rs, you need to be very sensitive, as a beginning teacher, to the existing arrangements in any class that you work in. These will have been established over a long period of time by the permanent class teacher. But you can begin by being conscious of how each of these 'Rs' can have a positive impact on your teaching.

Rights (R1) and responsibilities (R2)

Both rights and responsibilities are inextricably linked. They refer equally to teacher and pupils, and are the basis on which classroom relationships, teaching and learning are built.

- Teacher's responsibilities – you must seek to enable all pupils to learn, to seek out and celebrate improvements in learning, to treat pupils with respect and to create a positive classroom environment in which pupils feel safe and able to learn.
- Teacher's rights – you must be allowed to teach with a minimum of hindrance, to feel safe, to be supported by colleagues and to be listened to.
- Pupils' responsibilities – pupils must be willing to learn, to allow others to learn, to co-operate with teaching and other staff and peers and to do their best at all times.
- Pupils' rights – pupils should be treated with respect, be safe, be able to learn and be listened to.

Rules (R3)

These are the guidelines by which rights and responsibilities are translated into a set of desired adult and pupil behaviours. They are best constructed collaboratively, so that the views of all pupils are taken into account.

Routines (R4)

These are the structures that underpin the rules and reinforce the smooth running of the classroom. The more habitual the routines become, the more likely they are to be used. Pupils who behave inappropriately often do so because they are unsure of what is happening in the classroom at a given time. Consistent application of your classroom rules will constitute a major step in establishing a routine in your class.

In using the 4Rs as a basis for promoting positive pupil behaviour, you should be encouraged to provide opportunities for your pupils to make choices about their behaviour, thus allowing them to take responsibility for their own actions. Choice is guided by their responsibilities and leads to positive or negative consequences according to the choice made by the pupil.

Consequences

Pupils need to know the consequences of sensible or inadvisable choices. Responsible choices lead to positive consequences; conversely, a choice to behave inappropriately leads to a known negative consequence.

Now complete Task 3.3.5, which is designed to help you to focus on the ways you use encouragement, positive feedback and praise in the classroom.



Task 3.3.5 Monitoring your use of praise and encouragement in the classroom

A very useful starting point to promote the notion of positive approaches to behaviour is to examine the ways in which you provide encouragement, positive feedback and praise to your pupils. You can assess this by developing a log of praise and encouragement to use as a tool for measuring these positive interactions.

Add to the following list of positive pupil behaviours identified, which you can use to give praise. Underneath each one, note the words or actions you might use to convey to the pupil that your recognition carries value and meaning in that they are clearly directed towards a particular pupil and are linked to the positive behaviour that the pupil has demonstrated.

1 Queuing sensibly and quietly to enter the classroom.
2 Allowing another pupil to go first.
3 Lending an item of equipment to another pupil.
4 Putting waste paper in the bin.
5 Supporting another pupil's learning.
6 _____
7 _____
8 _____
9 _____
10 _____

During your observation of a lesson taught by a more experienced teacher in your placement school, record other ways in which that teacher acknowledges positive behaviour by pupils. Compare your notes on this topic with another beginning teacher working in a different setting.

Among the possible research questions you might wish to consider are:

- Do boys prefer different kinds of praise and encouragement than girls?
- Does the nature and type of praise change according to age?
- What types of praise do pupils prefer?
- Is praise evenly distributed among your teaching group?
- Is praise carefully targeted and in response to specific pupil actions?
- Does personal praise link closely with a whole-school approach?

In Task 3.3.6, you are asked to explore the links between behaviour and learning.



Task 3.3.6 Exploring further the links between behaviour and learning

To explore further the links between behaviour and learning, select one or more pupils who you currently teach in your placement school and who present you with a particular challenge on account of their unacceptable behaviour. You should explore the learning and behaviour interface by responding to the following key questions:

- Does the educational achievement of this pupil vary from one curriculum subject to another?
- What are the characteristics of those curriculum subjects in which the pupil appears to perform more effectively?
- Has the pattern of educational achievement been inconsistent over time? Are there any logical explanations for this?
- Do you know anything about the pupil's preferred ways of learning?
- What are your own views about the capabilities of this pupil?
- What do other subject teachers say about the educational achievements of this pupil?

Each of these questions can form the basis of a small-scale classroom enquiry. For each, you could: (a) gather evidence from the school's pupil data; (b) obtain information from key personnel (for instance, the pupil's form tutor, or the SENCO); and (c) secure inputs from the pupil directly (subject to the appropriate permissions).

On the basis of what you discover, try to formulate a theoretical model for both the unacceptable behaviours displayed and their relationship with more positive aspects of this pupil's school performance.

SUMMARY AND KEY POINTS

- Adopting a positive approach to managing pupil behaviour is crucially important.
- Become familiar with your school's policies regarding behaviour management.
- Take steps to understand what are the underlying causes of the problem behaviour in your class. This is the first step in taking positive action.
- Try to establish a clear set of classroom rules and rewards and sanctions and ensure that they are applied consistently and fairly.
- Focus on your relationships with pupils. Establishing an effective working relationship is crucial to a positive classroom 'ethos'.
- Always try to lead by example. Try to model the kinds of positive behaviours you want to see from your pupils.
- Seek guidance and support from more experienced teachers in your placement school, and make use of all opportunities for professional development.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Dix, P. (2017) *When Adults Change Everything Changes*, Carmarthen: Independent Thinking Press.
This book advocates an inclusive approach to promoting behaviour in schools and classrooms. In this, behavioural expectations and boundaries are exemplified by people, not by a thousand rules that nobody can recall. The book illustrates how traditional sanction- and exclusion-led methods are outmoded, that what is required is a stable platform of daily rules and routines and how these can help deal with even the angriest learners. Each chapter includes three checklists, specially designed to help you form your own behaviour blueprint.

Ellis, S. and Tod, J. (2018) *Promoting Behaviour for Learning in the Classroom*, Abingdon: Routledge.
This book provides a concise analysis of established behaviour management strategies that you can use in your own classroom. It recognises that no single approach will work for *all* pupils and that it is important to understand the individual needs, attributes and personalities of your pupils when deciding how best to intervene. The book covers a range of issues, including developing positive relationships in the classroom, understanding personal style and self-management, making use of effective feedback and rewards, individual differences and special educational needs and dealing with challenging behaviour.

Rogers, W. (2015) *Classroom Behaviour: A Practical Guide to Effective Teaching, Behaviour Management and Colleague Support* (4th edn), London: Sage.
This book explores some of the issues you will face when working in today's classrooms. It describes real situations and dilemmas and offers advice on dealing with the challenges of the job. Emphasis is placed on how to establish and enhance your relationships with pupils. The book also considers some more specialist aspects of teaching pupils who have additional needs, including sections looking at dealing with bullying, teaching pupils on the autistic spectrum in a mainstream classroom and working with very challenging pupils.

Key policy documents relating to pupil behaviour for England can be located at www on the companion website:

Any additional resources and an editable version of any relevant tasks/tables in this unit are available on the companion website:

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3.4 Conceptualising and theorising primary–secondary school transitions

Divya Jindal-Snape, Charlotte Louise Bagnall, Kate Evans and Calum MacGillivray

Introduction

Primary–secondary school transitions have been the focus of a range of research, practice and policy initiatives. This is not surprising as every year millions of children around the world start secondary school. Further, research around these transitions *claims* that they are problematic, with a negative impact on academic outcomes and wellbeing. These findings are concerning as the negative impact of primary–secondary transitions experiences can be long term and lead to pupils dropping out with implications for the workforce. More importantly, if pupils' wellbeing is negatively affected, it can lead to mental health problems if not resolved. However, there are gaps in this literature that might lead to some misunderstandings about primary–secondary transitions. This unit will look at these aspects one by one to support your developing understanding of primary–secondary transitions.

As secondary school teachers can play an important role in facilitating successful transitions, this unit focuses on the conceptualisation and theorisation of primary–secondary school transitions, the discourse surrounding them and shares some insights from research on primary–secondary transitions, focusing on what these mean for you when learning to teach in a secondary school. It provides you with some resources to use when supporting pupils' transitions.

OBJECTIVES

At the end of this unit, you should be able to:

- conceptualise and theorise transitions;
- be aware of how pupils' perceptions of what they are excited and worried about change over time, and might be affected by the discourse in their environment;
- understand how relationships can have an impact on pupils' transitions experiences;
- consider best practices in supporting pupils and significant others.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

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Conceptualisation of primary–secondary transitions

It is important to be clear how transitions are conceptualised. As a secondary school teacher your conceptualisation of transitions will have an impact on your transitions practice, whether it is for pupils starting secondary school, moving through secondary school or leaving it. Further, your conceptualisation might differ from that of colleagues. In our research, we have found a lack of shared conceptualisation and understanding of transitions even in the same primary and secondary school cluster/catchment area (Jindal-Snape and Cantali, 2019). If primary and/or secondary school's transitions policy and practice are based on different conceptualisations, it is possible that there will be a mismatch and they will not be effective in supporting pupils' transitions.

Similarly, the research literature that you might access as a secondary school teacher to inform your transitions practice might not have explicitly stated what transitions mean in their study or are using conceptualisations that might be problematic (Jindal-Snape et al., 2021). This literature might influence the national or your school's transitions policy, which will also have an impact on your practice and support provided to pupils. Undertake Task 3.4.1 to understand different people's conceptualisation of transitions.



Task 3.4.1 Conceptualisation of transitions

What does the term transitions mean to you?

Ask school professionals or other student teachers what transitions mean to them.

Read a paper about primary–secondary school transitions. How, if at all, have the author/s defined transitions?

Compare your and other's conceptualisations with the article by Jindal-Snape et al. (2021). What are the similarities and differences between them?

Based on her research over the last 20 years, Jindal-Snape (2016) defines transitions as an ongoing, dynamic process of adaptation resulting from some type of change, usually multiple changes at the same time. This change can be related to contextual changes (e.g. primary to secondary school), interpersonal (e.g. leaving old friends and making new ones) and/or identity (e.g. primary school pupil to secondary school pupil). It is important to remember that contrary to teachers'/headteachers' (Jindal-Snape and Cantali, 2019) and researchers' conceptualisations (Jindal-Snape et al., 2021), *change is not transition*; change *triggers* transitions. Further, individuals experience multiple transitions at the same time. Transitions are, therefore, the processes of *adaptation* to change over time, and these adaptations might be, amongst others, psychological, social and educational (Jindal-Snape, 2016, 2018). They can be exciting and worrying for the same individual in different contexts at the same time, and over time. For instance, a pupil might look forward to starting secondary school due to better access to sports facilities but be worried about having to form relationships with multiple teachers. In some instances, a pupil might be excited and worried about the same aspect, such as excitement about being taught by several teachers as well as concerns about managing relationships with several teachers (Jindal-Snape and Cantali, 2019).

Successful transitions

As a secondary school teacher, it is important for you to understand what successful transitions mean. Historically, most primary–secondary transitions literature has focused on good educational attainment and engagement at school as a sign of successful transitions (see Jindal-Snape et al., 2020, for examples, such as Benner and Graham, 2009). However, recently, successful transitions are considered to involve pupils feeling connected, having a sense of belonging and developing positive relationships with secondary school teachers and peers (Peters, 2010; Jindal-Snape, 2023). However, there can be a mismatch in what pupils and families consider to be successful transitions (e.g. focus on making friends, good relationships with teachers, no problems with settling in; see Ashton, 2008; Jindal-Snape, 2016) as compared to teachers, who give greater weight to pupils getting admission in the school of their and their parents' choice (more relevant in the English context) and/or no dip in attainment (Evangelou et al., 2008).

Discourse around primary–secondary transitions

Your discourse around transitions will influence how you provide transitions support and/or the messages you might subconsciously give to pupils and families.

It is important that you, as a secondary school teacher, have a balanced discourse about primary–secondary transitions so that your pupils do not become anxious about the move. Figures 3.4.1 and 3.4.2 are based on a study conducted by Jindal-Snape and Cantali (2019), with data collected at four time points in Scotland: once in primary school (P7), twice in the first year of secondary (S1) and once at the end of the second year (S2).

As can be seen from Figure 3.4.1, when in primary school, pupils were looking forward to several aspects of moving to secondary school, e.g. different subjects, bigger playground. When in

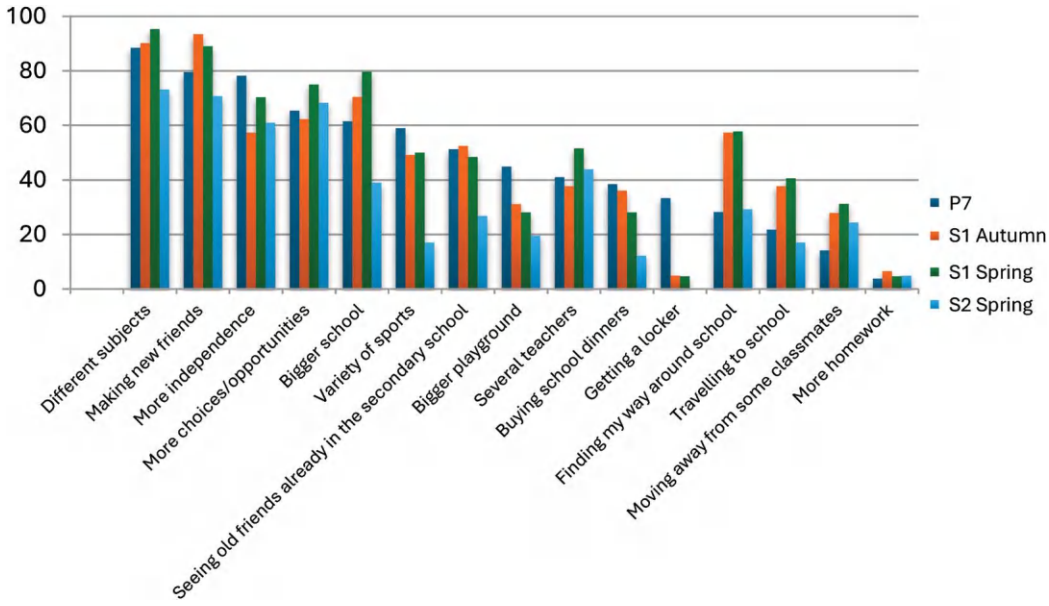


Figure 3.4.1 Expectations and reality: what pupils were looking forward to before moving to secondary school and what was good in S1 and S2

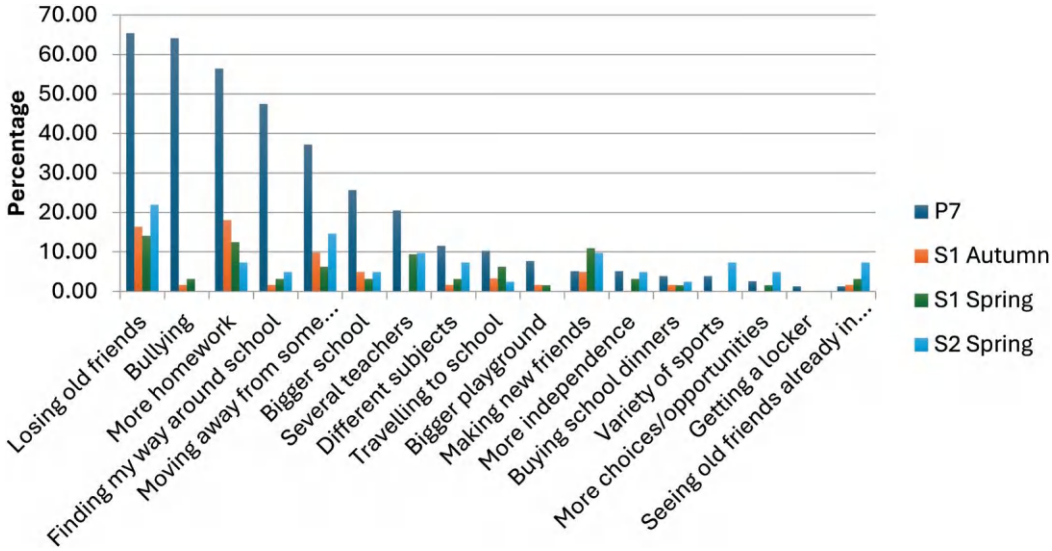


Figure 3.4.2 Expectations and reality: what pupils were worried about before moving to secondary school and what was still worrying them in S1 and S2

secondary school, they reported that these aspects had been good and there was a match between their expectation and reality of the positive aspects of transitions. However, it is worth noting that their views and experiences changed over time, not only across the two time points in the first year but also in the second year of secondary school.

On the other hand, as can be seen from Figure 3.4.2, the concerns that some pupils had before starting secondary school were found not to be a problem after a couple of months in the secondary school. It is important to note, however, that for some pupils there were continuing concerns and that they arose during transitions to S2. This highlights that as teachers you have to be mindful of ongoing transitions support needs of pupils.

These findings are positive but beg the question why pupils had such negative expectations. The answer might be linked to negative discourse in research literature. A review of 96 papers on primary–secondary transitions, found that most authors used negative discourse ($n = 60$), i.e. they viewed them as disruptive, leading to stress and anxiety, lowering of self-esteem, dip in educational attainment etc. (Jindal-Snape et al., 2021); whereas only two authors used primarily positive discourse. Some authors collected data just before and after the move when pupils had not had time to adapt to the new school, peers and teachers, with some using scales that were negatively worded, even in the name, e.g. The Impending Transition to Secondary School Perceived as a Challenge and Threat scale (see Bagnall and Jindal-Snape, 2023, for other examples). With a focus on research-informed practice, this could lead teachers and headteachers to believe that there is evidence that primary–secondary transitions are problematic for most pupils when they are not.

This seems to be a prevalent discourse in schools as well; pupils highlighted that primary school teachers became stricter to 'prepare them for secondary school' and said that they would *not get away with certain behaviours* in secondary schools (Jindal-Snape and Cantali, 2019). Pupils also highlighted the 'horror stories' (Delamont, 1991) they had heard from their families (Jindal-Snape

and Foggie, 2008). Further, negative discourse is also reinforced through media, including novels and films, e.g. Harry Potter or the Diary of a Wimpy Kid.

It is important that you are aware of pupils' potential negative perceptions and what you need to do to change them. Please complete Task 3.4.2 to explore the discourse around transitions in novels or films.



Task 3.4.2 Discourse around transition in novels or films

Read *Diary of a Wimpy Kid* (Book 1) or watch the movie based on it. Please make notes about the plot line, words and/or imagery used around starting a new school and/or their development stage.

What is the dominant discourse? What discourse is used around relationships with new peers, teachers and also their own family members?

Please reflect on what positive and/or negative messages there are for a young person.

Make a list of future actions that you can take to ensure a positive discourse.

It is worth noting that after analysing a large longitudinal dataset from when pupils (n = 2,559) in Scotland were in the penultimate year of primary school and then in the first year of secondary school, we found that most had experienced positive (36%) or moderately positive (42%) transitions, with a minority reporting negative transitions (22%) (Jindal-Snape et al., 2023a). Reduction in bullying was reported. However, 22% is still not a small percentage of pupils experiencing negative transitions, and transitions support needs to be improved.

Theorisation of transitions: Multiple and Multi-dimensional Transitions (MMT) Theory

According to Jindal-Snape's (2016, 2023) Multiple and Multi-dimensional Transitions (MMT) Theory, pupils moving to secondary school will experience multiple transitions in multiple domains and multiple contexts at the same time, triggered by their developmental stage, starting a new school, different pedagogical approaches, changes in language used, unspoken rules of the school, change in expectations of others, change in friendships, to name a few. Their multiple transitions interact with each other and can trigger transitions for other people (e.g. friends, parents, teachers) and vice versa, meaning that transitions overall are multi-dimensional processes (Jindal-Snape, 2016). Amongst others, these can require psychological, educational, cultural, social, spatial and financial adaptations. As a teacher, it is important for you to understand that pupils will be navigating multiple transitions and as mentioned before, their priorities might be different from those of the adults around them. The support you provide should take into consideration their unique and complex multiple transitions (Jindal-Snape, 2023).

Using the analogy of a Rubik's cube, if we consider each colour to represent one pupil and their significant others, any change for one pupil and their ecosystem (colour) will trigger changes across the Rubik's cube not only for that one colour but also for *all* colours and sides. This then leads to transitions for everyone in the connected ecosystems and these transitions will be experienced differently with dynamic adaptations over time. Additionally, these transitions are not taking place in

a vacuum and the wider environment, whether human, physical, economic, political or due to the recent pandemic, will be dynamic and lead to ongoing transitions.

Further, the pupil will not be the only one experiencing these multiple transitions. Their peers will be going through multiple transitions which are bound to interact with each other, leading to positive or negative transitions experiences. It is also important to remember that these transitions are ongoing and dynamic, requiring ongoing support from teachers over time.

Pupils' transitions to secondary school will trigger transitions for their families. For example, a parent who had started working part time after their birth, might consider going back to working full time after their child reaches secondary school age. Further, parents will be experiencing other transitions independent of their child, such as health or professional transitions. The parents' transitions will in turn trigger transitions for their child, e.g. the child having to take on more responsibility as a result of the parent now working full time, highlighting the multi-dimensional aspect of the theory. Therefore, as a teacher you might have to provide support not only to pupils but also to their families as the families' transitions will have an impact on your pupils' wellbeing and academic outcomes.

Similarly, pupils' transitions will trigger transitions for you, such as relationship transitions as you lose some pupils (and parents) and adapt to a new set, or educational transitions as you take a course to understand how to support transitions of an autistic pupil. Similarly, your identity transitions from beginning teacher to early career teacher to experienced teacher might influence your perceived self-efficacy over time in supporting pupils and parents with their transitions. Further, your professional transitions will have an impact on your home life such as moving to a new city and house when employed in a different city triggering transitions for the entire family. Therefore, it is important that you consider your own transitions support needs. If you are not supported with your transitions, it will be difficult for you to support pupils (or their families) with their transitions (Jindal-Snape, 2016, 2023). Being mindful of the MMT Theory might help you and your organisation understand the complexity of transitions and that to effectively support pupils, their significant others (e.g. families, teachers) should also be supported.

Impact of primary–secondary school transitions on educational and wellbeing outcomes

Planning and preparation for positive transitions is important as research suggests that they can have a negative impact on pupils' educational and wellbeing outcomes. Most transitions research has focused on the negative impact of primary–secondary transitions on educational outcomes, reporting reduction in grades (e.g. Benner and Graham, 2009; Benner et al., 2017), (negative) change in attitudes about particular subjects (e.g. Deieso and Fraser, 2019), decline in motivation and engagement (e.g. Madjar et al., 2018) and increase in absences and dropping out (e.g. Benner and Graham, 2009; West et al., 2010; Schwerdt and West, 2011).

In the context of wellbeing, four studies found that transitions had a positive impact, including positive change in social and emotional wellbeing of a small sample of autistic pupils (e.g. Fortuna, 2014). However, seven reported a negative impact (e.g. Witherspoon and Ennett, 2011; Lofgran et al., 2015). In particular, studies reported that anxiety scores of pupils with additional support needs were high (Hannah and Topping, 2012; Neal et al., 2016).

There is some evidence suggesting links between educational and wellbeing outcomes; however, it is not conclusive, and a causal relationship is difficult to establish (West et al., 2010). To provide

holistic transitions support, it is important for you to be mindful of supporting with, both, educational and wellbeing outcomes.

Planning and preparation for primary–secondary transitions

As primary–secondary transitions are ongoing processes, in this section we consider what transitions support is provided to pupils by primary and secondary school teachers. Jindal-Snape and Cantali (2019) found that all primary and secondary school teachers reported that they supported pupils in the same way. This included providing opportunities to discuss what to expect, and what pupils were excited or concerned about; and teachers meeting with their counterparts, especially in the case of pupils with additional support needs. Although primary school teachers invited secondary school teachers to visit their classes, this was not reciprocated. It is important for you as a secondary school teacher to consider benefits of reciprocal visits to ensure that any mismatch, for instance in terms of pedagogical approaches, can be minimised.

Pupils' views of the effectiveness of transitions planning and preparation changed over time, moving in favour of support provided by the secondary school. However, some pupils highlighted that this support was not provided in an ongoing manner, over the first year or for the transitions to the second year of secondary school. This might be due to teachers not conceptualising transitions as ongoing and seeing it as 'change' rather than *adaptation* to multiple changes over time. Of concern is that teachers do not get transitions-related training as part of teacher education programmes or as continuing professional development, despite the important role they play in supporting pupils with their transitions (Jindal-Snape and Cantali, 2019).

Factors having an impact on pupils' transitions experiences: relationships

Several factors were found to have an impact on pupils' transitions experiences, including relationships with peers and teachers, the school environment and curriculum, school and family characteristics and additional support needs which can lead to both positive and negative experiences (Jindal-Snape et al., 2020; Jindal-Snape et al., 2023a).

However, here we focus on relationships as this is where you can intervene most effectively to enhance pupils' transitions experiences, especially by involving families in the transitions process and by providing opportunities to build positive relationships with peers and you as a teacher (see Jindal-Snape et al., 2020 for details of other factors).

Relationships with family

Pupils consistently report that families are their strongest support network (Jindal-Snape and Cantali, 2019). Therefore, it is not surprising that relationships with family members could have both a positive and negative impact on their transitions experiences. Studies suggest that supportive and engaged parents, stable homes and siblings in the new school can all facilitate positive transitions (Mackenzie et al., 2012; Hammond, 2016), especially as they reduce anxiety about moving to a new school (Frey et al., 2009). Therefore, parental involvement in the transitions process is crucial and has been found to have a positive impact on their child's transitions experiences, highlighting the importance of you as a secondary school teacher involving parents in transitions planning and preparation.

Peer relationships

It seems that pupils had positive experiences of starting secondary school due to having a wider group of peers to choose from compared to in the primary school, and could re-invent themselves to make new friends (Farmer et al., 2011; Booth and Gerard, 2014; Neal and Frederickson, 2016; Symonds and Hargreaves, 2016). However, several studies also reported that relationships with peers were a cause of anxiety and negative transitions experiences, due to losing old friends, and worries about making new friends (Evangelou et al., 2008; Hammond, 2016). This has implications for you as a teacher in providing opportunities for development of positive peer relationships, for instance, through small group work, buddy system and/or early intervention if relationships deteriorate.

Relationships with teachers

Similarly, positive relationships with teachers were seen to support positive transitions, with some pupils highlighting the importance of knowledgeable and passionate subject specialists and some with additional support needs finding the clear structure and routine provided by secondary schools to be better for them (Neal and Frederickson, 2016). However, the majority of literature reports that pupils were worried about being able to form good relationships with teachers, which was related to a belief, sometimes based on what they had been told by primary school teachers, that secondary school teachers are stricter (Jindal-Snape and Cantali, 2019). Secondary school teachers were also considered to have inconsistent rules and expectations, did not trust or respect pupils and had different pedagogical approaches (Ashton, 2008; Tobbell and O'Donnell, 2013; Jindal-Snape et al., 2023a). Some longitudinal studies reported a change in pupils' perceptions once they started secondary school (Jindal-Snape and Cantali, 2019).

What seems to work for pupils?

Voice and opportunities to discuss excitement and concerns

Pupils can feel more in control if they are actively involved in decisions regarding transitions planning and preparation. For example, by creating a safe environment, both primary and secondary school teachers can involve pupils quite early on in the final year of primary school (or even earlier) in an exploration of what changes across primary–secondary might mean to them, what they might be looking forward to or worrying about. These discussions could be facilitated during reciprocal visits of primary and secondary school teachers and pupils. Further, you could encourage secondary school pupils to write a 'letter to my former self' to send to primary school pupils, in which they could share concerns they had when in primary school about the move, the strategies that worked for them and bust any myths. Similarly, you could encourage primary school teachers to take a note of the concerns that the pupils might have which you or secondary school pupils could respond to.

Familiarisation and knowledge of the new context

All secondary schools try to facilitate familiarisation of the new context. These attempts have been found to be most successful when familiarisation happens for a long period; for example, pupils starting to use the swimming pool in the secondary school a couple of years prior to the move. The Covid-19 pandemic led to virtual tours of the secondary school, and video messages from teachers

and secondary school pupils. In future, these can be made available for all pupils and families to be able to look at several months or years in advance of the move.

There is also a clear indication from pupils that they want to move to the new school with pupils they know. However, it is important to remember that moving with friends cannot, alone, guarantee successful transitions. Further, the link for some pupils is unclear, as some pupils who moved with friends reported difficulties and some who did not move with friends reported no difficulties with transitions (Jindal-Snape, 2010).

Rehearsing in a safe environment

The importance of rehearsing strategies for adaptation to the secondary school in a safe environment cannot be overstated. This can be done, for example, through creative drama by constructing a possible real-life scenario in which the 'actors' can depersonalise their actions and responses in the guise of 'playing the character'. This provides pupils with an opportunity to move from their concerns to 'Alex's concerns', a fictional character on whom everybody is comfortable projecting their own or other's concerns. Based on the Forum Theatre approach (see Jindal-Snape et al., 2011), Task 3.4.3 asks you to be involved in creative drama to experience how this might work.



Task 3.4.3 Using creative drama to construct a real-life scenario

Try this task with other beginning teachers. You or another beginning teacher could play the role of the facilitator.

Scenario

It is the first day at secondary school. Robert and Mary are amongst the 80 pupils who are starting today. They were together at primary school. Mary has an older brother at secondary school and is feeling quite confident about starting school. Robert doesn't know anybody in the school and is feeling concerned about the 'horror stories' he has heard from others.

In this scene, they are inside the school gates and have to be in classroom 1L10 to meet their mathematics teacher. They have been given a map but are having problems finding their way around school. They are overwhelmed by the size of the school and number of pupils around them. They had attended the induction event in the school before the summer holidays but everything feels different. They ask Sheila for help but she mocks them and starts teasing them because ...

Characters

- Robert (11 years old) is nervous and underconfident but ends up coming across as overconfident and aggressive.
- Mary (11 years old) is positive, cheerful, very sure of herself and tries to impress others by talking about her older brother in the same school, but this might lead to more teasing and name-calling.

- Sheila (13 years old) enjoys teasing other children and makes it difficult for Robert and Mary to find their way. *The character has to try to persistently block Robert and Mary in every way possible.*

Creative drama activity

Ask other beginning teachers to volunteer to play the part of each character. Those in the audience can suggest changes and can also come and replace the individual (Robert/Mary) and take over that character. (*The facilitator asks the actors to freeze and then asks questions such as: why do you think this is happening, what can be done, how do you think this person is feeling and so on.*)

The ultimate purpose is to try to resolve the conflict through audience participation and for Robert and Mary to find their way to class.

Note how this went and what you learned from it. What would you need to take into account if you used a similar approach with your own class?

We suggest, when ready, that you try this with pupils during your visits to primary school and/or when primary school pupils are in secondary school during reciprocal visits (in the presence of their primary school teacher), open days and induction days. However, it is important that you create a safe environment for pupils before doing this work and check that everyone is feeling safe and comfortable after you finish it. You could also create similar scenarios in which you use themes that might come from the pupils themselves; for example, making friends, bullying. This type of work can lead to a pupil being able to participate actively, as well as feeling in control by resolving any problems. It also gives pupils an armoury of strategies and responses they can use in similar real-life situations along with feeling more competent.

Active learning, learning agency, autonomy and choice

Learning agency, defined as 'a capacity for intentional and responsible management of new learning within and between school transitions' (Pietarinen et al., 2010, p. 144), is crucial for wellbeing. Research suggests that choice and autonomy can have a positive impact on a pupil's sense of self and ability to manage their transitions. Further, there is evidence that this is possible through active learning; for instance, through creative activities and a creative learning environment (Davies et al., 2013) (see Unit 5.2 on active learning). Jindal-Snape et al., (2011) reported on a study in which primary and secondary school teachers actively involved pupils from different primary schools going to the same secondary school in their preparation for transitions to and through the first year of secondary school. The focus was on curricular continuity and developing relationships with peers from different primary schools using a computer game. It was found that pupils benefitted from the opportunity to choose the game, as well as supporting their teachers in setting up the games consoles with pupils as 'experts' (see Bagnall and Jindal-Snape, 2026). Being involved in the game provided an opportunity for active learning and learning agency (Pietarinen et al., 2010).

To summarise your learning from this unit, especially in the context of your training needs to facilitate primary–secondary transitions and your own transitions and related support needs, please complete Task 3.4.4.

Task 3.4.4 Your training and transitions support needs

Reflect on the following:

1 Having read this unit, and more widely around the area, how prepared do you feel about your own transitions from Beginning Teacher to Early Career Teacher (ECT)? Draw up a table to identify your feelings about working as an ECT in a new school. Identify actions you might take to ensure you have successful transitions.
2 How well prepared do you feel to support pupils in their transitions from primary to secondary school? What professional development do you need to undertake to enable you to be prepared to support pupils and their families?
3 How might you meet these professional development needs?

You might want to discuss these questions with other beginning teachers. It is also advisable to revisit these questions at different times on your journey as beginning teacher and teacher; hence, we suggest you store your thoughts and answers.

SUMMARY AND KEY POINTS

- This unit emphasises that transitions are complex and dynamic processes of, amongst others, psychological, educational and social adaptation, with individual differences in when and how different pupils adapt to their new school and other changes taking place at the same time.
- It is important to have a holistic perspective of transitions and to understand pupils' concurrent multiple transitions.
- Pupils will trigger transitions for their family, peers and teachers, and vice versa.
- Teachers have an important role in changing the negative discourse around primary–secondary transitions.
- Positive relationships and opportunities for pupils to build them are crucial for successful transitions.
- Pupils' and families' meaningful involvement in planning and preparation for transitions should help them have voice and share their excitement and concerns in a safe environment, as well as promoting successful transitions.
- As everyone is experiencing transitions, all need to be supported.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

For bonus digital resources designed to enhance the book materials, head to routledgelearning.com/learningtoteach.

Further resources

Jindal-Snape, D. (2016). *A-Z of Transitions*, Basingstoke: Palgrave.

This book includes entries on various types of transitions, information on strategies that can be used to support transitions and provides clear implications for professionals.

Jindal-Snape, D. (2023b). 'Multiple and multi-dimensional educational and life transitions: conceptualization, theorization and XII pillars of transitions'. In R. J. Tierney, F. Rizvi, and K. Erkican (eds.), *International Encyclopedia of Education (Fourth Edition)* (4th edn, pp. 530–543), Elsevier, https://doi.org/10.1016/B978-0-12-818630-5.14060-6

This chapter provides more details on transitions theories, and on XII Pillars of successful transitions that can be used to facilitate successful transitions.

Jindal-Snape, D., Barlow, W., Goode, T., Hannah, E. F. S., Tooman, T., Ding, C., Burns, M., and Santiago, T. L. (2023b). *Multiple and Multi-dimensional Primary-Secondary School Transitions: Using Drama to Facilitate Transitions*, UniVerse, https://doi.org/10.20392/nx7a-bj66

This comic provides children's perspectives of their primary–secondary transitions, how they navigated them, and top tips for other children. It also provides information on how teachers and parents can use the comic as an educational resource.

Any additional resources and an editable version of any relevant tasks/tables in this unit are available on the companion website: www.routledgetextbooks.com

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4 Meeting individual differences

Introduction

Every pupil in your class is unique. A class of same-age pupils is likely to contain individuals at different stages of development owing to the combined differences in physical development, socio-emotional development, cognitive development and ethnic-cultural background. Significant differences arise in the achievements of members of a class of pupils, including pupils in a class where setting is practised. The cultural, religious and economic backgrounds of your pupils can affect their response to schooling. Some pupils respond to academic challenge while others see no point in such demands. Others are gifted and need special attention, as do many pupils with learning or behavioural difficulties. Some pupils are at ease with adults while others find the experience less comfortable.

This chapter, comprising six units, invites you to consider several aspects of the background and development of your pupils. In practice the features discussed interact, giving rise to the complex and varied behaviours that characterise human beings and a need to understand intersectionality. For ease of discussion, some factors are discussed separately; we hope this approach helps you subsequently to integrate better your understandings of pupils and their learning and meet their individual needs.

Unit 4.1 addresses pupil grouping for learning and goes on to discuss adaptive teaching and differentiation, focusing on ways to improve the learning of all pupils. Central to successful differentiation is the identification of pupil needs; thus, case studies invite you to enquire more deeply into the background and response of individual pupils and ways to plan adaptive teaching. You may want to return to this unit after dipping into other units.

Unit 4.2 focuses on the different aspects of adolescent development and implications for your practice, with a view to optimising pupils' wellbeing as they transition into adulthood. This unit considers the necessary personal development work for you as a practitioner, the best approach when working with pupils on an individual level and how you can contribute to whole-school drives to enhance pupil wellbeing. It looks at the concept of identity and intersectionality and a growing awareness around promoting wellbeing in children and adolescence, as mental health has gained more prominence during this period of change and transition in young people's lives.

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Unit 4.3 explores cognitive development in children; what this means and how it might be measured. The unit looks at the key features of cognitive development and some important research that helps explain how children develop in their assimilation of knowledge and thinking skills. Through the completion of specific tasks, you can explore cognitive demand and thinking skills in your own subject and think about how to support your pupils' learning. The unit also discusses programmes that have been designed to accelerate learning and the role of motivation and pupil interest in cognitive development. The unit has been updated to examine the role of 'grit' and 'flow' in the classroom and considers how teachers can harness and develop these attributes in learners. Central to grit is perseverance. This can be a challenge for pupils, especially when faced with subject matter and tasks they find boring or hard. Teachers can do much to help pupils see relevance in their learning and scaffold tasks to help. This unit also has further follow-up information and tasks on the companion website.

In Unit 4.4, 'Responding to diversity', the background of pupils is considered, including socio-economic class, gender identity, ethnicity and understanding how these can 'intersect' with one another. For example, a significant factor related to class is poverty, and while these factors are discussed separately, we recognise that for every pupil these factors combine in different ways. We highlight some differences in performance of groups of pupils from various backgrounds, using research evidence. We examine the barriers faced by marginalised groups, including those based on gender, ethnicity, socio-economic status and language proficiency and explore the importance of responding to diversity in your classroom, highlighting the social, cultural and economic factors that shape pupils' experiences and attainment. This includes increasing an understanding of the impact of unconscious bias in the classroom. We encourage you to critically reflect on your own practices, challenge stereotypes and foster inclusive learning environments. Ultimately, the unit aims to equip beginning teachers with the knowledge and tools to create an inclusive, supportive and effective learning experience for all pupils, through understanding intersectionality of identity, the significance of inclusive curricula and the responsibility of teachers in shaping a fairer education system.

In Unit 4.5, moral education and values education are introduced and explained, and their development linked to the curriculum for schools in England, including Citizenship and the importance of personal, social and health education (PSHE) courses. The emphasis is on the way that schools contribute to moral education and values education through both the overt and hidden curriculum. While not stressing differences between pupils, the focus of the unit does acknowledge the range of values and beliefs in our society and how schools not only have to respond to such differences, but also contribute to the spiritual, moral and cultural development of pupils in addition to their cognitive and physical development. Opportunities for teachers to address moral and ethical issues are provided through tasks, and ways to introduce discussion between pupils are suggested. The guidance in this unit is underpinned by value judgements about maintaining positive relationships with pupils, being fair, respectful and supportive. This is reiterated in sections of both the Initial Teacher Training Core Content Framework (ITT CCF), Early Career Framework and the Initial Teacher Training and Early Career Framework, which identify these as a fundamental professional requirement. All four constituent nations of Great Britain are explicit about the values that teachers must uphold.

Inclusion and Special Needs education is the focus of Unit 4.6, and it outlines the 2014 special educational needs and disability code of practice for schools in England, which has established a framework for the education of children and young people with special educational needs and

disabilities (SEND). The unit draws attention to significant changes in recent years in the way in which pupils with SEND are supported, and emphasises the importance of the classroom teacher in the identification of need and response to need. There is reference to sources of support and guidance, including the support of other professionals.

The unit also examines ableist classrooms and neurodivergent pupils and explores a model of a neuroinclusive school. In exploring neurodiversity, the unit embraces the idea of celebrating and nurturing difference by improving our current environment.



ONLINE RESOURCES

Watch the explainer video on understanding and meeting individual needs by scanning the QR code or visit routledgelearning.com/learningtoteach.





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4.1 Pupil grouping, progression and adaptive teaching

Alexandra Titchmarsh

Introduction

Effective planning and teaching requires you, among other things, to set suitable learning challenges, respond to pupils' diverse learning needs and help pupils to overcome potential barriers to learning.

This unit begins by briefly examining how different grouping arrangements at both the school and class levels can help address variations in pupils' performance. The way pupils are grouped may depend on the school's philosophy and the subject being taught (see, for example, the Education Endowment Foundation (EEF), 2018). The unit then explores how to achieve progression for all pupils across different ages and abilities through teaching and learning approaches that consider a range of learning needs. Tailoring learning to individual pupils' needs is referred to as either adaptive teaching or differentiation, depending on the context. Although there are distinctions between these terms (see, for example, Mould, 2021), differentiation is a key factor when adapting teaching to meet individual needs. In this unit, the terms 'adaptive teaching' and 'adaptation' are generally used, with 'differentiation' mentioned where appropriate.

OBJECTIVES

At the end of this unit you should be able to:

- understand the links between pupil grouping, progression and adaptive teaching;
- discuss progression and adaptive teaching and their relationship to learning;
- discuss teaching methods that allow for adaptation;
- begin to apply principles of adapting approaches to learning in lesson planning.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

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Grouping pupils across the school

Schools can address differences in pupil performance through various methods, such as streaming, banding, setting or assigning work at appropriate levels in wide or mixed-ability classes. Streaming involves grouping pupils by attainment for all or most lessons, placing the most able in one class and the least able in another, with graded classes in between. Banding groups of pupils into broad ability categories for all subjects, aiming to avoid classes composed solely of low-attaining or unmotivated pupils. Setting groups of pupils by attainment in specific subjects, like mathematics and English, rather than across the entire curriculum. Mixed-ability classes include pupils of varying abilities, with work adapted or differentiated to suit different levels. Streaming and banding support the idea of general intelligence, while setting recognises that pupil aptitude and attainment can vary across subjects and contexts. The evidence supporting one grouping method over another is not definitive. For example,

> The evidence suggests that setting and streaming has a very small negative impact for low and mid-range attaining learners, and a very small positive impact for higher attaining pupils. There are exceptions to this pattern, ... Overall the effects are small.
>
> (EEF, 2018, p. 4)

So, the advantages of grouping pupils one way rather than another in a school are not clear-cut; and the best way to group pupils remains a vexed question (for more on pupil grouping see, for example, Francis et al., 2017; Connolly et al., 2019; Mazenod et al., 2019; Hodgen et al., 2023).

Task 4.1.1 invites you to investigate pupil grouping in your school.



Task 4.1.1 How are pupils grouped in your placement school?

Find out the ways in which pupils are grouped in your placement school, the reasoning behind the grouping and how it works in practice. Grouping arrangements may change after pupils have been in the school for a term or a year, at the transition from Key Stage (KS)3 to KS4, as well as between subjects.

Are primary school records (for example, their achievement levels) used to group pupils?

Are other tests used to assess pupils and assign them to groups?

Write a summary of your findings and discuss it with your mentor.

Pupil grouping in classes

Within-class groupings may be by, for example:

- ability;
- mix of ability;
- gender;
- expertise;
- friendship;
- age.

Learning activities that make use of flexible and different forms of grouping include paired tutoring, jigsawing and rainbowing. In jigsawing, the class is split into groups to study a topic. Jigsaw is a co-operative learning strategy that enables each pupil of a 'home' group to specialise in one aspect of a learning unit. Pupils meet with members from other groups who are assigned the same aspect, and, after mastering the material, return to the 'home' group and teach the material to their group members. Rainbowing is similar to jigsawing. For example, pupils work in groups of four and discuss a problem or task. Each group member has a colour. Only four colours are used. After discussion, new groups are formed, getting together pupils with the same colour. This means that, in a class of 28, new groups of seven members are formed, and the findings or ideas from each group of four can be shared. Pupils can then return to their original groups of four armed with new ideas. You could, of course, begin with four groups of seven, each with a colour of the rainbow, and then form seven groups of four in the same way. Task 4.1.2 invites you to explore one of these strategies over a small number of lessons and evaluate the experience.



Task 4.1.2 In-class grouping and teaching strategies

Develop a plan to try out either jig-sawing or rainbowing with a class you teach. If you need further information before trying out these strategies, there are resources available on the Internet.

Suggested procedure:

- Try out one of the strategies with a class you teach (for example, one that would respond to a new grouping arrangement, or one with whom you are not making expected progress.
- Draft a rough plan for a lesson including a set of learning outcomes and including a topic suitable for subgroup working. Discuss the plan with your mentor or class teacher. Redraft and try out the plan on a small scale over one lesson to test out the groupings you form, the instructions you give, how you move pupils around groups and how you round off the lesson. Evaluate the lesson and decide whether to continue with this class or choose another.
- Develop a set of lesson plans covering three lessons, including learning outcomes for each lesson:
  - Lesson 1: identify the task, provide background for the pupils, including resources and perhaps homework. Brief your class for the second lesson.
  - Lesson 2: introduce the activity, indicate how groups will form and re-form and what pupils are expected to do in each group. Allow time for pupils to consolidate what they gained from the activity, for example, by a plenary session and introducing the final lesson.
  - Lesson 3: consolidate the work done, leading to the final product.

Check your plans with your mentor/class teacher and teach the lessons.

Evaluate your teaching against the outcomes, and any implicit aims such as enjoyment, enthusiasm, greater participation and improved behaviour pattern.

You may find it helpful for your evaluation to keep a pocket notebook handy to jot down events that occur, the good things and bad things that happen, including pupil responses and comments. Flesh out these notes after each lesson.

Identify pupil gains from the strategy, including knowledge and skills, both cognitively and affectively, the advantages and drawbacks of the strategies.

Discuss your evaluation report with your mentor and identify how you might develop this report into a piece of coursework.

Progression and adaptation

The greatest challenge for teachers is ensuring that all pupils in their class make progress in their learning. Each pupil is unique, whether they are in streamed, setted, banded, mixed or wide-ability classes. Regardless of how pupils are grouped at the school or class level, teaching must be adapted to meet individual learning and performance differences. What happens in the classroom, including the pedagogic models and teaching strategies used, often has a greater impact on achievement than the grouping method itself, although grouping is part of the overall strategy. Therefore, much of the debate focuses on strategies that target individual achievement, such as using data, assessment for learning, booster classes and adapted challenges. Approaches that maximise achievement for all pupils include emphasising thinking processes, learning to learn, intervention through questioning, collaborative learning, literacy across the curriculum and personalised learning, which aims to tailor the curriculum to individual needs. Each pupil brings unique knowledge, skills and attitudes to school, shaped by interactions with parents, peers, the media and their everyday experiences. Pupils are not blank slates on which new knowledge is written. Many pupils possess skills that the school may not be aware of: some care for siblings or animals, others play and adapt computer games and some work with their parents in the family business. Some pupils may know more arithmetic, as the following parody of stock market practice suggests:

Teacher: 'What is two plus two, Jane?'
Jane: 'Am I buying or selling?'

Your classroom is a reflection of your pupils' diversity of background and culture that interacts with their potential for learning. Each pupil responds to the curriculum in a different way. For example, some parents and their children may value a vocationally relevant education more highly because it is immediately applicable to earning a living and may not subscribe to the values placed by the school on a broad, largely academic education. The complexity of diversity is likely to be challenging for you as a beginning teacher.

As a teacher you must take account of personal interest, ability and motivation to design learning that challenges and interests pupils but, at the same time, ensures for each a large measure of success. Planning and teaching for progression in learning is the core business of teachers. It is important that you understand what progression means in your subject. In geography, for example, progression involves:

- an increase in the breadth of studies;
- an increasing depth of study to respond to pupils' growing capacity to deal with complexities and abstractions;

- an increase in the spatial scale of what is studied;
- a continuing development of skills to include specific techniques and more general strategies of enquiry, matched to pupils' developing cognitive abilities;
- increasing opportunity for pupils to examine social, economic, political and environmental issues;
- awareness and implications of interactions between people and their environments.

Now complete Task 4.1.3.



Task 4.1.3 Progression in your own and other subjects

This task is in two parts.

In relation to the progression for geography just mentioned, consider how these characteristics might relate to teaching in your subject.

Access a copy of the relevant documents for progression in your subject and summarise how they describe progression, writing a list like that for geography.

With another beginning teacher, discuss:

(i) similarities and differences between subjects; and
(ii) what progression means for your teaching in your subject.

For further guidance on progression and adaptive teaching you might want to start by referring to, for example, Addysg Cymru/Education Wales (n.d.), Bloom (2025), Eaton (2022), Huntingdon Research School (2023), Office for Standards in Education, Children's Services and Skills (Ofsted) (2011) and Times Educational Supplement/Education Endowment Foundation (TES/EEF) (2025).

Facilitating progression: approaches to adapting teaching and personalised learning

Planning learning for pupils, choosing learning objectives and learning outcomes based on knowledge of the pupils and of what constitutes progression in a particular curriculum area are critical in ensuring pupils' acquisition of the knowledge, skills and understanding underpinning progress. Progression and adaptation are therefore two sides of the same coin. The use and success of adapted approaches depends on teachers knowing their pupils, being secure in their own subject content knowledge and having access to a range of teaching strategies. You may think about planning for progression in relation to, for example:

1. the objectives you want pupils to achieve by the end of the lesson, the 'unit of work' or by the end of the year;
2. what pupils know, understand and are able to do at the start of the topic;
3. what sequence of learning activities may help pupils to progress from their current learning to enable them to reach the objective(s);
4. how you and the pupils know when pupils have reached the objective(s);
5. how you and the pupils know how the pupils can progress from here.

In planning for progression, you need to start from where the pupils are, not where you would like them to be. Not all pupils are at the same level, so some degree of adaptation must be built into your day-to-day lessons, units of work and the wider scheme of work for most pupils if you expect to achieve the five points just listed.

There is no one right way to adapt your teaching for pupils. Effective adaptation is a demanding task and is about raising the standards of all pupils in a school, not just those underachieving, with learning difficulties or the gifted. The purpose of a differentiated approach is to maximise the potential of each pupil and to improve learning by addressing each pupil's particular needs. But what exactly is meant by adaptive teaching (or differentiation) and how is it achieved?

Consider the following. Differentiation is:

- a planned process of intervention by the teacher in the pupil's learning;
- the matching of work to the differing capabilities of individuals or groups in order to extend their learning;
- about entitlement of access to a full curriculum;
- '"shaking up" what goes on in the classroom so that pupils have multiple options for taking in information, making sense of ideas, and expressing what they learn. In other words, a differentiated classroom provides different avenues to acquiring content, processing or making sense of ideas, and to developing products'.

(Tomlinson, 2014, p. 3)

However adaptive teaching or differentiation is described, the challenge begins with its implementation and practice, which in turn is affected by teachers' beliefs about the ability of their pupils, by expectations of particular groups of pupils and by an understanding of how people learn and of optimal learning environments. The nature of the subject itself and the kind of learning it involves may also affect adaptations to teaching.

Adaptive teaching or differentiation has sometimes had bad press because, at its worst, it has implied an unrealistic and daunting demand on teachers to provide consistently different work and different approaches at the level of the individual pupil or has, unwittingly perhaps on the part of teachers, placed a ceiling on achievement for some pupils (see, for example, Lieke et al., 2022).

At its best, given what is known about effective approaches to learning, the influence of high expectations and the potential of all pupils, adaptive teaching may be said to combine a variety of learning options that tap into different levels of readiness, interests, ability and learning profiles with more individualised support and challenge at appropriate times and in appropriate contexts. As relatively little is known about the potential of each pupil, adaptation should be used sensitively and judiciously. The EEF (n.d.) identified six aspects of high-quality teaching: scaffolding; cognitive and metacognitive strategies; chunking; explicit instruction; grouping; and technology. Although these were identified specifically for pupils with special educational needs and disabilities, they are equally relevant for all pupils.

A perspective on adaptation as liberating rather than constraining relies on a number of broader principles informing classroom learning and teaching:

- a focus on key concepts and skills;
- opportunities for problem-solving, critical and creative thinking;

- ongoing assessment for learning;
- a balance between flexible groupings and whole-class teaching;
- identifying pupils as active learners with whom learning goals and expectations are shared;
- collaborative and co-operative learning;
- achievable but challenging targets;
- motivating and interesting learning activities;
- supportive and stimulating learning environments;
- effective use of learning support assistants (LSAs) within the classroom.

Now complete Task 4.1.4.



Task 4.1.4 What do you understand by adaptive teaching?

How would you describe the term adaptive teaching?

You might like to read the blog by Mould (2021) to help you with this task.

Using this description, what do you think this means for practice in the classroom? Observe your mentor or another teacher teaching a class and write notes on:

- how they organised the class to enable learning needs to be addressed (e.g. seating arrangements, how pupils were grouped);
- whether learning outcomes were adapted (in which case, how different pupils knew which learning outcomes they were working towards) or whether different tasks were set to enable pupils to work towards the same learning outcomes;
- how work was scaffolded to support some pupils and what work some pupils moved onto in order to challenge them further;
- what teaching approaches they used to cater for different learning needs;
- how subject content was presented for pupils with different learning needs;
- how work built on pupils' interests, social and cultural experiences;
- whether the pace of work was appropriate for all pupils to have a chance to learn effectively and achieve success.

After the lesson, discuss with your mentor or teacher observed:

- if and how the lesson was adapted prior to the lesson so it was appropriate for this specific class and/or if it was adapted during the lesson for the needs of that class in that particular lesson;
- how effective the various adaptations to the lesson were for the needs of all pupils.

Managing adaptive teaching

Adaptive teaching starts with a clear view about what you want your pupils to achieve and what individual pupils may need as a particular learning goal as well as using and acting upon what you know about pupils' previous learning and achievement using assessment data from a range

of sources. Pupils should know what they are aiming for both in the short and long term. Realistic targets need to be discussed and set through both knowledge of the individual and the use of data. Then begins the consideration of appropriate adapting teaching strategies and how the process can be managed. It is unrealistic to expect one teacher to plan differentiated work separately for each pupil; it is perhaps better to identify groups of pupils who can work to a given set of learning outcomes using methods suitable to those pupils and the topic in question. However, you may find in some classes you do need to differentiate for individuals.

Adapting your teaching always needs to be included as part of your day-to-day lesson planning. Your lesson planning proforma should include a reminder about adapting work (see also lesson planning in Unit 2.2).

It may be helpful to have a framework or steps in which to plan work:

- **Step 1:** Your aims and short-term learning outcomes should be broad enough to encompass all pupils in your class. There are often multiple ways to achieve the same goal.
- **Step 2:** Consider which activities to assign to pupils, linking them to their prior knowledge and identifying outcomes. Achievable outcomes help maintain motivation but should also present a challenge, avoiding tasks that are too easy. By identifying achievable outcomes for different groups, you begin the process of adapting your teaching.
- **Step 3:** Select one or more activities. In addition to the factors in Step 2, check the availability of resources and necessary support, such as ensuring instructions are pupil-friendly, the language is appropriate, providing samples or examples and determining how any LSA will be utilised.
- **Step 4:** Planning must include formative assessment or assessment for learning. This can be done through various methods, such as question-and-answer sessions, small group discussions, responding to queries in class, asking questions during activities, listening to pupils discuss their work and marking books or short tests. The information gathered helps identify the next steps for each pupil. Assessments should reflect your objectives and align with your learning outcomes.

Some lesson plans, units of work or schemes of work plan for adaptation by identifying different priorities for activities, such as:

- pupils must/should/could;
- core/support/extension ideas/activities.

For example, an activity is selected that all pupils must attempt; it may contain the core idea of the lesson.

However, it is also important to recognise that pupils' learning needs may not be fixed or permanent and may relate to the learning context or topic at hand. Adaptive teaching may therefore involve support or challenge being given to different pupils at different times, for example, sometimes to:

- a whole group;
- a targeted group;
- those who work at speed.

Adaptive teaching strategies: stimulus–task–outcome

The outcome of any particular task depends on the way it is presented to the pupil and how they respond. Teaching methods can be restricted by teachers' own imagination; many teachers are inclined to present a task in just one way with one particular learning outcome in mind rather than to look for different ways to achieve the goals set for the pupils. One important, but limited, teaching goal is to ensure pupils remember things, which often involves rote learning, for example, learning Mark Antony's speech on the death of Caesar. Very simply, this exercise is:

- *Stimulus*: play the role of Mark Antony in a class presentation of excerpts from *Julius Caesar*;
- *Task*: learn by heart the relevant text;
- *Outcome*: complete oral recall.

Much learning depends on recall methods: learning the names of element symbols in science; preparing vocabulary in a language lesson; recalling formulae or tables from mathematics; learning to spell. Recall is necessary, if unexciting, when compared with creative forms of learning.

In order to help pupils recall and use knowledge, they need to move up a level, to consolidate and widen understanding. This situation opens opportunities to use a variety of contexts, including ones directly appropriate to pupils' needs, that is, adaptation by choice of stimulus. For example, to consolidate pupils' understanding of punctuation, you could:

1. ask pupils to punctuate a piece of text from which the punctuation has been removed;
2. as 1, using a written report of an interview;
3. as 1 or 2, but read it through first with the pupils;
4. engage in a taped discussion with pupils and ask pupils to write a short report of what was said, with verbatim examples;
5. ask pupils to interview other pupils, or staff, about a topic and write a report that includes a record of some interviews;
6. ask pupils to write a scene for a play.

Thus, for different stimuli, all pupils consolidate their understanding of punctuation, but the level of outcome is different according to the difficulty of the task and ability of the pupils, that is,

Now complete Tasks 4.1.5 and 4.1.6 which invite you to discuss and identify ways in which one stimulus might be used for different outcomes.



Task 4.1.5 Lesson planning for adaptive teaching: different tasks for a similar learning outcome

Choose a specific learning outcome for a topic you have taught or are about to teach. Identify two or three different tasks that allow pupils to achieve the identified learning outcome. In which ways are the tasks the same and different? Use the previous example, of developing understanding of punctuation, to help you.

Discuss the tasks with your mentor. If possible, try out the tasks with your class and review the outcomes of the lesson.



Task 4.1.6 Lesson planning for adaptive teaching: one stimulus with different tasks and/or outcomes

You have a set of photographs showing the changing landscape of London from 1850 to present. (If you do not like the choice of photographs, choose your own stimulus, for example, an astronaut working in a space lab, a contemporary image/photo from the news, or an environmental activist at work.) Describe two or more ways in which you could use these photographs to teach your subject. Confine your discussion to one to two lessons with a class you teach. For each example, identify:

- how you use the photographs;
- the activities you set your pupils;
- the objectives and learning outcomes;
- how you assess outcomes;
- the ways in which the activities are adapted.

Discuss with your mentor your plan, how tasks and/or outcomes have been adapted for your pupils and why your choice of task, learning outcomes and assessments (e.g. through different outcomes of different types of assessment) are appropriate.

Adaptive teaching through teacher input and support

Adaptation also takes place at the point of contact with the group or individual. The level and nature of your response to pupils is itself an act of adaptation, and includes:

- checking that pupils understand what they are supposed to do;
- listening to a discussion and prompting or questioning when needed;
- helping pupils to mind map or spider diagram an idea or problem;
- asking questions about procedure or techniques, suggesting further action when difficulties arise or motivation flags;
- giving pupils supporting worksheets or other written guidance appropriate to the problem in hand; the guidance might explain the topic in simpler terms or simpler language; you may, if appropriate, use word fills or add extra information;
- checking pupils' notebooks and noting progress;
- marking pupils' work;
- encouraging pupils by identifying success;
- setting achievable and realistic targets for improvement;
- increasing the demand of an existing task;
- noting unexpected events or achievements for a plenary session.

Now try Task 4.1.7.



Task 4.1.7 Adaptation: class–teacher interaction

With other beginning teachers, discuss the teacher support strategies from the previous list and add any other strategies you know about. Identify those appropriate to the teaching of your subject. Try them out in practice when you are able.

Knowing how to set adapted tasks depends on how well you know your pupils. The activity needs to be challenging, yet achievable. Other ways in which activities can be adapted include:

- the task's degree of open-endedness;
- the pupils' familiarity with the type of task (for example, pupils are often used to worksheet-based lessons where the information is given to them and they have to extract the information, and are not necessarily used to enquiry-based lessons);
- the pupils' degree of familiarity with the resources;
- whether the activity is a complete piece of work or a contributory part of a larger exercise;
- the amount of information you give pupils;
- the language level at which it is presented;
- whether the activity is set orally or by means of written guidance;
- degree of familiarity with the concepts needed to tackle the activity;
- the amount of guidance given to pupils, for example, in science lessons, the guidance given on making measurements, recording data or drawing a graph.

Now try Task 4.1.8.



Task 4.1.8 Adaptive teaching: how the task is presented and supported

Discuss the previous list with other beginning teachers in your subject. In groups, of say, two people, rewrite the list using strategies appropriate to your subject and the context of your teaching. Share your list with the group and go on to revise your own list.

Adaptation by outcome and how the activity is assessed

Differences in outcome may be recognised by the amount of help given to pupils and how the activity was set and supported. This aspect of adaptation was referred to in the previous task. In addition, your expectations of what counts as a satisfactory response to the activity lie in your assessment criteria. Your assessment strategy reflects your criteria. These criteria might include:

- the extent to which all aspects of the problem have been considered;
- the adoption of a suitable method of approaching the activity;
- the use of more difficult concepts or procedures in planning;

- the recognition of all the factors involved in successful completion of the activity and limiting the choice appropriately;
- thoroughness and accuracy of recording data in a quantitative exercise;
- appropriateness and selection of ways to present information and the thoroughness and depth of analysis;
- use of appropriate ideas (or theory) to discuss the work;
- accuracy and understanding of conclusions drawn from an activity (for example, are statements made appropriate to the content and purpose of the activity?);
- distinction between statements supported by evidence from speculation or opinion;
- the way the activity is written about, such as the selection of appropriate style for the target audience;
- the ability of pupils to express themselves in an increasingly sophisticated language;
- the use of imagination or insight;
- the selection of appropriate diagrams, sketches or pictures;
- sensible use of digital technologies supports a task;
- recognition of the limitations of the approach to a problem and awareness of ways to improve it.

By choice of assessment criteria, you differentiate the work set. Statements such as those listed help you construct your assessment strategy.

Knowing your pupils

Adaptation is good teaching and requires that you know your pupils. This knowledge enables you to judge the extent to which pupils have given an activity their best shot or settled for the easy option. Your role is to motivate your pupils and give support. Some pupils may present a greater challenge than others, as examples in the following case studies show. It is important to remember that you are unlikely to be successful with all pupils all the time. Read these case studies and then address Task 4.1.9 in the following. See Unit 4.6 on special educational needs and disabilities (SEND) and Pulsford and Rizvi (2019) for further discussion of adaptation of pupils with SEND.

Case studies of pupils

Hana

Hana joined secondary school with below average reading ages. Despite being 11 years old, Hana had a reading age of 5 years 3 months. Hana is confident and enjoys practical subjects, she has made friends easily and tries her best in all lessons. However, Hana struggles to complete long reading or writing tasks and this distresses her when she feels that she cannot keep up with her peers. Hana has high expectations of herself. She would like to be a doctor or dentist when she grows up.

Abbas

Abbas came to the UK from Syria when he was in Year 5. He is now in Year 8. When Abbas came to the UK he spoke basic English and could not read or write English. This is his third secondary school. Abbas is interested in the wider world and has excellent general knowledge. He can verbally express this knowledge well. His written communication is of a lower level than his oral communication, as

a result, in lessons if he cannot access the task, he can become disruptive and will start to attempt to entertain his peers. Abbas struggles to make friends and maintain relationships; he has already been moved classes once this academic year due to tensions in lessons.

Max

Max is a bright pupil. In Year 7, he was able to access content from the GCSE courses. He works very hard and finishes the class tasks quickly and efficiently, often well ahead of his peers. Max is an avid swimmer, he is hoping one day to compete in the Olympics. His dedication to his sport means that he has limited free time outside of school but always finishes his homework to an exemplary standard. Neither of Max's parents went to university but he hopes to be the first person in his family to go to university.

You may find it instructive to select one of the pupils described in these case studies and consider how their presence in your class would modify your lesson planning.

Now complete Task 4.1.9.



Task 4.1.9 Writing your own case study

Prepare a short case study of two pupils in one of your classes for whom further information would be helpful to you in lesson planning. Use the previous example case studies to help you identify the information you need to collect. Do not use the pupil's real name in any report you make or discussion outside the school; attention to ethical codes of conduct is part of your developing professionalism.

Collect information from the class subject teacher and the form tutor. The form tutor can give you background information about the pupils relevant to your study.

After collecting the information and writing your report, ask the subject teacher and/or form teacher to read it and comment on it. Finally, use the information to amend the last task or plan a new lesson, identifying some learning needs of these pupils and planning teaching strategies to take account of these needs.

SUMMARY AND KEY POINTS

This unit has considered different grouping arrangements at both the school and class levels and how to achieve progression for all pupils across different ages and abilities through adaptive teaching and learning approaches that consider a range of learning needs.

- This unit has explored various methods of grouping pupils for teaching, including streaming, setting, banding and mixed-ability classes. The benefits of each grouping method are not straightforward, and other factors, such as teacher expectations, significantly impact pupil achievement. The focus has shifted from grouping to in-class adaptation and personalised learning, although grouping remains part of the overall strategy.

- Progression and adaptive teaching have been discussed, with several examples provided. Tasks should be selected, set, supported and assessed to meet the diverse needs of pupils. Adaptive teaching is approached from multiple angles, starting with a simple planning model using stimulus, task and outcome. The teacher's role in supporting and guiding pupils is crucial. Strategies for managing adaptation emphasise that all classes have pupils with different needs, regardless of grouping methods. Adapting teaching is a responsibility for all teachers, not just those working with pupils with special educational needs.
- It has been suggested that tasks should relate to pupils' experiences whenever possible and that outcomes should be achievable. Discussions about Vygotsky and Piaget may help develop these ideas further.
- The skills of adapting teaching are gained through experience and a better understanding of pupils. While many beginning teachers change schools at least once during their ITT programme, it is essential to start understanding adaptive teaching approaches to enhance teaching and learning.

Check which requirements for your ITE programme you have addressed through this unit.



Further resources

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4.2 Adolescence, health and wellbeing

Understanding your role and PSHE

Sophie-Lauren McPhee

Introduction

Academic attainment does not exist in a vacuum. Although the Teachers' Standards require only 'an awareness of the physical, social and intellectual development of children' (Department for Education, 2011, p. 11), given that 'pupils with better health and wellbeing are likely to achieve better academically' (Public Health England, 2014, p. 4), it is incumbent upon us all to not only be aware of adolescent health, but actively promote it. For most beginning teachers, this will be through the role of form tutor, not least by contributing to the delivery of the tutor time curriculum in order to cover the statutory Health Education requirements (Department for Education, 2025). However, being knowledgeable about adolescent development and health serves a far greater purpose: to contribute to the cultivation of the whole human being. This unit invites you to adopt an understanding of your role as far more than inculcating knowledge for the purposes of an examination. As Wright (2024, p. 82) writes:

> We need to rethink the idea that school is a place whose sole purpose is to produce results that can be quantified. We need to recognise that what goes on in a school is about so much more than can be measured in an exam, and that the idea of education simply being an investment for the future leads to an endless deferral of the *now*. It's the difference between thinking of a house as a 'property' and thinking of it as a 'home'.

It is not an understatement to say that pupils' experiences of secondary school, not least the teachers they encounter, will stay with them for life. After their parents and caregivers, we are possibly the most influential adults they will form relationships with in their teen years. We must get it right.

OBJECTIVES

At the end of this unit you should be able to:

- Describe physical, social and psychological health differences in adolescence;
- Understand the importance of PSHE and your role and responsibilities;

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- Discuss what is meant by a 'whole-school' approach to promoting pupils' wellbeing;
- Decide on your own next steps with regard to training and development in this area.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

What is adolescence?

Adolescence is a transitional stage between childhood and adulthood that involves biological, social and psychological changes; however, we also need to understand the social factors influencing pupils' lives. For example, in an average class of secondary school pupils, a fifth will have a probable mental disorder, a third will have experienced a negative change in household circumstances in the past year … and one in six will have experienced in-person bullying in the previous 12 months (NHS Digital, 2023a). It is therefore important that teachers do not adopt a 'one size fits all' expectation of pupils, whilst simultaneously not imposing a glass ceiling on those enduring adverse circumstances.

Now do Task 4.2.1 to gain an understanding of circumstances affecting pupils.



Task 4.2.1 Profile of a year group: adverse childhood experiences

Select a year group you teach and find out about the adverse childhood experiences (ACEs) experienced by these pupils from the Designated Safeguarding Lead. For the purpose of this task, you need not find out which particular pupils have had such experiences, but rather the proportion of the year group.

Examples of ACEs are: physical, sexual and emotional abuse, living with someone who abuses substances, exposure to domestic violence, having a close relative in prison, living with someone with a severe mental illness or losing a parent.

Write a short report for your mentor which outlines your findings and considerations for your teaching practice.

Teachers also need to be aware of other key factors that can impact adolescents. Orban et al. (2022) suggest that the impact of social media use on life satisfaction differs between the sexes across the adolescent age range, but generally, increased social media use predicts lower life satisfaction at age 19. Additionally, over half of older adolescents (aged 17+) are now worried about climate change (NHS Digital, 2023a). Yet, according to Blakemore (2019, p. 7), 'adolescence is a formative period of life, when neural pathways are malleable, and passion and creativity run high'. It is therefore important that teachers take a balanced view of what is often deemed a turbulent time in a person's life.

Physical growth and development

By Year 7, some pupils will have already started puberty. 'It is not uncommon to have children in Year 4 … already showing the early signs' (Hunt, 2022, p. 87). However, just like with infant development, individuals go through these changes at different ages and at different rates. These differences may affect pupils' attitudes to and performance in academic work. For example, it can happen that some pupils who have developed physically earlier than their peers may feel self-conscious in front of their later-developing peers. If a pupil is not taking care of their hygiene or has acne, they may be subjected to unkindness. Attendance at school can be impacted; for example, nearly two-thirds of girls in the UK have missed school because of their period (Plan International, 2021). It is therefore important to observe closely whether pupils are being negatively impacted by physical changes and spot any patterns regarding school attendance.

Emotional development

When pupils feel good about themselves, they feel confident, ready to experience new things and resilient in the face of setbacks. Masten (2001, p. 235) writes that 'resilience does not come from rare and special qualities, but from the everyday magic of … the minds, brains and bodies of children, their families and in their communities'. As Kirkbride (2021, p. 258) suggests, it is important to 'offer opportunities for children and young people to develop a sense of their resilience in terms of challenges they have coped with and the skills they may have developed along the way'. Helping pupils feel good about themselves will help build strong relationships with your pupils and make behaviour management easier. During adolescence, emotions are particularly strong, so it is important for the beginning teacher not to take instances of 'emotional flooding' personally but rather approach them with curiosity.

Mental health

Although the majority of adolescents are unlikely to have a mental disorder (NHS Digital, 2023a), as Blakemore (2019, p. 158) states, 'this period of life does confer vulnerability to mental illness'. As someone working with young people, you are 'well placed to … identify those whose behaviour suggests that they may be experiencing a mental health problem or be at risk of developing one' (Department for Education, 2024a, p. 17). If you are a form tutor, you are in the frontline position to notice the subtle or significant behaviour changes that could be indicative of declining mental health, so it is important that you are proactive in this role, seeing it not just as an administrative duty, but as an opportunity to build relationships with and knowledge about the pupils who have been entrusted to your care.

It is essential that you familiarise yourself with the procedures in your school for raising mental health or safeguarding concerns. Many schools will use a secure online platform for this, and all schools should train their staff in safeguarding procedures on an annual basis. You should also be aware of the staff responsible for directly working with the pupils about whom concerns have been raised, such as the Designated Safeguarding Leads (DSLs), the Senior Mental Health Lead and any Social, Emotional and Mental Health (SEMH) mentors or school counsellors.

Now do Task 4.2.2 to familiarise yourself with safeguarding procedures in your school.



Task 4.2.2 Safeguarding

Obtain a copy of your school's safeguarding policy and familiarise yourself with the procedure for reporting concerns. Know who the Designated Safeguarding Lead(s) (DSLs) and Deputy Designated Safeguarding Lead(s) are and meet with them to learn more about their role and what classroom practitioners can do to support it.

Write a report for your mentor that summarises what you learned at the meeting about DSLs' roles and your responsibilities as a classroom teacher.

Identity

Identity development is the growth of a strong and stable sense of self and forms part of one of four 'qualities of our mind during adolescence', namely that of creative exploration (Siegel, 2014, p. 6), the others cited being novelty seeking, social engagement and increased emotional intensity (Crocetti, 2018). With access to the internet, the options for discovering and developing one's identity are vast, and, as Neufeld and Maté (2024, p. 9) assert, whilst in the past culture was 'always passed down vertically … the culture *our* children are being introduced to is much more likely to be the culture of their peers than that of their parents'; therefore what results may be a wonderful blend of past and present, or may be a source of internal struggle due to conflicting influences. As teachers, our role is to be interested in and support pupils in their identity development, being mindful of our duties regarding legally protected aspects of identity, and being alert to issues relating to identity that raise welfare or safeguarding concerns.

Racial and ethnic identities are an important part of how adolescents see themselves and how others see them. If pupils are aware of their race and ethnicity being seen, accounted for and celebrated, then this will contribute to wellbeing and a positive sense of self. Some schools run special events where pupils can dress in clothing and share food that is representative of their cultural heritage. There is also now a movement in education to 'decolonise the curriculum', because as McBean and Lynch (2022, p. 300) write, 'by dominating the curriculum with Eurocentric approaches, certain cultures become educationally maginalised'.

Thomas (2022) expounds on why diverse racial representation among teaching staff, within the curriculum and within teaching resources is important, and invites educators to consider how their own unconscious attitudes impact their interactions with pupils in the classroom. This idea of teachers taking responsibility for their own biases can be expanded to include all immutable characteristics, and for any protected under the Equality Act 2010, it is unlawful for a school to discriminate against a pupil by treating them less favourably (Department for Education, 2014).

Now do Task 4.2.3 to familiarise yourself with schools' duties under the Equality Act 2010.



Task 4.2.3 The Equality Act 2010

Read the Department for Education's 2014 document 'The Equality Act 2010 and schools' (listed in further resources section). Produce a one-page summary that covers the key points of your learning.

Despite this guidance, minority groups within schools are at higher risk of bullying. A 2021 report revealed that 42% of LGBT+ (lesbian, gay, bisexual, transgender and other minority sexual orientation and gender identities) pupils had experienced bullying – twice as many as their non-LGBT counterparts – and were twice as likely to contemplate suicide (Just Like Us, 2021). The same report found that Black LGBT+ pupils were three times as likely to contemplate suicide, which is a reminder to us not to see and treat each 'unit' of identity separately, but to account for how each aspect intersects with another. This is especially true for the intersection of religion or belief with sexual orientation and gender identity.

The issue of transgender-identifying young people is currently under public discussion, and Provisions to support gender-questioning children in schools (House of Commons Library, 2024) provides a helpful overview of the current guidance at whole-school level. However, it is important to consult your school's policy on supporting transgender pupils and consider additional training from organisations such as Brook. Indeed, any reading or research around any of the protected characteristics will contribute to the wellbeing of the diverse individuals in front of you. (See Unit 4.4 on responding to diversity and Unit 4.6 on inclusion, special educational needs and disability.)

Identity development, however, includes far more than just the protected characteristics. Some pupils will engage in political activism, and this may be an important part of finding their purpose and living their values. Whilst as educators we must remain politically impartial (Department for Education, 2022), we should encourage pupils to take an interest in the world around them with a view to figuring out who they will be in that world and how they will contribute to it, as research shows that purposefulness positively impacts subjective wellbeing in adolescents (Ratner, Li, Zhu, Estevez and Burrow, 2023). This journey of identity development may also include an accumulation of work experience and is likely to have implications for pupils' career choices, so teachers should encourage pupils to seek a range of opportunities outside of the classroom, including volunteering and employment.

Adolescence is a dynamic process in which identity can be iteratively formed and revised (Meeus, van de Schoot, Keijsers and Branje, 2012). You may notice pupils becoming depressed or anxious as they do not feel accepted as they explore their identities. They may be bullied for dressing or acting a certain way which will make them feel left out or discriminated against. This may cause them to become withdrawn and could lead to an increase in the risk of self-harm and eating disorders. It is therefore key for teachers to look out for and report to the school safeguarding team significant, protracted changes in a pupil's behaviour that could be indicative of any of the above or indeed other serious concerns. (See Unit 4.4 for further information on responding to diversity.) Another risk in the process of identity formation is radicalisation and extremism, so it is worth reading the government guidance on schools' statutory Prevent duty (Department for Education, 2023).

Although Siegel (2014, p. 8) acknowledges that 'searching for the meaning of life during the teen years can lead to a crisis of identity', on the upside, 'the brain changes of adolescence offer both risk and opportunity'. Whilst there is a developmental need to individuate, Siegel (2014), along with Neufeld and Maté (2024), advocate for parents to maintain attachment to their children during the potentially turbulent time of adolescence. Of course, being in loco parentis, this applies to us too. If 'the self – the human self – is a product of relationship' (Erskine, Moursund and Trautmann, 2023, p. 3), then the connections we maintain with our pupils and the time and care we put into them shape how they emerge into adulthood. Enjoy the process with them: take pleasure in observing their unfolding and the part you are playing in their journey.

A whole-school approach to wellbeing

In this section, we will consider how what we have covered above translates into the day-to-day operation of a school, although this is designed to be an overview and not necessarily what happens in every establishment.

Diet, exercise and sleep

It is essential that beginning teachers understand the implications of poor nutritional intake and lifestyle choices and how these may influence children's development, behaviour and conduct within the classroom. For children living in the most deprived areas, obesity prevalence is twice as high compared to those living in the least deprived areas (NHS Digital, 2023b). Contributing factors supporting these trends suggest that lower income groups struggle to maintain nutritionally balanced diets due to financial implications and knowledge of the benefits of healthy and active lifestyles. This suggests how important schools are in providing a network of support to children's families, which some do through breakfast clubs for pupils or information sessions for parents and carers. Your school may have an in-house family support worker who is responsible for coordinating such provisions.

Secondary school may be the first time pupils have had responsibility for choosing their own lunch from the canteen, and so knowledge about what constitutes a healthy diet is important. Such knowledge also forms part of the statutory content for Health Education (Department for Education, 2025). Although they will undoubtedly have been educated about this in primary school, it is important that messages get reinforced as pupils get older and they seek greater self-determination yet also approval from their peers. Cooking and nutrition forms part of the national curriculum for Design and Technology (Department for Education, 2013), but knowledge alone isn't enough to help pupils make the right choices. Therefore, Personal, Social, Health and Economic (PSHE) education should include a discussion of the many factors that come into play when we are choosing what to eat so pupils can have greater self-awareness when making decisions.

Each devolved authority has statutory guidance for food standards in schools. However, canteen staff also have a moral duty to monitor the pupils they are serving and reporting on any concerns they have. For example, a pupil may load their plate with just a potato-based item every day instead of a range of foods, another may be unable to pay for their meals day after day as more debt accumulates on their account or a third might have been caught stealing from the canteen several times, possibly indicating that there is not enough food at home. Maybe there is a pupil whose energy drink is peeking out of their blazer pocket, or another who only buys a fruit pot and has recently lost a lot of weight. You can see, therefore, that those working in school canteens could be the ones to spot a child in need.

As a form tutor, you too are well placed to observe physical changes in your pupils, which doesn't just cover significant weight loss or gain, but could be a general impression that a pupil is physically neglecting themselves, including not maintaining personal hygiene or getting enough sleep. It is also a good idea to make a list of the extra-curricular activities and hobbies your pupils are taking part in, not only in order to monitor the breadth of their personal development but also to encourage them to take part in more physical exercise if need be. Whilst schools are encouraged to embed PE and sport into the school ethos and culture, including by offering a minimum of two hours of PE

per week on the school timetable (Department for Education, 2024b), given that children and young people aged 5–18 are advised to undertake an hour of physical activity per day (NHS, 2021), the curriculum is again not enough to ensure healthy choices. In terms of your contribution, you may consider setting up a lunchtime club for your favourite sport, incorporate physical movement into some of your lessons, organise a trip to a local rock-climbing centre or deliver an assembly encouraging physical activities. It may be that you live close enough to role-model walking or cycling to school and back, or you could organise a staff vs. pupils football match. If you are lucky enough to have a school site with a lot of green space, you could take your form out on a nature or mindfulness walk, which has health benefits beyond the physical movement.

As with knowledge about nutrition, PSHE education can do a lot to extend the learning to add in critical thinking and develop self-awareness. With physical activity, this might be looking at the psychological benefits of exercise, or learning about how both plentiful exercise and high-quality sleep both boost learning power. Exercise 'stimulates the brain's release of a key growth hormone (BDNF), which … sharpens memory and concentration' (Ilardi, 2010) and in his very highly recommended book Why We Sleep, Walker (2018) tells us how good quality sleep can aid learning, with benefits available both before and after the learning window. Therefore, it is in every teacher's interest to promote a healthy lifestyle as it clearly has the potential to impact academic performance, and this is without considering how poor-quality food and too-little physical activity and sleep impact mood, and consequently behaviour and motivation in the classroom.

Working as part of a wider team

There are a lot of other staff in schools whose contribution to the wellbeing of pupils we have not yet accounted for, such as the school librarian, who can encourage greater engagement with reading not just for academic purposes, but for greater wellbeing, as research by the National Literacy Trust tells us that over half of children and young people find that reading helps them relax (National Literacy Trust, 2023). Librarians also help raise awareness of identity and diversity by creating book displays for Pride Month or Black History Month, for example.

Some schools also now have an attendance officer whose specific responsibility is to find out the reasons behind low attendance at school and find ways of supporting children and their families. Children and young people's mental health organisation Anna Freud asserts that 'reduced or non-attendance at school by a child or young person is often rooted in emotional, mental health or wellbeing issues' (Anna Freud, 2024, p. 2), and so in your role as form tutor it will be part of your responsibility to spot any patterns of absence among pupils in your form and work together with colleagues to implement agreed strategies for supporting and getting that child into school more often.

But what can you do as a subject teacher to contribute to a culture of wellbeing in your school? Apart from being observant and alert to behaviour and demeanour changes in your pupils that could be indicative of a mental health or safeguarding issue, you can take a proactive approach to creating a psychological safe space within your classroom. This means ensuring that all pupils are seen and valued, treated calmly and equitably, listened to and welcome. In doing so, you are creating the foundations from which their learning can take off.

Now try Task 4.2.4 to reflect on how you can contribute to a whole-school approach to pupil wellbeing.



Task 4.2.4 A whole-school approach to wellbeing

Identify what your school currently does to promote a whole-school approach to pupil wellbeing by interviewing other members of staff. Identify any gaps and therefore ways in which you might contribute to this area of school life by considering your own specialisms and interests.

Write a reflective journal piece conveying your thoughts and intentions, which can be added to your portfolio to discuss and submit to your mentor.

SUMMARY AND KEY POINTS

In this unit we have explored key ways in which you as a teacher working in a school can contribute to pupils' wellbeing in adolescence. It is important to remember that:

- Adolescents' health and wellbeing underpins their academic attainment.
- An understanding of adolescents' physical, mental, emotional, cognitive and social development is important foundational knowledge for all working in the secondary phase.
- All school staff must be aware of the process for raising safeguarding concerns, concerns about a pupil's mental health or significant behaviour changes.
- Teachers must be aware of schools' statutory duties under the Equality Act 2010 and issues relating to minority, marginalised or stigmatised identity groups.
- The role of form tutor is incredibly important for monitoring pupils' health and wellbeing.
- High quality, timetabled PSHE delivered by trained or specialist staff should form a part of a schools' approach to pupil health and wellbeing, alongside your understanding of your role in delivering PSHE.
- All staff are responsible for pupil wellbeing, whether teaching or non-teaching, and they must all work together as a team.
- The relationships that adults working in secondary schools form with pupils are significant contributors to pupils' self-regulation and their present and future wellbeing.



Further resources

Anna Freud – https://www.annafreud.org/, viewed 30 July 2025.
Anna Freud is a world-leading mental health charity for children and families. Their website offers research, resources and training.

Blakemore, S. *The Mysterious Workings of the Adolescent Brain* [video]. Available at: https://ed.ted.com/lessons/the-mysterious-workings-of-the-adolescent-brain-sarah-jayne-blakemore, viewed 22 March 2025.

British Nutrition Foundation – www.nutrition.org.uk, viewed 30 July 2025.
This site provides a wide range of resources for teachers, including activities for pupils that are downloadable free of charge.

Brook – https://www.brook.org.uk/, viewed 30 July 2025.
Brook operates a number of sexual health and wellbeing services across the UK, and its website provides advice and guidance for young people and those working with them.

McPhee, S. and Pugh, V. (2023) *Developing Quality PSHE in Secondary Schools and Colleges*. London: Bloomsbury Academic.

This book contains a comprehensive guide to planning and teaching PSHE, featuring contributions from a range of experts in the field.

The Association for Young People's Health – https://ayph.org.uk/. viewed 30 July 2025.

The Association for Young People's Health website offers data about the health and wellbeing needs of 10–25 year-olds, resources, and information about training courses.

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4.3 Cognitive development

Paul Davies

Introduction

During the secondary school years, pupils develop their knowledge and understanding of a wide range of subjects and also their ability to perceive, reason and solve problems. These skills are aspects of cognition (literally, 'knowing'). A key feature of cognition is that it involves us as learners in making sense of the world around us. As such, it is unlike more basic forms of learning such as memorising a song or rote learning multiplication tables. It includes skills that involve understanding, such as map reading, following instructions to make something, analysing data and solving problems (Willingham, 2021). Making sense, knowing, understanding, thinking and reasoning develop into adulthood and so cognitive development is an important feature of pupils' mental growth during the secondary school years.

Logical reasoning is one important aspect of cognitive development. Many school subjects require us to think and reason logically, e.g. when handling evidence, making judgements, understanding when and how to apply rules, untangling moral dilemmas or applying theories. Most Western societies in their schooling of children privilege logical, mathematical and linguistic abilities over other ways of knowing about the world. The tests of ability used by some schools to select new entrants or to allocate pupils to teaching groups are often problem-solving exercises involving pattern seeking, pattern recognition and pattern using, and the capacity to think logically and quickly.

This unit considers some of the ways in which pupils' cognitive abilities develop and are identified. The chapter also explores some of the cognitive demands made by activities in some curriculum subjects. Unit 5.1 addresses in more detail theories of how children learn and develop.

The evidence base for Unit 4.3

This unit draws mainly on theories and evidence from psychology in education concerning learning, thinking, reasoning and understanding and motivation. These topics are studied using systematic and rigorous psychological research methods, including observational studies and controlled experiments. With improvements in brain imaging and other techniques, research into learning and cognitive development is becoming integrated with neuroscience (see Unit 5.6) and revealing more detail about how the brain works and enables us to learn (Blakemore, 2012, 2019). Ideas are

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illustrated through a series of tasks while your observations of pupils as they grapple with learning are a key skill that provides insights into the difficulties they face.

OBJECTIVES

At the end of this unit you should be able to:

- understand some features of cognitive development and the cognitive demands made by curriculum subjects;
- explain and evaluate some ideas about the nature of intelligence;
- identify types of thinking and relate them to learning activities;
- begin to use tasks as a way of finding out about pupils' cognitive level.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

Differences between pupils

Differences between children are apparent from an early age. Even before they start school, some children pass developmental milestones such as walking and talking more quickly than others. Children may start reading and counting before they begin school or become very confident in their physical skills. When pupils start primary school, some are better than others at school tasks, and those who acquire good language and communication skills tend to be seen as more advanced and may be labelled as brighter than others, something which often leads to pupils being grouped by 'ability' in different subjects (Marks and Menter, 2025). In the secondary school, curriculum subjects call for the development and deployment of a range of cognitive skills. Each learning activity or task set by a teacher makes specific cognitive demands on the pupil, who may be well equipped to meet them or may have gaps in certain areas. At a given point in time, several factors may contribute to the pupil's capability, including their experience of similar tasks, their motivation to learn, characteristics and cognitive abilities.

Motivational beliefs and learning

One of the reasons that children may be more advanced in certain areas is that they are interested in the kinds of learning valued in school. It can be argued that schoolwork is a game that children have not chosen to play, but that others, teachers and society, have chosen for them. If this assumption is correct, it is likely that some pupils are not highly motivated by the content and focus of lessons, and, therefore, these pupils may be less successful in school. An alternative view is that we all have an intrinsic motivation to acquire competence and a tendency to protect our sense of self-worth, so those pupils who fall behind in their learning and feel they are not competent become demotivated and act in ways to protect their self-image.

Unfortunately, this reaction often involves maladaptive activity such as procrastination, denying interest or playing the class joker. This is challenging for teachers who often resort to extrinsic forms of motivation such as threats or praise. Such encouragement may be effective in the short term, but in the long run you are likely to find that it is beneficial to develop pupils' intrinsic

motivation by encouraging them to see the point of their work and emphasising their growing competence (Ryan et al., 2021). It is well documented that pupils work best at activities that they value (Wigfield and Eccles, 2000). See Unit 3.2 on motivation.

Pupils who do well in school subjects are sometimes thought to be 'more intelligent' than others or, more accurately, display more intelligent behaviour. However, Dweck (2012) found that some people view intelligence as a fixed capacity or 'entity' that sets a limit on what an individual can achieve, whereas others believe that intelligence is incremental and can grow with learning. These beliefs form 'Mindsets' that affect the way pupils and teachers approach learning (Dweck, 2012; Dweck and Yeager, 2020). An incremental view of intelligence carries with it the potential for change through teaching, whereas an entity view suggests that the effect of teaching is much more limited. Entity mindsets tend to limit pupils' achievements whereas young people and adults with growth mindsets thrive by taking advantage of learning opportunities; they believe that challenging tasks offer good opportunities for learning and that making mistakes is a natural part of the learning process (Dweck, 2012; see also Units 3.2 and 5.5 on personalising learning). Reflect back on your own experiences as a pupil at school. What mindset did you adopt when approaching challenges in your learning?

Schools commonly group pupils by attainment (see Unit 4.1), especially in some core curriculum subjects such as Mathematics. Although common practice, grouping by ability has been shown to make very little difference to overall pupil attainment (Jerrim et al., 2025). While ability grouping can make it more straightforward for the teacher to plan for a narrower range of pupils who have similar attainment in the subject, it can be detrimental to pupil progress, particularly for pupils of similar attainment who are placed in low ability groups (Ireson et al., 2005). Much research shows that grouping in this way can lead to the teacher not recognising the range within a group, and teaching as if the pupils are one, supposedly homogenous, group of individuals (Coe et al., 2014). This is a potential problem because, as discussed in Unit 4.1, pupils may require individualised support and failing to recognise this impacts on their attainment.

A question of intelligence?

Before reading on, complete Task 4.3.1.



Task 4.3.1 Intelligent people

Think of two people who you would say are intelligent – they could be adults or children. In what ways are they similar and how do they differ? Share your ideas with other beginning teachers in your group and make a note of the characteristics in your PP.

Although a wide variety of abilities are recognised in many cultures, within education the notion of ability generally relates to skills and competence in the performance of tasks that form part of the curriculum. Intelligence may be linked to a pupil's capacity to exercise linguistic and logical mathematical reasoning, though it is important to recognise that concepts of intelligence are complex and there is no simple definition that is universally accepted.

Existing tests of cognitive ability are based on a statistical assumption that if many people are tested, their scores form a bell-shaped distribution, similar to that for height or weight. Statistical methods have been used by researchers aiming to clarify relationships between the various cognitive abilities sampled by the test items, thus uncovering the structure of intelligence. The weight of

evidence from these studies suggests that there is a general factor underlying other more specific abilities (see Child, 2007, for a useful introduction). This body of work gives some support to classical theories of intelligence that suggest there is a general factor underlying our performance in a wide range of activities in school and work. However, there are some notable exceptions, too, where a person's intelligence test score may be unrelated to their level of success at particular problems that seem to demand considerable intelligence, such as solving mental arithmetic problems or remembering a very long list of unrelated digits. Thus, the classical position is open to challenge (Resnick, 2024).

Gardner (2021) and updated on his website, criticises classical theories for being concerned with only a very narrow range of human ability, namely language and mathematics. He argues that they fail to take account of many other aspects that are important in the real world. In his 'theory of multiple intelligences', he proposes that there are a number of relatively autonomous intelligences (see Figure 4.3.1) (Gardner, 2025a and 2025b). At one time, Gardner's theories played an important role in schools but some of the ideas behind the theory of multiple intelligences have received critical reviews (White, 1998, 2005) and it is important to note that there is little research evidence to support them, with what is reported in classrooms being mostly anecdotal. This raises an important issue in educational research where teachers 'feel' interventions and approaches in the classroom

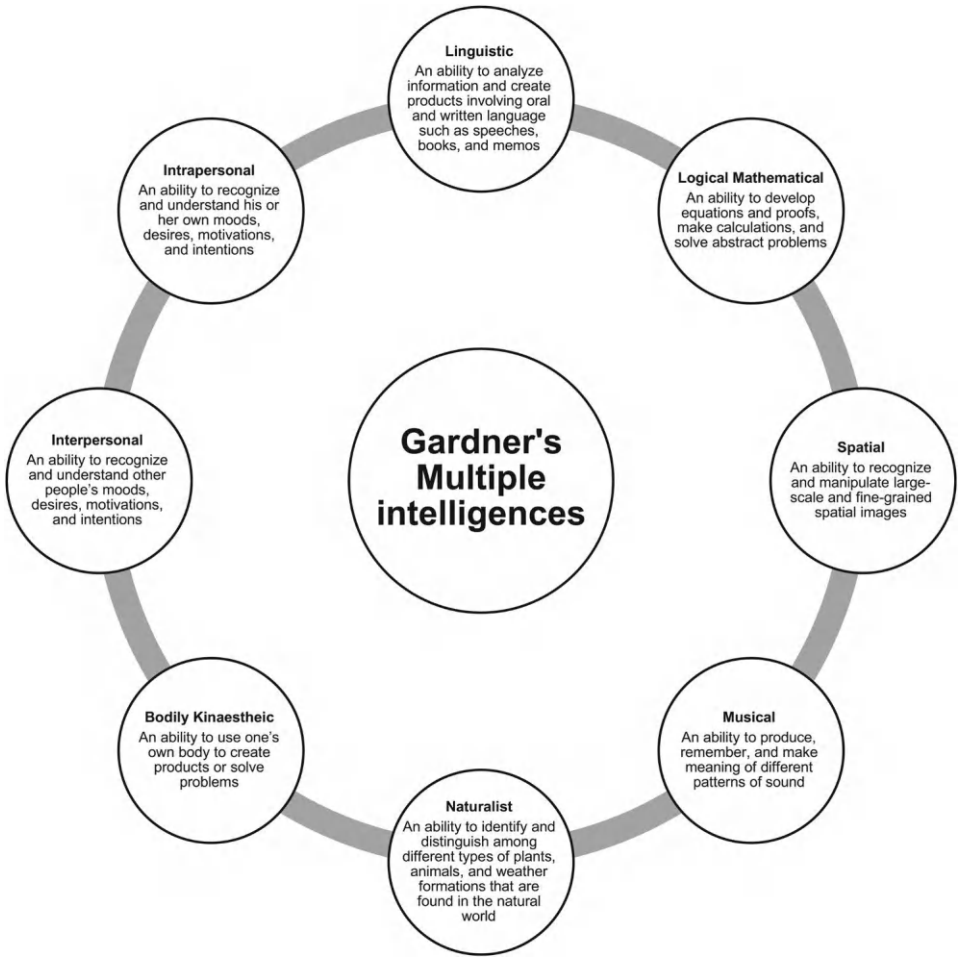


Figure 4.3.1 A summary of Gardner's multiple intelligences

produce reliable, and repeatable, changes while the education research community cannot produce robust evidence to support this. More and more, teachers are being expected to engage with research, both self-generated and published, to inform their practice, and they must be mindful of the persuasive nature of ideas that seem to 'make sense' but are not grounded in reliable research. Having said that multiple intelligence theory has had an important influence on classroom practice, for example, in designing different learning experiences catering for different learning dispositions and is still prevalent in some schools and teachers' thinking despite limited evidence of its use (Ferrero et al., 2021). For more information on Gardner's theories see the companion website.

It is now recognised that intellectual behaviour is a product of a general processing ability and a number of special abilities (Adey and Serret, 2010). Reflect back on Task 4.3.1 and consider how your ideas would map about a model of 'multiple intelligences' versus a general processing ability. A benefit of using some of Gardner's ideas might be that it can encourage you to think about the appropriateness of each learning activity for both the specific topic and individual pupils. Does this help you be more creative in your teaching?

Another problem associated with attempting to define 'intelligence' is the use of intelligence tests. There is a long history of how these have been applied in school, to both stream pupils into different ability groups and to determine entry to some types of school, notable grammar and many independent schools. The role of these tests and their use in school is complex school using data from national tests through to local testing in the classroom to group and, in some ways, define pupils. You will find it useful to research how such testing is used in your school and the impact it has on pupil motivation and progress. Further information about intelligence tests and how they are used by schools can be found on the companion website.

The role of character and cognitive ability

Character education has become an area of interest in education research (see Unit 4.5) an aspect of this specifically related to intelligence is the concept of 'grit' which Duckworth and Gross (2014, p. 2) define as 'perseverance and passion for long-term goals'. Duckworth and her colleagues have developed a 'grit scale' and found that people that scored highly for the attribute of grit were more determined and motivated, even when faced with challenges and failure. They have argued that grit is a better predictor of academic success than more traditional intelligence tests. Central to their ideas is that grit allows people to overcome setbacks and not give up (Duckworth 2013, 2017, 2025).

The idea of grit is beginning to have an importance in schools, particularly through curricula which are designed to develop pupil 'character' (e.g. University of Birmingham https://www.jubileecentre.ac.uk/character-education-/resources-for-schools/ and https://www.grit.org.uk/schools-colleges). In these programmes, pupils are encouraged to reflect on their attitudes towards learning and their beliefs about their abilities. The programmes show impressive change in pupils' academic outcomes as well as attendance and behaviour (Arthur and Harrison, 2023). Some of these interventions can be intensive while others are more straightforward and simply require a shift in how teachers talk about progress being careful to frame their language about academic outcomes in terms of effort, rather than natural ability or talent.

Central to grit is perseverance. This can be a challenge for pupils, especially when faced with subject matter and tasks they find boring or hard. Teachers can do much to help pupils see relevance in their learning and scaffold tasks to help them access complex ideas but, at the same time, pupils need to learn to not give up. One way of characterising this is through the concept of 'delay of gratification' where learners do not look for instant reward but are prepared to work hard, learn

from their mistakes and accept that they might not fully understand a concept for some time (for a review, including some of the problems of these approaches in school, see Audley and Donaldson, 2022). As children grow older, some of these attributes develop as the brain matures but they can also be taught. Classroom approaches such as teachers modelling what mastery of a concept looks like and talking through the steps involved to develop full understanding, as well as showing how they use their own mistakes to find solutions to problems can have a powerful impact on pupils.

The role that grit plays in developing skills and academic outcomes is not without criticism, the main one being that taking grit in isolation is problematic because motivation tends to be situation specific (Ericsson, 2009). Ericsson also argues that attributing ideas of grit and willpower to success can also be damaging for individuals who do not succeed to a high level. This might lead to learners giving up or feeling a failure. While this is important, it does not mean that teaching children to strive for greatest should not be encouraged, it is how teachers go about encouraging this that really matters. There is an opportunity to further explore these ideas for yourself now in Task 4.3.2.



Task 4.3.2 Grit and 'delay of gratification'

Think about a class that you teach and the how the pupils engage with their learning. Are there some which seem to give up more easily than others? What do you notice about those pupils who don't give up so easily? Do they find the concepts and skills they are learning easier to master or are they prepared to work harder and put more effort into their studies?

Think about a complex idea that you are going to teach. How will you break the ideas down to help pupils master the concept and skills? Consider how you will talk through your own thinking as you explain the idea and make reference to the trickiest aspects of the idea.

To encourage pupils to develop a 'delay of gratification' approach to their learning, create a task around one aspect of the concept and skills that you are teaching. The task should be:

- Well defined with a clear goal
- Carefully and clearly described and explained by you, the teacher
- Something the pupils can complete on their own
- Something where you are able to provide pupils with feedback as their progress though the task and provide information to help the pupils succeed
- Something that pupils could apply to different areas of their learning

After you have taught the lesson, record how the pupils responded. Consider especially whether the task allowed pupils to see their successes as the task progressed. Also consider how easy it was for you to plan the activity. What has this taught you about the need for careful and thoughtful planning seeing learning from pupils' perspective? You should discuss your responses with your peers and mentor.

Developing cognitive abilities

Nature and nurture

In general, pupils' cognitive abilities increase with age. Older pupils are more capable of abstract, symbolic thinking than younger pupils, who tend to use more concrete representations when solving problems (see the discussion of Piaget's (1957) theory of cognitive development in Unit 5.1). It is

also clear that in any year group there is considerable diversity, with some pupils able to cope with more demanding work, while others have great difficulty. Several different explanations have been suggested for this diversity and for the general pattern of development during the school years. Some of these are briefly outlined in the remainder of this section.

So, what might influence the development of cognitive abilities? Most answers to this question emphasise biological and environmental factors, or 'nature' and 'nurture'. 'Nature' refers to our inborn, genetic or inherited characteristics and how these unfold during maturation, while 'nurture' refers to the environment in which we grow and develop. Genetic factors undoubtedly affect development, but the extent of this influence is not entirely clear, as it is difficult to disentangle genetic and environmental processes. The most influential studies (e.g. Gustavson et al., 2022) compare twins raised together with twins separated at an early age and raised in different families. These studies show that inherited characteristics do have a strong effect, but they do not completely determine measured intelligence; the child's environment also makes an important contribution (see, for example, Freis et al., 2022).

A pupil's attainment on entry to secondary school is influenced by their experiences in the home and in primary school (their 'environment'). Evidence has accumulated to suggest that a lack of stimulation in early childhood limits the capacity of children to benefit from school and other learning situations (Smyth and Duta, 2023). Their research has involved following children from age 3 to 7 to develop a measure of 'home learning environment' (HLE) to describe activities such as reading with the child, playing with letters, numbers and shapes and taking children on visits to the library. A 'good' HLE contained these activities, regardless of the economic status of parents. Children from such homes have a head start when they enter primary school, as they have started to develop many of the cognitive, communicative and social skills needed in school. Later on, their parents may support them directly and indirectly, e.g. explaining homework or enabling them to participate in a wide range of activities. While the majority of parents are able to provide a good HLE, some may find this difficult, perhaps for economic, social or health reasons.

A pupil's performance in attainment in school is thus a reflection of their innate abilities and the influence of the environment. You should remember that things like attainment tests tell you what a pupil knows, understands and can do at a particular point in time; however, they are not necessarily reliable predictors of future attainment. Pupils who have had less support for their learning in the past, or who do not speak English as their first language, may be more capable than suggested by assessments (Hessel and Strand, 2023). For these reasons it is important to take care not to label pupils in a deterministic way as high, middle or low ability on the basis of attainment tests when they enter the school.

Knowledge base

Another important factor that is easily overlooked is the amount of knowledge an individual has acquired. It is sometimes assumed that the more advanced thinking of adults is due to biological maturation, when in fact it may be owing to increased knowledge. People who are experts are much better at recognising patterns in information and using principles to solve problems. This applies in domains as diverse as medicine, music, physics, computer programming and teaching (see Ireson, 2008). Experts build up a strong knowledge base and use it effectively to solve problems. For pupils, the development of a strong knowledge base in school subjects is important for remembering and thinking. It also makes learning more enjoyable and satisfying. Good teaching and active learning (see

Unit 5.2) should help pupils to build up a well-structured knowledge base, but this may be undermined by pressure to cover the curriculum, leaving insufficient time for understanding and consolidation.

One way of addressing this problem is through what Dehaene (2020) describes as 'Consolidation' (see also Unit 5.6). In all learning, there is a 'shift from slow, conscious and effortful processing to fast, unconscious and automatic expertise' (Dehaene, 2020, p. 272). In his work on making links between neuroscience and cognitive science, he argues that for knowledge to become implicit, learning must involve revisiting of concepts which allows the brain to move from a linear pattern of accessing information to a parallel one. For the learner, this means that knowledge is able to be used in new ways, making links across different knowledge. In schools, this is often accomplished through 'retrieval practice' where material is revisited in planned and explicit ways. Rosenshine's Principles of Instruction (Rosenshine, 2012) provides a useful summary of the elements of good teaching and learning, many of which are focused on how to 'chunk' information for pupils to make learning new information manageable and approaches which support retrieval practice (also see Sherrington, 2019). These ideas have become very important in schools in recent years.

Related to Dehaene's idea of learning moving to a fast, unconscious process is the psychological concept of 'flow'. First theorised by Csikszentmihalyi in 1977 (see Csikszentmihalyi and Csikzentmihaly, 1990 for a useful overview), in essence flow describes a psychological state where the person is fully immersed in an activity which they are also enjoying. A 'flow state' requires high levels of concentration and strikes a balance between the skills being used and the challenge of the task, if these are mismatched, then a flow state is not achieved, see Figure 4.3.2.

Flow theory proposes that to achieve a state of flow the activity must have a clear goal, there must be some aspect of immediate feedback which allows for adjustment to the way of working and the person must believe that they can achieve the goal of the activity. Various studies have shown that flow experiences are good for learning, not just in terms of learning concepts but also enhancing creativity and innovation (Katz-Buonincontro and Anderson, 2020). Planning sequences of lessons that support flow is not straightforward but something that you should have in mind. To help,

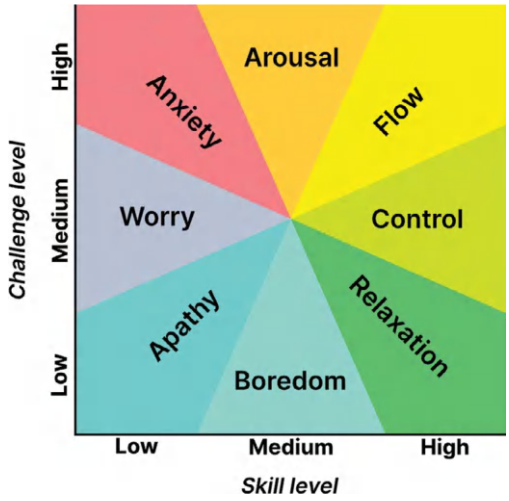


Figure 4.3.2 Csikszentmihalyi's Flow Model showing the relationship between skill and challenge level (based on image from Wiki Commons (2025): https://commons.wikimedia.org/wiki/File:Challenge_vs_skill.svg)

in your lesson, try to encourage your pupils to view learning as a process that takes time. You will also find it useful to ask your pupils to reflect on their learning and motivation, as discussed above.

Curriculum and development

Several ideas have now been introduced to explain pupils' performance in school. It has been suggested that performance is a product of inheritance and environment. Also, that, as they grow, develop and learn, pupils become more able to handle complex problems and situations and abstract (formal) thinking. You are not in a position to influence inheritance or the HLE, but you can influence classroom learning by providing activities that are aligned with pupils' current knowledge and understanding. To do this requires a careful analysis of the demands made by a piece of work and the current performance of the pupil or group of pupils with reference to relevant knowledge, skills and understanding needed to do the task.

Programmes of study for each of the national curriculum subjects set out aims for all pupils, in terms of their thinking and learning. You should review the programme of study for your subject and compare the aims with those for a different subject. You will notice that, even though each subject has different content knowledge and emphasis, there are also some similarities in learning and thinking processes across subjects. These include: thinking and reasoning skills required to draw inferences and make deductions from foundational knowledge, elements of problem-solving and reasoning and the use of evidence to develop and justify arguments and give explanations. Some subjects also place emphasis on pupil experience, imagination and creative thinking to generate and extend ideas. Now complete Task 4.3.3 to explore how this relates to your subject.



Task 4.3.3 Thinking skills in your subject and cognitive challenge

You should visit the Department for Education website for the Programmes of study by subject section (https://www.gov.uk/government/collections/national-curriculum#programmes-of-study-by-subject), find your subject and familiarise yourself with the 'Aims'.

Having identified the aims for your own subject, pick out statements that mention thinking skills. Which of these are likely to be the most and least challenging for pupils?

Now turn to a specific piece of work or topic you are planning to teach and identify the cognitive and other demands for pupils.

When you have clarified the task demands, carry out an assessment of pupils' current performance (through a process that represents an example of assessment for learning, see Unit 6.1; Procter, 2013).

Use the evidence from the assessment to inform your planning so as better to guarantee some success while offering a comfortable level of challenge.

This strategy depends on you being able to analyse the cognitive demands of the topic and associated learning activities and the relevant capabilities of pupils. See Unit 4.1, which covers processes of differentiating and personalising learning to meet the diverse needs of pupils.

While thinking about the cognitive demand of your subject, you should also consider the common misconceptions or naïve conceptions that pupils have in your subject. These are incomplete or inaccurate or intuitive ideas that pupils hold which can be barriers to learning. While these are

common in science, as shown in aspects of the pendulum task (see Task 4.3.5 on the companion website) they also surface in other subjects. Completing Task 4.3.4 gives you the opportunity to get to know about misconceptions and commonly held naïve conceptions in your own subject. This can help you see why pupils have difficulty with some new ideas and ways of thinking.



Task 4.3.4 Common misconceptions in your subject

Misconception and naïve conceptions can be barriers to learning. In this task, you should research that these might be for a topic that you are going to teach and consider ways to explore these ideas with pupils and encourage them to reflect on their thinking.

Thinking about a topic you are going to teach, identify common misconceptions by exploring the literature and speaking to more experienced teachers. You should then design a task (e.g. multiple-choice, open-ended questions, a concept cartoon) that allows pupils to identify and address the misconceptions.

Following these steps will help you:

1 Identify a Misconception
Research common misconceptions in the topic area you're focusing on. Consider what pupils often misunderstand or struggle with.

Examples:

- Science: 'Plants get food from the soil' (instead of photosynthesis).
- Maths: 'Multiplying always makes a number bigger'.
- History: 'All people in a certain time period were the same'.
- English: 'A summary is just rewording the text'.

Consider why it is important? Explain why addressing this misconception is crucial for pupil learning and understanding.

2 Design a Task to address the misconception
There are lots of approaches you could take, but here are some examples.

Multiple-choice:
Create questions with a correct answer and a few distractors that reflect the misconception.

Open-Ended Questions:
Pose questions that require pupils to explain their thinking and reasoning, allowing you to uncover misconceptions.

Concept cartoon:
An Internet search will return many examples of these where three-four alternative ideas about a problem are presented for pupils to discuss. They are similar to a multiple-choice question expect the task focuses on pupils talking about the ideas and deciding which are correct and, importantly, which are incorrect and why.

After you have taught the lesson, reflect on the misconceptions that the pupils struggled with and how their thinking developed. Discuss with your mentor how you might adapt the task you designed to further support pupils in the topic.

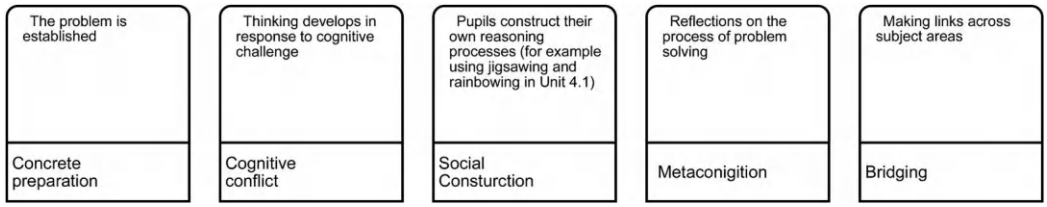


Figure 4.3.3 Five Pillar Model of the Cognitive Acceleration Programme

Cognitive acceleration

An important intervention that was designed to support pupils in their thinking about complex ideas is the Cognitive Acceleration Programme (Shayer and Adey, 2002). This researched-based programme incorporates several key components through what is termed the 'Five Pillar Model', see Figure 4.3.3. The programme was first developed for science education with mathematics and English programmes following soon afterwards. The emphasis in these programmes is on helping pupils to understand the context, to 'learn how to learn' and begin to construct meaning for themselves.

The programmes are highly prescribed with teaching needing specific training in how to teach each lesson. The structure of the lessons always follows the Five Pillar Model with pupils passing through each stage as their thinking develops (Shayer and Adey, 2002). A key aspect of the programme is that they help pupils become aware of their own learning strategies. This contributes to a pupil's sense of competence, something that is an important aspect of motivation and one that may help to raise pupils' aspirations (see Ryan, Deci, Vansteenkiste and Soenens, 2021).

Schools sometimes develop their own curriculum based on these cognitive accelerator programmes, or incorporate other approaches that draw on this thinking, for example, the Let's Think programme (https://letsthink.org.uk). These provide teachers with resources to support the use of the Five Pillar Model in the classroom as well as allowing them to tailor the lessons to the specific needs of their pupils. Visit the companion website if you are interested in cognitive acceleration programmes to find out more and explore a task which encourages you to reflect on how the programme might be applied in your own subject (Let's Think, 2025).

What does it mean to be 'good' at a subject?

There has been much debate in research education about the role of schools and, particularly the relationship between assessment and pupil ability (for example, Jacob and Rothstein, 2016). An important function of schools is to prepare pupils for public examinations, the results of which are high-stake for both the individual and the school. But does being good at passing examinations mean the same as being good that the subject? When considering this question, it is worth spending some time thinking about what drew you to your studies and your decisions to train to be a teacher. Was it that you found particular subjects easy at school or was it that you were particularly interested in the subject which led you to further study?

In the current system of assessment in most countries, the ability to 'perform' in examinations is the most important measure of success. But, in your classrooms you will come across pupils who are really interested in your subject and know lots about it but find the assessments very challenging. For example, pupils are often fascinated and knowledgeable about natural history, being able to name and classify a plethora of living things as well as explain their ecology and behaviour.

However, this type of knowledge is not valued in GCSE Biology, which can be demoralising for pupils, especially if they find the material they expected to learn is disengaging or too difficult. As a teacher, there is little you can do to change the assessment criteria of public examinations, but you have lots of control about what happens in your classroom. It can be useful from time to time to pause and reflect on why you are interested in your subject and share this with your pupils. This will often open out into discussions about their interests, which can be surprising. As you gain confidence as a teacher, you can then take these interests and weave them into your lessons, perhaps as a novel context when exploring a specific concept. Even better would be if you can ask reluctant learners to talk about their interests. To deepen your knowledge about the pupil experience of school you may wish to read Barbara Calvert's book *The role of the pupil* (Calvert, 2024). Although first published in 1975, it will give you insight into how children view school and the struggle they often face when their learning doesn't match their interests. For a more update analysis of pupil experience in school, you may like to read Daniel Willingham's (2021) *Why students don't like school*.

Linked to these competing views of how we value different types of knowledge in the curriculum is the idea of creative problem-solving. In a useful review of training approaches in schools, Sala and Gobet (2019) provide a compelling argument for not teaching knowledge skills at all but, instead, integrating their use into real problems within the curriculum. This is a radical shift from the content-heavy curriculum but ideas that are popular in some schools and straightforward to weave into your lesson. To find out more about these approaches, visit the companion website.

However, your career develops in teaching, it takes time, and the right school, for you to develop the confidence and skills to operate a classroom in a way that can support and inspire all learners. Learning is hard but try not to forget why you became a teacher in the first place. The answer to that is almost always, in part, because of your love of your subject and desire to instil this in children.

SUMMARY AND KEY POINTS

Cognitive development is described as a process through which pupils develop their knowledge, understanding, reasoning, problem solving and creative thinking. All these aspects of thinking develop into adulthood and so cognitive development is an important feature of pupils' mental growth through the secondary school years.

In thinking about cognitive development across various secondary school subjects, this unit brings together the following key ideas:

- Research reveals that there are differences between pupils in terms of the cognitive processes they use in, and the cognitive demands made by, different curriculum subjects;
- Carrying out the practical tasks in this unit encourages you to analyse the cognitive abilities involved in completing them as well as providing you with opportunities for finding out more about your pupils' capabilities;
- Key to this is listening to your pupils and finding out about their perspective on learning activities, through this you learn more about their understanding and thinking. This process helps you to design activities to assist their development.

We refer you to Units 3.2, 5.1, 5.2 and 5.6 for further discussion of related topics.

Check which requirements for your ITE programme you have addressed through this unit.

Further resources

Blakemore, S.-J. (2012) *The Mysterious Workings of the Adolescent Brain*, TED Talk, www.ted.com/talks/sarah_jayne_blakemore_the_mysterious_workings_of_the_adolescent_brain?language=en, viewed 30 March 2025.

Sarah-Jane Blakemore gives an excellent introduction to the workings of the brain and illustrates important changes that take place during adolescence. If this inspires you, you may wish to read the accompanying book, Blakemore, S.-J. (2019) *Inventing Ourselves: The Secret Life of the Teenage Brain*, Oxford: Blackwell.

Dehaene, S. (2020) *How We Learn: The New Science of Education and the Brain*, London: Allen Lane.

An accessible account of the brain's ability to learn and adapt to circumstances. Exploring the boundaries of cognitive psychology, neurobiology and education, the author suggests how educators might make best use of the brain's own algorithms.

Duckworth, A. (2017) *Grit: Why Passion and Resilience Are the Secrets to Success*, London: Vermilion.

In this book, Duckworth summarises her research into resilience and grit, bringing together ideas from business, industry, academia, education and sport. It provides a readable overview of that will provide you with tools to help pupils recognise that is important in their learning and how to develop attributes which will help both their academic progress and adult life. You might like to take the 'Grit Scale test' to see how gritty you are. Also see, https://angeladuckworth.com/grit-scale/ (viewed 30 March 2025) and listen to Duckworth talking about here work as 'grit' as a predictor or success: https://www.ted.com/talks/angela_lee_duckworth_grit_the_power_of_passion_and_perseverance?subtitle=en, viewed 30 March 2025.

Gardner, H. (2021) *Disciplined Mind: What All Students Should Understand*, New York: Simon and Schuster.

This brings together Gardner's previous work on his theory of multiple intelligences and re-examines the purposes and aims of education. Gardner provides a summary of his original theory and argues that the concept of intelligence should be broadened, but not so much that it includes every human faculty and value. He also describes how multiple intelligence (MI) theory has evolved. You might like to follow this up by watching this MIT Press talk where Gardner discusses the development of his idea of multiple intelligences and how his theories have evolved. Remember though, this theory is contested with respect to requiring more evidence to support its existence. https://www.youtube.com/watch?v=Q2_AQ6LtW5c&t=1s, viewed 30 March 2025.

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4.4 Responding to diversity

Helen Bowhay

Introduction

As teachers, we have the privilege of working with diverse pupils who bring unique knowledge, skills, and perspectives to the classroom. Each class reflects a multicultural society, encompassing varied cultural backgrounds, languages, identities, beliefs, and abilities. Recognising and celebrating this diversity is essential to fostering an inclusive, respectful, and supportive learning environment.

However, creating a more inclusive and equal society in the UK remains challenging (EHRC, 2023). Anti-immigrant rhetoric, prolific during the Brexit debate (2013–2020), media portrayals of refugees and asylum seekers, and rising extremism have contributed to a rise in racism, xenophobia and islamophobia with a resultant growth in far-right extremist parties. The 2024 UK riots, sparked by misinformation following the fatal stabbing of three young girls, serve to highlight the significant and damaging impact the media can have on society. Government policies like Prevent (HM Government, 2011), introduced to identify people who may be at risk of turning to extremism or terrorism, have further divided society (Thomas, 2019). While events like George Floyd's death, the Windrush scandal and debates over LGBTQ+ rights serve as a reminder of the inequalities that still exist in society.

To be a culturally sensitive and responsive teacher, you will need to understand how your pupils are different, so that you can adapt your teaching to their particular needs, addressing barriers to learning that may exist for some. This is easy to write but much more difficult to implement. Your own experiences of education and your perspectives on the educational, social, and political contexts of the day have a significant influence on the way in which you respond to diversity and approach inclusion.

This unit helps you to understand diversity in modern UK society, particularly how social diversity, educational opportunity and attainment are interrelated. It highlights the importance of inclusive education, the challenges faced by marginalised pupils, and the role teachers play in fostering change. As you read the unit, keep in mind that intelligence is not fixed, but incremental, depending on learning opportunities. (See Unit 4.3 on cognitive development.) While data primarily focuses on England, equivalent sources exist elsewhere.

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OBJECTIVES

At the end of this unit you should be able to:

- understand the history of diversity in the UK and its impact on society and education;
- access evidence about the relative academic performance of pupils from marginalised groups;
- discuss some of the barriers to education faced by marginalised groups of pupils;
- consider school policies and critique classroom practices in order to promote better opportunities for learning for all pupils.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

A history of diversity in the UK

Throughout the last century the population of England and Wales has steadily increased, from 37.9 million in 1921 to 59.6 million in 2022 (Office for National Statistics (ONS), 2022b). Minority ethnic population in England and Wales reached 25.6 per cent of the population in 2021, an increase from 12.5 per cent in 2001 and some studies suggest this may reach around 30 per cent by 2050 (ONS, 2022a). Post-Brexit migration from EU countries has declined, while non-EU migration has increased (ONS, 2020).

The increased diversity of our society does not just come from the range of ethnic and cultural backgrounds it represents. Advancements in health and social care mean more children are surviving with complex needs. In 2018/2019, the percentage of children reported as having a disability was 11 per cent, an increase of four percentage points since 2012/2013 (Department for Work and Pensions (DWP), 2024).

The Joseph Rowntree Foundation monitors poverty and social exclusion on an annual basis and highlights that British society is becoming increasingly polarised significantly exacerbated by the recent cost of living crisis. In 2020/2021, they reported 13.5 million people were living in poverty and of that 13.5 million, 3.9 million were children (Joseph Rowntree Foundation, 2023). Professor Philip Alston, United Nations Special Rapporteur on extreme poverty and human rights, produced a scathing report following his visit to the UK in 2018, decrying the austerity measures implemented by the Government since 2010 with the costs falling disproportionately on the poor, women, racial and ethnic minorities, children, single parents and people with disabilities (Alston, 2018). Austerity measures led to child poverty levels rising from 27 per cent in 2010/2011 to 30 per cent in 2018/2019. Indeed, in 2020 UNICEF, launched a domestic emergency response in the UK to help feed children affected by the Covid-19 crisis (Storer, 2020). In 2020/2021, child poverty levels decreased to 27 per cent largely due to increases in Universal Credit payments as a result of the Covid-19 pandemic (Ray-Chaudhuri et al., 2023). However, the cost of living crisis experienced post pandemic has seen child poverty return to the levels reported in 2018/2019 (Ray-Chaudhuri, Waters and Wernham, 2024).

Inclusion and educational equity

To respond to the growing diversity in UK society, and create a truly inclusive education system, we need to challenge the beliefs and attitudes that position difference as a threat rather than something to be valued and celebrated.

In 1994, a World Conference on Special Needs Education was held in the city of Salamanca leading to the Salamanca Statement and Framework for Action on Special Needs Education. While the main focus was on special needs education, the aim was to enact this through the development of inclusive education systems. The statement concluded:

> Regular schools with [an] inclusive orientation are the most effective means of combating discriminatory attitudes, creating welcoming communities, building an inclusive society and achieving education for all; moreover, they provide an effective education to the majority of children and improve the efficiency and ultimately the cost-effectiveness of the entire education system.
>
> (United Nations Educational, Scientific and Cultural Organization (UNESCO), 1994, p. ix)

Being inclusive entails unearthing and addressing all forms of exclusion and discriminatory attitudes and actions that may negatively impact marginalised groups. Inextricably linked to inclusion is the issue of equity. For many people it is self-evident that the implementation of equal opportunities policies reflects basic human rights, but how does the notion of equality of opportunity relate to a teacher's response to diversity? Should the same curriculum, teaching styles etc. be used consistently, or does such 'equality' in fact perpetuate inequality and so should we be calling for equity instead? Sociologists of education have argued for many years that if you assume that all pupils come to school equally prepared, and treat them accordingly, then, in reality, you advantage those who have been better prepared in their social milieu to succeed at school (Bourdieu, 1974; Bernstein, 1977; Willis, 1977 Coleman, 1988; Putnam, 2001; Major and Machin, 2018; Thompson, 2019; Hutchings, 2021).

Historically, underachievement has been blamed on pupils or their families. More recently, the focus has shifted to systemic factors, from government policies to classroom practices. This unit examines barriers to attainment linked to gender, ethnicity, socio-economic status, and bilingualism. When engaging with this unit remember that each pupil's identity is multidimensional and, therefore, there will inevitably be intersectionality between the different categories explored. Intersectionality was a term coined by Professor Kimberlé Crenshaw to describe how race, socio-economic status, gender and other individual characteristics 'intersect' with one another and overlap (Crenshaw, 1989). Thus, intersectionality is an important framework to understand how different aspects of a person's identity, e.g. race and gender can lead to discrimination and oppression.

Gender

Despite progress in gender equality, disparities remain. In 2023, GCSE results there was a 6.8 per cent gender gap with 73.7 per cent of females achieving a grade 4/C or above compared to 66.9 per cent of males. However, the gap does vary according to the subject with larger gaps in English with girls performing higher (13.3 per cent) but narrower in maths (1.5 per cent) with boys performing higher (Education Policy Institute, 2023).

In the past few decades, there has been a significant effort to promote STEM subjects to girls and for the first time in 2019 in England more girls than boys entered science subjects at A level, thus addressing some long-standing gender stereotypical biases in relation to subject choice. However, it could be argued why there has not been a similar focus on the lack of an uptake of arts, languages, and humanities subjects for boys.

Gender stereotypes persist and shape aspirations and self-perceptions of young people. Teachers can often, unintentionally, treat boys and girls differently which further exacerbates the issue (Skipper and Fox, 2021). In this sense each identity is socially constructed rather than one reflecting the views of individual boys and girls.

Campaigns such as #likeagirl served to highlight how stereotypical views of what it means to be female can influence and shape a young person's identity and, potentially, put limits on aspirations. Similarly, consider the slogans found on clothing for children with young boys depicted as 'little monsters' or 'trouble' in contrast to slogans referring to girls as 'little princesses'. More recently, the growth of online misogyny and 'toxic masculinity' in the 'manosphere' is influencing the attitudes of male pupils towards female pupils and staff thereby increasing the prevalence of sexual discrimination in schools (Over et al., 2025).

Responding to these issues has led to attempts to promote 'healthy masculinity' but this serves to reinforce gender binaries rather than promoting the recognition of all gender identities (Waling, 2019). As discussed in the next section, rather than seeing gender as a binary concept, we should consider gender identity and expression as more fluid concepts that may shift as children grow up. Similarly, there is a need to put a greater emphasis on examining the intersectionality between gender, attainment and other factors such as race or socio-economic status (Corry, 2017).

Gender identity

So far, we have been looking at the relationship between performance and biological differences between pupils; however, the notion of gender is more nuanced, including different masculinities and femininities. Traditional stereotypes are being broken down and, at the same time, becoming more fragmented. Such differences are increasingly celebrated, and this includes pupils' sexuality and gender identity. Thankfully, we have come a long way since the removal of Section 28 (the legislation that banned any discussion of same-sex relationships in schools) in 2003. However, nearly half of all LGBTQ+ pupils still face bullying due to their gender identity (Stonewall, 2019). Trans people in particular face inequality and discrimination in many aspects of their lives with myths and misconceptions frequently cited in the media (Stonewall, 2021). In 2020, Relationships and Sex Education and Health Education (RSHE) was made compulsory for all secondary school pupils and Health Education compulsory for all pupils in state-funded schools. Through the new curriculum all secondary pupils should be taught about sexual orientation and gender identity (DfE, 2019). The introduction of RSE to schools' curriculums was a positive move and it was hoped that LGBTQ+ young people and those from LGBTQ+ families would feel more valued and respected and all young people would grow up with more inclusive attitudes and values. However, in May 2024, the then Conservative Government brought forward the first review of the RSHE curriculum following alleged reports that some pupils were being taught inappropriate content in some schools. The proposals outlined in the 2024 review and put out for consultation include introducing age limits for topics such as puberty, sex education, sexual harassment and sexual activity. The guidance,

which will be statutory once the consultation is complete, also outlines that the concept of gender identity should not be taught. Many have seen this as a retrogressive step that could further isolate members of the LGBTQ+ community.

Ethnicity

A review of Britain's performance in relation to equality and human rights was undertaken in 2018 leading to the report 'Is Britain Fairer?' (Equality and Human Rights Commission (EHRC), 2019) and the follow up monitoring report in 2023 (EHRC, 2023). In terms of educational outcomes there has been recognition of some improvements with the number of young people who are NEET (not in education or employment) declining and access to university widening, particularly for ethnic minorities. However, progress has been slow. Educational outcomes vary across ethnic groups. In 2022/2023, Chinese and Indian pupils had the highest attainment levels, while Black Caribbean, Pakistani, and Gypsy, Roma, and Traveller pupils faced the greatest challenges. Clearly, there is a need for more research into the disparity in educational outcomes for some ethnic minority groups compared to others (Commission on Race and Ethnic Disparities (CRED), 2021).

Exclusion rates for different minority ethnic groups also present a worrying picture and suggest a failure to respond to the needs of vulnerable groups of pupils (Timpson, 2019). In 2018/2019, the national average for fixed-period exclusions in secondary schools was 5.3 per cent. Rates were lowest for Chinese (0.56 per cent) and Indian (0.88 per cent) pupils, while White British (6.01 per cent), Black Caribbean (10.37 per cent), Traveller (14.63 per cent), and Gypsy/Roma (21.26 per cent) pupils faced the highest exclusions. The reasons for high exclusion rates in schools are complex and multi-faceted but the 'performativity' agenda in the educational system no doubt plays a significant part (Ball, 2003).

The issues of underachievement of particular groups of pupils are an ongoing concern for government and there have been various initiatives over the years designed to address the problem. However, there is still a wide gap between the high and low achievers. The under-performance of many White working-class boys serves to emphasise the importance of considering how the intersection between socio-economic status and gender effects are embedded within the data showing attainments of ethnic groups (see Gillborn and Mirza, 2000; Strand, 2008). The data already presented, and research (e.g. Gillborn and Gipps, 1996; Smith, 2005; Archer and Francis, 2007; Strand, 2014; Gillborn et al., 2017; Farquharson, McNally and Tahir, 2024), shows that, although achievement has been raised for most pupils, underachievement still persists and is often linked to particular ethnic groups.

Archer (2008) argues that the dominant educational discourse in Britain contributes to excluding minority ethnic pupils from 'the identity of the "ideal pupil"' (Archer, 2008, p. 102). Indeed, our Eurocentric curriculum reflects the largely dominant social groups in our society and contributes to 'othering' creating a 'them' and 'us' environment. Seeing yourself represented in culture, curriculums, media etc. conveys a message that you are valued and respected. The recent expansion of the 'Black Lives Matter' movement has led to calls for the decolonisation of education at all levels acknowledging the 'power relations in the production and dissemination of knowledge' (Begum and Saini, 2019, p. 198).

English as an additional language

Twenty per cent of England's school population are defined as having English as an additional language (EAL) (DfE, 2023) and the recent exam performance of EAL pupils would seem to be a success story. In 2022/2023, pupils who use EAL outperformed their native speaking peers with an average

attainment 8 score of 48.5 compared to 45.9. However, it is dangerous to view EAL pupils as a homogenous group. EAL pupils encompass a diverse group including recent arrivals with little or no English along with those proficient in their use of the English language (Strand and Hessel, 2018), language skills in first language and prior educational experiences also contributing to academic performance. Consideration should also be given to refugee children who may face additional challenges in school. Many refugee children have come to the UK unaccompanied. Once in the UK they have had to navigate asylum legislation and the welfare system to secure a place to live before they can access education (McIntyre and Abrams, 2020). While access to education is an entitlement (SGD4 UNESCO, 2015), it is not straightforward with many schools reluctant to offer a place to refugee children, especially older pupils, due to the potential negative impact on the school's performance.

Closer inspection of the data highlights that some groups of EAL pupils are underperforming with proficiency in English the determining factor as shown in Figure 4.4.1.

For teachers the issue of supporting EAL pupils is important (Flynn et al., 2015). The issue of language goes far wider than classroom oracy. Many classroom resources are heavily text-based and their use of language requires not only a level of literacy but also cultural awareness. Teachers need to ensure they are adequately challenging pupils whose first language is not English, and they should carefully consider the language requirements that are made by textbooks, classroom talk and homework tasks to identify whether these are creating barriers to learning. There is evidence that giving pupils opportunities and time to use their first language to think about ideas and, where possible, to discuss concepts, allows them to operate at a higher cognitive level.

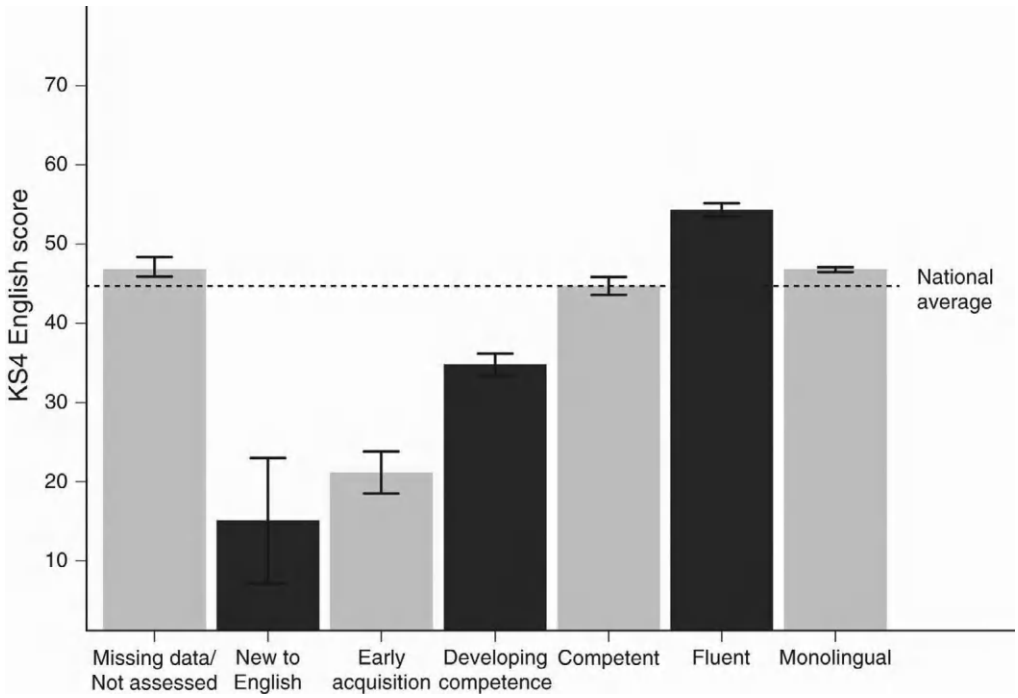


Figure 4.4.1 Mean KS4 attainment 8 score for EAL pupils at different levels of proficiency in English and for monolingual English speakers

Source: Strand and Hessel (2018)

Socio-economic status

The effect that pupils' economic circumstances have upon their education is very real (Strand, 2021). Latest data shows that the 'disadvantage gap index' is at its widest since 2011 with 25.2 per cent of disadvantaged pupils gaining a grade 5 in English and maths compared to 52.4 per cent of all other pupils (Social Mobility Commission, 2024).

Understanding social class goes beyond simply looking at economic capital (e.g. measures such as free school meals) but relates to other 'capitals' that pupils' families possess. This might be social capital or the cultural capital (Bourdieu, 1986) that includes having well-educated parents and ready access to books and digital media; and pupils may be exposed to a wider range of learning and cultural experiences than pupils of less well-educated parents; 'privilege begets privilege' (Slee, 2011, p. 42).

Furthermore, not all families can afford computers and mobile phones, which allows access to information, and therefore creating the concept of the 'digital divide'. The Covid-19 pandemic further exacerbated the performance divide between children from lower-income and higher-income families (Leask and Younie, 2022). Children from poorer households may have limited space in which to study and their Internet connection may be less reliable. In addition, pupils from low-income households may also have less access to a regular and healthy diet. Research has highlighted the links of poor diet to low educational attainment (Lanker and Parolin, 2020), see also Unit 4.2.

The challenge for teachers is, of course, that we cannot necessarily see who these pupils are, but they might well require different kinds of support. Social class works in many subtle ways to disadvantage those already disadvantaged. Through language, manners, cultural awareness etc., the middle classes have a better sense of the 'rules of the game' and so can capitalise better on their educational opportunities (Scherger and Savage, 2010).

A key policy development in England in relation to raising the attainment of disadvantaged pupils was the introduction of Pupil Premium funding in April 2011. In an attempt to close the gap between disadvantaged pupils and their peers, additional funding is given to schools for each of their pupils eligible for free school meals or who are classed as a looked-after child. Assessing the impact of Pupil Premium funding is complex as multiple factors influence pupil attainment. Schools use the funding in varied ways and wider socio-economic factors may affect outcomes. There is some evidence that Pupil Premium funding has been effective in primary schools but not in the secondary sector (Gorard, 2022).

Looked-after children

Children who are, or have been in care, otherwise known as looked-after children, are one of the minority groups with the poorest educational outcomes (Flynn et al. 2013), highlighting the need for more to be done to address the possible inequalities they face. In 2021/2022, there were approximately 105,400 children in care in the UK and attainment for this group of pupils falls significantly below the national average (National Society for the Prevention of Cruelty to Children (NSPCC), 2024). Looked-after children are also more likely to have one or more fixed period of exclusion from mainstream schools (Timpson, 2019). (See Unit 6.2.)

Looked-after children cannot be viewed as a homogenous group as they come from a range of different backgrounds and enter care for different reasons and at different stages of their life. However, the main reasons children are taken into care is abuse and neglect. The experiences of

abuse and neglect can lead to these children experiencing poor mental health and leave them vulnerable to exploitation. The employment prospects for young people leaving care are low and they are over-represented in prison and homeless populations (Sebba et al., 2015). Safeguarding these most vulnerable children requires that all professionals, in whatever field, have the appropriate knowledge, understanding and skills to effectively support and provide opportunities for success for this group.

In accordance with the Children Act 1989, local authorities have a duty to safeguard and promote the welfare of children in their care and this should be monitored by a senior manager, with most local authorities having a designated Virtual School Head. A primary role of the Virtual Head Teacher is to advise and support designated teachers who are responsible for promoting the attainment of looked-after children in their school (Driscoll, 2013). However, research has highlighted that for many young people in care it is the teachers and pastoral staff in schools with whom they interact, on a daily basis, which are the main determinants of their educational progress (Sebba et al., 2015).

This discussion reminds us that the performance of pupils is related to many factors some of which have been identified here. However, the variables used here are not causes; these variables hide causative factors, some of which have been identified, which contribute to underachievement.

Of note, another important area of consideration not discussed here is in relation to disability, it is important therefore that you engage with Unit 4.6.

School policy and classroom practice

In this section, we begin to examine both school policies and your classroom practice, in the light of the earlier discussion. If schools play a role in the ongoing structuring of inequality in society, albeit not deliberately, then teachers need to reflect critically upon their practice. This involves examining how your own position and action in a diverse society both help and hinder you from challenging unequal treatment in the classroom. See Task 4.4.1.



Task 4.4.1 Reviewing Your School's Equal Opportunities (EO) Policy

Policy Review

- Obtain and read the school's EO policy.
- Identify who wrote it, its age, and any updates or related documents.
- Assess which areas of school life it covers and whether any aspects are missing.
- Determine if it addresses gender, ethnicity, social class, or disabilities collectively or separately.

Awareness and Understanding

- Investigate how widely known the policy is among staff and pupils.
- Check if copies are displayed and how many have read it.
- Identify the person responsible for EO and find out if you can discuss their role.

Implementation and Impact

- Look for in-service training, departmental policies, and classroom practices reflecting EO principles.
- Assess the school environment (e.g. displays, reception area, website) for inclusivity.
- Analyse PSHE and RSE curricula for diversity and inclusion content.
- Review support for pupils with limited access to digital technology.

Summarise your findings, discuss them with your mentor, and file them in your professional portfolio.

When you first start teaching, your concern is to promote learning through well-ordered lessons. When you feel more confident, ask a colleague to observe one of your lessons, focusing on the questions in Tasks 4.4.2, 4.4.3, and 4.4.4. As you develop as a teacher you can consider wider issues of inclusion and diversity. Now complete Tasks 4.4.2, 4.4.3, and 4.4.4.



Task 4.4.2 Responses to diversity: classroom observation

Ask a beginning teacher to do parallel observation with you to find out which pupils participate in the lesson more than others, and which pupils you invite to answer questions or volunteer information. Share your findings confidentially with each other.

Keep a tally of the frequency of attention, and the time given, to different groups of pupils. Devise a recording sheet to collect information to explore one or two of the following:

- Who puts their hand up to answer a question?
- Who do you select?
- In class activities, how much time do you spend with different groups of pupils?
- How do you respond to certain pupils? With praise, criticism or further questioning?
- When pupils are reprimanded, is there any difference (gender, ethnicity) in your actions and in what you tolerate, or not.

Consider whether different messages are conveyed to different groups by your response to classroom interactions.

Identify and record the implications for your teaching and file in your professional portfolio.



Task 4.4.3 Responses to pupils with EAL: classroom observation

Redesign the record sheet you used in Task 4.4.2 to collect data about your response to pupils with EAL. Use similar questions as listed under Task 4.4.2 and observe the same protocols about observing classrooms (see Unit 2.1). You may wish to add to the list of questions:

- How do you support pupils whose first language is not English?
- How do you adapt resources for these pupils?
- How do you adapt their own language for these pupils?

Record your notes in your professional portfolio.



Task 4.4.4 Evaluating bias and stereotyping in teaching resources

Choose a commonly used resource in your subject—such as a PowerPoint presentation, textbook, worksheet, or online material—and critically analyse it for bias and stereotyping. Consider the following:

- How are gender roles represented? Are traditional stereotypes reinforced?
- Who is depicted in leadership or authority positions?
- Are professions or activities stereotyped based on gender, race, or social background (e.g. male scientists, female caregivers, Black athletes)?
- How are ethnic minorities in the UK portrayed? Are representations accurate and diverse?
- How are people and places from different global regions depicted? Are they shown through a deficit lens (e.g. poverty, underdevelopment) or with a balanced perspective?
- Does the resource reinforce outdated narratives or assumptions about different cultures, societies, or socio-economic groups?

Reflect on your findings and discuss key issues with your mentor and peers. Consider how you might adapt or supplement the resource to promote inclusivity and challenge stereotypes.

Responding to diversity in the classroom

Teacher expectations are shaped by both pupils' past performance and perceived social position, sometimes leading to unconscious bias (Noyes, 2003). You might have found yourself making assumptions about pupils from the time you first saw their names on the class register. It is crucial you take time to explore your own beliefs and values in relation to the social groups discussed here and explore your 'unconscious bias', that is what prejudices you may be subconsciously harbouring (Frith, 2015). It is through honest self-reflection that you are able to develop into a teacher committed to inclusive practice.

Some teachers may unknowingly favour certain students, such as asking boys more questions or responding differently to similar answers. Praise and criticism can reinforce stereotypes, potentially limiting pupils' progress. Reflecting on biases and comparing teaching behaviours across different contexts can help ensure more equitable and inclusive classroom practices. Now complete Task 4.4.5.



Task 4.4.5 Diversity and learning

Some schools have much more diverse populations than others; however, the issues of diversity are relevant for all schools and pupils. Find out from staff, school documents and inspection reports information about the following features of your placement school:

- the mix of the pupil population in terms of gender, ethnicity, socio-economic status;
- the different languages spoken by pupils;
- the number of looked-after children;

- the number of pupils needing and receiving support because English is not their first language;
- the academic performance of different groups of pupils, e.g. in-school progress tests, GCSE and GCE A level.

How does the school analyse and use data about their pupils?

Find out to what extent your subject department policy and practice relates to the whole-school equal opportunities policies. Focus particularly on the curriculum, for example:

- identify whether there is a commitment to multiculturalism at subject level? If so, identify how this influences teaching and learning;
- examine teaching and learning resources to see whether they reflect the school policies;
- explore whether there are different strategies for supporting pupils from different backgrounds.

Is there evidence of positive approaches to diversity in the broader life of the school? (For example, in display material, in assemblies, in the recognition of different cultural and religious practices at different times of the year, and in extra-curricular activities.) You may find other examples.

Summarise your findings and analyse the information. Draw out some key issues for discussion with your peers and mentor as a draft document. You may like to develop this enquiry into a piece of coursework.

See the companion text *'Surviving and Thriving in the Secondary School'* by Capel, Lawrence, Leask and Younie (2019) for advice about understanding your school and community.

SUMMARY AND KEY POINTS

This summary highlights key themes from the unit on diversity, exploring how social, economic, and cultural factors shape educational experiences, the challenges faced by marginalised pupils, and the role of inclusive policies and teaching practices in promoting equity.

- Promoting equity in schools requires understanding the factors that inhibit learning and finding ways to overcome them.
- Beyond awareness, teachers must act by reviewing and adapting their practices to create inclusive learning environments free from prejudice.
- This can involve changing resources, rethinking grouping strategies, and adopting varied teaching styles to challenge inequality rather than unintentionally sustaining it.
- Ignoring differences under the guise of treating all pupils the same is inadequate; recognising diversity without imposing limits enhances learning opportunities.
- Teachers should reflect on their biases and expectations, ensuring they do not restrict pupils based on assumptions about ethnicity, gender, or background.

- A deficit-based approach views difference as a problem, whereas true inclusion values what pupils can do and fosters a strengths-based perspective.
- An inclusive school, or society, is based on an appreciation, acceptance and, indeed, celebration of diversity. It welcomes into its cultures, policies and practices, the fresh input from diverse persons as full members within it. It does not try to make every person 'normal' (Bartolo et al., 2007 p. 49).

Check which requirements for your ITE programme you have addressed through this unit.



Further resources

Diversity, inclusion and social justice

The following YouTube clips are worth watching:

Adam, an Egyptian refugee, is a performer, an LGBTIQ advocate and an educator for transgender issues. https://www.youtube.com/watch?v=F-VngpJxw3k, viewed 16 February 2025.

Becoming an Anti-Racist – The Podcast: https://open.spotify.com/show/1SOJg5qKfJUJZvIAcBFfaI, viewed 16 February 2025.

Supporting refugee children in your school.

Peter Walker – Inclusive Education: a way to think differently about difference: https://www.youtube.com/watch?v=FoICetXtYGO, viewed 16 February 2025.

Riz Ahmed's Diversity Speech: https://www.youtube.com/watch?v=36bcxDVNr1s, viewed 16 February 2025.

You can access a free online teacher education programme focussed on supporting refugee children in the classroom via this link free training/programme.

Case studies from Schools of Sanctuary: https://schools.cityofsanctuary.org/the-schools-of-sanctuary-network/case-studies, viewed 16 February 2025.

NEU guidance: https://neu.org.uk/media/1936/view, viewed 16 February 2025.

Supporting EAL learners.

NALDIC offers ITE students free membership: https://naldic.org.uk/about-naldic/organisation/, viewed 16 February 2025.

The Bell Foundation has a wide range of resources: https://www.bell-foundation.org.uk/, viewed 16 February 2025.

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4.5 Values education

Discussion and deliberation

Ruth Heilbronn and Kate O'Shaughnessy

Introduction

This unit draws on research on moral development, character education and policy sources to suggest a foundation for values education. The unit focuses on you as a practitioner, on your pupils and their development, and lastly on classroom discussion.

As a member of a profession a teacher should behave with integrity and uphold professional values; these may be expressed in a code of conduct or in competency statements, such as the Teacher Standards in England (DfE, 2021) with its explicit *Preamble* about behaving with 'honesty and integrity'. The guidance is underpinned by value judgements about maintaining positive relationships with pupils, being fair, respectful and supportive. This is reiterated in sections of the Initial Teacher Training Core Content Framework (ITT CCF), Early Career Framework (DfE, 2024) and the upcoming Initial Teacher Training and Early Career Framework (DfE, 2024) which identify these as a fundamental professional requirement. All four constituent nations of Great Britain are explicit about the values that teachers must uphold (Hwb, 2017; GTCNI, 2018; GTCS, 2021).

'Values are complex and work to shape who teachers are as professionals' (GTCS, 2021). Teachers should exemplify their values in their practice, a requirement which is in many philosophy of education accounts (e.g. McLaughlin, 2004, pp. 339–353; Carr, 2023). When teachers stop a class because they have overheard a racist remark and tackle this sensitively, knowledgeably and tolerantly, we would say they dealt competently with the situation, drawing on their own values.

OBJECTIVES

At the end of this unit you should be able to:

- understand the place of values education and ethical deliberation in a subject and school context;
- understand the various legal responsibilities of schools in the area of moral, social and cultural development, including civic responsibility;
- identify core values for yourself and those of the school in which you are teaching;

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- identify opportunities to promote understanding and practice of these core common values in school;
- try out some methods of leading discussion and promoting ethical deliberation.

Check the requirements of your initial teacher education to see which relate to this unit.

Values education

The terms 'values education' and 'moral education' are often used interchangeably, but 'morality' involves wider notions of 'goodness' and particular views on the nature of right and wrong, than the term 'values' (see Robertson, 2013 for a fuller discussion and Pring, 2021). Any adult in a relationship with a child is in some sense teaching about values, in contexts such as one-to-one conversations, classroom teaching and parent interactions. Halstead and Taylor's definition of what values are still holds good. Values are

> the principles and fundamental convictions which act as general guides to behaviour, the standards by which particular actions are judged to be good or desirable. Examples of values are love, equality, freedom, justice, happiness, security ... and truth.
>
> (Halstead and Taylor, 2010, pp. 169–202)

Values education is a broad term and may carry a particular emphasis on 'character education' and may also be taught as related to spiritual, moral, social and cultural development. In many countries some form of civic education is a vehicle for promoting values and encouraging pupils to develop the ability to deliberate on moral matters which can help to develop resilience. This is particularly relevant to today's context (UNESCO, 2017; OECD, 2021). See the Council of Europe's *Report on Citizenship and Human Rights Education* (Council of Europe, 2021) for a fuller discussion.

Our values determine how we choose to act when faced with conflict and dilemmas concerning controversial topics. Value judgements on controversial issues differ from judgements of fact, where evidence can help us to make decisions. When faced with a moral dilemma, no amount of new facts will help with choices about what to do: we need to be able to deliberate on the issues involved. Deliberation is a kind of moral reasoning that goes on when making ethical judgements and having skills in deliberation can help to decide what to do when more facts cannot help.

Schools uphold values and express these in policies and practices, which often account for differences in ethos between schools. Research shows that a school's ethos can have a significant effect on pupils' wellbeing (Norwich et al., 2022; Barker et al., 2023; Clement, 2023; Long et al., 2023). Differences in the culture of the school are important to pupils' wellbeing (Barker et al., 2023); to teachers' ability to thrive and develop in school (Heilbronn et al., 2002, pp. 371–389; Day et al., 2009; Ainsworth and Oldfield, 2019; Gu and Coleman, 2023); to the pupils' expressions of respect for one another, and consequentially to their socially responsible behaviour in the classroom (Törnqvist, 2019; Hansen, 2021; Dally, 2023; Kelly et al., 2023). School ethos is largely created through leadership. Some expressions of a school's ethos might be found in the way parents are welcomed; the relationships between adults and young people outside as well as inside the

classroom; staff sensitivity to cultural and faith differences in the school, or how the school celebrates the success of its pupils. The ethos of a school contributes as much to values education as does the prescribed curriculum.

Values underlie all aspects of school life. In some schools there is a common understanding about how these values arise, as they are based on definable moral codes. In faith schools there is a clear understanding about 'right' and 'wrong', based on tradition and scripture. In secular schools this is not so clear. Nevertheless, even in faith schools there are areas where value judgements are made, which are not clearly based on the underlying faith, such as judgements about what is 'fair', which may involve choosing one claim above another. A policy may provide guidelines, for instance when dealing with racist or bullying incidents. Respect is usually a core value here, and many of the school rules and procedures may derive from this stated value. What cannot be prescribed in a policy is what any individual *ought* to say and do in response to any particular situation. As a teacher, it is a matter of judgement how to mediate the policy in the classroom where a teacher stands as a moral example. As an illustration a teacher might punish a pupil who steals from another pupil without much discussion; alternatively, the teacher might decide to talk to the pupil, to get them to understand that what they had done was wrong. Choosing how and when to have this discussion is also a matter of judgement (For a discussion on judgement and reasoning see Heilbronn, 2008; Orchard et al., 2016). Talking with pupils about the rights and wrongs of behaviour is important, as it acknowledges and respects the pupils' ability to develop an understanding of the consequences of their actions. Such talk encourages a sense of responsibility and agency and so helps pupils to develop moral understanding, autonomy and resilience. The policies of the school are in some sense a formal interface between the individual pupil and teacher. The teacher in the example could relate the discussion about why stealing is wrong, to the school's formal statement or rule relating to respect. In fact, 'everyday classroom life is saturated with moral meaning. Even the most routine aspects of teaching convey moral messages to student' (Hansen, 1995, para. 2).

Pupils judge their teachers on their relationships and believe that without good relationships they do not learn effectively (García-Moya et al., 2020; Platz, 2021). Pupils tend to have good relationships with teachers who are good at engaging in reasoned discussion and who are able to articulate the values that underlie the rules about acceptable behaviour. These teachers are more likely to develop trusting and respectful relationships than those who merely dictate what the rules demand (Lickona, 1983; Ellis and Tod, 2018). Although undertaken more than 40 years ago, Lickona's research is still as relevant today, as is Nucci's (1987) corroboration that:

> students rated highest those teachers who responded to moral transgressions with statements focusing on the effects of acts ('Joe, that really hurt Mike'). Rated lower were teachers who responded with statements of school rules or normative expectations ('That's not the way for a Hawthorne student to act'). Rated lowest were teachers who used simple commands (Stop it!'). (pp. 86–92)

It may not be possible to take all the time necessary to deal in depth with issues as they arise, but this can be acknowledged with the pupils. What is important is to reflect back to pupils the example of the kind of behaviour that is acceptable and the reasons behind the values expressed, rather than ignoring or suppressing what is not acceptable. It is important to attend to issues of values in

any particular classroom situation since pupil's experience lessons as individual young people and not as disembodied 'learners' of a particular curriculum area. Van Manen has written of 'the tact of teaching' and how good, trustful relationships rely on 'pedagogical thoughtfulness' on the part of the teacher (Van Manen, 1991). This is not always easy to achieve and at times deciding what to do may come to a difficult decision, for example, being fair to all while also taking individual needs into account. [Fairness became an ethical issue during the Covid-19 pandemic in relation to unequal access to resources (UK Parliament, 2020; Leask and Younie, 2021; Haelermans et al., 2022; Werner and Woessmann, 2023).]

Cultural diversity and common values

Quite young children can distinguish moral judgements related to justice and fairness from judgements about social conventions, such as how to address people or follow dress codes (Nucci and Ilten-Gee, 2021, p. 11). It's important in a pluralistic society to be able to distinguish fundamental common values that do not rely on specific cultural or religious foundations. Schools have a responsibility to promote values education in some form, as is evidenced in one of the Ofsted criteria for an outstanding school – 'relationships among learners and staff reflect a positive and respectful culture' (Ofsted, 2023).

Currently in England, schools are required to promote **'**fundamental British values', a term taken from the definition of extremism articulated in the Prevent Strategy, which was launched in June 2011 and is still in force in 2024. It includes 'democracy, the rule of law, individual liberty and mutual respect and tolerance of different faiths and beliefs' (DfE, 2014, p. 5). These values are fundamental to all liberal democracies so there is considerable debate about why they are classed as 'British' values (e.g. see Farini, 2019; Healy, 2019; Winter and Mills, 2020; Henshall et al., 2023). Because they underpin the Universal Declaration of Human Rights, they might usefully be called 'Universal Values'. The requirement for schools to show how they promote these values illustrates how at certain times some fundamental values might be seen to need reiteration and reinforcement and the belief that schools are places where this should happen. (See Unit 7.1 on the Aims of the Curriculum.)

Learning how to listen to the views of others and respect their rights to these views is fundamental to accepting a plurality of cultures, which can act to enable pupils to develop a critical faculty that helps to immunise them against extremism.

> This view of education, and the role of addressing extremism within it, rests on a foundation of strong protection of open inquiry and open expression … Thus, the fight against extremism, properly understood, can be seen as a dimension of the key educational mission of schools.
>
> (Ben-Porath, 2023, p. 378)

An unequivocal and open discussion in school about shared values can support this aim,

> An explicit treatment of values across the whole school as a subject of inquiry and agreement by all members of the school community can contribute to a healthy school environment and praxis for learners that serve the aims of preventing violent extremism.
>
> (Tibbitts, 2020, p. 79)

Settings and opportunities for values education

How a child develops moral reasoning is important in designing a curriculum. Building on Piaget's (1932) foundational work, Kohlberg developed his theory of moral reasoning. Both writers stressed maturation factors, arguing that mature moral reasoning depends on the development of the capacity for logical reasoning (Kohlberg, 1985, pp. 27–87). 'Moral development' is a term referring to children's increasing ability to use nuanced moral reasoning. (See Units 4.3 and 5.1 on maturation of reasoning.) The structured nature of Piaget's stage theory and the Kohlberg framework is a contested matter (see Gibbs, 2019; An, 2020; Rochat, 2023).

Some common features of schools seem to have a positive impact on the development of pupil values, such as participation in the communal life of the school and the classroom; encouragement to behave responsibly; provision of an orderly school environment, and clear rules that are fairly enforced. In such an environment pupils can experience the implementation of positive values which can help to develop their moral reasoning. In fact, many argue that moral reasoning only develops through experience (Dewey, 1909, 1916; Van Manen, 1991; Campbell, 2008) and that values are 'caught' not 'taught'. A school's explicitness about its values and the extent to which teachers practice shared values has an important influence on pupils' development of moral understanding and responsibility. Home influence is of course more significant than the school's, given the early nurture period and the time spent there. This reinforces the importance of a partnership approach between schools and their local communities. The work done by Larry Nucci is very relevant to today's context, although done years ago (Nucci, 1987, pp. 86–92. See also Lusse et al., 2019).

The curriculum is important in channelling moral development, in subjects such as religious education (RE) and civic education (discussed above), as in the citizenship curriculum in England. Citizenship education is inherently related to values. Political and social issues concern the question of how we should collectively live, which is at root an extension of the basic moral question 'How should I live my life?' (Haydon and Hayward, 2004, pp. 161–175). Many areas of public life remain value-laden and may often call for sensitive handling. Secondary schools must also provide sex and relationship education, usually as part of a PSHE course, which offers opportunities for values education. There are specific learning activities, such as resources for 'character education' (e.g. Jubilee Centre, 2022; Arthur and Harrison, 2023), and arguments in its favour (Curren, 2017). Some have challenged character education programmes on the grounds that there can be debate about which values should be promoted to children as good character traits and that the resources may be biased (Howard et al., 2004; Peterson, 2020; Jerome and Kisby, 2022). There are also resources specific to values education, published by the Values Based Education Trust which works in educational settings to 'create an environment and pedagogy that allows all to flourish as human beings' (Hawkes, 2023, p. 2. See also, DfE, 2019; Lovat, 2023; Toomey, 2023).

There remains the fundamental question as to whether values can be taught at all, or only caught by example. Kohlberg did comment on the artificiality of creating a specific curriculum area purely to foster moral development. In an evaluation of one of his own early projects he stated: **'**While the intervention operation was a success, the patient died. When we went back a year later, we found not a single teacher had continued to engage in moral discussion after the commitment to the research had ended' (Kohlberg, 1985, p. 80). This underlines the importance of personal experience and continued commitment to engaging with discussion about values. When we talk to a pupil about something we have both experienced, we engage in a process of talking about values in a direct and practical way.

Ethical deliberation through discussion

Discussion is a common means of raising moral and ethical issues and teachers need preparation for managing such discussions (Nganga et al., 2020; Sætra, 2020). In their discussions pupils may not reach agreement, but the deliberation involved can lead to a deeper understanding of the issues and the positions of others. Underlying these discussions are the values that participants hold, although they may not be able to articulate them fully. Taking part in well-managed and well-conceived deliberation on ethical matters enhances pupils' social interaction, because they need to listen to each other and respect each other's opinions. (See Hand and Levinson, 2011 on discussing controversial issues in the classroom. Also, Nucci and Ilten-Gee, pp. 126–137 have pertinent advice.)

The ability to make moral judgements wisely depends on weighing up fine balances of possible actions, which we expect to develop as children mature. This is 'a gradual process in which students actively transform their understanding of morality and social convention through reflection and construction' (Nucci, 1987, p. 90). The new understandings 'constructed' in peer exchange discussions can be powerful in developing pupils' moral understandings (Berkowitz et al., 1980, pp. 341–357; see also Berkowitz, 2011). Engaging in ethical deliberation through discussion involves respect for the opinions of others, sincerity in seeking to clarify what is said by others and skill in relating what is said to one's own situation and choices. Managing such discussions or explorations requires skill, to avoid them becoming merely ritualistic or worse, a confirmation of prejudices and an opportunity for the strongest characters to dominate (Bleazby, 2020). The SAPERE organisation trains teachers to undertake this kind of work in schools in philosophy for children (see https://www.sapere.org/ for more information).

Discussions need to be appropriately related to the pupils' own experience or imaginative capacities. At any given time there are likely to be 'hot topics' that concern young people in their lives outside the classroom. Some helpful advice for teachers to manage discussions on controversial issues comes from Meira Levinson (Ross, 2022). Often, the discussions can be based on issues pupils may raise such as relationships, gender/sexual identity, social networking, trust and privacy that draw on the ethics of a responsible online community. Other examples are the environment, endangered species, fair trade, equality of opportunity, or legal decisions, such as the debate on assisted dying taking place in many jurisdictions at the moment. Do clinicians have the right to help people suffering from terminal illness to die, without facing legal consequences? (Li Ying Quah et al., 2023). Proponents of this right cite the right to choice and to dignity of the terminally ill person. Opposers point to possible pressure that may be suffered by vulnerable people who may believe themselves to be a burden. The debate about conflicts on moral issues continues to be relevant as medicine's technical possibilities grow (Quintal et al., 2023; Munung et al., 2023).

> As medicine drives us past the limits of our settled moral deliberations, it becomes increasingly important to take stock of the rules and principles by which our lives, and those of others, are ordered.
>
> (Clucas and O'Donnell, 2002)

Digital computer games exist to increase values awareness, such as the *RealLives* simulation that enables players to learn more about different cultures and develop inter-cultural competence (Struppert, 2010). Younie and Leask (2013) argue gaming can help engender values and resilience in young people through learning to manage emotions (from frustration of losing to elation at winning)

by following rules and in team-based games, through co-operation to achieve shared goals. Citing Piaget (1932) they outline how games can help pupils develop their social and cognitive capabilities, especially boys (Hromek and Roffey, 2009). They point out that working with rules aids the development of self-discipline, which underpins society and social order.

The following five Tasks 4.5.1–4.5.5 are designed to develop reflection on values in teaching.



Task 4.5.1 Reflecting on teachers' values

The task draws on the Teachers' Standards in England (DfE, 2021): https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/665520/Teachers__Standards.pdf

Look particularly at the Preamble as well as the standards.

- Do you believe that these should be your core values as a teacher? Are there any you do not agree with?
- Are there any aspects that may be more difficult to apply than others?
- Are there any issues that you think you might experience in putting these values into place?
- Are there any aspects of the values relating to teaching that you think are missing here?

Produce a one-page summary that covers the key points of your thinking and reflections on this.



Task 4.5.2 Aims of the school and how they are interpreted

Carry out one or more sub-tasks, alone or in pairs and discuss your findings with other beginning teachers.

1 Collect the aims of your placement school and, if possible, that of another school. Compare and contrast the aims of each school.
   - Does the school in its aims express a responsibility for the moral development of pupils?
   - How are these aims expressed and in what ways do the school statements differ?

2 Many school curricular statements are prefaced with broad aims, relating to fostering pupils' self-esteem, valuing their contribution and helping them to develop independently, while also enabling them to work with peers. How might these aims of this school curriculum be translated into opportunities for the pupil to achieve them?

3 In what ways does the PSHE course contribute to the education of your pupils in the area of values? Select a topic, for example:
   - Identifying bias and stereotyping and developing strategies to deal with it
   - Care of the environment
   - Practices and attitudes towards adult relationships.

4 How might a school assembly contribute to the moral and values education of pupils? Some opportunities for reflection on this might arise by:

- Attending assembly – keep a record of its purpose, what was said and the way ideas were presented. Interview some pupils afterwards and compare your perception of the assembly with theirs. Did they understand the underlying point of any 'message'? If so, did they agree with it? Is the message relevant to their life and their family, and if so, how?
- Helping to plan an assembly with a teacher and with the support of other student teachers. Some questions that may be helpful are:
  - Does the school have a programme for assemblies? How is the content and approach agreed within your school?
  - How does the content and messages of assemblies relate to the cultural mix of your school?
  - Does assembly develop a sense of community or is it merely authority enhancing? How is success celebrated and whose success is mostly recognised?
  - What is the school's programme of continuing professional development (CPD)? Is there any provision relating to moral, ethical and values education? How is such work focused and by whom? (Your mentor can direct you to the staff responsible for CPD.)

Consider writing a short reflective piece conveying your findings, which can be discussed with your mentor or added to a portfolio to submit to your mentor.



Task 4.5.3 The place of subject work in promoting moral development and values education

Identify a social, ethical or moral issue that forms part of teaching your subject.

Review the issue as a teaching task in preparation for discussing it with other beginning teachers. Your review could include:

- a statement of the subject matter and its place in the curriculum;
- the ethical focus for pupils;
- a sample of teaching material;
- an outline teaching strategy, e.g. a draft lesson plan;
- any problems that you anticipate teaching the issue;
- any questions that other student teachers might help you resolve;
- resources to support the ethical focus.

Keep notes on the teaching task and produce a short summary that covers the key points of your learning.

Task 4.5.4 Classroom management

Class detention

Consider the following scenario:

> A class is reading from a set text and from time to time stopping to discuss points. Some pupils get bored and start to flick pellets at other pupils sitting near the front. The teacher knows it is happening but cannot identify the main culprits. One or two pupils take offence at being hit by pellets and start to complain. The teacher tries unsuccessfully to stop the disruption. The noise level rises and the whole class gets involved. The teacher says that the person who started throwing the pellets will be given a detention and needs to own up. No one owns up and the class is given time to sort it out among themselves if they can. No one owns up and the whole class is kept in after school for 15 minutes. This causes resentment among the pupils and some walk out and refuse to stay.

How might the teacher have dealt differently in each case with the following:

- the whole incident?
- those left in detention?
- those who walked out?
- complaints about unfair practice/injustice?

Reflect on – What issues are raised by the incident – e.g. the justice of collective punishment? Discuss this with your mentor. To add more depth, consider also citing relevant research and philosophical literature and writing a reflective piece for your professional portfolio that can be shared with your mentor too.

Critical incidents

Many incidents with which teachers deal in school have an ethical dimension. How we respond to them is to a large extent a function of our own ethical beliefs and what we do also models our underlying values. Whether we act judiciously or react inappropriately in haste, may determine whether we are just and fair or not, and pupils quickly pick up which is the case. Two examples of such incidents are presented in Task 4.5.5. Consideration of the incidents is designed to show the kinds of judgements that may need to be made. In the first incident an individual pupil has to decide what to do – a question of conscience – and a teacher has to act accordingly. In the second example the teacher is faced with a situation in which there is a conflict of responsibilities towards the parents on the one hand and the pupils on the other hand and is required to take a stand.

Task 4.5.5 Critical incidents

Read through the following extracts and discuss the two cases with another student teacher, or in a group. There are prompt questions to support your reflections.

1 Wayne is in school during break to keep an appointment with a teacher. On the way there, walking past his form room, he sees a pupil from his form going through the drawers of

the teachers' desk. The pupil is not facing Wayne but it seems apparent that the pupil is taking things out of the desk and putting them in his pocket. Wayne walks away unseen by the other pupil. Wayne decides to tell his class teacher.

2 This incident concerns a 13-year-old girl who attends a local girls' school. Her parents don't approve of her mixing with other pupils outside school hours and expect her to return home promptly after school. But she often goes home with other girls and part of the way with boys from the nearby boys' school. One day she arranges to meet one of the boys after school and gets another girl in her class, Serena, to give her an alibi. On the day of the meeting, Serena decides that she doesn't want to lie to Serena's parents. During the afternoon the two girls argue in class to the extent that the teacher keeps them back after school. The teacher then learns the cause of the argument and so finds out about the arrangements.

Some questions to support discussion:

- What would you do in each of these cases if you were the teacher?
- What advice would you give to the pupils?
- Can you imagine any difficult decisions you might have to take? If so, explain the considerations.
- If you didn't know what to do, do you know where to get the advice and support you need to help you manage challenging situations?
- How do the situations relate to the particular formulation of the ethical responsibilities of teachers as stated in your course requirements or accreditation standards?

Reflect on – What issues are raised by these cases. Discuss and justify your response with relation to theory and research. Consider discussing with your mentor and writing a short report that outlines your considerations for your teaching practice.

SUMMARY AND KEY POINTS

In this unit, we have explored what it might mean to you personally as a teacher to follow ethical guidelines and how your personal values might play out in your teaching role. Some key points of the chapter are listed below:

- Moral judgement and moral behaviour derive from personal, social, cultural, religious and political viewpoints and conventions.
- Societies with diverse groups have a potential for enrichment or friction.
- Young people develop an ability to reason about values and to make ethical choices. They are helped towards maturity in this respect by parents and carers, teachers and peers. The mature stage of development may lead to questioning the values they have grown up with.
- Some moral and ethical matters arise naturally in dealing with children but subject work may afford opportunities for raising them. While some subjects lend themselves more easily than others to these matters – such as English, RE, citizenship and civic education – all subjects can provide an opportunity. This can be done through using simulations and discussions, which are good ways of introducing moral and ethical matters.

- Teachers have a responsibility to promote pupils' development as ethically responsible people and to serve as role models of good ethical practice.
- Ethical dilemmas continually arise in practice and teachers need good judgement in these cases. It is important to know the school policy about the issues that arise and the expected response in your school.
- When it is difficult to respond by yourself you should refer to experienced staff for help. These dilemmas are part of being a teacher with responsibility for others.

Check which requirements for your initial teacher education you have addressed through this unit.

Further resources

NB – Philosophy books remain current because philosophy deals with conceptual matters and while the context may change, the underlying arguments and debates and questions do not change.

British Humanist Association: https://humanism.org.uk/education/teachers/, viewed 31 July 2025.

Citizenship Foundation: https://www.youngcitizens.org/importance-of-citizenship-education, viewed 31 July 2025.

Curren, R. (2017) *Why character education*? Impact – Philosophical Perspectives on Education Policy, 24, 1–44.

Curren explains why schools should create supporting environments and why cultivating good character implies promoting the values of democracy, the rule of law, individual liberty, and mutual respect and tolerance. IMPACT Pamphlet. Open source downloadable from https://onlinelibrary.wiley.com/doi/full/10.1111/2048-416X.2017.12004.x, viewed 31 July 2025.

Dewey, J. (1909) *The Moral Principles in Education. The Middle Works of John Dewey, 1899–1924. Volume 4: 1907–1909, Essays*, Carbondale and Edwardsville: Southern Illinois University Press.

This book is a classic and as valid today as in 1909. It is open source and freely downloadable. From http://www.gutenberg.org/ebooks/25172, viewed 31 July 2025.

Haydon, G. (2007) *Values in Education*, London: Continuum.

This book addresses the questions: what are the fundamental aims and values underlying education and how can education promote values in a pluralistic world. Also, it discusses morality and should schools teach it. In a secular society, how should schools treat the links between morality and religion?

Jubilee Centre for Character and Virtues: http://www.jubileecentre.ac.uk, viewed 31 July 2025.

Lovat, T., Toomey, R., Clement, N., and Dally, K. (eds.) (2023) *Second International Research Handbook on Values Education and Student Wellbeing*, New York: Springer.

A collection of articles dealing with the principles underlying values education and addressing some of the social and learning challenges in education today. A good overview of the field.

Nucci, L., Narvaez, D., and Krettenauer, T. (2025) *Handbook of Moral and Character Education*, 3rd edn, London: Routledge.

This edition includes updates on mental and cognitive psychology and moral and character education.

Values Based Education Trust (VbE): https://www.valuesbasededucation.com/home, viewed 31 July 2025.

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4.6 An introduction to inclusion, special educational needs and disability

Sana Rizvi and Helen Knowler

Introduction

In this unit, we will introduce you to key concepts, useful frameworks and challenges in working with pupils with Special Educational Needs and Disability (SEND) using examples from England. Whilst it is not possible to provide a comprehensive background within a short unit, we will include tasks that you will find relevant as a beginning teacher and signpost useful resources where required.

OBJECTIVES

At the end of this unit, you should be able to:

- Understand key historical and legislative developments that have shaped the provisions for SEND;
- Reflect on how inclusive thinking can support provision for learners with SEND in your classroom;
- Develop your knowledge of the areas of need and related SEND categories you are likely to encounter in your setting;
- Understanding the need for developing different ways of working with different colleagues such as Special Educational Needs Coordinators (SENCo) and Teaching Assistants (TA).

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Background

In England and Wales, a pupil with SEN is defined as those with, "... a learning difficulty or disability which calls for special educational provision to be made for him or her. A child of compulsory school age or a young person has a learning difficulty or disability if he or she:

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> has a significantly greater difficulty in learning than the majority of others of the same age, or has a disability which prevents or hinders him or her from making use of facilities of a kind generally provided for others of the same age in mainstream schools or mainstream post-16 institutions."
>
> (DfE and DfH, 2015, p. 16)

Some children with SEN also have a disability. Disability under the Equality Act 2010 is defined as,

> a physical or mental impairment which has a long-term and substantial adverse effect on their ability to carry out normal day-to-day activities.
>
> (DfE and DfH, 2015, p. 16)

Whilst England, Wales and Northern Ireland use the term SEND, Scotland has adopted the more inclusive term Additional Support Needs. This unit will support your knowledge of statutory regulations in England and Wales, helping you to manage SEND provisions in your own classroom as a beginning teacher. The examples in this unit are largely from England – it would be useful for you to find appropriate resources in your country or context (see Task 4.6.1).



Task 4.6.1 Special educational needs provision: comparing and contrasting provision

The European Commission Eurydice Network facilitates cooperation between countries and collates information on National Educational Systems. Use the link below to explore how SEND provision is organised across 40 European countries: https://eacea.ec.europa.eu/national-policies/eurydice/national-description_en

Reflect on the similarities and differences between these countries and the country you are working in.

There are approximately 1.6 million pupils currently identified with SEND in England (DfE, 2024). Following Covid-19 pandemic, the percentage of pupils meeting the expected standards has dropped across key stages for pupils with SEND.

In 2023/2024, Key Stage 1, 27% of pupils with SEND in England met expected standards in Reading and 30% met expected standards in Maths, compared to 84% and 85% of pupils without SEND (DfE, 2023a, 2023b). This attainment gap increased at Key Stage 2, where 19% of pupils with SEND achieved the standard in English and Maths compared to 70% of pupils without SEND (DfE, 2025a). At Key Stage 4 level, 23% of pupils with SEND attained 5 or above in GCSE English and Maths compared to 56% of pupils without SEND (DfE, 2025b).

With such significant attainment gaps, how can you as a beginning teacher ensure that all pupils are given the appropriate support and opportunities? As a beginning teacher working within diverse classrooms, this gap in teacher training is important because it influences how confident you will feel in working with pupils with SEND, and in becoming an inclusive educator (Zagona, Kurth and MacFarland, 2017; Pulsford and Rizvi, 2019).

Historical context to policies and statutory frameworks

This section discusses key SEND policies and legislations – also see Peacey (2019) on the companion website and Hodkinson's historical analysis (in Key Issues in Special Educational Needs, Disability and Inclusion, 2019) which provide a comprehensive policy explanation. The Warnock Report (1978) was a significant UK education milestone, replacing outdated categories of 'handicap' with the continuum of special needs. Warnock pushed to include parental input in the education of children with SEND, recommending where possible that children with SEND be educated in mainstream schools. The subsequent Education Act 1981 included many of Warnock's recommendations, such as educating children with SEND in mainstream schools, requiring every teacher to have responsibility for educating pupils with SEND, and the development of the Statement, a legal document for children with more severe SEND outlining provisions and entitlements after a statutory assessment. However, since the report's publication, many policy makers and academics (Farrell, 2010; Hornby, 2011) have critiqued the idea of 'full inclusion' and viewed problems in teacher capacity and preparedness to justify why inclusion is costly for schools. For instance, Mary Warnock in 2005 recalled her earlier position on inclusion, stating it was a mistake and that disabled children have poor experiences in mainstream schools (Warnock, 2005). The authors of this unit take the stance, that such critique of inclusion is not only rooted in deficit model that dismisses the rights of disabled children and young people and also presents problems facing meaningful inclusion as down to issues with the pupil rather than the failures in current schooling system (Loxley and Thomas, 2001).

Despite persistent critiques of inclusion, in England and Wales, the first SEND Code of Practice (CoP) was developed in 1994, mandating all local authorities to provide SEND provisions for pupils regardless of school placement settings (DfE, 1994). The CoP 1994 outlined five stages in SEND assessment and introduced the role of SENCos. The practical guidance in this CoP echoed the global push for inclusive education (e.g. Salamanca Statement by the United Nations Education, Scientific, and Cultural Organisation; UNESCO, 1994). Whilst the CoP 1994 made an important contribution in terms of SEND assessment, it incorporated little guidance about inclusive classroom practice. Consequently, schools increasingly relied on local authorities to fund provisions and statutory assessment referrals. A revised CoP (DfES, 2001) introduced 'School Action' and 'School Action Plus' in England, a staged approach requiring schools to demonstrate that they had used their own resources to work with a child before requesting local authorities for support.

In 2014, The Children and Families Act was introduced that updated SEND CoP (DfE and DfH, 2015) which superseded all previous CoP. This unit will reference the SEND CoP 2015 throughout. The CoP 2015 outlines a graduated approach to identifying and supporting pupils with SEND, involving four stages of action: assess, plan, do and review which is coordinated by the SENCo (DfE and DfH, 2015, p. 100). A pupil with SEND could avail two levels of support depending on ongoing assessment and review of their difficulties:

- SEND support: If a pupil is identified with SEND, schools should utilise their own resources to arrange provisions to support learning and remove barriers. This may entail working with external stakeholders and specialists. If the pupil does not progress as expected, the school or parents can seek referral for Education, Health and Care (EHC) assessment.
- Education, Health and Care Plan (EHCP): Usually formulated after an EHC assessment from the local authority involving different professionals. EHCP outlines statutory guidance for requisite resources and interventions, and the roles of different stakeholders to ensure the pupil makes sufficient academic and social progress.

You could begin by looking within your own school setting and gain a better understanding of how your school supports pupils with diverse needs (see Task 4.6.2).



Task 4.6.2 Diversity in the classroom

The CoP outlines your statutory responsibilities as a teacher in England in relation to pupils' SEND.

- Talk to your mentor about the range of diverse needs in your classroom, and the specific statutory duties that you have as a beginning teacher – e.g. what is the government guidance on the identification, assessment and diagnosis of SEND?
- What are the statutory documents that you need to maintain in order to evidence provision for a pupil – e.g. an individual education plan (IEP) or a pupil passport?
- Are there national guidelines for all schools to follow, or does your school have its own guidelines regarding the evidence of support you need to collate for each pupil?

Becoming an inclusive educator

Rouse (2009) argues that in developing inclusive approaches, we need to do some thinking about SEND in general. Rouse and Florian (2006) note that one must not assume that areas of needs are 'hard and fast categories' whereby pupils have similar attainment levels across all curriculum areas. It goes without saying that most children in your class will have uneven profiles of attainment (measures in a test) and achievement (gains in skills and knowledge from their own starting point). Working in inclusive settings often challenges our own ideas and unconscious biases about the 'ideal pupil' (Barton, 1986), and what constitutes success, achievement, progress and challenges. Rouse's model below (see Table 4.6.1) can help you reflect about different ways of working and devising an action plan for your own professional learning (see Task 4.6.3).

Table 4.6.1 Developing inclusive practice dimensions

Knowing about	*Doing*	*Believing*
Teaching strategies for Disability and special needs How children learn What children need to learn Classroom organisation and management Where to get help when necessary Identifying and assessing difficulties Assessing and monitoring children's learning The legislative and policy context	Turning knowledge into action Moving beyond reflective practice Using evidence to improve practice Learning how to work with colleagues as well as children Becoming an 'activist' professional	That all children are worth educating That all children can learn That they have the capacity to make a difference to children's lives That such work is their responsibility and not only a task for specialists

Source: Adapted from: Rouse (2009)

Now move onto doing Task 4.6.3.

Task 4.6.3 Reflection on the knowing about, doing and believing framework

- Using Table 4.6.1, reflect on the elements that you feel confident about as a beginning teacher, those that you feel you can develop on your own, and those that you need support to work on. You can use a colour coding approach (red, amber, green) to help you to self-audit. You can share this framework for dialogue with your mentor or your school's SENCo or lead professional.
- Following this dialogue, develop an action plan for your own learning and development – e.g. collate relevant government guidelines or up-to-date research on the aspects that you need to develop, or find out which professional development activities your school can provide.

Ableist classrooms and neurodivergent pupils

If we critically examine our schooling system we begin to see how in our school curriculum, teacher training education, physical spaces, the language used, school practices and our belief systems, that we view difference and diversity as less desirable (Broderick and Lalvani, 2017). This might seem a bold claim but think back to your own experiences of starting your teacher training and whether you were required to meet statutory 'Fitness to Teach' requirements. Whilst these requirements are intended to ensure pupil welfare and safeguarding, they also inevitably act as exclusionary mechanisms for disabled aspiring teachers with significant health concerns (Keane, Heinz and Eaton, 2018). These exclusionary mechanisms only become consequential to those individuals who fail to conform to standards of able-bodiedness and so are often termed 'ableist'. In 1997, Chouinard defined ableism as 'ideas, practices, institutions and social relations that presume able-bodiedness, and by so doing, construct persons with disabilities as marginalised … and largely invisible "others"' (p. 380). In your own classrooms, ableism may present in various ways: for instance, holding stereotypical beliefs about different disabilities, providing teaching resources in one format only that maybe inaccessible to pupils with dyslexia, ability groupings, rigid uniform policies, inflexibility around toilet breaks, not considering representation of disabled people as part of school staff and curriculum, or running PE lessons in ways which does not take into account how disabled pupils can participate. Some of these practices are not intentional as Broderick and Lalvani (2017) put it, rather they are part of 'dysconscious ableism' – an absence of critical reflection and the unquestioned acceptance of knowledge that conforms to ableist standards.

One way to challenge dysconscious ableism is to think how as a beginning teacher you might establish a relationship with pupils who think and approach learning differently. To put it another way, is your classroom prepared for a neurodivergent learner? Connolly, Constable and Mullally (2023) define neurodivergence as 'a term for when someone's brain processes, learns and behaves differently from what is considered "typical"' (p. 4). The term 'neurodivergent' is a positive example of how disabled people and more recently pupils in education, have reclaimed the diagnostic categorisation of SEND to highlight pervasive ableism and shift the attention from an individual deficit discourse.

In your classrooms, neurodivergent pupils could include pupils with Attention-Deficit/Hyperactivity Disorder (ADHD), autism, dyslexia or dyspraxia, or any other SEND label that suggests they learn differently. Initially coined by Judy Singer in 2016, neurodiversity embraces the idea of

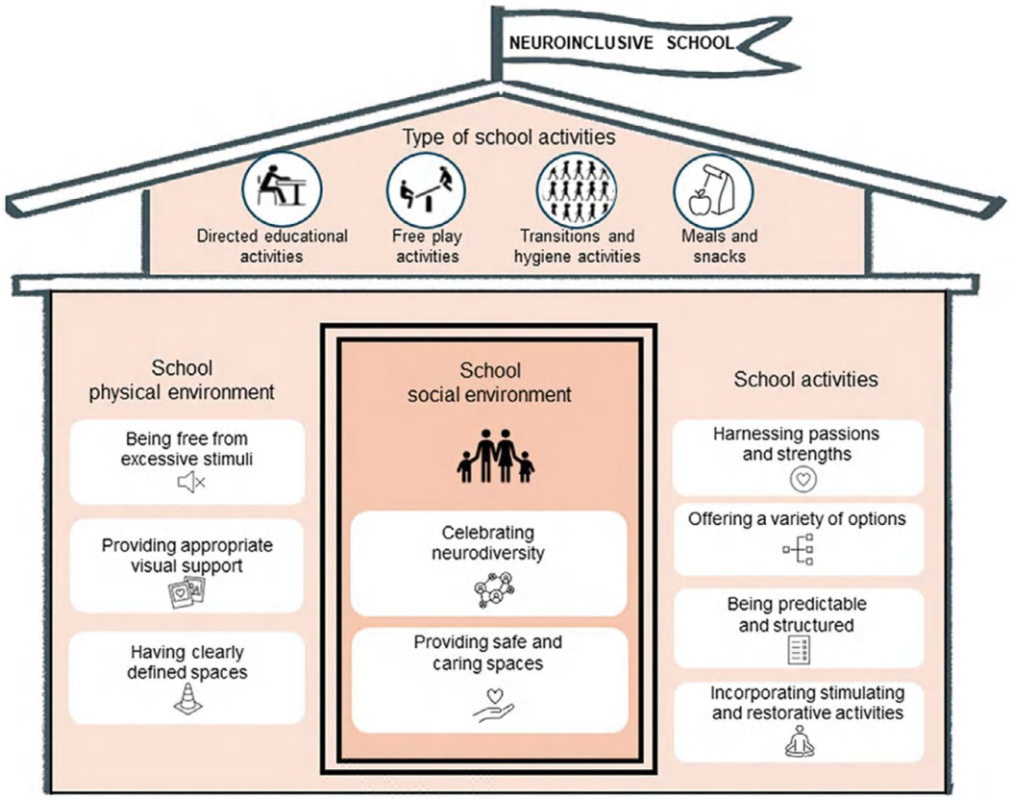


Figure 4.6.1 A neuroinclusive school model supporting meaningful participation and wellbeing for neurodivergent students and their peers (Rajotte et al., 2024)

celebrating and nurturing difference by improving our current environment. It is important to add that embracing neurodiversity does not mean ignoring a pupil's difficulties and only focusing on their strengths. Rather, it means changing our own practices, rules and structures at a school-wide level to ensure that neurodivergent pupils can participate in all aspects of school without necessarily 'fixing' themselves (Chapman, 2023). Rajotte et al. (2024) provide us with an excellent model of a neuroinclusive school (see Figure 4.6.1), which includes addressing the school's physical and social environment to enable meaningful participation for neurodivergent students and their peers. Key aspects of this model include ensuring it avoids excessive sensory stimuli, with the space designed to improve the engagement, learning and agency of pupils, and by focusing on strengths, restorative activities and making daily processes as predictable and structured as possible.

As a beginning teacher, can you think of small but inclusive steps that will help all learners in your classroom?

Developing knowledge and communities of support

The section above discussed how you can think inclusively as a beginning teacher. You also need subject-specific ideas and a deeper understanding of different areas of SEND to make effective current teaching strategies. For instance, the British Stammering Association offers excellent resources on including pupils who are hesitant to speak up in classrooms.

Developing knowledge

In England, the CoP 2015 identifies four main areas of SEND that you should consider when working with pupils. The categorisation was initially intended to support teachers in identifying and supporting pupils experiencing difficulties. Thomas and Loxley (2010) caution us that the creation of needs categories conceals other weaknesses that may exist within school structures, processes and policies. In a tightly resourced environment with pressure to perform to external metrics, teachers and school professionals are often expected to identify struggling pupils, 'examine the background, motivation and supposed traumas of the students' (Thomas and Loxley, 2010, p. 51), rather than reorganising school processes to ensure all learners are accommodated. Often, these assessments of needs are passed onto beginning teachers as established facts, when in reality many of these assessments take place in far from ideal classroom environments. A negative outcome of these categorisations that many scholars (Slee and Allan, 2001; Done and Knowler, 2022) point to, is the potential exclusion of children labelled as having 'SEND'. For instance, the most recent data on school exclusions and suspensions by DfE in 2022/2023 suggests that children with SEND and an EHCP were 2.8 times more likely to be permanently excluded than children without SEND provision; moreover, children with SEND but without an ECHP were 5.2 times more likely to be permanently excluded than children without SEND provision (DfE, 2024). It is due to concerns about exclusion that we encourage beginning teachers to develop a better understanding of how SEND areas can overlap or occur on a continuum (mild to severe), and how SEND can change over time. More importantly, experiences of disability intersect with pupils' ethnicity, gender, sexual orientation, social class, immigration trajectory, religion and other social markers. Below are the four areas of SEND outlined in the CoP 2015.

- communication and interaction;
- cognition and learning;
- social, emotional and mental health;
- sensory and/or physical needs.

Familiarity with different areas of need requires you to have an understanding of the impact of formal and informal labelling of pupils, including how it may adversely impact a pupil's self-esteem and overall educational experiences (Caslin, 2014).

Communication and Interaction

You may notice pupils who have difficulty in communicating what they want, understanding what others are saying, or difficulty in reading implicit rules in communication (e.g. turn-taking). Communication also encompasses eye contact, body language, gestures and facial expressions (Wearmouth, 2017). This area of need includes pupils with dyslexia, dyspraxia, hearing impairment, autism spectrum disorder (ASD) and speech and language delay. Pupils requiring support in communication and interaction in the classroom may also require support in developing peer and teacher relationships. Whereas English as an Additional Language (EAL) is not categorised as SEND, you may find that pupils with EAL also benefit from communication-based teaching strategies – see Table 4.6.2. As beginning teachers, you may find it easier to identify pupils requiring support in expressive communication, but more challenging to identify pupils with receptive language needs, i.e. pupils with difficulty understanding oral instructions which can be attributed to the pupil's personality.

Table 4.6.2 Teaching strategies for working with pupils with language needs

Peacey (2019, p. 283) provides useful teaching strategies for teachers working with pupils with language needs:

- Check understanding
- Use visual aids and cues to the topics being discussed
- Place the pupil so that they can hear and see
- Explain something in different ways if you have not been understood the first time
- Repeat what pupils say in discussion or question and answer sessions (in any case, others in the class may not have heard)
- Allow time for pupils to respond in question-and-answer sessions and, if necessary, ensure that they are pre-prepared for responding, perhaps by a TA

Source: For additional educational resources, you can visit: www.communication4all.co.uk, www.ican.org.uk, https://www.bdadyslexia.org.uk and the Dyslexia MESHGuides: https://www.meshguides.org//guides/node/612

Table 4.6.3 Teaching strategies for working with pupils with ASD

Inchley (2001 cited in Jones, 2013, p. 81) suggests considering:

- Having Clear Structure in delivering lessons
- Thinking of using visual images, text and props,
- Thinking of using Small Group Teaching to work on IEP targets
- Considering using different forms of communication and highlight non-verbal aspects of stories
- Songs and rhymes can used in a similar way to that used in musical interaction
- Incorporating turn-taking and paired working in your teaching practice

In addition, you can consider

- Using Social Stories
- Having Fixed lesson routines
- Reducing sensory clutter

Source: For additional resources, see: The National Autistic Society (www.autism.org.uk)

Pupils with ASD also benefit from support in social interaction, verbal and nonverbal social communication and social imagination. Pupils with ASD do not always require specialist resources and expertise; Inchley (2001) reports that pupils with ASD found features of Literacy Hour useful – as shown in Table 4.6.3.

Cognition and learning

As beginning teachers, you will be becoming familiar with various differentiation strategies (by learning outcome, grouping, task, process). However, some pupils will learn at a slower pace even with appropriate differentiation and adaptation (see Unit 4.1 on individual learning). For instance, pupils with Down Syndrome may require support in visual, hearing and speech and language, therefore, you should ask the SENCo to check if they have an EHCP outlining their provisions. Please see Table 4.6.4 which outlines some strategies that you can consider when working with pupils with Down Syndrome.

Table 4.6.4 Teaching strategies for pupils with Down Syndrome

The International Guidelines for the Education of Learners with Down Syndrome recommends the following practices. Notice how many of them are good teaching practices that would benefit all pupils.

- Visual supports to focus attention
- Visual scaffolds such as photos and pictures to support language teaching
- Common classroom aids such as number lines, calculators, letter charts, grids and diagrams (ready availability of these supports reduces the load on a student's working memory and visual supports such as letter and number charts can be affixed to a child's desk or kept in the back of an older student's workbook)
- Commonly available tools, such as using the calendar on smartphones for the classroom schedule (this supports attention as well as providing opportunities to become adept with devices in common usage)
- Models of completed work as a guide
- Explicitly plan the lesson, including the tasks to be undertaken by learning support assistants (LSA)
- Explain the learning intention of the lesson to the LSA prior to the lesson, including contingency plans such as alternative activities should they be needed
- Include the student with Down Syndrome in all phases of the lesson, including whole class instruction – then support these activities through planned adjustments
- Monitor the learning of all students in class, including the student with Down Syndrome by moving around the class during group and individual activities
- Make learning adjustments available to all learners
- Assess learning throughout the lesson through questioning, observing and gathering work samples
- Provide adjustments to tasks such as recording learning, to ensure the learner with Down
- Syndrome is able to take part
- Model respect and hold high expectations for all learners in the class

Source: Faragher, Robertson and Bird (2020, p. 24/26), https://www.ds-int.org/education

Social, emotional and mental health needs

Pupils with SEMH needs may:

> experience a wide range of social and emotional difficulties which manifest themselves in many ways. These may include becoming withdrawn or isolated, as well as displaying challenging, disruptive or disturbing behaviour. These behaviours may reflect underlying mental health difficulties such as anxiety or depression, self-harming, substance misuse, eating disorders or physical symptoms that are medically unexplained. Other children and young people may have disorders such as attention deficit disorder, attention deficit hyperactive disorder or attachment disorder.
>
> (DfE and DfH, 2015, p. 98)

As a beginning teacher, you should recognise that supporting pupils with an 'invisible' SEND is more challenging because initially it is harder to discern the pupil's barriers to learning. The stigma attached to SEMH needs can lead to the pupil (or their families) being blamed for their difficulties, being viewed as troublemakers rather than as someone in need of support. In some instances, the SENCo may refer the pupil to local Child and Mental Health Service (CAMHS) if they suspect the pupil might self-harm or harm others. Your classroom can provide a nurturing space that may not otherwise be available to pupils with SEMH needs. The Covid-19 pandemic, extended lockdowns and

Table 4.6.5 Teaching strategies for pupils with ADHD

As a teacher, you can consider:
■ The pace of tasks in lessons is appropriate ■ Check with pupil to see they understand the task set for them ■ Reward and celebrate their completion of task ■ Allocate a space in schools that pupils could use if they need to temporarily leave the class

Source: For further information see: Attention Deficit Disorder Information Service (ADDIS) http://www.addiss.co.uk/; Young Minds https://youngminds.org.uk/

remote schooling mean that your role is more critical in recognising pupils' SEMH needs and establishing appropriate support. The Children's Commissioner (2024) noted that during 2022–2023 almost 120,000 children were missing from education so creating mentally healthy classrooms for all children and young people is an important task for teachers.

In a busy and diverse classroom, developing skills in supporting neurodiverse learners is important. For example, pupils with attention deficit hyperactive disorder (ADHD) can require additional support for their SEMH needs. They can appear restless, inattentive, hyperactive and/or impulsive in class. Whilst pupils with ADHD are often supported with behavioural and medicinal interventions in accordance with the National Institute for Health and Care Excellence (NICE) guidelines, you can utilise different educational strategies as highlighted in Table 4.6.5.

Physical and/or sensory needs

> Many children and young people with vision impairment (VI), hearing impairment (HI) or a multi-sensory impairment (MSI) will require specialist support and/or equipment to access their learning, or habilitation support. Children and young people with an MSI have a combination of vision and hearing difficulties … Some children and young people with a physical disability (PD) require additional ongoing support and equipment to access all the opportunities available to their peers.
>
> (DfE and DfH, 2015, p. 98)

Many pupils with physical and/or sensory needs have no cognitive impairments and therefore the key role of the teacher is to remove barriers to access and participation in the curriculum. Talking to pupils with physical and/or sensory impairments and finding out about their strategies can be important to ensure that we do not assume that a physical and/or sensory impairment is necessarily a learning difficulty. For example, Hearing impaired pupils can experience varying levels of difficult in language processing skills in hearing classrooms (Wearmouth, 2017). About 78% of school-aged Deaf children in Britain attend mainstream schools without specialist provision, whilst 6% attend mainstream schools with resource provisions (National Deaf Children's Society, 2023). To support access to learning, you can employ radio aids, specialist TAs trained in British Sign Language (BSL), and a combination of different communication approaches (There are MESHGuides on BSL and also on using radio aids to support teachers' classroom practice, see https://www.new.meshguides.org). Practically, you can also utilise other inclusive strategies as highlighted in Table 4.6.6, but it is also important to talk to pupils about the effective strategies they use to support their own learning.

Table 4.6.6 Teaching strategies for pupils with hearing impairments

As a beginning teacher, you can:

- Seat child near the front of the class and away from sources of noise (e.g. outside traffic, noisy heaters, hum of overhead projector)
- Make sure that you have the pupil's attention before starting to talk, otherwise they may not have got the first part of the conversation or instructions
- Face the pupil when you are talking
- Try not to do any of these things:
  - cover your face with your hands or objects
  - walk around the room
  - standing with your back to a window as this creates a shadow
- Use whole sentences not single words – if a child does not understand a word, use a different word with the same meaning
- Do not talk at the same time as writing on the black/whiteboard
- Speak clearly, naturally and at a normal rate, shouting can distort lip patterns and also give the impression from facial expression that you are angry with the pupil
- Allow pupils time to read or look at visual aids and/or instructions before talking
- Encourage all children to say when they do not understand something as very often a HI child will nod and smile when in fact they do not understand
- When planning a lesson, it is vital that time is taken to brief support staff on subject matter, including the key points or new concepts to be covered, as well as the vocabulary to be used

Source: https://ccea.org.uk/sen-inclusion/inclusion-general-strategies/hearing-impaired-hi#section-3155.
For further educational resources, see: National Deaf Children's Society https://www.ndcs.org.uk

Developing communities of support

As you develop your experience of SEND educational practices, you will recognise that developing high-quality provision is dependent on closely collaborating with different stakeholders. For you to become a 'teacher of all children', you will need support and access to good signposting as you work with diverse learners. Your SENCo or similar lead professional will be an invaluable resource to this end.

Working with a lead professional: SENCos

The SENCo role in England was first proposed in the Warnock Report (1978) and became statutory in the 1993 Education Act and the resultant CoP 1994 (DfE, 1994). According to Oldham and Radford (2011) and Tissot (2013), the SENCo role was unclear in the earliest iterations of the CoP. However, since the implementation of the latest version of the code (DfE and DfH, 2015), the role is now firmly rooted in leadership and strategic decision-making. Currently, only SENCos in England must take a National Professional Qualification for Special Educational Needs Co-ordinators within three years of taking up the role of SENCo and the CoP lays out their responsibilities as follows:

- Overseeing the day-to-day operation of the school's SEN policy
- Co-ordinating provision for children with SEN
- Liaising with the relevant Designated Teacher where a looked after pupil has SEN

- Advising on the graduated approach to providing SEN support
- Advising on the deployment of the school's delegated budget and other resources to meet pupils' needs effectively
- Liaising with parents of pupils with SEN
- Liaising with early years providers, other schools, educational psychologists, health and social care professionals and independent or voluntary bodies
- Being a key point of contact with external agencies, especially the local authority and its support services
- Liaising with potential next providers of education to ensure a pupil and their parents are informed about options, and a smooth transition is planned
- Working with the headteacher and school governors to ensure that the school meets its responsibilities under The Equality Act 2010 with regard to reasonable adjustments and access arrangements
- Ensuring that the school keeps the records of all pupils with SEN up to date

(DfE and DfH, 2015, pp. 108–109)

Every school in England is statutorily required to have a SENCo. It would be a good idea for you to talk with your SENCo (or similar role) about the features of the SEND and/or additional needs provision in your school. SENCos are experienced and committed professionals and are likely to have a wealth of knowledge about the areas of needs, as well as a host of approaches that can help you to create an inclusive classroom and remove barriers to learning. You are not expected to be an 'expert' on SEND, indeed no one can be an expert in all aspects of the field, so knowing how and where to get support is crucial. Now read Task 4.6.4.



Task 4.6.4 Get to know your school

As a beginning teacher, you will have the remit to manage provisions in your own classroom. Find out who leads on SEND and/or additional learning needs provision in your country or particular context. This might be the SENCo or SENDCo, or it might be an Inclusion Coordinator. In Scotland, they are called Additional Support Needs teachers, whereas in Wales they are called Additional Learning Needs Coordinator. Arrange to meet with the lead professional in your setting to learn the following:

- How many pupils have been identified against each category of need in your school?
- Are there areas of need that are more represented than others?
- What specialist teaching resources does your school have?
- Which services does the SENCo/lead professional work with?
- How does the SENCo/lead professional work with teachers?
- Is there a well-resourced SEND or additional support needs team and how are they deployed throughout your school?
- How are TAs deployed and are there differences in how they work with learners?
- What is the process for accessing support if you are concerned about the progress of a pupil?

It would be useful for you to learn how SEND provisions are organised in the wider context of your school, so that you know how you can access support when you need it. Use the information you have collated to complete a gap analysis in the context of your own classroom.

Working with TAs

TAs are employed extensively in schools and are variously utilised to support teaching staff in the classroom, including being an invaluable resource for supporting learners with SEND. You will likely work with a TA at some point as they make up over a quarter of the school workforce (Sharples, Webster and Blatchford, 2015). It is important that you develop positive/productive relationships with TAs, as they play a vital role in ensuring that learners can access the curriculum, participate in learning activities, and experience success. Sharples, Webster and Blatchford (2015), reporting for the Education Endowment Fund (EEF), suggest key principles to consider ensuring you are 'making best use of TAs'.

- TAs should not be used as an informal teaching resource for low attaining pupils;
- Use TAs to add value to what teachers do, not replace them;
- Use TAs to help pupils develop independent learning skills and manage their own learning;
- Ensure TAs are fully prepared for their role in the classroom;
- Use TAs to deliver high quality one-to-one and small group support using structured interventions;

You should recognise TAs as assets who possess experience and expertise around specific areas of need. However, as Webster (2024) argues, in some cases TAs are pulled away from their support role to do cover support or 'backfill' teacher duties. Therefore, you need to be vigilant so that TAs are deployed effectively and that they are not diverted from supporting pupils with additional needs.

SUMMARY AND KEY POINTS

Every teacher has a responsibility to meet the needs of learners with SEND. This unit highlights key issues related to your statutory obligations as a beginning teacher regarding teaching children with SEND, and sources of support to help develop your practice.

- Most national education systems have a range of statutory policies to ensure that learners with SEND have a right to appropriate, accessible and inclusive education provision.
- Developing differentiated approaches to your teaching can be a creative and enjoyable process when working in diverse classrooms.
- There are various people in school who can help your development around supporting learners with SEND, such as the SENCo and a similar lead professional. They will signpost you when you need information.

Further resources

Frederickson, N. and Cline, T. (2015) *Special Educational Needs, Inclusion and Diversity Maidenhead*. New York: Open University Press.

This book provides a comprehensive and accessible approach for inclusive practice as well as an analysis of the implications for SEN of linguistic, cultural and ethnic diversity.

National Association for Special Educational Needs (NASEN): https://nasen.org.uk/resources

This is a charitable membership organisation that exists to support and champion those working with, and for, learners with SEND and learning differences. There are many useful resources on this site including training events, links to research, and tips for classroom teachers.

Special Needs Jungle: https://www.specialneedsjungle.com/

This is an extensive parent-led site that contains hundreds of resources from tips for working with children with specific needs to multi-professional working.

Whole School SEND: https://www.wholeschoolsend.org.uk/

Whole School SEND is a consortium which brings together schools, organisations and individuals who are committed to ensuring that every child and young person with SEND can maximise their potential. A free guide for teachers https://www.wholeschoolsend.org.uk/resources/teacher-handbook-send, viewed 30 July 2025.

MESHGuides for SEND, including for example, dyslexia, dyscalculia, BSL, hearing impaired, and more. See https://www.new.meshguides.org

MESHGuides for EAL: https://www.meshguides.org/guides/node/112

MESHGuides for Dyslexia: https://www.meshguides.org//guides/node/612

MESHGuides for SEND: https://meshagain.meshguides.org/category/send/

MESHGuides for Inclusion: https://meshagain.meshguides.org/category/inclusion/

MESHGuides for Deaf Education: https://meshagain.meshguides.org/category/deafeducation/

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5 Helping pupils learn

This chapter is about teaching and learning. As you work through these units, we hope that your knowledge about teaching and learning increases and that you feel confident to try out and evaluate different approaches.

Unit 5.1 introduces you to a number of theories of learning. Theories about teaching and learning provide frameworks for the analysis of learning situations and a language to describe the learning taking place. As you become more experienced, you develop your own theories of how the pupils you teach learn and you can place theories in the wider context.

In Unit 5.2, teaching methods that promote active learning are examined. How you use these methods reveals something of your personal theory of how pupils learn. At this point in your development we suggest that you gain experience with a range of teaching methods so that you are easily able to select the method most appropriate to the material being taught.

Unit 5.3 provides you with details about teaching styles. Again, we suggest that as you gain confidence with basic classroom management skills you try out different styles so that you develop a repertoire of teaching styles from which you can select as appropriate.

Unit 5.4 provides you with information about methods for finding out about the quality of your own teaching and that of others through reflection using action research techniques and drawing on the evidence base underpinning educational practice. During your initial teacher education course, you are using action research skills in a simple way when you observe classes. In this unit, we explain the key aspects of action research and reflective and evidence-informed practice.

Unit 5.5 explores the deepening challenge of closing the achievement gap in secondary education. Teachers play a key role in supporting all learners, by building strong relationships, fostering engagement and helping pupils realise their potential. Drawing on research and current policy initiatives, the unit outlines practical classroom strategies including adaptive teaching, self-regulated learning and direct instruction.

Unit 5.6 introduces you to research in neuroscience that is yielding knowledge about brain function and how teachers can maximise pupil learning. This area of work is possible because activity of the brain can be monitored more easily than before using the latest MRI scanners and other techniques.

In Unit 5.7, effective teaching is understood as a reciprocal exchange, where teachers learn from pupils' insights and perspectives, enriching their own understanding. Exploring varied pedagogical

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landscapes, educators experiment with different methods, technologies and assessment strategies to foster critical thinking and curiosity. In this unit, building connections is emphasised – bridging knowledge gaps and establishing trust and collaboration.

Unit 5.8 explores the dynamic relationship between thought and language and suggests that ideas do not merely translate into words but develop through linguistic interaction. Everyday experiences, such as struggling to articulate half-formed thoughts, illustrate this process. The unit postulates that opportunities for discussion, explanation and engagement with language are fundamental to learning, reinforcing the idea that mastering language underpins intellectual growth.

Unit 5.9 explores pedagogy as the language that helps us think and talk about teaching and learning. Pedagogy is not just what we see happening in classrooms, but includes the theories, values and justifications behind teaching decisions.



ONLINE RESOURCES

Watch the explainer video on helping pupils learn through blending teaching approaches by scanning the QR code or visit routledgelearning.com/learningtoteach.



5.1 Ways pupils learn

Diana Burton

Introduction

Research studies can often appear to have relevance to learning but we need to be critically enquiring of any research we access. In this unit, we consider perspectives that have longevity in terms of their influence on pedagogy as well as the quality of the evidence provided within the research. In addition, as you gain practical experience of teaching, pupils and learning, certain studies may resonate with you so it can be illuminating to work with these perspectives and reflect on them in practice, possibly to produce assignments.

For *teaching* to be effective we know that having excellent subject knowledge is very important but is by no means the whole story. For *learning* to be effective you have to help your pupils harness their individual skills, strategies and knowledge to fully engage with the *how*, *what* and *whether* of subject content. The interaction between the activities of teaching and the outcomes of learning, therefore, is critical. In order to develop presentation and communication techniques that facilitate effective learning, a teacher must have some understanding of how pupils learn. Initial teacher education (ITE) lectures and school placements add to and refine the ideas you already have about learning and reveal differences in how individuals learn. Relating this increasing expertise to research studies from psychology can further illuminate your understanding. Of course, human development and learning do not occur independently of environment so we must always be mindful of cultural and societal differences. Siann and Ugwuegbu (2024) draw critically on cross-cultural data to look at how different home environments and differing social values across the world can affect children's emotional, social and intellectual development.

OBJECTIVES

At the end of this unit you should be able to:

- appreciate the interaction between ideas about learning and pedagogic strategies;
- explain and differentiate between some psychological perspectives on learning;
- appreciate more about your own approach to learning and teaching.

Check the requirements of your ITE programme to see which relate to this unit.

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How ideas about teaching and learning interact

Decisions you make about how to approach a particular lesson with a particular pupil or group of pupils depend on the interplay between your subject knowledge and pedagogic knowledge, your knowledge of the pupils and your understanding about how learning happens. Let us imagine that you have spent ten minutes carefully introducing a new concept to your class, but a few pupils unexpectedly fail to grasp it.

You test the reasons for their lack of understanding against what you know about the pupils and their:

- prior knowledge of the topic;
- levels of attention;
- interest and motivation;
- physical and emotional state of readiness to learn, etc.

You also consider factors relating to the topic and the way you explained it, for example:

- the relevance of the new material to the pupils;
- how well the new concept fits into the structure of the topic;
- the level of difficulty of the concept;
- your clarity of speech and explanation;
- the accessibility of any new terminology;
- the questioning and summaries you gave at intervals during your explanation.

Finally, you draw on explanations from educational psychology that have informed your understanding of how pupils learn and provided you with questions such as:

- Does the mode of presentation suit this task?
- Has sufficient time been allowed for pupils to process the new information?
- Does the structure of the explanation reflect the inherent conceptual structure of the topic?
- Do these pupils need to talk to each other to help them understand the new concept?

Your ideas about learning usually derive from a number of theories rather than from one specific theory, which allows you to continually revise them as you gain more experience, through the process of 'reflecting on theory in practice'.

The influence of psychological research on education

Table 5.1.1 shows some of the key psychological and cognitive research theories that have influenced teaching and education.

A determining factor in lesson preparation is the knowledge that pupils already possess. Unfortunately, identifying this knowledge is not as simple as recalling what you taught the pupils last time, as each pupil's experiences of, attitudes towards and methods of processing prior knowledge are distinct. Psychologists are interested in how pupils actively construct this individualised knowledge or 'meaning', and different theories offer different notions about what constitutes knowledge. *Cognitive developmental theory* depicts knowledge as being generated through pupils' active

Table 5.1.1 Psychological and cognitive research influences on education

Theories	*Key propositions/ideas*	*Implications for learning*	*Main theorists*	*Units*
Behaviourist Learning Theory	Started with experiments on animals; emphasises external stimuli for learning; concepts of operant and classical conditioning	Instructional learning, guided discovery, behaviour reinforcement/ management	Pavlov (1897) Watson (1913) Skinner (1957)	3.3 5.7 5.3
Social Learning (later Social Cognitive) Theory	Emphasises the impact of observational learning on cognition – abstract rules extracted from observing behaviour without such rules being explicitly stated; bridges behaviourist and cognitive-developmental theories	Peer learning, e.g. 'laddish' ethos; modelling socially acceptable behaviour and norms	Bandura (1977)	1.1 5.7 5.3
Gestalt Theory	Perception driven by the brain's search for 'wholeness'; people reorganise information to impose order on it using principles of proximity, similarity, figure-ground, continuity, closure and connection	Provide big picture in advance of detailed information; ensure context is clear	Wertheimer (1923) Koffka (1935) Kohler (1940)	
Cognitive Developmental Theory	Developmental maturation through stages needed for learning; experience of world leads to *assimilation* and *accommodation* of new concepts Extrapolation of stage theory beyond cognition to social, emotional and ethical development	Pupils only learn certain concepts when ready; provide pupil-centred discovery learning; concrete examples assist abstract thinking. Discuss moral issues; be alert to adolescent sensitivities	Piaget (1932) Kohlberg (1976) Selman (1980)	4.3 4.5
Metacognition and Self-Regulation Theories	Learning to learn: understanding and controlling own learning strategies	Encourage techniques for remembering, ways of presenting information when thinking and approaches to problem-solving	Flavell (1979) Adey (1992, 2008) Shayer and Adey (2002)	4.3 5.5 6.1
Social Constructivist Theory	Knowledge is socially constituted and actively restructured through experience with the environment; language is central to the development of thinking	Use group work, pupil talk and teacher scaffolding/ structured intervention to promote higher level thinking	Vygotsky (1962, 1978) Bruner (1966) Wood (1988)	5.7 5.3

(Continued)

Table 5.1.1 (Continued)

Theories	*Key propositions/ideas*	*Implications for learning*	*Main theorists*	*Units*
Constructivist Theory	People make their own sense of things in a unique way by examining it in relation to prior conceptions and experiences and see to what extent it fits; existing conceptions are paramount	Avoid direct instruction – create situations which facilitate pupils constructing their own knowledge starting with their own mis/ conceptions	Driver and Bell (1986)	5.3
Information Processing Theory	Concept development involves reconstructing rather than recalling 'schema'; key memory and retrieval processes are attention, analysis and retention.	Link new knowledge to what is already known; move from general to specific using sequential procedures; use key words and ideas to cue the pupil; meaningful not rote learning.	Bartlett (1932) Baddeley and Hitch (1974) Rumelhart and McClelland (1986)	5.3
Cognitive Load Theory	CLT builds from IP theory; extraneous stimuli impede pupils' ability to process new information and to create long-term memories	Information overload and distractions must be avoided or limited	Sweller (1988)	

exploration of their world (Piaget, 1932, 1954). More complex ways of thinking about things are developed as the individual matures. *Metacognition* research shows how the potential for acquiring knowledge is increased if pupils understand and can control their own learning strategies (Shayer and Adey, 2002). *Social constructivism* explains pupils acquire knowledge by actively constructing their individual meanings as their experiences and interactions with others help develop the theories they hold (Brown, 1994; Rogoff, 1990). *Information processing* views knowledge as pieces of information that the pupil's brain processes systematically and stores as abstractions of experiences. These theories have been particularly influential in shaping pedagogy and are outlined briefly below.

Cognitive developmental theory

Piaget

Jean Piaget was a Swiss psychologist who applied Gessell's (1925) concept of maturation (genetically programmed sequential pattern of changing physical characteristics) to cognitive growth. He saw intellectual and moral development as sequential, with the child moving through stages of thinking driven by an internal need to understand the world. His theory implied an investigative, experiential approach to learning.

According to Piaget, the stages through which the child's thinking moves and develops are linked to age, as exemplified in Table 5.1.2.

The pupil's stage of thinking interacts with experience of the world in a process called *adaptation*. The term *operations* is used to describe the strategies, skills and mental activities that children use

Table 5.1.2 Piaget's stage model of child development

Stage	*Age*	*Features*
Sensory-motor	Birth to about 2	Understand world through feeling, seeing, tasting, etc.
Preoperational	About 2 to 6	Begin to understand that other people have a viewpoint; increasingly able to classify objects into groups and to use symbols.
Concrete operational	About 6 to 12	Still tied to specific experience but can do mental manipulations as well as physical ones. Internal mental operations become available, such as addition, subtraction and class inclusion. Logical thinking develops.
Formal operational	About 12 to adult	Manipulate ideas in their heads and consider events that haven't yet happened or think about things never seen. Organise ideas and events systematically and can examine all possibilities. Deductive thinking becomes possible.

in interacting with new experiences. Thus, adding 2 and 2 together, whether mentally or on paper, is an operation. It is thought that discoveries are made sequentially, so, for example, adding and subtracting cannot be learned until objects are seen to be constant. Progress through the sequence of discoveries occurs slowly and at any one age children have a particular general view of the world, a logic or structure that dominates the way they explore the world. The logic changes as events are encountered that do not fit with their *schemata* (sets of ideas about objects or events). When major shifts in the structure of children's thinking occur, a new stage is reached. Central to Piaget's theory are the concepts of *assimilation*, taking in and adapting experience or objects to existing strategies or concepts, and *accommodation*, modifying and adjusting strategies or concepts as a result of new experiences or information (see Boyd and Bee, 2014, for a full description of Piaget's theory).

The stress on the idea of 'stages' in Piaget's theory was far-reaching, especially the implication that pupils need to be at a particular developmental level in order to cope with certain learning tasks. However, research studies have substantially refuted such limitations on a pupil's thinking (Donaldson, 1978). Thus, the construct of a staged model of cognitive maturation probably has less currency than the features of development described *within* the various stages. Flavell (1982), a former student of Piaget, argued that, while stage notions of development are unhelpful, Piaget's ideas about the sequences that pupils go through are still valid. They can help teachers to examine the level of difficulty of topics and curriculum material as a way of deciding how appropriate they are for particular age groups and ability levels. The key task for teachers is to examine the progress of individuals in order to determine when to increase the intellectual demand on them. Bruner (1966) argued that difficult ideas should be seen as a challenge that, if properly presented, can be learned by most pupils.

Metacognition/self-regulation

It was Flavell who first proposed the notion of *metacognition*, arguing that becoming 'aware of one's own "cognitive machinery" is a vital component of intelligence' (1979, p. 907). Being metacognitively aware can be likened to having a commentator in your mind who analyses and comments upon the methods you use to learn a new concept or skill while the learning is happening.

Israeli researchers, Mevarech and Kramarski, have studied metacognition, particularly in relation to mathematics, for 40 years, describing it as 'a higher-order thinking process which involves active control over cognitive processes' (2014, p. 36).

Over a similar period in England, Adey and Shayer demonstrated through their development of Cognitive Strategies in Science Education (CASE) that learning potential is increased if pupils are metacognitively aware, i.e. if they understand and can control their own learning strategies (Shayer and Adey, 2002; Adey, 2008; Adey and Shayer, 2013). These strategies include techniques for remembering, ways of presenting information when thinking, approaches to problems and so on. So, if you cue pupils into the specific skill being learned and encourage them to reflect on its transfer potential, this will extend their strategies to other tasks and subjects.

Debra Myhill has researched metacognition in relation to language in the UK for many years. She explains that 'On one level, every act of writing involves metalinguistic activity: writing is a deliberative act, and, regardless of the age and expertise of the writer, is one that requires conscious engagement with shaping text' (Myhill et al., 2018, p. 336).

It has been widely acknowledged that better outcomes can be attained if pupils harness their own metacognitive strategies (see, for example, a huge meta-analytic review of studies by Australian researcher, John Hattie, 2012). A systematic review of 50 studies by Perry et al. (2019) endorses these findings and provides a good overview of the genesis of metacognitive theory, showing how it links to social constructivism, thinking skills and research into self-regulation.

It seems, then, that the higher-order cognitive skills of Piaget's stage of formal operations can be promoted and encouraged through a focus on metacognition. Pupils need, above all, to learn to learn, and it is your job to help them. The ready availability of instant information via the Internet requires pupils to have the confidence and skills to sift through it and judge its utility and relevance to their work. Now complete Task 5.1.1.



Task 5.1.1 Analysing learning activities: learning how to learn

Give pupils at least one opportunity during each lesson to think about the strategies they are using in their learning. Ask them, for instance, how they reached a particular conclusion, how they tackled the drawing of a 3D shape or how they would undertake a journey from point A to point B.

Relate your findings to research on metacognition as well as to other research approaches, for example, those covered in this unit and in academic journals, to write a report on the impact of metacognition in your lessons.

Discuss your findings with a peer or your mentor or tutor.

Social constructivist theory

The ideas of Russian psychologist Lev Vygotsky and of Jerome Bruner in the USA have influenced educators increasingly over the past 50 years. Vygotsky's work dates from between the 1920s and 1930s (he died in 1934). His most influential work, *Thought and Language*, was not available in English until 1962 (Vygotsky and Kozulin, 2012).

Vygotsky

Both Piaget's and Vygotsky's theories reveal a child-centred position and an emphasis on action in the formation of thought but, for Vygotsky (1978), psychological activity has sociocultural characteristics from the very beginning of development (see Keenan et al., 2016). Conceptual understanding can be generated from a range of different stimuli, implying a problem-solving approach for pupils and a facilitator role for the teacher. Whereas Piaget considered language a tool of thought in the child's developing mind, for Vygotsky, language was generated from the need to communicate and was central to the development of thinking. He emphasised the functional value of egocentric speech to verbal reasoning and self-regulation, and the importance of sociocultural factors in its development. In communicative talk, the development is not just in the language contrived to formulate the sentence but also through the process of combining the words to shape the sentence because this shapes the thought itself.

Vygotsky's emphasis on the importance of talk as a learning tool has long been absorbed into UK teaching practices, with seminal reports reinforcing the status of talk in the classroom (Bullock Report, 1975; Norman, 1992). Vygotsky believed that such communicative instruction could reduce a pupil's 'zone of proximal development' (ZPD), i.e. the gap between their current level of learning and the level they could be functioning at with adult or peer support: 'What a child can do today in cooperation, tomorrow he will be able to do on his own' (Vygotsky, 1962, p. 67, see Vygotsky and Kozulin, 2012). Social constructivism, therefore, stresses the necessity for group interaction and adult intervention in consolidating learning and extending thinking. Myhill (2006) found that teacher discourse can variously support or impede pupil learning, so it is vital that your own discourse assists cognitive and conceptual connections in pupils' learning. Cambridge University's Thinking Together project provides resources for teachers drawn from over two decades of research into a dialogue-based approach to the development of children's thinking and learning (see, for example, Mercer et al., 2019, Cook et al., 2021). Task 5.1.2 asks you to analyse pupil talk.



Task 5.1.2 Analysing learning activities: pupil talk

Review the advice for teachers on the University of Cambridge Thinking Together website and their MESHGuide (Cook et al., 2021). Consider how often you make opportunities for your pupils to talk in lessons, to each other, to you, to a digital or video recorder. Listen to how the talk develops the thinking of each member of a small group of pupils. Notice how well-timed and focused intervention from you moves the thinking on. Write up examples, showing the advantages of promoting pupil talk with your pupils; compare and contrast your findings with a peer.

Bruner

Bruner also placed an emphasis on structured intervention within communicative learning models. He formulated a theory of instruction, central to which is the notion of systematic, structured pupil experience via a spiral curriculum where the pupil returns to address increasingly complex components of a topic. Current and past knowledge is deployed as the pupil constructs new ideas or concepts. Thus, the problem of fractions in Year 5 (ages 9–10 in England) may be tackled via many more

concrete examples than when it is returned to in Year 7 (ages 11–12 in England). Learning involves the active restructuring of knowledge through experience; the pupil selects and transforms information, constructs hypotheses and makes decisions, relying on a developing cognitive structure to do so.

You should try to encourage pupils to discover principles by themselves through active dialogue with yourself. Bruner said the teacher's job is to guide this discovery through structured support, e.g. by asking focused questions or providing appropriate materials; he called this process 'scaffolding' (Bruner, 1983). The ideas of pupils emerging through their talk are scaffolded or framed by the teacher putting in 'steps' or questions at appropriate junctures. When hearing an idea emerge you can intervene by asking a question that requires the pupils to address that idea explicitly, which may help them find the solution. Bruner argued that the scaffolding provided by the teacher should decrease in direct correspondence to the progress of the pupil.

Information processing theory (IP theory)

Information processing (IP) theory originated within explanations of perception and memory processes and was influenced by the growth of computer technology. Psychologists saw the functioning of computers as replicating the behaviour of the brain in relation to the processing of information. The basic idea is that the brain *attends* to sensory information as it is experienced, *analyses* it briefly within the short-term memory (STM) and *stores* it with other related concepts in the long-term memory (LTM). This process sounds linear but there is a continual looping back and forth between the stages as *existing* memories are recalled to help understand *new* information which in turn updates and refines *existing* information in LTM. This process is more efficient if material can be stored as abstractions of experience rather than as verbatim events. Now look at Task 5.1.3.



Task 5.1.3 Analysing learning activities: rote/meaningful tasks

If I ask you to tell me the six times table, you recite, 'one six is six, two sixes are twelve', and so on. You do not explain to me the mathematical principle of multiplication by a factor of six. On the other hand, if I ask you the meaning of the term 'economic enterprise', it is unlikely that you will recite a verbatim answer. Instead, your STM searches your LTM for your 'schema' or idea of enterprise. You then articulate your abstracted understanding of the term, which may well be different from that of the person next to you. In terms of intellectual challenge, articulating the second answer requires greater mental effort, although knowing one's multiplication tables is a very useful tool.

Apply this reasoning to something you teach within your own subject, making notes on both rote and meaningful versions of the task.

You always need to be clear that there is a good reason for requiring pupils to learn something by rote because rote learning is not inherently meaningful so cannot be stored in LTM with other related information. Rather, it must be stored in its full form, taking up a lot of 'disk' space in the memory. Information stored in this way is analysed only superficially in STM; the pupil does not have to think hard to make connections with other pieces of information.

It can often be difficult in secondary schools for teachers to determine precisely the prior knowledge of their pupils. You will observe that 'good' teachers cue pupils into their prior knowledge by asking questions about what was learned the lesson before or giving a brief resume of the point reached in a topic. Such strategies are very important because, if previous learning has been effective, information is stored by pupils in their LTMs and needs to be retrieved. Seminal psychological work established that learning is more effective, i.e. more likely to be understood and retained, if material is introduced to pupils according to the inherent conceptual structure of the topic (Ausubel, 1968; Gagné, 1977). Concept maps reflect the conceptual structure of a topic. When preparing a scheme of work, you should identify the over-arching broad concepts that encapsulate the topic, then think about how these break down into more specific ideas and how they apply within different contexts. To understand, for example, momentum, you need to understand mass and velocity (which in turn requires an understanding of speed and direction), and so on. Information that is stored using a logical structure is easier to recall because the brain can process it more easily in the first instance, linking the new ideas to ones that already exist in the memory. As a teacher, this requires you to have thought through the structure in advance, and to know how the concepts fit together; hence the importance of spending time on schemes of work even where these are already produced for you.

This emphasis on structure and sequence can be found perhaps most readily in the teaching of modern foreign languages (MFLs) and mathematics. In MFL, for instance, teachers encourage recognition and acquisition of vocabulary first, followed by the construction of spoken and written sentences. This approach might be described as moving from the general to the specific. Thus, in mathematics, we start with number recognition, moving to the general concepts of addition, subtraction and multiplication, and on to more specific computations such as calculating area, solving equations or estimating probabilities.

Concept development

It is helpful to consider briefly how IP relates to concept development. The material held in the LTM is stored as sets of ideas known as 'schemata' or schemas. We hold schemas for everything so we can classify and understand our experience of the world. A schema is a mental structure abstracted from experience. It consists of a set of expectations with which to categorise and understand new stimuli. For example, our schema of 'school' consists of expectations of pupils, teachers, classrooms, etc. As teachers, our school schema has been refined and developed as more and more information has been added and categorised. Thus, it includes expectations about hierarchies, pupil culture, staff room behaviour and so on. It is probable that our school schema is different from, and more complex than, the school schema held by a parent, simply because of our involvement in schools.

When children are young, their schemata do not allow them to differentiate between pieces of information in the way that those of older pupils do. A one-year-old's schema of dog might include expectations about cats too because they have insufficient experience of the two animals to tell them apart. As they experience cats as furry, dogs as hairy, cats meowing, dogs barking, etc., greater differentiation is possible. Since the object of school learning is to promote pupils' concept differentiation in a range of different subjects, you should encourage comparison between objects or ideas and introduce new ideas by reference to concrete examples. Even as adults, while we *can* think abstractly, we find new ideas easier to grasp if given concrete examples of them. Teaching in the context of IP theories stresses the application of knowledge and skills to new situations.

Your role is to help pupils find new ways of recalling previous knowledge, solving problems, formulating hypotheses and so on. Critical thinking tasks in which pupils have to wrestle with new ideas and issues encourage connections between areas of subject knowledge or experience. Rote learning, copying information from a book or taking dictated notes from the teacher are very unsuitable for promoting concept development. Meaningful approaches to learning whereby information is processed (or practised) to a greater degree than everyday observations are required for long-term memories (or learning) to develop (Baddeley and Hitch, 1974).

Cognitive load theory

Cognitive load theory builds on IP models of memory processes, in particular the long-term or 'working' memory (Sweller, 1988). Cognitive load is essentially the amount of effort required to process (think about) and learn (memorise) a certain piece of information or task. When too much information is presented at once, working memory becomes overwhelmed and much of that information can be lost. We know how difficult it is trying to attend to two things at once, hence regular reminders to pupils to stop gazing out of the window or, more worryingly, at their mobile phones!

John Sweller, an Australian educational psychologist, showed that cognitive load during processing is higher if tasks are unnecessarily complex or if pupils are distracted from information they need to pay attention to; as a consequence learning is less effective. He said that cognitive load can be intrinsic, extraneous or germane (Sweller, 2010).

Intrinsic load depends on the complexity of the task or concept and a pupil's ability to understand the new information. New, high order tasks, for example, solving a simultaneous equation, are more challenging than simple tasks like adding or dividing numbers. However, the load can be reduced by breaking complex tasks down into a series of self-contained separate steps.

Extraneous load comes from the demands imposed by the teacher or the task instructions, outside the task itself, so it can be reduced by good teaching methods. Setting clear objectives and a precise focus for pupils' work is essential so they're not distracted or taken off at a tangent. Presenting information in a way that best suits the task means pupils can focus fully on what it is they must learn. Sometimes a diagram will be far more effective than a written or oral description. Referring to a diagram in which all the key points are visible at once may be better than having to keep the early parts of a piece of narrative information in the memory until the later parts appear. Similarly, a role-play or physical demonstration might assist pupils grasping certain material more easily.

Germane cognitive load is produced by developing schemas, which, as described earlier, are conceptualisations of a particular idea or object, comprising a set of expectations that come to mind when we encounter it in the future. Learning new things or encountering new experiences produces germane cognitive load as we are observing and exploring the new idea in order to help us anticipate and understand it in the future. It is thus at the heart of learning and what we strive for in our lessons since it leads to progress in our pupils' thinking.

You can try to reduce intrinsic and extraneous cognitive load in a number of ways, including:

- Using worked examples or step-by-step instructions for new or complex tasks;
- Merging multiple sources of visual information to avoid splitting attention;
- Using both visual and auditory modes to aid working memory.

We have discussed cognitive-developmental theories of learning, metacognition, social constructivism and IP. In each of these theories there has been an emphasis on the individual and the differences between them.

SUMMARY AND KEY POINTS

This unit emphasises:

- How important it is to develop and refine your own models of learning as you gain more experience of your pupils, your subject and how learning contexts interact with teaching processes.
- The symbiosis between theoretical positions and pedagogic practices, for instance, concept development is enhanced when pupils are introduced to new ideas via concrete examples and retention is aided if topics are taught according to their inherent conceptual structure and the cognitive load is not too great.
- The key features of cognitive developmental, social constructivist and IP theories as they apply to pupil learning.
- The benefits of adopting a facilitative, interventionist approach and of aiming for a variety of approaches in presentation, resource, task and assessment can be extrapolated from all the theories that have been discussed.
- Learning is likely to be most effective when pupils are actively involved with the material through critical thinking, discussion with others and metacognitive awareness of their own learning strategies.

Check which requirements for your ITE programme you have addressed through this unit.

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Further resources

Bartlett, S. J. and Burton, D. M. (2024) *Introduction to Education Studies* (6th edition), London, UK: Sage.
Chapters 8 and 9 outline major psychological theories and contemporary psychological research into how pupils learn. Chapter 11 considers the pedagogical implications of this.

Buckler, S. and Castle, C. (2025) *Psychology for Teachers* (4th edition), London, UK: Sage.
A textbook for teachers and those training to teach on applying research from psychology to education. It offers practical advice on 'what works' in the classroom and why.

Claxton, G. and Powell, G. (2019) *Powering Up Students: The Learning Power Approach to High School Teaching*, Carmarthen: Crown House Publishing.
This book is one of a series to help teachers boost their pupils' learning dispositions and attitudes. The parent website offers extensive practical strategies for helping pupils learn how to learn: https://www.buildinglearningpower.com (viewed 8 November 2024).

Darling-Hammond, L., Schachner, A. C. W., Wojcikiewicz, S. K. and Flook, L. (2024) 'Educating teachers to enact the science of learning and development', *Applied Developmental Science*, 28(1), 1–21.
This article draws on a range of research from the social sciences to provide a thorough overview of how children learn and the implications for teacher education.

Levine, L. E. and Munsch, J. (2023) *Child Development From Infancy to Adolescence: An Active Learning Approach* (3rd edition), London, UK: Sage.
This book introduces readers to real-world applications of psychological research to child development while emphasising diversity. It covers the latest topics shaping the field of child development including a range of cultural examples.

Rudman, D. (2018) *Learning and Memory*, London, UK and New York, NY: Sage.
Topics include the major behaviourist theories of learning, cognitive theories of memory, social learning theories, the roles of emotion and motivation in learning and the neurological underpinnings of these perspectives.

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5.2 Active learning

Chris Harris

Introduction

Important tasks for you are to encourage the development of pupils' ability to learn independently and their understanding of key curriculum concepts and skills. Vygotsky's (1978) idea of the 'zone of proximal development' explains the role teachers play. The concept refers to the difference between what a learner can do without help and what they can achieve with guidance from a skilled partner. This concept is helpful to teachers in ensuring that we judge where our use of transmission techniques can be switched to facilitation with pupils taking responsibility for their own learning. Teachers cannot do the learning for the pupil and for enhanced understanding to occur pupils should be given opportunities to be active in the learning process.

This unit addresses ways in which you can help pupils to become independent learners through providing active learning in your classroom. Active learning underpins meaningful learning because it enables pupils to develop knowledge of the subject taught, and skills for understanding including the ability to reflect on the processes involved in the learning.

In this unit, active learning is defined. There is a discussion of different forms of active learning: lifelong learning, deep learning, learning to learn, Growth Mindset, Thinking and Learning, discovery learning, rote learning and aids to recall. There follows a section on active learning methodologies including Rich Tasks, Interdisciplinary curricula, Directed Activities Relating to Texts (DART) and utilising digital technology in remote or blended learning contexts (see also Unit 1.4). The section ends with ways to develop higher-order thinking skills.

OBJECTIVES

At the end of this unit you should be able to:

- explain the term 'active learning' and discuss the advantages of active learning for teachers and pupils.

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- understand how to embed active learning in your classroom.
- consider resources which support active learning.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

Defining active learning and experiential learning

Active Learning was a pedagogical term created by two professors (Bonwell and Eison, 1991) who argued that teaching should be less about imparting information to pupils via lecture-based learning and more about developing skills, while also engaging pupils in higher-order thinking, Active learning draws from theories of learning that were outlined in Unit 5.1, perhaps taking the best from each:

- It occurs when a learner takes some responsibility for the development of the activity, emphasising that a sense of ownership and personal involvement is the key to successful learning.
- It includes purposeful interactions with ideas, concepts and phenomena and incorporates reading, writing, listening, talking or working with tools, equipment and materials, such as paint, wood, chemicals, etc. Put simply, it is learning by doing, in contrast with being told.
- It refers to teaching practices (pedagogies) that give pupils opportunities to work with concepts over and over, in a variety of ways and with opportunities for immediate feedback, so that knowledge can take hold in their own minds. Active learning activities include role-playing, problem-solving, polling, debates, group work, case studies and simulations (see Unit 5.9 on different pedagogies).

Active learning is linked to experiential learning. Experiential learning is also learning by doing but with the additional feature of reflection upon both action and the results of action; only where pupils are 'engaged actively and purposively in their own learning is the term experiential appropriate' (Addison and Burgess, 2007, pp. 35–36).

Both active and experiential learning benefit teachers and pupils. They enable you to spend more time with groups or individuals, which allows better-quality formative assessment to take place (see Unit 6.1). They also enhance your support for pupils with special educational needs (see Unit 4.6). It is important to realise, however, that learning by doing, by itself, is not enough to ensure learning. The essential step to learning is reflection through active discussions with others, especially the teacher; such discussions involve 'thinking' as well as recalling.

Matt, an early-career teacher, articulates his experience of trying to embed active learning in his classroom in Table 5.2.1.

Table 5.2.1 An early career teacher's view on implementing active learning

It was difficult to understand active learning when I began teaching. Of course, you want to plan lessons that engage pupils, but the English National Curriculum almost implies that you have to 'put on a show' in order to have exciting and stimulating lessons. What I realised very soon was that entertainment isn't the focus. If you take that approach, you cannot do it for every lesson you teach. Using the principles of active learning made a real difference. When I started to think about how I structure activities to maximise each pupil's engagement and activity in the learning, I noticed that their attainment and achievement also improved. It takes time to become familiar with all of the different active learning strategies and I think you need to introduce them carefully and build them up over time in the classroom. Now I have a range of strategies I can use, and nothing beats the buzz of seeing pupils actively engaged in their own learning.

Lifelong learning: the four Rs – resilience, resourcefulness, reflectiveness and reciprocity

You have a role in opening the door to lifelong learning for your pupils. Guy Claxton (2002, 2015) describes four characteristics of expert, lifelong learners, commonly referred to as the four R's:

- **Resilience** describes learners' enjoyment of learning and their ability to resist distraction and persevere with their learning, despite setbacks.
- **Resourcefulness** refers to learners' ability to look more deeply into issues and question taken-for-granted assumptions.
- **Reflectiveness** is about learners' ability to consider and adjust approaches and concepts in the light of changing understandings and information.
- **Reciprocity** encompasses learners' ability to contribute to and learn from others' understandings and approaches.

Task 5.2.1 asks you to apply these ideas.



Task 5.2.1 The four Rs: resilience, resourcefulness, reflectiveness and reciprocity

Think of a lesson you have taught and answer the following:

1 What opportunities were there for the pupils to display or develop skills in the four Rs?
2 How could you adapt the lesson, or series of lessons, to enable pupils to develop or improve one or two of these characteristics?

Discuss your answers with another beginning teacher or your mentor.

Active learning encourages the development of the four Rs. Hargreaves (2005) argues that the best schools personalise learning to ensure that 'deep learning' happens. Most schools recognise that enabling pupils to perform well in exams is only a part of their wider educational purpose (see Unit 7.1). Schools can develop their pupils as learners equipped for a changing world requiring them to be lifelong learners. 'Deep learning' is best developed through encouraging pupil voice, formative assessment (see Unit 6.1) and learning to learn (Teaching and Learning Research Project, TLRP, James et al., 2006). Hargreaves stressed the importance of 'deep experience', arguing that pupil engagement is the key to good staff/pupil relationships and is a prerequisite for the development of learners who possess independence, responsibility, confidence and maturity.

Growth mindset

Dweck (2012) argues that a growth mindset allows an individual to believe their talents can be developed through hard work, effective strategies and support from others. These contrast with an individual with a fixed mindset who believes their intelligence is fixed and innate. A pupil with a growth mindset is more likely to embrace active learning and teachers should foster a growth mindset in pupils to develop their passion for learning.

Thinking together

We often place pupils in groups or pairs to discuss. Mercer (2019) has written extensively about the need to explicitly teach ground rules for talk so pupils can utilise these opportunities effectively. Pupils learn most from talk in class when:

- the teacher develops pupils' reflective awareness of how to talk and work together;
- the teacher encourages them to express tentative ideas;
- the teacher 'models' good talk skills in whole class discussions;
- the teacher 'scaffolds' group work but mainly stays out of it;
- group work involves tasks that really require children to 'think together'.

Mercer's work is further developed by the Cambridge Educational Dialogue Research Group (see Cook et al., 2021).

Learning how to learn

Learning how to learn is essential for active learning (James et al., 2006). By promoting activities that engage pupils and require them to participate in the task from the outset, teachers encourage an approach to learning that is both skills-based and attitude-based. Active learning methods promote habits of learning that are valuable in the workplace, the home and generally enhance pupils' capacity to cope with everyday life. School can be a place where pupils learn to do things well and in certain ways, thereby developing skills used throughout life. For example, pupils learn to consult a dictionary or a thesaurus in book form, as part of a word processing programme or online in order to find meanings or to check their spelling and grammar. These skills become habits, capable of reinforcement and development. Reinforcement leads to improved performance.

Discovery learning

At its simplest, discovery learning occurs when pupils are left to discover things for themselves within a planned framework. This is a common approach taken with younger pupils, e.g. in sensory and fine motor skill development with paints, sand and manipulative, for example, blocks/shapes, educational games (see MESHGuides Numeracy for All YouTube videos). In the secondary setting, a structured framework ('guided discovery') was originally introduced by Bruner (1966) and its use has been developed (Takaya, 2013). The theory invites learners to build on past experiences and knowledge and use their intuition, imagination and creativity to search for new information, and discover facts and truths. The pedagogy works through three inquiry-based phases:

- Formulating questions or giving pupils a problem or goal;
- Exploration and analysis;
- Drawing conclusions.

It is linked to 'investigation-based inquiries' and 'Rich Tasks' that adopt an approach to teaching and learning where pupils are provided with opportunities to investigate a problem and search for possible solutions, ask questions and test out ideas.

Rote learning

Discovery methods alone are not enough. Rote learning is needed for essential facts backed up with the development of memorisation skills. You need to consider what you want to achieve and match the teaching style with the purpose (see Unit 5.3 on teaching styles and Unit 5.9 on pedagogical tools). Rote learning may occur when pupils are required to listen to the teacher. There are occasions when you need to talk directly to pupils, for example: giving facts about language, of spelling or grammar, formulae in science or health matters such as facts about drug abuse or safe practice in the gymnasium. Some facts need to be learned by heart, by rote methods. There is nothing wrong with you requiring pupils to do this from time to time, provided that all their learning is not like that. Such facts are necessary to prepare for active learning opportunities; they contribute to the subroutines that allow us to function at a higher level.

Use Task 5.2.2 as a starting point to evaluate the different types of learning your teaching currently provides.

Task 5.2.2 Identifying types of learning

Review lessons that you have taught and identify where you have used active learning processes. Critically reflect on the effectiveness of your choices. If you have not given the pupils opportunities to learn in these ways, develop some lesson plans incorporating these.

Share your reflections with another beginning teacher or your mentor.

Active learning in the classroom: aids to recall and understanding

Sometimes information cannot easily be committed to memory unless a structure is developed around it to help recall. Rosenshine's (2012) principles of instruction are particularly helpful in establishing a methodology to achieve this. Recall is constructed using words, summaries, tables, diagrams, flow charts, spider diagrams or other visual stimuli as prompts. The act of compiling a summary contributes to remembering and learning. Figure 5.2.1 shows the 'plan, do, review, learn,

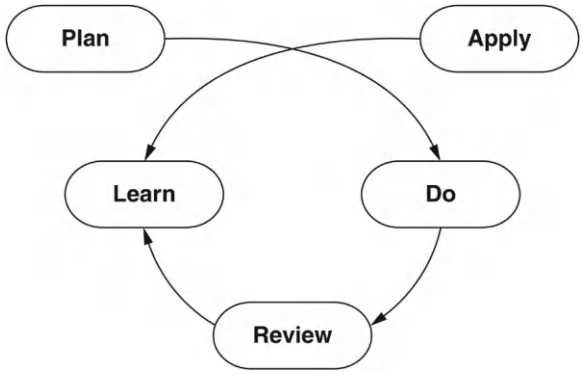


Figure 5.2.1 An active learning model: plan, do, review, learn, apply

Source: Watkins et al. (2007, p. 77)

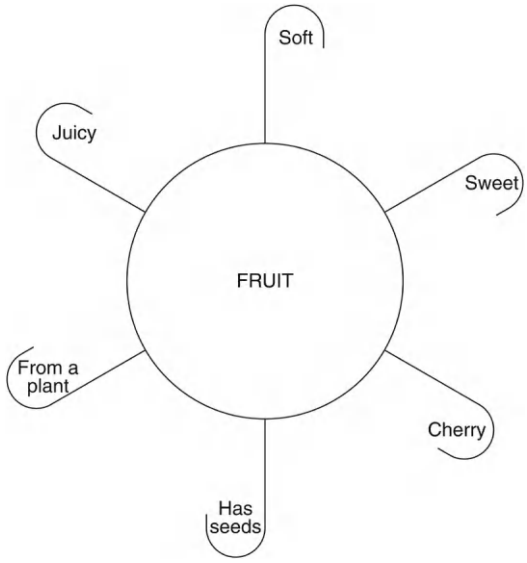


Figure 5.2.2 A pupil's meaning of fruit or a pupil's understanding of the concept of fruit

apply' model of active learning based on the idea that personal development proceeds best by reflection on your own actions. Reflection in the model is incorporated in the terms 'review' and 'learn'.

This model applies to pupils' learning as well as yours. To develop a more sophisticated model of active learning in relation to your own practice we recommend you carry out research in your own classroom. See Unit 5.4.

Mapping tools: spider diagrams and mind maps

It is often helpful for pupils and teachers to 'brainstorm' their understanding of an idea. One way to record that event is by a spider diagram (or burr diagram) or mind map in which the 'legs' identify the ideas related to the topic. Figure 5.2.2 shows a spider diagram constructed by a pupil of some ideas associated with 'fruit'.

Concept maps, more commonly called 'mind maps', are developed from spider diagrams and are used to display important concepts that occur in a topic and the links between them. See Figure 5.2.3. Concept maps can be made by pupils to summarise their knowledge. Making a concept map at the start of a unit of study helps to probe pupils' prior knowledge. In either case, you may need to provide a list of ideas with which pupils can work and to which they can add their own ideas. Task 5.2.3 asks you to prepare a concept map.



Task 5.2.3 Using concept mapping

Prepare a concept map for a unit of work that you are going to teach. Discuss this map with other beginning teachers and your mentor. Reflect on its value to enhance active learning in the lesson. Record your ideas for the use of concept maps in your lessons.

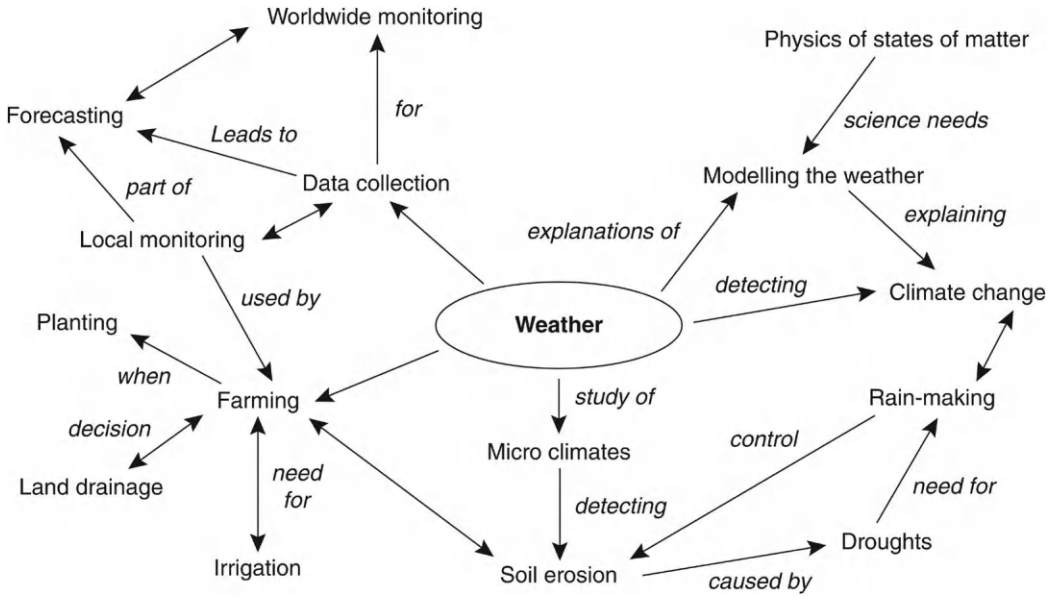


Figure 5.2.3 Concept map: Weather

Interdisciplinary learning and rich tasks

Another way to establish links between ideas and concepts is to explore topics and themes that connect different pools of knowledge. This is interdisciplinary learning, and it is instrumental in engaging pupils in active learning, moving beyond discrete subject areas to deeper learning opportunities. In secondary schools, colleagues are often willing to explore a theme with other subject areas and there is significant evidence in models like the International Baccalaureate (IB, 2021) or the Royal Society of Arts (RSA) Opening Minds project (RSA, 2021) and the XP School in Doncaster's 'expeditions' curriculum (XP, 2025) that this approach has positive learning impact. 'Rich Tasks' are the personalised way in which the pupils can respond to the interdisciplinary theme – through writing, presentation, drama, music, art, technology, etc. This form of active learning has been made easier to facilitate through teachers' and pupils' increased use and understanding of digital technology during the Covid-19 pandemic (Harris, 2021). (See also Unit 5.9 and Table 5.2.2.)

Directed activities related to text (DART) and pupil presentations

DART are ways of engaging pupils in active reading, writing and listening in order to foster their understanding of a text and their ability to reflect. These are long-standing classroom teaching strategies, which continue to be widely used (British Council, 2021). DART can involve the use of textbooks, but they use a variety of written and other visual materials, including resources downloaded from the web. DART are devised to ensure that pupils interact with a text. Interaction includes reconstruction activities for example: text completion, card sorts, sequencing and table and graph completion. Analysis activities include listing important words, text marking, drawing diagrams or reformulating a labelled diagram into continuous prose and re-ordering and restructuring of text,

Table 5.2.2 Examples from different subjects about how a study of rock 'n' roll in music could be integrated into different subjects to provide a deep learning experience for pupils

Theme: Rock N Roll and subjects contributing to cross-disciplinary working	*Broad content in the different subjects*
Music	Vocal techniques, forms and structures
History	Civil Rights movement to Luther King. Elvis's 'If I could dream'
Maths	Statistics – Elvis's record sales.
English	Creative writing inspired by images of Memphis in the 1960s or powerful speeches.
Art	Andy Warhol
Media	Photoshop a Mojo 1960s retro front cover including a 1950s/1960s rock n roll icon

Source: Adapted from Harris (2021)

for instance. The level of demand, i.e. differentiation, is adjusted by you to meet the needs of the pupils. A listening activity may be designed to help pupils understand instructions given by the teacher. DART and related types of learning activities emphasise the importance of language in learning and in assessing learning.

A productive way of gaining interest and involvement is to ask pupils to report their findings, summaries or interpretation of the text to the class. The summary could take one of the forms mentioned above. In addition, of course, a pupil could use the board, Interactive White board, a poster or a computer-assisted presentation such as PowerPoint. Pupils need to be prepared for this task and need to be given sufficient time. Public reporting is demanding on pupils, and it is helpful if a group of pupils draft the presentation and support the reporter. The presentation could be narrative, a poem, a simulation, diagram, play, etc. A suggested sequence of events is shown in the feedback loop, from 'Class discussion' to 'Task' in Figure 5.2.4, which can be introduced depending on the time available and the interest and ability of the class.

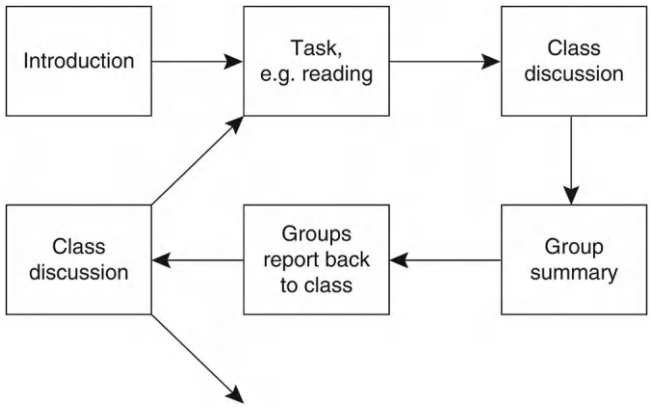


Figure 5.2.4 Reporting back

Lesson planning for active learning

It is important to distinguish between 'activities' and 'active learning'. It is relatively easy to fill a lesson with a series of activities that keep pupils busy and apparently enjoying them, yet these may provide an insufficient learning challenge. Such work may be well within the pupils' grasp and so they do not have to think much about what they are doing or why. Many pupils take seriously copying from the board, book or worksheet but such activities are often superficial and should be used very sparingly, even though they can keep a noisy class quiet.

Some lesson planning in the early stages of learning to teach may be focussed on ensuring that all your learning outcomes are addressed, or that your discipline is effective; this requirement may lead to a lesson that is teacher dominated. For example, you may have explained orally the lesson and its purpose and have asked some bridging questions; you may have then used a video and asked the pupils to complete a worksheet in response to watching the video; finally, you may have given out homework based on the class work. You, as the teacher, have been very active in the lesson and may well feel exhausted at the end of it! From the pupils' point of view, however, the lesson may have been quite passive because they were told what to do at every step, with little or no input into what they were to do and learn. In these circumstances many pupils may not fully engage in learning except in a superficial sense. Task 5.2.4 provides a springboard to explore these issues.



Task 5.2.4 Active classroom or active learning?

You may find this task best discussed with a group of peers.

A structured sequence for the learning of new vocabulary in a modern foreign language lesson is presented in Table 5.2.3.

The pupils seem to be busy but to what extent do you think the pupils are actively engaged in learning?

Identify any changes you would make to the lesson and explain why.

After you have carried out this task compare your response to that of an experienced teacher.

This short lesson sequence illustrates how to scaffold the learning at the initial stage (see scaffolding in Units 3.1 and 5.1). It provides opportunities for pupils to take part, but cognitive demands remain mostly at word or phrasal level and activities are teacher controlled. For pupils to progress further, they need to be led gradually towards the challenge of identifying patterns, generating sentences of their own and integrating this vocabulary into new structures and contexts. Furthermore, the memorisation techniques illustrated here may well be helpful in the short term but are unlikely to be sufficient to secure deep learning by way of retaining the newly acquired words in the longer term.

Some teachers might use the word *'hook'* to describe an activity planned at the start of a sequence of lessons that engages pupils in their learning. See Table 5.2.4.

Table 5.2.3 An example of a structured sequence for the oral introduction of new vocabulary for beginners in a modern foreign language lesson

Activity	*Pupils' physical involvement*	*Pupils' mental involvement*
Teacher presents new vocabulary items with visuals (e.g. downloaded pictures, flash cards, etc.)	Pupils repeat each word, in chorus and individually at random	Pupils try to work out the meaning of the word. Repetition helps them to commit the new vocabulary to their short-term memory
Teacher gives each visual a number on the board, then calls out the words at random (or this could be done with the use of a list on tape)	Pupils write the number that corresponds to the word they have heard	Pupils draw upon their short-term memory to identify the correct item.
Teacher asks question: 'What is this?' in the target language	Pupils put their hands up, selected individuals speak	Pupils draw upon their short-term memory to produce the words
Teacher asks pupils to work in pairs and ask each other what the new words are. Teacher encourages the class to answer in short sentences, e.g.: this is a …	All pupils involved in speaking	Pupils recall the words as well as the question. They begin to formulate a simple sentence.

Table 5.2.4 Examples of 'hooks' to engage pupils

'Hooks' can involve:

- 'do now' tasks (a short activity that pupils do at the beginning of the lesson);
- challenges;
- problems to solve;
- role play or simulations'
- visits and visitors;
- visual and auditory stimuli;
- digital technology;
- flipped learning.

Digital technology and flipped learning

Interactive whiteboards (IWBs), individual tablets and smartphones provide a multitude of ways to encourage digital active learning both for 'hooks' but also within a lesson (see Inspiring Learning, 2021). For instance, the IWB can be used to provide interactivity in pupils offering written suggestions to lesson objectives, content and plenaries to the screen through the pen facility.

Flipped Learning is a type of blended learning approach where the conventional notion of classroom-based learning is inverted or flipped. Pupils have prior access to the classroom materials (the PowerPoint, for instance) and the lesson's opening 'hook' will be a discussion or problem-solving based on the material already studied. Clearly, it is important to ensure that the flipped material is accessible to pupils and differentiated appropriately (Flipped Learning Network, 2016). A list of Digital Active Learning opportunities appears on the Flipped Learning website.

Task 5.2.5 invites you to consider the use of 'hooks' in one of your units of work.



Task 5.2.5 Using hooks for active learning

In one of your units of work, plan how you can involve pupils through the use of a 'hook' to promote active learning. Teach the lessons incorporating this 'hook' and then evaluate the effectiveness of this approach and how you might improve its use in future lessons by focusing on analysing learning outcomes.

Reflect on what you find with another beginning teacher or your mentor.

Task 5.2.6 provides strategies to activate pupils' prior knowledge.



Task 5.2.6 Activating pupils' prior knowledge

(See also short-term and long-term memory in Unit 5.1.)

Select a strategy for probing pupils' prior knowledge of a topic and use it in a lesson in which you are being observed. Evaluate the effectiveness of the activity yourself and also ask the observer to give you feedback on its effectiveness. Strategies you could use include a question-and-answer session, brainstorming in small groups (i.e. eliciting spontaneous recall of relevant information or randomly listing ideas/suggestions in relation to a particular topic or question) followed by a plenary session, asking pupils to prepare a spider diagram summarising what they know. You then use these to plan the next lesson.

Developing pupils' higher-order thinking skills

The explicit development of thinking skills needs to take place alongside teaching of factual content so the emphasis in learning is not just on outcomes, but also on processes. Teaching, therefore, needs to be designed to enable pupils to:

- develop logical reasoning in order to apply it to new contexts (formal thinking approach);
- deconstruct problems in order to find solutions to them (heuristic approach);
- reflect on, and evaluate, their own learning (metacognitive approach) (Muijs and Reynolds, 2005).

These three approaches are at the heart of active learning because they promote the learners' engagement with the task and encourage pupils to make sense of their learning (see also Unit 5.1). Bloom identified six levels of thinking of gradually increasing complexity, which make increasing demands on the cognitive processes of learners (Bloom, 1956). While there is the potential for active learning at every level, the last three levels can be linked to the higher-order thinking skills mentioned above.

Experienced teachers have the skill of asking questions beyond those planned in response to pupils' replies. They use the Assessment for Learning questioning Strategy – PPPB: Pose, Pause, Pounce, Bounce (Herts and Bucks Teaching School Alliance (TSA), 2018). You need to know the ideas/concepts that pupils find difficult and, therefore, need to probe their understanding/misunderstandings encouraging response to other pupils' responses, too.

Finally, complete Tasks 5.2.7 and 5.2.8.



Task 5.2.7 Developing higher-order thinking skills through the use of questioning

Observe a lesson and note the questions used by the teacher to promote learning. Try to classify against Bloom's (1956) taxonomy of educational objectives. How does the type and frequency of the question(s) used impact upon active learning?

Discuss your observations with colleagues and record them for future reference.



Task 5.2.8 Reflecting on your use of active learning

Read and make notes on the article: Powell (2005) 'Conceptualising and facilitating active learning: Teachers' video-stimulated reflective dialogues', *Reflective Practice*, 6 (3), 407–418.

Arrange to video-record a lesson, or a part of a lesson that aims to promote active learning. Review the video. If possible, discuss the video with colleagues, using Moyle's reflective framework (see Appendix 1(b) of Powell's article). Alternatively, write a critical analysis of your questioning technique using the information in the textboxes and other literature, and store this in your professional portfolio.

SUMMARY AND KEY POINTS

Active learning and Rich Tasks include an emphasis on co-operative learning and provide opportunities for pupils to take some responsibility for their own learning by, for example, active participation in the development of the task. Active learning requires pupil self-discipline which contributes to a wider goal of education. For the teacher, active learning opens up a wider range of teaching methods to develop personalised learning, supports resource-based learning and provides time for monitoring pupil progress and giving formative feedback.

- Teaching is an enabling process; teachers can guide pupils' learning but cannot do the learning for them.
- Active learning can be used to promote process skills and higher-order skills, as well as encouraging acquisition of knowledge and understanding.

- Pupils need to engage mentally with a task if learning is to take place; thus, you need to enthuse and motivate pupils, give purpose to their learning tasks and provide active learning experiences. Some of these activities introduce study skills which are important for pupils preparing for public examinations.
- Hands-on activity in practically based subjects, such as art and design, science and technology does not guarantee that learning takes place; both hand and brain need to be involved.
- Pupils' engagement in the design, execution and evaluation of practical work is as important as their active involvement in reading and listening.

Active learning pedagogies are a major part of your teaching repertoire and, as such, contribute significantly to professional standards.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

Further resources

The resources below provide additional information relating to this unit.

Burden, K. and Younie, S. (2014) *Using iPads Effectively to Enhance Learning in Schools*, MESHGuide, viewed 27 April, 2025, https://www.meshguides.org/guides/node/28

A MESHGuide summarising the findings of a large scale case study conducted over six months (January to July, 2012) in eight schools across Scotland (three primaries and five secondaries).

Kagan: http://www.kagan-uk.co.uk/

Kagan UK trains teachers to use Kagan Structures, a set of research-based instructional strategies that have been created to teach pupils co-operation and learning strategies to enhance achievement.

Leask, M. and Younie, S. (2021) *Education for All in a Time of Crisis*, Abingdon, UK: Routledge.

This book discusses the impact of the pandemic and opportunities for more active pedagogies and approaches moving forward.

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5.3 Teaching styles

Martin Fautley

Introduction

When you think of teaching styles you may well hark back to your own experiences as a pupil in secondary school. You might think, say, of Mr Smith, who spent a lot of time shouting and ensuring he was in control all the time, or Mr Singh, who was kind, friendly, and had a gentle voice, or Mrs Laws, who everyone – including the staff – described as being 'a bit mumsy' and who would be the first port of call for anyone who had a problem. There are many other teachers you will have encountered during your time at secondary school, and maybe the specialist teachers for the subject you are training to teach had a big effect on you too? Sometimes those embarking on initial teacher education (ITE) courses say that they want to be a teacher of their specialist subject as they were inspired by their own teachers of this subject; others say the complete opposite, that they are becoming a teacher so other pupils do not have to go through the same poor experiences they went through; others still fall somewhere between these two extremes.

This unit introduces the notion of teaching styles. It presents and discusses a range of ideas associated with this important aspect of pedagogy. Beginning with reflecting on your individual experiences, the unit then moves to a consideration of published taxonomies of teaching styles. It also brings in the important considerations of how theories of learning contribute to thinking about different teaching styles. The matter of having a repertoire of different styles available is also raised, and you are invited to reflect on how they might go about teaching similar things in different ways.

A key purpose of any consideration of teaching styles is to develop and improve learning in the pupils, and so this unit links thinking about teaching with thinking about learning.

OBJECTIVES

By the end of this unit you should:

- have thought about what teaching styles means both theoretically and in terms of your own classroom practice;
- have considered how teaching and learning are linked, and that different styles may be appropriate for different learning events;

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- have some knowledge of a number of the key classifications of teaching styles that have been published;
- have tried out some different teaching styles in your own professional practice.

Check the requirements of your ITE programme to see which relate to this unit.

Reflecting on your own education

Before reading further, do Task 5.3.1.



Task 5.3.1 Reflecting on your own education

Think back to your own secondary schooling experience.

- Which teachers can you recall?
- What subjects did they teach?
- How would you describe their teaching styles?
- Which of these would you like to model yourself on?
- Which of these have characteristics you would not wish to pursue in your own classroom?

Having thought back on what your own experiences entailed, think about what you can do straight away to put some of your ideas into practice. Discuss these with your mentor and see what they think about your ideas too.

Having done this task, it is important now to state that what we have been discussing so far are, in fact *personality styles*, rather than teaching styles in the way the term is understood in academic and professional circles. The notion of teaching styles has a very specific meaning. It is concerned with the ways in which pedagogy is organised so as to facilitate learning and maximise the potential learning outcomes for pupils in the classes being taught.

This may seem a complicated way of thinking about teaching styles, so let us take some concrete examples. If a Physical Education (PE) teacher wants pupils to be able to take a corner in football, they will, at some point, get them out on the field with some footballs to try it out themselves. In a similar fashion, if a music teacher wants their pupils to be able to play a chord of E-minor on the guitar they are very likely, at some stage, to give them a guitar and let them have a go. In neither of these instances will the most appropriate teaching style involve the pupils taking down written dictation from the teacher without ever actually trying the activity out for themselves (see also Unit 5.2 on active learning). This may seem obvious, but it is this idea which underpins the idea of teaching styles that is being applied in this unit.

The examples used in the previous paragraph may ring true for you, or they may seem a long way from your own subject area. This matters too, as each subject in the secondary school has its own culture and context:

> Every subject that is represented within our schools has a particular culture that underpins it. This culture informs the 'identifiable structures which are visibly expressed through classroom

> organization and pedagogical styles' (Goodson et al., 1998, p. 120). A subject's culture is what makes it unique and, in a simple sense, helps students sense that they are studying a particular subject at a specific moment in the school day. It is important to recognize that a subject's culture goes beyond its knowledge. It also incorporates ways of thinking, acting and being that inform the processes by which teachers teach (your pedagogy) and students learn.
>
> (Savage and Fautley, 2013, p. 187)

Your own specialist subject will have its own culture, but this does not mean that it carries within it the *only* ways in which teaching and learning can be organised, and that these are necessarily different from what is going on in other departments in the school. The American educational psychologist Lee Shulman has described an important notion for us in our consideration of teaching styles, and this has been one of his key contributions to the field, namely the notion of *pedagogic content knowledge* (PCK). This he described as being:

> ... the most useful forms of representation of those ideas, the most powerful analogies, illustrations, examples, explanations, and demonstrations – in a word, the ways of representing and formulating the subject that make it comprehensible to others.
>
> (Shulman, 1986, p. 9)

PCK for your subject is likely to be specific, but the ways in which you teach, in other words the teaching styles involved, will have elements which are recognisable and transferable across a range of teaching contexts.

Defining teaching styles

As ever with complex ideas in education, there is no simple list of teaching styles which can now be set out and discussed. What can be done, however, in this unit, is to present an overview of some key areas of teaching styles and ask you to reflect on how these are appropriate to you, in your subject context, in your school, and with the pupils you will be teaching.

Even when thinking about an overview of teaching styles, there are a number of these that have been published, and which describe in varying detail typologies, ideas, and structures. The publications cited in this unit, therefore, should not be taken as a definitive reference, but as a representative sampling of the materials commonly discussed in schools and on ITE courses.

Mosston and Ashworth

One of the most frequently encountered taxonomies of teaching styles is that of Mosston and Ashworth (2002). Although originally proposed for PE in schools, the Mosston and Ashworth classification is comprehensive, and contains a number of features which can readily be applied across a range of subjects. Mosston and Ashworth refer to this as a *spectrum,* which can be considered as being constructed from the perspective of learner autonomy along a continuum, ranging from less to more. The elements of the spectrum are presented in Table 5.3.1.

This is a complex and detailed set of classifications. It is important to note that the spectrum does not imply a hierarchy, wherein some teaching styles are considered to be better than others; rather there is a '... need for all styles to be experienced and not the valuing of one style over another' (Hewitt et al., 2020, p. 4).

Table 5.3.1 Taxonomy of teaching styles

Command Style	The defining characteristic of the Command Style is to exactly reproduce and replicate an anticipated response or performance in a short period of time so that the specific Command Style objectives in subject matter and behaviour can be developed. Cognitively, the Command Style emphasises automaticity.
Practice Style	The defining characteristic of the Practice Style is the initiation of independence by making decisions whilst practising a memory/reproduction task and receiving private and individual feedback from the teacher.
Reciprocal Style	The defining characteristics of the Reciprocal Style are social interaction whilst learning to give content feedback to a partner. This style's focus is social reciprocation whilst learning to make decisions that are inherent when giving and receiving feedback. Observers offer their partner content feedback using teacher prepared specific criteria. The teacher circulates and observes the partnerships for appropriate use of the criteria information and supportive verbal feedback interactions.
Self-check Style	The defining characteristic of the Self-Check Style is developing the decisions necessary for independence whilst practising a reproduction task and initiating self-checking skills using a prepared criterion. The teacher circulates among the pupils asking questions that elicit conversation (not teacher judgment) about the pupil's performance assessment. This style is not initially appropriate for the novice who is unfamiliar with and inexperienced in the content.
Inclusion Style	Tasks in this style are designed with multiple entry levels of difficulty so that continuous inclusion and participation can occur. Learners survey the options, select an entry-level of difficulty, practice, adjust the level choice if necessary, and self-assess performance against prepared criteria.
Guided Discovery Style	Content reinforces logical and sequential thinking. A series of questions are designed that sequentially lead a learner to discover a specific targeted predetermined concept, principle, relationship, or rule not previously known.
Convergent Discovery Style	Develops the cognitive capacity to individually discover a single anticipated, predetermined response to a new and unfamiliar stimulus by linking and connecting known information to produce a discovered unknown but anticipated response.
Divergent Discovery Style	Develops the emotional confidence and cognitive capacity to produce multiple solutions, to the same question, or to a series of unfamiliar situations, that seek to expand the parameters of the content beyond the known and expected.
Learner Designed Individual Program Style	Provides pupils with an opportunity to discover a process and procedure that solves a broad issue or problem that the teacher has generally identified. Each pupil examines the issue and selects the data to sequence, link, and construct their plan of action, including the performance criteria.
Learner Initiated Style	This is not a whole class teaching experience; rather this style acknowledges individual learner's motivation and cognitive desire to design his/her own learning experience. This style provides pupils, who independently initiate a request, with an opportunity to pursue the complexities inherent in making all decisions in the pre-impact, impact, and post-impact.
Self-Teaching Style	This style does not exist in the school; additionally, it is independent of any classroom teacher. This style's developmental opportunities are governed by the individual implementing the experience. The success in accomplishing the intended objectives relies on the tenacity, motivation, and knowledge of the individual to construct one's own learning experience.

Source: Mosston and Ashworth (2002); spectrumofteachingstyles.org

For practical and in-school purposes, the spectrum will require some unpicking, and it is worth you spending some time doing this. Now do Task 5.3.2.



Task 5.3.2 Putting the Mosston and Ashworth (2002) taxonomy to work in your classroom

Working with your ITE tutor and/or in-school mentor, think about topics you are teaching, or will soon be teaching.

- Which of the Mosston and Ashworth classifications do they think will be the most appropriate for you to engage with in order to achieve this?

A further question you can ask of both ITE tutors and in-school colleagues is:

- Whether they always teach the same topic in the same way each time.

These questions should elicit some fruitful discussions, and it will be worth you revisiting them after you have had some experiences of designing and operationalising teaching and learning experiences of your own in schools.

In Unit 5.1, you were introduced to the idea of learning theory and thinking about the different ways in which pupils learn things, and to the various terminologies employed in the science of learning. Thinking about teaching styles is inextricably linked with thinking about learning. Contemporary learning theory identifies a range of ways of delineating learning from a theoretical perspective. Teaching styles can be thought of in many ways as being linked to this, especially when anything other than a closed or command style is used. However, nuances of individual learning theories need to be thought through carefully in terms of the ways in which teaching styles are both conceptualised and used. It may well be the case that your placement school has a very specific view on learning, and you will obviously need to work within this paradigmatic framework, however, you do need to think about a whole range of factors, as you may well find yourself working in a school in the future which looks very different to your current one.

What can be said with a degree of certainty, though, is that any reflection on teaching styles needs to take account that teaching and learning are – or should be – very closely interlinked. Whilst an oversimplistic and naïve standpoint assumes that teaching simply involved telling pupils things, the reality is much more complex. Despite the information processing stance on learning, pupils do not have the equivalent of a USB socket in their heads onto which you can simply download a series of facts, then walk away, thinking 'job done'! This is where the Mosston and Ashworth (2002) spectrum – or many of the other taxonomies we consider in this section – comes into its own. This is because the *what* of what you want pupils to learn is closely entangled with *how* you want them to learn it. For example, if you are dealing with important health and safety issues, you do not under any circumstances want the pupils to undertake guided discovery learning to find out what happens if you stick your fingers in a plug socket, or stand in front of a javelin thrower! For matters like this the command style is likely to be the most important. However, if you are teaching a unit on Shelley's poetry, then there are likely to be many ways in which this can be done.

What this means is that when you are discussing with your ITE tutors and in-school mentors, they will be guiding you towards reflecting on both the *what* and the *how* of what you will be teaching. Indeed, one of things you will become more proficient at over time is reflecting on the teaching styles these people employ when working with you to develop your own individual teaching persona. Whilst for some mentors the phrase 'my way or the highway' might seem to be their philosophy, developing you as a clone of someone else is unlikely to prove successful in the long run! Now do Task 5.3.3.



Task 5.3.3 Teaching the same thing in a different way to different classes

Look back at Task 5.3.2. Now choose a topic you will be teaching to two more-or-less parallel classes in the same year group. Are there ways in which you could teach the same topic to these two different classes using different teaching styles identified from the Mosston and Ashworth spectrum?

- What happens if you do this?
- Ask your in-school mentor if they would be willing to do this, with you observing the results.
- If this is not possible in practice, think about it as an abstracted theoretical exercise. *Why* has it not been possible?
- What are the constraints that have prevented it happening.

If you are in an ITE cohort, you might be able do this as an exercise with other beginning teachers.

When you reflect on your teaching, try and go beyond the surface of what went well, or did not go so well, and really think about the effects on *learning* of teaching in different styles. Discuss this with your mentor, and find out why they do what they do, in the ways in which they do it.

Barnes et al.

The Mosston and Ashworth spectrum is not the only way of thinking about and classifying teaching styles. Barnes et al. (1987) have also helped establish a number of ways of talking about this. Table 5.3.2 shows the ways teaching styles can be classified according to the ways in which pupil participation in learning can be considered, in a similar fashion to Mosston and Ashworth, along a continuum. In the case of the Barnes et al continuum, this time it ranges from *closed,* where participation is tightly controlled by the teacher, via *framed*, and then finally to *negotiated*, where teacher and learners together discuss and make joint decisions.

This framework offers far fewer classifications than the Mosston and Ashworth, but in its simplicity lies much of its enduring appeal. The Barnes et al framework allows the teacher the vocabulary to plan for different teaching styles in a straightforward fashion, and tailor episodes within a single lesson to different aspects of it. Now do Task 5.3.4.

Table 5.3.2 Teaching styles

	The participation dimension		
	Closed	*Framed*	*Negotiated*
Content	Tightly controlled by the teacher. Not negotiable.	Teacher controls the topic, frames of reference and tasks; criteria made explicit.	Discussed at each point; joint decisions.
Focus	Authoritative knowledge and skills; simplified, monolithic.	Stress on empirical testing processes chosen by teacher; some legitimation of pupils' ideas.	Search for justifications and principles; strong legitimation of pupils' ideas.
Pupils' role	Acceptance; routine performance; little access to principles.	Join in teacher's thinking; make hypothesis, set up tests, operate teacher's frame.	Discuss goals and methods critically; share responsibility for frame and criteria.
Key concepts	'Authority': the proper procedures and the right answers.	'Access'; to skills, processes, criteria.	'Relevance': critical discussion of pupils' priorities.
Methods	Exposition; worksheets (closed); note giving; individual exercises; routine practical work. Teacher evaluates.	Exposition; with discussions eliciting suggestions; individual/ group problem solving; lists of tasks given; discussion of outcomes, but teacher adjudicates.	Group and class discussion and decision-making about goals and criteria. Pupils plan and carry out work, make presentations, evaluate success.

Source: Adapted from Barnes et al. (1987, p. 25; taken from McCormick and Leask, 2005, p. 280)



Task 5.3.4 Using the Barnes et al classification in lesson observation

Arrange with your in-school mentor to closely observe a lesson they will be teaching. Explain to them that you will be using the Barnes et al classification to delineate what they have been teaching in that lesson. Note the different episodes of the lesson, and ask yourself these questions:

- What was the learning episode?
- Was it closed, framed, or negotiated?
- How did the teacher deal with transitions between these episodes?
- Were different resources employed – worksheet; equipment, ICT, or what?

In reflection with the teacher afterwards, you could discuss:

- What do they think of your list of teaching episode classifications?
- Did they know they were doing this – was it a conscious decision?
- Would they have done it differently if you had not shared your observation parameters beforehand?
- Is this how would normally teach this lesson?
- How does the teacher know what learning took place?

Theories of learning and teaching styles

What is becoming apparent in these discussions of teaching styles is that what should be happening is that the teaching style utilised should have a clear link to desired learning. In order to delve into this more deeply it is useful at this stage to revisit learning theory, and see if it can be applied more directly to thinking about teaching. There is, though, a problem with this approach, as identified by James (2006):

> However, and this is an important point – they do not necessarily claim to have a view about the implications for the construction of learning environments, for teaching, or for assessment. This has sometimes created problems for learning theorists because practitioners and policy makers usually expect them to have a view on these matters, and if this is not the case then there are those who will try to fill the gap; some successfully and others less so. (p. 53)

James identifies three broad families of learning theories, *behaviourist*, *constructivist*, and *socio-cultural*. Whilst these do not have exclusive teaching styles associated with them, nonetheless they are worth considering for the ways in which they influence the pedagogic choices made by teachers.

Behaviourism

Whilst getting pupils to behave well is a *desideratum* for teachers and schools, this is not really what behaviourism refers to. Behaviourism can be defined as:

> … a theory of animal and human learning that focuses upon the behaviour of the learner and the change in behaviour that occurs when learning takes place. Learning in the context of behaviourism can be defined as the acquisition of a new behaviour or the modification of behaviour as a result of teaching, training or tutoring. Learning is demonstrated by the behaviour of the learner in their actions or reactions to further stimulus.
>
> (Woollard, 2010, p. 1)

What this means for a teaching style that takes behaviourism as its central point is that learning can be considered to have taken place when a change in behaviour occurs:

> According to behavioural theory, learning is totally influenced by external factors, and takes place when measurable, observable and permanent changes in particular behaviour are manifested. According to this theory, learners change, or repeat behaviour, based on the immediate and frequent rewards and reinforcement that they receive.
>
> (Zajda, 2023, p. 18)

What this means for your chosen teaching style is that when you are looking to effect a change in observable behaviour, you will choose a style which is most suited to this.

Constructivism

> The central idea here ... is that human learning is constructed, that pupils build new knowledge upon previous learning and, therefore, learning involves constructing one's own knowledge from one's own experiences. Pupils are encouraged to co-construct knowledge by working to solve realistic problems.
>
> (Lawes and Leask, 2022, p. 481)

Teaching styles for constructivism, then, are predicated on learners constructing knowledge and meanings. In other words,

> ... teachers cannot simply transmit knowledge to students, but students need to actively construct knowledge in their own minds. That is, they discover and transform information, check new information against old, and revise rules when they do not longer apply.
>
> (Bada, 2015, p. 66)

This places a different emphasis on the teaching styles involved in a constructivist stance on pedagogy than can be employed from those of behaviourism. Here, the emphasis is on learners constructing meaning for themselves. This means that teaching as telling is not the only way in which learning takes place.

Socio-cultural

> According to this perspective, learning occurs in interaction between the individual and the social environment. ... Thinking is conducted through actions that alter the situation and the situation changes the thinking; the two constantly interact. Especially important is the notion that learning is a mediated activity in which cultural artefacts have a crucial role. These can be physical artefacts such as books and equipment but they can also be symbolic tools such as language.
>
> (James, 2006, p. 57)

An implication of this view of learning for teaching is that the teacher may well want to set up a situation where authentic tasks are undertaken by the pupils. So, in PE, the pupils play netball matches, in drama they create their own plays, and in music they compose and perform their own songs. The teaching styles employed recognise that learning has a social dimension, and contexts are established where that is the case. This will be the case in many other school subject areas too. Now do Task 5.3.5.



Task 5.3.5 Thinking and discussing teaching styles

Discuss with tutors and mentors what the implications of thinking about teaching from the different perspectives offered by these classifications. What does it mean for both *what* you will be teaching, and *how* you will approach teaching it.

Direct instruction (DI and di)

Direct instruction as a teaching style is rendered complex to describe and analyse for beginning teachers because it has a number of separate meanings. In essence, and for the purposes of this section, these distinctions can be thought of as Direct Instruction (capitalised 'DI') deriving from the work of Siegfried Englemann (*inter alia* Gersten et al., 1987), and the view of direct instruction (lower-case 'di') described by Rosenshine and Stevens (1986), where they depict what happens when a teacher teaches what they refer to as a 'well-structured sequence of learning', which for them involves these aspects:

- Begin a lesson with a short review of previous learning.
- Begin a lesson with a short statement of goals.
- Present new material in small steps, providing for student practice after each step.
- Give clear and detailed instructions and explanations.
- Provide a high level of active practice for all students.
- Ask a large number of questions, check for student understanding, and obtain responses from all students.
- Guide students during initial practice.
- Provide systematic feedback and corrections.
- Provide explicit instruction and practice for seatwork exercises and monitor students during seatwork (Rosenshine, 2008, p. 1).

A third, and often problematic use of the term 'direct instruction' is simply that of overtly didactic teaching, or 'chalk and talk', as it used to be known. One of the issues with this is that some see it as being in opposition to what has often taken place in schools:

> It is now fashionable to extol the virtues of 'Direct Instruction', and to claim that empirical research often supports it, as opposed to methods associated for many decades with 'progressive' and 'child-centred' education. Unsurprisingly, characterizations of Direct Instruction vary significantly.
>
> (Davis, 2017, p. 295)

As the main concern in this unit is to introduce you to the terminology of teaching styles, it would be remiss not to include Direct Instruction, but should you hear the term being used in schools it is important for you to try to uncover exactly which meaning is being used, as this can affect both what is understood, and what is going on in teaching and learning contexts in the classroom. See also Unit 5.5. Now do Task 5.3.6.



Task 5.3.6 Attitudes towards scripted lessons

A legitimate critique of some forms of Direct Instruction (whether 'DI' or 'di') is that in one of its many versions its proponents argue for tightly controlled and scripted lessons (Eppley and Dudley-Marling, 2019). Many see this in itself being problematic on many counts.

- Discuss with your ITE tutors and mentors what they think of scripted lessons.
- What do you think?
- Have you been given a lesson script to follow with a class?

- How did it go?
- Was it helpful or problematic for you?

The issue of scripted lessons will vary considerably between schools, and also between subjects in the same school. Thinking about this issue in different placements, schools, or contexts can vary significantly, and so you should try to find out what others in different places and subjects think about this issue.

Teaching styles in practice

Undertaking the tasks outlined in this unit, and then consequently thinking about teaching styles from the pragmatic perspective of the classroom practitioner means that you are unlikely to encounter schoolteachers who declare themselves to be out-and-out behaviourists, say, although you might possibly come across some who are Direct Instruction proponents. Education has always been an eclectic profession, and teachers generally adopt what might be termed a 'pick-and-mix' approach to teaching and learning, adopting, trialling, and developing what works for them, in their school, with their pupils. For you, as a beginning teacher, you will want to be taking the opportunity to try out different ways of doing things. You will, however, want to maximise learning in your pupils, and you will want to teach in ways that enable this to happen. One of the things that careful systematic and structured reflection on teaching styles does is to take you beyond the simplistic 'teaching is telling', and into the utilisation of more reflective forms of pedagogy.

This does not mean that thinking about teaching styles should be downplayed in school practice, but as Atkinson and Claxton (2000) observe, the expert practitioner often operates seemingly intuitively, and cannot always say what they are doing with regard to teaching and learning at any given moment in the classroom. This is important as you observe teaching taking place, as you, a beginning teacher, are trying to identify what is going on, whilst the expert practitioner you are observing may be operating using years of PCK experiences to know how to teach a given subject or topic, and consequently may well move between different teaching styles during the course of a single lesson which you are observing. Equally likely is the possibility that if in the post-lesson discussion you say something like 'I noted that at 1035 a.m. you did …', the teacher may reply with 'did I?' This means they have moved from overt thinking to the application of tacit knowledge, in Polanyi's (1967, p. 4) much cited phrase 'we know more than we can tell'. However, as you are still operating at the novice stages of teaching, you will need to think about *what* you are doing, *how* you will be doing it, and, importantly, *why* you are doing it in this way. Knowledge of teaching styles will help you with this and equip you with the professional vocabulary to be able to discuss it.

SUMMARY AND KEY POINTS

In this unit, we have:

- Discussed what the notion of teaching styles entails;
- Considered how learning and teaching are inextricably linked;

- Reflected on different theories of learning, and how they impact on how things are taught;
- Considered different ways of teaching the same topic;
- Discussed with mentors to ascertain their views on teaching styles; and
- Reflected on your own emerging professional practice, and how to think about and enact teaching in your own ways.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Pollard, A. (ed.) (2014) *Readings for Reflective Teaching in Schools*, London, UK: Bloomsbury Publishing.
A general book for supporting teacher education, but which contains many useful chapters concerned with teaching and learning, and which provides a sound foundation for reflecting on your emerging professional practice.

Pritchard, A. (2017) *Ways of Learning: Learning Theories for the Classroom*, Oxford, UK: Taylor and Francis.
This book provides a helpful overview of learning, and is designed purposely to be helpful to teachers in being able to use in their own planning and teaching. Working from here provides a useful way of thinking about how teaching styles can be employed in the service of pupil learning.

SueSee, B., Hewitt, M. and Pill, S. (eds.) (2020) *The Spectrum of Teaching Styles in Physical Education*, Oxford, UK: Taylor and Francis Group.
Although targeted firmly at Physical Education teachers, nonetheless there is a lot of useful information in here from which teachers of all subjects can take something. This book really delves into Mosston and Ashworth's teaching styles research, and helps explain many of its finer points.

Waring, M. and Evans, C. (2015) *Understanding Pedagogy: Developing a Critical Approach to Teaching and Learning*, Abingdon, UK: Routledge.
The notion of pedagogy is multifaceted, and the ways in which teaching styles fits into it are complex. This book encourages and facilitates guided reflection on teaching and learning, and helps unpack some of the issues around teaching styles.

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5.4 Improving your teaching

An introduction to practitioner research, reflective practice and evidence-informed practice

Helen Lewis, Marilyn Leask and Tony Liversidge

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Introduction

As a beginning teacher, you might ask yourself why there is such an emphasis placed on reflective practice and the use of research and evidence to underpin your professional judgement in your initial teacher education (ITE) programme (see Units 1.1 and 8.1). The analysis of your practice is one of many activities that you are asked to do, such as lesson evaluations, lesson debriefs with mentors, critical reflection on aspects of school-based experience and taking part in lecture and seminar discussions.

In this unit, we explore why developing reflective skills is so important. We will discuss why understanding the role of research and enquiry is important. Being able to critically analyse what you are doing, why you are doing it and how to do it effectively, as well as systematically evaluating what you have done, particularly in terms of improving pupils' learning experiences, is a central part of your practice.

Reflection offers the opportunity to question, refresh and enliven what you do to avoid becoming stale and static, and reflective practice is also undertaken to ensure your teaching produces new pupil learning. It also helps you to decide whether your lesson went well. Observing that pupils are quiet, busy, happy or good, and look as if they are working industriously is no guarantee that the learning you have intended is taking place. Professor Robert Coe's Proxies for Learning provides advice on this issue (Sherrington, 2021). In relation to this, detailed advice about evaluation of your lessons is provided in Units 2.1 and 2.2.

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OBJECTIVES

At the end of this unit you should be able to:

- demonstrate an understanding of the terms: practitioner research, action research, reflective practice and evidence-informed practice;
- identify different forms of evidence on which you can draw to enable you to make an informed decision concerning elements of practice;
- apply research strategies to evaluate and improve aspects of your teaching;
- develop your ability to reflect on practice-based evidence from research to acquire higher levels of professional knowledge and judgement.

Check the requirements of your ITE programme to see which relate to this unit.

Reflective practice and evidence-informed practice

Wright (2008) argues that developing the habit of reflecting on your work is perhaps the most significant driver of your learning to teach, and your mentors and tutors will encourage you from the very start of your journey to develop these good habits. However, they recognise that you need constructive guidance on reflective practice and they give feedback on your teaching, suggest new strategies to try out, and encourage you to spend time and energy thinking about how to develop your teaching (see Unit 5.9). The time that you spend thinking about your practice will help you to develop your ability to reflect effectively. One useful way of going about this is to first focus on positive aspects of your practice, then areas for further development, and then think about specific targets that you want to achieve over the next period of your practice.

The importance of reflective practice is tied closely to broader notions of teacher professionalism, agency and identity (Anani Sarab and Mardian, 2023). As prospective teachers you will be expected to have 'a sound understanding of the "why" and "how" of teaching as well as the "what"' (Donaldson, 2015, p. 58). In this way, your evaluation of practice becomes more rigorous and can be regarded as being practical research into teaching with a view to enhancing the quality of learning and teaching at the same time.

It is also important to know how to use existing knowledge about education to support your development. Increasingly, teachers are able to access high-quality educational research and evidence to inform their decisions. This approach gives rise to the phrases 'evidence-informed', 'evidence-based' or 'research-informed' practice. It is no coincidence that governmental championing of evidence-based teaching has recognised the importance of teachers taking an evidence-informed approach and that there needs to be a strong evidence base to build arguments about effective practice.

You may find that you have the opportunity to join teacher–researcher networks, which have the goal of building the evidence base for practice. Schools may be part of networks and you may be part of your subject association specialist network or Teaching Council, LinkedIn, Facebook, Chartered College of Teachers or the developing MESH (Mapping Educational Specialist KnowHow) Initiative, an educational charity which aims to provide a sustainable system using resources already in the education system to generate, quality assure and update evidence-based summaries written

for educators (www.meshguides.org). The Chartered College of Teachers publishes its own journal, *Impact*, which contains a wide range of evidence, much of it conducted by teachers themselves. All beginning teachers can join the Chartered College for free (https://chartered.college/join/), giving them access to resources including the journal as well as EBSCO, an educational research database, and a useful research skills self-audit tool for teachers, where individuals can reflect on their awareness, engagement with, and use of research in their practice. In addition, the Education Endowment Foundation (EEF) (www.educationendowmentfoundation.org.uk) has been influential in the movement for research-informed practice in schools, and through its guidance reports it provides accessible materials that teachers can use (Gittner and Harrison, 2020).

Processes of reflective practice and practitioner research

Dewey (1933) introduced the notion of 'reflective thinking'. This term signifies how professionals, such as teachers, can analyse the effectiveness of their actions and develop different ways of working as a result. Thus, professionals are constantly learning about what they do and so improve their practice. Schön (1983) used the phrase 'reflective practitioners' to explain how enlightened professionals work in modern society. Within his description, he distinguished between 'reflection *in* action' and 'reflection *on* action'. *Reflection in action* involves reflecting on an incident whilst it can still benefit the situation and can be a particularly useful tool for a teacher who might have to react to a classroom event at the time it occurs. Alternatively, *reflection on action* involves reflecting on how practice can be developed after the event using knowledge of previous similar events, and how actions might have contributed to the successful outcome (or otherwise).

Drawing upon the concepts advanced by Schön and Dewey, as well as others, Zwozdiak-Myers (2009) identified nine dimensions of reflective practice. You may find it helpful to keep these in mind as you work through this unit and consider how you might collect data on your own practice to provide a foundation for improvement. Figure 5.4.1 sets out these dimensions.

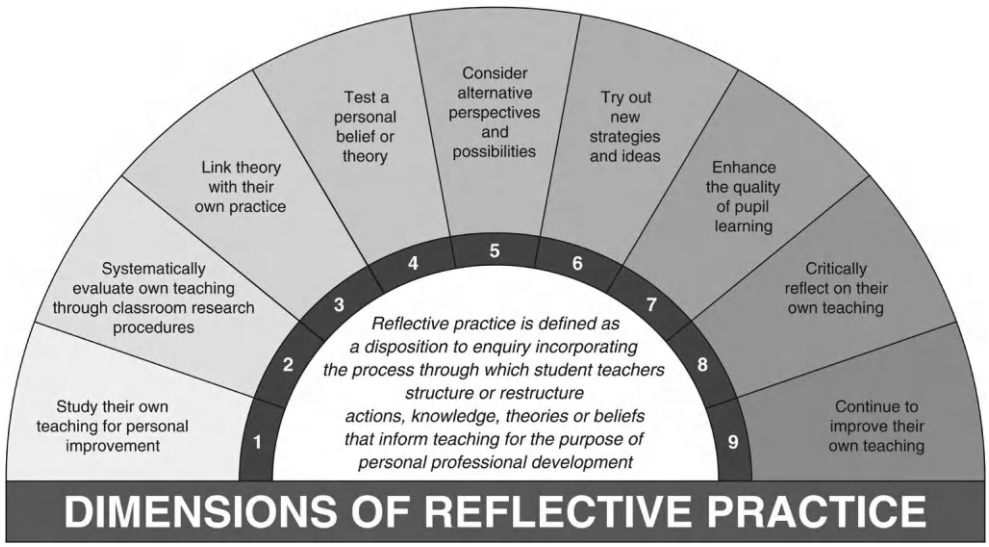


Figure 5.4.1 Dimensions of reflective practice

Source: Zwozdiak-Myers (2009)

It is important to bear in mind that reflection is not always easy. For example, how do you know if you are reflecting on the right things to improve the teaching and learning in your class? What if you are unaware of some of your practices, both the good and the not so good, and how can you ensure that you are reflecting on actual and not perceived events?

A number of approaches can help you reflect more effectively on your practice. Asking questions after the lesson about teaching and learning are useful starting points, these might include:

- Did all pupils benefit from this learning method/activity? How do I know and why?
- Why did I choose this method/activity to cover this topic? Could it be improved? How?
- Was my attitude towards my class today effective for student learning?

Some beginning teachers benefit from keeping a reflective journal or diary. Your programme may adopt a specific model of reflection to scaffold you through the reflective process. You may be encouraged to reflect on lessons using 'Brookfield's Lenses' (e.g. Brookfield, 2017), which are four complementary lenses that encourage you to explore the assumptions that frame how you teach, namely: the *lens of your own experiences as a learner*; the *lens of your students' eyes*; the *lens of colleagues' perceptions* and the *lens of educational literature*. Reviewing practice through these lenses helps to surface the ideas you might hold about pedagogic methods, techniques and approaches, and the assumptions you might make concerning the conditions that best foster pupils' learning.

Teachers and research

Merely thinking about what you do is not the same as researching it. A teacher researcher, whilst also being a reflective practitioner, adds to this by using research techniques to enhance and systematise that reflection to justify their actions and, thus, to improve their practice and continue to develop professionally. Indeed, engaging in such research activity offers opportunity to become more self-questioning, helps develop a shared professional language to talk about teaching and learning and may improve professional satisfaction. Such investigation into their practice by professionals themselves has come to be known as '*action research*', '*practitioner research*' or '*teacher enquiry*'. This can take a number of forms such as '*lesson study*' and is a type of professional learning that is centred on the classroom, is designed to change practice and is a collaborative endeavour.

Before embarking on practitioner research, you, along with other colleagues involved, need to have thought through and discussed the purpose of the enquiry. This enables you to construct a plan that identifies what is going to be investigated, what questions you want to try and answer, how the research will be carried out and what will be the expected outcomes in terms of data and analysis. These factors can then be put in a timeframe that offers targets to work towards. Whilst it is important to have a clear plan, as a researcher you must be prepared to adapt and change according to altering circumstances. In this way, planning and conducting classroom research mirrors the teaching process itself. The research encompasses some of the methods you may have already been using, such as observation, keeping a diary, obtaining the perspective of different interested parties (pupils, staff, parents) by the use of interview and questionnaire, and examining documents (Patterson, 2016a, 2016b, 2016c).

Table 5.4.1 Planning a practitioner research project

Research Focus
Be clear about the focus of the research: What question do you want to answer? It is a useful exercise to write a paragraph at the outset explaining what is being investigated and why. Then further, more precise, research questions need to be devised in order to identify what you wish to find out. These provide you with a clear trail to follow.
Research Question(s)
Writing a good research question is crucial. Clough and Nutbrown (2007, p. 41) suggest considering the 'Goldilocks' principle when writing a research question – considering if it is 'too big' to be tackled at a particular time, 'too small' to warrant investigation, 'too hot' or sensitive to ask or 'just right' for investigation in a particular context by a researcher. For example, the question 'Which teachers successfully include pupils with additional learning needs in their lessons?' might be considered 'too hot'; the question 'How do pupils learn best?' might be considered 'too big', whereas the question 'What do Y8 teachers do when they successfully include pupils with ASD in their PE lessons?' might be considered 'just right'.
Data Sources
It is important to *consider current research findings* and what has already been written about your focus. (Many schools have links with universities and so can access university libraries; increasingly, there are open online resources – see the websites at the end of this unit as well as the further reading. Alternatively, local libraries can usually order in articles and books from the national libraries.) This will add to your professional knowledge and inform your research. You then need to decide *what data* you need to collect *and where* that data might be held. The most likely sources of data for teacher researchers include *people*: pupils, teachers from the same or different schools, parents, other adults working with pupils; *documents* such as pupil reports, school records and written policies, inspection reports, government or similar publications; *observations*, *interviews* and *questionnaires*. Please also make yourself aware of the ethical guidelines governing educational research and use of sources (see Patterson, 2016a, 2016b, 2016c; BERA, 2024).
Timeline
Now that you have a clear research focus and research questions, the sources of data have been identified and the methods have been decided upon, you can develop a timeline for the collection and analysis of data. A clear plan helps ensure that nothing important has been missed out and gives you more control over the process by fitting the research into your work commitments. It is particularly important to have a clear timeline when evaluating initiatives where data has to be collected at specific points. For instance, the evaluation of a curriculum initiative on the teaching of a history module using drama techniques to a Year 7 class would have to be carefully planned beforehand as data could only be collected during the teaching and immediately after.

Table 5.4.1 outlines points to consider in planning a practitioner research project. Schools have access to benchmarking data and pupil data, which can provide a useful foundation for researching the impact of different forms of teaching on different types of pupils, e.g. data from the Centre for Evaluation and Management pupil tests (ALIS and YELLIS, MidYIS, SOSCA, see www.cemcentre.org/) or the GL Assessment Cognitive Abilities Tests (CATS, see https://support.gl-assessment.co.uk).

Evaluating evidence from practice and research is a professional skill. Task 5.4.1 is designed to give you practice in this.



Task 5.4.1 Evaluating research and evidence

Choose a research article from an education research journal on an area that particularly interests you and make notes as you read this, for example:

What was the context of the research and how was it approached? How were the data collected? What are the key findings; are they relevant in your context? Do they add to your knowledge or understanding in any way? What changes in practice might I consider because of reading this?

Some research techniques for use in the classroom

Often, methods of collecting data used are interviews (group or one-to-one), questionnaires, observations, diaries and analysis of documents. When used, these methods need to be carefully designed to ensure the information you want is obtained whilst also considering the feelings of those from whom it is being collected.

Ethical issues

An ethical approach is crucial in any research activity. Teachers and other professionals working in classrooms have a duty of care for their pupils and colleagues, with a need to respect others, share information as appropriate and maintain trust and individual confidentiality. Further, you should consider the role of the pupils in your research and how much to involve them. Thus, acting ethically means that you will think about who, how and what your project will affect at all times (Kara et al., 2021). You must be open about the purposes of your research and obtain agreement from those who are in a position to give it; your mentor/tutor may advise you to get the permission of the head teacher and others involved. It is also worth remembering that seeking ethical approval via a formal university process, such as the completion of the relevant form, is only one part of being an ethical researcher. The British Educational Research Association ethical guidelines (BERA, 2024 provide guidance on ethical issues). Table 5.4.2 outlines the key areas to consider:

Interviewing

Interviews take many forms depending upon whom you are interviewing, where and the focus. When conducting interviews, whether individually or as part of a small group, as part of practitioner research, it is important to consider beforehand what questions need to be asked, how they will be asked and how the data will be recorded. You may wish to record them; however, it may be easier to make brief notes under key headings during or immediately after the interviews. As a new researcher, semi-structured interviews are a valuable research technique due to their flexibility and depth. They combine structured questions with the freedom to explore topics in more detail, allowing researchers to gather rich, qualitative data. This approach may help you understand participants' perspectives and experiences more comprehensively. Semi-structured interviews also enable researchers to probe for deeper insights, clarify responses and adapt questions based on the interview flow. This adaptability makes them particularly useful for exploring complex issues and generating new ideas, providing a balance between consistency and the ability to uncover unexpected information.

Table 5.4.2 An ethical approach to practitioner research

You must take responsibility for the ethical use of data and for maintaining confidentiality. Before starting, check the ethical requirements at your university and/or school, we suggest that you should, as a matter of course:

1. Ask a senior member of staff as well as teachers directly involved with your classes for permission to carry out your project. Get ethics approval from your ITE provider. They will have a formal process for this.
2. Before you start, provide staff involved with a copy of the outline of your project which should include:
   (a) the area you are investigating;
   (b) how you are going to collect any evidence;
   (c) who you are going to collect evidence from;
   (d) what you intend to do with the data collected, e.g. whether it is confidential and whether it will be written up anonymously or not;
   (e) who the audience for your report will be;
   (f) any other factors relevant to the particular situation.
3. Consider if you need to ask for pupil and/or parental consent.
4. Think about how you want the pupils to be involved in the research.
5. Consider how you will monitor pupils' assent to continue to be involved and how you will manage any who may withdraw during the process.
6. Check whether staff expect to be given a copy of your work.
7. If you store data electronically, then you should check that you conform to the requirements of the Data Protection Act, and GDPR Guidelines. For example, you should not store personal data on computers without the explicit authorisation of the individual.

Questionnaires

A questionnaire is useful for surveying pupil, teacher or parental views. Questionnaires enable the collection of information from a large number of people comparatively quickly and anonymously. Questionnaires and surveys can provide both quantitative and qualitative data, making them versatile research tools. Quantitative data is gathered through closed-ended questions, such as multiple-choice or rating scales, which can be statistically analysed to identify patterns and trends. This type of data is useful for measuring the extent of a phenomenon and making generalisations. Qualitative data is collected through open-ended questions, allowing respondents to express their thoughts and experiences in their own words. This data provides deeper insights into participants' perspectives and the reasons behind their responses. Combining both types of data may offer a comprehensive understanding of the research topic, enabling researchers to validate findings and gain nuanced insights.

The wording and layout of the questions are significant. They need to be framed so that those being asked can understand and answer appropriately. Always test out your questions first with a small group to check they are understood and that the answers are likely to be relevant to the topics being researched.

PIT Stops

Sometimes beginning teachers can find it challenging to know what to reflect on, and to identify appropriate targets after a lesson. One approach to address this is a model known as 'PIT Stop' which involves working closely with your mentor, and reflecting during the lesson (Grigg and Lewis, 2018). PIT Stop is conceived as a form of ongoing professional enquiry rather than conventional

assessment of school-based teaching experiences. PIT Stop stands for **P**recise, **I**nsightful and **T**imely feedback (alternatively described by one of our beginning teachers as 'Pause in Teaching'). Such feedback takes the form of in-lesson learning conversations between the observing mentor and mentee, each lasting 30–60 seconds and occurring twice during a lesson. One is called by the mentor focusing on an agreed or 'contracted' target beforehand, and the other by beginning teachers on anything which surprises or puzzles them. Finally, a post-lesson discussion reflects on the impact of these conversations. PIT stop seeks to develop a relationship between observer and observed based on mutual respect and a commitment to learn from each other where feedback becomes a two-way process. Results show that all beginning teachers valued the dialogue as professional learning and almost all could act on the mentors' advice immediately (Grigg et al., 2024).

Additional research techniques for use in the classroom, such as observation schedules, paired observation and research diaries can be found on the companion website.

Participatory approaches

In recent years, there has been a shift from viewing research that is something that happens 'to' or is 'about' children and young people, to a process that acknowledges that they can be active participants in the process. They should be respected as being experts in, and able to report on, their own lives, behaviours and experiences. The United Nations Convention on the Rights of the Child (UNCRC) stresses the importance of children expressing their views on matters that affect them, and this has implications for how children are involved in shaping research and in how they are included in research that acknowledges their lived experiences.

Some of the issues regarding children's status are particularly acute for children who are marginalised, such as children in poverty (e.g. Redmond, 2008), children with additional learning needs or other groups of more marginalised children (such as ethnic, religious or linguistic minorities). As such, Participatory Action Research (PAR) 'nurtures two-way relationships between researchers and children that minimise negative effects and builds the capacity of child participants to take action' (Institute of Development Studies (IDS), 2009, p. 1). When planning and conducting your own research it is therefore important to reflect on whether the young people destined to benefit from your enquiry could have a voice in the design and research process.

Creative research methods

There are many more ways of collecting information and it is important to be creative as well as adaptable in your considerations. Emerging methods allow us to hear pupils' voices more creatively. For example, 'Walking Interviews' are based on the view that 'talking becomes easier with walking' (Kinney, 2017, p. 1). The participant selects the location and route, and the interview is conducted as they walk. This can be beneficial as it removes some of the potential power imbalances that can exist in more traditional interview contexts and also supports more spontaneous responses. For example, in one secondary school, Key Stage 3 pupils talked openly and honestly to a researcher about worries related to the coronavirus pandemic when walking the school's wellbeing dog around the school grounds (Lewis and Knight, 2021).

Developing rapport that enables participants to feel able to give honest responses can be challenging, but creative methods can help with this. For instance, Rainford (2020) used LEGO figures representing different jobs to create an enjoyable atmosphere when exploring career

aspirations. As part of the study, participants arranged the figures on a 'Ladder of Aspiration' and then had the opportunity to discuss their decisions and the thinking underpinning this with the researcher.

Cartoon storyboards can be an engaging and accessible approach to exploring research questions, as they do not rely on literacy skills in the same way as a questionnaire does. For example, Carlyle (see Lewis and Grigg, 2020) used a comic book approach to ascertain pupils' views and attitudes regarding whether or not a school dog would be a valuable addition to the school. This allowed the pupils to air their thoughts and opinions regardless of their ability to write these formally and decisions were subsequently made in a manner that valued pupil voice.

Task 5.4.2 is designed to give you practice in planning the kind of research project that beginning teachers are likely to undertake.



Task 5.4.2 Planning your research project

ITE programmes normally include a small research project as an assignment. Check your programme requirements with your mentor or tutor. The process is as follows:

- Identify an issue or problem associated with your teaching for further investigation (e.g. challenging the more able pupils or the development of active learning techniques).
- Outline the focus of the research and explain why this is an important area for you to investigate.
- Write a number of key research questions (about three) that identify what you need to find out.
- Undertake a literature review to check what has been done before (use the websites at the end).
- List the likely sources of data.
- Identify methods to collect the data.
- If they are available, have a look at some of the projects that have been done previously by people on your programme.

We suggest you discuss your findings with your mentor or tutor and other beginning teachers. Store your findings in your professional portfolio.

Analysing evidence about teaching and learning

As a practitioner-researcher, your strategy for data collection will often involve the use of both quantitative and qualitative data. Whatever you use, the onus will be on you to present your analysis in a way that can be understood, so that, for instance, patterns can be more easily identified. The evidence available for drawing conclusions about teaching and learning can take different forms. This evidence includes:

- Quantitative data in numerical form that can be collected from a range of sources, e.g. statistical returns, questionnaires and school management information systems. League tables of school performance are a good example of how quantitative data can be presented and used.

This type of data is useful for measurement and comparison on a large scale but often lacks explanation for individual differences and can feel very impersonal.

- Qualitative data, which are more descriptive and often include detailed personal explanations. These data are 'richer' and give a feel for particular cases but are sometimes harder to analyse and do not lend themselves easily to measurement. Such data are collected through observation, interview, analysis of documents, diaries, video, photographs, discussions and focus group brainstorming.

For instance, if you were investigating a problem of pupil truancy at your school, you would be likely to require quantitative data showing the extent of truancy and any relationship this has to factors such as age, gender, pupil performance, the time of day or particular lessons. You may also want qualitative data that gives a more in-depth and personal explanation of truancy from the perspectives of truants, their teachers and parents. Using both types of data helps to give a fuller understanding of the issues.

Task 5.4.3 gives you an overview of the performance data available in your school.



Task 5.4.3 Reviewing performance data at your school

Schools have information systems that provide data showing how pupils are performing. Subject departments collect such data as part of the annual reviewing process and to inform future development plans. Find out about the different types of evidence and the process used in your school to evaluate the effectiveness of teaching, pupil learning and the monitoring of individual pupil progress. Consider how this data informs the setting of future targets for pupils, teachers, subject departments and the school. Record your findings in your professional portfolio.

The aspects that will enable you to produce an effective and reflective research assignment (which on PGCE programmes will be marked using postgraduate level 7 criteria) will be the relevance of your writing to the topic chosen and an appropriate methodology and method(s) that enable you to answer your research question(s). The key to this is to demonstrate your critical understanding of, reflection on and analysis of the literature that you have sourced in relation to your own results and findings. In addition, you should have a clear structure for the work, that is written in a good academic style and is correctly referenced in both the text and the bibliography.

Finally, it is also important to think about how you will share your research (beyond your university assignment) for those who might be interested in it. For example, Mannay et al. (2019) conducted research about the educational experiences and aspirations of care-experienced children. When the project finished, the researchers produced a research report with three summaries (executive, child-friendly and for young people), but also produced short films, magazines, songs with music videos and posters in order to make key messages accessible and appeal to a diverse audience.

SUMMARY AND KEY POINTS

In this unit, we have provided a brief introduction into an important area of professional practice and accountability.

- Evidence-informed practice = professional judgement + evidence. Critical reflection aided by practitioner research provides the means by which the quality of teaching and learning in the classroom can be evaluated as a prelude to improvement.
- You should now have ideas about practitioner research and how to reflect on and evaluate the quality of your teaching through using a continuous cycle of critical reflection so that you can plan improvement based on evidence.
- Building your professional knowledge and judgement are longer-term goals, which are developed through reflection and further professional development.
- Revisit this process again later in your career because increasingly as a professional you can expect to be asked to provide evidence to underpin your approach to education. The application of practitioner research to your work at that later stage opens your eyes to factors influencing your teaching and learning that you may not have known existed.
- If you intend to develop your research skills, then we suggest that you join relevant communities of practice through a professional association or school network.

Check which requirements for your ITE programme you have addressed through this unit.

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Further resources

Bell, J. and Waters, S. (2019) *Doing Your Research Project: A Guide for First-Time Researchers* (6th edn), Maidenhead: McGraw-Hill Education.
This is an excellent text that gives step-by-step advice.

Costello, P.J.M. (2011) *Effective Action Research. Developing Reflective Thinking and Practice*, London: Continuum.
This book provides clear and accessible advice.

ERIC (the Education Resources Information Center): www.eric.ed.gov/ERICWebPortal/resources/html/about/about_eric.html.
This is the US government-sponsored online digital library for education.

Evidence for Policy and Practice Information Centre: https://eppi.ioe.ac.uk/cms/.
This has a list of systematic reviews of practice in education.

Hopkins, D. (2014) *A Teacher's Guide to Classroom Research* (5th edn), Maidenhead: Open University Press.
This provides very useful guidance for new researchers.

Leask, M. (2011) 'Improving the professional knowledge base for education: Using knowledge management and Web 2.0 tools', *Policy Futures in Education*, 9(5), 644–660. http://dx.doi.org/10.2304/pfie.2011.9.5.644.
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5.5 Closing the achievement gap

Helping learners achieve by responding to how they learn

Carrie Winstanley

Introduction

Achievement gaps between different groups of learners have persisted since the inception of formal schooling, in all levels of education, including secondary. Addressing these disparities is a complex issue with many of the possible solutions lying beyond the reach of classroom teachers. Nonetheless, there are some effective actions that teachers can take, and these are explored in this unit.

So, what do we mean by an 'achievement gap'? At first sight this seems obvious; consistently different outcomes in academic performance between groups of learners. However, although teachers, school leaders, policymakers and other stakeholders often discuss achievement, it is not always clear what is meant. Consider, for example, the language used by Ofsted, where 'attainment' and 'progress' are the key terms, rather than 'achievement'. Attainment refers to the level or standard of a learner's work, typically demonstrated through tests, examinations or by meeting benchmarks such as national Standard Attainment Tests (SATs) or public exams like GCSEs. These results can be compared nationally, or even internationally with measures such as PISA (OECD, 2022). 'Progress' is measured by comparing a pupil's attainment over time, showing where their performance has changed relative to previous levels and again, these measures can be compared across individuals, small groups, specific schools, areas or very large cohorts. 'Achievement' differs from both of these concepts in that it is a more personal and integrated concept, encompassing not only measurable progress and attainment but also broader experiences and accomplishments that may extend beyond standardised metrics.

OBJECTIVES

At the end of this unit you should be able to:

- show awareness of the complexities of closing the achievement gap;
- understand what is meant by self-regulated learning and by personalising learning (including adaptive teaching and direct instruction);

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- recognise instances of good practice in self-regulation and in personalising learning;
- be better equipped to support all pupils in developing self-regulation and to embed personalising learning in your own teaching.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Closing the achievement gap

When discussing the 'achievement gap', often it is attainment and progress that is considered significant within the wider social context, but for us, as schoolteachers, the broader, holistic conception of achievement should really matter. We are in a position to build relationships with learners through which we will encounter pupils with seemingly untapped potential who are underperforming in the classroom or in tests. Supporting these pupils will help improve their experience in school, likely contributing to a narrowing of the achievement gap. This is true of those who are low attainers, who may consequently make large improvements, but can also be true of pupils who are already high attainers. If we improve the performance of high attaining pupils, we may not contribute to narrowing the achievement gap, but we surely have a duty to help all pupils in our care.

Successive UK governments have focused on closing the achievement gap through different approaches to school improvement, general education policy and social initiatives. During the Covid-19 pandemic, periods of lockdown were experienced by almost all pupils globally, and these difficult times were exacerbated for some learners by lack of access to technology and other resources. Post-pandemic, attention has been focused on closing increasing gaps of equity and achievement, since pupils from disadvantaged backgrounds have been further limited. UK government researchers confirm that 'disadvantaged children and some others, for example, those with low prior attainment, were disproportionately affected' (Public Accounts Committee, 2023: Section 1, paragraph 3). This issue was recognised and addressed by the introduction of the Recovery Premium and is monitored through the 'disadvantage gap index' (a measure of differences in attainment – see Department for Education (DfE) (2024) for more information). The English disadvantage gap index figures reveal the impact of the pandemic. Indices for pupils entering secondary education demonstrate that any previous progress has been significantly undermined:

2011: 3.34
2018: 2.90
2022: 3.23

(Public Accounts Committee 2023: Section 1, paragraph 7)

Returning to face-to-face education has also been unequal, impacting the achievement gap:

> There have been widespread concerns about the dip in attendance levels since schools returned to usual patterns following lockdowns during the Covid pandemic. Particular concerns have been raised that increased persistent absence is concentrated among disadvantaged pupils, such as those who receive free school meals.
>
> (Long and Roberts, 2024, p. 17)

Overall, inequalities have been exacerbated and attempts to close achievement gaps have been even more challenging than pre-pandemic. The strategies in this unit are relevant in these ongoing challenging times since pupils will benefit more significantly from teachers understanding how to help them learn and schools will need to be more cognisant of the difficulties of each individual. Early secondary school education had already been identified as a significant point for the introduction of 'intensive catch-up programmes' to address problems persisting from early years and primary school contexts (Wellings and Wood, 2012, p. 7). Units in Chapter 7 consider issues around social initiatives and education policy. This unit is concerned more with pedagogic practices that can help in the struggle to raise achievement through matching learning tasks to pupils' strengths and supporting them effectively where necessary.

Personalised learning occurs when teaching, learning, activities and instruction are tailored to individual needs, interests and aptitudes. The aim is to ensure that each pupil achieves the highest standards possible for their abilities, regardless of their background, prior attainment levels and other circumstances. A further emphasis is on equipping pupils for more autonomy in their learning, which is considered vital to cope in the rapidly changing world of work. Closely related to this is the notion of self-regulated learning; pupils can be helped to learn about how they learn and to control and improve their learning.

We will focus on personalising education, and on helping pupils to improve how they understand their own learning through self-regulation. These pedagogic strategies are effective for all kinds of pupils, not only for groups of pupils who persistently tend to have low attainment, including, but not limited to, pupils with English as an Additional Language (EAL), pupils eligible for Free School Meals (FSM), those with Special Educational Needs and Disability (SEND) and Looked After Children (LAC). For these particular target groups, specific Pupil Premium (and 'Catch-up' Pupil Premium) projects are used, generally targeting literacy and numeracy. See DfE (2023a, 2023b) for details and also the Education Endowment Foundation (EEF) (2023a, 2023b) for a number of evaluations of these projects. Whilst these are effective (usually more so if implemented early on, in Year 7 (ages 11–12) for pupils who have not reached end-of-primary school Key Stage 2 standards), personalising and self-regulation approaches are more specific pedagogic techniques that benefit all pupils through providing positive and generalisable work habits that we can influence.

After a little more information around the context of achievement gaps, this unit explains concepts of personalising learning, adaptive teaching and direct instruction. All concepts in this unit are linked to aspects of good practice in learning and teaching, such as the importance of knowing pupils' interests, motivations, strengths and problems. The role of the teacher in ensuring consistent best practice, meeting pupils' needs and facilitating acceptable levels of achievement is therefore very important.

The achievement gap, the Pupil Premium and your responsibilities

In England, 'Pupil Premium' is a funding allocation to support pupils who need additional support. Schools are able to use this to directly meet pupil needs, but the funding goes to the school to be used as required (unlike Education, Health and Care Plan (EHCP) funds which are more closely linked to individual pupils). Schools are required to publish their Pupil Premium strategy, outlining their approaches and priorities. This information is therefore readily available online, allowing you to review a school's objectives and any relevant pupil progress tracking tools. As already noted, ensuring achievement for all is a key aim of the education system. Many of the methods and commitments

required to close the achievement gap extend beyond the teacher in the classroom, but you still have responsibilities in regard to improving achievement. Through effective systems and organisation, school structures need to create a supportive ethos, since a positive school culture promotes teachers and school leaders with high aspirations for all pupils, and facilitates significant levels of engagement. This, in turn, has the potential to maximise achievement for all. You can contribute to this (Winstanley, 2019). Practices like transition visits, collaboration with feeder primary schools are common, as are small-group teaching approaches, progress coaching, breakfast clubs, targeted literacy and mathematics support and additional support and progress tracking staffing.

Schools do not operate in a vacuum. It is incumbent on Local Authorities (LAs), Academy Trusts and government agencies to provide a supportive and flexible resource to help schools achieve their learning aims. Similarly, pupils' parents and carers should be encouraged to develop active partnerships with schools in order to help them understand, value and contribute to their children's education. By involving families and communities with their pupils' education, the high aspirations and support are not restricted to the hours in a day when the child is in school. Current Pupil Premium materials (DfE, 2024) are useful for summarising effective practices and build on earlier DfE recommendations, now integrated into recent publications. Strategies were initially outlined as beneficial for low attainers, but these can also be effective for all pupils, for example: early intervention; progress monitoring; tailored teaching; coaching teachers and teaching assistants in specific support strategies; cognitive approaches; one-to-one tuition; peer support; and home-school collaboration. These have been reviewed by the EEF, providing evidence of effectiveness, allowing for a more informed understanding of what might reduce the achievement gap. But take note that although the EEF provides some helpful research, their work is always under scrutiny, particularly by those who (quite rightly) question the application of relatively rigid scientific method to the social science of education and more specifically the practice of teaching and learning. See Wrigley (2018), for a fascinating and insightful analysis.

Task 5.5.1 asks you to reflect on the role of the Pupil Premium in closing achievement gaps for your pupils.



Task 5.5.1 Effective Use of the Pupil Premium

Access these documents (these can be accessed, for free, via an internet search engine):

- EEF Guide to the Pupil Premium for School Leaders
- EEF Tiered Model
- DfE Menu of Approaches

Consider which elements of the DfE 'Menu of Approaches' would be most helpful for supporting lower attaining pupils in your placement school. Consider all three aspects of the tiered model and review which are most relevant in your setting.

You will need to analyse the attainment gaps within your setting in order to help you make well-considered choices and this is possible by reviewing their Pupil Premium strategy which has to be published. Review the extent to which their strategy claims match the priorities that you have encountered during your time in school.

Cross-reference the schools' strategies with evidence-based recommendations from the EEF, or make use of additional DfE information, or other research papers.

What empirical support is there for the choices? How well do the strategies meet specific needs of pupils?

Record your responses in your professional record or portfolio and discuss with your mentor and/or other beginning teachers.

In England, your responsibilities have been enshrined in the Core Content Framework (CCF) for your training and early career development. Recommendations to improve the CCF emphasised SEND provision, noting that 'high quality teaching is the most important way to improve outcomes for pupils' (DfE, 2024, p. 5). Although this emphasis is specifically on SEND pupils, the principle of high-quality teaching is universally applicable to all learners. The first priority for you, as you learn to teach, is how to manage general classroom learning, and so, the remainder of this unit will focus on self-regulation and on personalising learning in practice. Teaching and learning techniques and strategies can help close the achievement gap if harnessed effectively.

Raising achievement for all pupils through self-regulation and personalised learning

Self-regulated learning

'Self-regulation' refers to the ability to understand and manage one's emotions and behaviour. 'Self-regulated learning' is applied to learning, employing strategies like active reflection about attitudes to learning, or generating suitable learning goals. Pupils can be explicitly taught techniques and strategies for self-regulation that work well for them individually. The process requires careful consideration of activities and behaviours whereby pupils evaluate what they have done and how well it has gone. Through repeatedly asking themselves if their chosen strategies have been helpful, they can try to reduce negative behaviours and increase positive behaviours. This active process of reflection equips pupils to take responsibility for their own learning thereby improving and developing their own performance. As pupils develop self-regulation, they become more independent. Pupils take on greater responsibility by actively planning their work, rather than responding to tasks; this helps them approach and manage resources with a more critical mindset. Less supervision is required as increased confidence is developed through learning effective techniques for tackling problems. Pupils build resilience and persistence through taking control of their learning. Focused group work and meaningful conversations with teachers help improve written and oral communication skills. This heightened engagement and encouragement to make creative responses can lead to increased motivation and in turn, to deeper learning.

Zimmerman (2002) describes self-regulated learning as a self-directive process, where learning is a proactive pursuit that pupils do for themselves. Pupils can be taught to self-regulate:

- behaviour – actively controlling resources for study (time, place of study, use of staff and peer support);
- motivation and affect – controlling feelings (reducing anxiety) and setting reasonable goals;
- cognitive strategies – aiming for deep learning by using effective processing strategies.

(Zimmerman and Schunk, 1989)

Among many other researchers, Bjork et al. reviewed self-regulation, concluding that 'research on learning, memory, and metacognitive processes has provided evidence that people often have a faulty mental model of how they learn and remember' (2013, p. 417). This reinforces the need for teachers to guide pupils in their reflective strategies. In addition, pupils are prone to lapse into counter-productive assumptions, such as 'over-attributing differences in performance to innate ability' and 'assuming that learning should be easy' (2013, p. 436). This links to Dweck's notion of a flexible, growth mindset (2012) in which pupils can be helped to foster a belief that their efforts can reap rewards, and that success is achievable as a result of hard work, rather than despairing by falsely attributing low attainment to an immutable lack of ability.

Now complete Task 5.5.2. This is your own reflective self-regulation activity. To increase the challenge, show you can adapt these questions for use with pupils, considering how they can be used to improve pupils' personal practices.



Task 5.5.2 Self-regulating how you study

Study time: How long do you study each day? Is it dependent on set tasks, or a fixed amount of 'work-time'? How often do you take breaks and how do you use them?

Study space: Can you access a dedicated study space? Is it shared? Personal? Do you enjoy studying there and how could you enhance it?

Personal timetables: Do you study best in the morning or evening? What other activities do you do? Which days are best for your studies? What about weekends?

Keeping on track: What distracts you? How can you manage distractions? How do you record what you've achieved and what's left to complete? How well do you stick to study plans?

Planning: Are you good at planning how long tasks will take? Have you tried timing yourself and how well do you manage deadlines? When working right up to a deadline, how does this make you feel? How good are you at prioritising and ordering your work and priorities?

Complete this task individually and then come together with other beginning teachers. Discuss both your factual responses and your emotional responses. Perhaps you are making excuses to yourself, or being overly harsh about some poor work habits. Discuss how this task could work with pupils. Adjust the questions to meet your group's needs and interests and try it out in school. Present the findings to the rest of the group. Record the outcomes in your professional portfolio.

Explicitly teaching pupils to set goals and self-regulate (see also Unit 3.2)

Goal setting is a key aspect of self-regulation and can be actively taught. Once cultivated, such skills empower pupils to be involved in setting realistic and negotiated learning goals in collaboration with teachers with pupils developing strategies for reviewing their own performance. The resulting clarity and transparency make subsequent steps explicit, helping pupils stay on target. Where pupils are progressing particularly well, there is also potential to compact curricula, allowing pupils to skip tasks that do not provide them with adequate challenge.

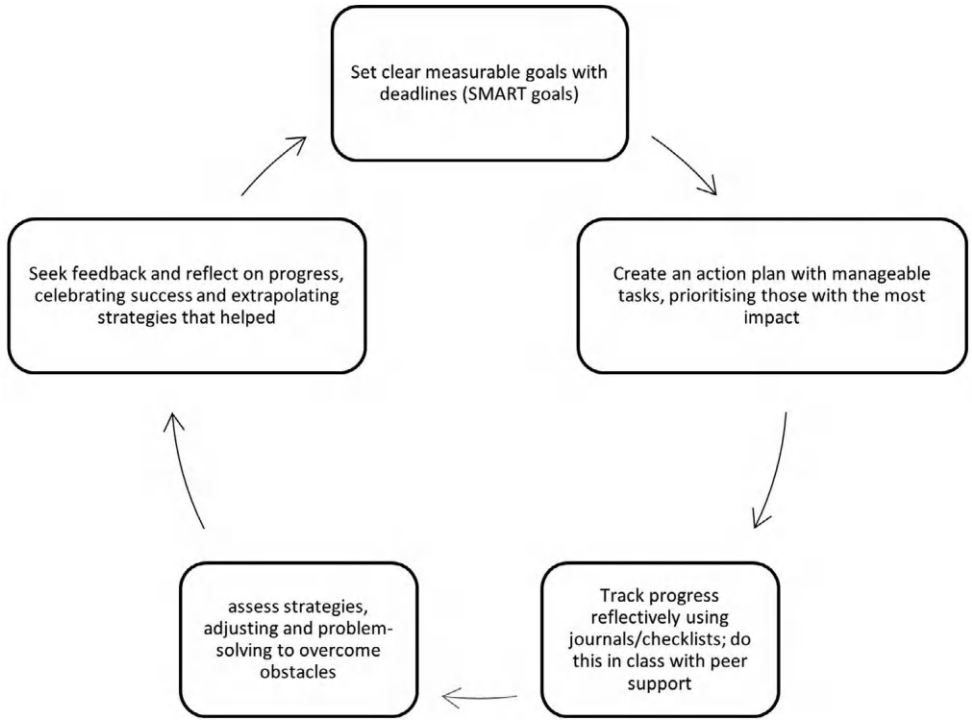


Figure 5.5.1 Goal-setting and self-regulation cycle

Integrated goal-setting and self-regulation can be approached in many different ways. Here, in Figure 5.5.1, you can see one of many examples that could be adapted to suit your circumstances.

All these steps are best undertaken together with the teacher until the pupil is able to take responsibility for their own learning. Pupils can be helped to evaluate their learning at the start of a task by reviewing how much they already know about a topic and through linking new experiences to their existing understanding. In breaking tasks into manageable learning goals, it helps learners consider task sequences, priorities, timing and expected pace of work. Remember that self-regulation is more than just applying effective study skills because it is on-going. It requires careful and thorough self-monitoring of pupils' own performance once they have attempted tasks, received feedback and explicitly noted potential improvements and progress so far.

Assessing and teaching for self-regulation

Pupils develop self-regulation through clearly expressed constructive feedback focusing on progress and what can be learned from errors (see also Unit 6.1). Avoid comparisons, and provide clear, task-focused feedback linked to attainable standards. Engage pupils in reflection and dialogue, recognising their efforts and helping them manage their work, discussing the completed tasks rather than the pupils' personal characteristics.

Actively helping pupils learn how to self-regulate can be done by teachers modelling reflection, through thinking aloud and encouraging pupils to practise the same processes. This can be done in groups, collaboratively with teachers and peers. Feedback can also be used to deepen pupils' learning, not just to correct the mechanics of presentation, writing and accuracy. Teachers can help pupils

make links between concepts, directing them to relevant information, drawing attention to relevant ideas and demonstrating how to develop an analytical approach. It is sometimes possible to incorporate a grade for how well pupils have responded to feedback. Here, teachers can provide some points of actionable feedback, and then evaluate how well the pupil has understood and incorporated the suggestions as improvements in subsequent tasks or drafts. This has the advantage of directly feeding back to the pupil on their understanding and implementation of feedback, helping them see their own development, making it easier to generalise their progress to other aspects of their work.

Pupils require direct and explicit teaching in order to develop effective self-regulation. This is worthwhile because such strategies have been shown to improve learning (Bjork et al., 2013). With goal-setting, for example, teachers can help pupils to tease apart the components of tasks, so they can be tackled effectively. This can be done explicitly at first, and then phased out as pupils learn how to break down the tasks for themselves. The intention is to shift responsibility for learning from the teacher to the pupil. Teachers should also encourage a range of pupils' learning strategies through allowing different responses to tasks where appropriate such as permitting audio and visual answers, not only confining responses to writing.

What is meant by personalising learning?

This section considers the value of strategies such as adaptive teaching and direct instruction for helping to close the achievement gap. See also Unit 5.3. These strategies fall under the umbrella term of 'personalised learning' which is a catch-all without a widely agreed definition. You might find different interpretations in official documents and in education publications, and you may see diverse definitions in policy documents in different schools. Generally, however, all types of personalised learning' refer to matching learning tasks to pupils' needs wherever possible. Teachers are shaping teaching and learning around the different ways that pupils learn so that every pupil's learning is maximised in the context of the current curriculum; it does not mean that each pupil has a personalised curriculum.

More sharply tailored learning (or personal curricula) can be designed for pupils with additional learning needs and these Individual Education Plans (IEPs) would fall under the requirements of an Education and Health Care Plan (EHCP). Pupils in receipt of these plans generally have additional learning needs, having been assessed by a psychologist or other specialist. In these instances, pupils can be provided with a detailed modified curriculum for each and every subject, going beyond the more straightforward personalised learning discussed here. (See Unit 4.6.)

The principles of personalising learning in the classroom reflect good practice. Pupils' individual needs must be assessed, and any set tasks should be responsive to these needs, creating an inclusive approach. Useful assessment supports learning, being linked to personal targets, and making use of effective feedback primarily from teachers, but also including peer and self-assessment. Peer support should be harnessed, including mentoring both for learning matters, and for socio-emotional concerns. As noted, learning strategies should be directly taught and linked directly to self-regulation. Effective ways to personalise learning in the classroom, include:

- using formative assessment (see Unit 6.1), i.e. focusing assessment on its value for learning and negotiated and realistic target setting;
- small group early interventions to minimise falling behind;
- developing learning strategies and independent learning skills;

- dynamic approaches to pupil grouping;
- attention paid to safe school environments where hindrances to learning and teaching (e.g. bullying, poor behaviour) are tackled promptly and decisively; ensuring all pupils (especially those from disadvantaged backgrounds) may study safely, e.g. supervised homework etc.;
- celebrate achievement; high expectations of pupils with appropriate challenge; oral praise; prompt feedback clarifying misunderstandings;
- (at a school level) augmenting/replacing parents' evenings with 'pupil review days' allowing personal pupil meetings discussing targets and progress; shifting to more active carer involvement.

Classroom approaches: teaching/pedagogical strategies

Since personalised learning is a broad concept, many of these approaches are discussed in other units. Because strategies have dedicated units they are briefly noted here, with an emphasis on how they specifically support improving achievement.

Grouping pupils and group work (see also Units 2.3 and 4.1)

Collaborative group work is helpful as it fosters peer support and allows for diverse perspectives, enhancing learning and motivation for all pupils. Through leveraging the strengths of each group member, pupils can address individual learning needs more effectively, developing skills through shared problem-solving and mutual encouragement. Flexibility of grouping is key however and needs monitoring to ensure positive pupil experiences.

Formative assessment (see also Units 6.1 and 6.3)

Formative assessment helps improve learning. Teachers are exhorted to understand how the design of educational activities can enhance pupil learning through considering 'assessment as part of pedagogy' (Black and Wiliam, 2018, p. 551). Formative and summative feedback should be conducted in ways that help pupils understand their strengths and areas for improvement, thus allowing them to make targeted adjustments to their learning strategies, promoting continuous progress.

The importance of questioning techniques (see also Units 3.1 and 5.2)

The quality of questions both to pupils and from pupils is important, and the 1956 work of Bloom is a still much-used (often adapted) way of thinking about higher-order questioning. Numerous different versions of Bloom's Taxonomy (Bloom et al., 1956) exist in the public domain, but a useful rule of thumb is to ensure that the higher-level aspects or types of thinking – analysis, evaluation and synthesis – are incorporated into all lessons so that no learners are limited by the questions they are being asked.

Problem-solving and investigations (see also Units 5.2 and 5.3)

Real-life problems and in-depth investigations are preferable to disconnected tasks, as they engage pupils in real-world challenges that require critical thinking and collaboration, thereby deepening their understanding and application of concepts. Such tasks can boost pupil interest and investment through offering choice and autonomy, making learning relevant and personal.

Cognitive issues: accelerated learning, learning styles and metacognition (see also Units 4.3 and 5.1)

Some popular ideas about harnessing children's cognitive processes have been controversial, such as the initially attractive notion of relying on 'learning styles', which has since been debunked and even shown as potentially detrimental (Bjork et al., 2013). 'Metacognition' concerns how we understand our own learning and ways to improve our learning. Useful metacognitive techniques incorporate self-regulation, which helps pupils analyse their own strengths and weaknesses and develop their self-understanding in relation to their work.

Emotional intelligence and motivation (see Units 3.2, 4.1 and 4.3)

Together with metacognition, knowledge of pupils' emotional intelligence affords insights to attitudes and responses to tasks, which can also assist you with pupil groupings. Personalising learning makes it easier to maintain motivation as pupils have more agency in choosing their learning.

Adaptive teaching, direct instruction and differentiation (see also Units 4.1 and 5.3)

All three of the terms in this sub-title can be considered buzzwords in education. As such, the concepts have both diehard supporters and serious detractors with claims for and arguments against their contested efficacy. It can be hard to navigate around education jargon as an experienced teacher and even more so as a novice. The point here is not to worry unduly about the terminology and instead to keep sight of the objective; supporting learners to achieve well.

For many years, teachers were urged to 'differentiate' in the classroom and when done skilfully, this can help reduce achievement gaps. However, this would entail differentiating a number of elements such as content, pace, assessment, ways of recording ideas, level of support, resources etc. These are rich and nuanced ways of tailoring and matching experiences to learners, but very complex and consequently impossible to maintain. Common criticisms of the notion of differentiation tend to refer to an impoverished understanding of differentiating by outcome. Many an educative blog emphasises the difficulties of restricting pupils by expecting a lower level of outcome than their peers. This is not a reason, however, to disregard all types of differentiation.

As an example, in an otherwise interesting discussion, on an EEF blog, this question is posed: 'Is it time for 'adaptive teaching' to replace differentiation?' (Eaton, 2022). By dichotomising adaptive teaching and differentiation, there's an assumption of differentiation as being associated predominantly with outcome, which does not have to be the case. The short blog has many merits however and some very useful links to deeper discussions, but the point being made here is that it's easy to find polemic and strong views that may end up dismissing some potentially positive options. Where poorly executed or using ossified practices, differentiation could involve extensive pre-prepared separate lessons, resources or tasks, adaptive teaching is more focused on adjusting instruction in real time, as learning is happening. Because prepared materials would be assigned to groups, there was the temptation of assuming a fixed notion of pupil ability and thereby limiting possibilities to progress as well as perpetuating existing achievement gaps. This could be damaging. It helps to explain why differentiation has fallen out of favour.

So – what is meant by adaptive teaching in this context, and how does it relate to differentiation? Adaptive teaching is when teachers 'adapt teaching to respond to the strengths and needs of all

pupils' (DfE, 2011, Teachers' Standards, no. 5). This requires sound knowledge and understanding of individuals and of their learning needs. Scaffolding is considered a key element of adaptive teaching, providing support at the early stages of an idea or task, but this must be dismantled as the learner progresses. Much like the ideas in self-regulation, feedback, monitoring and continuously reassessing goals is needed to ensure positive achievement.

Direct instruction (di – lowercase) is another pedagogy (see also Unit 5.3) that is sometimes lauded as a good way to support lower attaining learners and to help make up learning loss. It usually refers to a teacher-directed method in which exposition is key, followed by modelling and lots of practice. It sometimes includes immediate feedback, particularly using levelled programmes through technology allowing pupils to progress at their own rate once they have been taught an initial concept. Although there is some evidence that direct instruction can make a positive difference for SEND learners and for those seeking to make up lost ground in particular (Aubin, 2023). However, it is often dismissed as an outdated, didactic approach that has no place in the contemporary classroom.

As can be seen, any single method is going to have pros and cons and these will also depend on the interpretation of the pedagogy, which is often open to debate. As is often the case in education, a range of different approaches will be the only way to meet the complex needs of a range of learners. Combining problem-solving and investigative work with some use of direct instruction, for example, can provide both the targeted, explicit aspects and allow for development of independence and self-reliance.

Closing the gaps across the nations

The key references in this unit have been to England's DfE, often supported by the work of the EEF which is predominantly centred on English policies, although commonly relevant beyond England. If you are based in England you will be interested to see the national differences; considerable resources are available online to support curriculum and teaching in each country, reviewing these is good practice. If you are based in Northern Ireland, Scotland or Wales, it is essential that you familiarise yourself with the relevant resources. Task 5.5.3 asks you to consider these learning resources. In each country, our shared goal is for every pupil to succeed. Collegial support and collaborative working are important features of teaching, so you are encouraged to discuss your work with peers.



Task 5.5.3 Resources to support closing the achievement gap across the nations

Review the different approaches to closing the achievement gaps in the different nations within the UK. We suggest using these links as a starting point:

- **Department for Education Northern Ireland:** The key relevant NI programme is *Every CHILD* ('Every child and young person is happy, learning and succeeding'), supported in turn by the RAISE programme which targets specific geographical areas of most need, working as a whole community and place-based approach CHILD – https://www.education-ni.gov.uk/sites/default/files/publications/education/Every%20CHILD%20DE%20Corporate%20Plan%202023-2028.pdf RAISE – https://www.education-ni.gov.uk/publications/raise-programme-summary-document

- Scotland has a National Improvement Framework (NIF) aiming for both excellence and equity in education with a further targeted programme, the Scottish Attainment Challenge NIF – https://www.gov.scot/policies/schools/national-improvement-framework/: Scottish Attainment Challenge – https://www.gov.scot/publications/scottish-attainment-challenge-framework-recovery-accelerating-progress-2/pages/2/
- Since 2018, Wales has adopted the language 'Additional Learning Needs' (ALN) in a change to previously used SEND definitions, whilst also gradually altering strategies in order to create a more holistic support system. Alongside this has been the development of a new curriculum aiming to raise standards and also specifically designed to reduce the impact of disadvantage on educational attainment as a stated aim: Curriculum – https://www.gov.wales/additional-learning-needs
- Across the UK: The Education Endowment Foundation: https://educationendowment-foundation.org.uk/ has been funded by the UK government specifically to support closing the gap in attainment.

Compare what you have found with other beginning teachers and record your findings and responses in your professional record or portfolio.

SUMMARY AND KEY POINTS

Many of the ideas in the unit are already likely to feature in your plans for working in the classroom since they reinforce ideas widely recognised as best practice. The key points are:

- **Self-regulating learning:** Supporting pupils to reflect on, plan and monitor their own learning has been shown to significantly improve motivation and achievement. It encourages independence, persistence and a greater sense of responsibility for learning outcomes.
- **Personalising learning:** This involves adapting teaching to reflect the different ways pupils learn, using a range of pedagogical approaches. It means responding to learners' needs without diluting challenge or lowering expectations.
- **Important aspects of closing the achievement gap include:** getting to know your pupils as individuals, varying pedagogical approaches to meet diverse needs, helping pupils understand and develop their own learning strategies, and using assessment not just to measure attainment but to inform next steps and promote learning are all key strategies for supporting progress and closing the achievement gap.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.

Further resources

Cash, R.M. (2016) *Self-Regulation in the Classroom: Helping Students Learn How to Learn*, Golden Valley, MN: Free Spirit.

Despite its American context, this book has ideas that are relevant for every classroom. Covering theoretical and practical issues of self-regulation, the ideas on reflection, goal-setting and procrastination are helpful for you as a teacher, and also as a lifelong learner.

Wallace, B., Leyden, S., Montgomery, D., Winstanley, C., Pomerantz, M. and Fitton, S. (2009) *Raising the Achievement of All Pupils within an Inclusive Setting: Practical Strategies for Developing Best Practice*, Abingdon, UK: Routledge.

For really practical examples of proven effective classroom and whole-school practices, despite its date, this book remains extremely useful. Many of the recommended practices have bene enshrined in policies discussed in this unit. Based on case-studies of schools, each chapter combines theory with illustrative examples and covers areas including multiple exceptionalities, socio-emotional issues, leadership, inclusion etc.

Winstanley, C. (2020) *Breaking Down Barriers: Opening Doors to Achievement and Removing Barriers to Success for More Able Learners*, Oxford, UK: National Association for Able Children in Education.

This booklet shares ideas around reducing the achievement and attainment gaps through considering what is holding pupils back from success. Although the focus is around more able pupils, there is information that is relevant for all pupils and ideas around the responsibilities of schools.

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5.6 Educational neuroscience

Classroom practice and the brain

Paul Howard-Jones and Carolina Gordillo

Introduction

> Education is about enhancing learning, and neuroscience is about understanding the mental processes involved in learning. This common ground suggests a future in which educational practice can be transformed by science, just as medical practice was transformed by science about a century ago.
>
> (Opening lines of the Royal Society 2011, p. 1)

In recent years, there has been a step change in efforts to bring neuroscience and education together in dialogue. Excitingly, our scientific understanding of learning is now allowing us to consider everyday learning in the classroom in terms of underlying brain processes.

OBJECTIVES

At the end of this unit you should be able to:

- avoid some popular neuromyths;
- think about classroom practice in terms of three categories of learning process (engage, build, consolidate);
- draw on insights from cognitive neuroscience when planning your lessons;
- draw on insights from cognitive neuroscience to evaluate your lessons.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Neuromyths to avoid

Most teachers consider a knowledge of the brain is important when developing their practice and, given the brain's central role in learning, this belief appears justified. However, a lack of valid information about neuroscience in ITE has helped create a parallel world of pseudo-neuroscience within

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schools. Many concepts that claim a brain-basis are unscientific and educationally unhelpful, so let's first identify some of the most popular but unscientific beliefs about the brain (so-called neuromyths). These often originate from authentic science that has then become misinterpreted or overinterpreted. That makes it difficult, but also important, to separate the fact from the fiction. Task 5.6.1 introduces some common neuromyths.



Task 5.6.1 Neuromyths: true or false?

Carry out this quick TRUE/FALSE survey on some teachers, learners and friends. You may be surprised how common neuromyths are. The answers are in brackets.

1 We mostly only use 10% of our brains (FALSE – we use all our brains all the time).
2 Exercise can improve mental function (TRUE – see main text).
3 Co-ordination exercises can improve integration of left and right hemispheric brain function (FALSE – see main text).
4 Pupils are more attentive after sugary drinks and snacks (TRUE – although unhealthy in other ways, sugary snacks actually increase attention; Busch et al., 2002).
5 Regular drinking of caffeinated drinks reduces alertness (TRUE – regular caffeine, e.g. two cans of caffeinated soft drink a day, reduces children's cognitive function by initiating counter-regulatory changes in the brain; Heatherley et al., 2006).
6 Individuals learn better when they receive information in their preferred learning style (FALSE – see main text).

Using this knowledge, review the schemes of work you will be teaching to see how some of these ideas might be applied in practice.

Surveys of teachers in countries as diverse as the United Kingdom (UK), Turkey and China show over 90% of teachers believe teaching to pupils' preferred learning styles can improve outcomes (Howard-Jones, 2014). Learning preferences, however, do exist, in the sense that different individuals may prefer to receive information in different ways. For example, it is possible to categorise learners' preferences in terms of visual, auditory or kinaesthetic (VAK). However, there seems little educational value in doing so, despite the references to neuroscience by those promoting learning style theory. Different regions of the brain are more involved than others in processing visual, auditory and somatosensory (touch) information. But performance in most everyday tasks, including learning of information provided in just one modality (e.g. visual), prompts many regions in both hemispheres to work together in a sophisticated parallel fashion.

In reality, the brain is so massively interconnected that sensory experience is always cross-modal in nature. This means, for example, that visual experience can result in activity in parts of the brain often referred to as auditory. As Kayser (2007, p. 24) puts it, 'the brain sees with its ears and touch, and hears with its eyes'. This interconnectedness of the brain also means that neuroscience cannot be used to support teaching children according to a profile based on multiple intelligences theory, (as discussed in Unit 4.3), since such 'intelligences' are likely to involve distributed and overlapping neural networks. Scientific evidence from laboratory experiments shows that providing information in a learner's preferred VAK style is a wasted effort (Kratzig and Arbuthnott, 2006). At the end of their extensive educational review, Coffield et al. (2004) concluded there were no clear implications for pedagogy arising from any existing models of learning styles. However, our different

senses are a boon for learning because they offer us a diverse range of multimodal ways to communicate the same concept (through, for example, diagrams, videos, objects, oral explanations, text, etc.). Educational and scientific evidence suggests that, when pupils are encouraged to make links between these different forms of representation, learning can be improved.

Educational kinesiology (or Edu-K, also often sold under the brand name of Brain Gym®) draws on ideas that include integrating left and right hemispheric brain function through co-ordination exercises. Such ideas spawned several training programmes to remediate learning difficulties through exercises, but these were shown to be ineffective by numerous studies in the 1970s and 1980s (Arter and Jenkins, 1979; Bochner, 1978; Cohen, 1969; Hammill et al., 1974; Kavale and Forness, 1987; Sullivan, 1972). A major review of the theoretical foundations of Brain Gym® and the associated peer-reviewed research studies failed to support the contentions of its promoters (Hyatt, 2007). Despite this, many teachers remain enthusiastic about Brain Gym® and convinced that it supports learning, with reports of increased reaction times following its exercises suggestive of some positive effect on cognition (Sifft and Khalsa, 1991). Any positive effects may, however, relate to a more a general relationship between exercise and learning that would be better harnessed by more aerobic forms of exercise.

There is an emerging body of multidisciplinary research showing the beneficial effect of aerobic exercise on selective aspects of brain function, and some of these aspects happen to be particularly important for education (Hillman et al., 2008). For example, a study of adults revealed increased levels of brain-derived neurotrophic factor (BDNF) after two three-minute sprints (Winter et al., 2007). When compared to sedentary or moderate exercise conditions, participants showed a 20% increase in the speed of recall for words they learned immediately following their intense exercise. BDNF plays an important role in synaptic plasticity, the making of connections between neurons, which is thought to underlie our ability to learn. Neuroscience is shedding light on how exercise and fitness can promote classroom learning. But, it seems that aerobic exercise is what's needed, rather than co-ordination exercises.

Brain plasticity and learning

Whatever your age and stage of development, the brain remains plastic. That means that our experience, including our educational experience, can change the way our brain functions, the way its neurons connect with each other and even the shape and size of its various parts. Of course, the brain is more plastic during childhood but, as adults, it remains plastic and always able to learn. It is important that teachers and their pupils understand about brain plasticity. Many learners feel their brains limit their potential and prevent them from learning but, when a pupil decides they want to learn, they are choosing to change their brain's function, connectivity and structure – essentially building their learning capacity. Research has shown that when learners discover this, it can improve their self-concept and their academic potential. So, tell your pupils: their brain isn't something they're stuck with – they can change their brains – and the best way to change them is through learning!

Three broad categories of learning processes provide a convenient starting point for thinking about classroom practice in terms of underlying brain function. Recent research offers an understanding of how a pupil becomes *engaged* with a source of new knowledge prior to the building of new knowledge, how they first come to *build* new knowledge, and how this new knowledge later undergoes *consolidation*, causing it to become more permanent, accessible and useful. These three categories are broad headings that are helpful for breaking down learning into component processes. They are introduced in this order for the sake of convenience; in reality, and depending on the learning experience, they may occur in any sequence and even simultaneously.

Engagement of the learner

Every learner's brain is unique, and pupils differ in what most engages their attention and the extent to which they can control it, making it important for the teacher to monitor engagement and vary their approach according to pupil response. Scientific studies have revealed the role of *subcortical* structures deep below the cortex in the emotional states that encourage us to attend and learn. For example, praise is commonly used as an effective means to reinforce classroom behaviours conducive to learning (Becker et al., 1967; Dufrene et al., 2014; Reinke et al., 2008; Sutherland et al., 2000). Praise is a social reward but recruits similar subcortical regions of the brain's reward system as receiving money (Izuma et al., 2008; Knutson et al., 2001) or anticipating food (Farooqi et al., 2007). A neuroimaging study has shown increases in reward system activity during the answering of educational questions, in 'gamified' conditions that favoured engagement and educational learning (Howard-Jones et al. 2016). The anticipation and receipt of rewards can release neuromodulators capable of improving attention and memory, justifying your use of many teaching strategies known to stimulate this 'approach' response in the brain's reward system. These might include awarding points as part of a game, novelty, allowing pupils some choice or including activities such as peer or group work and presentation to peers that involve the sharing of attention. All of these can potentially improve the engagement of your pupils with learning.

Other types of subcortical activity, including activations within the amygdala, are implicated in the development of maths anxiety and negative emotions towards learning (Young et al., 2012). Therefore, in both positive and negative ways, pupil engagement is influenced by a two-way interaction between their subcortical structures linked to emotion and the cortical processes required for learning (Figure 5.6.1).

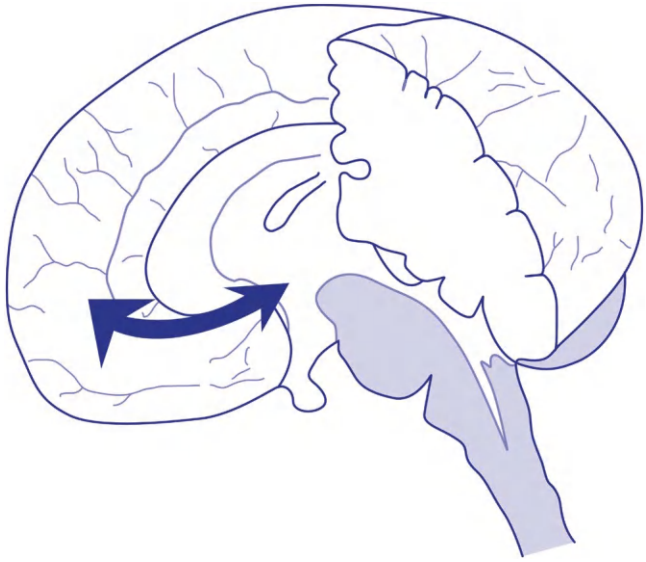


Figure 5.6.1 Engagement with a learning opportunity is influenced by interaction between subcortical structures (that are important for processing emotion) and frontal regions of the cortex (that are involved with attention and working memory)

Building of knowledge and understanding

Once a pupil is engaged with a source of new knowledge, such as an explanation or other stimulus provided by the teacher, a channel of communication can open to enable new learning to occur – although much depends on the quality of this communication. Pupils have to apply effort to hold and work on the new input in a temporary store called 'working memory'. This store, which has limited capacity, involves brain circuitry in the prefrontal cortex (Kane and Engle, 2002), as shown in Figure 5.6.2. For this reason, teachers (and the resources they use) should communicate clearly and concisely, with minimal irrelevant and distracting information that might place an unnecessary burden on working memory. Accounting for the role and limitations of working memory when teaching and learning (such as avoiding distractions that compete for its limited capacity) is sometimes referred to as Cognitive Load Theory (Sweller, 1988) (see Unit 5.1).

The brain is organised around our senses and movement, and instruction that encourages the making of links between different representations of concepts, using our different senses and our ability to enact ideas, can enhance communication and understanding. It may be helpful to remember that some communication is unconscious. When we move, or express emotion in our faces, the brains of those observing us activate as if they were moving similarly or experiencing similar emotions. The brain networks involved have been referred to as the 'Mirror Neuron System'. If gestures and faces communicate knowledge and emotions both consciously and unconsciously, this emphasises the importance of feeling genuinely confident and enthusiastic about what you are teaching – if you want your pupils to feel the same.

Pupils' learning benefits from connecting new information to older, prior knowledge that appears to 'fit' (i.e. is congruent) with the new learning (Shing and Brod, 2016). So, for example, you might choose to prompt pupils to reactivate appropriate prior knowledge from a previous lesson by using

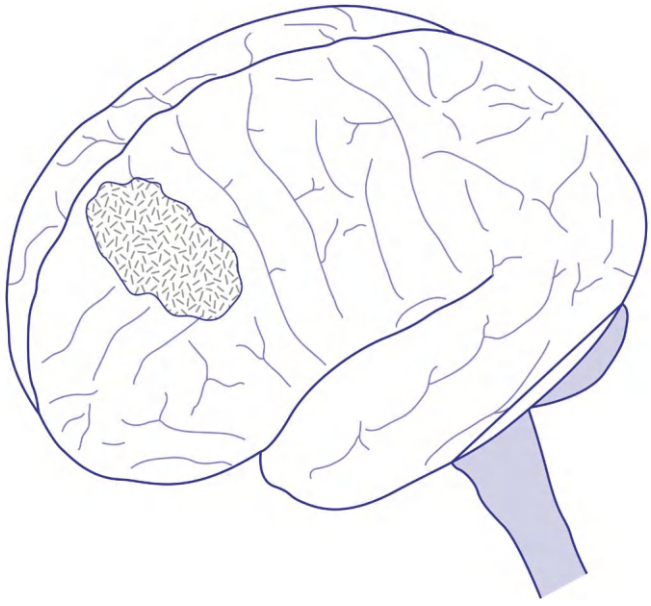


Figure 5.6.2 Building new knowledge and understanding requires effortful conscious processing and activation of a working memory network that includes prefrontal regions of the cortex

a revision question-and-answer before new information is presented. This will help your pupils bring the prior knowledge more readily to mind and may enable some learners to make appropriate connections automatically. Others may need more encouragement and direction (e.g. you might ask them directly what connections exist between today's topic and previous lesson content). One important form of prior knowledge is a pupil's personal experience. Connecting to personal experience can help pupils generate a more personally meaningful and memorable understanding about lesson content (e.g. grasping the meaning of the word 'clowning' by recalling the holiday memory of a clown). However, although a child's brain makes connections in much the same way as an adult's, children have more difficulty with finding an appropriate fit with what they already know, simply because they know less. Therefore, asking pupils to make their personal connections explicit through classroom discussion, and scaffolding their connection-making as needed, can help them develop the most helpful meanings and appropriate understanding. This modern scientific understanding of how knowledge is constructed echoes the emphasis placed by traditional theorists such as Vygotsky on the importance of prior knowledge for learning (see Unit 5.1).

Differences in prior learning, but also in the development of the working memory network, are two major factors that result in diverse individual differences in terms of children's capacity to build new knowledge.

Consolidation of learning

New learning is vulnerable to loss, and recalling and applying freshly learnt knowledge occupies the limited working memory available. Having just learnt something new, your pupils' working memory needs freeing up if they are to learn more. Fortunately, when pupils consolidate their learning, it not only becomes more permanent but also becomes easier and quicker to access (see Unit 6.3), demanding less conscious effort (Tham et al., 2015). Consolidation is usually helped by practising recall of the knowledge and understanding, which tends to shift activity away from working memory regions to those more involved with automatic unconscious processing (i.e. away from the front of the brain, as represented schematically in Figure 5.6.3).

This was demonstrated neatly in a study of adults undertaking mathematics training. Figure 5.6.4 shows how their brain activity when doing mathematical problems shifted after training from the front to the back of the brain. The left image (Scan 1) shows where activity reduced after training, and the image on the right (Scan 2) shows where activity increased. This shift arises from a decrease in working memory load (i.e. a reduced amount of information to hold in one's immediate attention – such as the steps that must be followed to complete a calculation) and an increase in more automatic processing (i.e. doing some parts of the mathematical processes without thinking about them).

Consolidation can be helped by practising the recall and application of new knowledge in different contexts. Questioning is often useful to assess what learning your pupils have achieved, but it is also an effective means to ensure that this deeper processing occurs, and to accelerate the rate at which their learning becomes consolidated. This emphasises the value of pupils being provided with interesting opportunities to apply and test their knowledge, but in low-risk tasks that are free of anxiety (unlike exams or formal assessments). Recent neuroimaging research suggests that repeatedly retrieving information causes it to become represented in the brain in different ways – essentially connecting it with different meanings and making it easier to retrieve in the future (Wirebring et al., 2015).

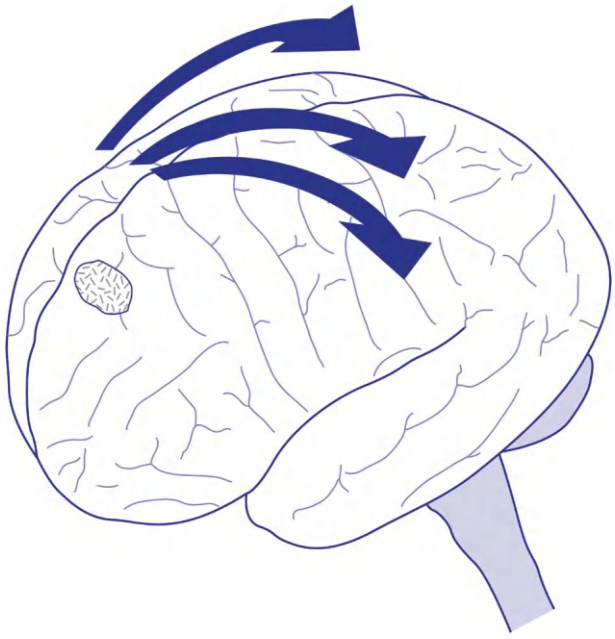


Figure 5.6.3 Consolidation of knowledge and understanding results in a shift in activation away from frontal regions – freeing up working memory for more learning

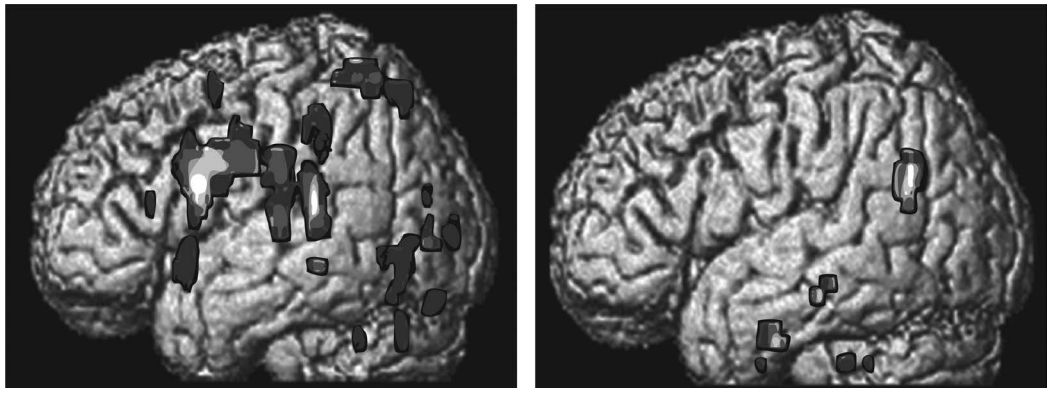


Figure 5.6.4 Changes in brain activity following training. Scan 1: activity decrease; Scan 2: activity increase

Source: Delazer et al. (2003, pp. 82–83)

Applying scientific insights when planning activities

The learning processes underlying a particular classroom activity can be broken down into those supporting engagement, the building of knowledge and the consolidation of learning (Figure 5.6.5).

However, it should be made clear that the categories 'engage', 'build' and 'consolidate' should not be used to partition a lesson. This would oversimplify many learning experiences encountered in a real classroom. In everyday teaching and learning, all three types of process might occur

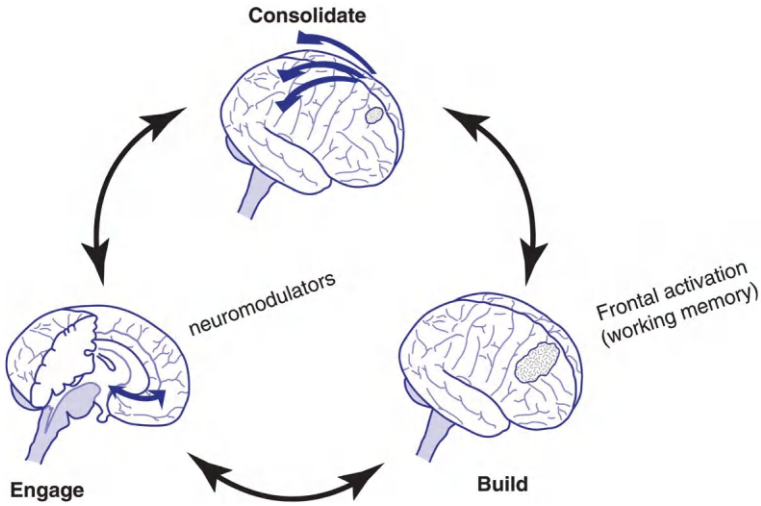


Figure 5.6.5 Engage, build, consolidate: the learning underlying classroom learning can be categorised into three types of process

simultaneously, or at least in such quick succession that allocating one type of process to any single stage in a classroom activity can be unhelpful. For example, consider the situation in which you use a quiz activity to introduce a new topic and link it to yesterday's learning. Based on a scientific understanding of the brain, you might use rewards (or points or praise) to promote engagement, verbally scaffolding the pupils' efforts to build their knowledge and link the new learning to old, while also using questions that require application of the prior knowledge in a range of novel ways. This simple activity would, therefore, involve concepts from all three categories of engage, build and consolidate.

Also, the science does not provide a list of 'top tips' or best practices. Rather, it offers concepts and a scientifically determined understanding of how learning generally works, based on concrete measurement of behaviour and brain function. This cannot tell you exactly what to do but it can valuably inform your decisions, help you reflect about why certain activities might not work as expected, and help you think more carefully about planning and evaluation. Now do Task 5.6.2.

Task 5.6.2 Discovering the science behind your lesson plans

Choose a typical lesson plan in which you will carry out a range of different activities. Which of the concepts discussed in this unit provide insight about why or how each activity might support learning? One way of approaching this task is to summarise and number the concepts above and then use this as a key to map concepts onto activities:

Engage

E1: Every learner's brain is unique, and pupils differ in what most engages their attention and the extent to which they can control their attention, making it important for teachers to monitor engagement and vary approach accordingly.

E2: Teachers employ many strategies known to stimulate an 'approach' response in the brain. These include rewards such as praise and tokens acknowledging achievement, novelty, provision of choice and shared attention. The anticipation and receipt of rewards can release neuromodulators capable of improving attention and memory.
E3: In contrast, fearfulness can avert attention, and anxiety can diminish a pupil's learning by reducing the brain's ability to process information.
Etc.

Build

B1: Being aware of pupils' prior knowledge is important for a teacher because this is the foundation on which the pupils' new knowledge will build.
B2: Teachers can help pupils think meaningfully about new ideas by encouraging them to make connections with their prior knowledge.
B3: The brain is multisensory, and understanding of new knowledge is helped by clear, concise instruction using the senses, embodiment and action to encourage links between different representations.
Etc.

Consolidate

C1: Practice and rehearsal of freshly learnt knowledge causes it to become automatically accessible. This frees up the brain's limited capacity to pay conscious attention, and so be ready for further learning.
C2: Answering questions, applying knowledge in new situations, discussing it with others or expressing it in new forms consolidate our learning through helping us to store it in different ways – making it easier to recall and apply it.
Etc.

Discuss your analysis with others and record your notes in your professional record or portfolio.

The science of learning and the reflective practitioner

Effective teachers tend to employ 'good' practices that have, in general terms, been identified by educational researchers as effective. However, research suggests simply applying best practices cannot assure the best possible levels of learning are achieved. This may be because teachers require an understanding of why, when and how each practice can be effective, in order to adapt it optimally for their pupils (Coe et al., 2014). Lesson evaluation is an important part of developing this understanding. Ensuring your evaluation is informed by the science of learning can help provide insight into the specific lesson and also help build your general understanding of classroom learning practices, so improving your ability to adapt these practices to different classes, individuals and contexts. Task 5.6.3 asks you to reflect on learning in your lesson evaluations.



Task 5.6.3 Reflecting on learning in your lesson evaluations

When evaluating the lesson you planned in Task 5.6.2, look back at the concepts you linked to each activity. Use these when reflecting on how the lesson went, and to prompt questions for thinking more deeply about learning processes and why learning may have occurred or not. For example, see Table 5.6.1.

Discuss this with other beginning teachers and/or your mentor.

We are still at the beginning of a long journey when it comes to introducing the science of learning into classroom practice, and some critics have suggested that psychology alone should inform education about learning. However, the field of cognitive neuroscience combines psychology with concrete data about brain function, providing a potentially more valid and often more visual understanding of learning than psychology alone. That said, there remain significant theoretical and cultural gaps between education and the sciences of mind and brain that will require investment, research and time to fill. While such work continues, this short introduction should allow you to begin introducing the science of learning into your planning and evaluation in a meaningful and insightful way.

Table 5.6.1 Questions to help you reflect on learning

Activity (referenced to concepts above)	*Questions that might be prompted by the scientific concepts …*
Q and A on content from last lesson using images	What difficulties did the pupils have when responding to your questions – was more support needed for some pupils? Did you spot individual differences in engagement – why might this be and what could you have done differently? Were contexts sufficiently novel, were there signs of anxiety, should it have been more collaborative requiring more shared attention, allowed for more choice? Where was support needed, e.g. were pupils having difficulties linking your verbal questions to your visual images? Did their responses suggest they had had sufficient practise on this topic using questions, applying knowledge in new situations, discussing, expressing in new forms?
Introduce new topic, discussion linking to prior knowledge	Did some pupils show lack of willingness to engage with the discussion – why might this be – and what could you do differently? Did pupils demonstrate that you had communicated the new concepts clearly? Did you make sufficient use of their senses, embodiment and action and encourage links across different representations of the concepts? Would the pupils have benefited from encouragement and support to connect the new concepts to their prior knowledge?
Pair work: investigating new topic …	Did the pupils share attention to the stimulus and tasks you provided them with?

SUMMARY AND KEY POINTS

- There is a lack of evidence for many supposedly brain-based ideas meant to improve learning, e.g. teaching to learning styles or through rehearsing co-ordination exercises.
- In contrast, a genuine understanding of learning from the sciences of mind and brain can provide insight into everyday classroom learning processes.
- Such understanding is helpful for developing a more secure rationale when planning your approaches to teaching in the classroom and, when evaluating, for critically reflecting on the effectiveness, or otherwise, of these approaches.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

Further resources

Howard-Jones, P.A. (2018) *Evolution of the Learning Brain: Or How You Got to Be so Smart*. Abingdon, UK: Routledge.
Written for non-specialists, this is an accessible journey through the most important brain processes for learning, explained through the engaging narrative of evolution.

Kim, M. and Sankey, D. (2022) *The Science of Learning and Development in Education: A Research-based Approach to Educational Practice*. Cambridge, UK: Cambridge University Press.
An expertly written guide for teachers who want to know whether and how neuroscience can be applied in the classroom.

Rogers, C. and Thomas, M.S.C. (2023) *Educational Neuroscience: The Basics*. Abingdon, UK: Routledge.
A very readable exploration of concepts from neuroscience that are relevant for teachers.

Science of Learning, viewed 16 July 2024, www.scienceoflearning-ebc.org
Contains videos explaining the concepts discussed in this unit created by Paul Howard-Jones, videos of mentoring sessions that focus on teachers reflecting on the scientific basis of their practice and some useful summaries of key concepts.

The International Mind, Brain and Education Society, viewed 16 July 2024, www.imbes.org
You can also find out here about the journal associated with this society.

ANNEX Unit 5.6 Evidence Basis for this Unit
The concepts set out in this unit derive from peer-reviewed literature review of the science and a two-way dialogue between scientists and lecturers on the PGCE (secondary education) team at the School of Education, University of Bristol, UK. Selection of concepts incorporated concerns around the priorities and needs of learners in the classroom, the need to provide explanatory value across curriculum areas and abilities, the provision of a foundation for further studies (anticipating future encounters with the science of learning mathematics and reading, and for understanding development and disorders), the need for parsimony, the avoidance of any sense that teaching approaches were being prescribed, and the anticipation and addressing of common misconceptions.

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5.7 Developing critical thinking

Chris Bolton

Introduction

Teaching, and learning to teach, can be considered as a journey without a particular fixed destination. It is one that involves continuous growth, exploration, and transformation, and the metaphor of a 'journey' is useful in thinking about how you might develop your own critical thinking and that of your pupils. In doing so, you will be developing as a professionally curious teacher with a view of critical thinking that can be empowering and creative. As you move through this unit you might consider different elements of a journey:

- **Navigating the terrain:** Like any journey, teaching has its ups and downs; a difficult class, curriculum changes, or personal setbacks. You will encounter diverse learners, varied backgrounds, and unexpected detours. Flexibility, adaptability, and resilience become your compass.
- **Learning along the way:** Effective teaching is not a one-way street. You will learn from your pupils – their questions, insights, and unique perspectives. Your understanding deepens as you engage in dialogue and exchange ideas.
- **Exploring pedagogical landscapes:** The teaching landscape is vast and transient. You will, no doubt, explore different teaching methods, technologies, and assessment strategies. You will seek ways to ignite curiosity and foster critical thinking.
- **Building bridges:** Teaching is about connecting – bridging gaps between knowledge and understanding, theory and practice, teacher and pupils. Hopefully, you will build bridges of empathy, trust, and collaboration.
- **Celebrating milestones and leaving footprints:** Along the journey, you should celebrate small victories – the 'aha' moments, the growth you witness, and the impact you make. These milestones will fuel your commitment. Your influence will extend beyond the classroom – shaping minds, inspiring futures, and contributing to a better world.

No doubt you will have had a unique and nuanced journey in getting to this point with a view to this continuing. As a beginning teacher you will bring a wealth of varied knowledge with you to your role. This knowledge can be thought of as professional knowledge, subject-specific knowledge, and pedagogical knowledge. Bringing these fields of knowledge to a school context will require you

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to reconsider and potentially recognise how your knowledge(s) can become enculturated into the school context, or not!

According to the World Economic Forum (2025), the top skills that will help humans thrive as they journey through the fourth industrial revolution (i.e., the age of machine learning and artificial intelligence) are critical thinking, curiosity, and creative thinking. Therefore, it is important that you are able to not only strengthen your own critical thinking, and the skills involved in this, but also the critical thinking skills of your pupils.

OBJECTIVES

At the end of this unit you should be able to:

- read the physical environment of your school (the architecture, signs, and symbols) with a critical eye;
- understand the learning spaces you construct from a critical perspective;
- approach the interactions and pedagogical encounters you have with your pupils from a critical perspective;
- develop critical thinking in your pupils.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Critical thinking

Critical thinking skills enable you to take a step back from any immediate situation that you might face and interpret it from a more objective position. This critical objectivity can help you to understand issues that might arise in relation to teaching, learning, or the introduction of, for example, a new curriculum specification document. Alongside this, critical thinking skills enable teachers to read text(s), situations, and contexts from an informed position.

Despite this, you will need to be wary of reducing critical thinking to a framework of skills for it is much more complex than that. Usefully, Ennis (1989, p. 4) defined critical thinking as 'reasonable reflective thinking focussed on deciding what to believe or do', with Bryan et al. (2010, p. 62) more recently describing critical thinking as 'an attitude, underpinned by curiosity'. This attitude can be motivated by looking for a deeper level of understanding beyond what might be visible. To do this, there are different approaches and models to support the development of critical thinking in relation to reflective practice. For example, Poulson and Wallace's (2004) six critical thinking considerations, Elder and Paul's (2012) *RED model* or Brookfield's (1998) *Reflective lenses*. For Brookfield, using reflective lenses enables teachers to inquire, discover, and research assumptions about teaching and learning, for example, to frame how they work. Here he suggested that

> Becoming aware of our assumptions is a puzzling and contradictory task … no matter how much we think we have an accurate sense of ourselves, we are stymied by the fact that we are using our own interpretive filters to become aware of our own interpretive filters.
>
> (1998, p. 200)

In other words, reflective practice can be like trying to see the back of your head in a mirror!

Whilst these and other established approaches are useful, such as Kolb (1984) and Gibbs (1988), the metaphorical approach to critical thinking offered in this unit – a journey – intends to support the development of your critical thinking in relation to your school culture, your own pedagogy, and your interactions within a teaching and learning environment.

Embarking on the journey

Before we leave on our journey, think about schools. Schools, in a literal sense, are buildings in which teaching and learning takes place. However, what else might these buildings mean and to whom? What signs and signals do school buildings, their design, architecture, and the framing of certain images and icons within them mean to a teacher journeying into the school building? What do these signs and symbols mean in relation to the quality of teaching and learning that takes place? Here you might adopt a hermeneutic approach (literally the study of interpretation and understanding) as part of your critical thinking development, to make sense of the physical building.

You might also view schools as sophisticated and complex sociological cultures containing their own rules, rituals, sign systems, and formalities in which education takes place, and thus the literal definition of a school given earlier becomes entwined with additional ideas around educational culture and the meaning of teaching and learning within that culture. Given the complexity and intersectionality of this, it is vital that beginning teachers can develop their own critical thinking perspective(s) to understand what education means in their current context. Developing an attitude of critical thinking, and the skills that might enable it, will go some way in preventing Halford's (1998, p. 33) characterisation of the teaching profession as one that 'eats its young' and give you the necessary resilience and flexibility to understand what education in a school might mean. This is important. Developing critical thinking will empower you to have agency in deciding how best to teach.

Not only does school culture affect pupils, it can also affect the thinking and practice of teachers who make schools function. Hall (1976, p. 9) usefully suggested that 'the natural act of thinking is greatly modified by culture' and it is from here that you might start considering how the cultural modifications of a school environment impact on teachers' thinking, their practice, and their interaction(s) within the teaching and learning environment. Using Hall's seminal work on culture you might begin to critically think about school culture in the following two ways:

1. Overt Culture – Visible and easily described: 'The school is challenging, tough, and the pupils are noisy!'
2. Covert Culture – Not visible and harder to describe: 'The pupils face a variety of socio-economic challenges made worse by a perceived lack of cultural capital'.

Thinking about a school's overt and covert culture helps beginning teachers understand its meaning. Considering what is visible and hidden allows you to develop critical thinking, strengthen curiosity, and increase awareness of the school environment. Essentially, it encourages objective and questioning perspectives and moves you towards a position of professional curiosity.

Let's imagine your journey to school on the first day of your teaching practice placement. Inevitably, as you arrive at the school, you will start to make interpretations about the school building, its appearance, the area in which it is located. Perhaps you have already made these

interpretations beforehand by reading recent school inspection reports. However, the architecture of the school building and its reception area will no doubt give you covert and overt signs and signals about the culture of that school. For example, a boys' grammar school built in the 1930s will have a different cultural provenance from a newly built academy commissioned in 2024. Thus, the overt and covert design of space will embody signs and potentially esoteric messages about what the school represents in relation to beliefs about learning, teaching, and the place of the school in its community. Now do Task 5.7.1.



Task 5.7.1 Consider your placement school from the outside

In this task, and the following tasks in this unit, record your notes and reconsider your observations when you know the school better. Then make notes about what you would like to see in the school in which you are working.

Reflection questions	*Critical thinking indicators*
What do you imagine the school building means to its pupils? What 'messages' does the building give?	Being open-minded to other perspectives (Poulson and Wallace, 2004).
In what way is this a school for its community? What do you interpret from this?	Challenging the importance of context (Scott, 2000).
How well does the design of the building work in relation to educating young people today? Imagine what could be better.	Imagining and exploring alternatives (Scott, 2000).

Navigating the terrain

Continuing your journey into the school building, you will inevitably notice visible signs and symbols. The school's reception area is a great example of where you will first notice these, particularly as you will need to sign-in via the reception area. Whilst these signs, symbols, or icons will be physical, such as a school crest or trophies, for example, the meaning of them and why they are there might not be so obvious at first. However, you will be making critical interpretations about the 'messages' that the signs give you. Here you begin navigating the world of semiotics (coined from the ancient Greek word for 'sign'). Semiotics, also known as semiology, was originally defined by Ferdinand de Saussure, a Swiss linguist, as the study of the life of signs within society (Bouissac, 2004). Essentially, objects, words, colours, symbols, and how they are arranged within a space can convey meaning and values. For example, a school with a highly polished sport-trophy cabinet might convey the message that this school values its sporting achievements, whereas a school that displays artwork from its pupils might convey a different message.

By learning to read the messages transmitted semiotically, you are developing a critical perspective on what the school is revealing about itself to visitors. In this sense, the reception area tells you how the school receives its visitors whilst at the same time how visitors receive messages about the school. As such, the reception area is a space where values about the school are expressed, and a critical reading of the space in this area provides you with a perspective about how the school wishes to be understood by visitors. Now do Task 5.7.2.



Task 5.7.2 Consider the reception area of your school

As suggested in Task 5.7.1, make some notes and reconsider your observations when you know the school better. You could discuss your impressions with colleagues.

Reflection questions	*Critical thinking indicators*
What artefacts, sign, or symbols are in the reception area? What is the relationship between these and the mission statement?	Being open-minded to other perspectives (Poulson and Wallace, 2004).
What are the badges of school success or sponsorship? What does this suggest to you in terms of the way the school values external validation?	Identify and challenge assumptions (Scott, 2000). Adopting an attitude of scepticism (Poulson and Wallace, 2004).
Is pupil work on display in reception? If yes, is it written work or only artwork? How long has the work been on display?	Adopting an attitude of scepticism (Poulson and Wallace, 2004).
What else would you expect to see in this school's reception area?	Being constructive by using your scepticism to find better ways or interpretations (Poulson and Wallace, 2004).

The learning space – learning along the way

Teachers teach in a variety of learning spaces: traditional classrooms, sports fields, or drama studios. As you journey from the reception area to your learning space, pause. On entry into your learning space, take a moment to consider it. If you are in a traditional classroom, start by looking at the layout of the classroom furniture. What do you notice about the layout of the desks and chairs? How does the physical environment reflect what you know about effective learning? What changes might be made to create different learning spaces for different purposes? Is there any flexibility for you to change the learning space? Similar questions might be asked for less traditional learning spaces such as drama studios. How might you manage the learning space given that there is likely to be a lack of furniture, for example?

Using a traditional behaviourist perspective (see Units 5.1 and 5.3), the teacher is placed at the centre of the learning process. Principally, the role of the teacher in a behaviourist-influenced classroom would be as director, instructor, and transmitter, with the role of her pupils being a passive recipient of a predetermined, firmly structured, linear curriculum. In a learning environment such as this, knowledge can be thought of as being under the control of the teacher or within selected texts, with pupils being rewarded for 'correct' answers or behaviours. In this context, the teacher can be viewed as a sage on the stage. Task 5.7.3 asks you to consider the learning space from a behaviourist perspective.

Conversely, using a constructivist approach (see Units 5.1 and 5.3) to view the learning space, places the pupil(s) at the centre of the learning process. The role of the teacher in a constructivist influenced learning space would be as a facilitator, scaffolder, and co-creator of learning. This approach could be developed into a more social constructivist learning space through the



Task 5.7.3 Consider the learning space from a behaviourist perspective

Discuss your responses with another beginning teacher, potentially one who teaches a different subject to you.

Reflection questions	*Critical thinking indicators*
Does the layout of the space and its furniture suggest a behaviourist approach to learning? If yes, why do you think the classroom is laid out in this way? What has been the guiding principle for the teacher whose layout this is?	Being open-minded to other perspectives (Poulson and Wallace, 2004). Imagining and exploring alternatives (Scott, 2000).
What does the teacher gain or lose by laying out the learning space in this way?	Identifying and challenging assumptions (Scott, 2000).
What do the pupils gain or lose in having the learning space organised in this way? What changes might you make?	Imagining and exploring alternatives (Scott, 2000).

introduction of group work and a heightened importance on the role of language in learning. Either way, the role of the pupil would be as an active participant of a roughly determined, loosely structured, spiral curriculum. In a social constructivist learning environment, knowledge can be thought of as being the responsibility of both pupils and teacher, as they democratically construct knowledge together through interaction(s) with text, language, and experience(s). Pupils are likely to be rewarded for curiosity, critical thinking, and asking insightful questions. In spaces such as these, pupils take part in a meaning-making process and become active participants in constructing their own understandings of the world. Of course, active meaning-making is a complex concept and faces particular challenges in an educational context that prioritises assessment of learning and tangible evidence. However, this can be countered by considering Moon's (2005, p. 106) position in that 'whether learning is meaningful can only be judged by the learner because meaningfulness is an expression of the relationship between the material of learning and the learner's existing understandings'. The role of assessment in meaning-making, and whether or not this justifies the approach taken, is considered in Units 6.1, 6.2, and 6.3.

However, as you may know from your reading (Unit 5.1, for instance), Vygotsky built upon Piaget's constructivist theory by strengthening the case for language within learning. Vygotsky's model of social constructivist learning emphasised the centrality of language in cognitive development, which is reinforced by considering that the use of language is a process of meaning-making in itself. Vygotsky (1978, p. 25) recognised this by stating that 'A child's speech is as important as the role of action in attaining a goal' with children being seen to 'not only speak about what they are doing' but that their 'speech and action are part of one and the same complex psychological function, directed toward the solution of the problem at hand'. In this sense, the teacher can be viewed as the guide on the side, using language to support pupils' learning. Task 5.7.4 asks you to consider the learning space from a social constructivist perspective.

Task 5.7.4 Consider the learning space from a social constructivist perspective

Discuss your responses with another beginning teacher, potentially one who teaches a different subject to you.

Reflection questions	*Critical thinking indicators*
Does the layout of the space and its furniture suggest a social constructivist approach to learning? If yes, why do you think the classroom is laid out in this way? What has been the guiding principle for the teacher whose layout this is?	Being open-minded to other perspectives (Poulson and Wallace, 2004). Imagining and exploring alternatives (Scott, 2000).
How might the space promote an active meaning-making role for the pupils?	Identifying and challenging assumptions (Scott, 2000).
How does the space promote a scaffolding role for the teacher?	Imagining and exploring alternatives (Scott, 2000).

Other modes of learning exist, such as imitation and learning from observation (Bandura, 1969) and 'living through' approaches (Bolton, 1998). Organising spaces around a belief in experiential learning can also be seen as a starting point for learning (Miller and Boud, 1996; Kolb, 1984). Task 5.7.5 invites you to critically evaluate learning spaces from an experiential learning perspective.

Task 5.7.5 Consider the learning space from an experiential learning perspective

Discuss your responses with another beginning teacher.

Reflection questions	*Critical thinking indicators*
How might first-hand experiences occur in this learning space?	Developing reflective scepticism (Scott, 2000).
How are pupils supported to reflect on their experiences in this learning space?	Being open-minded (Poulson and Wallace, 2004).
How might pupils reflect upon their learning in this space?	Imagining and exploring alternatives (Scott, 2000).
What do you imagine this learning space means to/for the pupils/teacher?	Imagining and exploring alternatives (Scott, 2000).

Interactions and building bridges

Whilst physical structures are important, learning conditions are primarily driven by teachers and their actions, and are enhanced by student interactions with each other, texts, and adults.

Behaviourist spaces emphasise teacher-pupil and text interaction. Constructivist spaces encourage interaction between pupils, teachers, and resources. Social constructivist spaces add interactions with a 'more knowledgeable other' (Vygotsky, 1978). Similarly, experiential learning also involves interaction with firsthand resources, including adults. Understanding these interactions connects them to learning theories.

Interactions within these diverse learning spaces, however, can be understood differently. The pedagogy of the teacher acting within those spaces creates different contexts and interactions. Pedagogy is the term that describes relationships and 'interactions between teachers, students and the learning environment and the learning tasks' (Murphy, 2008, p. 35), with others defining the term as the 'artistry' of teaching (Moon, 2005), collaborative practices (Pickering et al., 2007), and the 'science of teaching' (Bryan et al., 2010). Pedagogy might be viewed metaphorically as the teacher's involvement in building a bridge between the learning that takes place in a learning space and the reality of her pupil's lives. However you view pedagogy, it is complex. It is all this and more. Pedagogy goes beyond the practices undertaken by teachers to bring about specific learning outcomes, and you will realise that pedagogical practices can potentially oppress or liberate the learner (Freire, 1992). In this way, pedagogy can be understood to be political (see Giroux's [2011] notion of 'Critical pedagogy'). Task 5.7.6 invites you to critically evaluate interactions within your placement school.



Task 5.7.6 Consider the interactions

Discuss your responses with another beginning teacher.

Reflection questions	*Critical thinking indicators*
In what way do you construct your role in relation to your pupils and talk about your role to other beginning teachers in relation to pupils?	Scrutinising claims to see how convincing they are (Poulson and Wallace, 2004).
What are your expectations of pupil relationships?	Scrutinising claims to see how convincing they are (Poulson and Wallace, 2004).
Do the learning opportunities you create draw from a range of learning theories? Why? Why not?	Imagining and exploring alternatives (Scott, 2000).
To what extent do you base your pedagogical interactions on those you experienced as a pupil yourself?	Habitually questioning the quality of others' specific claims (Poulson and Wallace, 2004).
How might you interact with pupils differently? What might you gain or lose from this?	Being constructive by using your scepticism to find better ways or interpretations (Poulson and Wallace, 2004).

Engaging in these reflective questions so far will go some way in developing an inquisitive, consciously questioning state of mind in relation to interactions in the school culture. By being professionally curious, and strengthening your critical attitude, you will be in a more objective position. By using the 'journey' metaphor to consider the steps into school you have moved from considering the overt, literal school building, architecture, and signs, towards the more covert and esoteric school culture built on pedagogy. The activities you have undertaken using the idea of a journey

lead you towards a 'critical stance … [which] has the connotation of sharpness and precision' (Moon, 2000, p. 60). That is, a disposition that enables you to analyse, question, interrogate to engage in different 'ways of seeing' (Berger, 1972). The section that follows supports you in developing a critical disposition in your pupils in school.

Celebrating milestones and leaving footprints: creating a critical disposition in your pupils

Influentially, Carol Dweck (2000) explored the ways in which teachers understand children's capacity to learn and develop. Rather than teachers viewing their pupils' ability as fixed, she proposed that children are capable of change, which she referred to as taking an 'incremental' perspective in relation to learning. Dweck and Molden (2017, p. 263) further developed and built on the idea of 'growth mindset' when they proposed that a growth mindset leads to a desire to learn and therefore there is 'a tendency to embrace challenges, persist in the fact of setbacks, see effort as the path to mastery, learn from criticism, find lessons and inspiration in the success of others'. As an expert in the field, Dweck's ideas situate the importance of teachers viewing their pupils' learning potential as organic and capable of growth. This positions the teacher at the heart of the pupils' learning processes. Thus, the capacity for pupils to think critically is an invaluable disposition that can be developed through this view. The development of critical thinking in your pupils requires you to focus first upon the word 'thinking' and apply your pedagogical approach appropriately.

To develop critical thinking in students, consider these 10 steps: (1) Encourage inquiry (value questions); (2) Teach critical thinking skills (debates, problem-solving); (3) Model critical thinking (think aloud); (4) Use real-world problems; (5) Reflective practice (journals, discussions); (6) Feedback loop (constructive challenges); (7) Diverse perspectives; (8) Socratic method; (9) Collaborative learning; and (10) Media literacy.

Whilst these 10 individual steps can function alone, they can also be blended to deepen the critical thinking of your pupils. The suggestions in Table 5.7.1 provide you with approaches to the

Table 5.7.1 Some critical thinking approaches

Critical inquiry step	*Pupil activity*	*Implications for your practice*
Encourage Inquiry and Collaborative Learning: *Pupils are provided with a subject-specific text*	Pupils work in groups to read a text and generate and note down as many questions about it as they can.	You will need to find appropriately challenging texts within your subject. You might provide a questioning matrix to frame your pupils' question generation. You also need to be prepared to not know the answers to their questions.
Teach critical thinking skills and Diverse Perspectives: *Pupils are given a topic to debate such as deforestation*	Pupils are split into small groups and asked to consider particular sides of the argument that represent the various stakeholders in the deforestation dilemma.	You will need to outline the rules and expectations of debate, such as listening and respecting others' points of view. You will need to consider how the debate will be managed, when and how people speak, and the aim of the debate.

(Continued)

Table 5.7.1 (Continued)

Critical inquiry step	*Pupil activity*	*Implications for your practice*
Model Critical Thinking: *Pupils should be able to hear their teacher talking-through potential problems and their solutions*	Pupils are asked to listen to their teacher as they talk through the answer to a mathematical or scientific equation.	You will need to ensure that your vocabulary and thinking process is clear, concise, and easy to follow. You will need to allow thinking time for your pupils to ask questions about your explanation.
Use Real-World Problems and Media Literacy: *Provide pupils with extracts from current newspaper of issues that are controversial (e.g. global warming; politics). Provide pupils with the same story but presented in a range of newspapers*	Pupils should work in small group to discuss the issues and list the further questions they have. Pupils might suggest possible solutions and consider the consequences of each solution.	You need to find current events that are well written in newspapers. Pupils need plenty of time to engage in this activity. As above, but you need a range of newspapers offering the same
Reflective Practice: *Invite pupils to consider a personal critical incident at a time when they were faced with a dilemma that they are happy to discuss and explore*	Pupils work in small groups to first describe the dilemma. Pupils then move on to explain how the actions they took/ the dilemma made them feel. Finally, other pupils ask questions to encourage the pupil to explore other ways in which they might have acted.	You will need to set up this activity in a sensitive manner, and only involve pupils who wish to discuss a critical incident. You should ensure that the 'rules' are followed in terms of the supportive nature of this activity.
Develop a Feedback Loop: *Provide pupils with spoken and/or written formative feedback on their effort*	Pupils might be asked to prepare for a debate in small groups. As part of this they need to listen, analyse and apply their thinking as a group to formulate their argument	You will need to find appropriate times to feedback on the effort and progress of the group work. Comments such as 'I really like the way you are listening to each other' or 'The way that you are challenging each other's thinking is impressive' can be useful feedback to develop a growth mindset.
Socratic Method: Invite the pupils to create a presentation on a particular topic	Pupils work together to create a presentation on a topic that may include some of the work from the other steps listed in this table.	You will need to model an attitude of professional curiosity. Questions such as the following would strengthen this approach: ■ Why do you think that? ■ What has led you to think the way you are? ■ What if you are wrong? ■ Why have you not considered an alternative? Is there one? ■ How have you come to this decision about your presentation? ■ Why have you included this content and not other elements?

development of critical thinking in learning based on the 10 steps above. They offer the opportunity for pupils to work collectively in the development of critical reading, the development of a critical perspective and the start of critical writing.

Whilst this list is not exhaustive it goes some way in supporting you in starting to develop critical thinking skills in your learners. By integrating these approaches into your pedagogy, you can help your pupils to develop a critical disposition that will serve them well throughout their education at whatever level.

Exploring pedagogical landscapes

Artificial intelligence (AI) has clearly marked the rise of the fourth industrial revolution (i.e., the age of machine learning). Given the increasing influence of AI, what does the concept of critical thinking mean for you and your pupils in the future? How might this form of intelligence change, influence, or impact pedagogy and learning? Whilst beliefs about growth mindsets are founded on notions of dedication, hard work and effort, what might this mean as AI technologies evolve to make our lives easier?

AI is becoming increasingly integrated into secondary education in the United Kingdom (UK), with a focus on enhancing teaching and learning experiences. Evidence of this can be seen from the Department for Education's policy paper on Generative AI (DfE, 2023) and the potential benefits and limitations to this form of intelligence. The growing influence of AI in secondary education aims to support teachers, enhance learning outcomes, and prepare pupils for a future where AI is an integral part of society. Developing critical thinking and the skills necessary for this are increasingly important.

Perhaps here, the critical thinking developed in this unit might now be expanded to include new(er) ways of thinking. You might consider the following four key points on growth mindsets in the context of AI proposed by Tero Ojanperä, a Professor of Practise on Intelligent Platforms (2023) and add them to your own critical thinking skills:

1. **Nurturing AI:** Just like nurturing a pupil's learning, development, and critical thinking, AI systems, such as ChatGPT, can be developed effectively and safely when they are treated with a growth mindset. Using them as a tool to promote objective thinking is necessary.
2. **Continuous Improvement:** The journey towards developing effective AI in schools involves a commitment to continuous improvement, a readiness to confront challenges, and dedication to the responsible use of AI technology.
3. **Learning Culture:** Developing a growth mindset in the AI era means cultivating a learning culture within schools and learning spaces that embraces change and is able to adapt to new circumstances.
4. **Professional Development:** In an AI world, a growth mindset encourages teachers to view challenges as an opportunity for growth by embracing the learning curve that comes with new technologies. Are you prepared for this?

These principles highlight the importance of a growth mindset and critical thinking in adapting to and thriving in an environment increasingly influenced by AI. It is about leveraging AI as a tool for enhancement rather than seeing it as a replacement. As you journey into the future, this approach fosters a symbiotic relationship between human and machine intelligence.

SUMMARY AND KEY POINTS

This unit has introduced you to the concept of critical thinking as an empowering, creative disposition.

- You have been offered the metaphor of a journey to develop your skills of interrogation and analysis.
- You have begun to understand the benefits of stepping back from any immediate situation, asking questions, and looking for details. In developing a critical mindset in relation to education, you are enhancing your ability to engage in a 'critical confrontation with your problems' (Smyth, 1989).
- Developing such a critical disposition in relation to education is in line with Freire's notion that emancipation results from critical and evaluative engagement (Freire, 1992).
- You may have begun to notice throughout this unit that critical thinking and reflective practice are closely linked (Smyth, 1989). In essence, reflective thinking seeks to understand phenomena (Brookfield, 1998), whereas critical thinking is 'an attitude underpinned by curiosity' (Bryan et al., 2010, p. 62).
- This unit has also introduced you to the possibility of developing a critical attitude in the minds of your pupils and this is demonstrated in part in the learning space activities suggested earlier, where pupils are encouraged to arrive at solution, rather than simply debate issues or be passive recipients of knowledge.

Check which requirements for your ITE programme you have addressed through this unit.



ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.





Further resources

Bryan, H., Carpenter, C. and Hoult, S. (2010) ***Learning and Teaching at M Level: A Guide for Student Teachers*****, London, UK: Sage.**
This text is designed for all beginning teachers. It promotes the importance of enquiry in professional practice and, as such, makes explicit links between theory and practice.

Fitzpatrick, D., Fox, A. and Weinstein, B. (2023) ***The AI Classroom: The Ultimate Guide to Artificial Intelligence in Education: 1 (The Everything Edtech Series)*****, Teachergoals Publishing.**
This thought-provoking book provides practical strategies for incorporating AI tools into your teaching practices, whilst exploring the potential of AI to transform traditional models of teaching and learning.

Grigg, R. and Lewis, H. (2019) ***Teaching Creative and Critical Thinking in Schools*** **(1st ed.), London, UK: SAGE Publications Ltd.**
Research-based and highly practical, this book provides guidance on how to develop creative and critical thinking through your classroom.

Paul, R. and Elder, L. (2019) ***The Miniature Guide to Critical Thinking Concepts and Tools*** **(8th ed.), London, UK: Rowman and Littlefield.**
This small but powerful book contains core critical thinking concepts and principles distilled into a compact size.

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5.8 Creating a language-rich classroom

Annabel Watson, Ruth Newman and Debra Myhill

Introduction: understanding the role of language in learning

Language is fundamental to learning and fundamental to being human. Language distinguishes us from animals: although animals can communicate, only humans can use language to reflect on the past, to communicate complex abstract ideas or emotions and to shape or imagine new futures. Crucially, language enables us to learn from others, over time and space. Language shapes how we see and interpret the world: as Wittgenstein famously proposed: 'The limits of my language are the limits of my world. All I know is what I have words for' (Wittgenstein, 1961, pp. 5–6).

Wittgenstein emphasises the integral relationship between language and thought. What we can articulate is influenced by the language we have to express it, and language itself influences how we think. Think of the common metaphor of the sun setting over the horizon, creating a conceptual image of the sun moving rather than the correct scientific understanding that it is the earth rotating that causes the sun to disappear. When new learning or discoveries occur, we have to create new words to express them. Language is the most powerful tool we have as learning beings.

The relationship between thinking and language has been explored most comprehensively by Vygotsky, who argued that 'children solve practical problems with the help of their speech, as well as with their eyes and hands' (Vygotsky, 1978, p. 26). He explored the symbiotic relationship between language and thought: thoughts are not mentally 'translated' into words; rather thought and language interact together to generate new knowledge and understanding. It is a dynamic, constructive process:

> The relation of thought to word is not a thing but a process, a continual movement back and forth from thought to word and from word to thought. In that process the relation of thought to word undergoes changes which themselves may be regarded as development in the functional sense. Thought is not merely expressed in words; it comes into existence through them. (Vygotsky, 1986, p. 218)

Although this might seem an abstract philosophical idea, it is one we all experience many times in our own social interactions with others. Most of us have had the experience of beginning a sentence and being surprised at where we end up, or of trying to articulate a half-formed idea and getting

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tangled up in our own words as we struggle to bring thoughts and words together. So, opportunities, both spoken and written, to discuss our ideas, to explain our thinking and to share our viewpoints are central to the process of learning. Indeed, as Halliday claimed: 'When children learn language … they are learning the foundations of learning itself' (Halliday, 1993, p. 93).

In the classroom, then, it is important that we see opportunities to use language to talk, read and foster disciplinary literacy (Education Endowment Foundation (EEF), 2021). This unit sets out to encourage you to think how you can capture the affordances of language to make learning in your classroom a meaningful, dynamic and vibrant process.

OBJECTIVES

At the end of this unit you should be able to:

- begin to establish participatory and interactive classrooms;
- begin to set up effective exploratory, dialogic talk;
- begin to create tasks and activities that generate active reading;
- support the development of writing in your subject.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

A talking classroom

Talk is a powerful tool for learning (Booth, 2020). It allows us to formulate and express ideas, to reformulate or clarify thinking, to communicate ideas and receive feedback from others and to reflect. Language, therefore, is not only an *intra*personal 'psychological tool' that allows us to crystallise and represent thought to ourselves, but also an *inter*personal 'cultural tool' that provides 'a means for people to think and learn together' (Mercer, 1995, p. 4). It is crucial, therefore, that teachers foster effective and purposeful classroom talk which helps pupils to formulate, test and evaluate their own ideas.

Despite the clear benefits of classroom talk which supports pupils to verbalise and explore ideas, research over the past 40 years indicates that teacher-led talk is dominated by recitation (Michaels and O'Connor, 2015). Recitation is an interactional pattern often referred to as the IRE or IRF – Initiation, Response, Evaluation/Feedback. Typically, the teacher initiates discussion with a question, the pupil responds and the teacher offers some form of evaluation or feedback. Dominant use of this interactional pattern, particularly the delivery of the third evaluation/feedback 'turn', can limit pupils' involvement in the dialogue and halt their thinking. It has also been observed (Alexander, 2018) that teachers' questions mostly invite factual recall, that pupil responses are often brief, and that teachers tend to echo or repeat pupil answers, reinforcing the implicit message that the teacher is controller of dialogue and arbiter of knowledge.

Creating more space for learning, where pupils are given the opportunity to develop their understanding or ideas and articulate their thoughts, requires careful planning of whole-class discussion. One approach is to utilise a range of 'talk moves' (Michaels and O'Connor, 2015) that disrupt the IRE/F pattern described above and that encourage pupils to elaborate on and extend their thinking.

Particularly important is thinking about the way the 'third turn' is managed: instead of closing the interactional exchange, taking up pupils' responses and asking follow-up questions – for example, *can you say more? Why do you think that?* – can open 'cognitive doors' (Alexander, 2020, p. 126). Sequences of questions should invite pupils to use higher-order thinking skills of analysis, synthesis and evaluation, and invite pupils to elaborate on their ideas and explain their thinking. Pupils should also be prompted to ask their own questions about what they are learning. Using 'wait time' allows pupils space to think before they answer, whilst talking with a partner helps pupils to explore and clarify ideas before sharing with the class. Most importantly, pay full attention to pupils' responses, exploring their thinking and using mistakes as opportunities to collaboratively explore misunderstandings or miscommunication.

Task 5.8.1 is designed to help you with this.



Task 5.8.1 Observing and analysing whole-class discussion

Observe a five-ten-minute episode of whole-class discussion.

Make a note of:

- How many conversational 'turns' the teacher takes.
- How many turns pupils take.

Reflect and evaluate:

- How far does the pattern of interaction fit the 'IRF' pattern described above? What happens in the 'third turn', when the teacher responds to the pupil?
- What does the teacher do to try to encourage pupil participation in the discussion?
- How successful are they in doing this?

Discuss your reflections with a peer and/or your mentor. Now pair up with a colleague and undertake the same exercise in one of the lessons you are teaching. Were the results what you had planned for? If not, why not? Record the outcomes of your reflections and discussions in your professional portfolio.

Dialogic talk for learning

A 'dialogic' pedagogy is an approach to teaching which values and utilises classroom talk. In dialogic classrooms, pupils are offered wide-ranging opportunities to listen to and engage in different kinds of talk, disrupting the routine pattern of IRF exchanges and aiming to be more consistently searching and more genuinely reciprocal and cumulative (Alexander, 2020). It is widely believed that dialogic teaching makes a difference to student outcomes and engagement. Previous research has shown that where teachers use dialogic talk effectively, pupils' outcomes in Maths, English and Science are improved (Jay et al., 2017; Alexander, 2018). Curriculum mastery has also been associated specifically with extensive participation in discussion which involves elaborating on and querying contributions (Howe et al., 2019; Hennessy et al., 2023).

Alexander's (2020) evolved dialogic teaching framework includes six *principles* intended to guide the planning of classroom talk and serve as criteria for its 'dialogic' quality. According to these principles, dialogic talk is *collective, supportive, reciprocal, deliberative, cumulative, purposeful*

(Alexander, 2020). Teachers and pupils *collectively* address learning tasks together, in groups or as a whole class. They are *supportive* in allowing all members of the group to freely express their ideas without fear of embarrassment. Teachers and pupils *reciprocate* by sharing ideas, listening to each other and considering alternative views. The discussion is *deliberative* in exploring and resolving different viewpoints, and *cumulative* in that teachers and pupils respond to each other's ideas, building them into a coherent line of enquiry. The talk is also *purposeful* in that it is planned and steered by the teacher in order to reach a specific educational goal.

Another framework element captures the different *repertoires* that are at the heart of dialogic teaching, illustrating the potential breadth of talk for learning in a dialogic classroom. These include different organisational settings, for example, whole-class discussion, group or pair work, within which different types of talk may occur, for instance, exploratory, expressive and evaluative talk. A dialogic talk repertoire also acknowledges the need for teachers to command and model talk, suggesting that more 'authoritative input' is also an important part of classroom talk which supports learning.

To enable your pupils to succeed in talk, you need to explicitly teach a range of skills, including how to listen carefully, respect, manage and monitor contributions, and how to include others. This might involve teaching ground rules for group work or inviting pupils to reflect on the qualities of their talk (Newman, 2018; Sutherland, 2015). Interpersonal and communicative skills are essential for effective group work in all areas of the curriculum, and you should make opportunities to model good (and bad!) practice.

Now complete Task 5.8.2.



Task 5.8.2 Analysing pupil discussions

Read this extract (taken from Myhill and Newman, 2023, p. 145) in which four Year 9 (ages 13–14) pupils – Krissy, May, Miriam and Rose (pseudonyms) – are working on a task to design an iPad app. The transcript below shows pupils part way through the task when they are developing their ideas.

- What features of dialogic talk are evident?
- What would be a suitable topic of enquiry within your own subject area?
- How would you prepare your pupils to take part in an activity like this?

Miriam: So out of two options that we've really come up with are the cinematography and … but which one do people prefer?
Rose: I think the cinematography is quite a unique thing.
May: The (other) would be quite fun but it's more of a novelty … once you've played it a few times, it'd be, like, yeah.
Rose: Whereas the other one you could use for different things.
Miriam: I think that one would appeal to more ages.
May: Yeah.
Miriam: People would take it seriously, even if you're an adult doing photography, or children.

Rose: Yeah.
Miriam: But you'd have to make sure it's really simple.
Rose: I was thinking: how are we going to do it and how are we going to explain it?
May: Well, we can show it and explain …
Rose: You could include a little step-by-step guide.
Miriam: Krissy, maybe you should Google it when you get home so you know exactly what we're talking about.
[Miriam realises that Krissy has not contributed. The group get the teacher's laptop to research and explain the concept to Krissy]
Krissy: I get what you mean now … it's like still pictures but with one thing moving?
Miriam: Yeah … 'cause these are obviously more adult ones but we could change it … we could do different ones so that …
Krissy: Yeah, for like, children just to mess around with.

Record your ideas in your portfolio.

Reading the world

Active reading involves interpreting, questioning and meaning-making, rather than simply decoding words on a page or screen (Stuart et al., 2008). Successful reading underpins almost all facets of the curriculum – consider the comprehension demands of mathematical 'word problems', for example (Vilenius-Tuohimaa et al., 2008). To support reading development, you need to understand a little about the nature of reading. Constructing meaning from texts involves an interaction between the reader, the text being read, and context. As we read, we bring all our previous reading experiences, our own life experiences and our cultural knowledge and values: there are 'as many different possible readings of text as there are readers' (Stevens and Fleming, 1998, p. 64). What we already know influences how we interpret the text, so our ability to draw on personal experiences can be important: reading an explanation of volcanic craters, for example, could be influenced by a holiday trip to Vesuvius. We also read very much from our own cultural perspective. A description of a loyal dog has many cultural resonances in England, with our view of a dog as 'man's best friend', but many readers from other countries would have very different views of dogs, perhaps as scavenging pests to be deterred. Readers who have English as an additional language (EAL) or come from different cultural backgrounds may draw different meanings from texts.

Developing comprehension

When we comprehend a text, we 'interpret, integrate, critique, infer, analyse, connect and evaluate ideas (New South Wales (NSW) Education Department, 2010, p. 2). This process of constructing meaning is both reader-led and text-led: problems with comprehension are always the result of a mismatch between the knowledge and skills of the reader and the level of difficulty of a text. All of us can struggle with comprehension at times; common causes of difficulty include layout issues (e.g. font size), grammatical complexity, unfamiliar vocabulary, unfamiliar concepts or the text not

being in our first language (e.g. EAL pupils). Consequently, to ensure that pupils can access and understand the texts you use, you need to first determine what makes them difficult. You may need to break a text down into smaller chunks, or to help students to make sense of complex grammatical constructions.

Background knowledge is a crucial component in reading difficulty. The more a pupil knows about a topic, the easier it will be for them to understand a text about it. There is evidence that pupils can compensate for weaker reading skills if they have good background knowledge (Smith et al., 2021). Knowledge of vocabulary also correlates strongly with reading ability (Cain and Oakhill, 2014; Suggate et al., 2018). This relationship is reciprocal – having a good knowledge of vocabulary supports comprehension, and reading widely supports vocabulary acquisition. At present, evidence to support 'vocabulary interventions' that focus on teaching knowledge of words is limited. However, vocabulary interventions which also target cognitive and metacognitive reading strategies show more promise (Wright and Cervetti, 2016). You could devise some *pre-reading* activities to explore key concepts, explain key vocabulary or discuss culturally specific content before reading.

Reading motivation and interest also have a relationship to comprehension: it's important not to underestimate the impact that being interested and motivated to read can have on understanding of texts (Anmarkrud and Bråten, 2009). Consider ways to enthuse pupils about reading they're about to do. Contributing to whole school initiatives designed to promote reading for pleasure can also create longer term, cross-curricular rewards, as enjoyment and frequency of reading reinforces reading ability, which impacts on attainment across the curriculum (Cremin, 2014).

Confident readers make use of a range of strategies to support comprehension, and these strategies can be taught (Snow, 2002). Strong readers know instinctively that we don't read every text in the same way. Compare how you read a novel to a website, for example (Baron, 2021). The strategies that we use to make meaning from texts are both cognitive and metacognitive, that is, they involve both thinking (without a conscious decision to employ a strategy) and thinking *about* thinking (where we consciously and deliberately decide to use a strategy). If you flick back in a text to check a piece of information, or skim over unimportant information then you are using a cognitive strategy. If you deliberately decide to highlight key words in a text, or choose to summarise information in a table to help you to understand it, then you are using a metacognitive strategy. Teaching comprehension strategies is essentially about enabling pupils to develop better metacognitive control of their reading – it involves making explicit what good readers do instinctively so that all students can consciously choose to make use of different strategies to improve their comprehension. Over time, some of these may then become automatic.

One important strategy is comprehension monitoring (Yuill and Oakhill, 1991). This metacognitive strategy involves checking what you do and don't understand in a text, enabling you then to find ways to address what you don't understand (e.g. by referring back in a text, or looking for further information). It is an important skill for all readers, including those with Special Educational Needs (Berkeley et al., 2010). As well as comprehension monitoring, other strategies which have been shown to improve reading ability when they are taught explicitly (NSW, 2010) include:

- making connections: comparing ideas within a text, across other texts you've read and with life experiences you have had;
- predicting: anticipating what will happen next;
- visualising: creating mental representations of people, places, activities;

- questioning: asking questions of the text;
- inferring: reading between the lines;
- summarising: putting content into your own words.

Note that these are not *teaching* strategies; but are rather *reading* strategies that pupils need to be taught. You can refer to this list when devising a lesson which involves reading and consider what activities would be relevant to both the text and the development of pupils' reading strategies.

DART – Directed Activities Related to Text (see also Unit 5.2)

One approach to developing reading strategies is to use a DART activity. These were originally developed by Lunzer and Gardner (1979) and are particularly relevant to non-fiction texts. They can be used to support the development of reading strategies (e.g. doing a prediction activity, if you highlight to students what they are doing, will develop pupils' understanding of how to use prediction), although they are themselves *activities* rather than general *strategies*. They also often align well with opportunities for exploratory talk, if done in small groups or pairs. Choose a DART that naturally accompanies the kind of text you are using and the particular knowledge or understanding you are hoping to develop and clarify what the learning point of any given activity is. Through a successful DART activity:

> the processes of comprehension, of gaining meaning and drawing inferences from text, are brought out into the open, and it is from this that one reader can learn from another how to become a better or more thoughtful creator of meaning.
>
> (Harrison, 2004, p. 100)

The following list groups the DART activities into clusters and explains what reading skills each might develop. DART activities include the following:

Text marking: underlining, annotating or numbering the text to show sequence. Skills include: skimming or scanning to find specific information; deciding what is relevant; finding main ideas.

Graphic reorganisation: reading and remodelling the information in another format, such as flow charts, diagrams, grids, lists, maps, charts or concept maps. Skills include: identifying what is key and relevant; applying what is known in a new context; summary and prioritisation; synthesising.

Sequencing: reconstructing a text that has been cut into chunks. Skills include: analysing structure of the text; understanding link words; analysing the logic or organising principle of the text.

Cloze: gap-filling tasks that involve the reader in actively constructing meaning. Skills include: close monitoring of comprehension; applying grammatical knowledge; applying existing knowledge of the topic; inferring and deducing; re-reading.

Prediction: informed speculation about what happens next. Skills include: inference; drawing on reader knowledge of other similar texts; justifying decisions with textual evidence.

Text reformulation: re-presenting the information or ideas in a text in a different format. Skills include: selecting key information; deciding what is relevant; applying knowledge of genre conventions.

(Adapted from Department for Education and Skills (DfES), 2002, p. 31)

Now complete Task 5.8.3.



Task 5.8.3 Incorporating comprehension strategies into lessons

Take a text for your subject that you might use in a lesson and create a lesson activity that explicitly teaches one of the comprehension strategies outlined previously, perhaps using a DART activity.

Write a short paragraph explaining the rationale for the task you have developed.

Writing like an expert

In teaching your pupils to write 'like an expert', the aim is to enable them to communicate with authority in your subject – both to support their confidence in tackling written exams and assignments, and to raise the quality of their responses. This is inherent in the concept of 'disciplinary literacy' advanced by the EEF (2021). Pupils need to learn to think and to write in disciplinary ways that follow the conventions of their particular subject. To achieve this, you will need to spend some time analysing the conventions typical of writing within your subject, examining the key vocabulary and patterns of language which students need to learn, and considering how to interweave the teaching of writing within your subject content.

One important aspect of written disciplinary expertise is being able to understand and use both the content-specific vocabulary of the discipline and the academic vocabulary required by disciplinary genres (Durrant and Brenchley, 2019). Precise terminology is crucial for conveying expertise, and will often allow pupils to present their points more concisely. Consider the difference between 'the heart would go faster' and 'the heart rate increases' or between 'people who have moved to work in another place' and 'economic migrants'. As well as simply displaying vocabulary, you could teach pupils the etymology of key words in order to support their understanding of related words (e.g. photo-synthesis: light + putting together) or create word-webs of related words to underline relationships (e.g. geography; geophysical; geothermal). When starting to plan for a new topic, you might consider drawing up two vocabulary lists: *entitlement vocabulary*, the words you want all pupils to know, use and spell correctly; and *extension vocabulary*, the words you want your highest attaining pupils to know.

Appropriate use of vocabulary is, of course, only one part of understanding the expectations of the disciplinary genres of your subject. Genres in writing are 'how things get done, when language is used to accomplish them' (Martin, 1985, p. 250); they are a collectively and implicitly agreed set of conventions for writing and 'preferred ways of creating and communicating knowledge within particular communities' (Swales, 1990, p. 4). Ask yourself what kinds of writing you ask pupils to complete: it might be evaluations, reports, explanations, argument essays, instructions. Don't get bogged down with naming text types: there are many different ways of categorising genres (e.g. Christie and Martin, 2005) but it is more important that you can identify the features and conventions that pupils need to produce and explain how they work. One way to approach this is to consider the different command words used to frame writing tasks. These might be words like

'describe', 'explain', 'analyse', 'evaluate' or 'explore'. To some extent these words have standard definitions, but there can be variations in what is expected according to subject. An example of this is presented below, in an extract from a sample answer in a GCSE Geography examination mark scheme (Assessment and Qualifications Alliance (AQA), 2017, p. 22). The full answer responds to the question more comprehensively, including both advantages and disadvantages and coming to a conclusion.

Question: 'Transnational corporations (TNCs) only bring advantages to the host country.' Do you agree with this statement? Justify your decision.

Answer (extract): An example of a transnational corporation is Shell, who operate doing oil extraction in Nigeria, particularly in the Niger Delta where there are vast oil reserves. Shell have brought 65,000 permanent to Nigeria and 250,000 jobs in related industries. This has triggered the multiplier effect, because all of these workers will have money to spend on services and in shops for example. This cycle of monetary increase is known as the multiplier effect and increased GNI in a country effectively. Similarly 91% of Shell's oil contracts went to Nigerian companies which has also contributed to reducing the development gap.

The command word here is 'justify', which requires pupils to explain, with reasons and examples from a case study. We can look more closely to consider which linguistic features make this answer sound as though it were written by an expert. There is disciplinary terminology used to convey precise information in a concise way and demonstrate knowledge, including 'multiplier effect' and 'GNI'. There is also more generic sophisticated vocabulary such as 'operate' and 'triggered'. Cause and effect is signalled by use of 'This' to refer back to previous sentences, as well as the conjunction 'because', and the answer is expanded by comparators 'similarly' and 'also'. There are noun phrases in which a single noun is expanded with extra information, to provide precision and detail: 'vast oil reserves', 'This cycle of monetary increase', 'Shell's oil contracts'. Many pupils answering examination questions, such as this, are not confident in recognising the language expectations of a test question, and how *what* they know is strengthened by *how* they write about it.

These structures and patterns of language can be taught and help to demystify for writers what they are being asked to do in writing tasks. It is helpful for you as a teacher to model writing for the pupils. Modelling is a scaffolding strategy (Wood et al., 1976) teachers use that gives additional assistance in the early stages of learning, and that is later withdrawn. Whittaker et al. (2006) describe the modelling of written genres as a three-step process of deconstruction – examining a text – joint construction – writing an example together – and independent construction.

The benefit of modelling how a genre is constructed is that it makes visible things that writers might not have noticed otherwise and allows them to have more choice and control in designing their writing (Myhill, Watson and Newman, 2020). But one danger is teaching writers formulaic approaches to writing, rather like offering them a recipe that they must dutifully follow. Encourage pupils to appreciate that there are often a range of ways to fulfil the demands of the genre, and that sometimes being unconventional can be very successful.

Now complete Task 5.8.4.



Task 5.8.4 Deconstructing text

Take an example of a high-level extended writing from your subject and annotate with the genre and linguistic features you would want pupils to understand and be able to use. Use the following questions as prompts:

- How is this text structured and set out?
- Do visual, tabular or graphic information form part of this text?
- Is it written in the first person or the third person?
- Is it formal or informal?
- Does it use the active or passive voice?
- Do the sentences tend to be short or long, or have any typical characteristics?
- Are there particular conjunctions or adverbs used to link ideas?
- Is there a particular vocabulary associated with this kind of text?

Then create a teaching episode where you model this text, using the deconstruction, joint construction and independent construction approach.

Weaving together: the interrelationship of talk, reading and writing

We've considered talk, reading and writing separately in this unit, but you will have seen that these interrelate. Talk supports reading – for example, through discussion of texts or peer talk during DARTs activities. Talk and reading support writing, as analysis and discussion of models of writing help pupils to develop their own 'expert' writing style. In fact, evidence suggests strongly that development in reading and writing is reciprocal (Graham et al., 2018) and the EEF (2021) recommends that teachers in all subjects should combine reading activities and writing instruction. Following the recommendations in this unit will allow you to develop students' literacy in this interrelated way.

SUMMARY AND KEY POINTS

This unit has addressed the following points which will enable you to create a language-rich classroom:

- The primacy of language for learning, and the importance of language in creating learning classrooms;
- The relationship between thought and language, and how, through developing young people's language, we are simultaneously supporting their development as learners;
- The interrelationship of all three language modes – talking, reading and writing;
- The concept of disciplinary literacy – supporting literacy within the practices and expectations of the subject.

- The importance of classroom interaction patterns where pupils talk more than teachers, and engage in collaborative and constructive group work.
- The need to offer pupils activities that help them to become active and critical readers of texts.
- The value of identifying and modelling the written genres used in your subject.
- The importance of subject teachers understanding the power of language and recognising that language is the very heart of empowered learning.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Alexander, R. (2020) *A Dialogic Teaching Companion*, Abingdon, UK: Routledge.
This book builds on Alexander's sustained research on dialogic teaching. It draws on this research to explain the power of talk, and provides practical, evidence-informed guidance on implementation in the classroom.

Chen, H., Myhill, D. and Lewis, H. (Eds.) (2020) *Developing Writers across the Primary and Secondary Years: Growing into Writing*, Abingdon, UK: Routledge.
This book offers a range of contributions which focus on how writing develops and how teachers can support writing across school years, including looking making connections between reading and writing.

Myhill, D., Watson, A., Newman, R. and Dowdall, C. (2022) *Understanding Literacy and Disadvantage*, London, UK: Sage.
This book synthesises research into literacy and social background and offers evidence-informed suggestions for teaching.

Dialogic Teaching: https://www.educ.cam.ac.uk/research/groups/cedir/
Teaching Writing: https://www.exeter.ac.uk/research/centres/languageandliteracy/grammar-teacher-resources/
Teaching Disciplinary Genres: https://www.teachingenglish.org.uk/sites/teacheng/files/RW8_Reading%25

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5.9 Pedagogy

The language of teaching

Elizabeth Hidson, Andrew Carr and Marilyn Leask

Introduction

The Organisation for Economic Co-operation and Development (OECD) identifies poor pedagogical knowledge of teachers as a major hindrance to improving educational outcomes (Guerriero, 2015, 2017). To begin to address this, the way that we define 'pedagogy' is important, especially when trying to find consensus: either around the world, between the UK nations, or within a single school. Pedagogy, however, is far more than just the observable act of teaching. As Alexander (2009) eloquently defines it:

> Pedagogy is the observable act of teaching together with its attendant discourse of educational theories, values, evidence and justifications. It is what one needs to know, and the skills one needs to command, in order to make and justify the many different kinds of decisions of which teaching is constituted. (p. 928)

During initial teacher education (ITE), you are developing your knowledge and skills into a pedagogic 'toolkit'. This allows you to teach intelligently and make sense of teaching and learning; it underpins the decisions you make before, during and after lessons. It is important to be able to talk about these decisions so that you can have effective professional discussions about teaching and learning. This forms the discourse that Alexander referred to in the definition above and is one of the ways that you will demonstrate your theoretical understanding, critical reflection and intentional decision-making throughout your career as a teacher. In this unit, we will tend to use the term *pedagogical* to refer to broader theoretical aspects of teaching, and *pedagogic* in relation to specific teaching techniques or practices.

OBJECTIVES

At the end of this unit you should be able to:

- identify a range of pedagogic tools for different purposes;
- explain how and why you might choose different pedagogic tools;

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- consider the curriculum and assessment requirements in your subject and context, and how these might affect your pedagogic choices.

Check the requirements of your ITE programme to see which relate to this unit.

What is pedagogy?

Pedagogy is an umbrella term that helps us to think and talk about teaching and learning. Understanding pedagogy requires exploring how learning occurs. Ertmer and Newby (2013) consider several viewpoints broadly as behaviourism, cognitivism and constructivism, aligning with what learners are doing, thinking and creating (see Table 5.9.1 for a summary). When selecting pedagogic tools and strategies, you will find they align broadly to one or more of these approaches, but the way they are applied in practice will vary depending on the nature of your subject and context, and the learning objectives. In recent years, educational neuroscience has been presented as a way of thinking about memory and cognitive load as part of teaching and learning (see Unit 5.6 for more information on educational neuroscience). For a fuller discussion on learning theories, see Unit 5.1 or follow up with a general learning theories text such as Bates (2023), which covers summaries of classical learning theories and contemporary thinking on teaching and learning.

We know that these approaches can complement each other: helping pupils to learn is the goal, but learning happens in different ways, and for different purposes. What is most relevant to the learning objectives of a PE teacher or a languages teacher will be different. The skill of the subject specialist teacher is in the deliberate structuring of learning activities for best effect.

Table 5.9.1 Essential learning theories underpinning pedagogy

Behaviourism	*Cognitivism*	*Constructivism*
■ Observable performance: what learners must do ■ Stimulus and response ■ Environment, routines, cues, prompts habits, mastery, practice ■ Generalisation leads to application of learning ■ Rewards and sanctions based on response reinforce learning ■ Reactive learners	■ Cognitive: thinking, problem-solving, processing ■ Drawn from cognitive science; neuroscience ■ Development of knowledge: reception, memory, organisation, retrieval ■ Environment, explanations, think-aloud, modelling, feedback ■ Active learners	■ Learner makes meaning from experience ■ Links practice, knowledge and context in authentic and meaningful learning experiences ■ Problem-based applications ■ Focus on process as well as outcome ■ Metacognition, self-regulation ■ Social and collaborative learners

Source: Building on Ertmer and Newby (2013)

Now complete Task 5.9.1.



Task 5.9.1 Analysing learning activities

Take an existing written lesson plan or medium-term plan (or else observe and make notes if you are observing) and analyse it through the lens of learning approaches (behaviourist, cognitivist, constructivist) by:

1 Identifying specific teaching strategies used;
2 Mapping strategies to different learning approaches;
3 Examining how pupils engage with the lesson content and with each other (for example, memorisation versus collaboration, are pupils learning from their own experiences or from the ideas of others?);
4 Reflecting on wider skills development and learning approaches (are there any subject-specific strategies being used?);
5 Connecting observed strategies to relevant educational theorists.

The goal is to critically evaluate the lesson's pedagogical design, understanding how different learning theories manifest in practical teaching practices. Discuss your thinking with fellow beginning teachers or your mentor.

What is pedagogical knowledge?

When we use terms such as 'pedagogy', 'pedagogical knowledge' and 'pedagogic practices', for example, we are aiming to group together components of teaching and learning into different categories so that you find them easier to understand and work with as a beginning teacher. Therefore, we can describe various approaches as *pedagogies*: connecting practices that form a distinct approach to teaching and learning. Sharples (2019) presented a range of innovative pedagogies, well worth further exploration, grouped under the themes of personalisation, connectivity, reflection, extension, embodiment and scale.

Table 1.1.3 in Unit 1.1 lists Shulman's names for types of 'professional' knowledge for teaching: subject content knowledge, general pedagogic knowledge, pedagogical content knowledge (PCK), curriculum knowledge, knowledge of learners and their characteristics, knowledge of educational contexts, knowledge of educational ends (aims), purposes, values and philosophical and historical influences. Although Shulman uses the term 'professional knowledge' we see all of these types of knowledge as included in the umbrella term 'pedagogical knowledge'.

Developing your pedagogical knowledge

After five years of research, Entz (2006, p. 22) concluded that it is 'the classroom teacher, ultimately, who has the most significant impact on student achievement'. You, as the teacher, are central to all of these ideas around teaching and learning, and adopting approaches that have a solid research base will give you confidence for implementation and discussion.

Teachers often talk passionately about approaches they have tried in the classroom, and their enthusiasm may be infectious. One risk for you is thinking that there is one specific way that teaching has to happen in order for learning to be successful. In fact, the opposite is more likely to be true: there are many pedagogies, many ways of approaching teaching and learning. The more you are comfortable in trying out different approaches to respond to your pupils' challenges as they work to understand and acquire new concepts, skills, attitudes and knowledge, the better you will be as a teacher.

A common analogy is that of learning to cook. You will start out with basic recipes and over time you will learn more sophisticated recipes. If you then try out a new type of cuisine, you may need to learn a new range of basic skills. Over time, you will refine, experiment, find your own magic touch, and the result may be more of a fusion. Pedagogies can be like this. You will start with essential routines and systematic approaches: tried and tested essentials, then you try out new pedagogies. Over time, your teaching will become more sophisticated and, more importantly, you will have a reflective and evidence-informed basis for your beliefs about teaching.

You may have encountered debates about whether teaching is an art, a craft or a science (cf. Biesta, 2023; Hattie, 2023). It is possible to see elements of all three, plus there are performance and process elements:

- *Teaching as artistry:* developing teaching in the way that artists continually wish to enhance their ***art***, because teaching requires creativity, judgment and responsiveness to unique situations.
- *Teaching as a craft:* the ***craft*** part of the work: *knowing how* to construct positive learning environments and learning materials, being able to choose the appropriate pedagogical strategies for the pupil and subject context.
- *Teaching as science:* Teachers work with the mind and so should be knowledgeable about the ***learning sciences:*** theories of how the brain works, cognition, motivation, behaviour and development of positive attitudes are all relevant.
- *Teaching as performance:* The way you ***perform*** in the classroom – how you capture the attention of a group, explain, demonstrate and use language effectively to convey meaning; how you question, how you respond to pupils, even your gestures all contribute to creating your unique pedagogical style.
- ***Processes*** *shaping teaching:* the structure of the school day, term and year, whether your classes are mixed ability or streamed, lesson length, the space between lessons where you can give immediate post-lesson support and individual guidance, the assessment structures applied to your subject and classes.

Entz (2006) suggests that teachers use their *pedagogical knowledge* to draw on the other forms of professional knowledge for teaching when creating effective learning experiences and environments:

> The teacher must have not only a mastery of the content and curriculum, an appreciation of the various forms of standards, an awareness of assessment, and the ability to organize the lessons, but also be able to engage students – to know them well enough to make appropriate instructional decisions. It is through pedagogy, the science of teaching, that the skilful teacher ties these elements together. The ways in which a teacher interacts with students and organizes instruction are critically important aspects of helping each child learn. (p. 10)

In addition, Guerriero (2015) makes the case for including psychological knowledge as a category of general pedagogical knowledge because most models do not make this explicit. Psychological components account for the fact that learning occurs in a social context and learning success depends on the general cognitive and affective characteristics of individual students.

Pedagogic tools and your options

Pedagogic tools are what we use to turn theory into practice, so that we are less reliant on guesswork, intuition or even just trial and error. Many of these tools can serve a number of theoretical purposes, and some may become personal favourites of a teacher. As well as pedagogy, quality of teaching and classroom climate, your beliefs as a teacher contribute to great teaching (Coe et al., 2014). As an increasingly evidence-informed profession, teachers, now more than ever, are encouraged to make pedagogic decisions with an eye on what the wider research has shown to be effective. Close-to-practice research, with elements of action research and reflective practitioner enquiry (see Unit 5.4) allows you to build up a repertoire of pedagogic tools that you know through your own situational verification work well. Your own classroom becomes the source of data for analysis of your practice. Over time, you develop the confidence to work with a looser framework and be more innovative so that your teaching is responsive to the context you are in and the pupils you are responsible for.

Table 5.9.2 lists some of the terms and examples used to describe the pedagogic tools available to teachers. Which ones you choose depends on your understanding of cognition and neurosciences as they relate to your pupils' learning particular concepts, knowledge, skills and attitudes relevant to particular topics in your subject. The way that we teach different subjects differs: you may read about Shulman's (2005) idea of signature pedagogies. For example, Pascall (2021) highlights using case studies as a 'signature pedagogy' for business studies. Experimentation is a 'signature pedagogy' for the sciences. Debating, by contrast, is a pedagogical tool for all disciplines which enables pupils to develop cognitive, analytical and communication skills essential to adult life.

The role of the teacher as the central pedagogic tool

Another important aspect is to consider the teacher themselves. Silences, wait time, and techniques to grab attention or stimulate memory all originate in the physical presence and performance of the teacher. Transdisciplinary, authentic lessons, making use of co-operative learning, flipped classrooms, metacognition and mindfulness stem from the teacher's willingness to stimulate learning in different ways. The way that the teacher designs and orchestrates learning through formative assessment, pupil-friendly learning objectives, mastery learning, relationships and motivation, summarising, celebrating errors, whole-class feedback, and demonstrating higher-order thinking all point to the centrality and presence of the teacher at the front, at the side or in the centre of the lesson. See, for example, Heightman and Gardner (2003) where structuring, signposting what is to come, pausing and pacing to allow pupils to think, prompting and probing, developing thinking, recognising mistakes, misconceptions, listening to replies and responding, sequencing, directing and distributing talk to include all are highlighted in terms of the teacher being part of their own repertoire.

Table 5.9.2 Examples of terms describing Pedagogic Tools

General terms	*Examples and references*
Assessment tools: formative, summative, ipsative	Questioning, portfolios, peer assessment, self-assessment, multiple choice tests, quizzes, exit tickets, essays, Directed Activities Related to Text (DARTs), individual and team presentations, products, e.g. plays, poetry, stories, artefacts from art, design technology, food technology.
Technology devices; AR; VR; IWBs, iPads, voting devices; IT, software, online and social media and apps as tools	Simulations, scenarios and creative tasks; ways of teaching with technologies; flipped learning. Online learning, whiteboards, interactive environments, Google Earth for Geography, VR 'field-trips', Music composition software, blogging, video annotations.
Language- and oracy-based tools	Discussion and debate; dialogic teaching, think-aloud approaches, storytelling, performance.
Image or Picture Pedagogy – Visual resources	Dictionaries, comics, concept maps, pictures, diagrams, anchor charts, film, video, graphic organisers. See, for example, Duncum (2020) and Kuechel et al. (2015).
Teaching spaces; spatial design	Outdoor education, field-trips, forest schools, school gardens, structure of the classroom: layout, decoration, furniture and educational materials on view. See, for example, Jewitt (2005).
Arts-related tools	Music, dance, drama, film, role-play.
Body enactment	Gesture, intonation, routines, 'working the room', positioning within the room.
Playful and creative pedagogic tools; experiential approaches	Games and gamification Escape rooms; Lego, board games, 'unplugged' approaches in computing, DNA models in science.
Metacognitive tools	Self-regulation, repetition, spaced learning, speculation, imaging and modelling, critiquing Peer assessment, reflective writing, mnemonics, thinking skills, habits of mind. For memory: analogy, similes, metaphors, see Newton (2011) and Tokuhama-Espinoza (2014), for repetition and spaced learning, see Stables (2020).
Situated tools, tasks, rich tasks, the 4 Cs,	Group-work, interactive workshops, experimentation, problem-solving, Kagan structures, enterprise education, STEM education. Rich tasks (see Harris, 2021) for 'mastery' or deep learning approach.
Teaching materials; published classroom resources	Textbooks and worksheets. See, for example, developing pedagogy by proxy through shared lesson resources (see Hidson, 2021).
Teacher language, demonstration and modelling, direct instruction	Instruction, exposition, explanation Demonstration in D&T; live programming, 'I do, we do, you do', prompt and practice.
Text-based tasks	Writing, design, annotation, mapping, writing frames, case studies.
Physical artefacts	Collections, models, kits, maker technology.

Social interactions, context and pedagogy

The examples above are dependent on the social style of the teacher's interactions. Whether your own prior education experience has been more didactic or more inquiry-based will influence your teaching style, but developing your confidence and competence when interacting with pupils is paramount to designing learning. The importance of social interactions in learning was described by Tharp et al. (2000) as follows:

> The nature and quality of the social and emotional interactions between teacher and students is therefore central to any discussion of quality education. (pp. 2–3)

As teachers you will work with pupils in group situations. How will your planning and pedagogy take the social aspects of learning into consideration? Whether you use a collaborative approach like think-pair-share or an open-ended whole-class questioning session, thinking through the way that pupils will interact with each other and with you will help you to make good pedagogic decisions. One practical way to develop this skill is to have a class profile or seating plan in front of you when planning lessons. Each group will have different personalities, individuals and groups. Part of planning will be to consider how the lesson will work with each class. If you are teaching three groups of Year 8, for example, the nuances of each group will have an impact, so be sure to do a mental walk through of the various parts of the lesson and how you will interact and signal transitions at key points. Consider the context of these groups, in this school, and this topic.

Now complete Task 5.9.2. It asks you to make deliberate choices about what you want to try in your own teaching so that you can gauge the impact.



Task 5.9.2 Action research approaches to developing your pedagogy

Look carefully at your medium-term planning to select a topic that you can use to carry out a systematic enquiry into developing your pedagogic skills. Unit 5.4 on improving your teaching has plenty of prompts for this. Decide on one specific strategy that you are going to implement over a sequence of lessons. Formulate a research question something like this: What impact does [implementing this strategy] have [on what demonstrable outcomes] for [which pupils], and how will I know? In terms of data collection, consider what you will use to know how well a strategy is working. This could be observations done by your mentor, assessment materials, pupil work or field notes that you make. Review these after each lesson and decide what tweaks might need to be made so that this approach works more effectively next time? At the end of the sequence, being able to lay out the steps you took, and the impact of these actions on the outcomes you set for your pupils mean that you will be able to discuss, in line with McNiff (2017), how you have improved your practice as a teacher, describe what you have done and explain why you have done it.

Discuss your approach with fellow beginning teachers or your mentor. You may also be able to use this towards a formal assignment or portfolio of practice.

Blended learning approaches

Given the variability of funding for educational technology in schools, it is impossible to assume that there is a standard level of technology available. There are a number of debates around ICT, computing and educational technology (see Younie and Bradshaw, 2018) but it is important that you are confident with using technology. Technology for administrative purposes such as management information systems, lesson registration and report writing are ubiquitous. Schools will have a range of technologies and software subscriptions for teaching, learning and classroom management, including for visual display purposes. Beginning teachers on practical school placements will need to explore the technology that is available to them and learn how to use it. Unit 1.4 gives you several useful self-audit tools.

In a sense, all learning is always blended, whether it is the blending of pedagogical approaches, blending synchronous and asynchronous learning (e.g. classwork and homework) or blending face-to-face and remote teaching. Experienced educators in this sector have pointed to the need to consider the principles of instructional design, for example, Laurillard (2012). Laurillard points out that the theories of how pupils learn have not changed even though the technologies change. In terms of selecting digital tools to support learning, pedagogy still needs to be the starting point. Video may work well for introducing knowledge about a topic, but you will need to turn that into a more active learning experience, either through discussion (in class or through an online breakout room) or the production of a response (written notes in the classroom or answering an online set of questions). It really is about the learning and how that can be supported by the teacher's planning and use of pedagogic tools. For a much fuller discussion, see the work of Aubrey-Smith and Twining (2024) in changing the idea of EdTech to PedTech.

Of course, a real benefit in teaching is the willingness of other teachers to share their practice. You don't need to start re-inventing the wheel for every lesson. Teachers form communities of practice to share materials and ideas. Some of these will be freely available and may specify the pedagogic basis explicitly, such as the various widely shared internet versions of the *padagogy wheel* (a play on the word iPad), which structures mobile tablet app selection according to Bloom's (1956) taxonomy (later revised by Anderson and Krathwohl, 2001) from knowledge and understanding through to evaluation and creation. Others may write about their approaches to learning, for example, *Teach Like a Champion 3.0* (Lemov, 2021), which is based around the idea of a consistent set of strategies that distil the ideas we touched on earlier about the process and performance elements of teaching. Task 5.9.3 looks at designing a blended learning lesson.



Task 5.9.3 Designing a blended learning lesson

Using the Learning Designer website (https://www.ucl.ac.uk/learning-designer/), which is based on the Laurillard approach (above), design a classroom-based lesson with homework, thinking carefully about what your pupils will be doing, timing, grouping and resources. Refer back to Table 5.9.2 for ideas about different pedagogic tools you could incorporate. When the lesson is complete, design a second version where the same lesson is taught remotely. In a post-pandemic world, remote teaching might yet be the end of school closures for 'snow days' – how might you plan for that? Discuss the differences with a colleague.

Curriculum, pedagogy and assessment considerations in your context

The frameworks within which we teach tend to shape many aspects of what we teach, why, and to a variable extent, how. National curriculum expectations and government policies give a broad steer, which is then developed further by multi-academy trusts or school groups in terms of centralised or shared curricula. Inspectors, school improvement partners and consultants of various kinds will shape school policy and there is an expectation that the curriculum offered in each school is presented transparently for a range of internal and external purposes. Curriculum maps are a common expectation, for example. In addition, pupils who take part in national assessments or external exams (see Unit 6.2) will also need their learning aligned with the form of summative assessment they will experience.

For pupils to be able to demonstrate what they have learned for any assessments set by external bodies it is essential that curriculum, pedagogies and assessment are aligned: that what is taught (curriculum content) and how it is taught (pedagogy) match the expectations of the examination board or awarding organisation (see Bhatti and Jumani, 2019 for detail about alignment issues). Check your subject's exam specifications to make sure the pedagogic approaches you use ensure pupils are prepared for the forms of assessment used in your subject. It is often surprising to pupils when they realise that summative exams may take the form of written papers even in a 'practical' subject such as Food Technology or PE, or that non-exam practical assessment may account for a very small percentage of the total marks. You may need to balance the push to 'cover the curriculum' with your own beliefs about teaching and learning in your subject. You may be inspired by creative approaches, have learned of different ways that learning can take place, and feel that a deeper understanding of your subject is possible than the default position suggested by documents you have encountered so far. One area for debate with other teachers and mentors may be around the extent to which curriculum documents and exam specifications have the potential to prescribe or fix the pedagogic approaches taken. To some extent this is understandable in a high-stakes Key Stage 4 (ages 14–16) exam class, but there are often concerns about the impact this has on Key Stage 3 (ages 11–14). Is Key Stage 3 shortened, for example (i.e. to two years), and a 3-year Key Stage 4 programme put in place? Do your colleagues see Key Stage 3 as feeding directly into Key Stage 4, or do they see this as a distinct, possibly more creative stage? Are texts provided to pupils so that they get used to interpreting them for exams, or for other liberal, personal or enjoyable purposes? Are pupils encouraged to think creatively?

Looking ahead

As with all who ask professional questions about our practices, those who are interested in pedagogic research continue to push the boundaries through innovative practices. Since 2012, the Open University's (OU) Institute of Educational Technology (IET) has produced its innovating pedagogy reports to explore new forms of learning, teaching and assessment practice for our interactive, technology-enabled world. The Innovating Pedagogy 2024 report (Kukulska-Hulme et al., 2024) highlights challenges such as the climate emergency, the proliferation of Artificial Intelligence (AI) and the fragility of social trust. The 2024 report identifies ten proposed pedagogies including climate action pedagogy, exploring scientific models from the inside, pedagogies of peace, 'speculative worlds', immersive language and culture through games, 'intelligent textbooks' and several

AI- and VR-related approaches including learning in conversation with Generative AI, AI ethics, AI-enhanced multimodal writing and assessment through extended reality. Each of these is covered in depth in the report and we encourage you to read the summaries in order to extend your ideas about pedagogies.

Now complete Task 5.9.4, where you can reflect on how to incorporate some of these wider aims into your pupils' experiences.



Task 5.9.4 Rich task pedagogies versus knowledge-focused pedagogies

For this task, we ask you to consider what skills, knowledge and attitudes young people are likely to need for working lives that will continue through to the late 21st century.

What pedagogies are you going to use with your pupils? To what end? If you find you are in an institution that requires you to prepare the pupils for closed knowledge-based assessments then don't forget that homework and extra-curricular activities can support a broader education. Rather than setting individual homework tasks for each lesson, you might consider extended projects with presentations of progress to peers being part of the process. Additionally, if class time is not available and the technology is available, pupils can record their presentations for you to assess. You may also get involved in the competitions for pupils organised by different organisations.

Discuss your ideas with fellow teachers. What can you add to your pupils' experiences of learning in your subject?

We have provided a set of tasks in this unit that encourage you to develop your skills of reflection and systematic enquiry so that you can rationalise your own choices and that of others in determining your pedagogic approach. You will continue to develop your pedagogy throughout your whole career, so keep on the lookout for inspiring and innovative ideas that you can draw into your practice.

SUMMARY AND KEY POINTS

We hope this unit has helped you to think about pedagogy in general terms as well as in more practical ways.

- We have introduced the range of pedagogic tools you have at your disposal.
- We have guided you to reflect on the interdependencies between assessment, curriculum and pedagogy in your context.

We hope that this has given you the confidence to choose tools for particular purposes including the use of rich tasks to motivate and engage pupils.

Check which requirements for your ITE programme you have addressed through this unit.



ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.





Further resources

English-Speaking Union resources
Public speaking, debating and oracy tools from the English-Speaking Union
https://www.esu.org/resources/

MESHGuides
Research summaries and evidence to skill and support the professional judgement of teachers.
https://www.meshguides.org/

Sage: 10 Learning Theorists Every Trainee Teacher Must Know
A downloadable set of cards covering ten of the major learning theorists.
https://uk.sagepub.com/sites/default/files/learning_theories_cards.pdf

University of Sunderland Pedagogy Boost Podcasts
A curated series of podcasts covering a range of relevant topics, such as Rosenshine's Principles of Instruction, The Science of Learning and Teaching to the Top.
https://uosmentor.sunderland.ac.uk/?page_id=133

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6 Assessment

Assessment and its reporting are a central issue in teaching and learning throughout education. Assessment is a complex and widely debated aspect of schooling, serving diverse functions from national exams like GCSEs and A-Levels to continual, informal classroom checks. In England, since the 1988 Education Reform Act, national testing has taken centre stage in monitoring standards in schools. The role of national examinations is therefore analysed.

This chapter addresses the purposes of assessment, their relationships to teaching and learning, and changes that have occurred in assessment practice and reporting. The importance of formative assessment in raising achievement is discussed. Formative assessment is contrasted with summative assessment as "low stakes" vs "high stakes" testing. The concepts of validity and reliability in assessment are addressed, including the tensions between them. Learning intentions that are difficult to assess but educationally important are also considered.

Formative assessment provides information about an individual pupil's progress, helps you devise appropriate teaching and learning strategies and gives parents helpful information about their child's progress. Summative assessments are often used to compare pupils, schools and local authorities across the country. This chapter also discusses the extent to which any one test can be used to assess pupil progress, the quality of teaching and teachers, and school effectiveness in raising standards.

Unit 6.1, "Developing formative assessment practice for high-impact teaching", takes the position that assessment is best described as a procedure for drawing conclusions about learning where every test is an assessment but not every assessment is a test. It clarifies what formative assessment is, why it has not had the impact it has promised, and discusses key formative assessment strategies including: learning intentions, success criteria, feedback, self- and peer-assessment and self-regulation, all of which can be used on a day-to-day basis to develop high-impact teaching.

Unit 6.2, "External assessment and examinations", considers preparing pupils for public examinations, an important feature of a teacher's work; this unit considers assessment as exemplified by GCSE and GCE Advanced Level. This unit links national monitoring of standards with your classroom work, raising issues of accountability. Public examinations grade pupils on a nationally recognised scale and exercise control over both entry to jobs and higher education. This unit addresses how national standards are maintained and national grades are awarded. National developments in vocational education are discussed and the status of vocational education in relation to academic

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education is highlighted. Contrasts are drawn between the assessment methods used for vocational courses and academic courses. A recurrent theme that arises when discussing external assessment and examinations is validity and reliability. This unit explains the main purposes of external assessment and the impact of external accountability measures on schools and teaching.

Unit 6.3, "Using assessment data effectively: Making better decisions for teaching and learning", examines the issues relating to data management and workload and suggests how these might be overcome. It deliberates the often-confused and seldom-discussed concepts of validity and reliability and how, using practical strategies, they can better serve teachers to make enhanced decisions about where teaching and learning should go next. This unit explores how teachers can manage data effectively, use robust assessment systems to draw valid, reliable insights and apply formative assessment to inform teaching decisions. It aims to make data work for practice rather than overwhelm it. The focus is on practical strategies for integrating data into everyday teaching to support learning and enhance professional judgment across varied classroom scenarios.



ONLINE RESOURCES

Watch the explainer video on understanding formative assessment, external assessment and assessment data by scanning the QR code or visit routledgelearning.com/learningtoteach.



6.1 Developing formative assessment practice for high-impact teaching

Nikki Booth

Introduction

Assessment is complex, multifaceted and is one of the most debated areas of educational discourse. It covers a wide variety of purposes and uses ranging from nationally recognised external assessments, for example, GCSEs and A-Levels (see Unit 6.2), to day-by-day, even minute-by-minute, assessments. This unit focuses on how assessment can be used formatively to serve teaching and learning in both face-to-face and online learning environments. It is important to state from the outset, however, that every school may see and do assessment differently, and although this unit is research- and practitioner-informed, it is important that it is read in conjunction with your school's assessment policy.

OBJECTIVES

At the end of this unit you should be able to:

- explain what "assessment" is;
- explain what formative assessment is and understand why it has not had the national impact it promised;
- write clear learning intentions (learning objectives) and process success criteria;
- define and understand what research says about feedback;
- explain the issues surrounding marking workload and how they might be overcome;
- give quality and impactful feedback.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

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Understanding "assessment" and its purposes

Before reading this section, do Task 6.1.1.



Task 6.1.1 Experiences of assessment

Think back to when you were at school/college/university. What were your experiences of assessment? Were these positive experiences or negative? Why was this? Now think about the pupils in the classes that you are currently teaching. What are their views about assessment? If there are any negative views, how might you make assessment more of a positive experience for them?

Collect your findings and discuss these with your mentor(s).

There is a wealth of educational research, in a wide range of subject areas, that draws a distinction between formative and summative assessment. However, simply describing a learning activity as formative, on the basis that it takes place *during* the teaching and learning process, or summative, because it takes place *at the end* of the process, can be considered inaccurate and somewhat unhelpful for teachers. Instead, to deepen our understanding of assessment for high-impact teaching, this unit adopts Cronbach's (1971) definition where "assessment" is perhaps better described as a procedure for drawing conclusions. In other words, as Wiliam (2020) helpfully clarifies: "we give students things to do – such as tasks, activities, tests and so on – and we collect evidence from the students, from which we draw conclusions" (2020, p. 22). In practice, what this means is that there is no such thing as *a* summative or *a* formative assessment because the same activity, regardless of when it takes place in the teaching and learning process, can elicit conclusions that are summative, formative or both.

Following a learning activity, examples of conclusions you make as a teacher could be about an individual pupil's status such as "you are really good at solving equations" or "you are going to be a successful writer in the future". These conclusions might be considered *summative* because their purpose is to "sum up". Alternatively, or in addition, you might conclude that something needs to happen to improve learning further such as "the class need more practise learning their irregular present tense verbs" or "this group needs to improve their composing using the Blues scale on C". These examples could be considered *formative* because their purpose is to help shape the direction of where teaching and learning should go next.

By drawing on Cronbach's (1971) definition, thinking of assessment as a means for drawing conclusions can be important for two reasons. First, as stated previously, it means that the same assessment information you gather from an activity could be used for differing purposes including a pupil's status, informing what needs to be done next, or both. Second, it implies an important point that summative assessment is not just something which occurs during "assessment week" or the "assessment lesson" or through the giving of a mark, level or grade to pupils. As Booth (2023) found, it can, in fact, be very much present in every lesson. To be clear, this is not to say that summative assessment should be reduced *per se* because conclusions such as those mentioned above can be important for pupil motivation, especially if they are used wisely (Lipnevich et al., 2023);

rather we would do well to become more critically aware of the summative-formative balance in our practices and consider how an even greater formative teaching and learning environment can be embedded.

Now do Task 6.1.2.



Task 6.1.2 Identifying summative and formative assessment in teaching practices

Make your own version of this table:

Assessment inference	*Tally*	*Examples*
Summative assessment		
Formative assessment		

Observe your mentor (or another teacher) teaching. Apply Cronbach's (1971) definition of assessment to make a tally of the summative and formative comments the teacher gives to pupils during a lesson. Write examples of the types of things you hear. What was the balance between summative and formative assessment in this lesson? Was there a dominance towards one modality of assessment? Why was this? Discuss your findings with the person you observed.

Ask your mentor (or another colleague) to observe you with a class of your choice. Ask them to do the same task for you. What did the findings tell you about your current practice?

Discuss this information with your mentor(s).

Formative assessment

Defining "formative assessment"

In the United Kingdom, the term "formative assessment" tends to be built upon the work of Black and Wiliam (1998) who defined it as "all those activities undertaken by teachers and/or their students, which provide information to be used as feedback to modify teaching and learning activities in which they are engaged" (1998, pp. 7–8).

Despite this definition, follow-up research investigating formative assessment practice in schools found that there was a general lack of understanding of what formative assessment was and how it could be implemented successfully into the classroom (Carter, 2015; DfE, 2015; LKMco/Pearson, 2017) which is possibly a reason why formative assessment "has [had] no (or at best limited) effect on outcomes nationally" (Coe, 2013, p. 10). This understanding and effective implementation into teaching practices is important because there is over 50-years-worth of research evidence to suggest that when information is used *formatively* it can have a significant impact on teaching and learning (Booth, 2017). For example, upon its completion in 2018 and based on data from 140 schools, England's Education Endowment Fund (EEF) reported that pupils "in the Embedding Formative Assessment schools made the equivalent of two additional months' progress in their Attainment 8 GCSE score" (EEF, 2018, n.p.). A further scaled-up Embedding Formative Assessment programme project is due to take place from 2025.

Why formative assessment has not had the impact it promised

There are several reported reasons why formative assessment has not had the impact it promised in the United Kingdom. The first relates to confusion over the term "formative assessment". As the Assessment Reform Group (ARG, 1999) explain:

> The term 'formative' is open to a variety of interpretations and often means no more than that assessment is carried out frequently and is planned at the same time as teaching. It may be formative in helping the teacher identify areas where more explanation of practice is needed. But for the pupils, the marks or remarks in their work may tell them about the successes or failures but not how to make progress towards future learning. (p. 7)

This point is exemplified further by Wiliam who, in an interview published in the *Times Educational Supplement*, stated, "the big mistake Paul and I made was calling this stuff 'assessment' … because when you use the word assessment, people think about tests and exams" (Stewart, 2012, n.p.).

Second, Harlen and James (1997) posited that one of the main influences on teachers' difficulty in implementing formative assessment was the lack of understanding on how to differentiate assessment for summative and formative purposes. Due to this lack of understanding, the task of using it effectively in the day-to-day classroom became a challenge to them (Harris and Brown, 2009).

Third, it could be argued that in some Western countries, including those in the United Kingdom, many teachers have been taught and trained to become teachers when behaviourist approaches significantly influenced teaching, learning and assessment (Perumanathan, 2014). As such, many teachers have been caught in a paradigm shift from a behaviourist teacher-centred approach on the transmission of knowledge to a more facilitative, interactive and pupil-centred one (Clarke and Hollingsworth, 2002).

Finally, with reference to the influence of national examinations, Fautley and Savage (2008) acknowledge that, in secondary schools, there is pressure on teachers and pupils to produce high levels of attainment in the form of marks or grades from assessments. As such, what can happen in schools is that the end *products* of learning (for example, mini-quizzes and end-of-unit tests to determine pupil progression) are the main areas of attention, leaving the learning *process* to be largely ignored (Leon-Guerrero, 2008). To be clear, this is not to say that such products are not important – of course they are – but results from these types of assessment will only ever be a snapshot of a pupil's story and will not improve learning by themselves; something has to be done with the information. In other words, it needs to be used *formatively* – for a pupil's learning to improve (Booth, 2021).

Clarifying formative assessment

As stated previously, in the United Kingdom, the term "formative assessment" tends to be built upon the work of Black and Wiliam (1998). See above for the relevant quotation. Based on this oft-cited definition, Booth (2023) argued that formative assessment requires two key ingredients: *intention* and *action*, and that both are required for formative assessment to occur. These ingredients, their relation to Black and Wiliam's (1998) work, and how they are interpreted, are further clarified in Table 6.1.1.

Booth's (2023) *intention-action* distinction is important and could be of significant value to teachers when thinking about high-impact formative assessment practice. Although this is discussed later in the unit, let us take feedback as an example. When teachers give learners feedback to improve it is being given with the *intention* it will be acted on. This action by pupils might include, for

Table 6.1.1 Unpacking formative assessment

Formative assessment ingredient	*Citation from Black and Wiliam (1998) [italics as per Booth's (2023) research]*	*Interpretation (Booth, 2023)*
Formative *intention*	"… which provide information *to be used*" (Black and Wiliam, 1998, p. 7)	This suggests the collecting or sharing of information via a strategy of some sort with the intention it will be acted on.
Formative *action*	"… *to modify* teaching and learning activities" (Black and Wiliam, 1998, pp. 7–8)	This implies that something must be done with the information which arises from any such assessment.

instance, a group discussion to challenge the feedback given or, if the feedback is accepted, a pupil would engage in some sort of response to improve their work, or both. However, if, as was found in Booth's (2023) study, pupils become "off-task" or simply ignore the feedback (in other words, nothing is done with the feedback) then it is very unlikely that high-impact formative assessment to improve learning is going to take place. Booth's (2023) intention-action approach, therefore, is a key reason why formative assessment should be thought of as a process, not a "thing".

Figure 6.1.1 shows how *formative intention* (through the use of strategies to elicit information about current learning) requires *formative action* (a response of some sort in relation to the

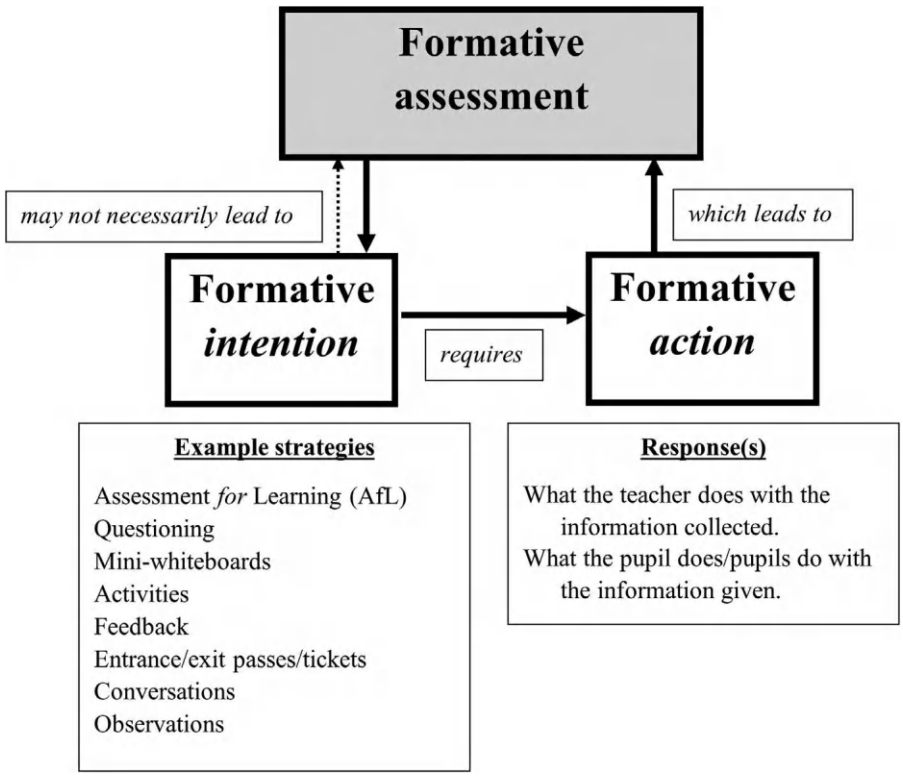


Figure 6.1.1 Unpacking formative assessment

Source: Adapted from Booth (2023)

information collected by the teacher and/or pupil(s)) for effective *formative assessment* to take place. This cycle continues throughout the teaching and learning process. The dotted arrow above formative intention signifies that the collecting of information, without any form of response, is unlikely to lead to high-impact formative assessment taking place.

Formative assessment in digital learning environments

Formative assessment in digital learning environments has become an important part of contemporary education (Bearman et al., 2020). Although the use of digital technology in teaching and learning practices was significantly accelerated during the COVID-19 pandemic, research in this area was taking place before this time. For example, Dalby and Swan (2019) found that existing pedagogical practices can be effectively implemented into digital contexts, However, others have established that pupils' active engagement online was problematic (Lervik et al., 2018) with subsequent research identifying that implementing formative assessment was particularly challenging in digital contexts (Vattøy and Gamlem, 2025).

The next section unpacks some core formative assessment strategies which can be applied to both face-to-face and online learning environments.

Some formative assessment strategies

There are a number of strategies which can be used in your teaching practice to help support you in developing effective formative assessment. In this unit, three are discussed: learning intentions, success criteria and feedback. Further strategies can be found in, for instance, Wiliam and Leahy (2015).

Beginning with the end: what do clear learning intentions look like?

A requirement within many classrooms today is the sharing of learning intentions (also known as "learning objectives" or "learning outcomes") with pupils at the beginning of the lesson. Although wording may be different between schools, fundamentally, these fall into three categories as shown in Table 6.1.2.

The clarification of the learning intention(s) can be considered essential so that pupils are able to hold a concept similar to that of you, the teacher (Sadler, 1989). Furthermore, as Wiliam and Leahy (2015) state:

> [b]efore we can find out what our learners are learning, before we can give feedback, before we can engage learners as resources for one another or as owners of their own learning, we have to be clear about where we are going. (p. 27)

Table 6.1.2 Categories of learning intentions

■ **To know ...**	A learning intention which shows **how much** a pupil knows about a topic.
■ **To understand ...** (sometimes used as **to know how** or **know why ...**)	A learning intention which shows **how well** (rather than how much) a pupil knows about a topic.
■ **To be able to ...**	A learning intention which shows how well a pupil can **apply knowledge** of a topic.

Writing good learning intentions is challenging, even for more experienced teachers, and there are two key aspects which we need to take into consideration.

The first relates to the work of Clarke (2005) and the problem of including the *context* within the learning intention itself. For example, in Key Stage 3 English, a learning intention of "to (be able to) write a letter to your local council to keep your local swimming pool open" could be considered confused. In Clarke's (2005) view, pupils who have been working on this, perhaps for several lessons, are likely to do well – but so what? As teachers, we may not necessarily be interested in whether pupils can just write about keeping their local pool open *per se*, but whether the knowledge they have acquired throughout the sequence of lessons can be transferred into other contexts, for example, their viewpoints regarding fox hunting. Instead, a slightly clearer learning intention could be "to (be able to) write a letter" where "to your local council to keep your local swimming pool open" would be the *context*.

Second, it is important to have a clear understanding not only of what the pupils will be *doing* (lesson activities), but what they will be *learning* by doing them. Our slightly clearer learning intention of "to be able to write a letter" is actually activity-focused, not learning-focused. What are pupils going to *learn* by writing this letter? Are they, for example, writing to persuade, or argue, is it formal or informal? Depending on where this particular learning intention falls within a unit of work, and what pupils' prior knowledge of writing letters is, possible alternative learning intentions could include:

- To know the key features of a persuasive letter; or
- To know (understand) how to use persuasive features; or even
- To be able to write persuasively.

Once the desired destination of the lesson is clear, the next thing to consider is what pupils need to *do* in order to arrive there. This is where sharing and modelling what success looks like becomes another important aspect of formative assessment practice.

What does success look like?

In many school policies, it is also a requirement for teachers to share success criteria with pupils at the beginning of a lesson. In most subjects, these might be presented on, for example, the lesson's PowerPoint slide, but for more practical subjects, such as Physical Education, it is probably more appropriate to communicate them orally.

In a lesson, when a pupil asks "'What exactly do I have to do?', or 'Did I do this right?', then this mostly shows that they have no clear idea what level of quality is expected of them. They are lacking an *understanding* or *sense of quality*" (Kneyber et al., 2024, p. 31), italics in original). Therefore, sharing what success can look like can be seen as a high-impact formative assessment strategy for supporting and scaffolding pupils during the lesson (Leahy et al., 2005; Hattie and Clarke, 2019; Clarke, 2021; Kneyber et al., 2024). This also has implications for the giving of in-lesson feedback (discussed later) because:

> [f]eedback to students is automatically focused around the success criteria, resulting in more specific and accurate targeting of [individual, group or whole class] needs.
>
> (Clarke, 2005, p. 36)

That said, it is also important to recognise that only focusing feedback on success criteria can be problematic because "we can find ourselves ignoring examples of excellence which do not match the success criteria" (Hattie and Clarke, 2019, p. 62).

The sharing of success criteria can support pupils in understanding what "quality" work looks (or sounds) like (Kneyber et al., 2024). As Sadler (1989) notes:

> The indispensable conditions for improvement are that the *student* comes to hold a concept of quality roughly similar to that held by the teacher, is continuously able to monitor the quality of what is being produced *during the act of production itself*, and has a repertoire of alternative moves or strategies from which to draw at any given point. (p. 121, italics in original)

In other words, as one pupil from Spendlove's (2009) research commented: knowing the success criteria for a task is "like knowing the teacher's secret" (Spendlove, 2009, p. 18). This is important for high-impact formative assessment to take place; "the purpose of success criteria is to bring about success" (Wiliam, 2018, p. 70).

On an international level, the work of Clarke (2005, 2021) has been particularly influential with regard to defining two types of success criteria: *process* and *product*. Process success criteria can be seen as a summary of "the key steps or ingredients the student needs in order to fulfil the learning objective" (Clarke, 2005, p. 29). Product success criteria, on the other hand, merely tell pupils what the end product would look like. With the third English learning intention example above, the product success criteria example could be "the reader will be persuaded by your letter". To some this may seem fine, but this raises the question of what is it *exactly* pupils need to do in order to be persuasive writers? To unpack this further, Table 6.1.3 shows how the last of the three learning intentions can be supported by process success criteria.

As stated above, in some practical subjects, such as Physical Education, sharing success criteria with pupils orally is probably more contextually appropriate than sharing them in a written format. Physical Education teachers are often very good at developing process success criteria, as Wiliam (2018) states:

> they break complex skills down into simpler ones and then reassemble them. To those who have never had to teach throwing a ball, it just looks like one action, but [teachers] know that most children learn to do this better when it is broken down into a number of steps that are practiced individually and then integrated into one fluid action. (p. 68)

Table 6.1.3 Example learning intention with process success criteria

Learning intention:	*To be able to write persuasively.*
Process Success Criteria:	You need to: ■ state your point of view ■ give reasons for your viewpoints, with proof ■ give alternative viewpoints ■ use subjective (personal feelings) language ■ use rhetorical questions ■ include a summary

Regardless, though, of whether process success criteria are shared orally or in a written format with pupils, Clarke (2005) makes it clear that:

> The purpose of students having [and using] success criteria is not to give them a simple fix-it list, but rather to remind them of those aspects of the task on which they most need to focus. (p. 34)

Now do Task 6.1.3.



Task 6.1.3 Writing clear learning intentions and process success criteria

Using the information you have just read, write a clear, context-free and learning-focused learning intention, with process success criteria, for a lesson within a unit of work. Give this to a colleague, who is not a subject specialist, and see if they understand what you propose pupils will be learning and what specifically they will need to do to be successful.

Reflect on the feedback you receive and consider how you can further develop your skills in this area.

How can feedback be made more impactful?

Defining feedback

Ramaprasad (1983) defined feedback as "information about the gap between the actual level and reference level of a system which is used to alter the gap in some way" (1983, p. 4). This definition was later extended by Sadler's (1989) formative assessment theory where, in order for a pupil to be successful, meaningful practice requires a "feedback loop" (1989, p. 121). The feedback loop has the potential to support high-impact formative assessment practice for several reasons:

- it informs teachers about levels of knowledge, understanding and skills attained or yet to be attained by the pupil;
- it aims to facilitate learners in being able to identify and amend a learning gap;
- it assists teachers in reflecting on and selecting suitable tasks or activities; and
- it allows teachers to modify their teaching in order to support the closing of the gap.

Andrade and Heritage (2018) make it clear, however, that simply knowing how work would be improved (defined by Booth (2023) as formative *intention*) is not feedback unless it is actively used to serve this purpose (defined by Booth (2023) as formative *action*). This is particularly important to elicit pupil engagement in the formative assessment process. Although feedback can, indeed, be said to be at the heart of the formative assessment process (Barnes and Marks, 2021), which will be discussed later in this unit, it is also important that you are aware of how to manage your workload with regard to feedback and marking.

Tackling excessive workload

In the report *Eliminating unnecessary workload around marking* (DfE, 2016) for England, the Independent Teacher Workload Review Group (ITWRG) stated that "marking had become a burden that simply must be addressed, not only for those currently in the profession but also for those about to enter it" (DfE, 2016, p. 3). As part of their published summary, they listed a range of ineffective marking practices, for example:

- extensive written comments in different colour pens;
- the use of "VF" on pupil's work to indicate when oral feedback had been given;
- marking which fails to help pupils improve their understanding; and
- marking that fails to encourage motivation and resilience.

The report also recognised that "teachers who mark work late at night and at weekends are unlikely to operate effectively in the classroom" (DfE, 2016, p. 7).

In 2019, a follow-up document entitled *Ways to reduce workload in your school(s)* (DfE, 2019) was published. Although this document is primarily aimed at school leadership teams, the recommendations offered can also be considered helpful for teachers, including those new to the teaching profession. These recommendations, and what they might mean for developing more effective feedback and marking practices, are shown in Table 6.1.4.

The advice provided by these two documents has been brought together with the clear aim of reducing teacher workload. If, during your teaching practice, you find your feedback and marking load becoming unmanageable, speak to your mentor(s) to see how they can best support you.

Table 6.1.4 Recommendations for managing feedback and marking workload

Recommendations	*What this can mean for developing feedback and marking practices*
"Consider why you are marking."	"Ignore myths that suggest that spending hours marking makes a better teacher, and that writing pages of feedback makes you more effective: concentrate on feedback which is meaningful, manageable and motivating."
"Consider a range of feedback techniques which are proportionate."	"Not all feedback has to be written. …. Consider live class feedback (e.g. audio, photo, video) which can be recorded and added to electronic versions of pupils' work."
"Build in live sampling."	"Select a sample of work and mark it in front of pupils, e.g. use a visualiser, so students see how work is corrected, enabling them to then make amendments to their own work."
"Use 'sampling for planning'."	"Focus on a sample of pupils' work after a lesson and use the understanding gained about progress to inform the planning of the subsequent lesson(s)."
"Use low stakes and self-marking tools."	"Prepared questions [using online tools] can be saved and shared, allowing teachers to re-use or adapt the content rather than create something from scratch."
"Use feedback codes where appropriate."	"These can be used to draw pupils' attention to common strengths and areas to develop to reduce the need for lengthy responses."
"Prioritise key pieces of work."	"Use schemes of work to identify which pieces of work should be prioritised for more detailed feedback."

Source: DfE (2019, p. 5)

What does the research say about feedback?

In their meta-analysis (an analysis of analyses) of 131 studies on feedback, Kluger and DiNisi (1996) found that, on average, feedback increased pupil performance. While this may not necessarily be a surprising finding, what is interesting, though, is that they also found that approximately one-third of the studies (38% – 50 studies out of 131) actually lowered pupil performance, with the remainder showing that feedback made no difference at all. What Kluger and DiNisi's (1996) research suggests, therefore, is that giving feedback is a lot more complex than it might seem at first. Although we need to be mindful that what works in one educational context may not necessarily work as well (if at all!) in another, it becomes necessary to look at some individual feedback studies for us to understand how feedback might better support the teaching and learning process. In this unit, two pioneering and oft-cited feedback studies are described, one by Elawar and Corno (1985) the other by Butler (1988), and both relate to the contentious issue of providing feedback containing a score, level or grade with comments to improve.

Comments versus grades

Elawar and Corno's (1985) study investigated the quality of feedback given on pupils' mathematics homework. The feedback given to one group of pupils (experimental group) included detailed comments on mistakes and errors, suggestions on how to improve, and at least one positive comment. The second group of pupils were split into two sub-groups, half of which received constructive feedback (experimental group), and the other half just scores (control group). A third group of pupils received their usual form of feedback on their homework – just the giving of scores (control group). The research found that the pupils who received the constructive feedback (the experimental groups) learned twice as fast as those in the control groups. What was also found was that in the lessons where pupils had been given constructive feedback, not only did achievement increase, but the gender gap between boys and girls was reduced and there was more of a positive attitude towards learning mathematics. Similarly, Butler's (1988) study found that pupils who received comment-only feedback made 30% more progress than pupils who just received a score and, interestingly, those who received both comments and scores.

Across these two studies, the reason giving scores and comments appears not to be a high-impact strategy is simple, as Wiliam (2012) explains: "When students receive both scores and comments, the first thing they look at is their score, and the second thing they look at is … someone else's score" (2012, n.p.). As such, what can happen in practice is that the comments teachers write, the part of the feedback which can help move learning forward, go largely ignored. To be clear, this is not to say that we should never give our pupils scores, levels or grades when we give feedback; rather we need to consider carefully *when* it is more appropriate to do so. For example, some research (for instance, Kuepper-Telzel and Gardner, 2021) suggests that giving constructive comments for pupils to act on first *before* they receive their score leads to greater gains in performance when compared to those who received their score first then the constructive feedback. Furthermore, it is important to note that feedback consisting of only comments can also be considered problematic; the desired impact of this can be very much dependent on the *quality* of the comments being given.

Establishing quality feedback for high impact

To better understand the impact of different qualities of feedback, Nyquist's (2003) meta-analysis considered findings from 86 studies. From this research, a typology of different kinds of feedback

Table 6.1.5 Nyquist's (2003) typology of feedback

Typology of feedback	*This means giving pupils …*
Weaker feedback only	■ only scores, levels or grades.
Feedback only	■ scores, levels or grades; and ■ clear goals to work towards; ■ or, feedback on the correct answers to the questions they responded to.
Weak formative assessment	■ information on which questions they attempted are correct; ■ and an explanation as to why they are correct.
Moderate formative assessment	■ information on which questions they attempted are correct; ■ an explanation as to why they are correct; and ■ some specific suggestions for gap reduction.
Strong formative assessment	■ information on which questions they attempted are correct; ■ an explanation as to why they are correct; and, ■ targeted activities on incorrect questions to put the feedback into practice and support the improvements.

was established. This is shown in Table 6.1.5. The table begins with the feedback type found to have the lowest impact on achievement leading to the greatest impact.

What Nyquist's (2003) typology shows, which can be of interest to teachers in developing high-impact formative assessment practice, is that just giving pupils feedback about their current level of achievement, even if this is through a comment-only approach, only produces moderate benefits at best. Where the impact seems to lie, however, is when pupils are given a specific and meaningful follow-up activity in which they can put the comments into practice. In other words, pupils are being provided with the appropriate time and space to act (formative *action*) on the feedback they are given (formative *intention*) (Andrade and Heritage, 2018; Booth, 2023).

Audio feedback

Research into the use of audio feedback has been conducted since the 1970s using cassette tapes (inter alia, Klammer, 1973). What many of the studies seem to suggest is that teacher audio feedback given to pupils is far richer when compared to written comments (inter alia Voelkel and Mello, 2014). However, these findings are not consistent. For example, Cavanaugh (2014), who investigated teacher and pupil perceptions between audio and recorded feedback, found that teachers tended to have negative feelings towards providing audio comments on the basis that they lacked sufficient quality. Pupils, on the other hand, disagreed with their teachers and voiced that the audio feedback they received was "a more valuable tool than written comments" (2014, p. 128). More recently, Vattøy and Gamlem (2025) found that "most teachers recognised the immediate advantages of digital feedback in terms of accessing, storing, and revisiting feedback regarding students' learning processes" (2025, p. 11) and that "students were better prepared to use previously provided formative feedback in various situations due to the convenience of storage options" (2025 p. 11).

Applying a formative assessment lens, Booth (2023) investigated audio feedback in several Key Stage 3 (ages 11–14) group-based classroom learning environments. Pupil participants, including those with a Special Educational Need and/or Disability, found the ability to listen back and, when needed, re-listen (formative *intention*) to their teacher's audio-recorded feedback a valuable aide memoire. For many, this helped provide a clear focus for what they needed to do (formative *action*)

in the next lesson – usually a week later. Teacher participants reported that recording audio feedback outside of the lesson was efficient in terms of their time with frequent comments being made that it provided more time for giving better quality feedback while still being able to maintain a positive workload balance. Booth (2023) posits that teachers using an audio recorder in this way allowed them the time to slow down the feedback process by being able to reflect on current pupil learning and think carefully about the best way to respond. However, Booth's (2023) research also makes clear that audio feedback should not be a replacement for face-to-face "live" teacher feedback. This became apparent in two of the case-study schools where groups listened to their teacher's feedback at the start of the next lesson but sometimes did not always fully understand it. As such, the desired response (formative *action*) based on the feedback given (formative *intention*) became hindered.

Some examples of feedback in practice

There are multiple methods in which effective feedback can be given. The following vignettes have been shared by teachers whose schools have reflected on their feedback practices with the aim of reducing teacher workload while maintaining a focus on delivering high-impact feedback practices.

Vignette 1

> When we identify a small number of pupils who seem to share the same misconception, instead of stopping each individual and giving each one feedback, we bring them all together and re-explain in one go. Sometimes, when the number of learners who seem to misunderstand is rather high, we also stop the whole class, look at some model examples, and work through the misconceptions together. OK, there will be pupils who do understand, but it's a good opportunity for those who do understand to help us clarify it to those who don't.
>
> (Music teacher, personal communication)

Vignette 2

> In our school, we use 'exit tickets' so that students can provide feedback to us at the end of the lesson. Outside of the lesson, we have a quick read of their exit ticket reflections and make or amend plans accordingly for their next lesson.
>
> (Spanish teacher, personal communication)

In this school, reflective prompts are provided (taken from Wiliam, 2018, pp. 184–185) for pupils who require a little assistance in the reflective process:

- Today I have learned ...
- I was surprised by ...
- The most useful thing I will take from this lesson is ...
- I was interested in ...
- What I liked most about the lesson was ...
- One thing I am not sure about is ...
- The main thing I want to find out more about is ...
- After this lesson, I feel ...
- I might have gotten more from this lesson if ...

Vignette 3

> We don't spend time marking and feeding back on each individual book. Instead, we create a PowerPoint slide containing positive aspects as well as a list of common errors (for example, spellings, misconceptions etc.) at whole class level as well. These errors, then, are next-step tasks for all learners to act upon at the start of the next lesson. The beauty of this approach is that it allows me to re-teach misconceptions from the front of the room and to further guide and scaffold their learning. Generally, this has led to improved learner responses and more closed gaps in learning. The learners have responded positively to this approach and feel supported through the learning process. I believe this encourages them to take on challenges and reduces the fear of making mistakes. This type of marking has helped create and prompt a culture of learning from our mistakes.
>
> (Geography teacher, personal communication)

Now do Task 6.1.4.



Task 6.1.4 Pupil perceptions of feedback

Arrange to meet with a group of 5–6 pupils including different genders as well as a range of abilities and backgrounds.

Create a list of about 10 questions and discuss with them what their perceptions of feedback at school are. Some areas for inquiry could include:

1. Do they like receiving feedback from teachers? Why? Why not?
2. Do they understand the feedback they are given?
3. In which subjects do they feel they get more useful feedback than others? Why is this?
4. Are they given time in lessons to read, ask questions about and respond to feedback given?
5. Do they feel that the feedback they are given is beneficial to their learning? Why is this?
6. Any other questions which you see as relevant.

Analyse the findings and, in the light of the research mentioned earlier, write a strategy for feedback to guide your work with your pupils. Discuss this with your mentor(s). Keep this strategy in your records or portfolio for future reference.

As has been implied throughout this unit, assessment should be valued as a means both to improve student learning and guide teaching. Seeing assessment as a procedure for making inferences opens up opportunities as to how assessment and pedagogical practices can be further, and deeper, understood. This is important not only so that assessment can be used as a faithful servant to better meet the needs of learners, but also to further support teachers in doing what they do best – teach great lessons.

Now do Task 6.1.5.



Task 6.1.5 Understanding the benefits of assessment tasks

Select one class that you teach and then some activities that you have given pupils to do in recent lessons.

1 Over a period of several lessons with the class, consider to what extent your activities address the points outlined in this unit.
2 Write a record summarising your analysis and implications this may have for planning future activities for this class.
3 Use your analysis and discussion to plan and implement a unit of work that includes all (or most) of the benefits you have identified from reflecting on assessment in this unit.
4 Finally, review the success or otherwise of your planned work, identifying the ideas and/or theories that have influenced your thinking about how to assess pupils' work and progress.

Discuss this with your mentor(s). Reflect on the feedback you receive and consider how you can further develop your skills in this area.

SUMMARY AND KEY POINTS

At the end of this unit, you should now have a greater understanding of how to develop formative assessment practice for high-impact teaching. Remember:

- Assessment can be considered as a procedure for making inferences.
  - Summative assessment: can relate to "status" and "sums up" learning.
  - Formative assessment: can relate to what to do next.
    - Formative *intention* (collecting information in order to make a difference).
    - Formative *action* (acting upon the information to move teaching and learning forward).
- Learning intentions (learning objectives or learning outcomes) can be made clear for pupils if they are context-free and focus on what pupils are going to *learn* rather than what they are going to *do*.
- Process success criteria reveal to pupils what the key ingredients are for a quality piece of work.
- Quality feedback, whether oral, written or recorded, can be shown through the clarification and focus of *how* pupils can improve their work. Feedback is only effective, though, if it is actively used.
- Workload must be considered when marking and feedback tasks occur.
- Marking and feedback should be consistent with the school's assessment and/or marking and feedback policy.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

In addition to the work cited in this unit, the following books will also be of use:

Assessment reform Group website: http://www.nuffieldfoundation.org/assessment-reform-group
Download research papers and buy resources.

Booth, N. (2020) *Assessment: Formative and classroom-based*. Available from: https://www.meshguides.org/guides/node/1713
A MESHGuide research summary on formative assessment in the classroom.

Fautley, M. and Savage, J. (2008) *Assessment for Learning and Teaching in Secondary Schools*, Exeter, UK: Learning Matters.
Targeted specifically at beginning teachers, this text links explicitly to the qualified teacher status (QTS) standards, and the tasks provide opportunities for reflection and for practising the range of skills involved in assessing pupils.

Fautley, M. and Savage, J. (2013) *Lesson Planning for Effective Learning*, Berkshire, UK: Open University Press.
Practical strategies for teachers in order to plan successfully and reflect on practice.

Kneyber, R., Sluijmans, D., Devid, V. and Wilde-López, B. (2024) *Formative Action: From Instrument to Design*, Woodbridge, UK: John Catt.
Guide for teachers who want to integrate formative assessment practices into their daily teaching methods.

Stobart, G. (2014) *The Expert Learner: Challenging the Myth of Ability*, Berkshire, UK: Open University Press.
This book demolished the idea that ability is fixed and brings together ideas on what being a successful learner is through effective teaching.

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6.2 External assessment and examinations

Cara McLaughlin

Introduction

Principles of assessment were introduced in Unit 6.1. This unit looks at the particular role, function and nature of external assessment and examinations. We suggest you read Unit 6.1 before studying this unit.

This unit aims to provide you with an overview of the framework for external assessment and examinations in secondary schools with a particular focus on England. Issues relevant to Northern Ireland, Scotland and Wales are covered in units in Chapter 7 on the companion website. Although you are familiar with the public examinations that you took in school, there are regular changes and significant developments in assessment methods and in the range of external examinations taken by pupils in secondary schools. In England, each phase of schooling has been the subject of various policy initiatives in recent years.

As well as knowing how pupils are assessed throughout their secondary education, it is important to be aware of the many purposes of external assessment and examinations. These purposes can be usefully divided into those associated with candidates and those that have more to do with educational establishments and public accountability. Two important, recurrent themes that arise when discussing external assessment and examinations are validity and reliability. These two concepts, together with the agencies, regulations and processes involved in ensuring consistency in these two areas, are defined in Unit 6.1. Teaching externally examined classes is a challenge for any teacher and demands particular teaching skills and strategies, in addition to the routine elements of good lesson planning and teaching. This aspect of teaching, and the constraints and influence of the accountability agenda on the work of schools, is discussed in the final part of this unit.

OBJECTIVES

At the end of this unit, you should be able to:

- identify the range of external assessments in secondary schools and access the national qualifications and credit framework;

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- understand the relationship between the national curriculum (NC) in a home country and external examinations;
- explain the main purposes of external assessment;
- understand the processes involved in the setting of external examinations and the organisations involved;
- understand the impact of external accountability measures on schools and teaching.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Your own experience

A good starting point for this unit is your own experience of external assessment and examinations whilst at school. What did you think was the purpose of sitting examinations, and did preparing for examinations impact on your motivation as a learner? Did the teaching strategies of examination classes differ from non-examination classes? Recalling these impressions may provide a good starting point from which to develop your understanding of the issues pertaining to external assessment and examinations.

Types of assessment

In Unit 6.1, we discussed formative and summative assessment and we remind you of those terms. Formative assessment can be defined as assessment for learning, and summative assessment as assessment of learning (Stobart and Gipps, 1997). External assessment and examinations are generally considered to be forms of summative assessment. There are two important methods used extensively in summative assessment that you need to be familiar with, namely norm-referenced assessment and criterion-referenced assessment.

Norm-referenced assessment has been used extensively throughout the British education system. In norm-referenced assessment, the value of, or grade related to, any mark awarded depends on how it compares with the marks of other candidates sitting the same assessment. The basis for this form of assessment is the assumption that the marks are normally distributed; that is, if you plot the marks awarded against the number of candidates, a bell-shaped curve is produced, providing the sample is big enough. This curve is then used to assign grade boundaries based on predetermined conditions; for example, that 80% of all those sitting the examination pass and 20% fail. In this way, an element of failure is built into the examination. The system of reporting by grades is essentially norm-referenced. For example, if you are awarded the highest grade in a subject in a public examination, this grade does not give any specific information about what you can do in that subject, simply that you were placed within the top group of candidates.

Criterion-referenced assessment, on the other hand, is concerned with what a candidate can do without reference to the performance of others, and so it provides an alternative method to address the limitations of norm-referencing. A simple example of criterion-referenced assessment is that of a swimming test. If someone is entered for a 100-metre swimming award, and swims 100 metres, then they are awarded that certificate irrespective of how many other people also reach this standard. Academic courses, such as General Certificate of Secondary Education (GCSE) Drama, use

criterion-referencing when assessing practical skills as part of the teacher-assessed element of the course, but the grades are awarded and the results are reported by comparing pupils with other pupils, that is, norm-referenced. Thus, the overall assessment in most examinations is a mixture of criterion-referenced marking together with norm-referenced grading and reporting of the level of achievement.

The framework of external assessment in secondary schools

External assessment in secondary schools is an area that has seen considerable change over a short period of time. We consider, in turn, Key Stage 3 (KS3), GCSE and post-16 education in England. Different arrangements are in place in Northern Ireland, Scotland and Wales. We suggest that you ensure you gather the information relevant to the context in which you are learning to teach. For example, in England, four assessment stages are referred to in this section; that is, Key Stage 2 (KS2), Years 3–6; Key Stage 3 (KS3), Years 7–9; Key Stage 4 (KS4) (that is, GCSE); and post-16 assessment.

Key Stage 3 assessment

On transfer to secondary school, pupils bring with them information obtained from the end of Key Stage 2 assessment. This information has historically been reported as teacher-assessed and externally assessed NC levels. The introduction of a new NC in September 2014 saw the removal of levels and no intention to replace them (DfE, 2014), see Unit 6.1. The stated purposes of the 2014 reform include to 'allow teachers greater flexibility in the way that they plan and assess pupils' learning' and a commitment to 'curriculum freedom' (DfE, 2014). In practice, this means that summer 2015 was the final year that an end of Key Stage 3 'level' was reported. New NC tests for Key Stages 1 and 2 were introduced in summer 2016, two years after the start of a new NC. One purpose of the new Key Stage 2 tests is to test that pupils are 'secondary-ready' by reporting: a scaled score, which will show whether a pupil has met the expected standard and is 'secondary-ready'; ranking in the national cohort (by decile); and the rate of progress from a baseline (DfE, 2014).

Key Stage 4 assessment (Years 10–11) GCSE

The GCSE examination was introduced in 1986 to be taken at the end of Key Stage 4 to replace the General Certificate of Education (GCE) Ordinary (O) level and Certificate of Secondary Education (CSE), and to provide certification for about 90% of candidates. Similar examinations are in place in Northern Ireland and Wales. The Scottish system has a different structure (see Unit 7.5 on the companion website).

In the *Importance of Teaching* White Paper (DfE, 2010), the government of the day signalled changes to 'restore confidence in GCSEs'. This statement heralded significant modifications to the existing GCSE qualifications. The first change, introduced in September 2012, means that all GCSE courses are assessed by summer terminal examination only; the only exception to this rule is the opportunity to retake examinations in English Literature, English Language and Mathematics GCSEs in November. Alongside these changes has been a return to the compulsory inclusion of marks for spelling, punctuation and grammar in GCSE examinations.

GCSE examinations are reported under a number grading structure. In this system, there are nine pass grades, 9–1; the 9 grade recognises exceptional performance and a grade 4 is recognised as a 'standard pass'. The number grading system has been implemented in line with the GCSE reforms and to allow better differentiation between pupils' abilities.

The GCSE examinations are designed to test recall, understanding and skills. The reformed GCSE system still has a tiered assessment pattern for many of the subjects. This enables pupils to be entered for the paper most appropriate for their current achievement. Teachers assess pupils' progress and potential, and advise pupils on which tier of the examination to enter. Tiered papers carry grade limits, thus narrowing the opportunities of pupils. For example, a higher paper may enable pupils to be awarded grades 9–4, whilst a foundation paper may allow only the award of grades 5–1. Furthermore, if pupils fail to achieve the marks required for the lowest grade in their tier, then they receive an unclassified (U) grade. Any updates you can add here?

Coursework has become progressively restricted by government legislation through the subject specifications and the amount allowed depends on the subject. In the 2015 GCSE reforms, for more practically based subjects (for example, Art and Drama), a proportion of teacher-based assessment contributes to the final mark. No subject can be assessed only by teacher-assessed coursework, and for most other subjects there is no coursework. English Language does have a non-examination assessment of spoken English but it does not contribute to the overall GCSE assessment. With the rapid rise in the use of generative artificial intelligence (GenAI) since 2024, concerns are present on the impact of this on non-exam assessments. Whilst the current rates of reported malpractice are low, the DfE and Ofqual are considering longer term interventions to ensure awarding organisations are managing and reducing the risks of GenAI malpractice (Ofqual, 2024).

It is important that you have a clear understanding of how and why GCSEs have changed in recent times. Task 6.2.1 asks you to address this issue.



Task 6.2.1 GCSE examinations

Obtain copies of GCSE specifications from before and after the reforms introduced in 2015. For each specification, read the general introduction and familiarise yourself with the aims, assessment objectives and assessment patterns of the specification. Now turn to the assessment section and compare for each specification:

- how the subject content differs;
- how the structure of the qualification differs;
- what form the assessment takes in each of the specifications;
- whether there is any assessment outside the final examinations.

Discuss your findings with an experienced teacher in your placement school and find out:

- their perceptions of how the GCSE assessment has changed in recent years;
- what impact the changes have had on workload;
- any impact the changes have had on pupils' motivation, engagement, achievement and attainment;
- how they expect GCSE assessment to change in the future.

A record of your work could be placed in your portfolio or equivalent.

Post-16 assessment

GCE A level courses were first introduced in 1951 and since then have been regarded as the academic 'gold standard' by successive governments. Substantial changes were made to the framework of the 16–19 qualifications in 2000 following The Dearing Review (1996), with further revisions introduced in 2007 (Ofqual, 2007).

Alongside the GCSE reforms outlined above, new GCE AS and A levels were introduced in schools in England from September 2015. AS assessment usually occurs after one year's study and A levels after two. The courses are no longer divided into modules and the AS qualification is now 'standalone' and does not contribute to the assessment of the full A level.

The AS- and A-Level specifications include the setting of a broad range of question types to ensure that a wide range of skills is assessed, and a requirement for extended writing to give candidates the opportunity to demonstrate the depth and breadth of their knowledge and understanding.

All GCE A-Level courses must include an element of synoptic assessment designed to test a candidate's ability to make connections between different aspects of the course. There is no synoptic assessment at AS level. GCE A-level pass grades range from A*–E, with A* the highest grade. The A* grade was introduced in response to concerns about the standard of GCE A level and the increase in the number of candidates achieving Grade A.

Vocational courses post-16

The main vocational qualifications traditionally encountered in secondary schools in England have been the General National Vocational Qualifications (GNVQs), which were introduced into schools in 1992 and developed from National Vocational Qualifications (NVQs).

GNVQs were introduced to provide pupils with an introduction to occupational sectors through school- or college-based courses. One of the main challenges to the uptake of vocational qualifications by schools was the difference in assessment practice and terminology between academic and vocational courses identified in the report on 16–19 qualifications (Dearing, 1996). The report identified the need for a coherent qualifications framework encompassing all national qualifications, and one that provides a status for vocational qualifications equivalent to the academic subjects at GCE A level. Following that report, vocational GCSE and vocational GCE A levels were introduced, accompanied by phased withdrawal of GNVQ courses. The vocational GCSE and GCE qualifications were renamed and known as GCSE and GCE in applied subjects with pupils at both levels maintaining their study of core curriculum subjects.

The vocational sector of qualifications was also subject to significant change in the 2015 reforms. At Key Stage 4, the only qualifications that are recognised in performance tables are technical awards that are 'broad, high quality level 1 and level 2 qualifications in non EBacc subjects that equip pupils with applied knowledge and associated practical skills not usually acquired through general education' (DfE, 2020, p. 9). For post-16 pupils, they follow one of three routes: an academic route, a vocational route or a combined route. Within the vocational route, there are two main types of qualification: applied general qualifications (including BTECS and Cambridge Technicals) and T Levels (DfE, 2023). As with the GCSE reforms, there is now a significant level of external assessment. Further discussion of vocational (applied) courses can be found in Wolf (2015).

Table 6.2.1 Framework of national qualifications (entry level to Level 3 only)

Level of qualification	*General qualifications*
3	GCE A-level grades A*–E
2	GCSE grades 9-4
1	GCSE grades 3-1
Entry	Certificate of (educational) achievement

Originating in 1984, BTECs (Business and Technology Education Council) were developed to provide a vocational qualification for those students who were interested in a specific area within industry. Assessment protocols vary between subjects, with some being assessed only through assignments and others through exams, projects and industry placements. BTECs can vary in A level equivalency, with courses ranging of 1–3 A levels equivalent.

Launched in 2020, T levels were introduced by the then Conservative government as a way to streamline level 3 qualifications (DfE, 2021). Developed in partnership with both industry and educational experts, these qualifications sought to provide students with the knowledge and skills necessary to enter employment, further education or an apprenticeship. Whilst similar to BTECs, T levels focus on practical subjects and are equivalent to 3 A levels (DfE, 2023). Where educational outcomes in a level 3 qualification are reported as overlapping with a T level, the government are withdrawing funding from these courses in order to streamline Post-16 progression (DfE, 2017a).

The National Qualifications framework in Table 6.2.1 shows how the three qualification strands discussed are intended to overlap. The framework for national qualifications has a total of nine separate levels of qualification (entry level to level 8) and shows progression across academic and vocational qualifications.

The purposes of external assessment

There is a long-standing history in the United Kingdom of externally examining pupils at particular stages in their education, which is quite different from the practice in some other countries (DfE, 2014). In recent years in England, this practice has extended to the external assessment of pupils at the end of Key Stage 2. If the time and resources spent on this form of assessment are to be justified, then it is important that the purposes of external assessment are fully understood. The purposes of external, summative assessment can be thought of in terms of certificating candidates and the public accountability of teachers and schools. Another purpose is to categorise candidates in order to select people for higher education, employment, or to recognise achievement. The outcome of national examinations provides pupils with a grade so that they and other people can compare them with other candidates.

One recognised function of external assessment is that of certification. If you hold a certificate, then it is evidence that competence in particular skills or a level of knowledge has been achieved. For example, if you hold a driving licence, this is evidence that in a driving test you successfully performed a hill start, completed a three-point turn, and so on. The significance to pupils, of both the categorising and certification purposes, is evidenced by the fact that an impending examination provides an incentive for pupils to concentrate on their studies and to acquire the relevant knowledge and skills required by the examination for which they are entered. Thus, a consequence of external examinations is to provide motivation for both pupils and teachers. Motivation is discussed in greater detail in Unit 3.2.

Public accountability

Assessment is high on the political agenda because it is inextricably linked with the notion of raising standards, narrowing the attainment gap between different groups of pupils, school improvement and making schools publicly accountable. Schools are statutorily required to publish information on public examination results and school statistics (for example, number on roll, gender balance and uptake of free school meals) each year in what are now known as the performance tables, often referred to simply as the 'league tables'. In 2011, the data reported on KS4 and GCE A level attainment were expanded further to include significantly more data on a school's performance. The tables now include:

- how well disadvantaged pupils perform in each school;
- whether previous high-, middle- and low-achieving pupils continue to make progress;
- how many pupils at each school are entered into the core academic subjects that make up the English Baccalaureate.

The English Baccalaureate is not a qualification in its own right; it was introduced in the 2010 performance tables as a measure of the number of pupils securing a grade 4 or better in a selected core of academic subjects (DfE, 2017b). The core subjects are closely prescribed, and at the time of writing include English, Mathematics, Science, a language (modern or classical) and either Geography or History. The reasons behind the introduction of this measure are discussed below.

There is a variety of statistical information published each year; currently, schools in England can be placed in rank order based on a number of variables. Once such variable is the percentage of pupils gaining a grade 5 or above in English and mathematics. The high profile given to the performance tables by the different stakeholders in the education process has led schools to implement strategies to increase their league table ranking. For example, schools often target pupils predicted to achieve grade 4s at GCSE for additional tutoring and academic support as part of the school's strategy to increase the number of pupils achieving grade 5s in their GCSE exams.

There has been a dramatic increase in the number of non-academic qualifications taken each year from about 15,000 in 2004 to 575,000 in 2010 (DfE, 2015). The Wolf Report (2015) explains how this shift was a consequence of two factors: the accountability regime becoming more rigorous and the simultaneous relaxing of NC requirements at Key Stage 4 (Wolf, 2011). However, by 2019, just 2% of KS4 candidates were following a purely vocational pathway (Vidal Rodeiro and Vitello, 2020). This large decrease in the number of vocational certificate students was explained by the DfE as being a result of the reforms to the list of approved technical certificate qualifications on offer to students (Ofqual, 2022).

It is against this backdrop that the English Baccalaureate was introduced, and the government announced a series of measures further tightening the qualifications eligible to be included in the league tables. Significant changes to the performance tables took effect from 2016, when schools were required to report four key measures on their website: pupils' progress across eight subjects from Key Stage 2 to Key Stage 4 to be known as Progress 8; an Attainment 8 measure; percentage of pupils achieving a grade 5 or higher in English and Mathematics; and the percentage of pupils gaining an English Baccalaureate. The Progress 8 and the Attainment 8 measures can include up to three recognised vocational qualifications.

The introduction of external assessment of pupils at the end of Key Stage 2 also had a significant impact on the teaching of pupils in this age range, providing further evidence of the effect that

so-called 'high stakes' external assessment has on classroom practice. This impact is reported as a concentration on the curriculum subjects to be assessed through SATs or GCSE at the expense of a broader curriculum (Lucas, 2023).

The term 'high stakes' is used to describe assessment that has significant consequences for either the candidate or the school. Although no longer required to set and publish targets for all pupils aged 11–16 to demonstrate year-on-year improvements, in practice, target setting on many different levels has become a feature of school culture. To assist schools in setting targets, from 2017 all schools in England have been able to access the programme Analyse School Performance (ASP). This programme provides interactive analysis of school and pupil performance data across the four Key Stages of the NC. This package contains benchmark data so that schools can compare whole-school performance with that of schools with similar intakes and profiles; also, it contains contextual value added (CVA) data, which aims to take into account a wide range of variables that may affect a school's performance, for example, number of pupils with SEN or number of pupils entitled to free school meals. The programme allows for primary schools to assess how well their pupils have performed in different aspects of the curriculum. This data then feeds into secondary schools who can then use the same data to identify strengths and weaknesses in their new intake of Year 7 pupils each year.

Validity and reliability

The concepts of validity and reliability are central to understanding an assessment process and are also discussed in Unit 6.1. For all external assessments and examinations, frameworks of regulations have been developed to ensure that the examination process and the results produced are both valid and reliable. To understand this framework, you need to be aware of the organisations and processes involved in this regulation.

Since October 2011, Ofqual is the government agency that approves all course specifications, as well as monitoring examinations through a programme of scrutinies, comparability exercises and probes, and is accountable to the DfE. There are three awarding organisations in England authorised by the government to offer GCSE, GCE A and AS and vocational courses. These are:

- Assessment and Qualifications Alliance (AQA): www.aqa.org.uk
- Edexcel: www.edexcel.com/
- Oxford Cambridge and RSA Examinations (OCR): www.ocr.org.uk/index.html

Teaching externally assessed courses

All teachers have to think beyond the particular lesson they are teaching to the end of the unit of work, to ensure that pupils can respond successfully to any assessment scheduled to take place. When pupils are assessed externally, the same considerations apply; that is, how to maximise pupils' achievement. However, you do need to be fully aware of the nature of the external assessment for which you are preparing your pupils. It is important not just to teach to the examination, but to hold on to the principles of good classroom practice.

In preparing your pupils for external assessment, you need to be familiar not only with the subject content but also with the particular demands of the assessment process, such as the types of questions set and the language used in setting questions. Questions are set that often employ words with a specific meaning; for example, they ask candidates to describe, explain or summarise.

Candidates need to know what these command words mean in examination conditions. Task 6.2.2 is designed to help you become familiar with types of questions currently set in examination papers in your own subject and the corresponding reports of examiners.



Task 6.2.2 Using examination papers in your teaching

Collect a number of examination papers (for example, GCSE or the equivalent in other countries) for your subject together with the mark schemes and specification. Where possible, obtain the relevant examiners' reports. Read through the specification for the examination arrangements. Then address the questions in the paper in the following way:

- answer the questions on the paper yourself;
- mark your answers using the mark scheme;
- evaluate your answers and marking and identify the key knowledge and concepts needed to gain maximum marks. Look back at the examination questions, identify the key words and phrases most often used in the questions;
- use the examiners' report to review and refine your findings;
- identify any ideas that might be useful to consider in your day-to-day teaching;
- repeat the exercise for another year of the same paper, or repeat the exercise using papers from a different level (for example, post-16 course).

The completed task should be placed in your professional portfolio.

Once you are familiar with the structure and language used in past papers, you can then integrate this information into your teaching throughout the course. Another important aspect to consider is the development of study skills both in your lessons and throughout the school (see Task 6.2.3).



Task 6.2.3 League tables and pupil grouping

Performance tables ('league tables') have been said to influence not only what is taught in a school but also how pupils are grouped and the subjects they are offered for study, thus influencing their final qualifications. Investigate whether pupil grouping in the school is affected by the league tables; that is, if teaching and learning are being driven by public accountability. You may find sharing this work with another student teacher helpful.

Check first your understanding of setting, banding, streaming and mixed-ability grouping (see Unit 4.1).

In your placement school (see Unit 4.1 on pupil grouping):

1 for each year cohort 7–11, identify how tutor groups are formed for pastoral purposes and the criteria used to do this;
2 identify in broad terms which subjects reorganise the tutor groups for teaching purposes;
3 for your teaching subject and either English, Mathematics or science in Year 7, find out how pupils new to the school are assessed and placed into teaching groups. Compare the criteria used for each subject;

4 how flexible is the placement? Can pupils move between groups? If so, how is this movement managed in practice?
5 for your teaching subject and either English, Mathematics or Science in Year 10, identify how pupils are grouped for teaching, and which courses the different groups are offered, and the criteria used to do this.

With the information gained from tasks 1–5 above, arrange to interview a number of pupils in Year 10 to elicit their views about subject and course choice and grouping. Choose your subject area and, if time permits, one other subject. How do pupils make their subject/course choices at the end of Year 9? Are pupils:

1 provided with information to help make choices?
2 guided towards particular courses by teachers at the school?
3 aware of the long-term consequences of the courses they select?
4 encouraged to choose courses that help them to achieve five 'good' GCSE passes in English Baccalaureate subjects?

Within the same subject areas, find out what pupils think about the groups they are taught in. Are pupils aware of:

1 the reasons they were allocated to a group in that subject?
2 any advantages to the grouping system?
3 any limitations to their particular group, such as limit on grade at end of their course (for example, GCSE)?

Finally, interview a head of department for one or more subject areas chosen to find out their views about pupil choice of subject in Year 9 and how pupils are grouped for teaching purposes in their subject. You could use the questions you have put to the pupils.

Write a summary of your findings for discussion with your tutors, indicating whether the importance of league tables has had any effect on the way pupils are grouped. Evidence should be kept anonymous. The final document can be filed in your professional portfolio.

SUMMARY AND KEY POINTS

Every teacher has the responsibility to understand the requirements of external assessment and examinations. At the end of this unit, you should as a beginning teacher:

- Be familiar with the framework for external assessment and have an understanding of the nature and purposes of summative assessment.
- Understand that external examinations utilise aspects of both norm-referenced and criterion-referenced methods.
- Recognise that norm-referencing and criterion-referencing are factors used in discussions seeking to explain the steady increase in the proportion of candidates achieving 9–4 grades.

- Understand that changes recently introduced to all sectors of education regarding external assessments aim to promote the academic attainment of all pupils.
- Be aware that there are likely to be further innovations in approaches to assessment and, as long as there remains a political focus on raising standards in our schools, external assessment and examinations will maintain their present high profile and powerful influence on educational practice.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Child, S. and Ellis, P. (2021) *The What, Why and How of Assessment: A Guide for Teachers and School Leaders*, London: Sage.

Ofqual's YouTube Channel: https://www.youtube.com/c/OfqualGovUk/featured

A comprehensive resource with a range of videos, including a selection of FAQs for pupils and parents.

Wolf, A. (2015) *Review of Vocational Education: The Wolf Report: Recommendations Final Progress Report*, DFE-00070-2015, viewed 12 August 2024, from https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/405986/Wolf_Recommendations_Progress_Report_February_2015_v01.pdf, viewed 12 August 2024.

A comprehensive review of the recommendations set out by Wolf in 2015.

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6.3 Using assessment data effectively

Making better decisions for teaching and learning

Nikki Booth

Introduction

The term "data" can be considered one of the most common assessment-related words used in schools today. It covers a plethora of information for teachers ranging from scores, levels, or grades from national examinations to ongoing conversations with and observations of pupils. This unit covers a range of topics relating to the day-to-day use of data including how to make data management manageable, how an effective assessment system can support teachers in making more valid and reliable inferences about data gathered, and how formative assessment can enhance practice to make better decisions for teaching and learning. Whilst this unit offers a wide range of information regarding data, it is important that it is read in conjunction with your school's assessment and/or marking and feedback policy.

OBJECTIVES

At the end of this unit you should be able to:

- explain the issues surrounding data management and workload and how they can be overcome;
- explain the concept of validity in relation to assessment data;
- explain the concept of reliability in relation to assessment data;
- know how to enhance the quality of assessment data; and
- understand the role of formative assessment in eliciting high-quality data.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

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Making data manageable

Tackling excessive workload

In the report *Eliminating unnecessary workload associated with data management* (DfE, 2016) for England, the Independent Teacher Workload Review Group identified two key reasons why data management can hinder, rather than help, teaching and learning:

> Firstly, when the **purpose** of collecting data has not clearly identified how it will be used to improve outcomes. Secondly, when the **process** of collecting data is inefficient. For example, when it is duplicated or requires too much time to complete. In both cases, a flawed understanding of the **validity** of different types of data contributes to burdensome practice. (p. 6, bold words in original)

As a starting point, the report states that anyone who is involved in the production and use of data should start by having clear answers to three questions:

1. Am I clear on the purpose?
   Why is data being collected, and how will it help improve the quality of provision?
2. Is this the most efficient process?
   Have the workload implications been properly considered and is there a less burdensome way to collect, enter, analyse, interpret, and present the information?
3. Is the data valid?
   Does the data actually provide a reliable and defensible measure of educational attainment?

(DfE, 2016, p. 8)

By keeping the data management workload down to a manageable level, not only does it mean a more balanced home-to-work life, but it also allows you to focus on what you do best – teach great lessons! What is required, however, is a deeper discussion on some of the issues surrounding assessment data and how these might be problematised.

Now do Task 6.3.1.



Task 6.3.1 Investigating data management practices

Design a staff survey investigating the data management practices at the school you are currently at. Your survey should be a mixed-methods design (quantitative and qualitative data). If possible, organise focus-group interviews with individual staff members to support your survey data. Analyse your findings.

1 What key themes emerged from your data analysis?
2 Do these themes differ for different subject areas?
3 What effective practices have you found regarding data management across the school?
4 What less-effective practices have you found regarding data management across the school? What implications do these have on practice? Write a strategy on how these might be resolved.

Discuss your findings with your mentor(s).

Understanding the quality of assessment data

Validity

There is no such thing as a valid test. This is because "[o]ne validates, not a test, but an *interpretation of data arising from a specified procedure*" (Cronbach, 1971, p. 447, italics in original). In other words, validity can be seen as a property of the *inferences* that are made from the assessment information and not from the tests themselves. This is an important distinction because the same data collected from an assessment might be valid for one purpose but not others. Although there are approximately 150 different types of validity (Newton and Shaw, 2014), assessment data used to predict a pupil's future performance is a good example here.

Data used for predictive validity

Data aimed at predicting pupils' future achievement (for example, providing schools, teachers, parents, and pupils with GCSE target grades) is a complex mix of how different groups of pupils within the cohort performed previously (prior attainment) in national tests (for instance, in England, Key Stage 2 national tests – taken at age 11), their socio-economic status, as well as other information. The Fischer Family Trust (FFT) is one well-known example within the United Kingdom of an agency which, upon subscription, provides this information to schools. Predictions (for example, GCSE grades) provided by the FFT usually come with probable percentages. This means that each predicted grade for a pupil is accompanied by a percentage indicating the probability of the pupil achieving each grade. The higher the percentage, the more likely it is that they are going to achieve that particular grade according to the collection of information.

Although providing schools, teachers, parents, and pupils with predicted grades based on a range of factors can be useful, it can also be problematic. For example, since part of the mix of information contains data regarding, for example, Key Stage 2 national test results, this information only factors in a pupil's achievement in English and mathematics. This becomes an issue of validity for some subjects, such as music, art, or physical education; a pupil's previous achievements in these areas are not taken into consideration at all.

Further to this, there is also the question of the likelihood that each pupil will actually achieve their target grade. For example, in Education DataLab's (2015) oft-cited report, it was revealed that only 9% of pupils made the expected linear pathway through Key Stage 2 (ages 7–11), Key Stage 3 (ages 11–14), and Key Stage 4 (ages 14–16). When each Key Stage was analysed separately, it was found that 55% of pupils made the predicted linear progress to reach the Key Stage 2 standard that is anticipated from their Key Stage 1 (ages 5–7) achievement. The remaining 45%, however, either overperformed or underperformed. Furthermore, the accuracy of predicting a pupil's achievement fell in the secondary context; with 45% of learners making the expected linear progress between Key Stage 2 and Key Stage 3, and only 33% of them made the anticipated linear progress between Key Stage 3 and 4. Although the report's findings can be considered somewhat dated with the inclusion of National Curriculum levels and sub-levels, and letter names for GCSE grades, the important point from a predictive validity stance is that when attempting to use a model of linear progression to predict future achievement, most pupils will perform better or worse than their expected Key Stage achievement on one or two occasions, whereas some frequently overperform or underperform throughout their school career. This notion is represented diagrammatically in Figure 6.3.1.

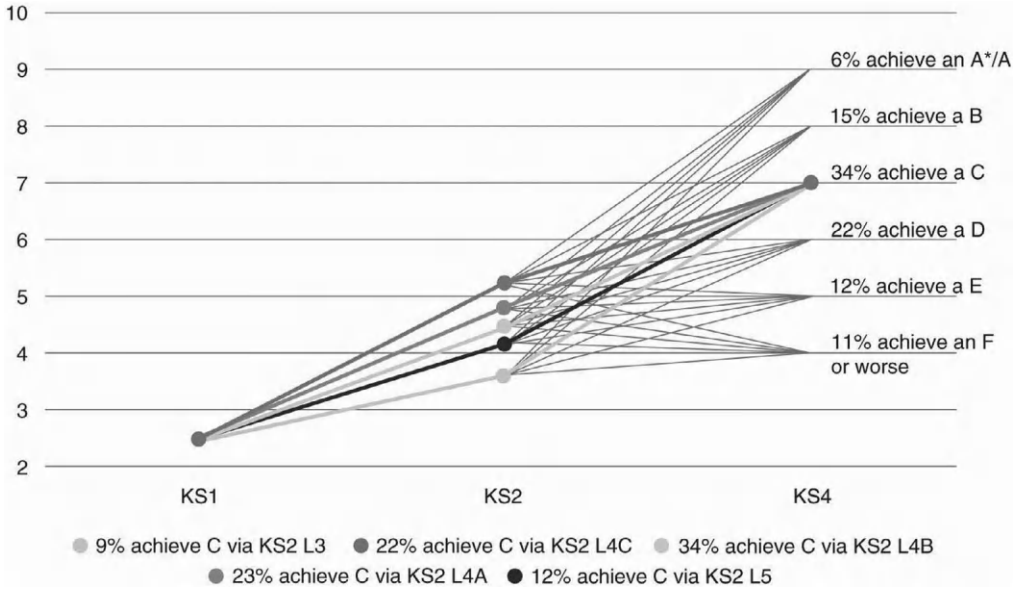


Figure 6.3.1 Model of consistent linear progression

Source: Image taken from Education DataLab (2015, p. 13), used with permission

Threats to validity

Exposing the threats to validity is important when discussing assessment data because it can affect the inferences that are made. There are two important threats worthy of discussion: *construct-irrelevant variance* and *construct under-representation*. A visual representation of these is shown in Figure 6.3.2.

Construct irrelevant-variance, as the name suggests, can be observed when a test is assessing irrelevant variants within the construct of interest (what is taught and learned). In other words, the assessment procedure is "too big" (Wiliam, 2020, p. 25). For example, in a mathematics test,

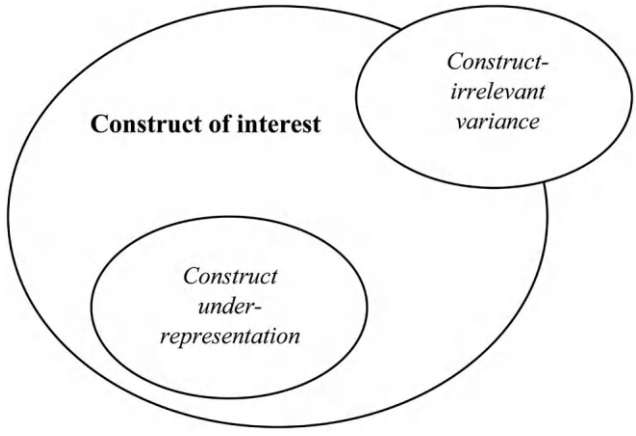


Figure 6.3.2 Threats to validity

the scores pupils attain should be related to their mathematical ability. This becomes problematic, however, when questions with a high reading demand affect the pupil's ability to answer the question. As such, attempting to make valid inferences from the assessment data as to how pupils did (summative assessment) and where teaching and learning should go next (formative assessment) would be difficult because pupils who scored low might have done so because of reading issues not necessarily *mathematical* ones.

Construct under-representation, also as the name suggests, can be seen when an assessment is under-representing the construct of interest. In other words, the assessment can be seen as "too small" (Wiliam, 2020, p. 25). For example, any in-school end-of-unit test or nationally standardised examination will only ever be able to assess a sample of the domain taught and learned (Koretz, 2017). This can be said to be an issue with all timed tests and examinations because "no test can cover all the learning that is set out in the curriculum" (Harlen, 2007, p. 23). As such, making valid inferences regarding how well pupils have learned, for example, the unit in the final test (summative assessment) and informing where teaching and learning should go next (formative assessment) could be problematic because the test has not been able to assess other equally important content taught and learned throughout the, say, six weeks.

Reliability

Reliability often refers to pupils' "consistency of outcomes that would be observed from an assessment process were it to be repeated" (Newton, 2009, p. 183). What this means is that, like with validity, there is no such thing as a reliable test and classifying pupils (by grading them, for example) will always consist of a certain amount of error. It is important to be open about these errors within educational assessment because they can have implications on the conclusions made about teaching and learning.

A Margin of Error considers the degree of uncertainty that a single test or examination might represent between pupils' "observed scores" (the scores gained from taking a test) and their "true scores" (the average of the individual's scores if the same test was administered several times) (Koretz, 2008). For example, after completing an end-of-unit test, Jon was awarded 93%, Jessica 90%, Shimra 88%, and Phil 87% and it was revealed that there was a margin of error of, say, +/- 3 marks. At face value it would appear that Jon is better than Phil, but when the margin of error is taken into account this may not necessarily be the case; Jon may well be as low as 90% and Phil as high as 90%, so we cannot conclusively say that Jon is better; he could, in fact, be equal to Phil. Furthermore, all of these pupils' scores might even be equal! The notion of the margin of error is also commented on by Wiliam (2017) who states:

> A typical school test would have a margin of error of 10 marks or more, and many, perhaps most, teachers would be uncomfortable saying to a parent that the pass mark for a course was 70, and their child scored 65, plus or minus 10. The parent might ask whether their child passed, and all we could say is, 'Probably not. But they may have done.' At this point the parent would probably ask, 'Why don't you know?' to which the only sensible reply would be, 'Because no test is perfectly reliable.' Parents may well ask, 'Can't you make the test more reliable?' and of course the answer is yes. We can make the test more reliable, but this involves making the test longer. Much longer – and we have better things to do with the time, such as teaching. (pp. 1–2)

The notion of the margin of error becomes somewhat problematic when, in the case of England, for example, GCSE grades are awarded to classify how well pupils performed, particularly for those who are close to the grade boundaries. This has been acknowledged by Ofqual who state: "[c]learly, candidates with scores near the grade boundaries are going to be more at risk of misclassification than those with scores in the middle of the band" (Ofqual, 2013, n.p.). Furthermore, Ofqual (2013) also reports that "the more grades that are available for an exam [for example, tiered versus non-tiered subjects], the greater the unreliability associated with the grade because the grade is covered by a smaller mark range and so is more susceptible to misclassification" (Ofqual, 2013). In short, "assessment is not an exact science, and we must stop presenting it as such" (Gipps, 1994, p. 167). Although the literature cited above might be considered somewhat dated, the important point, as Booth (2021) comments, is that we need to be more open and honest about the limitations of assessment and "every effort should be made to communicate this to all those involved in education" (2021, p. 160) including senior leaders, fellow teachers, learners, and parents/carers.

Enhancing the quality of assessment data

To help remedy some of the issues presented in the previous section, there is an argument to establish better assessment systems for enhancing the quality of assessment data.

Separating performance from learning

Figure 6.3.3 shows what could be considered a typical procedure for teaching and assessing pupils' learning using mathematics as an example. For instance, a unit of work (lasting several weeks) would be taught on Pythagoras followed by an end-of-unit test. Following this, a new unit of work on, say, statistics would be taught leading to a test, and so on with a third module. At a particular point in time, the teacher might be required to enter a grade into the school's assessment manager system (for example, SIMS) to determine where pupils are at in their learning. The inferences the teacher makes about pupils' learning could be considered problematic following this type of assessment procedure.

There is often confusion, in some schools, that when pupils perform well on an end-of-unit test this means that they have learned the material well. This is not necessarily the case. With reference to the mathematics example above, if the teacher has spent the last few weeks teaching

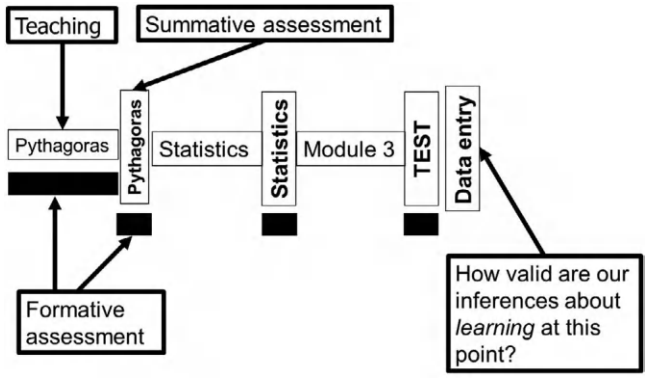


Figure 6.3.3 A blocked (or atomistic) approach to assessment

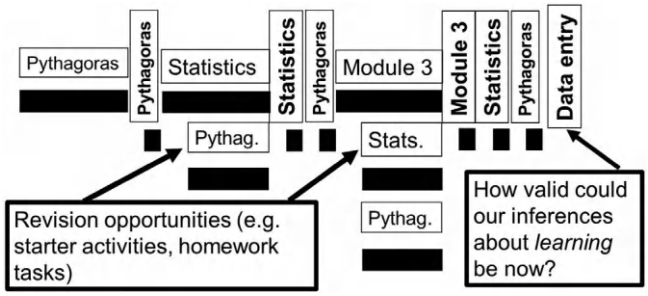


Figure 6.3.4 A more holistic (or synoptic) approach to assessment

pupils about Pythagoras, providing meaningful activities to practise, giving feedback which has been acted upon, and then tests the pupils, they are likely to do well. Some cognitive scientists (for instance, Bjork and Bjork, 1992) would consider this *performance* not *learning*. It is *performance* because the material has just been taught and is still likely to have high retrieval strength (how accessible, or retrievable, something is). If the teacher moves on to the next unit of work (statistics), and then re-tests the pupils on some Pythagoras content six weeks later, would they get the same result? This is unlikely; now, several weeks later, the Pythagoras material has not been practised so pupils' retrieval strength has become low. Furthermore, the storage strength (how well something is learned) is also low because some pupils cannot remember how to answer questions. Considering the separation of performance and learning can have huge implications on the notion of tests used, and the inferences made from the data, in schools. As such, a more holistic (or synoptic) assessment system is required. What this might look like is shown in Figure 6.3.4.

In an oft-cited article, Kirschner, Sweller, and Clark (2006) state that learning requires a change in long-term memory. They go on to say that, "[t]he aim of all instruction is to alter long-term memory. If nothing has changed in long-term memory, nothing has been learned" (2006, p. 77). Although learning is, indeed, very messy (Yardell, 2017) this is important; when you teach pupils new material, naturally forgetting the previous material is likely to occur. Meta-analysis research (for example, Cepeda et al., 2006) has shown, however, that the use of synoptic assessment (also called spaced practice or distributed practice) can have positive effects on summative tests/examinations and long-term learning. What this means in practice is that instead of solely testing the content covered within a single unit of work (as shown in Figure 6.3.3), the addition of questions covering previous material (as shown in Figure 6.3.4) allows pupils the opportunity to practise retrieving the information from their long-term memories. To be clear, this does not mean that tests have to be increasingly longer throughout the school year to account for the vast amount of content covered. It is a process to engage pupils in retrieval practice as a high-impact learning strategy, which then helps speed up the retrieval strength so the information is more quickly accessed in future, as well as increasing the likelihood that the information will develop higher storage strength for its potential future use. See Unit 4.3 on cognitive development, too.

Furthermore, when pupils are given regular opportunities to answer questions on previously taught content (for further examples see *inter alia* Hunton, 2018; Weinstein et al., 2019) the construct under-representation as a threat to the validity of assessment information (discussed earlier in this unit) is becoming remedied. Although this model for assessing will help you make better inferences about pupils' learning, there is also the need for high-quality formative assessment to gather meaningful evidence of learning on a lesson-by-lesson basis (see Unit 6.1).

Now do Task 6.3.2.

Task 6.3.2 Designing and implementing an effective assessment system to make better valid and reliable inferences about pupil learning

Design and implement a holistic (synoptic) assessment model (as shown in Figure 6.3.4) for a series of units of work you are going to teach. When designing the model, consider the following:

1 What will the sequence of your units of work be? Why in this order?
2 What opportunities will there be for pupils to retrieve previously taught material (e.g. starter activities, homework tasks etc.)?
3 What will your synoptic end-of-unit assessments look like?

When your holistic assessment model is being implemented:
During a series of lessons:

1 How have pupils responded to the opportunities to retrieve previously taught material?
2 Have these opportunities resulted in a change of direction for your lesson planning? Why/why not?

Following the end-of-unit tests:

3 What valid and reliable inferences can you draw from the assessment data about how your pupils' long-term learning?
4 What does this mean for where you are going next with this class? How might your teacher practice change as a result of the data gathered?

Discuss your model with you mentor(s) and your school's assessment lead. Retain a copy of your model for your professional development.

Enhancing formative assessment practice to elicit high-quality data

A full discussion on how effective formative assessment practice can support teaching and learning was dealt with in Unit 6.1. What was not discussed, however, is the role of teacher questioning. See Units 5.2 and 5.8 for teacher talk for learning, also.

Teacher questioning

Although asking pupils questions during the lesson is indeed an important formative pedagogical strategy (for more information on teacher questioning and classroom-based talk, see Booth, 2020) it can also be problematic due to its apparent construct under-representation. For example,

what tends to happen in many learning environments is the following sequence: the teacher asks a question, a small number of pupils raise their hands, the teacher chooses one pupil, the pupil gets the answer right, the teacher says "good", and then moves on with the rest of the lesson. Although there might be slight variants in this data collection sequence, the point is that the teacher cannot be sure about the learning of the other, say, 29 pupils. In other words, questioning in this format can be said to suffer from construct under-representation as a threat to making valid inferences about pupils' learning. This is important; as Harford (2021) makes clear, when interpreting any data "we should ask who is missing from the data we're being shown, and whether our conclusions might differ if they were included" (2021, p. 265). To be clear, this is not to say that teachers should stop asking questions to individual pupils; this is still an important concept for learning. What needs to happen more often, though, is for the teacher to build up an evidence base on where *all pupils* are in their current learning. This can be a powerful pedagogical strategy because better evidence collected, can mean better decisions, which leads to better learning (Wiliam, 2021). The point is that when teachers are doing this more frequently, they are able to make better inferences about pupil learning, establish whether the whole class is ready to move on, and, where needed, intervene swiftly. There are some useful formative strategies that can be used to support this.

"Exit tickets"

At the end of a lesson, a teacher could use an "exit ticket" to check whether pupils have understood the lesson. In practice this means that, at the end of the session, pupils respond to a question or small number of questions relating to the learning intention(s). When the exit tickets are collected in, the teacher looks at the pupils' responses and, crucially, uses this information to inform their planning for the next lesson. This quick check is not only a strategy that can be useful for reflecting on whether teaching has had the desired effect or not, but in relation to managing workloads, also helps reduce the amount of marking and individual feedback.

"Hinge-point" questions

It is also highly beneficial to check whether learning is heading in the right direction *before* the end of the lesson. Effective strategies for this could include formative conversations with pupils working in groups, for example, where teacher-initiated oral feedback would impact on all members. Depending on the length of the lesson, groups may well receive at least two opportunities of teacher feedback to respond to. Similarly, in some lessons, the use of quick diagnostic "hinge questions" (see Wiliam, 2018) which require every pupil to respond would elicit useful data before proceeding with the lesson. When 100% of pupils respond to a teacher question, via a mini-whiteboard, for example (or a poll in an online learning context), and show their response, they are showing whether they have understood the concept or whether they need further support. Regardless of whether it is one way, the other, or both, it is at this point that formative assessment takes place; not only has data been collected (i.e. I am looking around the room to see how many pupils have got this), but, perhaps more importantly, the data then drives decision-making as to what happens next. As such, it is this teacher decision-making which drives pupil achievement.

Now do Task 6.3.3.



Task 6.3.3 Using data effectively to promote quality teaching and learning

Collect a range of data for a class you are going to teach and use this information to inform your planning.

Before the lesson:

- What do the target grades and any prior attainment data tell you about this class? How can you use this data (as well as other data including those who are SEN or EAL etc.) effectively to plan for differing learning needs?

During the lesson:

- What sorts of data can you collect during the lesson that will inform whether learning is on track? What will you do if you identify a group of pupils who have not quite got it yet?

End of the lesson:

- How will you check that the Learning Intention(s) for the lesson have been met by all pupils?

Post-lesson:

- What does the range of information you gathered both during and at the end of the lesson tell you about where to start with these pupils next time?

Record your analysis in your portfolio and repeat the exercise periodically with different classes.

SUMMARY AND KEY POINTS

At the end of this unit, you should now have a greater understanding of how to use assessment data effectively, to enable you to make better decisions for teaching and learning. Remember:

- Key questions for tackling the data management workload problem include:
  - Am I clear on the purpose?
  - Is this the most efficient process?
  - Is the data valid?
- There is no such thing as a valid test.
  - Whilst target grades are often given and used in schools these are also problematic due to the information they are based on.

- Two important threats to validity are:
  - *Construct-irrelevant variance* (when an assessment procedure is "too big"); and
  - *Construct under-representation* (when an assessment procedure is "too small").
- There is no such thing as a reliable test.
  - A "margin of error" considers the degree of uncertainty that a single test or examination might represent in pupils' scores.
- There is often a confusion, in some schools, between *performance* and *learning*:
  - *performance* relates to how a pupil has done on a test on a particular day; whereas
  - *learning* is seen, by some, as a change in long-term memory.
- Teachers can make more valid and reliable inferences about pupils' learning when a holistic (or synoptic) assessment model is used.
- Teacher questioning can be said to suffer from construct under-representation.
  - This can be remedied through using 100% pupil response systems including, for example, "hinge-point" questions and "exit passes".

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

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Further resources

Independent Teacher Workload Review Group (ITWRG) (2016) *Eliminating unnecessary workload associated with data management*. Available from: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/511258/Eliminating-unnecessary-workload-associated-with-data-management.pdf, viewed 25 July 2025.

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A comprehensive deconstruction of the complexities of educational assessment and testing.

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An in-depth book on how scores from educational tests can be misunderstood and misused.

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7 The school, curriculum and society

If education is 'what is left after most of what you have learned in school is forgotten', a quote attributed to Einstein, then what is education for and who decides? This chapter takes you away from the immediacy of teaching to consider these fundamental questions about the aims of education and how the curriculum reflects those aims.

Young people spend a substantial part of their formative years in school, and a significant slice of the national budget is channelled into education. In the day-to-day urgency of teaching, it is easy to push the 'why' into the background and simply get on with the 'how'. However, as a beginning teacher you can expect to be held accountable for what you teach. You need to be able to explain to pupils and parents the benefits of learning your subject, and understand why your school and country approach education in particular ways while other nations take different approaches.

This accountability extends beyond simply delivering subject content. As a professional educator, you need a broader understanding of the educational landscape: why certain decisions are made about curriculum content, how different stakeholders influence these decisions and how your practice fits within wider educational, social and political contexts. Teachers who understand these frameworks are better equipped to justify their pedagogical choices, engage meaningfully with educational debates and adapt their practice as systems evolve throughout their careers.

Crucially, this understanding enhances your teacher agency – your capacity to act purposefully and reflectively within your professional context. While curriculum frameworks and policies may appear constraining, teachers with deeper knowledge of educational systems are better positioned to exercise professional judgement, identify spaces for innovation and influence change from within. Understanding the 'why' behind educational decisions empowers you to be an active agent in your profession rather than a passive implementer.

Since devolution in 1999, the education systems across the UK have increasingly diverged, with each nation developing distinct approaches to curriculum, assessment and teaching standards. This creates both opportunities for innovation and challenges for teacher mobility and research sharing across the UK.

The chapter addresses how the four nations organise their education systems and the different emphasis they place on various aspects of education: the types of schools and grouping of pupils, national assessments and the role of public examinations, and inspection and accountability arrangements.

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Unit 7.1, 'Aims of education', takes a comparative and analytical approach to examine assumptions about education and the different purposes of schooling. It discusses the origin of national aims and how they translate into school aims, contrasting the aims of society with those of the individual.

Unit 7.2, 'The secondary school curriculum', examines the secondary school curriculum in terms of aims and purposes. The unit draws attention to the formal and informal (or hidden) curriculum, discusses how the formal curriculum develops from broad educational aims, and asks 'Who decides?'. The debates around decolonising the curriculum are also explored.

Unit 7.3, 'Secondary schools in England', examines the structure and governance of education in England, including distinctive features of the English system, the National Curriculum, assessment arrangements and teacher education requirements.

Unit 7.4, 'Secondary schools in Northern Ireland', focuses on the unique characteristics of Northern Ireland's education system, including its segregated school structure, the Northern Ireland Curriculum, academic selection issues and teaching profession requirements.

Unit 7.5, 'Secondary schools in Scotland', explores Scotland's distinctive approach to education, including 'Curriculum for Excellence', the structure of schooling, professional standards for teachers and ongoing educational reform.

Unit 7.6, 'Secondary schools in Wales', examines the Welsh education system following devolution, including implementation of the new 'Curriculum for Wales', bilingual education and initial teacher education provision.

As you develop as a teacher, this comparative understanding will help you appreciate why certain decisions are made about curriculum content, teaching methods and assessment practices in your particular context, and how you might contribute to shaping educational experiences and outcomes.



ONLINE RESOURCES

Watch the explainer video on the similarities and differences between approaches to the curriculum across the UK by scanning the QR code or visit routledgelearning.com/learningtoteach



7.1 Aims of education

Ruth Heilbronn and Janet Orchard

Introduction

Schools are busy places which place many demands on teachers. Nevertheless, it is still important to find time to step back from the daily routine at various points in a teaching career to reflect on *why* teaching matters. Teachers, whether at parents' consultations or addressing challenging questions in classrooms from disaffected pupils, find themselves needing to justify what they're doing. Furthermore, teachers come into teaching motivated by certain values which may not be the same for everybody. You might ask yourself where you stand regarding the aims of the curriculum for the context in which you are planning to teach and this unit is written in the spirit of engaging with reflection on the values and aims of education.

While you may have your own views about the aims of the curriculum you are required as a teacher to follow the curriculum for the context in which you are teaching. Schools will determine aims and values which they expect their teachers to follow. In addition, you can expect to have some broader aims, for example, in England where they are included in the national curriculum. The most recently revised statement is from 2013 as follows:

> The national curriculum provides pupils with an introduction to the essential knowledge they need to be educated citizens. It introduces pupils to the best that has been thought and said, and helps engender an appreciation of human creativity and achievement.
>
> (DfE, 2013, section 3.1)

Do you agree? Why? Why not? Thinking through such questions has both a practical and intellectual value, for example, when the going gets tough. At such times, teachers may consider the point of it all and wonder whether teaching is for them. Aims matter. Personal clarity at this deeper level as to what you are trying to achieve as a teacher and why, can help you to decide whether or not to keep going.

This unit considers the aims of education in some detail, demonstrating why this is not a matter that can be completely side-lined by practical concerns about *how* to teach. The 'how' and

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'why' are inter-related. For example, in secondary schools you may need to justify the aims of schooling and the subjects you teach in accessible language, when anybody asks, 'What's the point of this?'

OBJECTIVES

At the end of this unit you should be able to:

- list a variety of actual and possible aims for education;
- reflect on and formulate your own aims in being a teacher;
- discuss aims of education with other teachers and with parents.

You should also check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Thinking about aims

Beginning teachers work in contexts in which many expectations about aims are already in place, whether explicitly stated or not. You are not free to pursue any aims you like. You might wonder then whether there is much point in thinking about the aims of education. This unit assumes that good teachers *will* consider the aims of education and know what they endorse.

We suggest two reasons why your own thinking about aims is relevant and you may well think of others. First, your own thinking about aims influences the way you teach your subject, given the constraints of your individual school, curriculum and circumstances. Second, while you have the same right as any other citizen to form and express your own view about the aims of education in general, being a professional teacher with a duty of care towards young people, you are in a better position than others to make your views clear and should be prepared to argue for them responsibly.

You may find the following established distinctions in the kinds of possible aims helpful. Are they intrinsic, extrinsic, instrumental or a combination? The aims of education might be directed towards things of intrinsic value where, regardless of any practical benefit, they bring satisfaction for their own sake, for example, the love of reading or solving logical puzzles. By contrast, an extrinsic aim will be concerned with doing something either for external rewards or to avoid negative consequences; one educational example might be to pass the exams at the grades required to achieve a place at university; or to access jobs with certain qualifications. Instrumental aims are pragmatic reasons for gaining knowledge and skills, whether these are individual (e.g. to earn a living) or societal (e.g. which prepare people for the world of work) including learning to live together constructively in schools and communities.

Bear in mind that education and schooling are not the same thing. Education is a broader idea than schooling, derived from the Latin word '*educare*', which may mean either 'to draw out' or 'to train or to mould'. There are both formal and informal ways of being educated, with informal kinds including learning from peers, life experiences, independent reading and engagement with other kinds of 'text', including social media and film. Schooling refers specifically to formal education taking place in institutions, i.e. 'schools'. Now complete Tasks 7.1.1 and 7.1.2, which encourage reflection on the nature and variety of aims in education.



Task 7.1.1 School aims: a comparison

1 What do you consider the aims of education in general to be? Some prompt questions may help:

'What is education for?'
'Who is education for?'
'What kind of person do we wish to develop and why?'

2 Think about two schools you know, e.g. the school in which you (or perhaps your children) received your secondary education (or most of it) and your placement school.

Answer the following questions as far as possible:

- Did your school have an explicit aims statement?
- Were you as a pupil aware of the school's aims?
- Roughly what kinds of educational aims were they (e.g. academic learning, promoting community relations)? In what ways did the aims of your school impinge on your experience as a pupil?

For your placement school, ask:

- Does your school have an explicit statement of aims – if so, what does it say?
- Are the pupils you're teaching aware of the school's aims?
- Does the existence of these aims appear to make any difference to the pupils' experience in the school?

Compare your findings for the two schools. Do you find that aims have a higher profile in one school or the other? Is there evidence that the existence of an explicit policy on aims enhances the education the school is providing?

3 Compare your answer to question 1 on the wider aims of education to your answers to the specific school aims you identified in question 2. Are they divergent or similar and what does this imply?

Keep your notes in your professional portfolio.



Task 7.1.2 The aims statement in your placement school

1 If your placement school has an aims statement, evaluate how far the aims of the school fit with your own view of the broader aims of education. Refer back to the questions you answered in 7.1.1. The statement might be in the form of a 'vision' or a 'mission' statement.

2 What do pupils understand the aims of education to be? Reflect on comments pupils may have made and discuss with other beginning teachers.

Store this information in your professional portfolio.

The social and political context of aims

Whether or not explicitly stated, education systems are governed by underlying aims, decided at different political or administrative levels. Most countries have a national education system that is at least partly state-funded and state-controlled, although exact arrangements vary from one jurisdiction to another. For example, the People's Republic of China, the largest education system in the world, has a state-run system, with relatively few private providers of schools; and is increasingly decentralised. The Ministry of Education monitors provision at a macro level, steering education reform via laws, plans, budget allocation, information services, policy guidance, while local governments take primary responsibility for governance and delivery at a micro level.

By contrast, England offers a mixed picture of state funded, partially funded and independently funded or private schools with different level of responsibilities depending on governance arrangements. Many English state-maintained schools are now part of a Multi-Academy Trust (MAT) arrangement with one or two areas (for example, Hampshire), retaining a local education authority.

Even where governance is devolved, schools across a country's education system tend to draw on a widely shared sense of what the aims of education should be, revealing certain national and cultural values about education. In the early part of the twentieth century in the US the problem of reconciling different beliefs about the aims of schooling led to an influential argument that a national education system ought to forge a single nation out of its diverse indigenous and migrant communities, what John Dewey famously promoted as the 'common school' (Dewey, 1916).

By contrast, through much of the twentieth century in Britain, schools have enjoyed relative autonomy in setting and pursuing their aims. Local Authorities (LAs) in England had some control over the curriculum until changes to school governance from the 1980s onwards reduced their involvement. The National Curriculum was introduced in the 1988 Education Reform Act (ERA) since when it has been revised, e.g. 2010 and 2014 when it was last updated. Further moves to decentralise since 2010 have encouraged groups of academy schools to organise on the model of charter schools established in the US and Sweden. The national curriculum and its aims remain statutory for local authority schools, but the content has been slimmed down considerably in 2014 compared with earlier versions.

The social and political context of aims also proves significant in the way in which governments use the school curriculum to enforce society's values. Hence in England today, the aim to promote common values through schools is seen in the requirement to demonstrate and promote values that are essential to liberal democracies, requiring 'Fundamental British Values' to be taught (DfE, 2014). This has generated considerable comment, on the grounds that the values promoted seem not exclusively those of one particular nation, but rather general values concerning democracy and 'the rule of law' (e.g. Farini, 2019; Healy, 2019; Winter and Mills, 2020; James, 2022). Regardless, the requirement itself illustrates how at certain times governments identify values which they insist need to be reiterated and reinforced for the greater good and believe schools are places where this should happen. (See Unit 4.5 on values in education.)

Statements about 'fundamental British values' offer one example of a prompt for professional reflection about aims and values. These statements, commonly available on school websites, illustrate how government-led imperatives are interpreted and enacted by schools, including out of school activities such as sporting events and visits. They might reference preparing pupils for life in modern Britain, envisaging the curriculum as a vehicle for understanding the values related to democracy and the rule of law. Some people may support these aims; others may support the

sentiments but question the extent to which they are typically 'British'; others again may question whether a duty of this kind is one best left to families and communities. Your own views will reflect a position on the aims of education, informed by your educational values and principles.

In short, aims for education and schooling do not arise in a vacuum. Statements regarding the aims of education and what should be taught in school form part of a long-standing debate about who ought to determine aims for state-funded schools and what they should be. (See Reiss and White (2013) and Marples (2014) for a fuller discussion of the aims of education.) One common view is that schools could be governed far less by a national imposition of aims and focused instead on devising and making clear their own aims, respectful of the wishes of local parents and relevant to pupils. Others challenge this, concerned for the welfare of children and young people unless standard entitlements are nationally agreed, rather than decided on at a community level. Do you think that there are 'right' curriculum aims for the national context in which you are situated? Should these be insisted on for the equal benefit of all? Or do you think the aims of education are a matter of opinion to be determined freely, depending on context and circumstances? Is your view somewhere in between? We avoid here the radical claim that the state has no role to play in determining the aims of education and suggest there are at least some which all schools should follow. That being the case, what should these be?

There has been a shift in England since 2010 towards a more traditionally academic curriculum directed by national policy makers, which some citizens have welcomed, supporting their case with appeals to the work of E.D. Hirsch's idea of the importance of 'cultural literacy' and 'core knowledge' that every school pupil should know (Core Knowledge Foundation, 2021). Others have made a more progressive argument for academic learning by arguing for 'powerful knowledge', which all children and young people should encounter. Young (2013) argues that 'curriculum theory must begin not from the learner but from the learner's entitlement to knowledge' (p. 101). Transmitting this knowledge aims to empower learners (see Young, 2013; Eaglestone, 2021; Muller, 2023).

A further shift since 2010 has been made towards testing attainment through standardised tests, implying that the success of a school is best judged in relation to knowledge acquired by pupils which they can demonstrate in examinations and tests. Others have questioned the wisdom of this development, concerned that by focusing too intensively on academic attainment measured by examination results, we risk losing sight of other potentially valuable educational aims, including promoting 'wellbeing' or human flourishing. An exclusive focus on targets and examination results can be said to side-line the wider aims of education, such as preparing young people to take their place as future citizens (Ravitch, 2013; Ayers, 2020; Wahlström 2020; Norwich, Black and Norwich, 2022). Some other wider aims which schools may include might be developing critical skills and digital literacy to arm pupils with the ability to discriminate on information sources. An example which arose during the Covid-19 pandemic was the prevalence of anti-vaccination propaganda masquerading as scientific knowledge in social media.

While schools are judged by inspection regimes and required to compete for a place on the national league tables according to specified criteria enabling governments to make comparisons, it isn't clear that schools do in fact enjoy much autonomy. This in turn limits the degree of professional freedom teachers can expect to exercise. You will need to be clear where you stand on educational debates of all kinds, including those concerned with the aims of education, and be prepared to understand others and defend your own views with recourse to reasonable arguments.

Politicians may not often explicitly express their views on the aims of education. However, political statements frequently reflect educational aims implicitly; for example, when politicians say that

schools should enable Britain to compete economically with other nations, or that schools should inculcate 'British values' or should promote active citizenship. In each case, they are recommending certain aims. At this broad level in political rhetoric, it is generally assumed that the same aims are shared by schools, creating a potential tension between the possibility of a diversity of aims in different schools to reflect the needs of particular communities and the promotion of common aims across a whole school system.

Societal and individual aims

Government education policies are inevitably influenced by political priorities which tend to be driven by the promotion of economic stability and growth. Parents and pupils may see the primary point of going to school to be getting qualifications for employment or further study. These instrumental aims are important but as we have already pointed out there is room for other aims, therefore other conceptions of education. Indeed, there are problems with tying educational aims to 'economic growth' as this may not continue year on year. For example, the global pandemic of 2020–2021 severely impacted worldwide economic prosperity (UK Parliament, 2021). We may argue from this that a government's responsibilities to promote education must extend beyond a concern with material conditions and there is some evidence to suggest that, after a minimal level, increasing affluence does not produce increasing happiness (James, 2007). Governments also have a responsibility to protect and promote citizens' wellbeing in a broader sense, although policies do not spell such matters out; teachers need to be able to think about these more precisely.

Governments also have a responsibility to consider the ways, both positive and negative, that people's lives impinge on each other, and this implies an explicit concern with values education (see Unit 4.5). We might say that education should do more than enable people to gain qualifications and 'transferable skills' for the global economy, because qualifications are only part of the preparation for becoming an adult in any society. As technological changes are bringing about radical social change (Johnson and Wetmore, 2021; Volti and Croissant, 2024), we can no longer predict what kind of employment and challenges young people will face as adults. It follows that education might need to be broadly based, enabling people to adapt what they know and enjoy what they are able to do, as preparation for life in uncertain times. As well as a core of skills, a deep familiarity with the arts and humanities and a critical and curious approach to problem solving and discussion might be valuable aspects of education alongside more instrumental, economic ones (Greene, 1981; Nussbaum, 2010; Eaglestone, 2021; Donahue and Stuart, 2024).

On this approach, the aims of education might be broadly considered starting with the question: 'what should count as an educated young person in this day and age?', i.e. to ask questions about which human qualities we wish to nurture and develop and how education may foster them (Oakeshott, 1972). These might cover the knowledge and understanding required for work, but also for managing one's life and relationships. It is important too that people develop a practical capacity and an ability to make sensible and grounded decisions, given changing economic and social conditions. 'Moral seriousness' (Haydon, 2007; Pring, 2012; Greenwell, 2023) is another quality that has often been highlighted as important for both individuals as well as wider society. This might involve having a sense of responsibility for the community and include kindness and respect towards others, highlighted by the Covid-19 pandemic in such actions as wearing a face covering to protect others. On a wider scale, how we best live in a time of climate change/emergency and the implications for schools and the curriculum is another issue of moral seriousness.

Often, and perhaps increasingly, the language of 'skills' and knowledge tend to dominate (Ball, 2003; Gleeson and Husbands, 2004). This is not to question the value of education that leads to skills and subject knowledge; indeed, there are pragmatic reasons for gaining knowledge and skills which may be individual (including earning a living) or societal (including the filling of occupational roles that the society needs). There is also an intrinsic value to some knowledge and skills: it may just be interesting to know about certain things, or satisfying to be able to do certain things, regardless of practical benefit. (See The Organisation for Economic Cooperation and Development (OECD), 2020 for an international perspective.)

It is arguable that education should be at least partly about giving people such intrinsic sources of satisfaction. However, it is questionable whether all the capacities and qualities we might want pupils to develop can be captured by the idea of skills and knowledge alone (see Green, 2011; Pring, 2012). We also need to consider the kinds of qualities and dispositions we think pupils need to develop.

There are then individual aims of education and societal aims. One example of where they converge is health education where a school may influence a person's chances of having a healthy life, for example, by promoting healthy eating. Reflecting on the aims of education is important, to enable us to understand how far and in what way the school curriculum has a special role to play in promoting wellbeing and limiting negative effects for the individual and society. This is another example of why aims matter and why reflecting on aims is important. There are political and educational choices to be made about resources of time, money and the role of schools and teachers which follow in these cases. What sort of society a person lives in may be largely outside the individual's control, but education may influence how far someone fits into that society or may seek to change it. Now complete Task 7.1.3.



Task 7.1.3 Individual and societal aims

For individual reflection and group discussion.

(a) There are some combinations of individual and societal aims that seem to fit neatly together: e.g. the societal aim of a society of full employment and the individual aim of giving each individual the knowledge and skills to equip them to do one of the jobs available.

(b) There are other combinations where the relationship between societal and individual aims may be problematic: e.g. a societal aim in one characterised by traditional ways of doing things, and an individual aim of young people being independent and critical.

Can you think of other combinations of societal and individual aims that seem to be 'made for each other'? And other ways in which there may be conflicts between societal and individual aims?

- Do you think societal aims should always have priority over individual ones? Or vice versa?
- Do you think that it is easier for societal and individual aims to fit together in some kinds of society than in others? (You could think, for instance, about how authoritarian or democratic a society is).

Make notes of your discussion and summarise your views about aims. Keep your notes in your professional portfolio.

Recognising the diversity of aims

In a free society like England the differences in people's views on the aims of education are legitimate, there being no 'right' answers. This can make a practical difference to how:

- a whole school system is organised, e.g. the movement towards comprehensive education which began in the 1960s was driven at least partly by an explicit aim to break down class barriers and distribute opportunities for good education more widely. (The debate about whether or not it is desirable to have selection by ability in education continues.)
- an individual school is run, e.g. various aspects of a school's ethos and organisation may be motivated by the aim that pupils should respect and tolerate each other's differences.
- curriculum content is selected and taught (see Unit 7.2).

This then raises questions such as on how much overall coherence and agreement about aims is needed; how far there is room for internal variation; are some views unacceptable? For example, in some parts of the world there is one official textbook to support the curriculum and which everybody follows. Attempts to introduce this to England, where no such tradition previously existed, in the early part of the twenty-first century, met with resistance from other educational stakeholders.

How do philosophers seek to navigate these differences and to come up with arguments to justify some aims over others? One way of dealing with this difficulty is to argue that certain aims are fundamental to the concept of education itself. One argument is that education can be understood as initiation into practices relating to being human in the world (Peters, 1964; Oakeshott, 1972; Pring 2012; Cuypers, 2021; Huen, 2023). This differentiates 'education' from 'training'.

Training has been defined as *conditioning* in contrast with *education,* which is about developing autonomy (Thiessen, 1985; Hocutt, 2005; Snook, 2010). Teaching may be connected to training when it imparts a certain mode of thinking, or a canon, which cannot be criticised. The distinction between 'education' and 'training' is a live issue, returning to the question about whether, or not schooling is largely about passing exams or being educated into how to live well with others in community.

Perhaps there are certain beliefs or habits or skills that teachers should be trying to inculcate in their pupils in an unquestioning way, but this needs to be justified. Conversely, perhaps teachers should be aiming to turn pupils into autonomous adults, who can think critically about everything they are told and make up their own minds about what to think and do. Again, if so, why?

Given considerable diversity in views about aims it might be helpful to identify some common ground as far as possible. One fundamental and common view of education is that it should in some way be improving the quality of individuals' lives or of society in general, otherwise, why bother? This general aim leaves room for more specific discussions over what priority to put on aims for each individual, as against aims intended to improve society.

Equal aims for everyone?

Through much of the history of education, it would have been an unquestioned assumption that the aims of education should be different for different people. For example, in Victorian Britain the expansion of education was driven in part by the aim that the mass of the population should be educated to form a productive workforce, but not so well educated that they might rebel against

the differently educated ruling classes. Until quite recently, it was usual for the education of boys to have different aims from the education of girls: boys for certain sorts of occupation, dependant on their social position; girls for home-making futures. In the mid-twentieth century in England, selection for different schools was routinely by ability through an 11+ test which required children to have access to knowledge and skills (not to be confused with ability) and there were different aims behind the education offered in different types of schools: secondary modern, technical and grammar.

Despite these examples, today it is largely undisputed that the aims of education should be the same for everyone, which underlies developments in the promotion of equal opportunities, e.g. the debate stemming from the Warnock Report (DES, 1978) over the integration of pupils defined as having Special Education Needs (SEN) in mainstream schools. (See Unit 4.6 for the current position on SEND.) Today, when people argue for single-sex schooling or for dividing teaching groups according to gender, it is usually on equal opportunities terms. (See Unit 4.4 for further discussion of gender issues.) At the same time, group members may have special aims they would like to see pursued for their own children, for example, in the case of parents choosing a school of a religious character for whom bringing children up within the faith of their community while at school is a pressing priority. These examples illustrate the potential for controversy when aims are considered in more detail and why more general statements are useful, even if they appear at first sight rather bland, even platitudinous. Unit 7.2 raises several questions about the relationship between stated aims and curriculum content in this regard.

To conclude the present unit, we suggest you address Task 7.1.4 that asks you to reflect on any single published statement of aims.



Task 7.1.4 Aims and education – a statement of aims

Take any statement of educational aims – a former statement, one from another country, or your school, or a particular subject, or from the curriculum for schools as developed by the governments in England, Northern Ireland, Scotland or Wales – see Units 7.3, 7.4, 7.5 and 7.6 on this book's website and the Further Resources at the end of the unit.

Analyse and evaluate the statement addressing questions such as:

- Does the statement comprise a list of discrete aims in no order of priority, or does it convey that some aims have priority over others?
- Where there are subordinate aims under a broader aim, is there a clear rationale for the particular subordinate aims listed?
- Do the aims listed seem to you to fit together into a coherent idea of what education is about?

If the answer to the last question is 'Yes', can you write a sentence summarising this conception of education?

Finally, consider whether there is anything in this statement of aims that you would not have expected to see. Is anything likely to prove controversial? Is anything omitted that should be mentioned?

Discuss your responses with other beginning teachers, or with your mentor.

SUMMARY AND KEY POINTS

We hope this unit has helped you to think about the aims of education. This is similar to the summary box for all the other units.

- In working as a teacher, you necessarily have some aims, and these are more likely to be coherent and defensible if you have thought them through.
- At the same time, you are operating within the context of aims set by others.
- Aims can exist at different levels, local and national.
- In England in 2014, there was a revision of the National Curriculum that enabled schools to plan much of their own curricula. The question of aims is therefore open to discussion where planning occurs, and this leaves room for you to form your own view on the most important priorities for education, and to discuss with others how these aims can best be realised.
- It is always possible to raise questions about the justification of educational aims. Ultimately, our aims for education rest on our values – our conceptions of what makes for a good life both for individuals and for our society as a whole.
- Because we do not share all of our values with each other, there is always room for debate about the aims of education.

Check which requirements for your ITE programme you have addressed through this unit.

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Further resources

Carr, D. (2023) 'Personal and Professional Values in Teaching', in Lovat, T., Toomey, R., Clement, N. and Dally, K. (eds.) *Second International Research Handbook on Values Education and Student Wellbeing*. Springer International Handbooks of Education. Cham: Springer. https://doi.org/10.1007/978-3-031-24420-9_20

A chapter written by a noted philosopher of education, exploring the conceptual and ethical complexities of what it means to be a good teacher. Carr claims that many of the so-called technical aspects of teaching may be fundamentally based on teachers' characters and values. He acknowledges that teaching is a deeply contested notion and its aims should be debated and analysed.

Greene, M. (1995) *Releasing the Imagination: Essays on Education, the Arts, and Social Change*. The Jossey-Bass Education Series.

This has become a classic text most relevant to our times as it champions education through the arts that fosters what Greene calls social imagination – the capacity 'to invent visions of what should be and what might be in our deficit society, on the streets where we live, in our schools' (p. 5).

Marples, R. (2014) 'What Is Education For?', in R. Bailey (ed.) *The Philosophy of Education: An Introduction*. London: Bloomsbury Publishing.

This chapter discusses three curriculum theories relating to the aims of education, i.e. curricula based on an academic curriculum, a vocational curriculum and one in which pupil wellbeing is fore-fronted.

Reiss, M. and White, J. (2013) *An Aims Based Curriculum*. London: IoE Press.

This book, relevant to both this unit and Unit 7.2, argues that curriculum design should start from a consideration of aims rather than from the more familiar approach based on subject divisions. It starts by identifying some possible overarching aims of education that might equip learners to lead personally fulfilling lives and help others to do so too. From these, more specific aims are derived, which cover the personal qualities, skills and understanding needed for a life of personal, civic and vocational wellbeing.

Sundberg, D., Nilholm, C., Sundberg, D., Forsberg, E., Hirsh, A. and Román, H. (2021) 'The Aims and Meaning of Teaching as Reflected in High-impact Reviews of Teaching Research', *Teaching and Teacher Education*, 107, p. 103488, https://doi.org/10.1016/j.tate.2021.103488

This paper reviews the aims and meaning of education and teaching as a contested field. The authors draw on 75 reviews of research about teaching and discern 4 types of aims – knowledge/cognitive aims, social aims, aims encompassing the development of personal characteristics, and democratic aims. The implications of the empirical findings are discussed with relation to John Dewey's philosophy of education.

Young, M., Lambert, D., Roberts, C. and Roberts, M. (2014) *Knowledge and the Future School: Curriculum and Social Justice*. London: Bloomsbury.

The authors explain recent ideas in the sociology of educational knowledge, particularly drawing on Michael Young's research with Johan Muller. The book explicates three models of the curriculum in terms of their assumptions about knowledge and link the idea of 'powerful knowledge' for all pupils as a curriculum principle for any school. The question of knowledge, it is argued, is closely tied to the issue of social justice.

National Curriculum documents

England (2024) *National Curriculum in England: Framework for Key Stages 1 to 4*. Available at: https://www.gov.uk/government/publications/national-curriculum-in-england-framework-for-key-stages-1-to-4 (Viewed: 31 July 2025).

Northern Ireland (2024) *Curriculum*. Council for the Curriculum, Examinations and Assessment (CCEA). Available at: https://ccea.org.uk/about/what-we-do/curriculum (Viewed: 31 July 2025).

Scotland (2024) *Curriculum for Excellence*. Available at: http://www.gov.scot/Topics/Education/Schools/curriculum (Viewed: 31 July 2025).

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7.2 The secondary school curriculum

Elizabeth Hidson and Sarah Younie

Introduction

Unit 7.1 asks you to consider the aims of the school-based education system. These aims then directly influence decisions about the school curriculum in terms of the content of lessons and what a society considers their young people need to know and be able to do.

This unit considers how teachers might engage in shaping the curriculum. In order to do this, it is first worthwhile to take a step back and examine what is meant by 'the curriculum', its centrality to education and what influences and informs curriculum content.

OBJECTIVES

At the end of this unit you should be able to:

- appreciate the central importance of the curriculum in education;
- identify competing views and approaches to the curriculum and its content;
- consider how teachers might contribute to the development of the secondary school curriculum.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

What is 'the curriculum'?

Over time, we have come to consider that 'the curriculum' encompasses everything taught and learned in schools. This includes the timetabled traditional subject disciplines that we still see in the secondary school, plus all the activities promoting intellectual, personal, social and physical development of pupils. These elements have become increasingly formalised in secondary schools. Wider learning opportunities may also be called 'extra-curricular', 'co-curricular' or 'super-curricular', for example, with a focus on developing pupils as young people. You may also encounter the term

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'the hidden curriculum' (see Unit 5.3), which relates to the unspoken social rules, expectations and cultural norms that pupils are expected to understand and follow at school, but which aren't explicitly taught.

The curriculum content: who decides?

The school curriculum is under regular revision as societies develop, as national priorities change and as new knowledge is created. But who decides what is in the curriculum? How the curriculum is shaped and set out is now very much a product of how political decision-makers and society view the value and purpose of education. This was not always the case. From a historical perspective, there have been numerous curriculum ideologies and models that, to a greater or lesser extent, reflected the historical or social context in which they were developed (see the Education in the UK website curated by Gillard, 2018). These curriculum models were traditionally influenced by the academic disciplines taught in universities because that was largely where knowledge was produced and categorised.

In the UK, the departments of education in England, Northern Ireland, Scotland and Wales lead on curriculum decisions for state-maintained schools in their nation. Units 7.3, 7.4, 7.5 and 7.6 on the companion website provide details of the different approaches to curriculum development for each of the four nations. Broadly speaking, England's 'National Curriculum' follows a subject-based, knowledge-focused approach. The curriculum has detailed statutory requirements, with high stakes testing and accountability measures. Northern Ireland emphasises key skills and competencies over content knowledge. Scotland's 'Curriculum for Excellence' promotes learner-centred flexibility from birth to 18, encouraging teacher autonomy along with flexibility for schools to adapt the curriculum to meet local needs. Wales is implementing a new curriculum emphasising adaptable life skills and digital competence across all areas. The implications of devolution and differing national political priorities create practical challenges for the UK: one of which is that it fragments the research base for education and makes it harder to share effective curriculum practices.

Decisions about the curriculum may be taken following formal and inclusive consultations involving employers, teachers and parents or alternatively decisions may be driven by political ideology. The freedom given to schools to develop their own curricula varies between countries, and also the type of school. As the curriculum can be used for political ends and to give or deny voice to particular histories and knowledge, you may find as a teacher that your work is affected by controversy over what you should be teaching. See, for example, Section 28 of the Local Government Act 1988, which banned local authorities and schools from 'promoting homosexuality' (UK Government, 1988). This was repealed in 2003 and in response to changing views in society, was followed in 2004 by civil partnerships legislation providing legal recognition of same-sex relationships (UK Government, 2020) and then, in 2013, by legislation recognising same-sex relationships through the Marriage (Same Sex Couples) Act 2013 (UK Government, 2013). More recent examples include the requirement from 2014 for all schools to actively promote 'fundamental British values' of democracy, the rule of law, individual liberty and mutual respect and tolerance of different faiths and beliefs (DfE, 2014). This directive emerged following concerns about extremism in some schools but has generated debate about what constitutes 'British' values and whether such requirements risk marginalising certain communities. In addition, the 2022 guidance on political impartiality in schools, issued following concerns about activist teaching, further illustrates how contemporary political sensitivities around topics such as climate change, social justice and historical interpretation continue to shape what teachers feel able to discuss in their classrooms (DfE, 2022).

In most subjects, you will find there are controversies over what should be taught, such as specific set texts in English, creationism in science and so on. The 2020 statutory introduction of Relationships and Sex Education (RSE) in secondary schools has similarly demonstrated how curriculum content becomes politically charged, with ongoing tensions between inclusive education policies and parental rights to influence their children's learning about sexuality and gender identity (DfE, 2019a). Recent years have also seen debates about 'decolonising' the curriculum, especially in English and History, with discussions about which authors should be studied and how topics should be presented.

In England in state-maintained schools, teachers have become accustomed to government-prescribed curricula, though this centralised approach only began with the 1988 National Curriculum. Previously, teachers and examination boards decided curriculum content. The 1970s saw elements of experimentation in a 'progressive' direction where the value and relevance of an academically orientated education for all pupils was called into question. Over the decades following the 1988 National Curriculum, the content of the secondary school curriculum, as well as its aims and purposes, have been increasingly prescribed and managed by government. Since 2010 the EBacc (English Baccalaureate) has been presented as a suite of GCSE subjects in England designed to provide students with a broad and balanced education, but is not a qualification itself, merely a performance measure used by schools. League tables detailing the assessment statistics for each school are compiled, intended 'to encourage a "free market" by allowing parents to choose schools based on their measured ability to teach the National Curriculum' (Bartlett et al., 2001).

In England, ministerial announcements have led to increased emphasis being placed on subject knowledge (Gibb, 2017), while other pressures on the curriculum were brought about by the Covid-19 pandemic (2020–2022) which put pressure on schools and examination boards to slim the curriculum and associated assessments because of the learning time that had been lost (Leask and Younie, 2021). The Year 11 cohort of 2025 who were in Year 6 when the Covid-19 pandemic disrupted their final primary school year represent the worst-hit cohort: their outcomes have attracted much attention.

International organisations may also influence curriculum design. For example, the international body representing the wealthier nations, the Organisation for Economic Co-operation and Development (OECD) in 2008 sparked a debate about 21st-century skills essential for all learners. Fadel (2008) discussed essential 21st-century skills including basic knowledge (literacy, numeracy, sciences, humanities) and applied skills (critical thinking, communication, collaboration, creativity, digital literacy and ethical responsibility). See also van Laar et al.'s (2020) update on 21st-century skills thinking. More recently, the DigComp project, implemented by the Joint Research Centre on behalf of the European Commission to focus on digital competence specifically, is now in its fourth iteration (Vuorikari, Kluzer and Punie, 2022). The EU sees being digitally competent as a task for the 21st-century citizen. In England, digital literacy has been a strand of the computing programmes of study since 2013.

Curriculum priorities vary between schools, reflecting their distinct educational visions and values. Nevertheless, while an emphasis on academic attainment remains, the social aspects of education have gained increasing importance in the curriculum in recent times. These may be formalised through Personal, Social and Health Education (known as PSHE – see www.pshe-association.org.uk) and/or Citizenship (see https://www.teachingcitizenship.org.uk/). In England, see, for example, the guidance on 'character education' (DfE, 2019b), which contributes to the requirements to address pupils' spiritual, moral, social and cultural development (SMSC). The effectiveness of SMSC education is inspected and reported on.

Now complete Task 7.2.1, which asks you to think about your own experience.

Task 7.2.1 Personal curriculum audit

Reflect on your own secondary education to create a baseline for thinking about curriculum issues.

1 **Subject inventory** – list all the formal subjects you studied, as well as any broader personal or social education elements.
2 **Initial thoughts** – what seemed to be emphasised? Looking back, does anything seem missing or underdeveloped?
3 **Questions for later** – are there any aspects you are uncertain about and want to revisit later?

Store your initial reflections in your portfolio – you will return to them after the next section.

It would not be too strong a claim to make that a school's curriculum offer fundamentally defines it, serving as the primary expression of its beliefs, values and educational philosophy.

Curriculum theory

Curriculum theory is a substantial academic body of knowledge that describes, examines and explains the principles that underpin the curriculum and seeks 'justification for the pursuit of an activity' (Barrow and Woods, 1988). The study of the curriculum is traditionally informed by the 'foundation disciplines' of the philosophy, sociology, psychology and history of education. Kelly (2009, p. 23) emphasises that it is 'an academic and intellectual exploration of all the factors we need to take account of in order to devise an *educational* curriculum'. If the curriculum is the concrete manifestation of what education is for, the design of the curriculum involves deep thought about the nature of knowledge and its value to learners and society. It is not surprising, therefore, that there are a number of competing arguments about how the curriculum is established and what it should be in practice (see Kelly, 1990, pp. 22–42, for an interesting historical summary).

For example, one long-standing debate has been between 'academic' and 'vocational' education, often presented as opposing approaches, that resurface in different forms over time. The notion of 'parity of esteem' (Pring, 1995) remains elusive, with policy makers in England again attempting to elevate vocational education through T Levels alongside A Levels and apprenticeships post-16 (DfE, 2023). Education is now viewed primarily in terms of economic benefit, but it was not always seen this way. Unlike in the past, education and the pursuit of knowledge for its own sake and the value of university education are now questioned by many.

For our purposes, perhaps a good starting point for exploring some of the arguments would be to ask the following key questions, and then consider the different ways of answering them:

- What should be taught in schools?
- Why should it be taught? To whom should it be taught?
- What does it mean to be an educated person?

What should be taught in schools?

If we start from the premise that schools teach what cannot be learned elsewhere, we can focus the curriculum on intellectual transformation rather than everyday knowledge gained through experience. This approach supports a 'knowledge-based' curriculum where pupils are entitled to encounter the intellectual achievements of humanity through distinct subject disciplines with clear boundaries and conceptual structures.

The theory of 'powerful knowledge' (Young, 2008; Young and Muller, 2016; Hudson and Shelton, 2021) argues that subject-based curricula need not be elitist but can provide all children with access to specialised knowledge that transcends their immediate experience. This approach is evident in England's current emphasis on subject knowledge in GCSE reforms and the focus on academic content over generic skills.

However, this knowledge-focused approach contrasts sharply with 'progressive education' influenced by John Dewey and constructivist learning theories. The central idea here, drawing on the body of work by Lev Vygotsky (see Vygotsky, 1978), is that human learning is constructed, that pupils build new knowledge upon previous learning and, therefore, learning involves constructing one's own knowledge from one's own experiences. Pupils are therefore encouraged to co-construct knowledge by working to solve realistic problems. Advocates like White (2004) argue that the curriculum is for all children, not an academic élite, and should be planned accordingly.

While there is renewed support for this approach to designing the curriculum, it is a contentious position to take. The trend over the last 30 years or so has been towards a blurring of subject boundaries, cross-curricular themes and a greater focus on the social development of the child. John Dewey, the famous American philosopher and educational reformer of the late 19th century, emphasised the importance of education as a means of social reconstruction. Dewey (2016), like many other educators since, saw the purpose of education as being essentially about the development of human beings in society and so the focus of the curriculum should be based on the interests of the child with due regard to useful knowledge and its application. The effect of these ideas on the content of the curriculum is to have a much more fluid view of subjects and the organisation of knowledge, and to elevate experience over substantive knowledge.

The expansion of the curriculum since the late 1990s, explicitly to include a greater responsibility for the social and mental health of the child, expressed an increasing emphasis on the psychological wellbeing of pupils, which manifested itself through a whole range of interventions concerned with self-esteem, wellbeing, resilience, character-building and 'personal capital'. Dubbed by critics as 'the therapeutic turn' (Ecclestone and Hayes, 2009), these additions to the curriculum may be seen to shift the emphasis of the curriculum from developing an understanding of the world to an understanding of the self: to elevate therapeutic goals over academic aims.

Politics, as we have already seen, has always steered the direction of education to a greater or lesser extent, but for some educationalists, 'The politicisation of education, including the curriculum, often means that positions on knowledge are conflated with particular political positions or broad cultural outlooks' (Sehgal Cuthbert and Standish, 2021, p. 8). The content of what we expect learners to learn during their schooling is far more contentious than in the past and curriculum planning is thus more complex.

Whatever the debate about and balance between the 'subject, social and cultural', it is still the case that the range of subject disciplines in the secondary curriculum has remained relatively static since the introduction of comprehensive education in the late 1960s. Nevertheless, the importance

placed on certain subjects has undoubtedly changed, as has the content of the curriculum. The challenge for teachers lies in navigating what 19th-century cultural critic and poet Matthew Arnold called 'the best which has been thought and said' (Arnold, 1869) and the contemporary demands for relevant, inclusive and skills-focused education. These tensions translate into daily decisions for teachers – should a history teacher prioritise chronological knowledge or critical thinking skills; should English focus on literary heritage or helping pupils to respond to contemporary diverse texts? Understanding these philosophical tensions will help you to make informed curriculum choices in your subject area, recognising that those decisions reflect deeper beliefs about the purpose of education and the value society places on it.

Why should it be taught? To whom should it be taught?

Clearly, *what* should be taught requires justification and different curriculum models are based on differing views of the value and purpose of education. The notion that education is a good in itself with no external aims was once dominant, but this is no longer the case.

Today's emphasis focuses on ensuring young people gain qualifications and skills needed for their place in society and economic contribution, with personal wellbeing also important. Education's instrumental value now dominates curriculum decisions, with a direct relationship assumed between education and economic needs. Words like 'relevant' and 'useful' are frequently applied to curriculum content, certain subjects are privileged over others, and instrumentalism is often expressed through 'employability' agendas.

However, how an 'all-around education' is achieved is another area of discussion. Some teachers and policy-makers argue that decisions about *who* learns *what* are necessary for all pupils to achieve their potential while maintaining educational standards. Here, 'relevance' is sometimes applied, often with the best intentions, but mostly with the effect of limiting access to some subjects.

This becomes evident in secondary education when options choices are made by pupils about which subjects to continue to learn at examination levels, raising implicit questions about whether all pupils benefit from disciplinary knowledge and can develop conceptual understanding. While we cannot easily answer this here, it is an important issue to consider and then to ask: what happens if children miss out on academic knowledge as has happened with the Covid-19 pandemic and the subsequent disruption of schooling? Is it only through the curriculum that schools can provide all children with the chance to appreciate the intellectual achievements of humanity? How far is it the role of the school to prepare young people for work?

These questions might lead us to think about the subjects that make up the curriculum in more detail. Task 7.2.2 asks you to compare your current and past experiences of curricula.



Task 7.2.2 School curriculum analysis

1 **Revisit your personal experience notes from Task 7.2.1.**

(a) Does your experience resemble any of the four nations' approaches?

(b) Which curriculum theory perspectives do you recognise in your own experience (e.g. knowledge-based versus skills-based)?

(c) Do you recall aspects similar to the OECD 21st-century skills framework?

2 **Compare to a current school (e.g. placement school).**
   (a) Schools usually publish their curriculum details on their websites. How does your current school's curriculum offer differ from your personal experiences?
   (b) Which theoretical framework best explains these differences?
   (c) What factors might account for the differences (e.g. policy shifts, assessment pressures, school type etc?)

3 **Critical evaluation**
   (a) How do you see political or social pressures influencing curriculum choices?
   (b) What implications does this have for your role as a teacher?
   (c) What implications does this have for your subject area?

Store your reflections in your portfolio, and discuss them with your colleagues, a mentor or head of department to compare notes.

The subject curriculum

Given that school subjects evolved out of university subject disciplines, the way that knowledge has been classified, categorised and structured over time may be something that we now take for granted. It is important for teachers today to recognise that as our knowledge of the world is developing all the time, the curriculum cannot remain static. We must recognise the importance of the foundational knowledge – both conceptual and factual, as well as the 'threshold concepts' – that have endured over time, and contribute to our understanding of what it means to be educated. The question arises: should 'curriculum development' mean organising the curriculum around prevailing social and political or economic concerns? Looking back over the last 150+ years since our education system was formalised, these concerns have shaped the evolution of the curriculum, with different emphases at different times. Global issues include climate change and environmental issues, pollution, geopolitics, ethical consumerism, food politics and artificial intelligence to list a few: how might they now influence what is taught in schools?

We often refer to 'core' and 'foundation' subjects, indicating some agreement about basic curriculum entitlement. In the 19th century, the 'three Rs' (Reading (W)riting and (A)rithmetic), i.e. 'basic literacy' and 'numeracy' – dominated mass education, partly due to economic needs in an increasingly industrialised society requiring a literate workforce. Over time, changing social attitudes and democratic values led to sciences, arts and humanities being introduced for all children.

While the range of secondary subjects has remained relatively stable over recent decades, their relative importance and content have shifted significantly. In the 1970s, partly due to recent membership of the European Economic Community, modern foreign languages (MFLs) were prioritised but in some girls' schools in England sciences were not on offer. By 2005, MFLs became optional in the secondary curriculum. Currently, STEM subjects (Science, Technology, Engineering and Mathematics) are favoured, while Latin, Classical Studies and Philosophy are rarely taught in state schools today, perhaps reflecting the privileging of 'useful' knowledge over the supposedly 'useless' knowledge of the past. In 2000, IT became ICT (Information and Communications Technology), now referred to as Computing since 2014. Design Technology has also replaced woodwork and metalwork, showing how technological change can drive curriculum evolution.

The selection of knowledge within subjects, what is seen as important and why, is central to the values, aims and purposes of education and is also susceptible to change. Curriculum content is no longer sacrosanct and that must surely be a good thing as long as democratic values continue to hold firm and we have a solid belief in the value and meaning of education. But as Pinar (2012) warns from an American perspective, when schools are too closely aligned with political agendas and teachers become overwhelmed by present circumstances and cultural pressures, they lose control of their professional purpose (Pinar, 2012). At this point, we might ask, how robust and reflexive is the school curriculum shaping your teaching?

In our fast-changing world, we cannot predict future knowledge needs, hence the focus on 21st-century skills to better prepare young people for uncertainty. This fundamental tension between knowledge and skills has gained renewed urgency with the recent emergence of freely available generative artificial intelligence technologies. The capacity of AI systems to write, to solve maths problems, and to synthesise information challenges traditional assumptions about what constitutes valuable human learning.

What does it mean to be an educated person?

It was the philosopher of education, Professor Richard Pring, who, in the 2009 Nuffield Review of 14–19 Education and Training (Pring et al., 2009, p. 3), posed the question, 'What counts as an educated 19-year-old in this day and age?'. The question is entirely relevant to our discussion of the curriculum and for you to consider in developing your own educational philosophy as a beginning teacher. We have already highlighted the tendency to see educational success in narrow instrumental terms, primarily through exam results, and have pointed to the need for the curriculum to go beyond such confines.

You may find that curriculum planning is built around 'outcomes' and some sort of tangible evidence of learning is expected, often at the end of every lesson. But if we take the 'long view' of education and consider that the accumulation of knowledge in a subject and the development of conceptual understanding cannot be entirely measured, then the school curriculum will reflect longer-term aspirations and the belief in knowledge as personal enrichment. While exam success remains important for pupils' future opportunities, the curriculum should provide an enriched experience of subjects that seeks to inspire young people and enliven their curiosity to understand the world through knowledge.

Perhaps the one principle that unites almost everyone involved in education is the desire to see all young people succeed to the best of their abilities and to lead fulfilled and successful lives. If we return to the notion of a 'broad and balanced' curriculum, are we in fact supporting the traditional liberal education for the 21st century that Michael Young argues for? Achieving a genuinely knowledge-based curriculum requires teachers to believe firmly that all pupils deserve access to knowledge-rich learning and are capable of acquiring foundational subject knowledge when taught by confident, well-prepared teachers.

Not all 19-year-olds will achieve the same things, nor have the same aspirations, opportunities and goals, but they can all be inspired by knowledge. Anderson's (1980) image of young people as 'heirs to all the ages' suggests that education should connect students with the broader sweep of human knowledge and cultural development, while simultaneously preparing them for contemporary challenges that previous generations could not have anticipated.

Teachers as curriculum makers

While statutory programmes of study are a natural starting point for beginning teachers to get a sense of the intended curriculum they need to teach, you may find when you join a school that your department's own long-term and medium-term plans represent the actual curriculum you will teach. Often these plans are developed and resourced over long periods of time and fit together carefully forming a curriculum journey across each school year. This can be thought of as the 'enacted' curriculum. One further step, the 'received' curriculum, can be considered as what the pupils actually learn. A common misconception for beginning teachers is to confuse what they think they have taught as they aim to cover the curriculum, and what the pupils have learned!

As a beginning teacher you may feel wary of experimenting and may be discouraged from attempting to stray from the tried and tested schemes of work that schools have been using for years. This is understandable, but this is also the very time to be thinking about what different subject content might be introduced in the future, and make some small steps to experiment and challenge yourself and your pupils. A beginning teacher can bring fresh thinking and new ideas into a department. A related concept here is that of 'teacher agency' (Biesta, Priestley & Robinson, 2015; Priestley & Drew, 2019), a teacher's ability to exercise their professional judgment to impact their own teaching practice and the broader educational system. The journey from statutory requirements to classroom reality involves numerous decisions about content, pedagogy, sequencing and assessment that will shape what pupils learn. When teachers have the professional expertise to adapt and transform curriculum documents, this will result in meaningful learning experiences for their pupils.

Teachers as curriculum makers consider not just what to teach but how to inspire curiosity, develop critical thinking and connect learning to their pupils' lives and the broader societal challenges we have discussed. This might involve making connections across subject boundaries, incorporating contemporary issues into traditional disciplines, or adapting teaching approaches to reflect diverse student backgrounds and learning needs. Such decisions require teachers to understand both their subject's contribution to pupils' overall education and how their curriculum choices affect pupil engagement and achievement. This will mean something different for each curriculum subject, but we should never underestimate what our pupils are capable of if they are inspired by the knowledge that teachers can share. Developing an educationally defensible curriculum is predicated on a clear vision of the value and purpose of education. Task 7.2.3 asks you to think about the intended curriculum for your subject area.



Task 7.2.3 The enacted versus the received curriculum in your subject area

1. Examine the curriculum for your subject. In England, this may be the national curriculum programme of study.
2. Review your department's schemes of work for one year group or key stage.
3. What links can you see between the intended curriculum and the curriculum as it is taught in your department?
4. What additional content or activities are included beyond statutory minimums?
5. Are there any omissions that surprise you, or things that you expected or did not expect to see?

Discuss your findings with your head of department to understand how the curriculum has been sequenced, and why those decisions have been made. Discuss which areas of the curriculum could be enhanced in the future. Write a summary for your professional portfolio.

In order to develop your teacher agency as part of a career-long professional trajectory (Campbell, 2023) and begin to see yourself as a curriculum maker, not just a curriculum deliverer, you should embrace opportunities that arise to put your own stamp on sequences of learning. Many programmes of ITE expect you to develop a project, or a piece of action research based on your classroom practice (see Unit 5.4 on practitioner research and reflective practice). Over time, this skill will lead you to continue developing the enacted curriculum in your school to ensure best fit with the pupils you teach. Beginning teachers who start their careers with an open and questioning mind and a desire to develop a vision of education are the curriculum makers of the future. We hope you enjoy the professional challenge!

SUMMARY AND KEY POINTS

Through examining what constitutes 'the curriculum' and exploring your own experiences, this unit has highlighted the central role that curriculum plays in shaping educational experiences and outcomes.

- The curriculum encompasses everything taught and learned in schools, from traditional subject disciplines to activities promoting intellectual, personal, social and physical development of pupils. This includes the formal timetabled subjects, personal and social education elements, and the 'hidden curriculum' of unspoken expectations and cultural norms.
- Teachers play a crucial role as curriculum makers, not just curriculum deliverers. While national curricula provide frameworks, as beginning teachers you have significant opportunities to exercise professional agency in adapting content, selecting pedagogical approaches and making connections that inspire curiosity and critical thinking in your pupils.
- Curriculum decisions reflect broader educational philosophies and societal values. Different approaches emphasise knowledge versus skills, academic versus vocational preparation and individual versus social development. Understanding these competing perspectives helps you develop your own educational vision and make informed decisions about your practice.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

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8 Your professional development

The focus of this chapter is on life beyond your initial teacher education (ITE). The chapter is designed to prepare you for applying for your first post, to be aware of opportunities available to support your continuing professional development as a teacher after you have completed your ITE and to understand the various accountabilities governing schools and teachers.

Getting a job at the end of your ITE is important, time consuming and a high priority for beginning teachers. Units 8.1 and 8.2 are designed to help you at every stage of the process. Unit 8.1 takes you through the stages of considering, once again, why you want to enter the teaching profession; deciding where, and in what type of school, you want to teach and looking for suitable vacancies. It moves on to think about how you send for further details of posts that interest you and then how to make your application. Unit 8.2 looks at how you prepare for and attend an interview – including teaching a lesson as part of an interview, and how you go about accepting a post.

The success of any school depends on its staff. However, although you have successfully met the requirements to qualify as a teacher at the end of your ITE, you still have a lot to learn about teaching to increase your effectiveness. Further, society is constantly changing and so the demands on teachers also change. Consequently, as your career progresses, you will find you need new skills and knowledge about teaching and learning (pedagogy), so you can expect to continue to learn through your career. Professional development is therefore a lifelong process that is aided by regular reflection on practice and continuing education.

Unit 8.3 considers your further development as a teacher. It starts by looking at becoming a 'professional', including being a learner of your own teaching. It then encourages you to look at developing a teacher identity. The third section focuses on understanding that developing professional judgement is highly complex and requires developmental tools. Having looked at your early career development, it concludes by considering professional learning within a school community, including the impact of school cultures on professional learning.

Unit 8.4 focuses on developing your resilience as a teacher, which is what sustains and enables you to thrive rather than just survive in the profession. It helps you identify the signs that current levels of resilience are not necessarily up to the demands being placed on you. The unit then helps you to identify when you are stressed and looks at potential causes of stress in order that you may prevent stress. It then provides advice on coping with and managing stress proactively. Major causes of stress are workload and time constraints. The unit considers these, focusing on developing ways

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to manage your time and workload effectively, both inside and outside the classroom, across a term and a year. Although approaches to managing your stress, workload and time are individual, being successful in managing these gives you time to enjoy your work and have time for leisure.

Unit 8.5 is designed to give you an insight into the accountabilities governing schools and teachers. The unit is organised around moral and ethical accountabilities: your accountability as a professional to the people you serve and to your colleagues. Initially focusing on the Nolan Principles of Public Life, it moves to focusing on legal and regulatory accountability; the state-determined infrastructure of accountability comprising legislation; inspection; and regulation. These are considered in relation to safeguarding and child protection, Prevent Duty, health and safety and equalities. The unit then focuses on employer and contractual: your accountability to your employer, focusing on safer recruitment, your contract of employment, and progression/promotion, performance management and performance-related pay.



ONLINE RESOURCES

Watch the explainer video on professional development by scanning the QR code or visit routledgelearning.com/learningtoteach.



8.1 Getting your first post

Julia Lawrence

Introduction

Applying for your first teaching post is a very important undertaking. The application process involves a number of important stages. You need to be clear about why you want to continue your teaching journey, as well as about where and in what type of school you want to teach. You need to look for suitable vacancies, select a post that interests you and request further details. You will need to prepare your application including, for example, a curriculum vitae (CV), write a letter of application as well as contacting potential referees. Once these have been completed, you need to complete your application.

Being organised and prepared is key to your success. You will need to allocate time to complete these tasks and may need to seek support from others.

The aim of this unit is to help offer advice and guidance to support you with the process of getting your first post within the teaching profession.

OBJECTIVES

At the end of this unit you should be able to:

- articulate why you want to enter the teaching profession;
- follow the procedure for, and the process of, applying for your first teaching post;
- make a written application.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Getting started

Before you start applying for jobs and feel inspired by a job advert for a specific post in a specific school, there are a number of things for you to consider. Task 8.1.1 identifies questions for you to consider before you start your application.

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Task 8.1.1 Things to consider prior to applying for a teaching post

- In preparation for your first teaching application, consider the following questions;
- Where were you in your life and career before you decided to become a teacher?
- Why did you choose to learn to teach?
- Reflect on your ITE and school placements – what have you learnt, what have you achieved?
- Think about your teaching experiences to date; how can you apply this experience within your job application?

You might find it useful to record your responses and thinking around these questions.

Deciding where, and in what type of school, you want to teach

Deciding on where you want to teach is possibly the most important decision. For example, you may be learning to teach in the same university/area you studied for your university degree and you want to settle and stay within that area. You may want to return to the area where you were brought up or teach in a new area entirely. You may even want to consider teaching overseas. Regardless of the reason you want to teach in a particular area, take time to look at a number of schools in your chosen area. Review their websites which will provide details around their values and ethos as well as their policies and expectations. Many also advertise posts on these sites. Even if they are not recruiting at the time, this may change as the year goes on.

If you are looking to teach abroad as a first teaching post, international schools offer many benefits. However, you need to check if they can support your statutory induction period, as this might have implications for you when you return home. Guidance around where induction periods can be completed can be found on your relevant government Department for Education site (see, for example, Department for Education (2024), if you are/have undertaken your ITE in England). You also need to consider contributions to pensions, payment of taxes and so on. Further, many countries require experience prior to appointment, so you may be better served completing your post-qualification induction before heading overseas.

Deciding where you want to teach is an important first step. Task 8.1.2 provides some suggested activities to support you with this.



Task 8.1.2 Where do you want to teach?

Think about where you would like to teach and how flexible you can be in where you can look for a post. List all areas in which you would consider working and find out as much information about these areas as you can; for example, transport networks, travelling time, local facilities, house rent or purchase prices. Talk to anyone you know who lives in these areas. If it is an area you do not know, if possible, visit the area so that you can get a 'general feel of the place'.

As well as location, you should also consider the type of school in which you wish to work. In England, there is a vast array of different types of schools. See 8.1a on the companion website for a list of these.

Other considerations about a school include its values and ethos and whether these match yours, the support provided for Early Career Teachers (ECTs) and professional development opportunities offered. Task 8.1.3 asks you to consider what type of school you might wish to work in.



Task 8.1.3 What type of school do you want to teach in?

Think back to the types of school you have experienced on school placement during your ITE programme or in other contexts. List positive aspects about working in the different types of school, opportunities you are/were able/not able to experience in these different types of school and identify any experiences you felt you would like to focus on in your first teaching post.

From this, identify the types of school you would be happy to teach in (for example, state, independent, specialist school, academy, free school or sixth form college). Identify why you want to teach in that type of school so that you are prepared to write or speak about this in your application or interview.

Looking for suitable vacancies

Any advert for a teaching job gives you some insight into the roles you may be asked to undertake. Check closely who the application is open to, for example, does it 'encourages applications from ECTs'. There are times when schools are not able to appoint an ECT. This could be due to, for example, the school's capacity to support you in the department, or they simply need a teacher with a bit more experience at that particular time. In England, schools placed in 'special measures' following an Ofsted inspection may not be able to host ECTs between inspections, so you are encouraged to check your relevant government Education Department for more guidance on this (see, for example, Department for Education (2024) if planning to teach in England). Adverts may highlight different aspects of the school or the role; some adverts encourage you to visit and make contact, some focus on what support they can offer you as a teacher.

Be proactive and take your job hunting seriously. As an ECT you are usually looking for teaching posts to start in the September of the year in which you gain qualified teacher status. Start looking for such teaching posts from around December onward. Teachers usually need to give two- or three-months' notice to leave their post at the end of term, to allow time for recruitment. Your placement schools are likely to mention possible jobs coming up at their school, and may even want to interview you quickly. This is common in secondary shortage core subjects. If this happens to you, take time to think carefully about whether the school is right for you, and matches what you are looking for. Try to avoid being pressured into something before you are ready.

Teaching jobs continue to be advertised right up until the end of the school year in July. Posts advertised after 31st May (the resignation deadline for teachers to take up a new post in September), usually focus on attracting ECTs.

Start searching – where to look?

You probably want to register your interest on the following job boards, as this ensures you receive email alerts of teaching jobs that are posted, which saves you time scrolling through job listings. You would not want to miss out on what could be your dream job in your dream school! These include:

Across the UK

- The TES: this is currently the number one place to look for a teaching job, particularly in secondary schools. It allows you to create your own job account and save your application details online. This saves time when you are applying for a number of jobs. See https://www.tes.com/jobs/
- Your University/ITE provider: some schools let university staff know of jobs.
- Social media platforms: Most schools use 'X' (formerly known as Twitter) and other social media platforms, for example, LinkedIn to post vacancies.
- Teacher recruitment agencies: Most agencies deal with both permanent and supply teaching posts. Many schools work with a preferred supply agencies, who support them to recruit staff for both short- and long-term posts.

In England,

- Department for Education (DfE) Teaching Vacancies: this is a relatively new job board that is free to schools and teachers and is part of the DfE strategy to help school recruit. See https://teaching-vacancies.service.gov.uk/
- Local authority (LA) job boards: Many local authorities have proactive job boards for teaching vacancies. Regardless of school type (e.g. England LA maintained, Academy, multi-academy trust, MAT), this is a good place to look.

(Note: Outside the UK you need to check where teaching posts are advertised, but TES also has international vacancies, as do the main international school organisations).

Although teaching posts are advertised all year round, the bulk of adverts for jobs to start in September occur after 31st May. If you do not secure a post for a September start, there are other opportunities, including supply work (although you need to check your formal induction requirements and your further professional development). Some schools manage their supply cover themselves and you can always request to join their supply pool direct. Other schools have a preferred supply agency. You may want to join a supply agency that operates in the area in which you want to teach (available from the LA, schools and teaching supply agencies). However, do your research. For example, some agencies will support with finding you more permanent work and provide you with professional development opportunities whilst you are working for them.

However, it is important to remember that whilst working for the supply agency you are their teacher and representing them whilst at school. Check the contract you have with them – specifically in relation to whether you are allowed to work for more than one agency. Also, be clear as to what happens if you secure a permanent post whilst working for them in relation to any fees that might need to be paid to release you from the contract you have with them.

Check your online professional identity

Issues around appropriate use of social media will have been highlighted throughout your ITE programme. You therefore know that potential employers check your online identity (for example, through X, Facebook and LinkedIn). You need to consider whether the way you present yourself online is appropriate for the professional you wish to be. Where possible, check your privacy settings,

and if you have any concerns about any of your social media posts, then you should remove them. Remember, it is not just potential employers who are able to see your posts; they can also be seen by the press, other teachers, pupils, parents and governors.

Selecting a post that interests you

As previously indicated, you should be actively seeking out appropriate posts. If an advertisement interests you, read the application pack if one is provided and contact the school for any further information not included in the pack. You can phone or email the school (the school secretary/administrator is a good starting point if the advertisement does not specify a point of contact); for example:

Dear Sir/Madam (or name, if given in the advertisement)

I am interested in the vacancy for a (subject) teacher (quote reference number if one is given) at ABC school, advertised in (publication; for example, *Times Educational Supplement*) of (date) and would be grateful to receive further details of this post.

Yours faithfully (if you use Sir/Madam) or sincerely (if you use a name)

Making an application

If you decide to apply, remember that your application is the first formal contact you have with the school, and first impressions count. An application is an early stage in the selection process, and the decision to call you for interview is based on the quality of this application. Hence, the importance of presenting yourself effectively in your application cannot be over emphasised. You need to make sure that you have as much information available to you as possible, both about the school and the particular post, so you can develop a better understanding about the school (for example, its values, ethos, pupil intake and curriculum). The school website provides information about the general activities of the school. It should include links to the latest school inspection report, which allows you to develop an impression of how the school is performing. If possible, organise a visit to the school; this allows you to get a better feel for the school and the area and to personalise your application.

Any job advertisement includes a description of the post (job description) and, in most cases, an overview of what they are looking for in the person applying (sometimes referred to as a person specification), which focuses on the main knowledge, skills and experiences the school are looking for. You need to consider whether you have the knowledge, skills, experience and qualities the school is looking for and whether the school meets some or all of your requirements.

When making your application make sure you highlight your strengths providing, where possible, examples of where you have demonstrated the skills and experience they are looking for. For example, if the advertisement identifies strong behaviour management as a requirement, you should include examples of when you have demonstrated this in your teaching: state what you did and why and how you knew it was successful. You should also demonstrate other qualities you have that would contribute to the school's values and ethos (which is why it is important to access as much information about the school as you can) and what makes you distinct from other applicants.

Plan the content of your application before you complete the application form or write a covering letter for a specific post and finalise your CV. If you are making multiple applications, do not be tempted to use the same form, letter or CV. Whilst many applications require the same basic

information (for example, previous experience, prior employment), each application needs to match your experience and qualifications to the specific requirements of the post. For each application you will need to highlight different points and varying the amount of detail you provide, according to the specific requirements of the post and the school. If you use some of the same information, make sure you have changed all references specific to other applications (this is a common mistake and does result in some applications being rejected). To support you with this, you may want to store relevant information in the form of a Word document, from which you draw from for a specific application.

As well as demonstrating why you want the job, a customised application also shows you have taken the effort to find out about a specific post in a specific school, which should help your application to stand out from the others. An application that fails to explain why you are interested in the specific post in the specific school is unlikely to be considered further. It is also important that you identify your referees at this stage.

The application process is more than just form filling. To do it properly takes time. Two hours is probably the minimum time to complete an application properly without rushing it, if you have prepared beforehand and have all the information available. Most applications have a closing date about two weeks after the original advertisement is placed. However, in some instances, some application windows might close earlier if the employer receives sufficient applications of the quality they need to create a shortlist (although this should be clear in any information about the post you see). Therefore, do not leave it to the last minute. You need to be organised and start the application early. Remember to proofread it before you send it off; you do not want to lose out on a job because of poor grammar, punctuation or spelling.

Referees

Most employers ask for the names and contact details of at least two people who are willing to act as referees. These are people who have a good knowledge of you as an individual as well as your teaching ability and academic work. Check whether there is one particular person within your ITE provider whom you should name as the first referee and, if not, ask someone who has knowledge of your teaching abilities. This might include, for example, a mentor from a placement school, a member of staff from your ITE provider or a previous employer. Your second referee should be someone who knows you well and is able to comment on your character, qualities, achievements and commitment to teaching as a career. Family members must not be used.

Make sure you contact your referees before making any application to check they are willing to act for you and to confirm their contact details. You are likely to need their telephone number(s) and email address. Many schools require references to be taken up prior to interviews, but if not, any offer of employment may be withheld until these have been received.

You have selected your referees because you feel they will be supportive of your application, so try to give them as much information as possible to help them write it. This might include a copy of your application, your CV and information about the post you are applying for, including the job description. Whilst the school will send information about the post to your referee, giving them time to prepare the reference in advance means that they can respond quickly when contacted.

You should be aware that some schools have a policy of open references; that is, a reference is shown to the applicant in certain circumstances. The referee knows this at the time of writing the reference.

Methods of application

Schools normally require applicants to submit a completed application form/letter of application and CV. Most frequently, you complete a form plus a letter or a personal statement. In these cases, the form acts as your CV.

Modern technology means that most applications are made online. You need to check what process is expected for the post for which you are applying. Whilst increasingly rare, if a postal application is required, you need to build time into the process (including getting access to the post office) so that it arrives before, or on, the due date.

An application form

The information required on an application form closely matches that identified for a letter of application and CV. See 8.1b on the companion website for more information on this.

It is likely that one page of the form will be blank for you to explain why you are applying for the post, add additional information and elaborate on the skills and experience that equip you for it. As with a letter of application (see below), unless otherwise stated, this should focus on the specific job you are applying for, on the experience, skills and qualities you have to be able to do the job, and any additional skills you might bring to the post. This section should be written in continuous prose as if it is a section of a letter, following the suggested format and containing the type of information in a letter of application. Adding headings related to the person specification and job description before the relevant text can help those who are shortlisting candidates to see easily that you meet the specification for the post.

When you submit the application, you may need to include a covering letter (this may be included in an email, if appropriate) indicating the post you have applied for and where you accessed information about it. This will generally be very short, but written formally, indicating that you have included your application for the post of (subject) teacher as advertised in (publication).

Letter of application

Some schools require you to write a letter of application. This should state clearly your reasons for applying for the post, matching your qualifications, experiences, particular skills and personal qualities to the post as described in the information sent to you from the school. The letter should normally be no more than two sides of A4 in length. See Figure 8.1.2 within 8.1c on the companion website for a suggested format. When completing your letter of application, it is important to check grammar and punctuation. Applications are frequently rejected if either of these are poor.

Your curriculum vitae (CV)

A CV summarises your educational background, qualifications, teaching and other work experience and any other relevant information (for example, interests and activities you enjoy outside of school), as well as any other relevant qualifications (for example, a clean driving licence) or any course you might have attended as part of your development as a teacher (for example, coaching qualifications, first aid). It should always accompany a letter of application; it provides detail and

further information to support your application. Remember, it is a supportive tool. Figure 8.1.2 provides an editable version of a sample template for a CV. This can be found in 8.1d on the companion website, along with some notes and reminders about applications to help you.

To consolidate your learning from this section, Task 8.1.4 provides you with an opportunity to compile both a CV and letter of application/form.

Task 8.1.4 Your CV/application

Draft a specimen letter of application and CV and obtain and complete an application form.

Ask your mentor or tutor to check these for you and give formative feedback on any changes they feel are appropriate. Use these as the basis for all your job applications.

SUMMARY AND KEY POINTS

This unit is designed to help you realise that, just as with your teaching, you must prepare for obtaining your first teaching post; you cannot leave it to chance. In this unit, you have been led through the steps, skills and techniques you need to prepare actively for obtaining your first post, that is:

- why you want to enter the teaching profession, as well as about where and in what type of school you want to teach;
- looking for suitable vacancies;
- selecting a post that interests you and requesting further details;
- preparing your CV, writing a letter of application or completing an application form and contacting potential referees;
- completing your application.

To be successful, you need to:

- be proactive;
- be prepared;
- practise your techniques;
- be confident.

We wish you the best in obtaining your first post.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.



Further resources

Griff, T. (2021) *Applying for a Teaching Job 2021: The Definitive Guide to Writing a Great Application for a Job as a Teacher, HoD, HoY, AHT, DHT or Headteacher* (Kindle edition).

Written by a consultant on staff appointments and based on experiences over many years, this text identifies the main reasons for applications being rejected. It provides step-by-step guidance on how to write your application so that it stands out.

The Chartered College of Teaching's Early Career Hub, viewed 25 May 2025, https://earlycareer.chartered.college/?s=job

The Chartered College of Teachers has several articles on the topics of looking for a job, applying for a job and interviewing for a job etc.

***Times Education Supplement*, viewed 25 May 2025, https://www.tes.com/jobs/careers-advice**

This provides career advice across your teaching career.

Reference

DfE (Department for Education) (2024) *Induction for Early Career Teachers (England) Statutory guidance for appropriate bodies, headteachers, school staff and governing bodies*. Available from https://assets.publishing.service.gov.uk/media/6629237f3b0122a378a7e6ef/Induction_for_early_career_teachers__England__statutory_guidance_.pdf [accessed 25 May 2025].

8.2 Preparing for interview

Julia Lawrence

Introduction

So, you've successfully applied for and been invited for interview. Taking time to prepare for and attend an interview is just as important as applying for the post. Being organised and prepared is key to your success. Allowing time to rehearse and discuss the interview process is as important as the application process.

The aim of this unit is to help offer advice and guidance to support you with the process of preparing for and attending your interview and what to do if you are successful and offered the post.

OBJECTIVES

At the end of this unit you should be able to:

- prepare for an interview for a teaching post;
- understand the processes involved in accepting a teaching post.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

The interview

If you are invited to interview confirm that you will attend, confirming the post, date and time of interview and the requirements for any lessons you are asked to teach and/or presentations you are asked to give. If your situation changes (for example, because of a clash, such as another interview on the same day, or you have had another interview and been offered that post) and you are not going to attend the interview, let the school know as soon as possible (do not just not turn up). This will allow them to invite another candidate if they wish.

For any change, where possible, speak directly to the school contact rather than leave an answerphone message or email, and be honest and explain the situation. Try to avoid ringing on the day of interview unless your problem is illness related.

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Preparing for the interview

When preparing for your interview you might find it helpful to reflect on why you applied for this particular post, so that you can put across the relevant information convincingly at interview. Read through the advertisement, job description, any other information about the post and school and your application again so that you can communicate effectively the information and evidence you consider to be relevant to the post. It also helps to avoid any contradictions between what you say and what you wrote in your application, as each member of the interview panel has a copy of your application so can compare answers.

It may be that someone you know has knowledge of the school; if so, talk to them. Access recent inspection reports on the school. Write down any questions you have about the post or the school. While many might be answered during the course of the day, it is always useful to have one or two to ask in the interview.

Make sure that you have all the information you need for teaching a lesson on the day of interview (see teaching a lesson as part of the interview, below). You should be told this in the letter of invitation to interview, but, if not, contact the school and ask. Prepare any resources you might need for your lesson in advance and remember to take them with you! If you are giving a presentation, offer to send it through prior to the interview so that it can be uploaded onto the school system (also, this means that you have it in your emails should you forget your memory stick).

As a professional, you are expected to keep as up to date as possible on your subject and education, generally. Read around your subject, as well as about education more generally, so you can discuss the latest educational issues and debates, as well as changes in your subject area. If appropriate, prepare a portfolio of your resources, or take the professional portfolio you are keeping to evidence attainment against the teachers' standards towards which you are working. Many ITE providers encourage beginning teachers to complete e-portfolios that allow you to showcase your work and progress over the duration of your ITE programme. On the interview day, you may not have time to show these, but reflecting on them prior to the day may well help relax you and give you good examples of practice to use during the interview itself.

Plan what you are going to wear at the interview. You should dress professionally, smartly and conservatively, but with appropriate clothing for teaching your lesson. And, if you have not already done so, you might wish to visit the school (always organise an appointment) or at least travel to it so that you know where you are going and how long it will take on the actual day.

Attending the interview

The interview starts as soon as you walk into the school and does not finish until you leave the school at the end of the day. Hence, the initial impact you make is very important as interviewers tend to form an overall impression early. You are assessed throughout the day, so your performance, including your verbal and non-verbal communication in each activity, is important and could make the difference between being offered the job or not.

An interview is a two-way process. The fact that you have been called for interview demonstrates you have the qualifications to do the job. By attending, the school is identifying whether you are what they are looking for, but you are being given the opportunity to identify whether you want to teach at that school. Thus, at the same time as being interviewed, you are, in effect, interviewing the school and deciding whether this is a school in which you could work and, therefore, whether

this is a post for you. Take the opportunity to learn as much as you can about the post, the school and the working environment. This requires you to be alert to what is being said and to be prepared to ask as well as answer questions. If at any stage during the day you feel this is not the right post/school for you, you can withdraw, and it is fair to do so.

It is difficult to generalise about interviews because they vary considerably. When you are offered an interview, you should expect to be given information about the format of the day. This might include a tour of the school, a teaching episode (a lesson/part of a lesson), an informal talk or interview with the head of department, a senior teacher or other staff in the department, group interview, individual interview and interview/talk with pupils. It might also include a presentation, group discussion or an activity such as an 'in-tray' exercise – for example, how you interpret date or drafting a letter to parents.

For many interviews, all interviewees are required to attend at the same time, while others specify a time for your interview, so you do not meet the other candidates. In some interviews, there is a sifting process, whereby some interviewees are sent home part-way through the day while the others remain for a formal interview. While in other interviews, some interviewees teach in the morning and some are interviewed in the morning, then vice versa in the afternoon. You need to understand the processes for the day, and if you are in any doubt, contact the school for clarification. Because of these variations, the format and length of an interview day varies.

Teaching a lesson as part of the interview

It is regular practice for candidates to be asked to teach a lesson or section of a lesson as part of the interviewing process. You should be notified of this when you are called for interview and be given specific information about the age and size of the class, what you are expected to teach and the pupils' ability and prior knowledge, the length of the lesson, what facilities, resources and equipment are available (i.e. all the information you require before teaching any class). If you require further information (for example, the prior learning of the pupils, their current level of progress, where the lesson fits into the wider unit of work), then you should contact the school as soon as you receive your initial information. You might also want to do some research on how the school currently delivers lessons (for example, do they have a prescribed format in relation to the retrieval of prior learning, how do they share learning outcomes, how do they assess progress) so that you can reflect aspects of this in your own delivery or offer activities that develop these principles further.

Plan the lesson carefully, giving attention to learning outcomes, purpose and content of activities, teaching strategies including questioning technique, resources and using as many of the principles you have developed during your ITE programme as possible. It is useful to have copies of the lesson plan available to give to those observing you. This is an opportunity to show the quality of your preparation and planning. Lessons taught as part of the interview process also provide you with the opportunity to demonstrate the level of your subject content knowledge and implementation of appropriate pedagogy so, again, prepare well, particularly if you have been asked to teach a topic with which you are not totally familiar. It is probably best to try to base your interview lesson on something that has been successful on a previous occasion with similar classes, if appropriate. The lesson is also an opportunity for you to show your enthusiasm for teaching, pupils' learning and your subject. For example, how you develop a rapport with your classes, as well as how you organise and progress lessons. Try to appear confident and relaxed, even though those observing you understand that you probably feel a little nervous! If possible, practise the lesson before you deliver it. This allows you the opportunity to evaluate and modify your plan based on feedback.

Be prepared to discuss your lesson during your formal interview; for example, how you feel it went, what the pupils learned, how you know they learned this and what you might change. It might be worth making a few notes about the lesson once you have taught it to help you with this. Be honest and do not be anxious about mentioning things that might have gone to plan. For example, your timing might have gone astray, or your instructions were not as clear as you had anticipated. Where this is the case, offer examples of developments/improvements you might make. In essence, you are providing a verbal evaluation of the lesson itself. In doing so, you demonstrate your ability to analyse why it happened and how you might change this in the future. This shows your level of reflexivity, but also that you have given serious thought to what you have just done, and hence about your practice.

The formal interview

As the format of interviews varies across schools, so does the composition and size of interview panels. In some interviews, you are faced by a panel comprising anything between two to seven people; in others, you have a series of interviews with different people. In either case, these people normally include the following: the head teacher, a (parent) governor, another senior member of the school staff and head of subject department. The length of the panel interview also varies, from about 30 minutes to 90 minutes. Commonly, interviews are scheduled to last for about 45 minutes, with time for you to ask questions yourself.

In the formal interview, the initial impression you make is very important. For example:

- How you greet the panel when you arrive.
- Do not sit down until you are invited to do so.
- Think about your body language; for example, how you sit (sit comfortably on your chair, looking alert; do not sit on the edge of your chair looking anxious or slouch in your chair looking too relaxed).
- Look and sound calm, relaxed and confident (even if you are not).
- Try to be yourself.
- Try to smile and direct your answers to the person who has asked the specific question, as well as the panel, during discussion.
- You may want to change your body position slightly to face the person asking the question so that you maintain eye contact when answering questions.
- Try not to speak too quickly and to keep your answers to the question posed.
- If you are unsure about how much information to give when answering questions, it is probably better to keep an answer brief and then ask the panel if they would like further information.
- If you are not sure what the question means, do not hesitate to ask for clarification. If your mind goes blank, ask them to repeat the question.
- Avoid repetition, but do not worry if you repeat information included in your application, as long as you do not contradict what you wrote.

Interviewers have various degrees of specialist knowledge and understanding. Avoid jargon in explanations but assume interviewers have some knowledge and understanding of your subject area. Aim to provide a balanced picture of yourself, being, on the whole, positive and emphasising your strengths, while acknowledging areas of potential development. Interviewers are trying to

form an impression of you as a future teacher and as a person, and so have a number of things they are looking for. These include:

- *Your knowledge and understanding of your subject and your ability to teach it.* Interviewers assess your ability to discuss, analyse, appraise and make critical comment about ideas, issues and developments in your subject and subject curriculum, your personal philosophy about and commitment to the teaching of your subject(s);
- *Your professional development as a teacher.* This is based partly on your school placements. Interviewers assess your ability to analyse observations of pupils' learning, behaviour and development, your own development and your involvement in the whole life of the school-on-school placements, and your ability to discuss, analyse, appraise and make critical comment about educational issues.
- *Your ability to cope with the post.* Interviewers assess your understanding of the different roles you are required to undertake as a teacher and how you would cope (for example, how you have coped or would cope, in a number of different situations, such as disciplining a pupil or class experiencing difficulties, dealing with a concerned parent or with teaching another subject).
- *Your ability to fit into the school and the staffroom and to make contact with and relate to colleagues and pupils.* Interviewers assess your interpersonal skills, verbal and non-verbal communication skills (your written communication skills have been assessed from your application).
- *Your commitment to the specific post.* Interviewers assess the interest and enthusiasm you show for the post, to try to find out whether this is a post you really want or whether you see this post as a short-term stop-gap before you can find a post in an area or school where you really want to teach.

After introductions and preliminaries, interviewers normally ask why you have applied for this specific post. They then focus on the information in your application, including your education, qualifications, teaching skills gained from school placement and other teaching and/or work experience, your interests and activities and other qualifications. Throughout your ITE programme, you should have been encouraged to gain as much wider experience as possible; it is now that you can demonstrate how this has contributed to your own personal development as well as how it will benefit the school as a whole. In addition, you are normally asked what you feel you can contribute to the school and general questions about professional or personal interests, ideas, issues or attitudes.

Remember that the interview provides you with an opportunity to expand upon information contained in your original application. Consequently, you can predict some of the questions that are likely to be asked based on the job description or person specification. Look back at the initial information you received to identify the parts of the job on which they expect to get information at the formal interview. Try to give practical examples to support your answers. If asked about your approaches to teaching, give examples of how you have used different approaches in your own practice. Talking through an example tends to allow you to demonstrate a deeper level of understanding than writing it down. However, try to remain focused. Make your point as clearly and succinctly as possible and then stop. If the panel want more information, they will probe further.

Therefore, think about areas you want to emphasise or any additional evidence of your suitability for the post that you did not have room to include in your application. Draw on both the university-based part of your ITE programme (for those on an ITE programme in partnership with a university) and school placement, other teaching experience and other experience; for example, other work

with young people, such as work in a youth club or voluntary work allowing you to demonstrate your commitment to working with young people.

You also need to show that you realise you still have things to learn and that you are committed to continuing your development as a teacher. As you come to the end of your ITE programme, you are expected to develop some form of early professional development/transition planning, so use this to support your answers. You should come prepared with examples of potential short-term and longer-term targets for development. For example, you need to consolidate your learning in your first year, perhaps gain further experience of other areas, such as teaching Years 12 and 13 or taking on a tutor group. You might also wish to consider undertaking an additional academic qualification, for example, studying at master's level (see also Unit 8.3). It is helpful to have a career plan, but not to appear so ambitious that you give the school the impression that you will leave at the first opportunity.

Possible interview questions

Questions asked at interview vary considerably; therefore, it is not possible for you to prepare precisely for an interview. However, it is helpful if you identify possible questions in your preparation and prepare some possible outline responses to such questions. It is useful to give a general response to the question to show you are aware of some of the principles and issues and also to refer to examples of your own practice. For example:

Interviewer

How did you set about adapting your teaching to meet the needs of all pupils in your classes?

Candidate

My starting point is to have a clear understanding of the pupils in my class. I do this by speaking to my mentor and support staff in school as well as accessing pupil records to identify specific needs. I then look at what I want my pupils to learn during the lesson and consider careful in my planning the sequencing of my activities. As part of this process, I look to break tasks into small steps and provide scaffolds and modelled examples. For example, in a recent lesson I taught I adopted a 'I do, we do, you do' approach that allowed me to model what I wanted the class to do. We then did an example as a class and then, if the students felt confident, I asked them to complete the next example on their own. For those who were less confident or who I had identified as needed more support, I completed another 'we do' example before giving them the opportunity to do it on their own. I then monitored their progress, using live marking and questioning.

See 8.2a on the companion website for some examples of possible interview questions.

Other questions

At the start or end of the interview you may be asked if you are still a firm candidate for the role or if you would accept the post if offered. This is an opportunity for the panel to confirm that you are still interested in the job. It is not a trick question.

At the end of the interview, you may be asked whether you have any questions. Asking one or two questions shows a genuine interest in the post and the school. You are likely to forget the questions you wanted to ask if you are nervous; therefore, do not be afraid to prepare a list of questions

and take this with you to interview. It is quite acceptable to refer to this at this stage of the interview. You may also wish, at this stage, to seek clarification on any issues that might have arisen during the day, such as whether you would be expected to have a tutor group or deliver across subject areas. You should enquire what arrangements there are for induction of ECTs (see DfE (Department for Education), 2024) in the school and what you might expect. You may also want to ask questions about the contract, for example, if this is not included in the application information you were sent, to check about the type of contract that is being offered (such as fixed term or temporary contract) or, if the school is a MAT, if you are only expected to teach in this school. You can also ask when you would be expected to start (some schools employ staff during the summer term or offer part-time work once you have finished your commitments to your ITE programme).

At some point, you may want to ask about your starting salary (unless this has already been indicated at the application stage). If you are an ECT, there may be little room for negotiation unless you have something the school really needs/wants, for example, if you have been a cover supervisor previously and/or have specialist behaviour management training above and beyond what you would have covered in an ITE programme. How you describe experience in an application and at interview, therefore, is very important as it may be used to support any claim for increments above the starting salary.

Avoid asking questions just because you think you should or just to impress. If all your questions have already been answered and you do not have any questions, just say politely that all the questions you wanted to ask have been answered during the course of the day (or in the interview).

Earlier, we identified the benefits of practising your lesson prior to delivering it at interview. Many ITE providers provide opportunities for you to experience a mock interview. You are encouraged to take up any such opportunity (see Task 8.2.1).



Task 8.2.1 A mock interview

Arrange for a mock interview with your mentor or tutor. If possible, either have an observer or video the interview so that you and the interviewer can use this to analyse your verbal and non-verbal communication after the interview. Identify ways you could improve and practise these. If possible, organise another mock interview.

Accepting a post

The process of being notified about the outcome of the interview should be explained to you at the end of the interview. If you are not going to be told prior to leaving the school, confirm the best form of contact, for example, your mobile number or landline. You may have to wait several days before being told whether or not you have been successful, but you should be told this in advance.

Where all candidates are invited for interview at the same time, you may be offered the post on the same day as the interview. You are normally expected to verbally accept or reject the offer at this time. Schools rarely give you time to think about an offer; therefore, it is important that you consider all the implications of accepting the post before you attend for interview. On rare

occasions, it may be that you feel you really need some time to think about the offer. You may want to ask whether you can think about the offer; in which case, be clear about when you will inform the school of your decision (for example, you want to think about it overnight and telephone first thing in the morning). Be prepared for such a request to be rejected. If your request is refused, you have to make the decision there and then or be prepared for the post to be offered to another candidate. Your decision depends on how much you want a particular job and how strong a position you think you are in.

If you are offered a post verbally at or following the interview, the offer of a post must be followed by a written confirmation and a contract. Also, it is normal practice for you to be asked to confirm your verbal acceptance of a post in writing. This might be via the completion of a letter issued by the school or in a formal letter written by yourself.

Check carefully your contract, your salary and the induction programme. Offers of a post are made on certain conditions and therefore may be conditional until those conditions are met (for example, until you have passed a medical, in which case the school or the LA sends you details), until references have been received or until you have met legal requirements (for example, undertaken an enhanced check with the Disclosure and Barring Service (DBS) – although you will have completed one of these as part of your ITE programme you may have to complete a new one: the school or HR department will advise you).

Once you have signed your contract, you are tied to it, so spending time seeking clarity is important. It is unprofessional to continue to apply for other teaching posts after you have verbally accepted a post, even if you see one advertised that you prefer.

The transition from beginning teacher to ECT is not easy. Being a teacher involves you coming to terms with your new role within a new institutional context with an awareness of the complexities of its organisation, structures and routines, as well as ethos and expectations. This transition may be easier if you have gained employment in one of the schools in which you completed your school placement. If not, make arrangements (with your future mentor, for example) to visit the school again to make preparations to start your post. This is not dissimilar to the preparations you make when you go on school placement, that is to meet members of the department and find out about facilities in the school and collect information about your teaching timetable as well as classes and what you will be teaching. You may find it useful to make a few visits to get to know the school culture. For example, you may become involved in some school activities such as sports day or perhaps observe some classes or teach some lessons before you start your post.

If you are not offered a post

It is disappointing when a post is offered to another candidate. However, try not to think of this in terms of failure on your part. There may be many legitimate reasons why the post was offered to the other candidate in preference to you. For example, the other candidate might have relevant teaching experience (which you did not have) in an aspect of the curriculum required for the post. If you are not successful, build this into your learning experience. Most interview panels routinely offer feedback to candidates. If not, you can ask if this is possible, as it helps you identify strengths and areas to develop in preparation for your next interview. Try to be reflective about the decision and acknowledge where you might need to make improvements next time. For example, can you gain any further experiences to strengthen your application? Can you take up any further opportunities to practise your interview technique?

Also, if you are still looking for a job in September, do not be too disappointed and give up. Ask yourself whether you have been too set on obtaining your ideal job and need to reassess your search criteria in terms of area, type of school or age range, for example. You may also want to consider other options, such as part-time work, a job-share, temporary post or supply work. These can give you valuable experience while you continue to search for a permanent full-time post and can sometimes lead to permanent work in a school. You can find out details about supply work in your chosen area from the LA, schools and teaching supply agencies (see above for information about supply teaching as an ECT). It is also worth maintaining contact with your ITE provider to see whether they are aware of any opportunities. You might even start to consider other teaching-related opportunities. Tasks 8.2.2 and 8.2.3 are designed to help you to reflect on the change from beginning teacher to ECT.



Task 8.2.2 Moving from beginning teacher to ECT

How do beginning teachers learn to become teachers?

Identify similarities and differences between being a beginning teacher and being an ECT. Discuss your perceptions with some other ECTs. Critically compare their biographical stories with your own considerations for entry into the teaching profession.



Task 8.2.3 Entering and staying in the profession

Carry out an initial review of journal articles on the topic of entering and remaining in the teaching profession. Design and conduct a piece of small-scale empirical research with selected established teachers in your placement school about the factors that: (a) influenced them to become teachers; and (b) make them stay in the teaching profession. Report critically on your analysis, identifying issues related to the literature and the philosophical assumptions underlying the methods of data collection and analysis. Give a critical evaluation of their choice, design and effective use, as well as a critical analysis of the factors identified and comparison with the literature and with your own personal circumstances.

SUMMARY AND KEY POINTS

This unit is designed to help you realise that, just as with your teaching, you must prepare for obtaining your first teaching post; you cannot leave it to chance or rely on your innate ability to perform well at interview. In this unit, we have tried to lead you through the steps, skills and techniques you need to prepare actively for obtaining your first post, that is:

- attending an interview in order to secure a full-time post;
- accepting a post.

To be successful, you need to:

- be proactive;
- be prepared;
- practise your techniques;
- be confident.

We wish you the best in obtaining your first post.



ONLINE RESOURCES

For bonus digital resources designed to enhance the book materials, head to routledgelearning.com/learningtoteach.





Further resources

The Chartered College of Teaching's Early Career Hub, viewed 26 May 2025, https://earlycareer.chartered.college/?s=job

The Chartered College of Teachers has several articles on the topics of looking for a job, applying for a job and interviewing for a job etc.

Reference

DfE (Department for Education) (2024) *Induction for Early Career Teachers (England) Statutory Guidance for Appropriate Bodies, Headteachers, School Staff and Governing Bodies*. Available from https://assets.publishing.service.gov.uk/media/6629237f3b0122a378a7e6ef/Induction_for_early_career_teachers__England__statutory_guidance_.pdf [accessed 26 May 2025].

8.3 Developing further as a teacher

Polly Glegg and Mark Quinn

Introduction

During initial teacher education (ITE) you receive a considerable amount of advice, observe many experienced teachers, become familiar with 'requirements' of the role and develop your teaching. However, none of this, and nobody, can tell you exactly how to teach to enhance pupils' learning. That is because teaching is a highly complex and relational *professional practice*; teachers develop ethical, social and intellectual behaviours that make them increasingly attuned to pupils as young individuals and affect the likelihood of pupils being able to learn. While teachers must, of course, become competent in a repertoire of teaching practices for use in their classroom, the complexity of their work means that they also need a sound conceptual framework – a set of connected ideas, concepts and theories that represent their understanding of teaching and learning – to guide their choices about which approaches are most appropriate to use at any given time. To develop as a teacher during ITE, on transition to an induction period and beyond, therefore also means examining your assumptions and beliefs about how pupils learn and about the teacher's role in this.

You may find yourself teaching with considerable energy during your ITE school experiences, but when you look deeply at what your pupils are learning, you may identify some significant gaps between what you set out to teach and what your pupils actually learn. This is completely normal in the early stages of learning to teach. It is important that you resist interpreting this as a fault of your pupils. This can be called a 'deficit analysis', where the teacher pays less attention to their own development needs and identifies 'problems' within pupils as the reasons why lessons do not go as anticipated. For example, when pupils' slow progression is attributed to a lack of 'ability' (a very common misconception) or their own behaviour issues (see Chapter 4.4). This is a major obstacle to beginning and early career teachers' development. As professionals, teachers accept that they are responsible for the conditions they bring about in the classroom to foster learning. Your development as a teacher therefore depends on how you are able to interrogate deeply what has happened in your lessons, including your own actions, and to reject simplified explanations for why pupils do not always learn what you intend. Development means that you become more able to adapt your teaching based on deepening knowledge of your pupils, within the context of your particular classroom and the subject(s) you teach. For this, you need good *professional judgement*.

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This unit introduces you to ideas around becoming a professional in the early years of teaching, developing your identity as a teacher and continuing your professional learning over time. Within this, it introduces you to core ideas about how teachers learn, in relation to your development as a professional. Exploring these ideas is not just important at the start of your career; they underpin the ways in which teachers develop into expert practitioners and continue learning throughout their professional lives.

OBJECTIVES

At the end of this unit you should be able to:

- understand what it means to become a 'professional' as you transition beyond your ITE programme;
- reflect on your developing teacher identity;
- understand the importance of learning throughout your teaching career and that developing professional judgement is highly complex;
- identify strategies to foster your ongoing professional learning as you move from ITE into employment (critical reflection, enquiry, the use of evidence, engagement with theory, collaboration, mentor feedback, engaging with professional communities).

Check the requirements of your ITE programme to see which relate to this unit.

Becoming a 'professional'

When we say that no one can tell you exactly how to teach, we are referring to a particular view of what it means to be a teacher, underpinned by an understanding that teaching is a professional practice. The term 'professional' is used in a variety of contexts, often meaning 'paid for' as distinguished from 'amateur', or as shorthand for a job well done. However, it also has a specific meaning as a distinctive form of occupation.

Common characteristics in different approaches to defining the term 'profession' are that it is a particular category of occupation which plays a specific, beneficial, role in society and which requires specialist expertise not available to a lay person. Each profession comes with a set of assumptions about the nature of the work and the responsibilities and activities of a professional. Professionals exercise technical skill with sound judgement and act under an ethical code (see Unit 8.5). In performing a specific role, with specific expertise, comes trust, prestige and status in society.

The distinction between professions and other spheres of work is evident in a categorisation suggested by Orchard and Winch (2015). Focusing specifically on teaching, they refer to three ways in which teachers and teaching are viewed, connected to the forms of knowledge needed to do their work (see also, for comparison, Shulman's (1986, 1987) categories of knowledge in Unit 1.1). Understanding these categories should help you when reflecting on your personal understanding of the components of professional practice, and when articulating the different aspects of your expertise.

The three forms of knowledge identified by Orchard and Winch (2015) are: situated understanding (also called 'tacit knowledge'), technical knowledge and critical reflection. Situated understanding is the sort of knowledge that is accumulated through experience and is often hard to quantify. Technical knowledge is well-codified, often through theory or research. Critical reflection refers to teachers' capacity to 'review thoughtfully and systematically what they have done in the past with a view to sustaining or improving their practice in the future' (Winch, Oancea and Orchard, 2015, p. 206).

The first category sees teachers as 'craft workers' who are heavily oriented towards situated understanding and who become more expert as they accumulate situational knowledge through experience and apprenticeship learning from expert colleagues. Such an approach to professional practice, if followed dogmatically, would be unsatisfactory because it would discount the wealth of theory and research which is available to help teachers make informed decisions about their work, alongside their situational knowledge.

In the second category, teaching is characterised as a highly skilled activity for which teachers should rely on technical knowledge to learn 'what works'. Orchard and Winch (2015, p. 11) argue that this continues to be a fairly impoverished view of teaching, as it positions teachers as 'executive technicians', who are 'told prescriptively by others what to do, without needing to understand why they are being told to do it'. This is also unsatisfactory since it underplays the importance of the teacher's situational knowledge and capacity for critical reflection on practice.

Orchard and Winch (2015, p. 14) instead promote a conception of teachers as professionals:

> The teacher who is able to engage with theory and the findings of educational research shares with the craft worker teacher a capacity for self-direction. By contrast, though, the professional teacher is able to judge right action in various school and classroom contexts from a more reliable basis for judgment than intuition or common sense. A teacher who is able to make good situational judgments does not rely on hearsay or unreflective prejudice. She draws on a well-thought-through and coherent conceptual framework, on knowledge of well-substantiated empirical research, and on considered ethical principles, to arrive at decisions in the classroom context.

The distinction, then, of teaching as a profession takes into account a more expansive view of teachers' knowledge and work, acknowledging the complexity of teaching and learning and the number of judgements teachers have to make in the course of their professional practice. It prompts teachers who are committed to developing as professionals to keep learning across the different knowledge bases identified – technical knowledge, situated understanding and critical reflection – without privileging or underplaying any of them.

Being a learner of your own teaching

As well as developing your knowledge bases through professional learning, you need to learn about your own teaching. This means focusing predominantly not on what you do as a teacher, but on how well pupils learn as a result of what you do. You can see how this connects to the description above of Orchard and Winch's (2015) 'professional teacher', because to do this well you will need to engage in deep critical reflection. It can be quite uncomfortable to reflect on your impact and acknowledge that pupils are not yet learning as you'd like them to, but this may be a necessary step in helping

you to recognise where and how you can develop. It is therefore important that you actively seek ways to keep developing professional knowledge in all its forms. Embracing this expectation from the start of your career will help to orientate you as a member of the teaching profession. Now complete Task 8.3.1.



Task 8.3.1 Being a learner of your own teaching

According to Hattie (2009), who carried out analysis of over 800 studies of effective teaching, '... the biggest effects on student learning occur when teachers become learners of their own teaching' (p. 22).

Before reading further in this unit, reflect on this statement, then discuss with other beginning teachers and your mentor/tutor:

- What do you think it means for teachers to become 'learners of their own teaching'?
- What activities and processes can help you to become a 'learner' of your own teaching?
- What challenges might you encounter to maintaining a focus on being a learner of your own teaching during your time as a beginning teacher and beyond?

Developing a teacher identity

Growth as a professional is deeply connected with growth as a person. That is why becoming a teacher can be unsettling as well as exhilarating; you are developing a teacher identity that involves far more than the acquisition of a set of skills (Beijaard et al., 2023). It involves considering the values and beliefs you currently hold about how pupils learn and re-evaluating your own attributes that will help you to become a teacher. You are forming your identity as a teacher at the same time as learning how to teach, to the extent that both of these processes can be seen within a reinforcing framework of what Beijaard et al. call 'professional identity learning' (2023, p. 764). This necessarily involves some discomfort and most beginning teachers find themselves having moments of doubt during their ITE – about their subject knowledge, skills, capacity to make relationships with learners and ability to manage the sheer complexity of teaching. It is extremely important to be aware of this process and to maintain a balanced perspective. Becoming a teacher is a life-changing choice. You will no doubt face many challenges, become extremely tired, possibly emotional at times. Hopefully you will also discover just how resilient you can be (see Unit 8.5) and also how immensely rewarding the job of teaching is.

It is not unusual to hear beginning teachers use acting metaphors when referring to classroom practice: the classroom becomes a 'stage', they worry about their 'performance' in a lesson or in their interactions with pupils. In more reflective moments, new teachers might recall occasions when they did (or did not) really 'feel like a teacher'. These metaphorical notions of performance can harden, particularly in schools where performance (of pupils, of the teachers, of the school) is measured frequently and in high-stakes ways. As you develop your teacher identity, you might be attracted to the role of performer – particularly so if your specialism (say, drama or music) is classed as a performing art. Some research (e.g. Englund and Frostenson, 2017) would caution

against striving towards a performance identity, as it can clash with the performance management methods operating in the school and lead the teacher into self-identifying as a 'struggling performer'. In any case, the more you realise that your focus ought to be less on your own actions and more on the impact of your actions on your pupils, the less helpful it becomes to think of yourself as being the performer on the stage. As you continue to develop as a teacher, consider the metaphors that best reflect your identity as a teacher: how else might you represent your work, if not as 'performance'? What does this communicate about your evolving conception of your professional self and work?

Developing professional judgement

> in education, "what works?" is rarely the right question, because everything works somewhere, and nothing works everywhere, which is why in education, the right question is, "Under what conditions does this work?"
>
> (Wiliam, 2014, n.p.)

This quotation from Dylan Wiliam begins to highlight the importance of teachers' professional judgement in making decisions about how to approach their teaching.

We have already introduced Hattie's (2009) idea of becoming a learner of your own teaching, which you will recognise is entwined with developing professional judgement because of the critically informed stance it encourages you to adopt. To do this requires a range of professional learning strategies or 'tools' that enable you to develop the judgements that form the basis for professional action. In England since 2021 new teachers' induction has been underpinned by the Early Career Framework; from 2025, the ITE and induction years are brought together under the Initial Teacher Training and Early Career Framework (ITTECF) (DfE, 2024a). Some commentary (e.g. Hordern and Brooks, 2023) criticises such frameworks for being too technicist and for promoting a narrow, ideological view of what teaching is, instead of foregrounding teacher judgement. However, the ITTECF – which sets out what it believes all new teachers should know about and how to do – also stipulates that teachers must be supported to develop their knowledge and practice through a selection of approaches that include feedback through mentoring, observation and 'seeking challenge and critique from expert colleagues with beginning teachers taking the lead in their own development' (DfE, 2024a, p. 8). Professional learning tools such as these are complex and underpinned by extensive research and theory (some are explored in other units in this book; see particularly the units in Chapter 5) but are also implicit throughout discussions of all aspects of developing as a teacher. These professional learning tools also include:

- critical reflection on your actions and behaviours as a teacher;
- enquiry into your own practice and its effects on your pupils (Unit 5.4 looks in more detail at practitioner enquiry/research);
- judicious use of a range of evidence that can inform decisions about teaching approaches;
- the use of theory about learning and teaching to inform your developing practice;
- collaboration and interaction with a range of other professionals;
- mentor dialogue.

These learning tools underpin your developing effectiveness, increasing your capacity to make careful and informed judgements about your behaviours and actions. This requires systematic focus because developing your judgement in a deep way does not occur naturally and cannot be left to chance in the very busy environments of schools. These learning tools are vital to overcome both 'common sense' approaches on the one hand (which may not actually be very sensible), and rigidly prescriptive approaches to becoming a teacher on the other. It is essential that you learn to adopt tools for teacher development that move you beyond focusing on 'survival', or 'easy solutions' that everyone can copy because they reduce the requirement to think deeply about your actions, your rationale for them and their consequence for pupils.

Support for you in being such a learner is available in the ways your ITE programme has been designed: through reading material and tasks such as those contained in this book, mentor/tutor input and feedback, discussions with peers, etc. A critical source of support for your development throughout your school experience is the school-based mentor, or cooperating teacher (see Unit 2.4). Professional organisations and associations such as subject associations (www.subjectassociations.org.uk) also provide support and are often excellent ways to learn about relevant research and learning opportunities. You also have a responsibility, as a professional, to proactively engage with opportunities available to you, so make a point of creating time to read, to research and to collaborate with others as your career progresses.

Early career development

Having met the requirements to become a teacher at the end of your ITE programme, you will work towards meeting a further set of expectations during your induction period. Although arrangements are different depending on the country in which you are teaching, most education systems set minimum expectations for support for teachers' early career development and the entitlements of new teachers to professional learning. A European Commission/EACEA/Eurydice (2021) report found that formal induction programmes are mandatory in most European countries and that more experienced teaching staff were commonly responsible for supporting (mentoring) new teachers in their first year. This reflects approaches commonly found in other countries, too (OECD, 2014). Each school has different support systems and ways of working with wider provision for induction (e.g. local authorities or school networks), but each should provide you with an induction mentor/tutor and protected development time. In addition, in England and many other countries, you should also expect to be observed and to have your performance assessed.

Beyond induction, schools have appraisal and performance management arrangements that continue to provide support. In the United Kingdom, each of the countries has its own standards, expectations and requirements for continuing professional development (General Teaching Council for Northern Ireland (GTCNI), 2011; General Teaching Council Scotland (GTCS), 2021; Welsh Government, 2023; Department for Education (DfE), 2024b).

Moving from your ITE programme into your first teaching post is an exciting, but possibly daunting, prospect. However, you are not expected to demonstrate the attributes of an experienced teacher and your school should provide you with a support programme. Inevitably, you will be anxious about this phase of your career and you will have expectations based on your current and past experiences in schools. All of this is very normal; most new teachers embark on induction with concerns that are varied but often include behaviour management issues (Hobson et al., 2009). The concept of 'practice shock' (e.g. Stokking et al., 2003) describes the challenging impact on new

teachers of moving from their ITE programme to the full commitment of working in a school. While this may be lessened if you have completed employment-based ITE, you should still be realistic about what you can achieve in your early days as a qualified teacher and acknowledge that you may feel less competent or confident for a while as you settle in. Recognise that each school is different; any difficulties you have experienced in one school may not be present in another school. Of course, the reverse is also true, and when you start your induction period, circumstances in a different school may present new challenges.

Research has explored the resilience of teachers. Gu and Day (2013, p. 39) define this as 'the capacity to manage the unavoidable uncertainties inherent in the realities of teaching'. They identify the range of influences on teachers' capacities to cope with the everyday demands of the job. These can range from personal to organisational influences and also the impacts of relations with others. Beltman et al. (2011), in their review of literature on teacher resilience, suggest that self-efficacy and intrinsic motivation, together with collegial and mentor support, are key protective factors. Your goal should be to sustain an approach to teaching that ensures you successfully meet induction requirements and, at the same time, maintain a proactive and planned approach to development. You are more likely to achieve this goal if you are a *learner of your own teaching*, seek support from more experienced teachers and, most importantly, take responsibility for identifying your own development needs.

Professional learning within a school community

As you may have experienced during your ITE programme, challenges come not only in the form of pupils: they also occur in your relationships with colleagues. As a new member of a subject department, you may be expected to understand the department ethos and work in particular ways before you are accepted as a valued, full member of the team. Lave and Wenger (1991) called this legitimate peripheral participation; that is, as a new member of the community, you are expected to watch from the sidelines and learn the rules before participating fully in the community. This concept can be seen as both problematic and productive. Working collaboratively as a member of a professional learning community, such as a subject department, is a very useful form of professional development. However, if membership of that community simply means adopting existing community practices or externally-driven agendas, it can be problematic (Kennedy, 2016). As a professional, you are responsible for upholding high standards (in England, this is set out in Part 2 of the Teachers' Standards (DfE, 2011); you also bring to your workplace your own expertise, ideas and experiences, which you arguably have a duty to use to contribute to the continued improvement of your community. When you are a qualified teacher, you will need to consider how to balance independence and innovation with collaborative working, and how to fit into the department while at the same time making an active contribution to the development of practice.

As you settle into the job some staff may forget that you are new. This can also be a problem. As the term and year progress, they may treat you as any other member of staff and not offer help and advice. Initially, this might be flattering, but this will not help to extend your expertise. Your learning is not just about addressing 'issues' – it is about becoming the best teacher you can be and making the greatest contribution to the learning of your pupils. Keep asking questions and seeking help from others – aim to maintain your identity as both a learner and a teacher.

You may also find it helpful to form a support group with other early career teachers in the school or within your subject community, through which you can share your concerns and problems, and support and learn from each other. Some schools create 'buddy' networks (Daly et al., 2021, p. 4) between recently qualified teachers and those beginning induction. Task 8.3.2 is designed to help you think through different forms of 'buddy' network that you could engage with.



Task 8.3.2 Learning from early career teachers

Set aside some time to reflect on, and research, the different ways that you could form or join a 'buddy' network to support you as you move into your induction year:

Speak to some early career teachers that you know, either in your ITE placement school or, if you are already in your induction year when you read this, in your school. Ask them how they access support networks, and what their experience of induction has been. What ideas can you gain from this?

Explore virtual networks that you could engage with, such as Facebook groups or other online forums for teachers of your subject(s). What are the benefits of communicating online, as well as forming in-person networks? What challenges might you encounter in doing this?

The impacts of school cultures on professional learning

As a teacher, you are a member of a community, and learning in that community has its advantages and challenges for your further development, depending on the cultures of professional growth adopted within the school.

Teachers bear their own responsibility to improve, but they cannot do this in isolation from their school or the wider educational community. Gu (2023) reminds us that schools, through high-quality leadership, must provide a range of resources to enable their teachers to grow – including intellectual, technical and social resources. Schools, school groups and education initiatives that take a purely 'human capital' approach – focusing only on the knowledge, skills and attitudes of individual teachers – fail in their missions to improve outcomes for children. This is because they fail to take into account the need to nurture the devotion and energy of committed teachers. Gu (2023) identifies school organisation, structure and culture as key mediating factors which help teachers to thrive in their chosen settings.

Certain modes of professional learning – for example, instructional coaching, whole-staff training, being sent on a course – are, despite their differences, associated with a human capital approach. By contrast, a 'social ecological' mode of teacher development emphasises the intellectual, social and organisational conditions in school workplaces and recognises the moral and political agency of teachers; prevalent forms of professional learning in this mode include collaborative action research, teacher networking and professional learning communities.

It can be helpful to do your research about schools you are thinking of applying to work at, to be confident that the culture of the school aligns with your own wants and needs in relation to professional learning.

Task 8.3.3 asks you to consider the forms of professional learning that you have encountered during your ITE.



Task 8.3.3 Reflecting on professional learning

Think about how you have learned during your ITE this year. You will likely have experienced a mixture of self-study, lectures, seminars, groupwork, mentoring and systematically trying things out in the classroom. Which of these might be termed 'human capital', and which could you think of as 'social ecological'?

Many argue that different forms of professional learning are more/less appropriate at different points as you develop as a teacher. Thinking ahead to the next year or two, which forms of professional learning do you anticipate being most appropriate for you?

What steps could you take to ensure that these forms of professional learning are available to you as you move into the next phase of your career? Discuss your ideas with other beginning teachers.

SUMMARY AND KEY POINTS

Learning to teach continues throughout your career. Your ITE is naturally the focus of your concerns right now, but it is important to consider this in the context of a teacher's complete professional life. Teachers promote curiosity in those they teach and this must be equalled by their own, ongoing, curiosity about how learning happens and how they can develop their practice to enable that learning. An underlying message throughout this book is that knowledge about how to teach is made by the individual, supported by engaging with pupils and colleagues and by the ways you work with the existing body of professional knowledge about teaching your subject: the knowledge base. This is facilitated by established professional learning strategies and tools that support you to be critical and enquiring in how you work with evidence to support your judgements. In conclusion:

- your experiences as a beginning teacher should establish you as a learner of your own teaching;
- this means taking responsibility for seeking support for your learning and being prepared for challenges to your current beliefs and assumptions about what makes an effective teacher;
- becoming a professional means developing your teacher identity;
- it also means developing the capacity to make informed judgements that affect your behaviours and actions;
- these judgements are individual but not simply dependent on your personal characteristics or circumstances; they involve others in exploration of your teaching and critical reflection on it;

- your development is facilitated by professional learning tools (critical reflection, enquiry, the use of evidence, engagement with theory, collaboration, mentor dialogue). These are explored in other units in this book, especially units in Chapter 5.
- school cultures impact on forms of professional learning you can experience.

Your further professional development is therefore a complex mixture of your own history and values as a learner, working within the unique social context of your classroom and the wider school, supported by sources of expertise in understanding and teaching your subject.



ONLINE RESOURCES

Visit routledgelearning.com/learningtoteach for a suite of digital resources designed to enhance your learning.





Further resources

***Andrew Pollard – Reflective Teaching*: https://www.reflectiveteaching.co.uk**

Andrew Pollard has written a substantial amount of print-based and online material on the subject of 'reflective teaching'.

Daly, C., Gandolfi, H., Pillinger, C., Glegg, P., Hardman, M.A., Stiasny, B. and Taylor, B. (2021) *Early Career Framework: A Guide for Mentors and Early Career Teachers*, London: Centre for Teachers and Teaching Research, UCL Institute of Education, viewed 25 October 2024, https://discovery.ucl.ac.uk/id/eprint/10124277/.

This is a brief guide to effectively meeting the needs of early career teachers in schools. It is intended for both early career teachers and mentors, setting out practical ways to support professional development during the induction phase. It is based on research into the pilot support programmes for new induction arrangements in England but the suggestions apply to all induction contexts.

Pollard, A. (2014) *Readings for Reflective Teaching in Schools*, London: Bloomsbury.

This book is a collection of extracts from the most important and influential thinkers and researchers into how teachers develop. The readings are short and grouped thematically. They have been chosen for their high relevance to teacher development. Teachers are introduced to relevant theory in a manageable way. The companion text is:

Pollard, A., Daly, C., Burn, K., Higgins, S., Kennedy, A., Mulholland, M., … Yandell, J. (2023) *Reflective Teaching in Secondary Schools*, London: Bloomsbury. This book provides a comprehensive collection of reflective activities.

These two books should support long-term career development. Extracts from these longer works, including visual summaries, can be found at Andrew Pollard's website: http://reflectiveteaching.co.uk/. Brief activities are provided that support beginning teachers to develop reflective practice, many of which expand the themes discussed in this unit, for example, 'To reflect on your own decision to become a teacher'.

***Reflecting on Outstanding Teaching, featuring Dr Russell Grigg. The Teachers TV ITE Lectures*, viewed 31 July 2025, www.youtube.com/watch?v=Wg32-PmpROA**

In this film (40 minutes), the speaker leads a discussion with beginning teachers about what it takes to teach effectively and to manage expectations as a qualified teacher about being 'outstanding'. It includes what pupils say about what makes great teaching from their point of view and is a powerful reminder that teachers learn a huge amount by listening to them. The first ten minutes are especially relevant to those in the early stages of learning to teach, while the whole film is helpful following a period of school experience.

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8.4 Developing your resilience

Managing stress, workload and time

Madeleine Findon and Sue Johnston-Wilder

Introduction

Teaching is rewarding, challenging, demanding and pressured. The most recent Teaching and Learning International Survey (TALIS) (Organisation for Economic Cooperation and Development (OECD), 2018) showed that over 90% of teachers world-wide were satisfied with their job and their performance. However, between one-quarter and one-third of teachers reported being very or extremely stressed. Positive psychology suggests when you meet pressure with resilience, you thrive (Seligman, 2003). However, when pressure increases beyond your current level of resilience, life becomes stressful, which if managed poorly can cause mental and physical harm.

Education Support (2023) surveyed over 3,000 UK teachers for their annual Teacher Wellbeing Index, finding 78% of teachers felt stressed, an increase of 6% since the previous year. Further, that 51% of staff were experiencing insomnia or difficulty sleeping (a 6% increase from 2022). Becoming a teacher, you face one of the biggest challenges of your life, perhaps exceeding your existing levels of resilience. The good news is that resilience can be developed. This unit starts by considering how to develop personal resilience and how to help ensure your school promotes resilience.

Throughout the unit, we use analogies of physical endurance activities. Whilst such activities may not be something you would ordinarily do, the motivation, preparation and pacing required to achieve such goals have parallels with the teaching challenge. We argue that effective teaching requires preparation, self-knowledge and self-care, noticing warning signs and taking breaks or seeking support when required; connecting with a wider community can provide expertise, assistance and motivation. With the right coaching, preparation and support, people can improve performance and achieve more than they currently imagine.

Like running marathons, teaching can use up inordinate amounts of time and energy, inside and outside the classroom, including beyond the school day. School days often leave little time to breathe deeply or eat and drink properly. In order to manage personal wellbeing, and to preserve time for yourself and your friends or family, you need to use time and energy effectively. We advise you to plan at least one day and one evening a week free from schoolwork, even during term time, for proper rest and self-care.

We encourage you to reflect regularly upon your practice and utilise available resources from your school, such as existing lesson plans and homework instructions, or quality-assured resources

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and networking opportunities from your subject association. Likewise, be a team player within your school. The practice of teacher leadership is growing in popularity: teachers can lead each other by sharing resources and expertise, being a catalyst for change and committing to life-long learning (Harrison and Killion, 2007; Schott, van Roekel and Tummers, 2020). Being proactive in these areas can help develop skills, resilience and a healthy work environment.

The unit looks at how you can build your resilience as a teacher, manage stress, workload and time proactively to preserve your wellbeing, optimise your effectiveness, avoid stress-related harm and contribute to an environment that benefits all participants in the school community. We distinguish between 'pressure' as a source of motivation, as in 'I work well under pressure', and 'stress' as in 'I feel stressed and I am not coping', indicating pressure is becoming excessive and harmful. Our bodies and minds generally function well under pressure and short bursts of stress, but they are not able to sustain long periods of stress without harm, indeed the Health and Safety Executive (HSE) (n.d.) definition of stress is 'the adverse reaction people have to excessive pressures or other types of demand placed on them'; similarly, physical exertion can build strength and fitness, but too much physical exertion can cause illness and injury.

OBJECTIVES

At the end of this unit, you should know how to develop your resilience as a teacher by:

- identifying when you are stressed;
- identifying factors that may cause you stress;
- developing methods of managing stress proactively;
- developing ways to manage time and workload effectively.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

What is resilience?

Resilience can be thought of as positive adaptation in the face of adversity: not a personality trait, but a phenomenon encouraged or restricted by internal and external factors (Luthar, 2006). You are probably already familiar with schemes that aim to promote 'growth mindset', 'resilience' or 'grit' in the pupils you teach. More recent research (Liebenberg and Moore, 2018; Heshmati, 2020) has considered resilience in adults, drawing upon positive psychology that is concerned with adult wellbeing (Luthar, 2006).

Seligman (2003) described resilience as a combination of: confidence (to start a task), persistence (to keep going when things get tough) and perseverance (to recognise when to recruit support and seek alternative approaches). It is important that you build up your resilience to manage the pressures of teaching. Periods when pressure builds too high are times to proactively manage stress, rather than passively coping. The next section looks at building your resilience.

Building your resilience

Resilience sustains and enables teachers to thrive rather than just survive under challenge and pressure (Kitching et al., 2009). It can be thought of as mental 'fitness'. Resilience can be developed

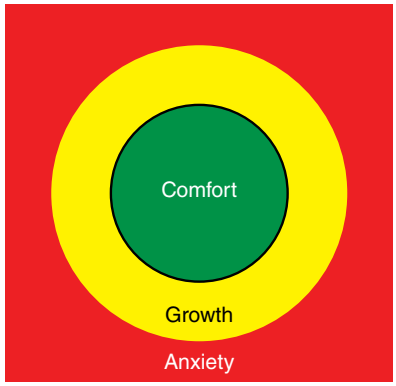


Figure 8.4.1 Growth Zone Model
Source: Johnston-Wilder and Lee (2008)

in times of challenge or adversity, where there is appropriate autonomy and support (Masten, 2014). In considering how resilience is built, it is helpful to consider connections between existing experiences and dealing with new ones, just as prior involvement in sport can equip us with the strength and coordination for new sporting challenges. A framework known as the growth zone model (Johnston-Wilder and Lee, 2008) (Figure 8.4.1), initially developed from research around growth mindset, agency, support and inclusion in the context of mathematics anxiety and avoidance, has been found useful in other stressful situations.

This model depicts three 'zones' or ways in which an individual experiences and deals with situations, using a psychosocial model of perceived risk. The comfort zone is characterised by feeling safe and confident, being able to use existing knowledge to good effect and not experiencing stress – like an easy run that is well within an individual's capabilities. Conversely, the danger zone is experienced as a place of danger or threat to wellbeing. In this zone, a 'fight, flight or freeze' reaction may be experienced: perhaps battling against (rather than engaging with) the scenario, fleeing or experiencing an inability to react cogently. Danger zone reactions occur because the primitive part of the brain reacts to physical, emotional and social threat; the amygdala ('the alarm system') triggers the release of chemicals such as adrenaline and cortisol. This useful reaction to physical danger is not amenable to clear, higher-order thinking (Siegel, 2010).

The zone between these two areas is where optimal growth can be experienced (see also Zaretskii, 2009): there is enough challenge to learn and develop one's skills, including managing risks and learning from mistakes. For the would-be athlete, this is where physical limits are stretched under the watchful eye of the coach and teammates. In becoming a resilient teacher, growth zone activity is optimised by the support of a learning community that encourages questioning and exploring strategies, thus helping to mitigate feelings of threat that can prevent healthy engagement. Being in a new school and unsure of the people around you, access to such support may be difficult at first; however, taking the initiative to develop co-operative networks, as with Harrison and Killion's (2007) teacher leaders, will help you and others to thrive and be resilient under pressure.

Some attributes and skills can promote resilience: altruistic motives, strong intrinsic motivation for teaching, high self-efficacy, feeling confident and competent, having coping strategies, taking credit for and drawing sustenance from accomplishments, collegial support and learning from mistakes (Ryan and Deci, 2000; Bandura, 2006; Hinds et al., 2015; Johnson et al., 2016). None of these are present in finite quantities; but all can be acquired and developed to an extent through

specific strategies or positive environmental conditions. Conversely, risk factors include: inability to ask for help (Fantilli and McDougall, 2009; Jenkins et al., 2009); perceived conflict between personal beliefs and school practices (McCormack and Gore, 2008); personal challenges or difficulties, including poor self-belief or confidence (Kitching et al., 2009); emotional avoidance (Hinds et al., 2015); difficult interpersonal interactions (Burke et al., 1996); lack of self-care and not noticing you have moved into your danger zone.

Day et al. (2011), amongst others, identified ways to build resilience and promote growth zone activity both in ITE and early in a teaching career:

- formal mentoring programmes and collegial support in the workplace;
- a mutually respectful, supportive mentoring relationship including open, honest, challenging yet sensitive communication;
- understanding roles and responsibilities and establishing realistic expectations of mentors;
- working together with mentors to improve teaching and learning;
- devising challenging targets for development that also recognise success;
- recognising the challenges in learning to teach, the reasons for these, and establishing collaborative approaches to seeking solutions;
- developing a mutually supportive, trusting network of peers and colleagues;
- improving self-evaluation, e.g. through use of video analysis of teaching or feedback from pupils;
- critiquing personal beliefs, values and practice;
- developing social skills, assertiveness, self-regulation and empathy (Tait, 2008);
- having autonomy and control over key aspects of work;
- valuing personal wellbeing.

Naturally, not all of these are necessarily within the control of a beginning teacher. Research (Heshmati, 2020; Heshmati, Johnston-Wilder and Findon, 2020) suggests the importance of attending to aspects of your situation that *are* within your control to build resilience. To achieve this, it is helpful to consider the principles and techniques of reflective practice. The points raised above have much in common with Schön's (1983) view that good professional practice requires rather more than application of technical knowledge: it requires making sense of complicated matters and engaging in careful inquiry about what has happened and what needs to happen next. Such reflection may take place in the moment, or some time afterwards: reflection-in-action and reflection-on-action (Schön, 1983) (see also Unit 5.4). Now complete Task 8.4.1.



Task 8.4.1 Building your resilience

Watch neurologist Dr. Daniel Siegel presenting the hand model of the brain on YouTube: https://www.youtube.com/watch?v=gm9CIJ74Oxw

Reflect on how the hand model may explain why you or a pupil may feel stupid or unable to think when confronted with something that relates to a prior bad experience.

Identify times you have been in your comfort, growth and danger zones. Create your own growth zone diagram. Add in keywords that help you notice what you experience in each zone. See example in Figure 8.4.2.

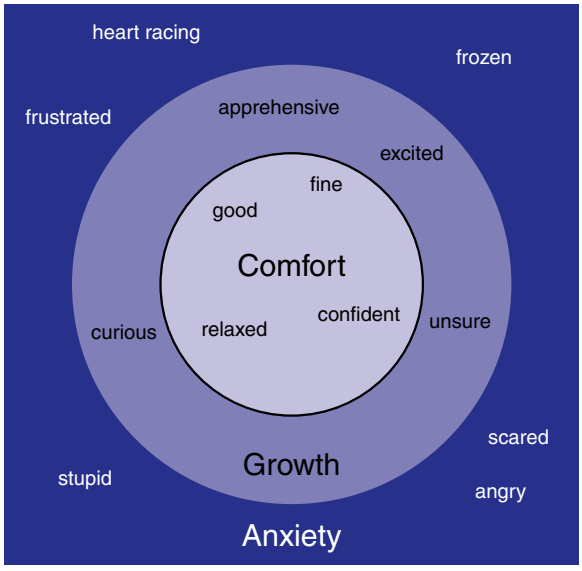


Figure 8.4.2 An example of an Annotated Growth Zone Model

Select some ways to build your resilience from the previous list. For each, reflect on your own situation and how you might improve it. If you select devising challenging targets, what are your current targets for development? Are they sufficiently challenging or could the challenge be increased but still be attainable?

Store your reflections, work on them and review your progress in a month.

Signs that current levels of resilience are not up to current demands

Our bodies and minds cope well with short periods in the danger zone; in fact, being occasionally in the danger zone is a good sign that you are undertaking as much challenge as you can. However, you need to be able to identify when you are in the danger zone and address the situation. Taking a break, breathing 5/7 (i.e. breathe in for 5 counts and out for 7 – for a more complete description go to ISMA https://isma.org.uk/isma-free-downloads?page=2, under 'The 60 Second Tranquiliser') or doing some strenuous exercise (Pittman and Karl, 2015) can all help.

Prolonged exposure to the danger zone may result in not thinking clearly, not sleeping well, irritability, tearfulness or stomach pain (Mind, n.d.). These are indicators of problem stress; you are unwell and you need to visit a doctor, just as you would if you had persistent pain. Historically, teachers we have spoken to have felt shame at this stage, but there should be parity between mental and physical wellbeing (see Royal College of Psychiatrists, 2024).

Preventing, managing and coping with stress

Teacher stress and teacher attrition are often connected (Harmsen et al., 2018). Data from 2024 showed the proportions of UK teachers still in post five and ten years after entering service were

67.6% and 57%, respectively (DfE, 2025). This wastage is not necessarily stress-related; there is no benchmark for 'normal' attrition rates in teaching and certainly career change or hiatus may occur for all kinds of reasons.

In line with broader cultural norms, the mental wellbeing of teaching staff in the UK has historically not been treated as equivalent to physical wellbeing (Education Support, n.d.). High levels of stress can be related to workload, pupil behaviour, organisational culture and inspections (Education Support, 2019, 2023). Furthermore, stress is 'contagious'; if 'the other person seems anxious about something we may conclude that this something may threaten us too (or that the other person's reaction itself is a threat to our well-being)' (Parkinson and Simons, 2012, p. 464); thus, beginning teachers working with teachers who are already stressed may find that impacts on their experiences.

There has been increased emphasis on self-care and wellbeing in recent years (El-Osta et al., 2019; Education Support, n.d.). You and your employer have joint responsibility to safeguard you; consequently, schools are progressively taking more care, especially regarding mental wellbeing of staff and pupils. However, in law, it is not self-evident that you are overloaded unless you have raised concerns in writing or returned to work after an absence documented as caused by work-based stress (a significant case is Walker v. Northumberland County Council, 1994). Thus, it is important that you identify causes of unnecessary stress for you and develop proactive strategies. You might go on a Mental Health First Aid course and certainly should inform yourself about when to seek help or medical advice.

Causes of stress

Causes of stress vary from teacher to teacher or even for the same teacher at different times, depending upon their current level of resilience and demands. Aside from the factors listed by Education Support (2019, 2023), personal experiences external to school, including physical health, life events or personal finances can all affect your level of resilience. A large-scale survey of teacher workload published by the DfE (2017) reported that teachers in England worked an average of 54.4 hours a week, suggesting that this significantly contributed to increased stress levels. This led to the initiation of the Workload reduction taskforce, which has set an aim of a 5 hour reduction per week for teachers and leaders within the next 3 years (DfE, 2024).

Other causes of pressure on teachers have been identified (Akhlaq et al., 2010; Klaasen, 2010; Klaasen and Chui, 2010; Maas et al., 2021), including:

- time pressure, including delivering unfamiliar material under pressure of time;
- motivating pupils and maintaining their interest;
- addressing the attainment range within a class;
- managing the class, including behaviour;
- dealing with conflict and confrontation;
- relationships with parents/carers.

In addition, there may be particular causes of stress for beginning teachers, for example:

- developing practical skills of teaching and lesson preparation;
- having unrealistic expectations of the profession;
- having unrealistic expectations of personal teaching performance;

- not being regarded as a real teacher;
- disagreement with the mentor or tutor;
- observation, evaluation and assessment of teaching by the mentor or tutor: feedback on classroom competence;
- role ambiguity, role conflict and role overload;
- lack of support in the growth zone.

Some beginning teachers feel they should know everything, be strong, not make mistakes and, in other words, be constantly in their comfort zone. However, some of the time you should expect to be in your growth zone; you can model being an adult learner to your pupils, sometimes vulnerable, sometimes making mistakes, being resilient. The role of the beginning teacher may thus feel ambiguous, but as a developing teacher you are required to function as both learner and educator. When you are in your growth zone, you need unambiguous, non-judgemental (unconditional), positive regard (Rogers, 2007). It may not come from a teacher-mentor with an assessment brief, but could come, for example, from a member of the pastoral team. Now complete Task 8.4.2.



Task 8.4.2 Identifying the causes of stress for beginning teachers

What causes you to go into your own danger zone at present?

Write a list of factors that cause you stress; whether in your role as a beginning teacher or outside. Compare these with another beginning teacher. Store the list and use it to complete Task 8.4.3.

Reflect on what works for getting you from the danger (stress) to growth (learning) zone.

Reflect on the possibility that it may be better to take one day off with stress symptoms than be ill for more days later in the year: why might teachers not do this in practice?

How can you cope with the pressures on you?

Long-term stress may result in you being less effective, having more issues with pupils' behaviour, taking time off or becoming burned out (Fernet et al., 2012; Agyapong et al., 2022). Different ways of coping with stress are appropriate for different people, and for the same person at different times.

Ways of coping have been classified in different ways. For example, Clunies-Ross et al. (2008) referred to proactive and reactive strategies, Lee and Johnston-Wilder (2015) used the term agency and Benson (2001) used the term autonomy. Personal strategies for coping with pressure may be classified as cognitive, physical, behavioural/organisational and emotional. Some examples of specific coping strategies are identified in Table 8.4.1, drawn from several sources (Crothers et al., 2010; Leung et al., 2011; Titchmarsh, 2012). Some of these strategies will help you to become more resilient and to promote learning and resilience in your pupils.

It is also important to pay attention to your aspirations and hopes as a teacher rather than focus on your concerns and fears. Conway and Clark (2003) suggested that focusing on resolving immediate concerns can result in 'an unduly pessimistic understanding of teachers and teaching' (p. 470).

Table 8.4.1 Stress coping strategies

Cognitive	■ *Have a good knowledge of what you are teaching*. Read around a topic as necessary prior to the lesson. ■ *Have a good knowledge of the structures, organisation and culture of the school*. Environmental understanding can help you navigate sources of pressure more easily. ■ *Identify where you can get help when needed*. Feedback is useful, but what sources of support are there? ■ *Take account of the amount and variety of work you are doing to reduce both role overload and conflict*. Try to take work home less often or take on fewer extracurricular activities. ■ *Develop attention to now (mindfulness)*. ■ *Recognise and try to develop your strengths as well as your weaknesses*. Rely on your strengths as you work on improving any weaknesses.
Physical	■ *Take regular exercise*. This releases endorphins in the body, relieving pain and making us feel better about ourselves. ■ *Eat regularly and well*. Stress is a drain on the immune system. Look after your immune system by eating well. ■ *Do some proactive relaxation*.
Behavioural/ organisational	■ *Proactively prepare for stressful situations when you are not under pressure*, e.g. prepare lessons days ahead. ■ *Actively prepare for a situation*. If you are anxious about a particular lesson, prepare it even more thoroughly: How can you reduce the likelihood of a problem occurring? ■ *Role play a situation that is causing you anxiety and/or visualise what you can do to overcome the problem*. This helps you to rehearse how you are going to cope. ■ *If you are asked a question which you cannot answer, you can praise the question, encourage pupils to look it up and find the answer before the next lesson*. You should then do the same: modelling being a growing learner. ■ *Plan how to work with a teaching assistant (TA) effectively*. Like anyone else, TAs appreciate knowing what they are doing beforehand. It is important to communicate well so that you work as a team in the classroom. ■ *Teach pupils to take responsibility for their own learning so that you are facilitating*. Provide them with the resources they need to learn so you do not have to orchestrate everything. This takes the pressure off you and is also effective pedagogically. ■ *Address any behavioural issues as early as possible* because this creates stress and puts pressure on time (see also Unit 3.3). Why does a particular pupil appear not 'ready to learn'? Is someone at home unwell? Remember that aggression often signifies fear. ■ *It is good to consider pupils outside the classroom*. It can help to support learning if you know something about pupils' lives. ■ *Develop effective self-management techniques*. Establish routines so that you can do things automatically, particularly when tired.
Emotional	■ *Develop social support systems*. Different people can provide help with different problems. Other beginning teachers can provide mutual support: discussing anxieties, developing shared understandings of problems and providing alternative solutions to address problems. ■ *If you worry about incidents that have happened in school, try to keep problems in proportion*. Try not to take problems home. ■ *Try to differentiate between feelings and facts*. You may feel that something has gone dreadfully wrong. If there is no evidence, no need to worry; if there is evidence, that can help identify how to improve things. ■ *Try not to worry about things that you cannot change*.

This is a pertinent observation for many areas of life, including our analogy of endurance activities: it is good to make regular checks on yourself, but to keep your focus mostly outwards and enjoy yourself (Latta, 2003). You might find this difficult as a beginning teacher but if you can do this, you are likely to have a more balanced view, be able to put things into perspective and therefore reduce your stress. Now complete Task 8.4.3.



Task 8.4.3 Coping with your stress

In Task 8.4.2, you listed causes of stress for you. Now identify ways that you can cope with this stress. Try out different coping methods and reflect on whether these are effective for you. Store your methods and evaluations; adapt or try new approaches until you find those that work best for you in different situations.

Managing your time and workload

As Amos (1998) emphasised, everyone has the same amount of time. However, experience of time is relative, as described in this popular story of unknown provenance.

> Imagine a teacher standing in front of her class. She picks up a large empty glass jar and fills it with rocks roughly 5 cm cubed. She asks the pupils if the jar is full. The pupils say yes.
>
> Then the teacher pours small pebbles into the jar, shaking the jar slightly so the pebbles roll into the spaces between the rocks. Again, she asks the pupils if the jar is full. The pupils say yes, more cautiously this time.
>
> The teacher pours sand into the jar. The sand fills the spaces between the pebbles.

The teacher's message is that the jar represents available time. The rocks represent the most important things, whatever they are to you, perhaps family, health and relationships, or sense of purpose: the things that make your life meaningful or are critical to your wellbeing. The pebbles are other things that matter in your life, such as work. The sand signifies the least important things. If you put sand into the jar first, there is no room for the rocks or the pebbles. Similarly, if you put the small stuff in your life first, you have insufficient time for important things.

Set priorities. It is important to take control of workload where you can. Some people seem to work long hours but achieve little, whereas others achieve a great deal but still appear to have plenty of time for things other than work. One of the key strategies for reducing stress and increasing productivity is to develop good time management; we next discuss strategies for managing time in the classroom, across a half-term, and then across a whole year. You may also want to refer to DfE (2018) for some tips on reducing workload.

Managing your time in the classroom

Sir Mo Farah (former British Olympic Gold medallist): 'As a distance runner, you learn that it is important to rest more and do less. That may sound counterproductive, but it's all about quality over quantity' (Farah with Andrews, 2013, p. 173). By cutting out unnecessary tasks and relinquishing

some responsibility to others, including pupils, you can improve the quality of what you do whilst contributing to the holistic development of the pupils. Good time management should also enable pupils to:

- spend a high proportion of time engaged on learning tasks;
- experience being part of a challenging, supportive learning community;
- develop self-efficacy and resilience through becoming more autonomous;
- experience a high degree of success during engaged time;
- demonstrate less adverse behaviour, which is a major stressor for teachers.

You can think of strategies to use classroom time effectively and economically under five headings: pre-emptive, culture-setting strategies; supervisory strategies that increase pupil autonomy and responsibility; in-the-moment strategies; balance of time strategies; and reflective strategies. Table 8.4.2 explores some of these approaches; you will develop your own approaches with experience.

Table 8.4.2 Examples of ways of using classroom time effectively and economically

Pre-emptive, culture-setting strategies ■ Spend time (especially in the first lesson) collecting pupils' ideas and establishing shared ground rules, routines and a safe, inclusive learning environment. This saves time on organisation and management, and on addressing affective barriers as you proceed. ■ Create an expectation that pupils seek answers themselves and amongst peers rather than putting their hand up as soon as they get stuck. ■ Keep a section of your mark book for coded comments about progress. Make brief notes as you see pupils' work in class or when you are marking. ■ Create an expectation and an environment where pupils settle down and pack up promptly and manage their behaviour and attention in the lesson. ■ Offer transition tasks and productive tasks for pupils to work on outside lessons. ■ Devise simple, fast procedures for routine events and dealing with recurring problems.
Supervisory strategies: delegation and increasing pupil autonomy and responsibility ■ Use TAs or pupils to do routine tasks, such as distribute and collect books or equipment, mark straightforward homework tests in class and tidy the classroom. Involving pupils develops autonomy and ownership of their learning environment. ■ Get pupils to help you; they can stick their own worksheets into their books, make wall-posters and videos, make helpful look-up tables or revision guides. ■ Use peer-, self- or group-marking for class tests or homework. Pupils can be perceptive when marking work; once they learn how, they will learn from the process and save you time. ■ Ask pupils to ensure that work is dated, each lesson's work is ruled off, and homework is clearly identified; this makes it easy to check what has been done and what is missing.
In-the-moment strategies ■ Carry a marking pen as you move around the class checking work. After negotiating with the pupil, make brief notes on each pupil's work as you go. This provides formative feedback, and saves you going back to the work at a later stage. ■ Collect in books that are open at the page where you should start marking or where the pupil would particularly like comments.

(Continued)

Table 8.4.2 (Continued)

Balance of time strategies
■ Maintain a good balance of time on teaching, supervisory, organisational and management activities, allocating a high proportion of available time for academic work. ■ Spend a high proportion of time in 'substantive interaction' with pupils (listening, observing, providing feedback, explaining, questioning, describing, illustrating).
Reflective strategies
■ Regularly review the effective use of your own and pupils' time in lessons. ■ Work to eliminate unnecessary routines and activities.

Now complete Task 8.4.4 which encourages you to look at how you use your time in the classroom.



Task 8.4.4 How you spend your time in lessons

Observe how experienced, effective teachers use their time in lessons. How do they divide their time between teaching, supervisory, organisational and management activities? How much time is spent listening, observing, providing feedback, explaining and questioning, managing pupils' behaviour, collecting homework or giving back books? What procedures are evident? What is delegated?

Ask another beginning teacher or your mentor to observe how you use your time in class. Discuss with the observer the findings and possible ways of using lesson time more effectively and economically.

Invite feedback from pupils, such as undertaking a brief survey.

Store your findings; try these ideas out systematically in your teaching.

Managing time outside the classroom effectively across a term

Teacher workload is high in term-time. Indeed, 94% of teachers cite workload as a factor in considering leaving their role (IFF Research and IOE, UCL's Faculty of Education and Society, 2024). A high workload can result in poorer work, working very long hours to get the task done and not having enough time to relax mentally or physically for work the following day. All elite athletes recognise the importance of self-care and rest time to recuperate; so should you. Thus, it is important to focus on doing 'a good enough' job with excellent elements: not heroic, but providing the necessary conditions for learning (Britzman, 1998).

To use your time effectively outside the classroom, you need to plan and prioritise your work. Preparing as far in advance as possible and keeping everything up-to-date means that you do not have to chase around at the last minute, before a mentor or tutor visit. Keeping records of activities can help, for example, keeping a file of activities for the week: lessons to plan, marking to do, assignments for your ITE programme, and specific records of your work, including how you have met certain standards. You should also plan to reflect on your teaching and your development as a teacher.

Time might be lost through being unsystematic in planning, handling paperwork or responding to emails: try not to put off work for later; learn to delegate and to say no. Task 8.4.5 is designed to help you review your use of time.



Task 8.4.5 Reviewing your use of time in term time

Record the time you spend in a week on schoolwork outside the classroom: planning, preparation, marking, record-keeping, extracurricular activities, meetings. You might use a grid such as the one below (a downloadable copy can be found on the companion website (insert link here).

- Did you take one day off?
- Did you take one evening off?
- Is the balance of time sustainable for you?
- Are you using time effectively?
- Do you need to spend more time on some activities?
- Could you reduce time on some activities?
- What can you delegate?

Compare your record with other beginning teachers. If time spent is unsustainable, plan what action you will take. Store this and recheck your use of time outside the classroom by repeating the log for a different week. What further action will you take? Might you use an automatic response on your email outlining when you will respond? If all else fails, you might request fewer classes, possibly with the support of your mentor/tutor. This would be better than becoming unwell with stress.

Day	*Work undertaken (and time for each activity)*	*Total time*
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		
Total time for one week		

Using time effectively helps you to be more efficient and productive, better able to plan long term, more satisfied with your work and less stressed. In addition, you will have more time for yourself and more opportunity to switch off from work.

Some specific examples of ways of using your time effectively include:

- Make a daily list of activities to complete. If there are activities left on the list at the end of the day, ask yourself why this is. Are you spending too much time on each activity? Are you unrealistic about how much you can achieve in a day? Spend five minutes at the end of a day identifying how you will save time the next day.
- Utilise your non-contact periods effectively (plan in advance what you are going to do).
- Plan to complete your work at time(s) best for you. Some people stay late at school, then do not work during the weekend; others set aside one day of the weekend and do all their work on that day. Whichever best suits you, remember the glass jar of time, and be strict with yourself; otherwise, you will be working all the time. Set yourself things to look forward to, such as attending a sports event during the weekend. This helps with time management as it prevents you from working through the weekend.
- Get to know your strengths and your weaknesses. Seek advice and suggestions from different people and adopt what works for you.
- Organise files and other work so that you can easily locate them. It is as important to organise electronic files and delete those you do not need again as it is to organise paper files and throw away paper you do not need. This may need dedicated days when you are not in school.
- You may find that doing more than one job at a time or moving from one thing to another can be disruptive, partly because you may not be concentrating fully on one task, which may be inefficient. Try switching off your phone so that you can focus on the task at hand; set aside time each day to respond to emails, rather than trying to respond immediately.
- Consider carefully how you keep up with colleagues, share resources and save time.

Managing time effectively across a whole year

Teaching is a profession with an uneven profile across the year. This allows for lifestyle choices; you can choose to work longer hours in term-time, or you can set aside 'holiday' time for some tasks, such as updating subject knowledge and long-term planning. The UK Government (Gov.uk, n.d.) advises that you should not work more than 48 hours a week, averaged over any period of 17 weeks including term-time and school vacation. Vacations can be used for an effective balance of recuperation, reflection and preparation, as well as for some larger tasks that may get put on one side during term time, such as home decorating or learning a new skill. You might notice a task that could wait and would be helpful to complete longer term; you might diarise that task to do in a 'holiday'. Task 8.4.6 asks you to look at your work-life balance.

It may be that, when you have done all you can, with advice and support, your workload or stress level is still unsustainable. Your wellbeing is important, not just to you but to your colleagues, your pupils and the profession you seek to join; it is a shared responsibility between you and your employer. If you do not do your part in keeping yourself out of harm's way, you cannot be the asset you wish to be to the teaching profession. 'Do not put yourself at risk' (Barraclough, 2015, p. 20) is the motto of the first-aider; it should apply equally to anyone whose work consists of helping

Task 8.4.6 Coping with your stress through work-life balance

Think about your year as a whole.

What are your own rocks, pebbles and sand? How would you classify:

- personal administration;
- family commitments;
- domestic duties;
- social life and hobbies;
- holiday;
- reviewing lifestyle – moving house, employment, etc.;
- upskilling.

How might you decide to spend your year in order to promote your longer-term wellbeing and achieve an overall balance between work and other time?

and supporting others. The more teachers assert their need for wellbeing, the better it is for all concerned. Your employer is not expected to notice if you develop stress symptoms such as not sleeping. Talk to your mentor, tutor, union representative or a more experienced colleague, read union or HSE guidance on stress management, go and see your GP, and get signed off before you become seriously unwell.

SUMMARY AND KEY POINTS

- As a beginning teacher on school placement, you will sometimes be tired, even exhausted. You may feel as though you do not have enough time to do everything. You might become anxious when someone comes in to watch your lessons, particularly if that person has a say in whether you become a qualified teacher, or worried about other aspects of your teaching or school placement.
- You are not alone in this; many conditions are the same for other beginning teachers.
- You can develop effective techniques to manage your time and cope with stress. Other people can help and support you with this, but nobody else can do it for you: what works for someone else may not work for you. Work at managing your time and stress.
- Build your resilience to enable you to thrive as a teacher. Support from colleagues at work, a mutually respectful and supportive relationship with your mentor (with open, honest and sensitive communication) and a mutually supportive network of trusted peers are all important.
- Focus on the positive aspects of teaching, your motivation to become a teacher, developing your self-efficacy, confidence and competence.
- Stress is contagious; your pupils and colleagues are vulnerable to stress. Consider how a 'stressed you' responds in everyday classroom occurrences and the impact such responses are likely to have upon those around you.

Teaching is challenging, rewarding and exhausting. It is a long game; the skills you gain along the way will help you achieve fulfilment. If insufficient support is in place, aspects of the role may become harmful, meaning that experienced teachers are lost, or standards suffer. Fortunately, it is possible to develop resilience and safeguard your physical and mental wellbeing. This will enable you to perform at your best, respond to crises and change and inspire generations of young people.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Head to routledgelearning.com/learningtoteach for free digital resources to take learning further.

Further resources

Bethune, A. and Kell, E. (2020) *A Little Guide for Teachers: Teacher Wellbeing and Self-care*, London: SAGE.

Thom, J. (2020) *Teacher Resilience: Managing Stress and Anxiety to Thrive in the Classroom*, Woodbridge: John Catt Educational, Limited.

Education Support (https://www.educationsupport.org.uk/)

Provides information, research evidence and a range of services, including counselling, to help all teachers.

My College. Early Career Hub (https://earlycareer.chartered.college/)

This resource from the Chartered College of Teaching is for beginning teachers (including early career teachers), and those supporting them. There are a range of articles on the website that will help you to develop your resilience, ranging from behaviour management to managing your workload effectively.

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8.5 Accountability

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Introduction

When you are learning to teach, the term 'accountability' may appear to be distant, relevant to structures and people further up the hierarchy, for example, senior leadership teams and head teachers. However, accountability is the professional obligation of an individual as well as an organisation to account for their actions and activities, accept responsibility for them and disclose the results in a transparent manner. Teaching is a profession. As such, the highest form of accountability is your own personal and professional accountability for the quality of your own work and to the children, young people, parents and carers whom the profession serves.

Governments set national accountability frameworks for schools in the form of legislation and statutory guidance which reflect the interests of the pupils, parents and the wider community, incentivising policies and behaviours that contribute to a high-quality education for all pupils. Governments then tasks specific bodies to oversee certain aspects of accountability. You should be aware of what these are in relation to where you are learning to teach.

You are also accountable to your employer. Your contract of employment sets out your contractual responsibilities, but employers may have their own accountability frameworks in addition to the government's accountability framework.

It is important that you understand the ways in which your placement school will be held to account, as well as your own professional accountabilities. This is because school or organisational accountability can have implications for an individual teacher. In addition to legal, regulatory and contractual accountabilities, as a teacher you are ethically and morally accountable.

This unit focuses on general aspects of accountability: moral and ethical; legal and regulatory; and employer and contractual. However, you need to understand the specific framework relevant to the system in the country in which you are learning to teach. Specific details in relation to accountability across the four nations of the UK; England, Northern Ireland, Scotland and Wales can be found on the companion website (insert link here).

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OBJECTIVES

At the end of this unit, you should be able to:

- be aware of the full range of accountabilities governing schools and teachers and how they relate to each other;
- know and understand moral and ethical, legal and regulatory and contractual accountabilities governing teachers and their practice.

Check the requirements of your initial teacher education (ITE) programme to see which relate to this unit.

Before reading further, in Task 8.5.1 you are asked to look at the governance of your school.

Task 8.5.1 The governance of your school

Focusing on your placement school, answer the following questions:

- What type of school is it? (see types of schools on the companion website for this book: https://www.gov.uk/types-of-school);
- Who is on the governing board?
- If you are part of a school trust (previously known as a Multi-Academy Trust MAT) who are the trustees/directors on your Trust board?
- Does your school have a local governing body?
- To whom is your school/governing board accountable?
- Who is your employer?

Create a diagram showing these relationships. Check that you are aware of any key documents that are relevant to your work by asking your mentor and checking key national websites.

Accountabilities governing schools and teachers

At its simplest, the term 'accountability' describes a relationship whereby one party – sometimes interpreted as an individual, sometimes an institution – has an obligation to account for their actions or performance to another (Brundrett and Rhodes, 2011). Accountability is 'answerability'.

The key purpose of school and teacher accountability is to improve the education system. According to Earley and Weindling (2004), schools have responsibilities for and have to account to:

- pupils (**moral accountability**);
- colleagues (**professional accountability**);
- employers or government (**contractual accountability**);
- the market, where parents and pupils have a choice of school (**market accountability**).

This assumes that accountability to the government is contractual, and to a degree, it is. Arguably, however, contractual accountability is primarily through the employment relationship. One might argue that market forces increasingly have a *determining effect* on schools, but this is not the same as arguing that school and teachers are accountable to the market. It is by no means clear that either schools or teachers are accountable to the market.

Collins and Leslie (2016) proposed a different taxonomy for accountability:

- **Organisational:** your responsibility to work collaboratively in ensuring compliance with relevant government targets, policies and standards.
- **Legal:** your responsibility to do all that is reasonable to protect the health, safety and welfare of pupils.
- **Professional:** standards defining the minimum level of practice expected of teachers and beginning teachers, from the point of being awarded qualified teacher status.
- **Moral:** your own values and beliefs and the expectation that you will demonstrate consistently high standards of personal and professional conduct.

This is a more helpful articulation of the different frameworks governing teachers' professional practice.

For this reason, in this unit, accountability is explored under the following headings:

- **Moral and ethical accountability:** your accountability as a professional to the people you serve and to your colleagues.
- **Legal and regulatory accountability:** the state-determined infrastructure of accountability.
- **Employer and contractual accountability:** your accountability to your employer.

We next explore each of these in turn.

Moral and ethical accountability

Teachers enter the profession for a variety of reasons, but many become a teacher in order to make a difference in the lives of children and young people. At the beginning of this unit, we said: the highest form of accountability is the teacher's professional accountability for the quality of their own work and to the children, young people, parents and carers whom the profession serves. This is first and foremost, an *ethical* position.

Teachers are trusted by parents and carers (and, indeed, the wider public) to educate children and to form them into the best they can be. It is therefore reasonable that the public expect teachers to uphold moral and ethical principles. These must be evident in everything you do as a teacher: how you behave in the classroom and with your peers, how you conduct yourselves in the teacher-pupil relationship, the standards of learning and behaviour expected and what you do when difficult decisions need to be made.

Teachers are *in loco parentis* during the time they are responsible for children and young people. This is the Latin for 'instead of a parent'. There is a statutory basis to this in the UK. The Children Act 1989 (UK Parliament, 1989) says that teachers have a duty of care towards the children and young people under their supervision, as well as promoting the safety and welfare of the children and young people in their care. The way this is measured is by what a 'reasonable parent' might do.

Personal and professional conduct; The teachers' standards

All teacher standards have statements on personal and professional conduct. You should check what the standards towards which you are working say in relation to personal and professional conduct or accountability.

For example, according to the Teachers' Standards in England (Department for Education (DfE), 2011) as a teacher you are 'expected to demonstrate consistently high standards of personal and professional conduct' (p. 14) (further details regarding the minimum expectations for teachers can be found on the companion website insert link).

The Nolan Principles of Public Life

In 1995, Lord Nolan chaired the first Committee on Standards in Public Life and set out a ministerial code (Nolan, 1995). The code, '7 principles of public life', now apply to anyone who works as a public office-holder in the UK. This includes all people appointed to work in education. These principles are summarised in Table 8.5.1 (see also, The Seven Principles of Public Life – GOV.UK, www.gov.uk).

The Nolan Principles offer a very clear statement of ethical behaviour and are arguably a better guide for teachers when having to make difficult ethical decisions than standards for teaching towards which you are working. Standards normally outline minimum expectations of conduct. Thus, teachers are accountable not just for your good conduct, but also for the ethics of your decision-making.

Task 8.5.2 asks you to consider moral and ethical accountability and Task 8.5.3 asks you to consider an ethical dilemma.

Table 8.5.1 Summary of the seven principles of public life

Principle	*Explanation*
Selflessness	Holders of public office should act solely in terms of the public interest.
Integrity	Holders of public office must avoid placing themselves under any obligation to people or organisations that might try inappropriately to influence them in their work. They should not act or take decisions in order to gain financial or other material benefits for themselves, their family or their friends. They must declare and resolve any interests and relationships.
Objectivity	Holders of public office must act and take decisions impartially, fairly and on merit, using the best evidence and without discrimination or bias.
Accountability	Holders of public office are accountable to the public for their decisions and actions and must submit themselves to the scrutiny necessary to ensure this.
Openness	Holders of public office should act and take decisions in an open and transparent manner. Information should not be withheld from the public unless there are clear and lawful reasons for so doing.
Honesty	Holders of public office should be truthful.
Leadership	Holders of public office should exhibit these principles in their own behaviour. They should actively promote and robustly support the principles and be willing to challenge poor behaviour wherever it occurs.

Task 8.5.2 Moral and ethical accountability

Consider what being morally and ethically accountable means to you and for the way you approach your work as a teacher. Discuss this with other beginning teachers or your mentor.

Task 8.5.3 An ethical dilemma

A pupil in your classroom suddenly hits out at another pupil. The second pupil is hurt. The first pupil runs out of the classroom, throwing a chair on his way out, which narrowly misses a third pupil. On his way out, he damages school property. You know this pupil has been in and out of the care system. He is currently accommodated in a children's home. He has been the victim of physical and emotional abuse by his father, who is now in prison.

It is not your decision about whether to exclude this pupil; that is for the head teacher to decide. However, you are asked to write a report on the incident. Further, the pupil has a very good relationship with you, perhaps better than any other teacher in the school, so you have also been asked to advise.

How do you balance the needs of the individual pupils with the rights of the rest of the class to a safe and orderly learning environment? Is it possible to serve the interests of both?
How do you interpret *in loco parentis* in this case?
Consider the Nolan Principles. How do they help?

Ethics is to do with right and wrong, the moral principles that govern our behaviour and decision-making, but there is no right answer. Ethics requires professional judgement, and you are also accountable for the decisions you make. But your ethical decisions are guided by statutory frameworks.

What pieces of legislation and/or statutory guidance do you need to take into account?
Discuss this case with your mentor and check key national websites.

The State and its agencies: legal and regulatory accountability

The State and its agencies

A government determines the national accountability framework for education. In the UK, it does this through:

- **legislation:** the process of making or enacting laws;
- **inspection:** the process of inspecting providers to reach judgements about the quality of provision and check they meet legal requirements;
- **regulation:** the process of taking enforcement action where requirements are not met.

It is important to understand not just to whom you and your school are accountable, but ultimately who can take enforcement action against you or your school when certain standards are not met.

The government sets the overall framework for accountability, usually through the prescription of **performance measures** for schools. In the UK, each government publishes tables of school performance against the measures it sets out.

Task 8.5.4 asks you to research the performance measures in the current accountability framework relevant to where you are learning to teach.



Task 8.5.4 Performance measures

Use government websites to find out the headline measures in the accountability framework for your placement school. Using the most recent data (they will have this if you cannot find it), discuss this information with your mentor. Talk to colleagues who lead on data to help you identify websites and performance measures and assess school performance.

Consider Goodhart's law (McIntyre, 2001): 'When a measure becomes a target, it ceases to be a good measure'. Discuss with your mentor: What are the unintended consequences of the current accountability framework?

Governments task certain agencies or individuals to oversee specific aspects of accountability. One aspect relates to inspection.

In Task 8.5.5, you are asked to consider the current inspection for your current placement school.



Task 8.5.5 Inspection

Download the current inspection framework relevant to your education system and your placement school's most recent inspection report.

Talk to the colleague responsible for data on how the school is performing against the inspection framework and what changes have been made in light of the report.

We now consider other aspects of legal and regulatory accountability for teachers.

Safeguarding and child protection

One of the most important aspects of legal and professional accountability for schools and teachers is safeguarding and child protection. You should be aware of the legislation relevant in your placement school. Child protection training is a mandatory requirement for all teachers in the UK.

As a teacher, part of your professional responsibility is to help your school create and maintain a safe learning environment for every pupil. You are expected to be able to identify pupils with special educational needs and demonstrate a sound understanding of the procedures for seeking advice and support from appropriate agencies, such as social workers and healthcare professionals. You must be fully conversant with the policies and procedures in your school that address issues of children protection, health and safety and bullying (including cyberbullying). As a beginning teacher, if you have any concerns about any safeguarding issues at any time, talk to your mentor. Task 8.5.6 focuses on the relevant document for safeguarding and child protection.



Task 8.5.6 Keeping children safe in education

During induction at your placement school, read the relevant documents for safeguarding and the child protection policy, staff behaviour policy (or code of conduct) and information about the role of the designated safeguarding lead in your school.

Ask your school's designated safeguarding lead if you have any questions. Also, talk to them about their role.

Prevent Duty

In the UK, the 'Prevent Duty' requires that teachers identify children and young people who may be vulnerable to radicalisation and know what to do when they are identified. Protecting pupils from the risk of radicalisation is part of a school's wider safeguarding duties and is similar in nature to protecting pupils from other harms (e.g. drugs, gangs, neglect, sexual exploitation). In Great Britain, the non-statutory guidance to schools on the Prevent Duty advises that schools can build pupils' resilience to radicalisation by providing a safe environment for debating controversial issues. Make sure you are aware of the Prevent Duty (for England and Wales, see HM Government, 2023 and for Scotland, see HM Government, 2024).

Health and safety

As a member of the school community, you have a responsibility for the arrangements made to meet the health and safety needs of pupils. The main legislation is the Health and Safety at Work Act 1974 (UK Parliament, 1974) and regulations made under that Act. Your employer is primarily responsible for health and safety; however, teachers have a general duty to look after their own and others' health and safety. Teachers also have a duty under common law to take care of pupils in the same way that a prudent parent would do so (*in loco parentis*) (see above). It is very rare for school staff to be prosecuted under criminal law with regard to accidents involving pupils, as long as they are not negligent.

Task 8.5.7 is designed to support you in understanding health and safety and risk assessments in your school and subject.



Task 8.5.7 Health and safety

Read the relevant health and safety and risk assessment documents in your school and subject. Make sure you are familiar with these and discuss any points you are not clear about with your mentor.

Ask for a copy of a recent risk assessment undertaken either for a particular activity in a lesson or prior to a school visit or trip. Read this carefully. Talk to the teacher who undertook the risk assessment. How did they go about identifying risk? What factors did they consider in how to support pupils to take risks safely?

Equalities

Schools and teachers are accountable for their actions in relation to equality and discrimination. In Great Britain, the Equality Act 2010 (UK Parliament, 2010) replaced all equality legislation such as the Race Relations Act, Disability Discrimination Act and Sex Discrimination Act. A separate shorter version of this Act applies to Northern Ireland. This and other Acts deal with the way that teachers and schools treat their pupils and prospective pupils. It is important for you to be familiar with, know and understand responsibilities and accountabilities under this legislation or legislation which applies where you are learning to teach, because, in practice, as a teacher you can be held liable for your ***own*** discriminatory actions.

The relationship between one pupil and another is not within the scope of the Act. It does not therefore bear directly on such issues as racist or homophobic bullying by pupils. However, if a school treats bullying which relates to a protected ground less seriously than other forms of bullying – for example, dismissing complaints of homophobic bullying or failing to protect a transgender pupil against bullying by classmates – then it may be guilty of unlawful discrimination.

In Task 8.5.8, you are asked to focus on the Equalities Act (or equivalent) and discrimination.



Task 8.5.8 The Equality Act and discrimination

In England, the departmental advice on equalities (DfE, 2014) gives the following examples of valid complaints that the curriculum is being delivered in a discriminatory way:

- a teacher uses the fact that *The Taming of the Shrew* is a set book to make derogatory generalisations about the inferiority of women, in a way which makes the girls in the class feel belittled. Or, in teaching *The Merchant of Venice*, he encourages the class to laugh at a Jewish pupil;
- in class discussions, black pupils are never called on and the teacher makes it clear that she is not interested in their views;
- girls are not allowed to do design technology or boys are discouraged from doing food technology. This is not intrinsic to the curriculum itself but to the way in which education is made available to pupils;
- the girls' cricket team are not allowed equal access to the cricket nets, or the boys' hockey team is given far better resources than the girls' team. This would be less favourable delivery of education rather than to do with the specific sports per se.

Study the definitions of direct discrimination; indirect discrimination; harassment and victimisation on page 9 of the guidance. Consider which definitions fit the vignettes above.

Employer and contractual accountability

Safer recruitment

As discussed above, all schools have responsibility for safeguarding children and young people. These inform the procedures for selection and pre-employment vetting for schools that provide the framework for all schools' recruitment policies. Schools have explicit employment policies that

take account of national and local safeguarding requirements. Make sure you familiarise yourself with these before writing and submitting an application for a post, to ensure that you comply with all requirements. The school or organisation to which you are applying will require you to apply for a disclosure check at the level appropriate to the post (in England, Northern Ireland and Wales through the Disclosure and Barring Service (DBS) and in Scotland through Disclosure Scotland) (see Unit 8.1 on applying for your first post).

Your contract of employment

Your contract of employment will set out specific responsibilities and accountabilities you have to your employer. Your contract of employment is legally binding. You should review it carefully, and ask any questions before signing it.

Progression/promotion, performance management and performance-related pay

In assessing teachers' performance for any progression/promotion of teachers, including any performance-related pay, schools take into account evidence from a range of sources in relation to:

- impact on pupil progress;
- impact on wider outcomes for pupils;
- contribution to improvements in other areas;
- professional and career development;
- wider contribution to the work of the school.

Progression/promotion, performance-related pay and pay policies are the focus of Task 8.5.9.



Task 8.5.9 Progression/promotion, performance-related pay and pay policies

Ask your placement school for a copy of the appraisal, progression/promotion and pay policies. Talk to your head of department or other school leader to find out more about factors and evidence that the school uses as part of teachers' appraisal. Consider the evidence you need to start collecting to help you demonstrate your performance (and your development) as a teacher (please note, you will not be subject to appraisal until after you have completed your induction period).

Accountability to your employer

You have a general responsibility to your employer to perform your duties as a teacher in the context of the wider accountability frameworks outlined above and detailed in documents relevant to

where you are learning to teach. You must have regard for the ethos, policies and practices of the school or organisation in which you teach and maintain high standards in your own attendance and punctuality.

As a beginning teacher, and, later, your contract of employment, will require you to work within the school or organisation's policies and procedures. If you do not follow policies or procedures or you fail to meet the standards or responsibilities in the accountability frameworks, it is your employer who will normally take first action.

SUMMARY AND KEY POINTS

Because teaching is a profession, like other professions, the job is rightly highly accountable. The accountability frameworks guiding a teacher's practice are complex. As the Teachers' Standards in England (DfE, 2011) indicate, a teacher is expected to demonstrate consistently high standards of personal and professional conduct, uphold public trust in the profession, have proper and professional regard for the ethos, policies and practices of the school or organisation and act within the statutory frameworks that set out professional duties and responsibilities. All teachers are responsible for the education, health, welfare and safety of the pupils in their care. It is right that teachers are both responsible and accountable.

This unit has considered:

- your moral and ethical responsibilities and accountabilities; the role of the State and its agencies;
- legal and regulatory accountability frameworks governing your practice;
- your accountability to your employer and your contractual responsibilities.

Please see the companion website for specific details in relation to accountability across the four nations of the UK; England, Northern Ireland, Scotland and Wales.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

Engage with the materials covered in this chapter with further resources at routledgelearning.com/learningtoteach.

Further resources

Berry, J. (2013) *Teachers' Legal Rights and Responsibilities: A Guide for Trainee Teachers and Those New to the Profession* (3rd edn), Hatfield: University of Hertfordshire Press.

This guide provides an overview of the legal issues relevant to teachers to help them understand their rights, responsibilities and professional duties.

Read and be familiar with the Acts, legislation and statutory guidance on accountability in England, or in the country in which you are learning to teach, as well as how these are translated into policy and practice in your placement school.

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Nolan, M. (1995) *Standards in Public life: First Report of the Committee on Standards in Public Life*, London: HMSO, viewed 23 July 2025, https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/336919/1stInquiryReport.pdf

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UK Parliament (1989) *The Children Act*, London: HMSO.

UK Parliament (2010) *The Equality Act*, London: HMSO.

9 And finally

Marilyn Leask and Sarah Younie

Introduction

The introduction to this text sets you the challenge of considering what you want to achieve by becoming a teacher and what it means to be a teacher. No one can predict what any learner will achieve through life and there are many instances of people who have had positive national and international impact not complying to 'the norm' in school. We ask you to stay alert to your professional and moral responsibility to build the confidence of each learner – in themselves and their ability to learn which will be different to others. Your interactions will show them they have value and can make a positive contribution to their community and society as a whole. Your learners may find it helpful to know that neuroscientists have demonstrated that ability is not fixed and that the human brain has capacity to learn new things throughout life. They may also find it helpful to know public examinations which they have to take are not indicators of what they can achieve in the wider world.

OBJECTIVES

At the end of this unit you should be able to:

- Reflect on the issue of how your values and beliefs shape your work as a teacher;
- Consider how your values affect the pupils you teach;
- Consider what you think the aims of education are, beyond teaching your subject, and how this can inform your practice going forward.

Check the requirements for your initial teacher education (ITE) programme to see which relate to this unit.

Developing the values which underpin your teaching

Here, in the last unit, we return to the issue of how your values and beliefs shape your work as a teacher. Unit 7.1 challenged you to think about the aims of education. Units 1.1 and 4.5 asked you to consider your values to adopt a code of conduct to guide your work as a professional. Over your

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teaching career, you can expect demands on teachers to be able to demonstrate evidence-informed practice to become more the norm. Would you expect your doctor to practice without reference to the latest evidence? You, as a teacher, work with the mind – the development of an individual's self-belief and sense of worth may be shaped by you. So, do you not owe it to your learners, their parents and carers, to provide research-informed up-to-date practice? In Unit 5.4, we made the point that research-informed practice = professional judgement + evidence: of course, research is never adopted without reference to the learner and the environment and subject context. Joining your subject association provides information to enable you to keep up to date in your curriculum discipline (see Appendix 2).

So, what are your core values? Schools are the one place in modern diverse societies where young people may come across others with entirely different backgrounds and expectations to their own. Will your learners learn from you to accept and value differences and to combat prejudice which sows division in societies? The effective functioning of our societies can never be taken for granted. History shows that the apparently most stable of countries can descend into chaos through internal division. Teachers have a role to play in shaping learners' values as do schools: the values underpinning school structures and procedures including rewards and punishments demonstrate to learners how they are valued.

Learning how to teach

Instructors deliver sets of facts and develop learners' skills. Teachers have the more complex task of working with individuals to develop their mental abilities, their skills of critical analysis and independent thinking as well as their knowledge of facts. This text has drawn on the evidence base for educational practice in providing information and background, then exploring these through linked tasks. Some tasks provide opportunities to reflect on and examine the practice of other teachers, of yourself and the organisation of schools; others are enquiry-based and generate data or ideas for reflective thinking upon which an understanding of, and an explanation for, the complex world of teaching and learning in schools are built. Each unit has been designed to help you understand the different elements of teaching. This 'whole-part-whole approach' to introducing you to teaching through this text mirrors effective practice in teaching – introducing the whole, undertaking work on the parts, then returning to review the learning in the context of the whole.

The relationship between explanation and practice is a dynamic one; explanations are needed to make sense of experience and inform developing practice. Some explanations will be your own, to be tried and tested against the theories of others, often more experienced teachers. At other times, you may use others' explanations directly. Explanations, in turn, generate working theories, responsive to practice and experience. Theory is important, as it provides a variety of ways of thinking about the complex world of the classroom and directs further personal research into improving the quality of teaching and learning. It also provides a basis for judging both personal and institutional change. The interplay of theory and practice develops your professional judgement and underpins the notion of the reflective and evidence-informed practitioner.

What values will you pass on?

To teach young people is to contribute directly to shaping the society of tomorrow. What values will you pass on to the young people you teach?

Table 9.1.1 'Dear teacher': A letter to new teachers

Dear teacher,

I am the survivor of a concentration camp. My eyes saw what no man should witness.
Gas chambers built by learned engineers,
Children poisoned by educated physicians,
Infants killed by trained nurses,
Women and babies shot and burned by high school and college graduates.
So I am suspicious of education.
My request is: Help your students become human.
Your efforts must never produce learned monsters, skilled psychopaths, educated Eichmanns.
Reading, writing and arithmetic are important only if they serve to make our children more human.

Table 9.1.2 'Children learn what they live'

If a child lives with criticism,
he learns to condemn,
If a child lives with hostility,
he learns to fight,
If a child lives with ridicule,
he learns to be shy,
If a child lives with shame,
he learns to feel guilty,
If a child lives with tolerance,
he learns to be patient,
If a child lives with encouragement,
he learns confidence,
If a child lives with praise,
he learns to appreciate,
If a child lives with fairness,
he learns justice,
If a child lives with security,
he learns to have faith,
If a child lives with approval,
he learns to like himself,
If a child lives with acceptance and friendship,
he learns to find love in the world.

Source: Dorothy Law Nolte

At a time when young people who are using social media are being targeted by those who may wish to manipulate their thinking, your role in helping young people to understand how to protect themselves and to adopt values that support stability in society through respect for others is crucial. As a last task, we ask you to consider the messages in the letter and poem in Tables 9.1.1 and 9.1.2.

The letter, 'Dear Teacher' (Table 9.1.1), was written by a US high school principal, a World War II concentration camp survivor who was separated from her sister and mother whom she never saw again. She gives the letter to all new staff at her school (Pring, 2004). The poem, 'Children learn what they live' (Table 9.1.2, Dorothy Law Nolte), is often displayed on staffroom walls. Read the letter and the poem and then complete Task 9.1.1.

Task 9.1.1 What will your pupils learn from you?

Read the poems in Tables 9.1.1 and 9.1.2. Consider the values that you as a teacher will pass on through your teaching and through the relationships you develop with your pupils. Write a note to yourself about what you hope to achieve, and revisit and review your aspirations for yourself as a teacher at the beginning of each term of your teaching. Store this in your professional development record, portfolio or equivalent.

Also, consider whether a contemporary analysis of the gendered language in the poem in Table 9.1.2 is now requiring a more pressing critique, as reference to only 'he' appears indicative of a previous era when 'he' was taken to be generic, but as women continue to fight for equal rights and recognition, we note that as female authors this includes our (women's) representation in language; not to be subsumed within the word 'he'.

SUMMARY AND KEY POINTS

At the end of this unit we hope you now have a greater understanding of the key points related to your values as a beginning teacher that will guide you across your career.

- As a teacher, you will have more impact than you will ever know – on the lives of the pupils you teach, their parents and carers, and the communities and societies within which they live.
- We hope that you will help pupils to build the personal self-confidence and skills to cope with adult life and to become autonomous learners and caring members of society.
- We hope, too, that what your pupils learn from you will help them make positive contributions to their world.

To achieve these goals, you should expect to carry on learning throughout your professional life.

- As a first step, joining your subject association will ensure you receive publications outlining pedagogic practice in your discipline, and attending your annual subject conference will introduce you to the network of educators taking thinking in your subject forward. You can find your subject association by asking colleagues and tutors or through the Council for Subject Associations (www.subjectassociation.org.uk). See also Appendix 2.
- Teachers are expected to continue to undertake reflective practice and learning beyond their ITE. Teachers in some countries are required to have master's level qualifications; in those countries where this is not required, many teachers study part-time to achieve this level of qualification to aid their professional development.

The issues raised in this unit about the purpose of education and how children learn are areas that you may wish to follow up through engaging with networks of other educators to develop and share the evidence base for professional practice in education.

Check which requirements for your ITE programme you have addressed through this unit.

FREE INSTRUCTOR & STUDENT RESOURCES

ONLINE RESOURCES

For bonus digital resources designed to enhance the book materials, head to routledgelearning.com/learningtoteach.

Further resources

Council for Subject Associations: www.subjectassociations.org.uk, viewed 31 July 2025.
This site provides contact details for the major professional associations. Associations run conferences, networks and professional development to help you develop as a teacher. Some are also trade unions.

Hawkes, N. (undated) *What is Values based Education*. https://www.valuesbasededucation.com/about/about-vbe, viewed 31 July 2025.
This site introduces Values Based Education.

Lansdown, G., Covell, K., Vaghri, Z. (2022) Article 29: The Aims of Education. In: Vaghri, Z., Zermatten, J., Lansdown, G., Ruggiero, R. (eds.) *Monitoring State Compliance with the UN Convention on the Rights of the Child*. Children's Well-Being: Indicators and Research, vol 25. Springer, Cham. https://doi.org/10.1007/978-3-030-84647-3_27, viewed 31 July 2025.
This chapter is freely available online and will introduce you to the UN Convention on the Rights of the Child.

Ledbury School (undated) The Enhanced Values-based Education Quality Mark. https://www.ledbury.hereford.sch.uk/values-based-education/, viewed 31 July 2025. This school site explains how the school practices values based education.

Packham, R. (2024) The Importance of Values Based Education. https://www.routledge.com/blog/article/the-importance-of-values-education?srsltid=AfmBOoqE2EIsi6n318cG-s64TIZY78NLVOY8eE7DzKMCISPE6eIOx9LN, viewed 31 July 2025. This blog introduces the concepts underpinning Values Based Education. You may wish to incorporate these into your practice.
Teaching Councils: Membership is by subscription and confirmation of qualifications. Normally, Councils have a code of conduct to which members are expected to adhere.

UNICEF (2024) *The Right to Education*. Introducing Articles 28 and 29 https://www.unicef.org.uk/rights-respecting-schools/the-rrsa/the-right-to-education/, viewed 31 July 2025.
You may wish to test your values and beliefs about the aims of education against those espoused by the UN.

Texts of academic and historical interest

The texts below remain relevant as they provide an introduction to key debates and thinking that have shaped educational practice in the UK over more than 100 years.

Aldrich, R. (2006) *Lessons from History of Education: The Selected Works of Richard Aldrich*, London: Routledge.
This text provides a useful introduction to recurrent issues in education.

Bruner, J.S. (2006) *In Search of Pedagogy Volume 1 and 2: The Selected Works of Jerome S. Bruner*, London: Routledge.
Bruner's work on pedagogy is seminal – changing thinking and practice.

Dewey, J. (1933) *How We Think*, Boston, MA: Houghton Mifflin.
Dewey's ideas endure. This text will challenge you to think to what extent these ideas are now spread across the education system.

Marples, R. (2012) *The Aims of Education*, Abingdon: Routledge.
This provides a useful introduction and overview of aims of education.



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Appendix 1

Glossary of terms

A level	See **GCE**
AS level/A2 level	See **GCE**
Academies	A central government political initiative in England to bring sponsors from business, faith or voluntary groups into school management removing local accountability and local authority oversight. See also Multi-Academy Trust, **State-Maintained Schools in England** and **Other State Schools in England**.
ACCAC	(Awdurdod Cymwysterau, Cwricwlwm ac Asesu Cymru) formerly the Qualifications, Curriculum and Assessment Authority for Wales. Merged with **DCELLS** in 2006. Now DfES.
Adaptation or adaptive teaching;	See **differentiation**
AEB	See **AQA**.
Analyse School Performance (ASP)	DFE system to analyse school performance data. Analysing school performance data is a key part of understanding the school's attainment and improving teaching and learning. Formerly known as **RAISEonline**.
Annual Review	The annual review of an Education, Health and Care Plan (EHCP) in England, which an **LA** must make within 12 months from issue or a previous review. See also EHCP.
AQA Assessment and Qualifications Alliance	An Awarding Body for GCSE, GCE A and AS levels and vocational qualifications. Online http://www.aqa.org.uk. Formed in 2000 by a merger of City and Guilds GNVQ, Associated Examining Board (AEB), Southern Examining Group (SEG) and Northern Examination and Assessment Board (NEAB).
Assessment	Assessment covers all those activities that are undertaken by teachers and others to measure the effectiveness of their teaching and of pupils learning. See also **Assessment for learning, Assessment of learning, Criterion-referenced assessment, Formative assessment, Ipsative assessment, Norm-referenced assessment** and **Summative assessment**.
Assessment for learning	Assessment for which the first priority is to promote pupils' learning. It allows teachers and pupils 'to decide where the learners are in their learning and encourages pupils to take ownership of their learning'. See also **Assessment, Assessment of learning, Formative assessment, Ipsative assessment, Norm-referenced assessment** and **Summative assessment**.

Assessment of learning	The periodic, summative assessment of pupils' attainment and progress in a variety of ways and for a variety of purposes. See also **Assessment, Assessment for learning, Criterion-referenced assessment, Formative assessment, Ipsative assessment, Norm-referenced assessment** and **Summative assessment**.
Assessment as learning	Assessment as learning is a subset of Assessment for Learning and sees pupil involvement in assessment, using feedback, participating in **peer assessment** and self-monitoring of progress, as moments of learning themselves.
Assessment criteria	The defined basis on which pupils' work will be judged. They are for use by teachers in assessing pupils' work and are usually provided to pupils to guide their work. May also be known as **success criteria**.
Attainment targets (ATs) and assessment levels of NC for England	The knowledge, skills and understanding that pupils of different abilities and maturities are expected to have by the end of each **Key Stage**. Attainment targets previously consisted of eight **level descriptions** of increasing difficulty, plus a description for exceptional performance above level 8. The latest national curriculum (2014) has removed all level descriptions. See also **Programmes of Study**.
Attainment 8	A government accountability measure of attainment across the same 8 qualifications as used in **Progress 8**.
Awarding Body	The main Awarding Bodies that set public examinations in England: AQA (the Assessment and Qualifications Alliance); CCEA (Council for the Curriculum, Examinations and Assessment); OCR (Oxford, Cambridge and RSA Exams); Pearson Edexcel (formerly EdExcel); WJEC – the Welsh Joint Education Committee). Also encompasses **exam boards** and **awarding organisations**.
BA/BSc (QTS) Bachelor of Arts/ Bachelor of Science with QTS (Qualified Teacher Status)	A teaching qualification awarded in England – a combined course with route to QTS. Note your teaching qualification may not be recognised in other countries including in others of the four countries in the UK. See also iQTS.
Banding	The structuring of a year group into divisions, each usually containing two or three classes, on grounds of general ability. Pupils are taught within the band for virtually all the curriculum. See also **Mixed ability grouping, Setting** and **Streaming**.
Baseline Testing	Any process that sets out to find out what the learner can do now in relation to the next stage of learning. For example, the assessment of practical skills and familiarity of pupils with equipment and tools prior to a **D and T** course, or the assessment of pupils in Year 1 and reception classes for speaking, listening, reading, writing, mathematics, social skills. See also **Benchmarking**.
BEd	Bachelor of Education, a teacher training qualification in England leading to **QTS**.
Benchmarking	A term used to describe a standard against which comparisons can be made. Can be used by schools, e.g. to measure success of the school in public examinations relative to a national norm.
BESD	The revised SEND Code of Practice in 2014 replaced the old category of 'Behavioural, Emotional and Social Difficulties' (BESD) by 'Social, Emotional and Mental Health' (SEMH) needs. A group of pupils with special educational needs. The term is often applied to pupils whose behaviour is consistently poor and not obviously related to the circumstances and environment in which pupils find themselves. See also **SEN, SEND**, **SEMH**.

BTEC	Business and Technician Education Council. Part of Pearson (**EdExcel)**, which offers a suite of vocational courses called BTECs, e.g. BTEC Nationals.
C and G	City and Guilds; see **AQA**
Career Entry and Development Profile (CEDP) or Career Entry Profile	Prior to 2021 changes, providers in England were required to provide newly qualified teachers with a document to help support **induction**. The initial teacher training and early career framework (ITTECF) sets out a package of support. See **ECT** and **ITTECF**.
Careers Education	Designed to help pupils to choose and prepare for opportunities, responsibilities and experiences in education, training and employment. See National Careers Service and statutory guidance for schools online in the list of websites.
Church and Faith Schools	A faith school is a British school teaching a general curriculum but with a particular religious character, or having formal links with a religious organisation. Regulations differ in detail among constituent countries of Britain. In England, the curriculum, admissions criteria and staffing policies may reflect their religious foundation. Similar to other **State-Maintained Schools** but follow a locally agreed religious education curriculum and have religion-centred admissions criteria and staffing policies. May also be referred to as schools with a religious character. See **Other State Schools in England**. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Citizenship	A compulsory subject of the English **NC** at **Key Stages** 3 and 4. See also **Cross-curricular** elements.
Collaborative group	A way of working in which groups of children are assigned to groups or engage spontaneously in working together to solve problems; sometimes called co-operative group work.
Comprehensive school	A type of state-maintained secondary school which admits pupils of age 11 to 16 or 19 from a given catchment area, regardless of their ability. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Community and Foundation special schools	For children with specific special educational needs, such as physical or learning difficulties. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Community of practice	Groups of people who share a concern for something or have knowledge and skills to share. For example, a subject association network or a network of teachers working on solving a particular problem.
Community School	A school run by the **LA**, which employs the staff, owns the land and buildings and decides admission criteria. Develops links with community. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Continuity	A feature of a curriculum and of lesson plans that ensure that learning builds on what has already been taught and experienced and prepares pupils for what is to come. See also **Progression**.
Core subjects	Foundation subjects that are taught at both KS3 and KS4 comprising English, Mathematics and Science in the National Curriculum for England.
Coursework	Work carried out by pupils during a course of study marked by teachers and contributing to the final examination mark. Usually externally moderated. May also be referred to as **non-exam assessment** (NEA)

CPD	Continuing professional development for teachers.
Criterion-referenced assessment	A process in which performance is measured by relating candidates' responses to pre-determined criteria. See also **Assessment**, **Assessment for learning, Assessment of learning, Formative assessment, Ipsative assessment, Norm-referenced assessment** and **Summative assessment**.
Curriculum	The full range of learning experiences in a pupil's time in education. It is planned and sequenced with a view to knowledge, skills and understanding and usually themed into subject groupings.
Curriculum guidelines	Written guidance for organising and teaching a particular subject or area of the curriculum. See also **Programmes of Study**.
DCELLS	Department for Children, Education, Lifelong Learning and Skills of the Welsh Assembly. http://www.accreditedqualifications.org.uk/department-for-children-education-lifelong-learning-and-skills-dcells.html
DES, DfE, DfEE, DfES, DCSF	Various names for the Ministry of Education in England. See **Government education departments and chronology**
Differentiation	The matching of work to the differing capabilities and learning needs of individuals or groups of pupils in order to extend their learning. Also known as **Adaptive teaching.**
Disapplication	Arrangement for lifting part or all of the **NC** in England requirements for individuals or for any other grouping specified by the Secretary of State.
Early Career Framework	(ECF). A statutory 2-year core induction programme (from Sept 2021) for all early career teachers. See also NQT, RQT, ECT/ECTE, ITTECF.
ECT/ECTE	The term early career teacher (ECT) replaced newly qualified teacher (NQT). See the Early Career Framework entitlement (ECTE): https://www.gov.uk/government/publications/early-career-teacher-entitlement.
EAL/ESL	English as an Additional Language. English as a second language.
EBacc	The EBacc, or English Baccalaureate, is a performance measure in England that recognises students who achieve a strong set of GCSE results in specific subjects. It is not a qualification itself. To gain the EBacc, pupils must take up to eight GCSEs across five subject 'pillars'. Numbers of students entering and achieving the EBacc are used as government accountability measures.
EdExcel/Pearson	An awarding body. See https://qualifications.pearson.com/.
Education welfare officer (EWO)	An official of the **LA** concerned with pupils' attendance and with liaison between the school, the parents and the authority.
Education, Health and Care Plan (EHCP)	Document setting out the education, healthcare and social care needs of a young person for whom extra support is needed in school, beyond that which the school can provide. Formerly known as a 'statement of special educational needs' so the legacy terminology of 'statemented' pupils may survive.
Exclusion	Headteachers of **state-maintained schools** and **other state schools** in England are empowered to exclude pupils temporarily or permanently when faced with a serious breach of their disciplinary code. The exclusions are either a fixed term or permanent. Schools may send pupils to a **pupil referral unit (PRU)**.
EYFS	Early Years Foundation Stage (of the **NC** for England).

Feedback	Includes information to pupils on their achievements, the purpose of which is to help to improve their performance. Feedback can be part of a formal assessment system (when it might be called formative assessment) or it can be informal and available from multiple sources.
Floor target	A government-set measure below which schools are considered underperforming. Also known as minimum or floor standards, e.g. floor standards for Progress 8 score.
Formative assessment	Also, assessment for learning, linked to teaching when the evidence from assessment is used to adapt teaching to meet pupil's learning needs. See also **Assessment, Assessment for learning, Assessment of learning, Criterion-referenced assessment, Ipsative assessment, Norm-referenced assessment** and **Summative assessment**.
Forms of entry (FE)	The number of forms (e.g. of 30 pupils) that a school takes into its intake year. From this can be estimated the size of the intake year and the size of the school.
Foundation schools	The governing body of these schools employs the staff and sets the admissions criteria. The school land and buildings are owned by the governing body or a charitable foundation. See **State-Maintained Schools** and **Other State Schools in England**.
Functional Skills	Functional skills are those core elements of English, Mathematics and ICT that provide individuals with the skills and abilities they need to operate confidently, effectively and independently in life, their communities and work. They can be examined individually.
Foundation subjects	Subjects that **state-maintained schools** are required by law to teach. In England, three **foundation subjects** are designated **core subjects**. See https://www.gov.uk/national-curriculum.
Free Schools	In England, a political initiative to set up independent, state-funded schools, non-selective and outside **LA** control, established in 2010 under the **Academies Act**. Free schools are set up by parents, teachers, charities or businesses. Grants are available to support the initial setting-up process. They are subject to the Schools Admissions Code of Practice, but priority is given to founders' children. In time, subject to **Ofsted** inspection. See also **State-Maintained Schools in England** and **Other State Schools in England** and https://www.gov.uk/types-of-school.
GCE	General Certificate of Education – Also called Advanced Level of the GCE. An award after two years of study usually post **GCSE**. May comprise two awards; an AS level which can be taken after one year of study and A2 level after two years of study or a linear 2-year A Level.
GCSE	General Certificate of Secondary Education.
GNVQ	General National Vocational Qualifications, last awarded in 2007.
Government education department titles	**and chronology** For England and Wales to 2006, then England.
DES	(Department of Education and Science) Pre-1992
DfE	(Department for Education) 1992–1995; title reappeared 2010–present.
DfEE	(Department for Education and Employment) 1995–2001.
DfES	(Department for Education and Skills) 2001–2007.

DCSF	(Department for Children, Schools and Families) 2007–2010.
Grade-related criteria	The identification of criteria, the achievement of which are related to different levels of performance by the candidate.
Grammar schools	State-maintained or independent schools that select all or almost all of their pupils based on academic ability, usually through the 11 plus examination. Parents often pay for extra tuition to give their children a higher chance of being admitted. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Group work	A way of organising pupils where the teacher assigns tasks to groups of pupils, to be undertaken collectively; although, the work is completed on an individual basis.
Gypsy, Traveller, Roma, Showmen and Boater (GTRSB)	A term used to cover those communities, some of which have minority ethnic status, and either are or have been traditionally associated with a nomadic lifestyle, and include gypsy travellers, fairground or show people, circus families, New Age travellers and bargees. The term Gypsy, Traveller, Roma, Showmen and Boater (GTRSB) may be used as an umbrella term. See **Traveller education**.
HEI	Higher Education Institution.
HMCI	Her Majesty's Chief Inspector of Schools in England.
HMI	Her Majesty's Inspectors of Schools in England.
House system	A structure for pastoral care/pupil welfare within a school in which pupils are grouped in vertical units, i.e. sections of the school that include pupils from all year groups. Alternative to the **year system**.
IB	International Baccalaureate. A post-16 qualification designed for university entrance. See https://www.ibo.org/.
iQTS	International qualified teacher status (iQTS) is a teaching qualification that's the equivalent of English qualified teacher status (QTS). iQTS enables beginning teachers to train to teach in the country where they live and work, with no need to visit England.
ITTECF	Initial teacher training and early career framework (ITTECF) – A framework to support trainee and early career teacher development. See https://www.gov.uk/government/publications/initial-teacher-training-and-early-career-framework
In-class support	Support within a lesson provided by an additional teacher, often with expertise in teaching pupils with special educational, disability or language needs. See also **Learning support, Learning support assistant, Partnership teaching, Withdrawal**.
Inclusion	Inclusion involves the processes of increasing the participation of pupils in, and reducing their exclusion from, schools. Inclusion is concerned with the learning participation of all pupils vulnerable to exclusionary pressures, not only those with impairments or categorised as having special educational needs.
Independent school	A private school that receives no state assistance but is financed by fees. Often registered as a charity. See also **Public school**.
Induction	For teachers, the first stage of **continuing professional development** (**CPD**). A statutory requirement for Early Career Teachers (**ECTs**), formerly known as newly qualified teachers (**NQTs**) in England in the first two years of teaching (formerly one year). Successful completion of induction requires ECTs to meet standards set by the regulatory body in the country in which they wish to practice.

Intervention	Additional support put in place in schools to address wider learning needs, and ameliorate educational impact of disadvantage. Usually evidence-based and possibly linked to socioeconomic status indicated by eligibility for **Pupil Premium Grant**.
Ipsative assessment	A process in which performance is measured against previous performance by the same person. See also **Assessment, Assessment for learning, Assessment of learning, Criterion-referenced assessment, Formative assessment, Norm-referenced assessment and Summative assessment**.
ITE	Initial Teacher Education
ITT	Initial Teacher Training.
ITTE	Initial Teacher Training and Education.
Key Stages (KS) England	The periods in each pupil's education to which the elements of the **NC** for England apply. There are four **Key Stages**, normally related to the age of the majority of the pupils in a teaching group. They are: Key Stage 1, beginning of compulsory education to age 7 (Years R (Reception), 1 and 2); Key Stage 2, ages 7–11 (Years 3–6); Key Stage 3, ages 11–14 (years 7– 9); Key Stage 4, 14 to end of compulsory education (Years 10 and 11). Post-16 is a further Key Stage.
LA	see **Local Authority**
Learning objectives	What pupils are expected to have learned as a result of an activity, lesson or topic.
Learning outcomes	Assessable learning objectives: the action or behaviour of pupils that provides evidence that they have met the learning objectives.
Learning support	A means of providing extra help for pupils, usually those with learning difficulties, e.g. through a specialist teacher or specially designed materials. See also **Learning support assistants, In-class support, Partnership teaching, Withdrawal**.
Learning support assistants	Teachers who give additional support for a variety of purposes, e.g. general learning support for pupils with **SEND** or **ESL**; most support is given in-class although sometimes pupils are withdrawn from class. See also **Learning support, In-class support, Partnership teaching, Withdrawal**.
Lesson plan	The detailed planning of work to be undertaken in a lesson. This follows a particular structure, appropriate to the demands of a particular lesson. An individual lesson plan is usually part of a series of lessons in a **Unit or sequence of work**.
Local Authority (LA)	An LA has responsibility for local services, including education, libraries and social services. It has a statutory duty to provide education in their area. Many schools have opted out of LA control, see, e.g. **Academies** and **Free schools**.
Maintained boarding schools	State-funded schools that offer free tuition but charge fees for board and lodging. See also State-Maintained Schools in England and Other State Schools in England.
Middle school	A school that caters for pupils aged from 8 to 12 or 9 to 13 years of age. They are classified legally as either primary or secondary schools depending on whether the preponderance of pupils in the school is under or over 11 years of age.
Minority ethnic groups	Pupils, many of whom have been born in the UK, from other ethnic heritages, e.g. those of Asian heritage from Bangladesh, China, Pakistan, India or East Africa, those of African or Caribbean heritage or from countries in the European Union.
Mixed ability grouping	Teaching group containing pupils representative of the range of ability within the school. See also Banding, Setting, Streaming.

Moderation	An exercise involving teachers representing an awarding body external to the school whose purpose is to check that standards are comparable across schools and teachers. Usually carried out by sampling coursework or examination papers. Informal moderation between teachers of the same subject in the same school is common.
Moderator	An examiner who monitors marking and examining to ensure that standards are consistent in a number of schools and colleges.
Module	A definable section of work of fixed length with specific learning objectives and usually with some form of terminal assessment. Several such units may constitute a modular course.
Multi-Academy Trust	An academy chain or trust that operates more than one academy school, see, e.g. **Academies**.
NC (National Curriculum)	for some schools only in England. See https://www.gov.uk/national-curriculum. The core and other foundation subjects and their associated attainment targets, programmes of study and assessment arrangements of the curriculum
National Curriculum Tests	formerly Standard Assessment Tasks (q.v.).
National Qualifications	A framework for qualifications. GCSE qualifications are level 2; A Levels are Level 3. See https://www.gov.uk/what-different-qualification-levels-mean.
NEA	Non-exam assessment. See **Coursework**.
NEAB	See AQA.
NFER	National Foundation for Educational Research. Carries out research and produces educational diagnostic tests.
Non-contact time	Time provided by a school for a teacher to prepare work or carry out assigned responsibilities other than direct teaching. May be known as PPA: 'planning, preparation and assessment' time.
Norm-referenced assessment	A process in which performance is measured by comparing candidates' responses. Individual success is relative to the performance of all other candidates. See also Assessment, Assessment for learning, Assessment of learning, Criterion-referenced assessment, Formative assessment, Ipsative assessment, Summative assessment.
NQT	Newly qualified teacher. From 21 September, the induction period for an NQT doubled to two years. See also RQT and Early Career Framework. Now referred to as an Early Career Teacher (ECT).
OCR	Oxford and Cambridge awarding body.
OFQUAL	Set up in 2008. Regulator of examinations and tests in England, taking over that aspect of QCA. Independent (of DfE ministers) and responsible directly to parliament. https://www.gov.uk/government/organisations/ofqual.
OFSTED	Office for Standards in Education. Non-ministerial government department established under the Education (Schools) Act (1992) to take responsibility for the inspection of schools in England. OFSTED inspects pre-school provision, further education, teacher education institutions and local authorities. Her Majesty's Inspectors (HMI) form the professional arm of OFSTED. See also OHMCI.
OHMCI	Office of Her Majesty's Chief Inspector (Wales). Non-ministerial government department established under the Education (Schools) Act (1992) to take responsibility for the inspection of schools in Wales. Her Majesty's Inspectors (HMI) form the professional arm of OHMCI. See also OFSTED.

Other State Schools in England	These include academies, community and foundation special schools, church and faith schools (see Voluntary schools), free schools, grammar schools, maintained boarding schools, specialist schools and pupil referral units. See Directgov website. See also State-Maintained Schools in England.
Parent	Under section 576 of the Education Act 1996 a parent includes any person who is not a parent of the child but has parental responsibility (see Parental Responsibility), or who cares for the child.
Parental responsibility	Under section 2 of the Children Act 1989, parental responsibility falls upon: all mothers and fathers who were married to each other at the time of the child's birth (including those who have since separated or divorced); mothers who were not married to the father at the time of the child's birth; and fathers who were not married to the mother at the time of the child's birth, but who have obtained parental responsibility either by agreement with the child's mother or through a court order. Under section 12 of the Children Act 1989, where a court makes a residence order in favour of any person who is not the parent or guardian of the child that person has parental responsibility for the child while the residence order remains in force.
Partnership teaching	An increasingly common means of meeting the language needs of bilingual pupils in which support and class teachers plan and implement together a specially devised programme of in-class teaching and learning. See also **Learning support, Learning support assistants, In-class support, Withdrawal**.
Pastoral care	Those aspects of a school's work and structures concerned to promote the general welfare of all pupils, particularly their academic, personal and social development, their attendance and behaviour.
Pedagogical content knowledge (PCK)	The skills to transform subject knowledge into suitable learning activities for a particular group of pupils.
Peer assessment	Peer assessment is a process whereby pupils assess others' work based on a teacher's benchmarks. Self-assessment is where pupils do the same for their own work. The aim in both cases is to improve students' understanding and their metacognitive skills.
PGCE	Post Graduate Certificate in Education. The main qualification for secondary school teachers in England and Wales recognised by the **DfE** for **QTS**.
Policy	An agreed school statement relating to a particular area of its life and work.
Profile	Samples of work of pupils, used to illustrate progress, with or without added comments by teachers' and/or pupils.
Programme of study (PoS) for NC in England	The subject matter, skills and processes that must be taught to pupils during each **Key Stage** in order that they may meet the objectives set out in **attainment targets**. They provide the basis for planning **schemes of work**.
Progress 8	A secondary school accountability measure, based on progress across 8 qualifications. See also Attainment 8.
Progression	The planned development of pupils' knowledge, skills, understanding and attitudes over time. See also **Continuity**.
Project	An investigation with a particular focus undertaken by individuals or small groups of pupils leading to a written, oral or graphic presentation of the outcome.
PSE	Personal and social education.

PSHCE	**PSHE** with a specific, additional **citizenship** component.
PSHE	Personal, social, health and education, a non-statutory subject in the English National Curriculum. See also **PSHCE**.
PSHEE	is **PSHE** with a specific additional economic component.
PTA	Parent-teacher association. Voluntary grouping of parents and school staff to support the school in a variety of ways.
PTR	Pupil Teacher Ratio. The ratio of pupils to teachers within a school or group of schools (e.g. 17.4:1).
Postgraduate Teaching Apprenticeship (PGTA)	An initial teacher training (ITT) route that combines paid work with off-the-job training and leads to qualified teacher status (QTS).
Public school/ Independent school	Not state funded. So-called because at their inception they were funded by public charity.
Pupil Premium Grant	Additional funding for schools to use to improve education outcomes for eligible disadvantaged pupils in schools in England. Schools are accountable for the way the money is spent and strategies must be research-informed. See also **intervention**.
Pupil referral units (PRUs)	For children of compulsory school age who may otherwise not receive suitable education, focusing on getting them back into a mainstream school.
QCA	The Qualifications and Curriculum Authority. Initiated in 1997 by the merger of **SCAA** and National Council for Vocational Qualifications **(NCVQ)**. Was responsible for overview of the curriculum, assessment and qualifications across the whole of education and training, from pre-school to higher vocational levels. QCA advised the Secretary of State for education on such matters. Aspects of assessment were delegated by QCA to the **NAA**. Dissolved in 2010 the responsibilities shared between **QCDA** and **Ofqual**. The Welsh equivalent of QCA was ACCAC, later DCELLS.
QTS	Qualified teacher status. This is usually attained by completion of a Post Graduate Certificate in Education (**PGCE**) or a Bachelor of Education (**BEd**) degree or a Bachelor of Arts/Science degree with Qualified Teacher Status (**BA/BSc (QTS)**). There are other routes into teaching leading to QTS such as SCITT, School Direct.
RAISEonline	Closed 2017. Reporting and Analysis for Improvement through School Self-Evaluation. In England, it provided interactive analysis of school and pupil performance data. Was replaced by Analyse School Performance (**ASP**) but the term may be in legacy use.
Reliability	A measure of the consistency of the assessment or test item; i.e. the extent to which the test gives repeatable results. See also **Validity**.
RQT	A Recently Qualified Teacher or early career teacher. Typically, someone who has completed their induction. See also NQT, ECT and Early Career Framework.
SACRE	The Standing Advisory Council on Religious Education in each LA to advise the LA on matters connected with religious education and collective worship, particularly methods of teaching, the choice of teaching materials and the provision of teacher training.
SATs	See **Standard Assessment Tasks**.

Scheme of work	A planned course of study over a period of time (e.g. a **Key Stage** or a Year). In England, it contains knowledge, skills and processes derived from the **programmes of study** and **attainment targets** together with **units of work** and **lesson plans**.
SCITT	School-Centred Initial Teacher Training. A postgraduate route to becoming a qualified teacher in England. SCITT programmes are school-led rather than university-led, and emphasise practical, hands-on experience in the classroom from the start, leading to QTS with or without a PGCE.
School Improvement Plan (SIP)	A coherent plan, required to be made by a school, identifying improvements needed in curriculum, organisation, staffing and resources and setting out action needed to make those improvements.
SEG	See **AQA**.
SEN (Special Educational Needs) and Disabilities (SEND)	Children have special educational needs if they have a *learning difficulty*. Children have a *learning difficulty* if they: have a significantly greater difficulty in learning than the majority of children of the same age; or have a disability that prevents or hinders them from making use of educational facilities of a kind generally provided for children of the same age in schools within the area of the local authority; are under compulsory school age and fall within the definition at (a) or (b) above or would so do if special educational provision was not made for them. Very able or gifted pupils are not included in SEN. See also **SEND, SENCO**.
SENCO/SENDCO	Special Educational Needs Coordinator in schools. May also be known as SENDCO (Special Educational Needs and Disability Coordinator). See also **SEN, SEND**.
SEN Code of Practice	Act of Parliament describing and prescribing the regulations for the support of pupils with SEN. Chapter 6 relates to schools.
Setting	The grouping of pupils according to their ability in a subject for lessons in that subject. See also **banding**, **mixed ability grouping** and **streaming**.
Sixth Form College	A post-16 institution for 16–19-year-olds. It offers **GCSE, GCE A level** and **vocational courses**.
SLD/SpLD	Specific learning difficulties.
Special school	See **Community and Foundation special schools**.
Specialist schools	Previously a programme whereby a school could focus on a specific specialised area. They would still teach the whole curriculum but with a focus on one subject area such as arts; business and enterprise; engineering; humanities; language; mathematics and computing; music; science; sports; technology. The funded programme ended in 2010, but many schools retain the title. See **Other State Schools in England**.
STA (Standards and Testing Agency)	An executive agency of the **DfE** set up in 2011. It is responsible for the development and delivery of all statutory assessments from early years to the end of Key Stage 3, which formerly were carried out by **QCDA** (dissolved in March 2012).
Standard Assessment Tasks (SATs)	Externally prescribed **National Curriculum for England** assessments that incorporate a variety of assessment methods depending on the subject and **Key Stage**. The term SAT is not now widely used, having been replaced by '**National Curriculum Tests**', overseen by **STA**.
State-Maintained Schools	All children in England between the ages of 5 and 16 are entitled to a free place at a state school. State schools receive funding through their local authority or directly from the government. You may see the terms community school, voluntary school, academies, free schools, grammar schools, special schools. See https://www.gov.uk/types-of-school.

Statutory order	A statutory instrument which is regarded as an extension of an Act, enabling provisions of the Act to be augmented or updated.
Streaming	The organisation of pupils according to general ability into classes in which they are taught for all subjects and courses. See also **Banding**, **Mixed ability grouping**, **Setting**.
Summative assessment	Assessment linked to the end of a course of study; it sums up achievement in aggregate terms and is used to rank, grade or compare pupils, groups or schools. It uses a narrow range of methods which are efficient and reliable, normally formal, i.e. under examination conditions. See also **Assessment, Assessment for learning, Assessment of learning, Formative assessment, Ipsative assessment and Norm-referenced assessment**.
Supply teacher	Teachers appointed temporarily to fill vacancies, usually from an agency. Schools may employ their own internal cover supervisor to cover daily planned or unplanned teacher absences.
Support Teacher	See **In-class support** and **Learning support**.
Teacher Assessments	Assessments made by teachers alongside **National Curriculum Tests** at some **Key Stages** in England.
Teacher Degree Apprenticeship (TDA)	A new route into teaching. It offers a salaried route to become a qualified teacher and gain an undergraduate degree and QTS. See also Postgraduate Teaching Apprenticeship (PGTA).
TRA (Teaching Regulation Agency)	The Teaching Regulation Agency (TRA) is an executive agency of the Department for Education (DfE). The TRA was formally known as the National College for Teaching and Leadership (NCTL). They have responsibility for the regulation of the teaching profession, including misconduct hearings and the maintenance of a record of teachers, trainee teachers and those who hold a teacher reference number (TRN).
Teacher's record book	A book in which teachers plan and record teaching and learning for their classes on a regular basis. May be known as a teacher's planner.
Team teaching	The teaching of a number of classes simultaneously by teachers acting as a team. They usually divide the work between them, allowing those with particular expertise to lead different parts of the work, the others supporting the follow-up work with groups or individuals. See also **In-class support, Learning support, Partnership teaching**.
T Levels (Technical Levels)	2-year courses taken after GCSEs and are broadly equivalent in size to 3 A Levels. Launched in September 2020, these courses have been developed in collaboration with employers and education providers so that the content meets the needs of industry and prepares students for entry into skilled employment, an apprenticeship or related technical study through further or higher education https://www.gov.uk/government/publications/introduction-of-t-levels/introduction-of-t-levels.
Traveller education	The development of policy and provision that provides traveller children with unhindered access to and full integration in mainstream education. See **GTRSB**.
Trust schools	A **foundation school** supported by a charitable foundation or trust, which appoints school governors. A trust school employs its own staff, manages its own land and assets and sets its own admissions criteria.
Tutor group	Grouping of secondary pupils for registration and pastoral care purposes.

Unit of work	Medium-term planning of work for pupils over half a term or a number of weeks. The number of lessons in a unit of work may vary according to each school's organisation. A unit of work usually introduces a new aspect of learning. Units of work derive from **schemes of work** and are the basis for **lesson plans**.
Validity	A measure of whether the assessment measures what it is meant to measure – often determined by consensus. Certain kinds of skills and abilities are extremely difficult to assess with validity via simple pencil and paper tests. See also **Reliability**.
Vocational courses	Programmes of study leading to vocational qualifications that are work-related, preparing learners for employment. **Awarding Bodies** offer vocational courses.
Voluntary-aided school	Often religious schools. The governing body, often a religious organisation, employs the staff and sets admissions criteria. The school land and buildings are also owned by a charitable foundation. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Voluntary-controlled school	Mainly religious or 'faith' schools but run by the LA. The land and buildings are often owned by a charitable foundation, but the LA employs the staff and has primary responsibility for admission arrangements. See also **State-Maintained Schools in England** and **Other State Schools in England**.
Voluntary school	School that receives financial assistance from the **LA**, but owned by a voluntary body, usually religious. See **State-Maintained Schools in England** and **Other State Schools in England**.
Withdrawal	Removal of pupils with particular needs from class teaching in primary schools and from specified subjects in secondary schools for extra help individually or in small groups. In-class support is increasingly provided in preference to withdrawal. See also **Learning support, Learning support assistants, In-class support, Partnership teaching**.
WJEC	(Welsh Joint Education Committee) provides examinations, assessment, professional development, educational resources, support for adults who wish to learn Welsh and access to youth arts activities. It also provides examinations throughout England.
Work experience	The opportunity for secondary pupils to have short experience of a work environment, usually within school time, during which a pupil carries out a particular job or range of jobs more or less as would regular employees, although with emphasis on the educational aspects of the experience.
Year system	A structure for pastoral care/pupil welfare within a school in which pupils are grouped according to years, i.e. in groups spanning an age range of only one year. An alternative grouping is the **house system**.
Years 1–11	Year of schooling in England. Five-year-olds start at Year 1 (Y1) and progress through to Year 11 (Y11) at 16 years old. See **Key Stages** for details.

Appendix 2

Subject associations

Teachers' councils

England

The Teaching Regulation Agency (TRA) is responsible for regulating the teaching profession
https://www.gov.uk/government/organisations/teaching-regulation-agency

Northern Ireland

The GTCNI is the statutory body for the teaching profession in NI
http://www.gtcni.org.uk/

Scotland

The GTCS is the independent regulator for teachers in Scotland
http://www.gtcs.org.uk/

Wales

The Education Workforce Council is the professional regulator for the education workforce in Wales
https://www.ewc.wales/

Subject Associations

The Chartered College of Teaching is the professional body for teachers https://chartered.college
UK Council for Subject Associations
https://www.subjectassociations.org.uk/
The following associations are members of the UK Council for Subject Associations, which acts as a voice on behalf of member subject associations for government, the press and the public.
All the URLs here were checked in July 2025.

Art, Craft and Design	
AccessArt	http://www.accessart.org.uk/
National Society for Education in Art and Design (NSEAD)	http://www.nsead.org/
Citizenship	
Association for Citizenship Teaching (ACT)	https://www.teachingcitizenship.org.uk/

Classical Studies	
The Classical Association	http://www.classicalassociation.org/
Computing and Educational Technology	
Association for Learning Technology (ALT)	http://www.alt.ac.uk/
Computing At School (CAS)	http://www.computingatschool.org.uk/
Technology, Pedagogy and Education Association	http://www.tpea.ac.uk/
Dance	
One Dance UK	http://www.onedanceuk.org/
Design and Technology	
The Design and Technology Association	http://www.data.org.uk/
Drama and Theatre	
National Drama (ND)	http://www.nationaldrama.org.uk/
National Association for the Teaching of Drama	https://natd.co.uk/
English	
The English Association (EA)	https://englishassociation.ac.uk/
United Kingdom Literacy Association (UKLA)	http://www.ukla.org/
National Association of Writers in Education (NAWE)	http://www.nawe.co.uk/
Geography	
The Geographical Association (GA)	http://www.geography.org.uk/
History	
The Historical Association (HA)	http://www.history.org.uk/
Languages	
Association for Language Learning (ALL)	http://www.all-languages.org.uk/
The National Subject Association for EAL (NALDIC)	https://naldic.org.uk/
Mathematics	
Joint Mathematical Council of the United Kingdom	http://www.jmc.org.uk/
Music	
Independent Society of Musicians (ISM)	https://www.ism.org/
Music Mark	http://www.musicmark.org.uk/
Music Teachers Association	http://www.musicteachers.org/
Physical Education	
Association for Physical Education (afPE)	http://www.afpe.org.uk/
PSHE	
PSHE Association	https://pshe-association.org.uk/

Psychology	
Association for the Teaching of Psychology (ATP)	http://www.theatp.uk/
Religious Education	
National Association of Religious Education (NATRE)	http://www.natre.org.uk/
Science	
Association for Science Education (ASE)	http://www.ase.org.uk/
Early Years	
Early Education	http://www.early-education.org.uk/
National Primary Teacher Education Council	http://www.naptec.org.uk/
Associate Members	
LitdriveUK	http://www.litdrive.org.uk/
National Association for Able Children in Education	http://www.nace.co.uk/
Educational Visits Hub	http://www.evolveadvice.co.uk/

Appendix 3

List of websites

Providers of web resources for teachers do so for a range of purposes. Some sites are professional, such as subject associations sharing knowledge between professionals, others explicitly support government policy, so advice may be changed or withdrawn on ideological grounds, and others are designed to sell you products. The list below includes websites from:

- Professional Associations, Teaching Councils and Unions (see also Appendix 2);
- Charities and University Research Centres and social enterprises;
- Government-funded organisations and;
- Private companies.

For the most part, we have excluded websites apparently linked with just one individual. Exceptions are where the individuals have clearly researched and published widely in the area.

There are formal and informal networks on various social media sites.

You are advised to check the reliability of any advice – on the web or in print. For teaching to be an evidence-informed profession, teachers need to know the strength of evidence for any pedagogical intervention. By strength of evidence we mean:

- Methods and ethics: has the advice been gathered by ethical (see the BERA ethical code www.bera.ac.uk) and reliable research methods (see Unit 5.4 and Patterson's MESHGuides). Usual research instruments are interviews, questionnaires, documentary analysis and observation but there is huge variation of options within each instrument.
- Independence: were the researchers independent? Who funded the research? Were researchers free to publish adverse findings?
- Quality assurance: has the advice been independently peer-reviewed? Peer review, by an independent panel of educators, is the normal form of quality assurance used for professional association and professional journal sites. Materials from other sites may or may not be peer reviewed.
- Sample: what is the size and type of the sample used to provide the evidence? What confidence does this give you in the results?

- Transferability: how transferable is the advice likely to be? How similar is the research context to your context? This is not at all to say you reject research and evidence from contexts different to your own, but just that you need to bring your professional judgement to bear in applying the findings. Teachers in many countries face similar challenges in maximising the learning of young people and there is a lot to learn from solutions elsewhere.

All websites listed here were accessed in July 2025. A list is available on the companion website. Most of the units in the book will have links to websites within them, especially in the further resources sections.

Further information is given only where it is not obvious what the website offers.

The list starts with generic websites followed by a list of sites grouped alphabetically by theme, e.g. Behaviour, Neuroscience, Subject Associations, Unions.

Generic websites (covering a wide range of areas)

- Repository of British Education Research Association conferences research papers: www.bera.ac.uk
- EEF (Education Endowment Foundation): https://educationendowmentfoundation.org.uk/
- ERIC – USA government Education Resources Information Center: http://eric.ed.gov/
- European SchoolNet: www.eun.org
- Evidence for Policy and Practice Information Centre: https://eppi.ioe.ac.uk/cms/ This has a list of systematic reviews of practice in education.
- Khan Academy – Teaching videos: www.khanacademy.org/
- MESHGuides (Mapping Education Specialist knowhow): www.meshguides.org
- National STEM Centre: https://www.stem.org.uk/resources
- Oak National Academy: https://www.thenational.academy/
- Open University, OpenLearn Create: https://www.open.edu/openlearncreate/
- Seneca Learning (www.senecalearning.com) free homework and revision platform
- Stanford University Teaching Commons – Resources tab: https://teachingcommons.stanford.edu/
- TED-Ed Lessons Worth Sharing: http://ed.ted.com/
- Times Educational Supplement: https://www.tes.co.uk/
- YouTube: Teachers: https://www.youtube.com/teachers

UK Government

- **England:** Department for Education: www.education.gov.uk/ and inspection: Office for Standards in Education (Ofsted): www.ofsted.gov.uk
- **Northern Ireland:** Department of Education Northern Ireland: https://www.northernireland.gov.uk/ and Inspection: The Education and Training Inspectorate Northern Ireland: https://www.etini.gov.uk/
- **Scotland:** Education Scotland: https://education.gov.scot/ and Education Scotland Inspection Reports: https://education.gov.scot/education-scotland/inspection-reports/
- **Wales:** Welsh Government: https://www.gov.wales/education-skills/ and Inspection: Estyn inspects education and training in Wales: https://www.estyn.gov.wales/

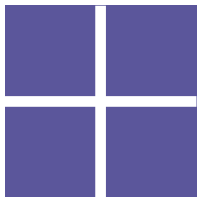
Additional websites by theme

a	
Assessment for learning:	■ see specialist MESHGuides www.meshguides.org and, for example, Wiliam, D.: https://www.dylanwiliam.org/
Autism:	■ National Autistic Society: https://www.autism.org.uk/
b	
Behaviour:	See DFE advice on behaviour and discipline: ■ Behaviour in Schools: Advice for headteachers and school staff https://www.gov,uk/government/publications/behaviour-in-schools--2 ■ Improving behaviour in schools https://www.gov.uk/government/collections/improving-behaviour-in-schools
c	
Careers	– See statutory guidance ■ DFE: https://www.gov.uk/government/publications/careers-guidance-provision-for-young-people-in-schools/careers-guidance-and-access-for-education-and-training-providers ■ National Careers Service: https://nationalcareers.service.gov.uk/
Citizenship:	■ British Humanist Association: https://humanists.uk/education/ ■ Young citizens: https://www.youngcitizens.org/ ■ Jubilee Centre for Character and Virtues: https://www.jubileecentre.ac.uk/432/character-education ■ Oxfam: https://www.oxfam.org.uk/education/who-we-are/global-citizenship-guides/ ■ Equal and Human Rights Commission (EHRC): www.equalityhumanrights.com ■ Joseph Rowntree Foundation (focus: poverty and injustice): www.jrf.org.uk
Code of practice for teaching	– See Teaching Councils
Curriculum – National requirements:	■ England: https://www.gov.uk/government/publications/national-curriculum-in-england-framework-for-key-stages-1-to-4/the-national-curriculum-in-england-framework-for-key-stages-1-to-4 ■ Northern Ireland: http://www.nicurriculum.org.uk/ (Note: Major curriculum reform underway – see Strategic Review June 2025: https://www.education-ni.gov.uk/articles/review-northern-ireland-curriculum) ■ Scotland: https://www.gov.scot/policies/schools/school-curriculum/ (see also: https://scotlandscurriculum.scot/) ■ Wales: https://hwb.gov.wales/curriculum-for-wales
d	
Debating	■ The English Speaking Union (https://www.esu.org)
Deaf and Hearing Impaired:	■ National Deaf Children's Society: www.ndcs.org.uk; ■ British Association of Teachers of the Deaf (BATOD) https://www.batod.org.uk/
Dialogic teaching:	■ Robin Alexander http://www.robinalexander.org.uk/dialogic-teaching/ ■ University of Cambridge http://www.educ.cam.ac.uk/research/projects/dialogic/whatis.html

E	
English as an additional language:	■ The Bell Foundation: https://www.bell-foundation.org.uk/ ■ Flynn, N., Pim, C. and Coles, S. (2015) Teaching English as an Additional Language MESHGuide. University of Winchester, UK: http://www.meshguides.org/guides/node/112
Ethics:	■ Professional – See Teaching Councils ■ Research – See BERA
Europe:	■ European SchoolNet (EUN): www.eun.org
g	
Gifted and Talented:	■ National Association for Able Children in Education www.nace.co.uk ■ UK Parliament briefing: https://commonslibrary.parliament.uk/research-briefings/cbp-9065/ ■ Potential Plus UK: https://potentialplusuk.org/
h	
Handwriting:	National Handwriting Association: www.nha-handwriting.org.uk
Health:	■ Healthy Schools: https://www.healthyschools.org.uk/ ■ British Nutrition Foundation: www.nutrition.org.uk ■ Food Standards Agency: https://www.food.gov.uk/ ■ NICE (National Institute for Health and Care Excellence) https://www.nice.org.uk/guidance ■ A Germ's Journey https://www.germsjourney.com/
I	
Lesson Study:	■ Dudley, P. https://lessonstudy.co.uk/
Learning theories	See the specialist sites in this list: ■ Claxton, G (2015) Building Learning Power http://www.buildinglearningpower.co.uk/ ■ Sage 20 learning theorists: https://us.sagepub.com/sites/default/files/learning_theorists_flashcards_25.pdf ■ Bob Bates: https://www.book2look.com/book/vvotT7TnIe ■ Gardner, H. https://www.multipleintelligencesoasis.org/resources
n	
Names (remembering):	■ TeacherVision: https://www.teachervision.com/teaching-methods/classroom-management/6708.html
Neuroscience:	■ Centre for Neuroscience in Education led by Professor Usha Goswami https://www.cne.psychol.cam.ac.uk/ ■ Centre for Educational Neuroscience (CEN) https://www.educationalneuroscience.org.uk/ ■ Neuroscience for Education https://www.neuroscienceforeducation.com/research
m	
MESHGuides:	www.meshguides.org: research summaries to support evidence-informed teaching.

p	
Philosophy:	■ Philosophy of Education, https://www.philosophy-of-education.org/for-teaching/ ■ Philosophy for Children https://www.sapere.org.uk/
Projects:	■ Collaborative projects across Europe – E-twinning: https://school-education.ec.europa.eu/en/etwinning ■ Apps for Good: https://www.appsforgood.org/
q	
Qualification levels	https://www.gov.uk/what-different-qualification-levels-mean
r	
Risk:	■ CLEAPSS http://www.cleapss.org.uk ■ NEU Risk Assessments: https://neu.org.uk/advice/health-and-safety/health-and-safety-schools-and-colleges/risk-assessment ■ Health and Safety Executive https://www.hse.gov.uk/risk/classroom-checklist.htm
Research	■ British Educational Research Association (BERA): http://www.bera.ac.uk ■ Education Endowment Foundation (EEF) https://educationendowmentfoundation.org.uk/ ■ MESHGuides www.meshguides.org
Research ethics:	■ British Educational Research Association (BERA) Ethics: https://www.bera.ac.uk/publication/ethical-guidelines-for-educational-research-fifth-edition-2024
s	
Safer Internet Day:	■ https://www.saferinternetday.org
SEND:	■ see also deaf, dyslexia: NASEN (National Association for Special Educational Needs): www.nasen.org.uk/ ■ Special Educational Needs and Disability: https://meshagain.meshguides.org/category/send/ ■ The Professional Association of Teachers of Students with Specific Learning Difficulties (PATOSS): www.patoss-dyslexia.org ■ Royal National Institute for the Blind: www.rnib.org.uk ■ ADHD Foundation https://www.adhdfoundation.org.uk/ ■ MindMate https://mindmate.org.uk/neurodiversity-information-hub/ ■ Makaton https://makaton.org/ ■ Anna Freud https://www.annafreud.org/ ■ The Noun Project https://thenounproject.com/
Subject Associations:	Member subject associations are represented by the Council for Subject Associations: www.subjectassociations.org.uk
t	
Teacher Support:	■ https://www.educationsupport.org.uk/ (mental health and wellbeing of teachers)

Teacher Standards:	■ England: ■ https://www.gov.uk/government/publications/teachers-standards ■ Initial teacher training and early career framework (ITTECF) https://www.gov.uk/government/publications/initial-teacher-training-and-early-career-framework ■ Northern Ireland: https://www.education-ni.gov.uk/topics/teaching-staff ■ Scotland: https://www.gtcs.org.uk/knowledge-base/articles/professional-standards-and-professional-learning ■ Wales: https://gov.wales/teaching-and-leadership
Thinking	– See also Dialogic teaching: ■ Teaching Thinking Skills: https://education.gov.scot/resources/a-framework-for-thinking-skills/ ■ University of Cambridge Thinking Together: http://thinkingtogether.educ.cam.ac.uk/
Transitions:	EEF https://educationendowmentfoundation.org.uk/news/eef-blog-supporting-pupils-through-transitions-a-trio-of-challenges
u	
UK Council for Child Safety (UKCCIS):	https://www.gov.uk/government/groups/uk-council-for-child-internet-safety-ukccis
Unions:	■ National Education Union https://neu.org.uk/ ■ National Association of Schoolmasters/Union of Women Teachers, England, Wales, Scotland and Northern Ireland: NASUWT www.nasuwt.org.uk/ ■ The Educational Institute of Scotland (EIS), Scotland: www.eis.org.uk/ ■ Ulster Teachers Union, Northern Ireland: www.utu.edu/ ■ Undeb Cenedlaethol Athrawon Cymru (UCAC) https://www.ucac.cymru/index.php/en/ ■ Irish National Teachers' Organisation, Northern Ireland: www.into.ie/NI/ ■ Scottish Secondary Teachers' Association, Scotland: www.ssta.org.uk ■ Voice, previously the Professional Association of Teachers, England, Wales and Northern Ireland: www.voicetheunion.org.uk/



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